
MARROW ED8

Obstetrics & Gynaecology

Comprehensive Question Bank

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Anatomy of Female Pelvic Organs

Question 1:

Which of the following is not a part of the vulva?

- a) Labia majora
- b) Vestibule
- c) Greater vestibular glands
- d) Cervix

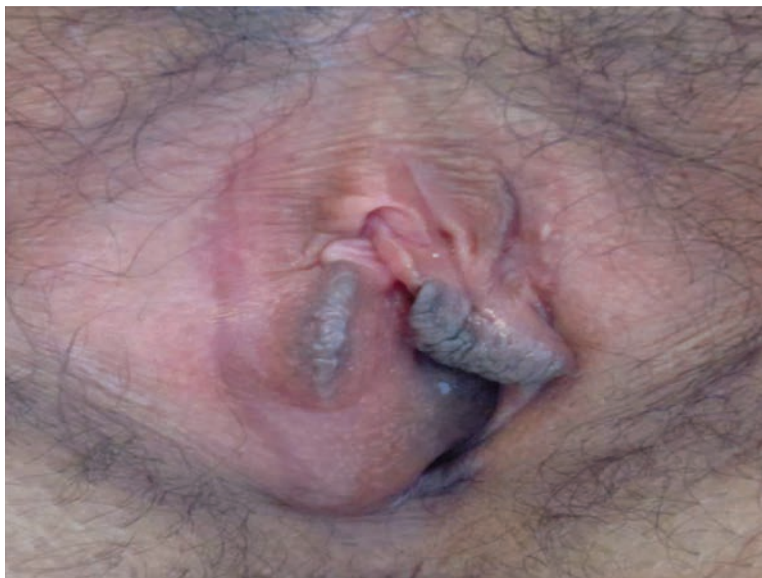
Question 2:

Which of the following statements is false regarding labia majora?

- a) They join posteriorly to form the posterior commissure
- b) It is homologous to scrotum in male
- c) The round ligament terminates at its posterior end
- d) It contains sebaceous glands and apocrine glands

Question 3:

A patient presented to the gynecology OPD with the following finding. Which structure in men is homologous to the affected structure?



- a) Prostate gland
- b) Cowper's gland
- c) Glands of Littre
- d) Meibomian glands

Question 4:

What is the pH of the vagina before puberty?

- a) 7
- b) 4.5
- c) 6
- d) 4.2

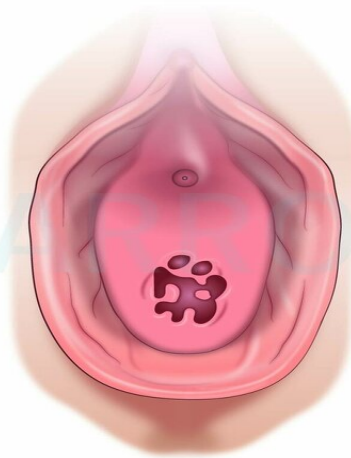
Question 5:

Which of the following is incorrect about relations of the vagina?

- a) Bladder - anterior
- b) Bartholin's gland - posterior
- c) Levator ani - lateral
- d) Pouch of Douglas - posterior

Question 6:

Identify the type of hymen.



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- a) Cribriform hymen
- b) Parous hymen
- c) Fimbriated hymen
- d) Septate hymen

Question 7:

A patient with uterine prolapse and urethrocele has weakness in the superficial and deep perineal muscles. What level of support is lost here?

- a) Level 1
- b) Level 2
- c) Level 3
- d) Level 4

Question 8:

Anteflexion of the uterus is at the level of ____.

- a) External os
- b) Internal os
- c) Upper part of vagina
- d) Lower part of vagina

Question 9:

A G3P2L2 woman presents in her third trimester with vaginal leaking. On PV examination, the anterior lip of the cervix can be palpated. What is the position of her uterus and the ligament that maintains this position?

- a) Anteverted uterus - Round ligament
- b) Anteflexed uterus - Round ligament
- c) Anteverted uterus - Uterosacral ligament
- d) Anteflexed uterus - Broad ligament

Question 10:

A multiparous woman presented with a urethrocele, due to weakness of the urogenital diaphragm. What is this structure also known as?

- a) Rectangular ligament
- b) Triangular ligament
- c) Pyramidal ligament
- d) Pear shaped ligament

Question 11:

What is the cavity of the uterus lined by?

- a) Single layer of ciliated columnar epithelium
- b) Double layer of ciliated columnar epithelium
- c) Single layer of ciliated cuboidal epithelium
- d) Double layer of ciliated cuboidal epithelium

Question 12:

A woman with regular menstrual cycles is undergoing an ultrasound on the 14th day of her cycle. What will the endometrial thickness be?

- a) 0.5 mm
- b) 4 mm
- c) 11 mm

d) 14 mm

Question 13:

Which of the following statements is false regarding uterine anatomy?

- a) Round ligaments receive their blood supply from Sampson's artery
- b) Transverse cervical ligaments are also called cardinal ligaments
- c) Uterine innervation is by the pudendal nerve plexus
- d) Lymphatic drainage is primarily to the obturator, internal and external iliac nodes.

Question 14:

A gynecologist is performing a hysterectomy. She accidentally nicks the ureter in the ureteric canal as it pierces the Mackenrodt's ligaments. Which vessels are likely to be damaged here?

- a) Uterine artery above and the uterine plexus of veins below
- b) Uterine plexus of veins above and the uterine artery below
- c) Ovarian artery above and the uterine artery below
- d) Uterine artery above and the ovarian artery below

Question 15:

You take a culture swab from the portio vaginalis of a woman having foul smelling vaginal discharge. What is this region also known as?

- a) Supra vaginal part of cervix
- b) Vaginal part of cervix
- c) Fornix of the vagina
- d) Introitus of the vagina

Question 16:

What is the cervix: corpus ratio before puberty?

- a) 1:2
- b) 2:1

- c) 1:3
- d) 3:1

Question 17:

The remnants of Wolffian ducts in females are located in_____.

- a) Pouch of Douglas
- b) Leaves of broad ligament
- c) Uterovesical pouch
- d) Iliac fossa

Question 18:

Which of the following is false about the contents of the broad ligament?

- a) Ovarian vessels in its infundibulopelvic part
- b) It contains the uterine arteries
- c) Ureter is present the lower portion
- d) Ovarian ligament in its anterior fold

Question 19:

Where are Peg cells seen?

- a) Vagina
- b) Vulva
- c) Ovary
- d) Fallopian tubes

Question 20:

What is the longest part of the fallopian tube?

- a) Interstitium
- b) Isthmus
- c) Ampulla

d) Infundibulum

Question 21:

Where is the tunica albuginea of the adult ovary derived from?

- a) The surface epithelium
- b) The sub epithelial connective tissue layer
- c) The parenchymatous zone
- d) Zona vasculosa

Question 22:

During the histopathological examination of an ovarian mass, Call-Exner bodies are seen. In which of the following layers are these structures seen?

- a) Granulosa cell layer
- b) Theca interna
- c) Discus proligerus
- d) Cumulus oophorus

Question 23:

Where is the sympathetic nerve supply of ovary derived from?

- a) T10-T11
- b) T9-T10
- c) S2-S4
- d) L4-L5

Question 24:

Bleeding during normal menstruation occurs due to rupture of ____

- a) Basilar artery
- b) Spiral artery
- c) Arcuate artery

d) Radial artery

Question 25:

A uterine artery embolization is being planned for a patient with uterine fibroids. Blood flow in which of the following arteries will not be affected by this procedure?

- a) Spiral artery
- b) Sampson's artery
- c) Vaginal artery
- d) Arcuate artery

Question 26:

A patient with stress incontinence required surgical management. While performing a Burch colposuspension, the surgeon approached the space of Retzius. What is this space also known as?

- a) Prevesical or presacral space
- b) Presacral or prerectal space
- c) Presacral or retropubic space
- d) Prevesical or retropubic space

Answer Key

Question No.	Correct Option
1	d
2	c
3	b
4	a
5	b
6	a
7	c
8	b
9	a

10	b
11	a
12	c
13	c
14	a
15	b
16	b
17	b
18	d
19	d
20	c
21	b
22	a
23	a
24	b
25	c
26	d

Detailed Explanations

Solution to Question 1:

Cervix is not a part of the vulva.

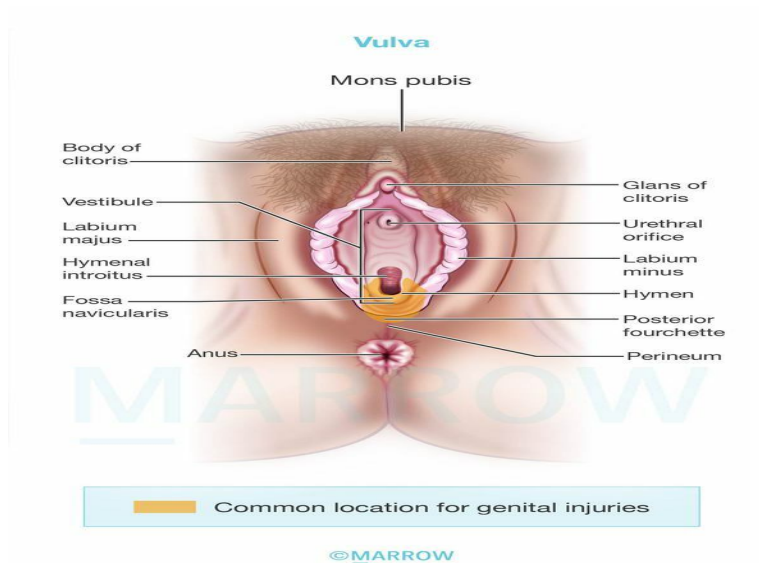
The vulva includes:

- Mons veneris
- Labia majora
- Labia minora
- Clitoris
- Vestibule
- Vestibular bulb
- Greater vestibular glands.

The vulva is posteriorly related to the perineum.

The triangular space bounded by clitoris, labia minora and fourchette is the vestibule. It is present between the labia minora anteriorly and the introitus posteriorly. The posterior part of the vestibule between the fourchette and the vaginal introitus is called fossa navicularis. The vestibule has 4 openings:

- Urethral opening
- Vaginal orifice opening
- Bartholin's duct on either side
- Ducts of paraurethral glands



Solution to Question 2:

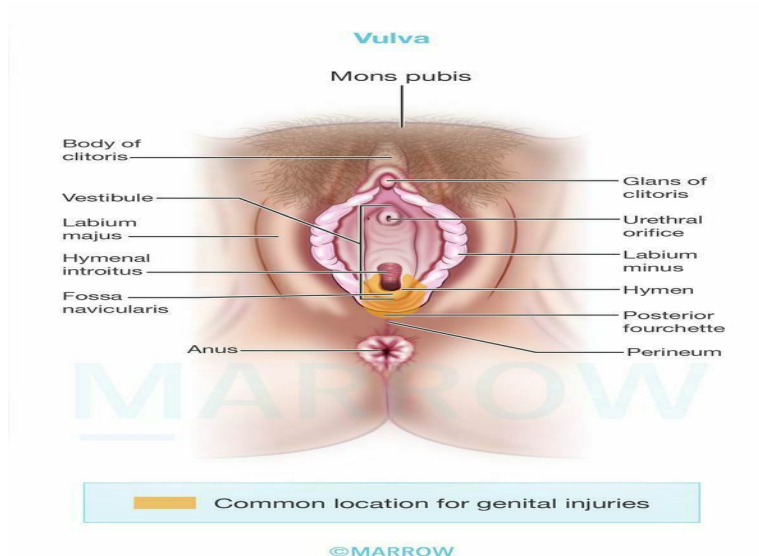
The round ligament terminates at the anterior end of the labia majora.

The labia majora lie on either side of the vulva and join posteriorly to form the posterior commissure. The inner side is hairless. Labia majora are homologous to the scrotum in males. The labia majora and the mons veneris contain:

- Hair follicles
- Sebaceous glands
- Modified sweat glands, known as apocrine glands

The labia minora are two cutaneous folds between the labia majora. Anteriorly they enclose the clitoris and unite with each other in front and behind the clitoris to form prepuce and frenulum respectively. The labia minora are homologous to the ventral aspect of the penis.

The clitoris is a small erectile body lying in the anterior-most part of the vulva. The clitoris is homologous to glans penis.

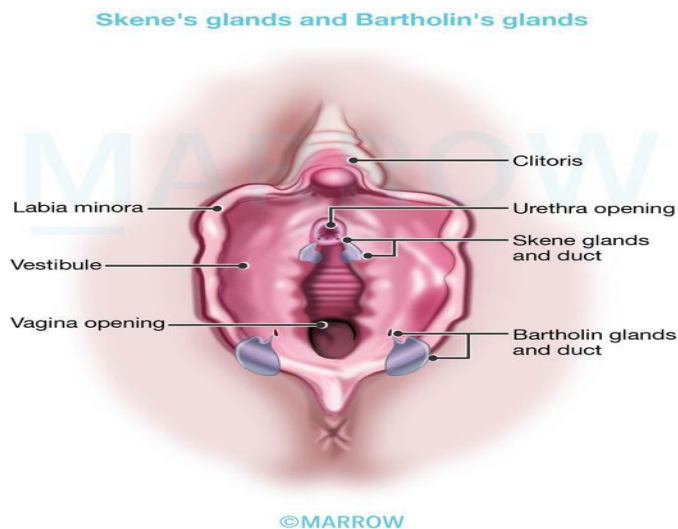


Solution to Question 3:

The given image shows a Bartholin's cyst, which involves the Bartholin's glands. These glands are homologous to the male bulbourethral or Cowper glands.

The ducts of Bartholin's glands open in the groove between the hymen and the labia minora at approximately 5 and 7 o'clock positions. They are stimulated by sexual arousal. The glands contain columnar cells that secrete mucus with lubricant properties. Contraction of the bulbospongiosus muscle expresses gland secretions.

Bartholin's cyst occurs due to obstruction of the ducts of the Bartholin's glands by proteinaceous material, or due to infection. Symptomatic or recurrent cysts may require marsupialization or gland excision. An infected Bartholin's cyst can lead to an Bartholin's abscess. This requires surgical drainage.



Solution to Question 4:

The pH of the vagina before puberty is 7. In adult females, the pH is acidic.

Doderlein's bacilli (*Lactobacillus* sp.) are part of the normal vaginal flora. They produce lactic acid from glycogen in the vaginal epithelium and make the vaginal pH acidic. They are present in newborns and disappear in 10-14 days. They reappear at puberty and disappear after menopause. Vaginal flora also includes microbes such as *Clostridium welchii*.

Vaginal defence mechanisms are influenced by estrogen. In newborns, maternal estrogens maintain the defence mechanisms for 10-14 days. After that, the levels wane. During puberty, the circulating estrogen maintains the defence mechanism.

Note: The pH is higher in the upper part of the vagina due to mixing with alkaline cervical mucus.

Period	Vaginal pH	Doderlein bacteria	Types of cells
Birth-2 weeks	4-5	+	-
2weeks-puberty	>7	-	Parabasal cells
Puberty	4-4.5	sparse	Superficial & intermediate
Reproductive	4-5	plenty	Superficial & intermediate
Pregnancy	3.8-4.1	plenty	Intermediate
Postmenopause	>7	-	Parabasal

Solution to Question 5:

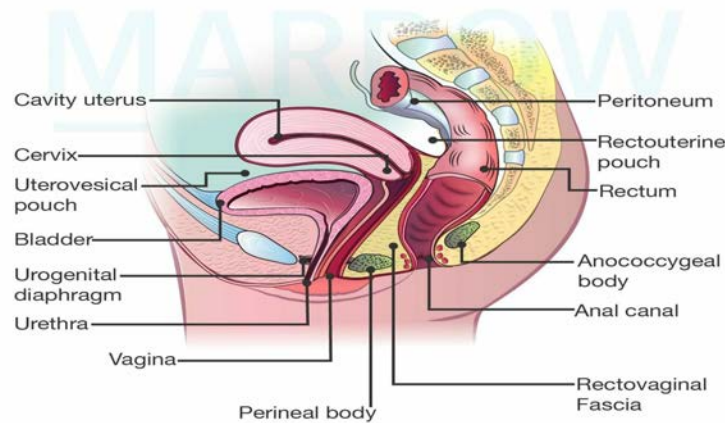
Bartholin's gland is present lateral to the vagina.

The vagina is related to the following structures:

- Anteriorly:
 - Bladder (upper 1/3rd)
 - Urethra (lower 2/3rd)
- Posteriorly
 - Pouch of Douglas
 - Ampulla of the rectum
- Perineal body
- Laterally

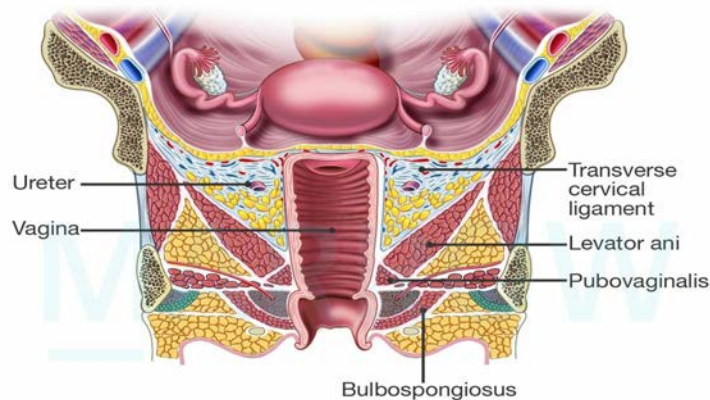
- Mackenrodt's ligament
- Levator ani
- Bartholin's glands

Anterior and posterior relations of vagina



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Lateral relations of the Vagina



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Solution to Question 6:

The given image shows cribriform hymen because it has multiple small openings.

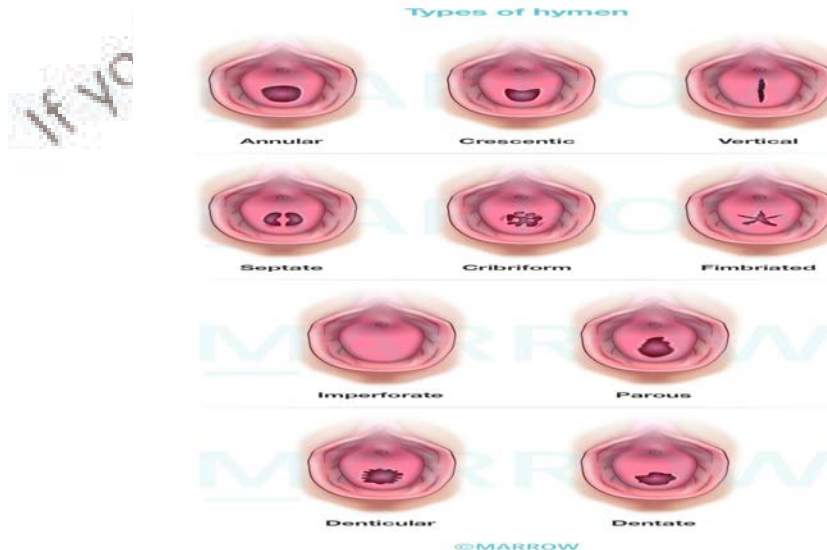
Based on the shape of its opening, the hymen may be classified as:

Other variants of the hymen:

The hymen is lined by stratified squamous epithelium on both sides. The hymen is most commonly torn posterolaterally or posteriorly. It is replaced by tags after childbirth which are called as carunculae myrtiformes.

Hymen	Description
Annular	Oval central opening
Crescentic	Semilunar central opening
Vertical	Vertical slit-like opening
Septate	2 lateral openings partitioned by a bridge of hymenal tissue
Cribriform	Multiple small openings

Variants	Description
Absence of hymen	Congenital deformity.
Infantile	Small central linear opening.
Imperforate	No opening
Fimbriated	Free margin has symmetrical indentations anteriorly
Ring or fringe-like	A narrow ring of thick membrane is present all around the vaginal orifice. It can easily distend during sexual intercourse without rupturing the hymen.



Solution to Question 7:

In the given case scenario, level 3 supports including the superficial and deep perineal muscles are lost. This causes urethrocele.

DeLancey's levels of support are used to describe the supports of the genital organs.

Solution to Question 8:

Anteflexion of the uterus is seen at the level of the internal os.

(Mnemonic - V for version, V for vagina)

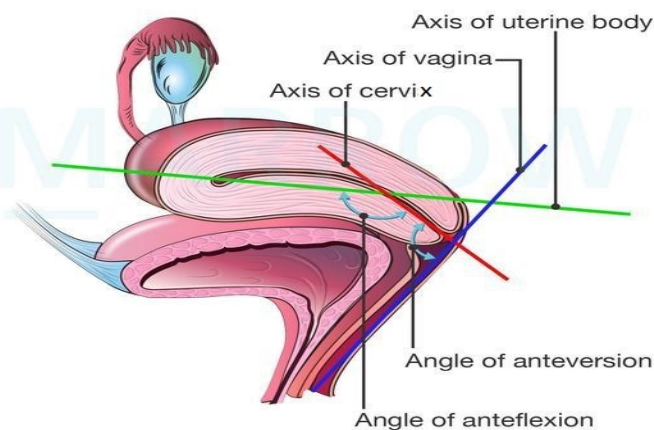
The uterus normally lies in a position of anteversion and anteflexion.

In the anteverted position, the external os is directed downward and backwards. On vaginal examination, the fingers come in contact with the anterior lip of the cervix as it is the lowest part.

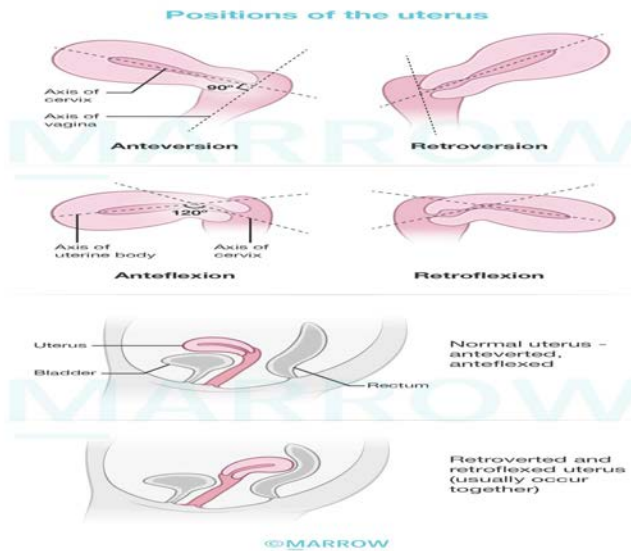
When the uterus is retroverted, the cervix is directed downward and forwards. The lowest part of the cervix becomes either the external os or its posterior lip.

Angle of Anteversion	Angle of Anteflexion
The angle between the cervix and vagina	The angle between cervix and uterus
90°	120°

Anteversion and anteflexion of the uterus



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Solution to Question 9:

In the given clinical scenario, the uterus is anteverted as the anterior lip of the cervix was palpable. The round ligament helps maintain the anteversion of the uterus during pregnancy.

However, the anteversion in the non-pregnant uterus is maintained by both the uterosacral ligament and round ligament.

(Mnemonic - V for version, V for vagina)

The uterus normally lies in a position of anteversion and antelexion.

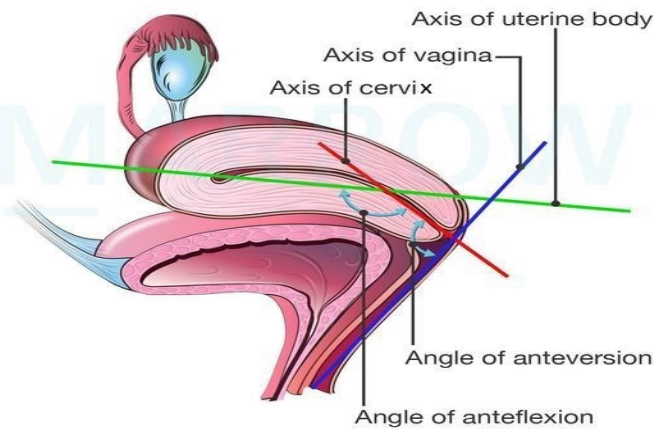
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When the uterus is retroverted, the cervix is directed downward and forwards. The lowest part of the cervix becomes either the external os or its posterior lip.

Antelexion is seen at the level of the internal os.

Angle of anteversion	Angle of antelexion
Angle between the cervix and vagina	Angle between cervix and uterus
90°	120°

Anteversion and ante flexion of the uterus



Solution to Question 10:

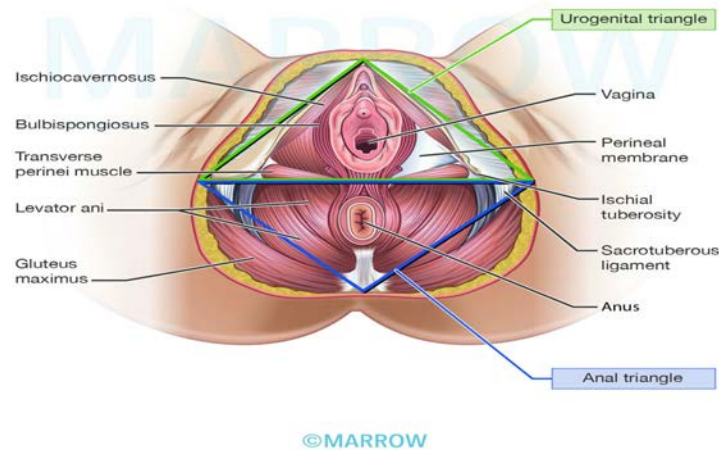
The urogenital diaphragm is also known as the triangular ligament.

The urogenital diaphragm extends from the pubic arch anteriorly to the central point of the perineum posteriorly and consists of two layers of fascia through which pass the vagina and the urethra. The central point of the female perineum lies between the vagina and the rectum.

Urogenital triangle:

- Present between 2 ischiopubic rami
- Stretching transversely across the rami there are 3 membranes.
- From above downwards the membranes are as follows:
 - Superior fascia of urogenital diaphragm
 - Inferior fascia of urogenital diaphragm
 - Membranous layer of superficial perineal fascia

Perineum - urogenital and anal triangles



Solution to Question 11:

The uterine cavity is lined by a single layer of ciliated columnar epithelium.

The wall of the uterus consists of 3 layers:

- Perimetrium is the serous coat adherent to the underlying muscle.
- Myometrium - it consists of thick bundles of muscles that form 3 distinct layers during pregnancy:
 - Outer longitudinal
 - Inner circular
 - Middle interlacing called living ligature
- Endometrium - It is the mucous lining of the cavity. It consists of lamina propria and surface epithelium which is a single layer of ciliated columnar epithelium.

Solution to Question 12:

During ovulation, the thickness of the endometrium is around 11 mm.

Phase	Endometrial thickness
Immediately after menstruation	1-2 mm
Early proliferative phase	5-7 mm
Secretory phase	7-16 mm

Phase	Endometrial thickness
Preovulatory phase	up to 11 mm
Postmenopausal	<4 mm
Postmenopausal - abnormal, requires biopsy	>4 mm

Solution to Question 13:

The uterus is innervated by the fibers of the uterovaginal plexus, also known as the Frankenhauser ganglion.

Option A: Round ligaments receive their blood supply from Sampson's artery which represents the anastomosis of the uterine artery and ovarian artery.

Option B: The cardinal ligaments are also called transverse cervical ligaments or Mackenrodt ligaments. They consist primarily of perivascular connective tissue. They attach to the posterolateral pelvic walls near the origin of the internal iliac artery and surround the vessels supplying the uterus and vagina.

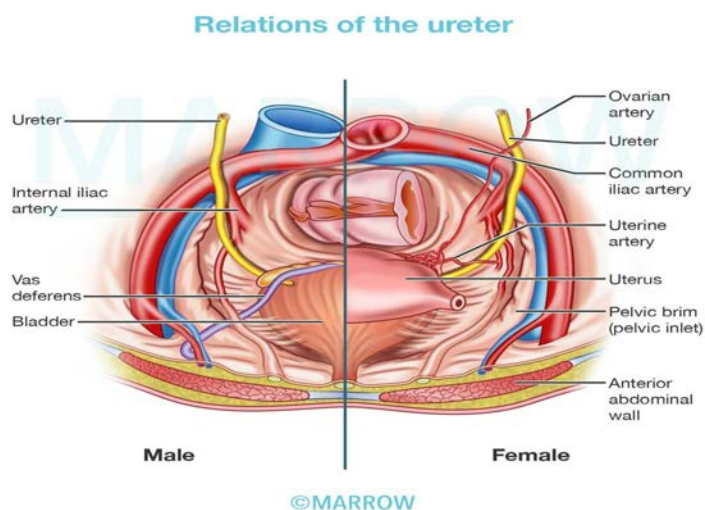
Option D: Lymphatic drainage of the uterus is primarily to the obturator and internal and external iliac nodes. One theory of endometriosis, states lymphatic spread as cause for seeding in the pelvis.

Solution to Question 14:

The uterine artery above and uterine venous plexus below are likely to be damaged here, as the ureter is present between these vessels.

The ureter pierces the Mackenrodt's ligament, forming the ureteric canal. In this canal, the ureter is crossed by the uterine artery above and the uterine plexus of veins below, thus being present in between the uterine vessels.

The ureteric canal provides the necessary room for normal peristalsis, without pressure from the surrounding structures. After leaving the ureteric canal, the ureter passes forwards and medially to reach the bladder. It is 1-2 cm away from the cervix.



Solution to Question 15:

Portio vaginalis or exocervix is the other name for the vaginal part of the cervix.

Cervix is the lowermost part of the uterus extending from the histological internal os to the external os. It is cylindrical in shape measuring 2.5 cm in length and diameter.

Cervix is made up of 10% muscle and the rest is collagen, elastin, and proteoglycan. The lining of the cervix has the following parts:

- Exocervix - also known as portio vaginalis. It is lined by non keratinized stratified squamous epithelium.
- Endocervix - it is lined by high columnar, ciliated epithelium.
- Endocervical glands - secrete alkaline mucus. When enlarged, they form swellings called Nabothian follicles/cysts.
- Transformation zone - the epithelium gradually changes from columnar to squamous in this region.

Solution to Question 16:

The cervix:corpus ratio before puberty is 2:1.

The relationship between the length of the cervix and that of the body of the uterus varies with age.

Age	Cervix/Corpus ratio
Before puberty	2:1
At puberty	1:2

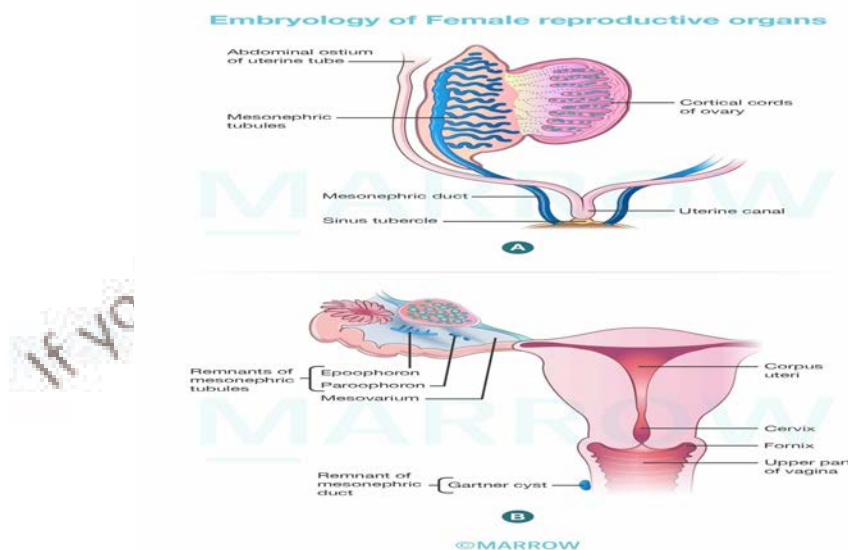
Age	Cervix/Corpus ratio
Reproductive age	1:3 or 1:4
After menopause	2:1 - however, the uterus and cervix are both atrophied

Solution to Question 17:

The Gartner's duct is a remnant of the Wolffian ducts in women. These are located in the broad ligament and the vagina.

The Gartner's duct is an embryological remnant of the mesonephric duct/Wolffian duct formed during the development of the urinary and reproductive organs. It runs parallel to, but below, the fallopian tube in the mesosalpinx. The duct passes downwards by the side of the uterus to the level of the internal os where it passes into the tissues of the cervix. It then runs forwards to reach the anterolateral aspect of the vaginal wall and may reach as far down as the hymen.

The duct can sometimes form a cyst in the broad ligament or in the vagina. It is called Gartner's cyst. It is a benign condition and can be treated with marsupialization.



Solution to Question 18:

The ovarian ligaments are contained in the posterior fold of the broad ligament.

The broad ligament contains the following parts:

- Mesosalpinx - it lies between the fallopian tube and ovarian ligament. It contains the pelvic vessels - anastomoses between uterine and ovarian arteries. The ovary is attached via ovarian ligament through the posterior folds to the mesovarium.

- Mesometrium - it contains the ovarian vessels within the infundibulopelvic ligament.

The ureter is present in the inferomedial part of the broad ligament. It is related to the uterine artery here and may be damaged during hysterectomy.

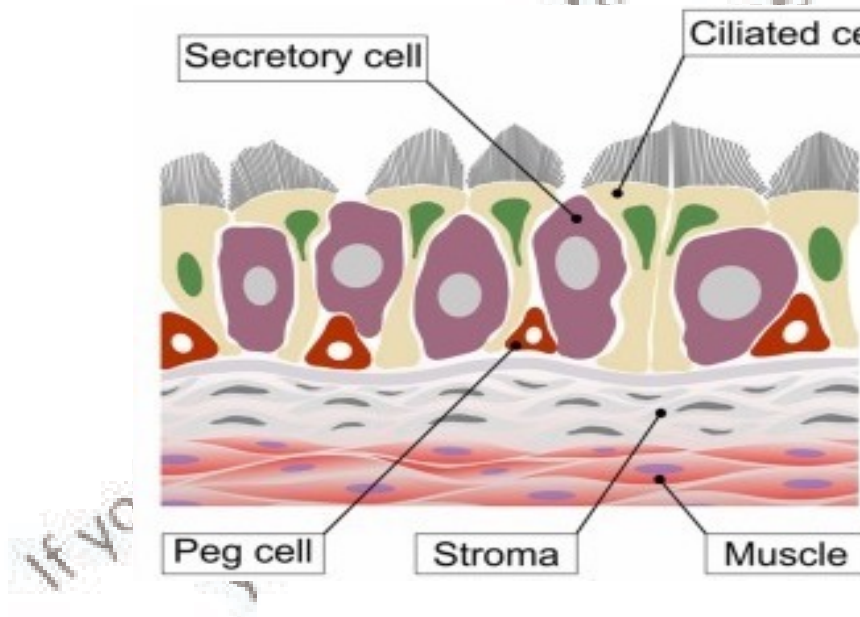
Note: Broad ligament is not a support of the uterus.

Solution to Question 19:

Peg cells are seen in the fallopian tubes.

A peg cell is a non-ciliated epithelial cell within the fallopian tube. These cells are also known as intercalated or intercalary cells whose function is not yet known.

The image below shows pale ciliated cells (arrows) and narrow, dark, non-ciliated peg cells (arrowheads).



Solution to Question 20:

The longest and widest part of the fallopian tube is the ampulla (5 cm).

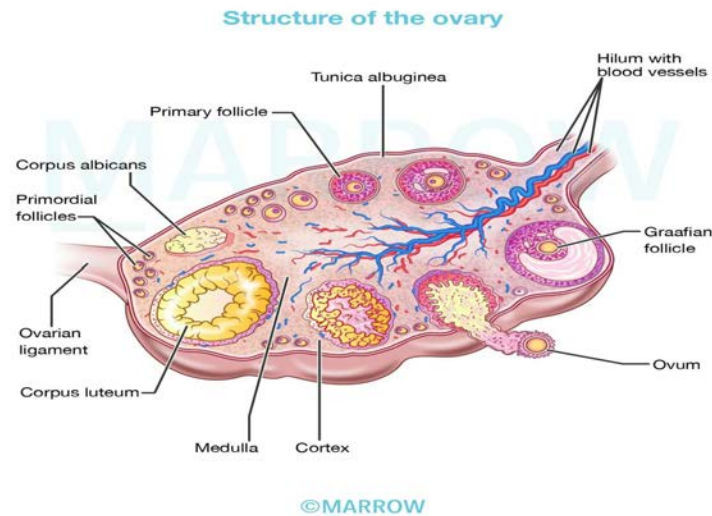
Solution to Question 21:

Tunica albuginea of the adult ovary is derived from the sub-epithelial connective tissue layer.

Structure of the ovary:

- Surface epithelium - a single layer of cuboidal cells is present, which later gives rise to the surface epithelium of the adult ovary. It is morphologically continuous with the mesothelium of the peritoneum.

- The subepithelial connective tissue layer - gives rise to the tunica albuginea of the adult ovary and the basement membrane beneath the surface epithelium.
- The parenchymatous zone - becomes the cortex and contains the egg nests and ovarian follicles.
- Zona vasculosa - becomes the medulla and contains the blood vessels. Hilar cells present here are homologous to interstitial cells of testes. They can rarely produce hilar cell tumors of the ovary.

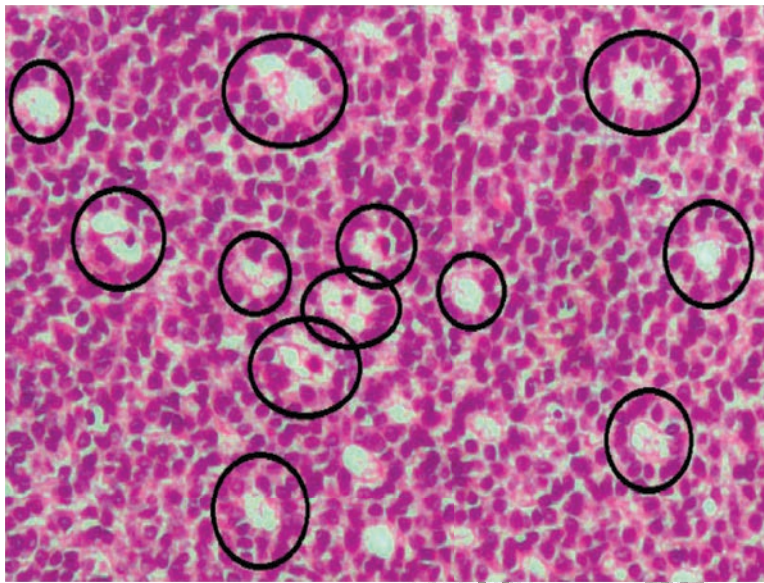


Solution to Question 22:

Call-Exner bodies are formed by granulosa cells present within the granulosa cell layer of the ovary.

They are small, distinctive, glandlike structures filled with acidophilic material, resembling immature follicles. They are characteristically seen in granulosa cell tumors of the ovary.

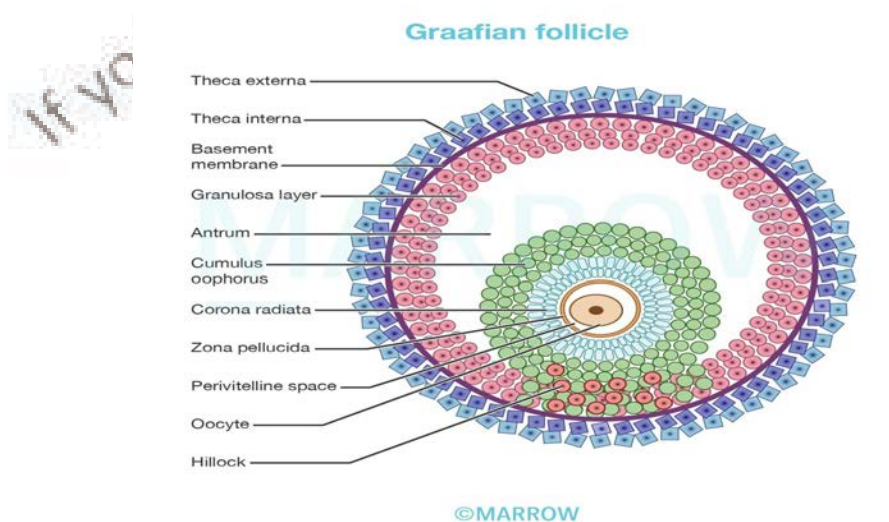
The image below shows Call-Exner bodies (circled in black).



Graafian follicle:

The mature Graafian follicle is spheroidal or ovoid in shape. It contains pent-up secretion known as the liquor folliculi. The lining consists of:

- Theca externa
- Theca interna - contain the theca cells
- Granulosa cell layer
- Cumulus oophorus - projection of granulosa cells into the cavity of the Graafian follicle. It is also called the discus proligerus.



Solution to Question 23:

The efferent preganglionic sympathetic fibres to the ovary are derived from the T10-T11 spinal levels.

These fibres travel to the prevertebral ganglia, where they synapse. The postganglionic sympathetic fibres arising from these ganglia contribute to the ovarian plexus, which innervates the ovary.

The ovarian plexus is a network of nerves comprising:

- Postganglionic sympathetic fibres
- Preganglionic parasympathetic fibres from the sacral outflow (S2-S4)
- Visceral afferent fibres (responsible for transmitting sensory signals)

The ovarian plexus has two parts based on its connections:

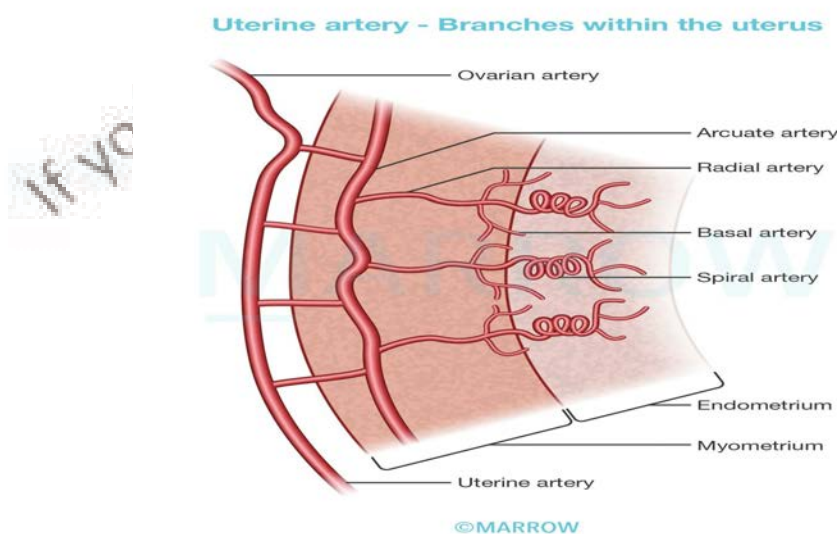
- Upper part: Derived from branches of the renal plexus and aortic plexus
- Lower part: Derived from the superior and inferior hypogastric plexuses.

Solution to Question 24:

Rupture of the spiral arteries causes bleeding during normal menstruation. This is a branch of the uterine artery.

The blood supply of the uterus is as follows:

Uterine artery → Arcuate artery → Radial artery → Basal artery → Spiral artery



Solution to Question 25:

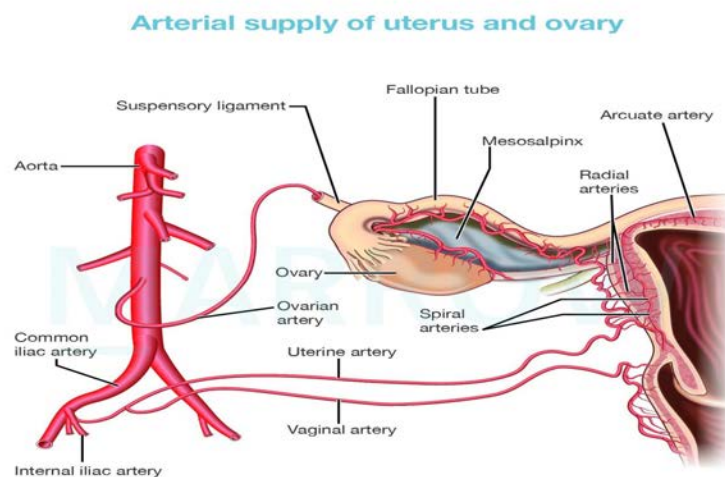
In the given case scenario, blood flow in the vaginal artery will not be affected, as it is not a branch of the uterine artery. The vaginal artery is a branch of the anterior division of the internal

iliac artery.

It is homologous to the inferior vesical artery in males. It supplies the vagina and the base of the bladder in women. It should not be confused with vaginal branches of the uterine artery.

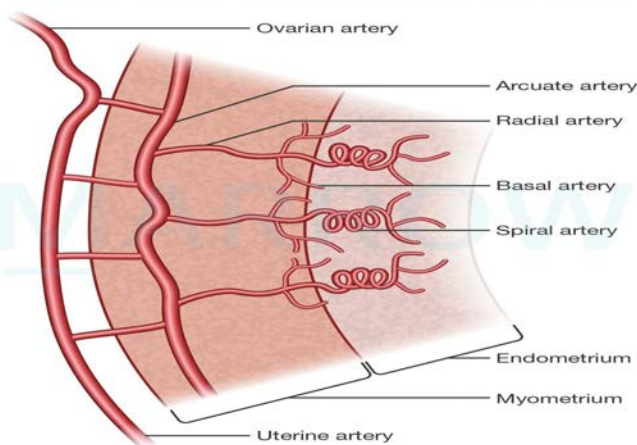
Branches of the uterine artery:

- Ureteric branch
- Vaginal branches (descending) - these unite to form the anterior and posterior azygos arteries of the vagina
- Circular cervical branch
- Sampson's artery - it runs in under the round ligament. It represents the anastomosis of the uterine artery and ovarian artery.
- Arcuate--& Radial--& Basal--& Spiral and straight arterioles of the functional layer of the endometrium



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Uterine artery - Branches within the uterus



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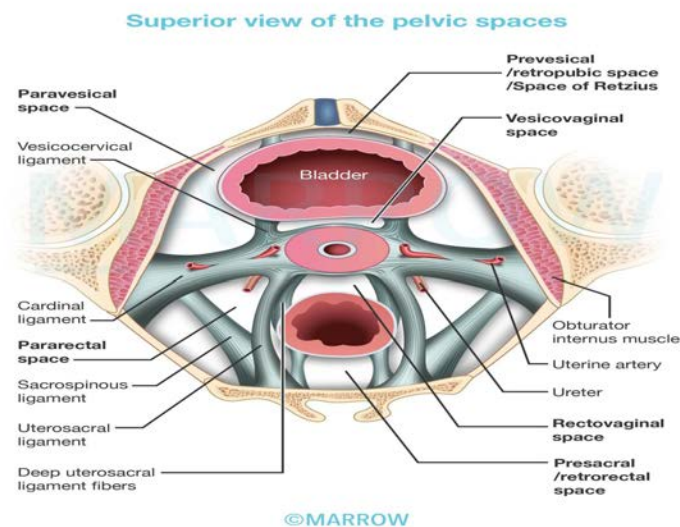
Solution to Question 26:

The space of Retzius is the other name for prevesical or retropubic space.

Burch colposuspension is a procedure done to relieve stress urinary incontinence. If an extraperitoneal approach is selected, the retropubic space is entered by perforating the transversalis fascia layer of the anterior abdominal wall.

Once the retropubic space is dissected, the bladder neck and the vaginal fascia underneath are identified. The bladder is mobilized to the opposite side. At each side of the bladder neck, two to four, preferably nonabsorbable sutures are placed in the space of Retzius.

Note: The presacral space is a retroperitoneal space that lies between the peritoneum of the posterior abdominal wall and the sacrum. It contains the superior and inferior hypogastric plexuses. The sacral sympathetic trunk lies against the sacrum's ventral surface.



The Physiology of Conception

Question 1:

When do germ cells appear in the yolk sac?

- a) 3 weeks
- b) 7 weeks
- c) 2 weeks
- d) 5 weeks

Question 2:

When does the ovary contain the maximum number of oogonia?

- a) 5th month of intrauterine life
- b) 7th month of intrauterine life
- c) At birth
- d) 3rd month of intrauterine life

Question 3:

A man with oligospermia has completed his course of hormonal treatment. After how many days will you test his sperm count?

- a) 50-60 days
- b) 60-70 days
- c) 80-90 days
- d) 70-80 days

Question 4:

What is the time taken for capacitation of the sperm?

- a) 2-4 hours
- b) 4-6 hours

- c) 6-8 hours
- d) 8-10 hours

Question 5:

Which of the following statements is true about the resumption of meiosis-I?

- a) It is not hormone dependent
- b) Resumed due to LH
- c) Resumed due to FSH
- d) Resumed due to GnRH

Question 6:

In which of the following steps does reductional division occur?

- a) Primary to secondary spermatocyte
- b) Secondary spermatocyte to spermatid
- c) Germ cells to spermatogonium
- d) Spermatogonium to primary spermatocyte

Question 7:

Which of the following is incorrect about the primary oocyte?

- a) It is formed after meiosis I
- b) It forms the primordial follicle
- c) It gets arrested in diplotene
- d) It is diploid

Question 8:

An embryologist is preparing embryos for implantation. When is zona hatching expected to occur?

- a) 2 days after fertilization
- b) 5 days after fertilization

- c) 7 days after fertilization
- d) 9 days after fertilization

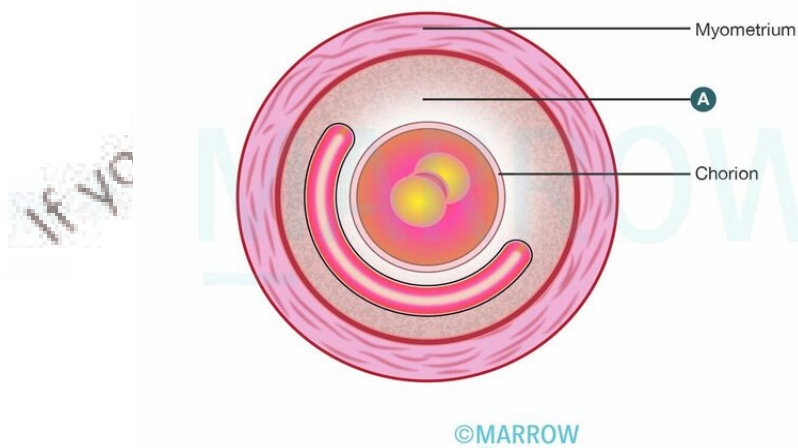
Question 9:

You are performing intrauterine insemination during the ovulatory period in a patient with infertility. When will you expect implantation to occur?

- a) 6–10 days after fertilization
- b) 3–5 days after fertilization
- c) 14–19 days after fertilization
- d) 1–2 days after fertilization

Question 10:

The following figure shows the transverse section of the uterus with an implanted zygote. Identify structure A.



- a) Decidua basalis
- b) Decidua parietalis
- c) Decidua capsularis
- d) Uterine cavity

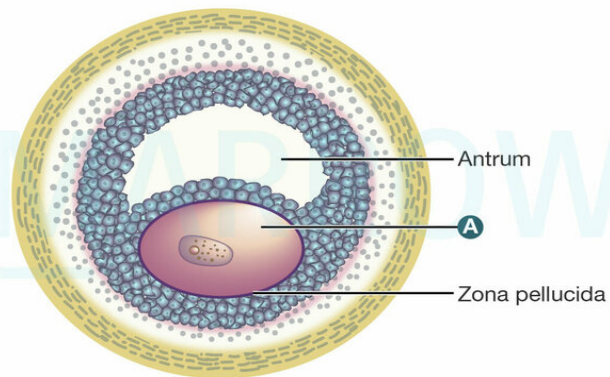
Question 11:

What do the Langhans cells form?

- a) Inner cell mass
- b) Morula
- c) Cytotrophoblast
- d) Syncytiotrophoblast

Question 12:

Identify the structure marked A.



- a) Primordial follicle
- b) Primary oocyte
- c) Theca interna
- d) Granulosa cells

Question 13:

In which part of the fallopian tube does fertilization occur?

- a) Ampulla
- b) Isthmus
- c) Interstitial
- d) Infundibular

Question 14:

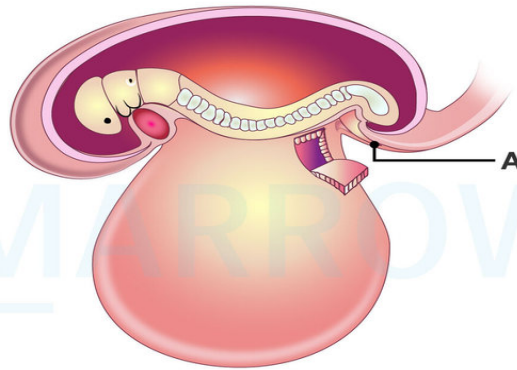
A newborn is admitted to the NICU with the following teratoma. Which embryonic layers is this derived from?



- a) Endoderm and ectoderm
- b) Endoderm and mesoderm
- c) Mesoderm and ectoderm
- d) Endoderm, mesoderm, ectoderm

Question 15:

Identify the structure marked as A.



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- a) Yolk sac
- b) Allantois
- c) Future umbilical cord
- d) Tail end

Answer Key

Question No.	Correct Option
1	a
2	a
3	c
4	c
5	b
6	a
7	a
8	b
9	a
10	a
11	c
12	b
13	a
14	d

Detailed Explanations

Solution to Question 1:

Germ cells appear in the yolk sac in the 3rd week.

The sequence of events in primordial germ cell formation and migration are:

- 2nd week - Germ cells are formed in the epiblast.
- 3rd week - Germ cells move through the primitive streak and migrate to the wall of the yolk sac.
- 4th week - Germ cells begin to migrate from the yolk sac to gonads.
- 5th week - Germ cell migration is complete.

Gametes are derived from these primordial germ cells.

Solution to Question 2:

The ovary contains the maximum number of oogonia at the 5th month of intrauterine life. The number of oogonia is estimated to be 7 million.

Oogenesis is the process wherein oogonia differentiate into mature oocytes. It begins before birth. At birth, there are approximately 2 million primordial follicles. At the initiation of puberty, the follicle count is less than 400,000, of which less than 500 are destined to ovulate. Every month, 15–20 follicles are released, out of which one matures to form the ovum.

Solution to Question 3:

The sperm count is tested 80-90 days after hormonal treatment, as the average time is taken for spermatogenesis is 74 days, and approximately 14 days to reach the ejaculatory ducts.

The effect of hormonal therapy on the early phase of spermatogenesis is reflected in the ejaculated sperm concentration after three months.

Spermatogenesis is the process of the formation of sperms from spermatogonia. Approximately 100 million sperms are produced daily.

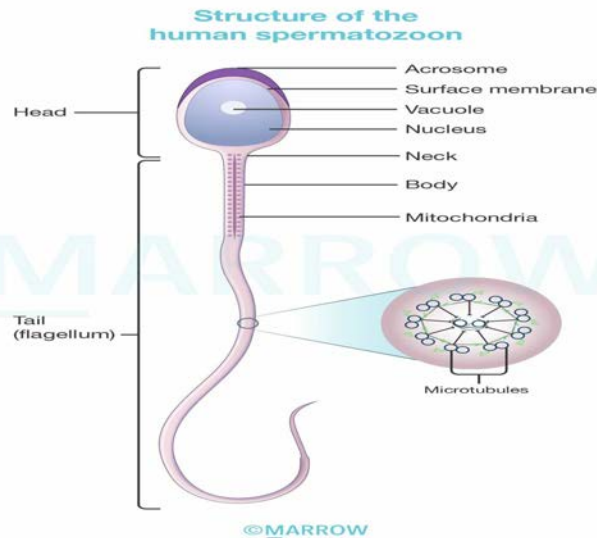
In patients diagnosed with oligospermia, hormonal treatment is carried out using FSH, hCG, or testosterone. Improvements in sperm count can take a few months, hence the course is usually for 3-6 months. Normal spermatogenesis occurs after this time period.

Spermiogenesis refers to the series of changes that occur in the formation of spermatozoa from spermatids. These are:

- Acrosome formation

- Condensation of the nucleus
- Formation of the neck, middle piece, and tail
- Shedding of cytoplasm as residual bodies

Complete motility is obtained in the epididymis.



Solution to Question 4:

The time taken for the capacitation of sperms is 6-8 hours (approximately 7 hours).

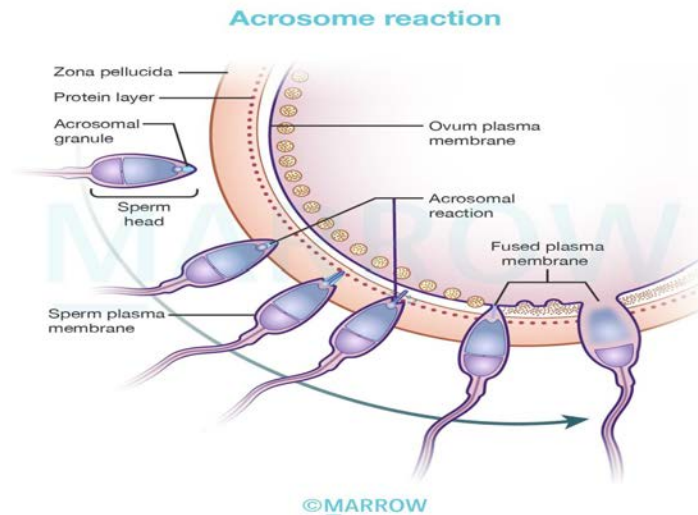
Sperms have to undergo two changes in the female reproductive tract to be able to fertilise the ovum:

1) Capacitation

- It is the functional maturation of the spermatozoa.
- It occurs in the fallopian tube and involves epithelial interactions between the sperm and the mucosal surface of the tube.
- The glycoprotein coat and seminal plasma proteins are removed.
- Only capacitated sperms can pass through the corona cells and undergo the acrosomal reaction.

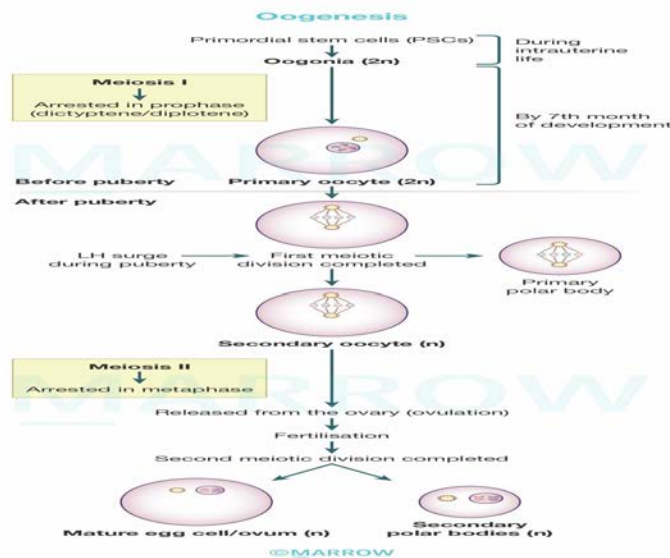
2) Acrosomal reaction

- It occurs after binding to zona pellucida, induced by zona proteins.
- Enzymes such as acrosin and trypsin-like protein are released to penetrate the zona pellucida
- Ca^{2+} plays an important role.



Solution to Question 5:

Meiosis-I in the primary oocyte is resumed after LH surge.

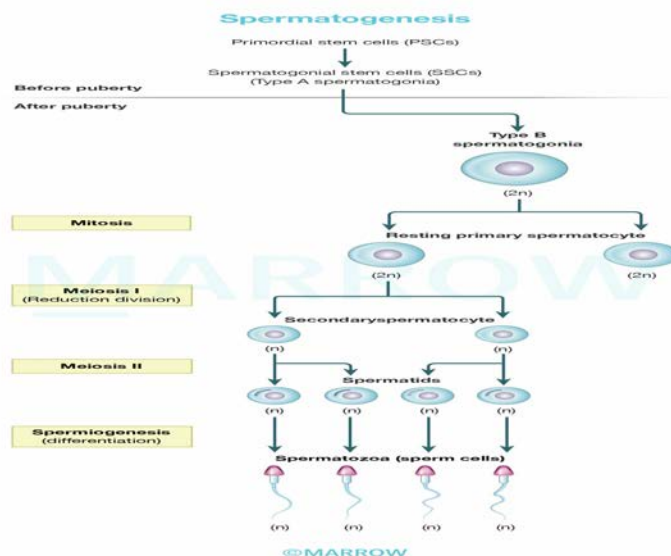


Solution to Question 6:

Meiosis I or reductional division occurs in the transition from primary spermatocyte to secondary spermatocyte.

Meiosis takes place in germ cells to generate male and female gametes. Meiosis involves two stages of cell division, namely, meiosis I and II.

Meiosis I produces two haploid cells. Because the ploidy is reduced from diploid to haploid, meiosis I is referred to as a reductional division. Meiosis II is an equational division analogous to mitosis, resulting in the creation of four haploid daughter cells.

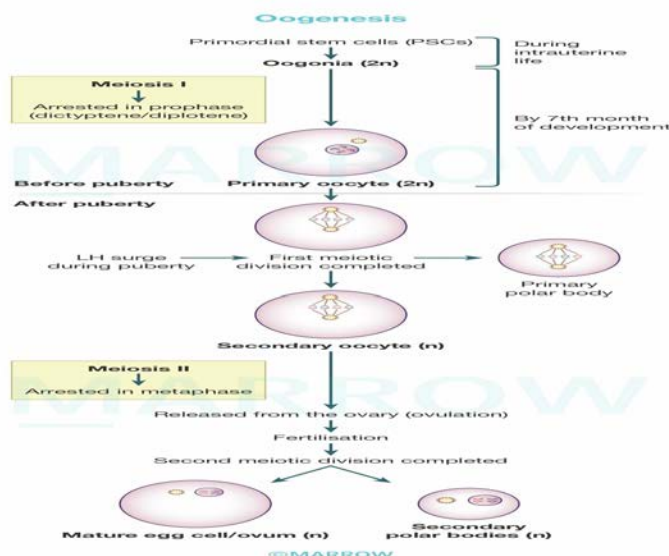


Solution to Question 7:

The secondary oocyte is formed after meiosis I.

Primordial germ cells divide by mitosis to form oogonia. The majority of the oogonia divide by mitosis, however, some enter meiosis I and are arrested in the diplotene stage of prophase and are now called primary oocytes. Primary oocytes have not yet completed meiosis, hence they are diploid. The primary oocyte along with surrounding epithelial cells forms the primordial follicle.

Meiosis I is completed after LH surge during puberty. Hence at the completion of meiosis I, secondary oocyte is produced.



Solution to Question 8:

Zona hatching occurs at around 5 days after fertilization.

Fertilization is divided into the following phases:

- Penetration of the corona radiata
- Penetration of the zona pellucida
- Fusion of the oocyte and sperm cell membranes

After fertilisation, the zygote differentiates to form the morula and then the blastocyst. The zona pellucida disappears at this stage, which is called zona hatching. The trophoblasts (outer cell mass) are now able to associate with the endometrium during implantation.

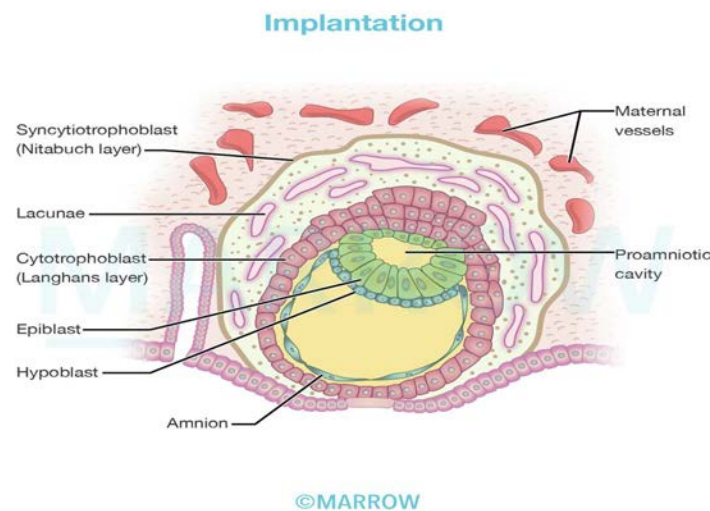
Solution to Question 9:

The fertilized ovum implants by 6–10 days in the uterine cavity.

Implantation involves:

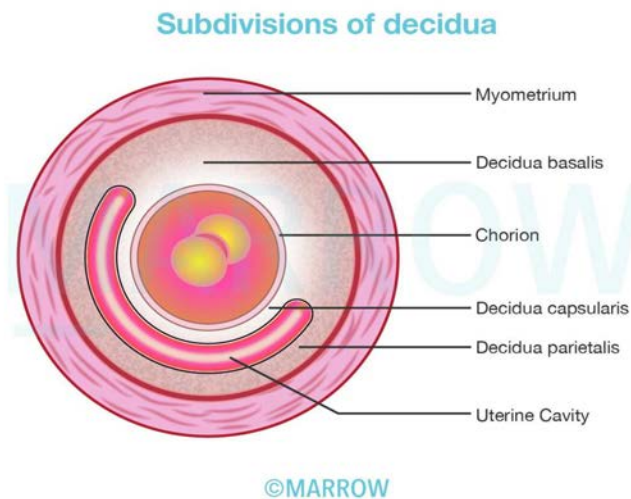
- Apposition: Initial contact of the blastocyst to the uterine wall
- Adhesion: Increased physical contact between the blastocyst and uterine epithelium
- Invasion: Penetration and invasion of syncytiotrophoblast and cytotrophoblasts into the endometrium

A successful implantation requires receptive endometrium primed with estrogen and progesterone released by the corpus luteum.



Solution to Question 10:

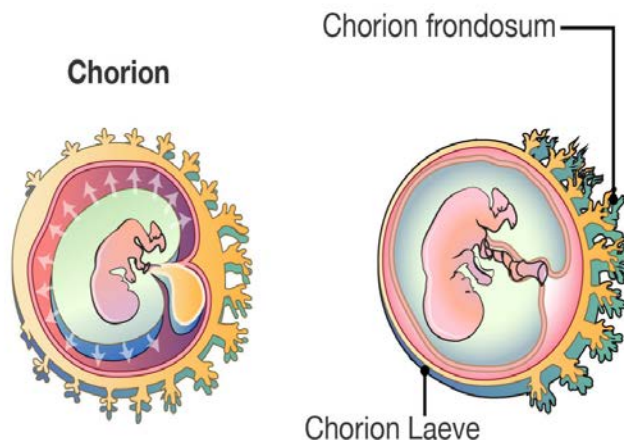
In the given image, the decidua basalis is marked as A. This part of the decidua forms the placenta.



Implantation occurs 6 to 7 days after fertilization. After implantation, the endometrium is called the decidua. It is of the following types:

- Decidua basalis - present over chorion frondosum. Decidua basalis and chorion frondosum form the placenta.
- Decidua capsularis - present over abembryonic pole.
- Decidua parietalis - present on the opposite side of the uterus.

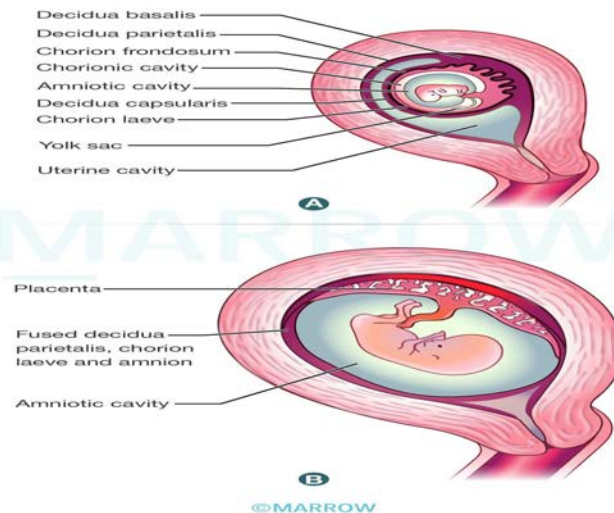
The villi cover the entire surface of the chorion during the early weeks. As the fetus grows, the villi on the embryonic pole continue to grow, giving rise to the chorion frondosum. At the abembryonic pole, the villi degenerate. By the third month, this side of the chorion becomes smooth, and is called the chorion laeve.



As the fetus grows, the chorion laeve comes into contact with the decidua parietalis on the opposite side of the uterus, and fusion occurs. Thus, the uterine lumen is obliterated. The amnion and chorion fuse, to form the amniochorionic membrane. Rupture of the amniochorionic membrane during labor is called 'breaking of waters'.

Note: Superfetation can theoretically occur till the time the uterine lumen is not obliterated.

Development of placenta and amniotic cavity



Solution to Question 11:

Langhans cells form the cytotrophoblast.

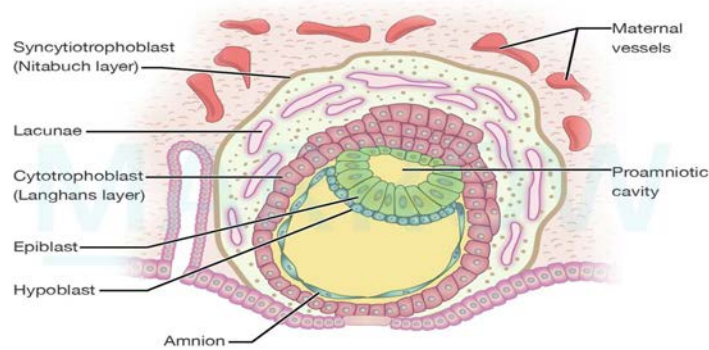
The cells of the blastocyst differentiate into an outer trophectoderm and an inner cell mass.

The trophectoderm is further differentiated into:

- An inner mononuclear cellular layer called cytotrophoblast or Langhans layer made of Langhans cells.
- An outer layer of multinucleated syncytium called syncytiotrophoblast.

The blastomere further divides to form the morula.

Implantation



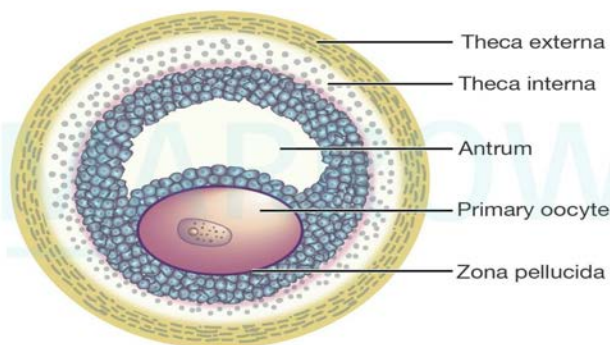
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Solution to Question 12:

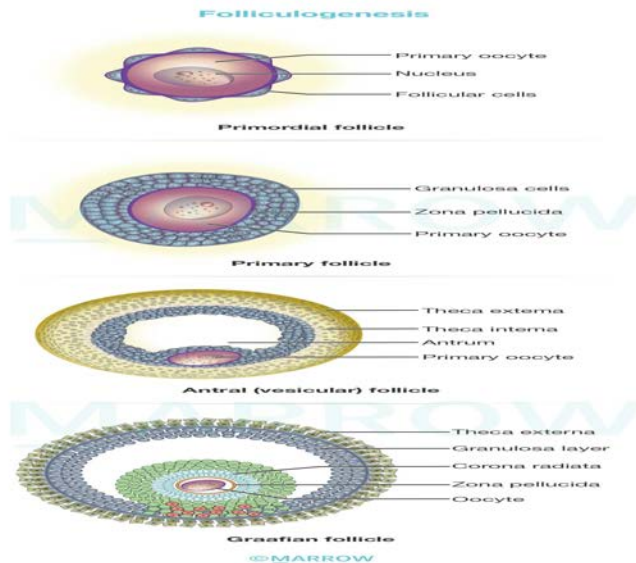
The structure marked A is primary oocyte. Primary oocytes are the germ cells seen in the medullary region of the ovary. These enlarge at the beginning of the growth phase.

During this cell growth period, many oocytes undergo degeneration, both before and after birth. A single layer of flattened follicular cells that were derived originally from the germinal epithelium soon surrounds the primary oocytes. The structure that is formed is now called a primordial follicle. These are seen first in the medulla and later in the cortex.

Antral (vesicular) follicle



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Solution to Question 13:

Fertilization usually occurs in the ampulla of the fallopian tube.

During ovulation, the ovum is released and it enters the fallopian tube. It is propelled towards the uterine cavity by the peristaltic action of the fallopian tubes, prostaglandins in the male seminal fluid and oxytocin. It can only be fertilized within 24 hours of ovulation. If coitus takes place during this period, the spermatozoa swimming up into the tubes meet the ovum in the ampulla.

Solution to Question 14:

The given image shows an oropharyngeal teratoma, which contains derivatives from all three germ layers.

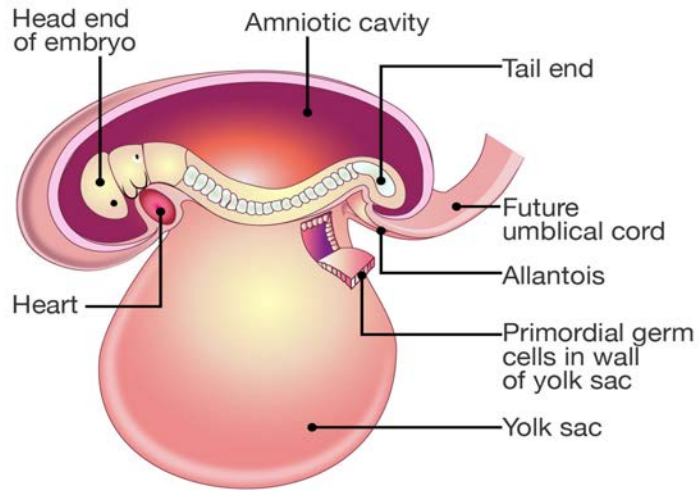
The tumor may include gut, bone, skin, and teeth. These tumors arise from pluripotent stem cells that differentiate into any of three germ layers forming tissues of all three germ layers.

Solution to Question 15:

The structure marked A is the allantois.

The posterior wall of the yolk sac forms the allantois on the 16th day of development. Its distal portion forms the urachus which forms the median umbilical ligament.

The given image shows an embryo at the end of the third week.



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Maternal Pelvis and Fetal Skull

Question 1:

The anatomical classification of the pelvis based on shape was given by _____.

- a) Caldwell and Moloy
- b) McAfee and Johnson
- c) Page and Sher
- d) Burns and Marshall

Question 2:

An obese woman with dysmenorrhea and irregular periods is diagnosed with dystocia dystrophia syndrome. What type of pelvis is seen here?

- a) Android pelvis
- b) Platypelloid pelvis
- c) Anthropoid pelvis
- d) Gynecoid pelvis

Question 3:

Which of the following is false about anthropoid pelvis?

- a) Sacral angle is more than 90°
- b) Ischial spines are not prominent
- c) Side walls are convergent
- d) Long and curved pubic arch

Question 4:

A 28-year-old term primigravida gave birth to a male baby by face to pubis delivery. Which type of pelvis is this mode of delivery commonly seen with?

- a) Gynecoid pelvis
- b) Anthropoid pelvis

- c) Android pelvis
- d) Platypelloid pelvis

Question 5:

An obstetrician performing clinical pelvimetry finds that the smallest transverse diameter is 8 cm. Which of the following parameters has she assessed?

- a) Interspinous diameter
- b) Obstetric conjugate
- c) True conjugate
- d) Intertuberous diameter

Question 6:

During clinical pelvimetry, the obstetrician determines the diagonal conjugate to be X cm. What is the obstetric conjugate?

- a) $X+1$ cm
- b) $X+2$ cm
- c) $X-1$ cm
- d) $X-2$ cm

Question 7:

A diabetic primigravida is suspected to have cephalopelvic disproportion. What should her obstetric conjugate be for a trial of labor to be conducted?

- a) 8.5 cm
- b) 9 cm
- c) 9.5 cm
- d) 10 cm

Question 8:

Which of the following parameters cannot be viewed in lateral plate X-ray pelvimetry?

- a) Diagonal conjugate

- b) True conjugate
- c) Bispinous diameter
- d) Inclination of the pelvis

Question 9:

A P1A1 woman with a history of obstructed labor is being evaluated for a contracted pelvis. CT pelvis shows both sacral alae are absent. What is the type of pelvis seen here?

- a) Naegele pelvis
- b) Robert pelvis
- c) Kyphotic pelvis
- d) Rachitic pelvis

Question 10:

A patient with prolonged second stage of labor is referred to the district hospital. You suspect her to have a contracted pelvis. Which of the following is true about this condition?

- a) Transverse diameter of inlet is 10 cm
- b) Anteroposterior diameter of inlet is 12 cm
- c) Transverse diameter of inlet is 12 cm
- d) Anteroposterior diameter of inlet is 14 cm

Question 11:

Which of the following is the smallest diameter of the fetal head?

- a) Biparietal diameter
- b) Suboccipitofrontal diameter
- c) Suboccipitobregmatic diameter
- d) Bitemporal diameter

Question 12:

A 32-year-old woman in labor is referred to a tertiary care center. On PV examination, the resident notes that the fetus is in brow presentation, and the largest diameter of the fetal skull

is engaged. Identify this diameter.

- a) Occipitofrontal
- b) Biparietal
- c) Submentovertical
- d) Mentovertical

Question 13:

A G6P5L5 woman presented to the casualty with abdominal pain. The PV examination findings are given below. What is the engaging diameter?

- a) Suboccipito-bregmatic diameter
- b) Suboccipito-frontal diameter
- c) Occipito-frontal diameter
- d) Occipito-posterior position

Question 14:

How many fontanelles are present in a newborn?

- a) 5
- b) 2
- c) 4
- d) 6

Question 15:

A 24-year-old primigravida delivered a full-term male baby by assisted forceps delivery. The next day, you note a swelling over the infant's head that is limited by suture lines. Which of the following is true regarding the swelling?

- a) Tension in the swelling varies on crying
- b) It may cause jaundice
- c) It contains edematous fluid
- d) Usually present at birth

Question 16:

The pediatrician notices a diffuse, reddish scalp swelling in a newborn born by vacuum extraction. Which of the following is not a characteristic of this condition?

- a) It crosses the midline.
- b) It crosses the suture line.
- c) It does not disappear within 2-3 days.
- d) It is a diffuse edematous swelling of the soft tissues of the scalp.

Answer Key

Question No.	Correct Option
1	a
2	a
3	c
4	b
5	a
6	d
7	d
8	c
9	b
10	a
11	d
12	d
13	a
14	d
15	b
16	c

Detailed Explanations**Solution to Question 1:**

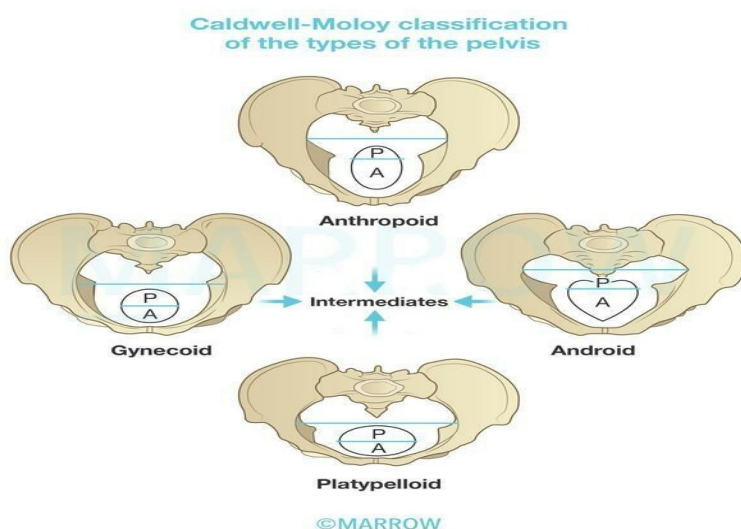
The anatomical classification of the pelvis based on shape was given by Caldwell and Moloy.

The greatest transverse diameter of the pelvic inlet divides the pelvis into anterior and posterior segments. The shapes of these segments help classify the pelvis as follows:

- Gynecoid
- Anthropoid
- Android
- Platypelloid

The various differences between these types is summarised below:

	Gynecoid	Anthropoid	Android	Platypelloid
Shape	Round	Anteroposteriorly oval	Triangular/ Heart-shaped	Transversely oval / Flat bowl shaped
Sacral angle	>90°	>90°	<90°	>90°
Side walls	Straight or slightly divergent	Straight or divergent	Convergent	Divergent
Ischial spines	Not prominent	Not prominent	Prominent	Not prominent
Subpubic angle	Wide (85°)	Slightly narrow	Narrow	Very wide (>90°)
Fetal Position	OccipitolateralOblique occipitoanterior	Direct occipitoanteriorOccipitoposterior	OccipitolateralOblique occipitoposterior	Occipitolateral
Engaging diameter	Transverse / oblique	Anteroposterior	Transverse / oblique	Transverse
Delivery	No difficulty	Face-to-pubis - most common.	Deep transverse arrest is common. Difficult delivery with an increased chance of perineal injuries	No difficulty



Option B: McAfee and Johnson's regimen is for the conservative management of placenta previa.

Option C: Page's and Sher's classification system is used to grade abruptio placentae.

Option D: Burns and Marshall maneuver is used in breech delivery.

Solution to Question 2:

In dystocia dystrophia syndrome, an android pelvis is seen.

Dystocia dystrophia syndrome:

- The patient is stockily built with a bull neck, broad shoulders, and short thighs. She is obese with a male distribution of hair.
- Subfertile, with dysmenorrhea, oligomenorrhea, or irregular periods.
- There is an increased incidence of pre-eclampsia and a tendency for postmaturity.
- The pelvis is of the android type. The occipito-posterior position is common.
- During labor, there is a tendency for deep transverse arrest or outlet dystocia, leading to difficult instrumental delivery or cesarean section.
- There is a chance of lactation failure.

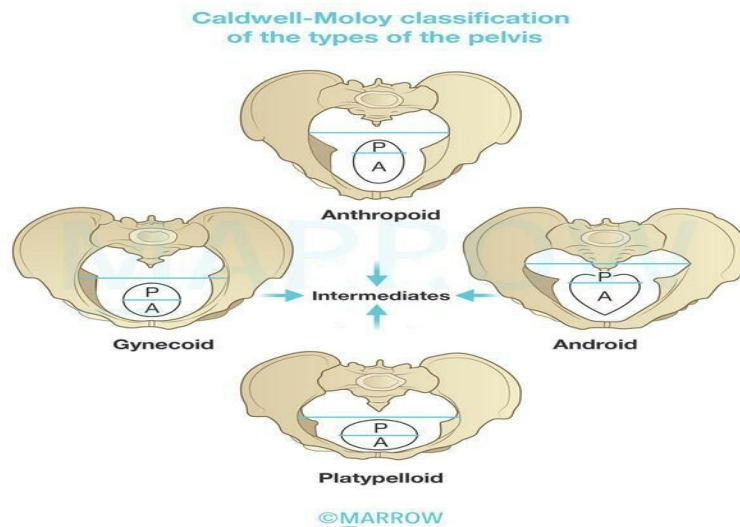
Solution to Question 3:

The anthropoid pelvis has straight side walls, not convergent side walls.

Given below are the differences between various types of pelvis as per Caldwell-Moloy classification.

	Gynecoid	Anthropoid	Android	Platypelloid
Shape	Round	Anteroposteriorly oval	Triangular/ Heart-shaped	Transversely oval / Flat bowl shaped
Sacral angle	>90°	>90°	<90°	>90°
Side walls	Straight or slightly divergent	Straight or divergent	Convergent	Divergent
Ischial spines	Not prominent	Not prominent	Prominent	Not prominent
Subpubic angle	Wide (85°)	Slightly narrow	Narrow	Very wide (>90°)
Fetal Position	OccipitolateralOblique occipitoanterior	Direct occipitoanteriorOccipitoposterior	OccipitolateralOblique occipitoposterior	Occipitolateral
Engaging diameter	Transverse / oblique	Anteroposterior	Transverse / oblique	Transverse

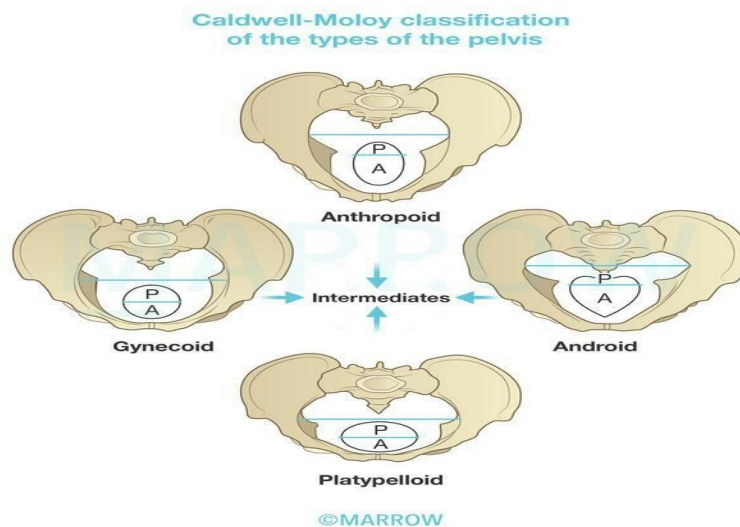
	Gynecoid	Anthropoid	Android	Platypelloid
Delivery	No difficulty	Face-to-pubis - most common.	Deep transverse arrest is common. Difficult delivery with an increased chance of perineal injuries	No difficulty



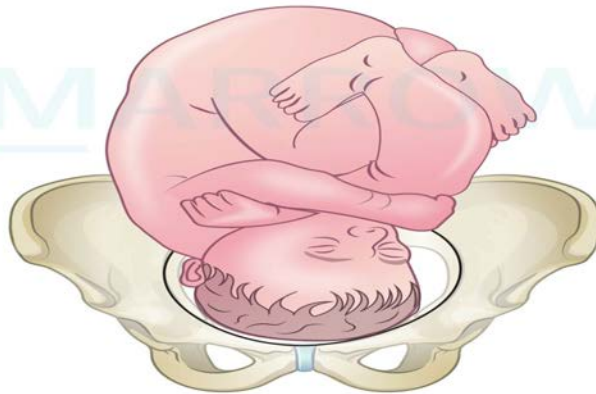
Solution to Question 4:

Face to pubis delivery is most commonly associated with anthropoid pelvis.

As the anthropoid pelvis is wider posteriorly, the fetal occiput occupies the posterior segment of the pelvis. Thus, the fetus is present in the occipitoposterior position which leads to a high chance of face-to-pubis delivery.



Occipitoposterior position



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Solution to Question 5:

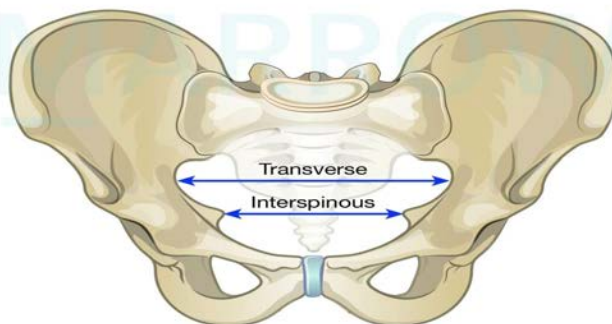
The interspinous diameter is the smallest transverse pelvic diameter.

The interspinous diameter or bispinous diameter is measured in the mid pelvis. It is the distance between the tips of two ischial spines. Normally, it is 10-10.5 cm.

It is useful in predicting obstructed labor. If the interspinous diameter is too small, the baby's head may get stuck at a higher station.

Note: The obstetric conjugate is the smallest anteroposterior pelvic diameter measuring 10 cm.

Transverse and interspinous diameters



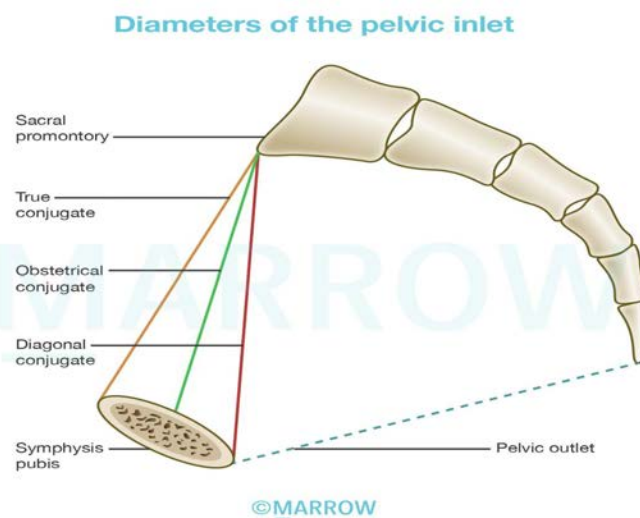
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Solution to Question 6:

The obstetric conjugate is $X-2\text{cm}$ where X is the diagonal conjugate.

Anteroposterior diameters of the pelvis:

- True conjugate or the anteroposterior diameter, extends from the upper most margin of the symphysis pubis to the sacral promontory. It is normally 11 cm.
- Obstetrical conjugate is the shortest distance between the sacral promontory and the symphysis pubis. It is normally 10 cm. It is estimated indirectly by subtracting 1.5–2 cm from the diagonal conjugate.
- Diagonal conjugate is determined by measuring the distance from the lowest margin of the symphysis to the sacral promontory. It is normally 12cm. It is measured clinically.



Solution to Question 7:

In the given case scenario, a trial of labor or vaginal delivery can be allowed if the obstetric conjugate is 10cm or more.

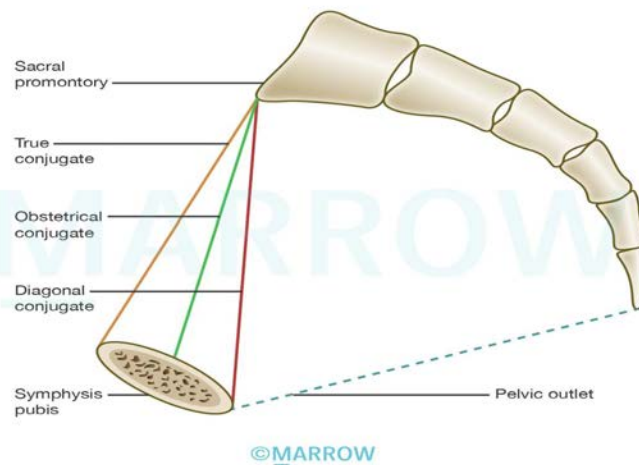
The obstetric conjugate is the shortest distance between the sacral promontory and the symphysis pubis. Normally, it is 10 cm or more.

The diagonal conjugate measures 12cm and the true conjugate measures 11cm.

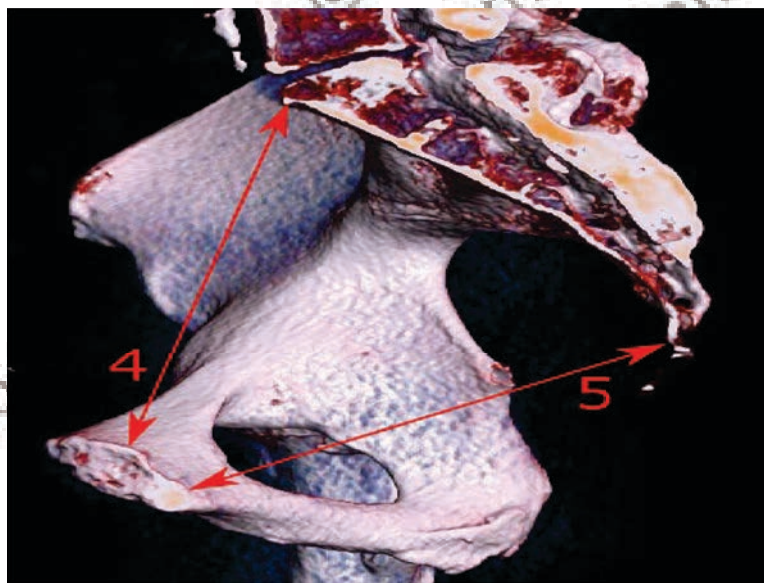
Clinically, only the diagonal conjugate is assessed by measuring the distance between the base of the examiner's fingers and the fingertip touching the sacral promontory. This determines the lowest margin of the symphysis to the sacral promontory. The obstetric conjugate is estimated indirectly by subtracting 1.5–2 cm from the diagonal conjugate.

Note: Overt diabetes can cause fetal macrosomia during pregnancy. Thus, it may lead to cephalopelvic disproportion.

Diameters of the pelvic inlet



Given below is an image of the lateral plane in CT pelvimetry, showing the anteroposterior diameters of the pelvis. Line no. 4 is the obstetric conjugate and line no. 5 is the pelvic outlet.



Solution to Question 8:

The bispinous diameter can be viewed in the anteroposterior view and not the lateral plate pelvimetry.

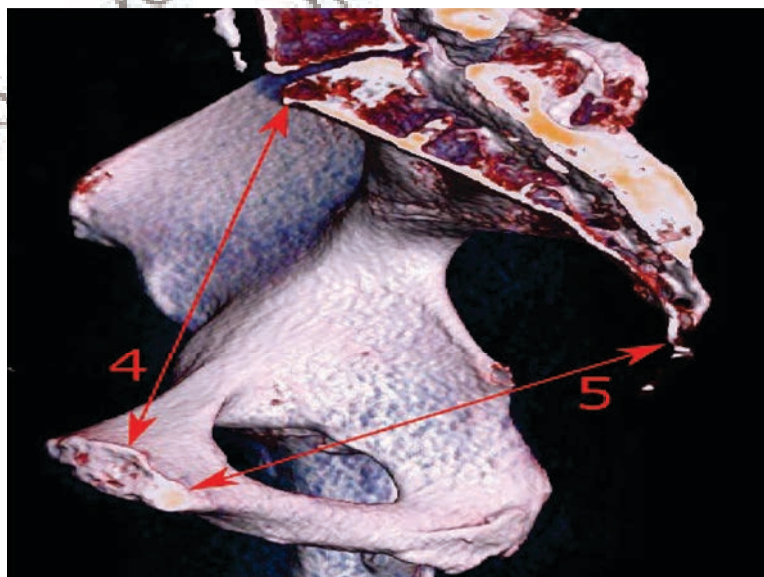
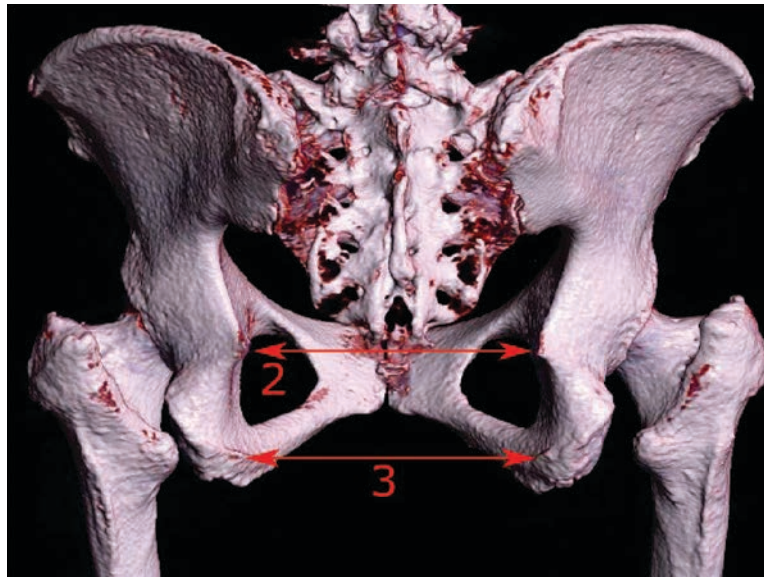
X-ray pelvimetry is done using 3 views:

- Anteroposterior - Transverse diameter and bispinous diameter is determined.
- Lateral - Obstetric conjugate, sacral curve and pelvic inclination are seen.
- Outlet - Outlet diameters are determined.

Note: CT pelvimetry is preferred over X-ray pelvimetry due to the lower dose of radiation delivered.

Given below are images of CT pelvimetry, anteroposterior and lateral views respectively. The marked diameters are as follows:

- 2 - Interspinous diameter
- 3 - Intertuberous diameter
- 4 - Obstetric conjugate
- 5 - Pelvic outlet



Solution to Question 9:

The type of pelvis seen here is Robert's pelvis. In it, the pelvis is transversely contracted and the ala on both sides are absent and the sacrum is fused with the innominate bones.

Option A: In Naegele's pelvis, there is arrested development of one ala of the sacrum.

Option C: Kyphotic pelvis occurs secondary to the kyphotic changes of the vertebral column either following tuberculosis or rickets.

Option D: In rachitic pelvis, due to softening of bones, flattening of the pelvis occurs.

Solution to Question 10:

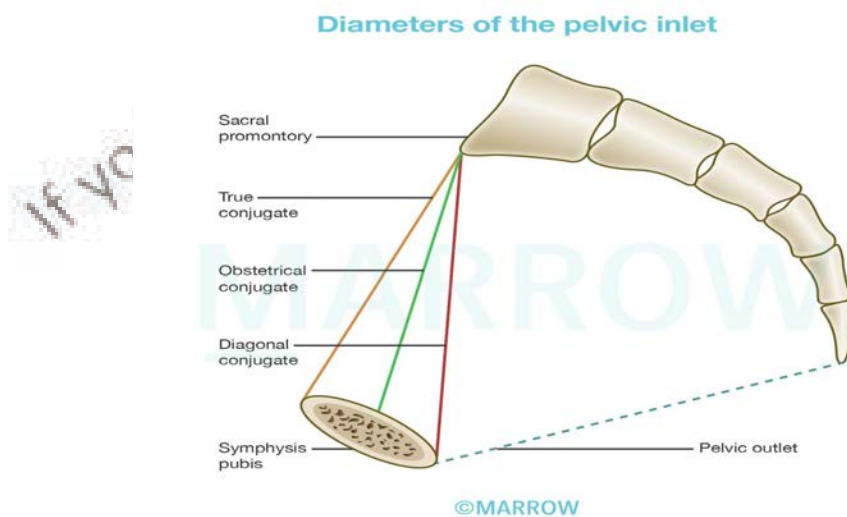
To make a diagnosis of contracted pelvis, the transverse diameter of the pelvic inlet should be 10cm.

The pelvic inlet is considered to be contracted if:

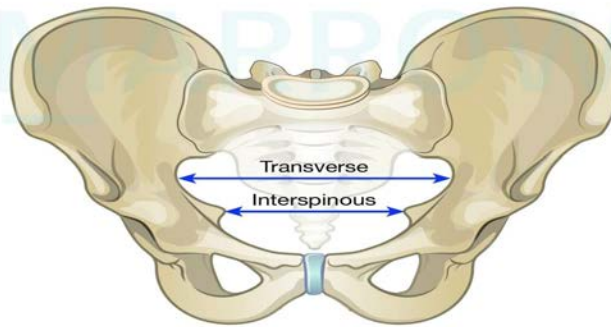
- the greatest transverse diameter is < 12 cm or
- the shortest anteroposterior diameter is < 10 cm.

The anteroposterior diameters of the pelvic inlet are the true conjugate, diagonal conjugate and obstetric conjugate. These extend between the sacral promontory and symphysis pubis. The obstetric conjugate is the shortest diameter, normally measuring 10 cm.

The transverse diameter of the pelvic inlet is at a right angle to the obstetric conjugate. It extends between the linea terminalis on either side. Normally, the transverse diameter is about 13 cm.



Transverse and interspinous diameters



©Marrow

Solution to Question 11:

Among the given options, the bitemporal diameter is the smallest. It is the distance between the anteroinferior ends of the coronal suture and usually measures 8 cm. This is the smallest engaging diameter.

Option A: The biparietal diameter is a transverse diameter between two parietal eminences, measuring 9.5cm.

Option B: The suboccipitofrontal diameter is an anteroposterior diameter, measuring 10 cm.

Option C: The suboccipitobregmatic diameter is an anteroposterior diameter, measuring 9.5 cm.

Note: The bimastoid diameter is a transverse diameter. It is the smallest diameter in the fetal skull at about 7.5 cm. However it is not clinically relevant as it does not engage.

Solution to Question 12:

In brow presentation, the mentovertical diameter is engaged which is the longest diameter of the fetal skull. Normally, it measures 14 cm.

The mentovertical diameter extends from the midpoint of the chin to the highest point on the sagittal suture.

Engaging diameters of the fetal skull vary with the attitude of the fetal head as given below:

Diameters	Attitude of the head
Suboccipitobregmatic	Complete flexion

Diameters	Attitude of the head
Suboccipito-frontal	Incomplete flexion
Occipitofrontal	Marked deflexion
Mento-vertical	Partial extension
Submentovertical	Incomplete extension
Submentobregmatic	Complete extension

Solution to Question 13:

In the given clinical scenario, the engaging diameter is the suboccipito-bregmatic diameter. It is engaged when the fetus is in vertex presentation with an attitude of complete flexion of the head.

The table given below shows the engaging diameter with each attitude of the fetal head.

Diameters	Attitude of the head
Suboccipitobregmatic	Complete flexion
Suboccipito-frontal	Incomplete flexion
Occipitofrontal	Marked deflexion
Mento-vertical	Partial extension
Submentovertical	Incomplete extension
Submentobregmatic	Complete extension

Solution to Question 14:

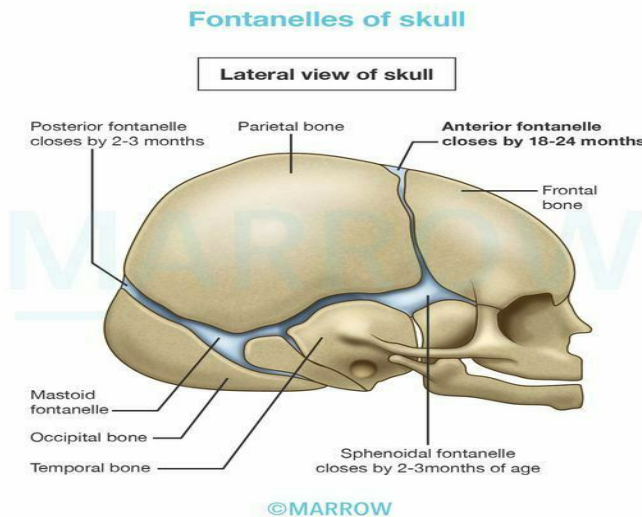
Newborns have a total of 6 fontanelles. These are fibrous, membrane-covered gaps between cranial bones.

The fontanelles seen in newborns are:

- Anterior fontanelle:
- It is also called bregma.
- Diameter ranges from 0.6-3.6 cm, with a mean of 2.1 cm.
- It completely closes by 24 months of age.
- Posterior fontanelle:

- It is also called lambda.
- Diameter ranges from 0.5-0.7 cm.
- It completely closes by 2 months of age.
- 2 sphenoid fontanelles
- 2 mastoid fontanelles

Note: A third fontanelle between the anterior and posterior fontanels is associated with hypothyroidism and Down's syndrome.



Solution to Question 15:

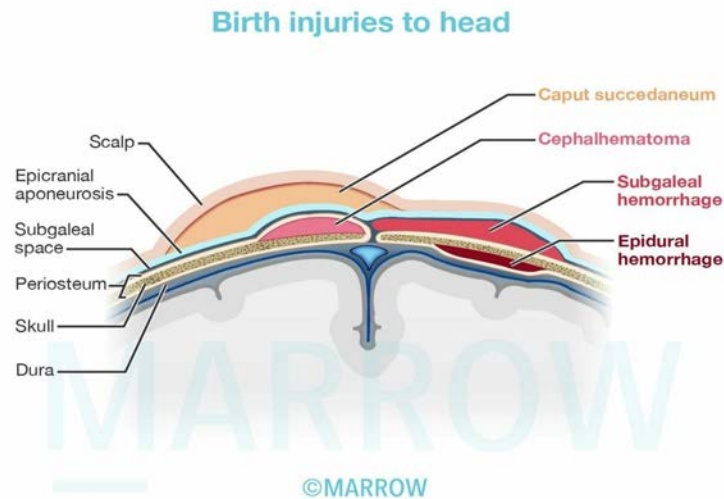
In the given clinical picture, the swelling over the infant's head is likely to be a cephalhematoma. Infants with cephalohematomas are at a heightened risk of developing jaundice because as the blood breaks down, the levels of bilirubin increase.

Cephalohematoma is a collection of blood in between the pericranium and the flat bone (subperiosteal) of the skull, usually unilateral and over a parietal bone. It is never present at birth but gradually develops after 12–24 h. The swelling is limited by the suture lines of the skull as the pericranium is fixed to the margins of the bone.

Option A: Tension in the swelling varies with crying in meningocele.

Option C: Edematous fluid collects in caput succedaneum.

Option D: Caput succedaneum may be present at birth.



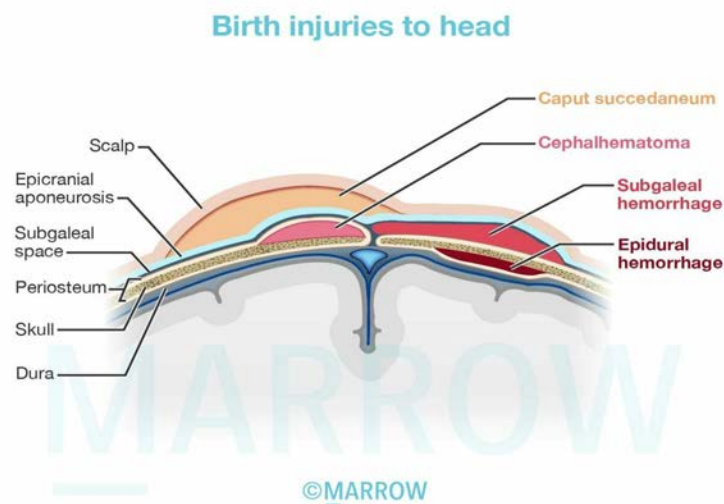
Solution to Question 16:

The given case scenario is suggestive of caput succedaneum which disappears within 2-3 days.

In vertex presentation, labor forces alter the shape of the fetal head. In prolonged labor, before complete cervical dilatation, the portion of the fetal scalp immediately over the cervical os becomes edematous. A swelling is formed due to the stagnation of fluid in the layers of the scalp beneath the girdle of contact. This is known as caput succedaneum.

The swelling is diffuse, boggy, and not limited by the suture lines. It resolves spontaneously within 2-3 days after birth. It is a result of the static position of the head for a long period of time.

Chignon is a type of artificial caput succedaneum, which is a swelling produced on the fetal head because of the scalp getting sucked into the cup during vacuum extraction.



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you may have been scammed.

Placenta and Fetal Membranes

Question 1:

How is the normal placenta described?

- a) 1 and 2
- b) 2 and 3
- c) 1 and 3
- d) 1, 2 and 3

Question 2:

What does a placental cotyledon comprise of?

- a) Portions of the basal plate divided by clefts
- b) Portions of the chorionic plate divided by cleft
- c) Area supplied by multiple truncal chorionic villi
- d) Area supplied by one spiral artery

Question 3:

Which of the following does not contribute to the prevention of rejection of fetus?

- a) Decidual natural killer cells
- b) Decidual stromal cells
- c) Trophoblasts
- d) Maternal leucopenia

Question 4:

At what period of gestation are the weights of the fetus and placenta equal?

- a) 14 weeks
- b) 15 weeks
- c) 17 weeks

d) 21 weeks

Question 5:

What is the uteroplacental blood flow at term?

- a) 350-600 ml/min
- b) 250-450 ml/min
- c) 500-750 ml/min
- d) 800-1050 ml/min

Question 6:

Which of the following is false about human chorionic gonadotropin?

- a) Alpha subunit is identical to LH, FSH, TSH
- b) It causes involution of corpus luteum
- c) Maximum level is seen at 60-70 days of gestation
- d) Detected in serum 8-9 days after fertilisation

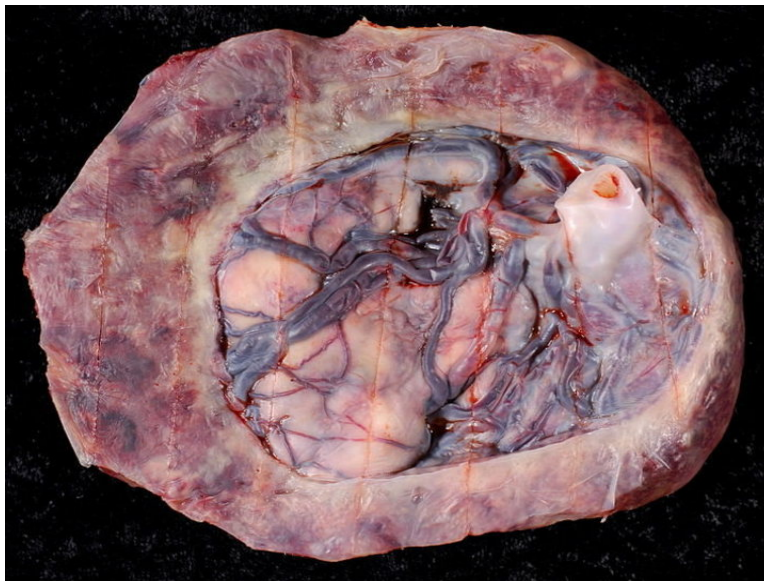
Question 7:

Which of the following cannot cross the placenta?

- a) 1 and 2
- b) Only 3
- c) 2 and 3
- d) 3 and 4

Question 8:

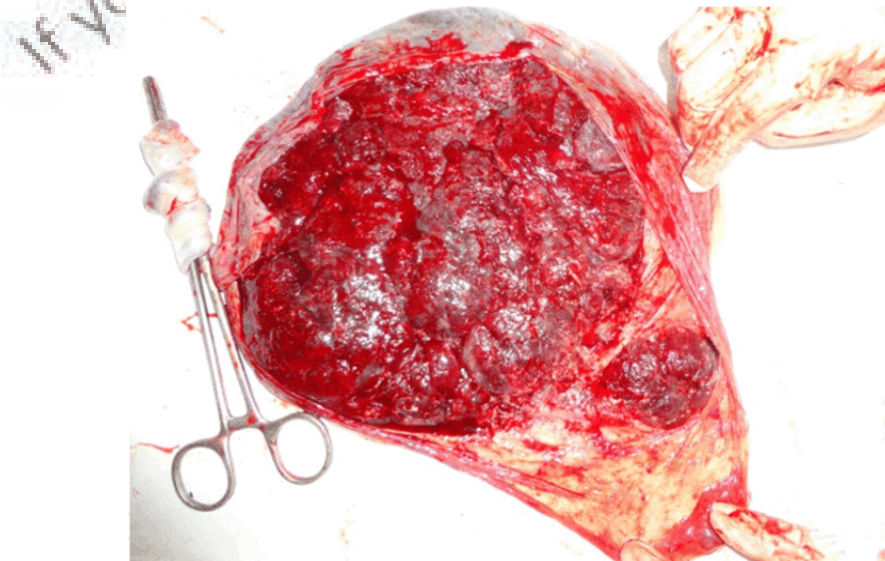
A primigravida delivered a baby at term with a birth weight of 1.9 kg via normal vaginal delivery. The placenta delivered appears as shown in the image below. What is this condition known as?



- a) Placenta succenturiata
- b) Placenta marginata
- c) Circumvallate placenta
- d) Circummarginate placenta

Question 9:

Following a normal vaginal delivery, the placenta delivered by controlled cord traction appeared as shown below. What is this condition known as?



- a) Bilobate placenta

- b) Placenta succenturiata
- c) Placenta membranacea
- d) Placenta marginata

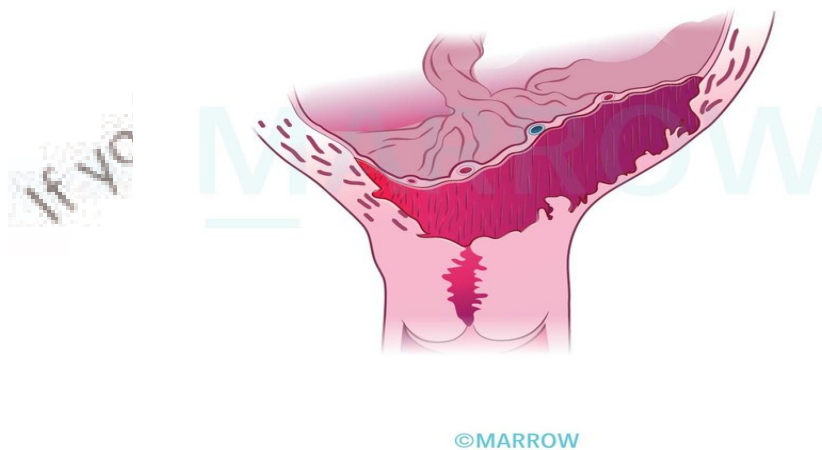
Question 10:

The USG scan of a multiparous woman at 30 weeks of gestation showed the presence of an accessory placental lobe. Which complication will you not anticipate in her?

- a) Preterm delivery
- b) Postpartum haemorrhage
- c) Vasa previa
- d) Subinvolution

Question 11:

Which of the following types of morbidly adherent placenta is shown in the given image?



- a) Placenta increta
- b) Focal placenta accreta
- c) Placenta percreta
- d) Total placenta accreta

Question 12:

What contributes to the formation of amniotic fluid during early pregnancy?

- a) Fetal urine
- b) Fetal extracellular fluid
- c) Ultrafiltrate of maternal plasma
- d) Ultrafiltrate of fetal plasma

Question 13:

While performing the USG assessment of a woman at 28 weeks of gestation, you note her amniotic fluid index to be 12.8. When does this parameter reach its maximum value in a normal pregnancy?

- a) 32 weeks of gestation
- b) 33 weeks of gestation
- c) 34 weeks of gestation
- d) 35 weeks of gestation

Question 14:

Which of the following is not a function of amniotic fluid?

- a) Maintain temperature
- b) Bacteriostatic property
- c) Provide nutrition in the form of fat and protein
- d) Prevent ascending infection to uterine cavity during labor

Question 15:

When does keratinization of fetal skin occur?

- a) 12-18 weeks
- b) 22-25 weeks
- c) 26-30 weeks
- d) 16-20 weeks

Question 16:

Which of the following statements is false about amniotic fluid?

- a) Composition remains same throughout pregnancy
- b) Major source is fetal urine after 20 weeks
- c) It is primarily reabsorbed by fetal swallowing
- d) It is alkaline in nature

Question 17:

A multigravida presents at 34 weeks of gestation with preterm premature rupture of membranes. On examination, the amniotic fluid is golden-colored. Which condition will you suspect here?

- a) Rh-incompatibility
- b) Fetal death
- c) Fetal distress
- d) Intrauterine growth restriction

Question 18:

Ultrasound findings of a few patients are given below. Which of the following patients have oligohydramnios?

- a) 1 and 2
- b) 2 and 3
- c) 2 and 4
- d) 2, 3 and 4

Question 19:

A 24-year-old primigravida at 30 weeks of gestation came for a regular antenatal checkup. On examination, size of the uterus corresponds to 26 weeks of gestation and the fetal heart rate is 130 beats/min. On ultrasound, AFI is 4. Which fetal anomaly would you suspect?

- a) Renal agenesis
- b) Neural tube defect

- c) Oesophageal atresia
- d) Cleft palate

Question 20:

A 30-year-old G3P2L2 woman with multifetal pregnancy is diagnosed with polyhydramnios. What is the amniotic fluid volume required to diagnose it as this condition?

- a) >1000 ml
- b) >1500 ml
- c) >2000 ml
- d) >2500 ml

Question 21:

A lady at 30 weeks of gestation came for an antenatal check-up. On examination, the uterus corresponds to 34 weeks of gestation and FHR is normal. Ultrasound revealed an AFI of 26 cm. Which fetal anomaly is most likely to be seen with this condition?

- a) Posterior urethral valve
- b) Cleft palate
- c) Polycystic kidney disease
- d) Bladder exstrophy

Question 22:

A routine ultrasound of a multigravida at 28 weeks of gestation revealed an AFI of 27cm. Which of the following cannot cause the given condition?

- a) Diabetes mellitus
- b) Preeclampsia
- c) Esophageal atresia
- d) Open spina bifida

Question 23:

A 28-year-old G2P1L1 at 35 weeks of gestation is complaining of difficulty in breathing. On examination, flanks are full, and abdominal girth is 160cm. Ultrasound determined that the SDP is 10cm. What is the next step in management?

- a) Intravenous furosemide
- b) Indomethacin therapy
- c) Amniocentesis
- d) Artificial rupture of membranes

Question 24:

A female presents with 35 weeks and 3 days of gestation, AFI = 5 and there is no end-diastolic blood flow. What is the management?

- a) Immediate termination
- b) Give surfactant and wait till 36 weeks
- c) Do Doppler everyday to check for reversion
- d) Terminate after 37 weeks

Question 25:

Which of the following statements is false about the umbilical cord?

- a) It has one artery and two veins
- b) It contains a gelatinous substance called Wharton's jelly
- c) Presence of a single umbilical artery is associated with congenital anomalies
- d) Normal average umbilical cord length is 50-60 cms

Question 26:

Ultrasound of a 24-year-old primigravida at 20 weeks of gestation shows the presence of a single umbilical artery. No other gross fetal anomalies were detected. Which of the following is true regarding this finding?

- a) It is an insignificant finding
- b) Occurs in 10% of newborns
- c) It is an indicator of increased risk of malformations of the fetus
- d) Equally common in newborns of diabetic and non-diabetic mothers

Answer Key

Question No.	Correct Option
1	d
2	a
3	d
4	c
5	c
6	b
7	c
8	c
9	b
10	a
11	a
12	c
13	c
14	c
15	b
16	a
17	a
18	b
19	a
20	c
21	b
22	b
23	c
24	a
25	a
26	c

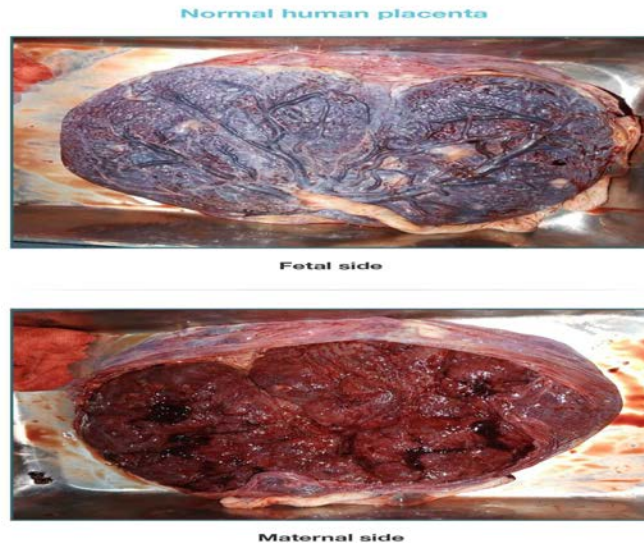
Detailed Explanations

Solution to Question 1:

The normal human placenta is discoid, deciduate, and hemochorial.

The term deciduate means that the placenta is shed at the time of parturition. Hemochorial means that the maternal blood directly comes in contact with the fetal trophoblasts.

The image given below shows a normal human placenta.



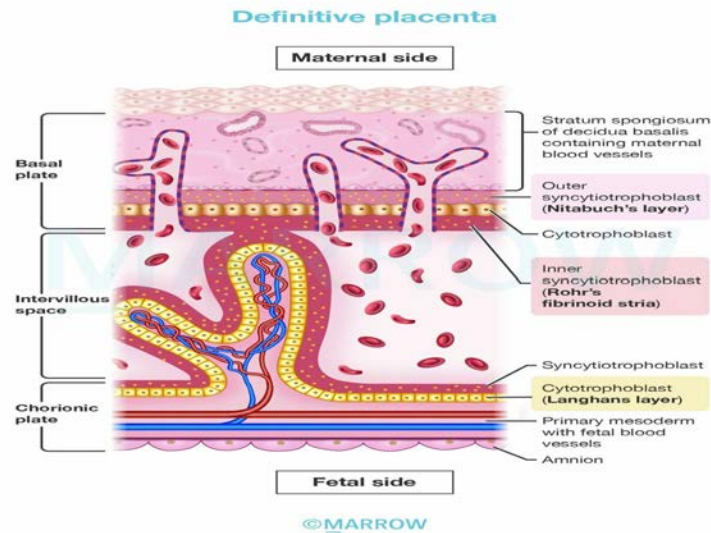
Solution to Question 2:

A placental cotyledon is comprised of portions of the basal plate divided by clefts. It is the functional unit of the placenta.

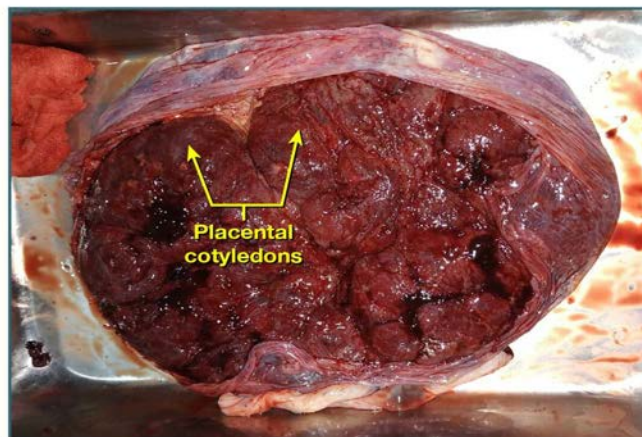
A placental cotyledon or placental lobule constitutes one truncal or main stem villi and its ramifications (rami). Each lobule is supplied by a single truncal branch of the chorionic artery. Each lobule also has a single vein. The lobules constitute functional units of placental architecture.

The fetal surface is the chorionic plate. The umbilical cord inserts in the center of the chorionic plate.

Note: The entire placenta is supplied by chorionic villi while each cotyledon is only supplied by a single main stem villus with its branches.



Maternal side of placenta



Solution to Question 3:

Maternal leucopenia is not seen in pregnancy, instead leucocytosis is seen. This plays no part in the prevention of rejection of fetus.

The fetus is a semiallogenic graft inside the maternal body as half of its genes are from the father. Uterine transplantation immunity contributes to the survival of the fetus in spite of it being semiallogenic. This occurs due to the following factors:

- Decidual natural killer cells are inefficient, and therefore are unable to attack the 'foreign' fetal cells.
- Decidual stromal cells work with natural killer cells to allow trophoblastic invasion.
- Villous trophoblastic cells don't express HLA antigens and therefore, they are non-immunogenic.

Solution to Question 4:

At 17 weeks, the weights of the fetus and placenta are equal.

In the first trimester, the placental growth is more rapid than that of the fetus. By approximately 17 weeks, the placental weight and fetal weight are almost equal. And by term, placental weight is approximately 1/6th of the fetal weight.

Solution to Question 5:

The uteroplacental blood flow at term is 500-750 ml/min.

Solution to Question 6:

Human chorionic gonadotropin (hCG) does not cause involution of the corpus luteum, but it maintains corpus luteum function.

hCG is produced by the syncytiotrophoblasts of the placenta. It is a glycoprotein hormone with two subunits. The alpha-subunit is similar to LH, FSH and TSH. The beta-subunit is specific and is measured by pregnancy tests.

hCG can be detected in maternal blood 8 days after fertilisation. Thus, it can help in diagnosis of pregnancy even before the period is missed. Its levels nearly doubles every 48 hours in early pregnancy reaching a maximum of 50000 - 100000 IU/L by 60-70 days of gestation. Then the level starts falling reaching a minimum at 16 weeks, which continues throughout pregnancy.

The various placental hormones and their functions are given below.

Other placental hormones include:

- GnRH, TRH, GHRH, Growth hormone variant (hGH-V)
- Parathyroid hormone-releasing protein (PTH-rP)
- Inhibin
- Activin
- Relaxin

Placental hormones	Functions
Human chorionic gonadotropin (hCG) Structurally similar to - LH, FSH, TSH	Maintains corpus luteum function Regulates testosterone secretion from fetal testes Stimulates maternal thyroid

Placental hormones	Functions
Placental lactogen (PL) Structurally similar to - GH, Prolactin	Helps the mother adapt to fetal energy requirements
Corticotropin-releasing hormone (CRH)	Vascular and myometrial smooth muscle relaxation Initiation of parturition Stimulates placental ACTH production
Placental adrenocorticotrophic hormone (ACTH)	Stimulates fetal and maternal adrenal glucocorticoid production

Solution to Question 7:

Albumin and IgM cannot cross the placenta.

High molecular weight substances such as proteins (e.g., albumin, IgM, IgA) cannot cross the placenta, except IgG.

Substances which can cross the placenta are as follows:

Note: Maternal hormones like insulin, steroid hormones, and thyroid hormones cross the placenta very slowly, and only in amounts enough to support fetal development. However, calcitonin and parathormone do not cross the placenta.

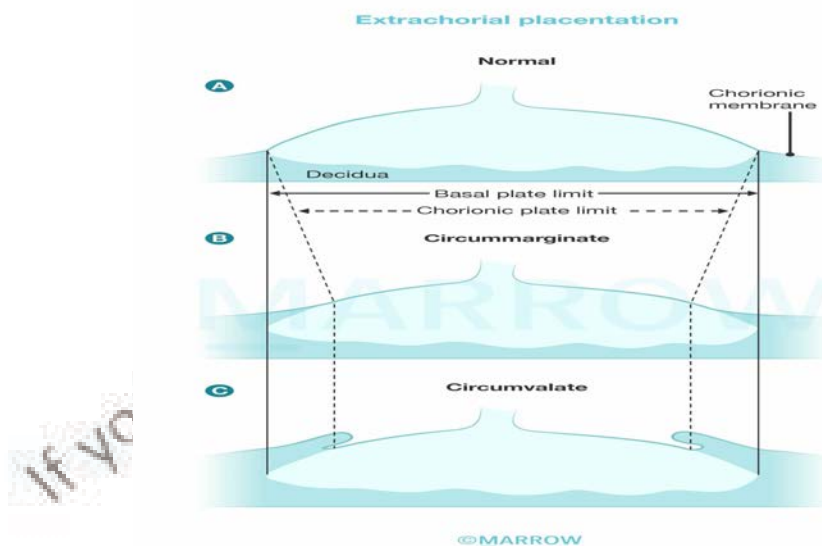
Substances	Mechanism
O ₂ CO ₂ WaterElectrolytesAnesthetic agents	Simple diffusion
Immunoglobulin G(molecular weight 160,000 Da)	Endocytosis via Fc receptors in the trophoblast
Glucose	Facilitated diffusion via glucose transport proteins (GLUTs)
Amino acids	Transport-protein-mediated active transport
Lipids:GlycerolEssential fatty acidsLDL	Endocytosis via specific receptors
Minerals:IodineIron (unidirectional from mother to fetus)CalciumPhosphorus	Carrier-mediated active transport
Vitamins:Vit AVit CVit D	Carrier-mediated active transport

Solution to Question 8:

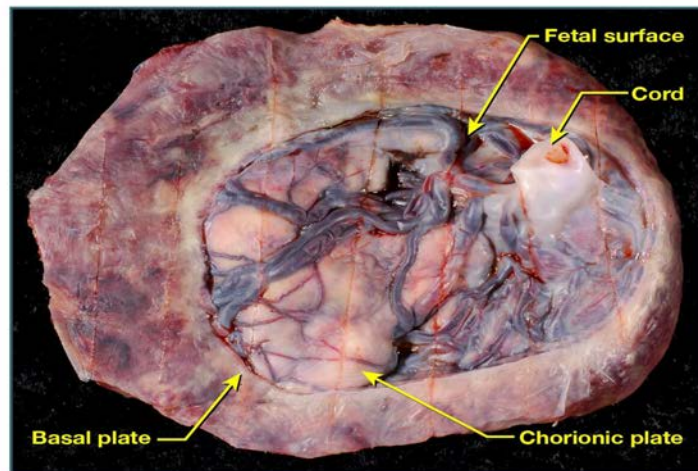
The given image shows a circumvallate placenta. The membranes at the periphery are thickened and consist of a double fold of chorion and amnion.

Normally, the chorionic plate and basal plates are of the same size. In conditions of extrachorial placentation, the chorionic plate is smaller than the basal plate. There are two types of extrachorial placentation:

- Circummarginate placenta - In this type, the space between the fetal membranes and the basal plate is filled up by fibrin and old hemorrhage.
- Circumvallate placenta - In this type, the peripheral chorion is thickened into a ridge composed of a double fold of chorion and amnion. Complications associated with circumvallate placenta are:
 - Preterm delivery
 - Abortions
 - IUGR
 - Antepartum hemorrhage
 - Retained placenta and membranes



Circumvallate placenta



Option A: Placenta succenturiata has one or more small lobes of the placenta placed at varying distances from the main placental margins.

Options B & D: Placenta marginata or circummarginate placenta - a thin margin is present between the chorionic and basal plates.

Solution to Question 9:

The given image shows a placenta with a small, accessory lobe. This condition is known as placenta succenturiata.

A succenturiate placenta contains single or multiple accessory lobes. Accessory lobes are attached to the main lobe by a membrane that has a leash of blood vessels.

Option A: Bilobate placenta - it consists of two placental lobes. The cord inserts between the two lobes.

Option C: Placenta membranacea - entire surface of the fetal membranes are covered by functioning villi and the placenta develops as a thin membranous structure occupying the entire periphery of the chorion.

Option D: Placenta marginata - chorionic plate is smaller than the basal plate, and a thin margin is present between the chorionic and basal plates.

Solution to Question 10:

The given scenario of the presence of an accessory placental lobe is suggestive of placenta succenturiata. Preterm delivery is not a complication seen in this condition.

Complications of placenta succenturiata include:

- Retained placenta

- Postpartum hemorrhage (PPH)
- Subinvolution of uterus
- Sepsis
- Placental polyp formation
- Vasa previa - vessels overlying the accessory lobes can cross the cervix. Their rupture can lead to hemorrhage from the fetus

Note: Preterm delivery is a complication associated with circumvallate placenta.

Solution to Question 11:

The given image shows placenta increta.

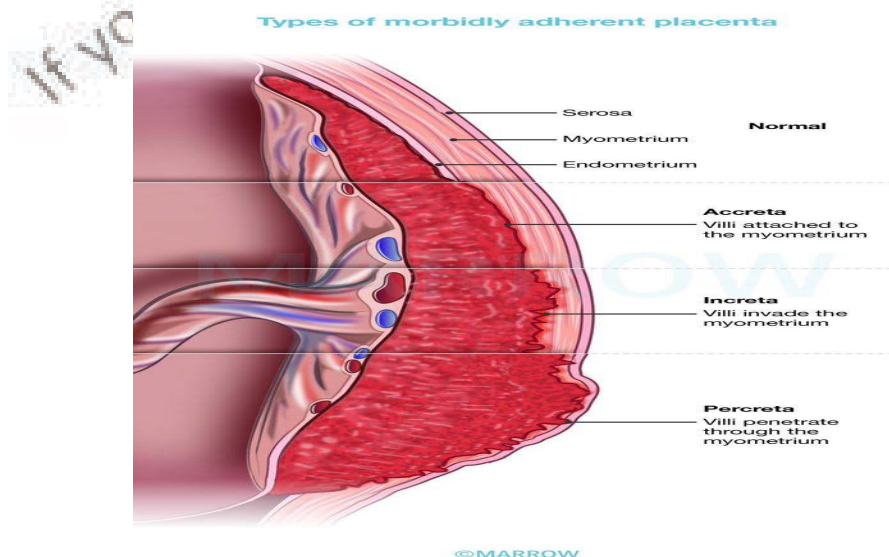
Morbidly adherent placenta occurs due to the partial or total absence of the decidua basalis and imperfect development of the fibrinoid or Nitabuch layer. It is of three types:

- Placenta accreta - villi are attached to the myometrium.
- Placenta increta - villi invade deep into the myometrium.
- Placenta percreta - villi penetrate through the entire myometrium and to or through the serosa.

Based on the adherence of placental lobules, all three types of morbidly adherent placenta can be:

- Total - involving all lobules of the placenta
- Focal - involving a single lobule or part of a lobule of the placenta

Morbidly adherent placenta can lead to torrential post-partum hemorrhage. It is managed via C-section and peripartum hysterectomy.



Solution to Question 12:

Ultrafiltrate of maternal plasma contributes to the formation of amniotic fluid during early pregnancy.

Amniotic fluid is formed by:

- Early pregnancy - ultrafiltrate of maternal plasma.
- Second trimester - the extracellular fluid that diffuses through the fetal skin and thus reflects the composition of fetal plasma.
- After 20 weeks - keratinization of fetal skin prevents diffusion across the skin. Hence, fetal urine forms amniotic fluid.

Note: Fetal kidneys start producing urine at 12 weeks (8–11 weeks), and by 18 weeks, they produce 7 to 14 mL per day.

Solution to Question 13:

The amniotic fluid index (AFI) is a measure of the amniotic volume which is maximum at 34 weeks of gestation.

The volume of amniotic fluid at each week is variable. The volume gradually increases till 32-34 weeks of gestation. At 34 weeks, it is maximum and thereafter there is a gradual reduction till term.

Amniotic fluid volume is a passive reflector of the fetal condition. Its volume is determined by the following:

Any alteration in fetal homeostasis, caused by fetal and maternal conditions, affects the amniotic fluid volume. Thus, AFI is measured as a part of the biophysical score.

Pathway	Volume in ml
Fetal urination	+1000
Fetal swallowing	-750
Fetal lung fluid secretion	+350
Intramembranous flow across fetal vessels on the placental surface	-350

Solution to Question 14:

The nutritive value of amniotic fluid is negligible because amniotic fluid is predominantly water. It contains the following:

- Water - 98–99%

- Solids 1–2% - Organic, inorganic, suspended particles

Functions of amniotic fluid during pregnancy:

- It acts as a shock absorber and prevents trauma to the fetus
- Temperature regulation of the uterine environment
- It distends the amniotic sac and thereby allows for growth and free movement of the fetus.
- It has no nutritive function, as it is mostly formed by water.
- Bacteriostatic property

Functions of amniotic fluid during labor:

- The amnion and chorion are combined to form a hydrostatic wedge which helps in dilatation of the cervix.
- During uterine contraction, it prevents any interference in placental circulation so long as the membranes remain intact.
- It protects the umbilical cord by providing a cushion between the fetus and the umbilical cord during contraction
- It flushes the birth canal at the end of the first stage of labor and by its aseptic action protects the fetus and prevents ascending infection to the uterine cavity.

Solution to Question 15:

Keratinization of fetal skin occurs at 22–25 weeks. Therefore, diffusion across fetal skin does not occur after this period. So fetal plasma does not contribute to amniotic fluid formation after this time period.

Solution to Question 16:

The composition of amniotic fluid is not the same throughout pregnancy but changes later in pregnancy due to contamination by fetal urinary metabolites.

In early pregnancy, it is identical to plasma in composition as it is mainly composed of a transudate of plasma. In late pregnancy, as amniotic fluid is primarily formed by fetal urination, the composition is altered. Disorders of the fetal kidney can lead to oligohydramnios.

Fetal swallowing is the primary method of resorption of amniotic fluid. Disorders of swallowing in the fetus can cause polyhydramnios.

Amniotic fluid is alkaline in nature and this helps in differentiating it from vaginal secretions which are acidic. This forms the basis of the nitrazine paper test used for the diagnosis of premature rupture of membranes.

Solution to Question 17:

Golden-colored amniotic fluid is seen in Rh incompatibility. This appearance is due to presence of excess bilirubin in the amniotic fluid due to hemolysis of fetal RBCs.

Color of amniotic fluid	Conditions
Pale straw	Near-term due to exfoliated lanugo hairs, exfoliated cells from fetal skin
Green	Meconium Listeria infection
Golden-yellow	Rh-incompatibility
Saffron	Post-maturity
Blood-stained	Low lying placenta Vasa previa
Dark maroon	Altered blood in accidental hemorrhage
Brown (Tobacco juice)	Intrauterine fetal demise

Solution to Question 18:

Among the given options, an AFI of 4.5cm ($<5\text{cm}$) or an SDP of 1.8cm ($<2\text{cm}$) indicate oligohydramnios.

Oligohydramnios means an abnormally decreased amount of amniotic fluid (liquor volume is $<200\text{ml}$). It is diagnosed using ultrasound parameters such as:

- Single deepest pocket (SDP) - The largest vertical pocket of amniotic fluid identified in the sagittal plane on ultrasound is the SDP. The normal range of SDP is between 2 to 8cm. Oligohydramnios is diagnosed if SDP $<2\text{cm}$.
- Amniotic fluid index (AFI) - It is the sum of the deepest pocket of amniotic fluid from each quadrant of the abdomen. The normal range of AFI is between 5 to 25 cm. Oligohydramnios is diagnosed if AFI $<5\text{cm}$.

Solution to Question 19:

The given case scenario is suggestive of oligohydramnios. It is associated with renal agenesis.

Oligohydramnios has the following features:

- Fundal height less than period of gestation
- Fetal parts easily palpable
- AFI $<5\text{cm}$
- SDP $<2\text{cm}$

Solution to Question 20:

Polyhydramnios is an abnormal increase in amniotic fluid. It is defined as the presence of amniotic fluid volume $\geq 2000\text{ml}$. Ultrasound parameters used to diagnose polyhydramnios are:

- AFI $\geq 25\text{ cm}$
- SDP $\geq 8\text{ cm}$

Solution to Question 21:

The given clinical picture is suggestive of polyhydramnios. It can be seen with a fetus having a cleft palate as fetal swallowing is impaired by this anomaly.

Polyhydramnios has the following features:

- Fundal height greater than period of gestation
- Fetal parts are not well palpable
- AFI $\geq 25\text{cm}$
- SDP $\geq 8\text{cm}$

Solution to Question 22:

The given case scenario is suggestive of polyhydramnios. Preeclampsia is a cause for oligohydramnios, and not polyhydramnios.

Solution to Question 23:

The given case scenario is suggestive of polyhydramnios associated with respiratory distress. In such cases, therapeutic amniocentesis termed amnioreduction is done.

Amnioreduction helps in polyhydramnios by:

- Relieving respiratory distress by reducing mechanical pressure on the diaphragm
- May improve obstructive uropathy
- May lower the preterm delivery risk if the patient has preterm labor or premature rupture of membranes

Option A: Diuretics have no role in this condition.

Option B: Indomethacin therapy used to be given earlier as it reduced amniotic fluid volume. However, it is not given anymore due to the risk of premature closure of ductus arteriosus.

Option D: Artificial rupture of membranes (ARM) is avoided as:

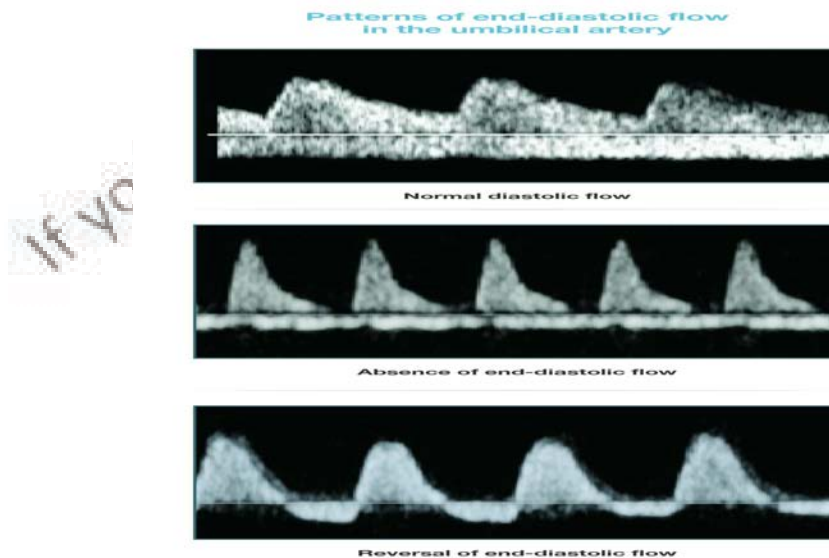
- It cannot be employed in a patient with an unfavourable cervix.
- Once ARM is done, delivery needs to be performed. In this case, preterm delivery would occur, hence ARM is avoided.
- ARM should be used with caution in patients with chronic polyhydramnios. It can lead to sudden decompression of the uterus causing placental abruption. Thus, controlled ARM is done in these patients.

Solution to Question 24:

The given case scenario is suggestive of fetal compromise. The management in this patient is immediate termination of pregnancy. Since this pregnancy has crossed 34 weeks, it can be terminated by cesarean section.

Abnormal flow patterns in the umbilical artery:

- Reduced end-diastolic flow indicates uteroplacental insufficiency (S/D ratio increases)
- Absent end-diastolic flow suggests fetal compromise, hence the pregnancy is terminated by C-section if ≥ 34 weeks.
- Reverse end-diastolic flow suggests impending doom of death, hence the pregnancy is terminated by C-section if ≥ 32 weeks.



Solution to Question 25:

The umbilical cord normally has two arteries and one vein. The presence of a single umbilical artery is the most common vascular anomaly of the cord. It is associated with congenital anomalies such as genitourinary and cardiac anomalies.

The umbilical cord is a tube that connects the fetus to the placenta. It contains umbilical vessels and an obliterated allantois duct, all surrounded by Wharton's jelly and contained within an outer layer of amnion. Maternal blood enters the fetus through the umbilical vein, and blood exits the fetus via the two umbilical arteries. Wharton's jelly is a gelatinous mucopolysaccharide that protects the umbilical vessels. Local collections of Wharton's jelly produce false knots in the cord.

The average length of the umbilical cord is between 40-70 cm at term. A long cord can form a loop around the fetus' neck and can cause asphyxia. A short cord can result in cord rupture during labor.

Solution to Question 26:

A single umbilical artery is the most common aberration of vessel number in the umbilical cord. It indicates an increased risk of congenital malformations in the fetus.

A single umbilical artery is seen in the following conditions:

- Twin pregnancy
- Gestational diabetes/Overt diabetes
- Epilepsy
- Oligohydramnios
- Polyhydramnios
- Preeclampsia
- Antepartum hemorrhage

It is frequently associated with congenital malformations of the fetus (20–25%) such as:

- Cardiovascular anomalies
- Genitourinary anomalies
- Trisomy 18

It increases the risks of adverse obstetric outcomes like:

- Abortion
- Fetal aneuploidy
- Prematurity
- Intrauterine growth restriction (IUGR)
- Perinatal mortality

Sexual Development, Puberty and Adolescence

Question 1:

Which of the following determines the sex of the fetus?

- a) Yp11
- b) Yp10
- c) Yq11
- d) Yq10

Question 2:

When does the sexual differentiation of the fetus occur?

- a) 5th week of development
- b) 7th week of development
- c) 8th week of development
- d) 6th week of development

Question 3:

Where is the lower part of the vagina derived from?

- a) Genital ridge
- b) Urogenital sinus
- c) Mullerian duct
- d) Urogenital folds

Question 4:

What is the vaginal epithelium derived from?

- a) Epithelium of urogenital sinus
- b) Epithelium of mesonephric duct
- c) Endoderm of genital ridge

- d) Epiderm of genital ridge

Question 5:

Which of the following structures in women is homologous to gubernaculum testis in men?

- a) Round ligament
- b) Broad ligament
- c) Uterosacral ligament
- d) Cardinal ligament

Question 6:

Which of the following is not a derivative of the paramesonephric duct?

- a) Appendix of testis
- b) Fallopian tube
- c) Uterus
- d) Gartner's duct

Question 7:

What is the most common developmental abnormality of the uterus?

- a) Didelphys
- b) Arcuate
- c) Bicornuate
- d) Septate

Question 8:

Given below is the hysterosalpingogram of a woman who presented with infertility. Identify the uterine anomaly seen.



- a) Uterus didelphys unicollis
- b) Bicornuate uterus
- c) Complete septate uterus
- d) Cannot be determined

Question 9:

A woman with unilateral renal agenesis presents to your clinic with a history of recurrent abortions. You suspect uterine didelphys. Which of the following would be the next best investigation?

- a) Laparoscopy
- b) Intravenous pyelography
- c) Hysterosalpingography
- d) Ultrasonography

Question 10:

A 24-year-old woman visits the infertility clinic with a complaint of recurrent abortions. Upon investigations, she is found to have a septate uterus. Which of the following corrective procedures has the best obstetric outcome?

- a) Jones metroplasty
- b) Strassmann metroplasty
- c) Tompkins metroplasty

- d) Transcervical resection of septum

Question 11:

What is the sequence of pubertal changes in girls?

- a) Thelarche→Pubarche→Menarche
- b) Pubarche→Thelarche→Menarche
- c) Pubarche→Menarche→Thelarche
- d) Menarche→Thelarche→Pubarche

Question 12:

Which of the following effects of puberty are not mediated by estrogen?

- a) Hair growth
- b) Cervical mucus secretion
- c) Vaginal cornification
- d) Menstruation

Question 13:

You are examining an 18-year-old woman and it appears she has mature adult breasts. What is the Tanner stage of breast development in her?

- a) B3
- b) B4
- c) B5
- d) B6

Question 14:

An adolescent girl is brought to OPD with a history of delayed menses. On examination, you note the following developmental changes. What is the Tanner staging?

- a) Tanner 2
- b) Tanner 3

- c) Tanner 4
- d) Tanner 5

Question 15:

Until which age is the growth rate of boys and girls similar?

- a) 10 years
- b) 10.5 years
- c) 11 years
- d) 11.5 years

Question 16:

A 13-year-old girl presents to the OBG clinic with complaints of heavy menstrual bleeding. She attained menarche 4 months back. What pattern of bleeding would you expect in this patient?

- a) Irregular and often shortened
- b) Irregular and often prolonged
- c) Regular and often shortened
- d) Regular and often prolonged

Question 17:

What is diagnostic criterion for considering puberty to be delayed in girls?

- a) Absence of menstruation by the age of 14 years
- b) Absence of secondary sexual characters by the age of 14 years
- c) Absence of menstruation by the age of 15 years
- d) Absence of secondary sexual characters by the age of 12 years

Question 18:

A 13-year-old girl is brought to the pediatrician with complaints of short stature and absence of menses. On examination, breasts are Tanner's stage 1. What is the most common cause of her condition?

- a) Idiopathic
- b) Constitutional
- c) Polycystic ovarian disease
- d) Turner's syndrome

Question 19:

A 16-year-old girl with primary amenorrhea has the following features. What is the most probable diagnosis?

- a) Testicular feminizing syndrome
- b) Gonadal dysgenesis
- c) Kallmann syndrome
- d) Turner syndrome

Question 20:

In which of the following conditions do secondary sexual characters appear to be normal?

- a) 4 and 5
- b) 1, 3 and 5
- c) 2, 3 and 4
- d) 1, 3, 4 and 5

Question 21:

A 16-year-old girl has come to your clinic with primary amenorrhea. On evaluation, height and secondary sexual characteristics are normal for her age. Ultrasound showed ovaries but no uterus was seen. Her karyotype is 46XX. What is the diagnosis?

- a) Mullerian agenesis
- b) Androgen insensitivity syndrome
- c) Kallman syndrome
- d) Swyer syndrome

Question 22:

A 14-year-old girl presented with cyclical abdominal pain and a midline hypogastric lump. She has not achieved menarche yet. On examination, a bluish bulge is noted on parting the labia. What is the likely cause of her condition?

- a) Bartholin cyst
- b) Transverse vaginal septum
- c) Imperforate hymen
- d) Endometrioma

Question 23:

A 16-year-old girl is brought to the clinic by her mother with complaints of failure of onset of menses. On examination, her height is 150 cm, her breasts are underdeveloped, and sparse pubic hair is present. Ultrasound shows a small uterus and no gonads are visible. Which of the following statements is incorrect about her condition?

- a) Karyotype may have a Y chromosome
- b) LH and FSH levels will be low
- c) Pregnancy can occur with ovum donation
- d) Estrogen and progesterone supplementation is the treatment

Question 24:

A 16-year-old girl with primary amenorrhea has come to your clinic. On examination, thelarche is absent. She has a normal stature, and her karyotype is 46XX. Which investigation would you prefer to perform first?

- a) Serum testosterone
- b) USG pelvis
- c) LH, FSH levels
- d) Hysterosalpingogram

Question 25:

A 25-year-old female came with complaints of 4 months of amenorrhea. Her FSH and LH were elevated above the normal value, thyroid function tests were normal. Which is the next step in management ?

- a) Estimate serum estradiol values

- b) Give progesterone and stop after 10 days for withdrawal bleeding
- c) Check USG for polycystic ovaries
- d) Do a urine pregnancy test

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	a
5	a
6	d
7	d
8	d
9	d
10	d
11	a
12	a
13	c
14	a
15	b
16	b
17	c
18	b
19	b
20	d
21	a
22	c
23	b
24	c
25	a

Detailed Explanations

Solution to Question 1:

The sex of the fetus is determined by Yp11, which is a region present on the short arm of the Y chromosome.

It contains the SRY gene, which produces the protein called the testis determining factor. This causes the gonads to develop into the testes. The target gene for SRY which favours testis differentiation is SOX9.

If SRY gene or Y chromosome are absent, the gonads develop as ovaries. The genes that favor this are:

- WnT4
- Rspo
- DAX1

Solution to Question 2:

Sexual differentiation of the fetus occurs only after the 7th week of development.

The chromosomal and genetic sex of an embryo is determined at fertilization by the kind of sperm that fertilizes the ovum (X or Y). However, male and female morphological characteristics begin developing after the 7th week. At the end of the 6th week of development, it is impossible to distinguish between the two sexes.

Solution to Question 3:

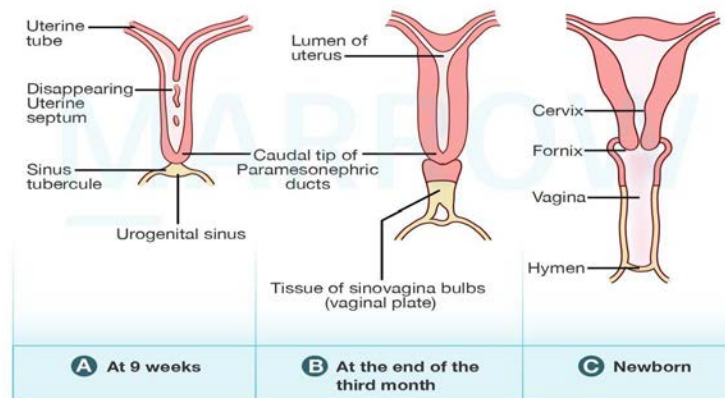
The lower part of the vagina is derived from the urogenital sinus.

Option A: The ovary and the testis are derived from the genital ridge.

Option C: The upper part of the vagina is derived from the paramesonephric duct or Mullerian duct. The uterus develops from the fusion of the right and left paramesonephric ducts in the midline.

Option D: The labia minora and the ventral aspect of the penis are derived from the urogenital folds.

Formation of the uterus and vagina



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Solution to Question 4:

The vaginal epithelium is derived from the epithelium of the urogenital sinus.

The epithelium of urogenital sinus is formed by the endoderm.

Note: Urogenital sinus epithelium stains positive for FOXA1, and Mullerian duct epithelium stains positive for PAX2. In 2017, immunohistochemical testing done on embryos revealed vaginal epithelium showed positive immunostaining for both FOXA1 and PAX2.

Solution to Question 5:

The round ligament in women is homologous to gubernaculum testis in men.

Homologous structures are those that have the same embryological origin.

Solution to Question 6:

The Gartner's duct is not derived from the paramesonephric duct. It is a remnant of the mesonephric/ Wolffian duct.

Solution to Question 7:

The most common developmental abnormality of the uterus is septate uterus, followed by bicornuate uterus.

Solution to Question 8:

In a hysterosalpingogram, the uterine anomaly cannot be determined as the external contour of the uterus cannot be seen.

Hysterosalpingography (HSG) provides excellent information about the interior contour of the uterine cavity. However, it doesn't show the external contour, so it cannot distinguish between uterine anomalies such as didelphys, bicornuate, septate or arcuate uteruses.

Transvaginal 3-D ultrasonography (3D TV-US) is very accurate for the diagnosis and differentiation of uterine anomalies. It is the first and only mandatory step in the assessment of the uterine cavity in patients with a suspected uterine anomaly.

Pelvic MRI without contrast provides excellent delineation of both internal and external uterine contours. It should be reserved for patients in whom 3D TV-US is not possible.

Solution to Question 9:

Ultrasonography (USG) is performed to confirm a diagnosis of uterus didelphys.

Uterine didelphys is a congenital condition that occurs when the incomplete fusion of Mullerian ducts that form the uterus during embryonic development leads to the development of two separate uterine cavities and two cervices, each with its own set of endometrial lining.

It is commonly associated with renal anomalies such as unilateral renal agenesis, severe renal hypoplasia, etc, and is often associated with recurrent pregnancy loss, preterm labour, malpresentation, and prematurity.

3D ultrasound, a non-invasive test, visualizes both the external (serosal) and internal (endometrial) uterine contours, effectively identifying anomalies. Hysterosalpingography (Option C) shows only the internal contour, making it unable to distinguish between conditions like didelphys, bicornuate, septate, or arcuate uterus.



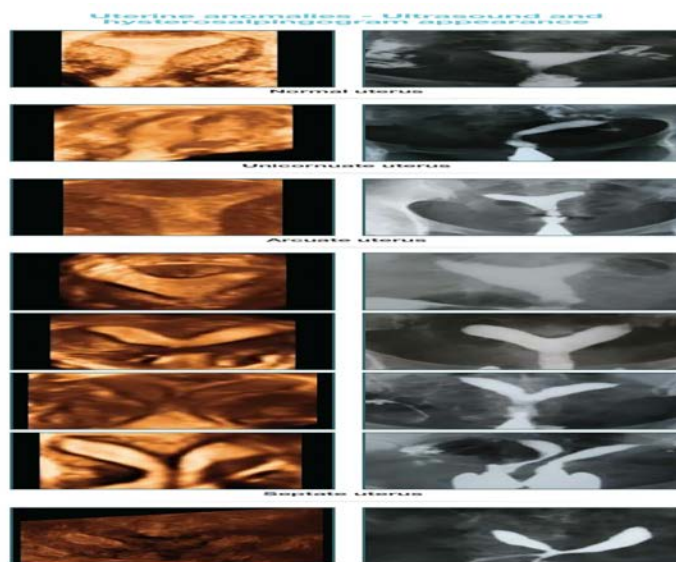
Uterus didelphys

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The USG findings in didelphys uterus are two widely separated uterine horns and a deep fundal indentation.

Laparoscopy and hysteroscopy, being invasive diagnostic procedures, should be reserved for cases where simultaneous surgical treatment is planned following a comprehensive non-invasive assessment of a Mullerian anomaly. (Option A)

Intravenous pyelography is used to diagnose congenital renal anomalies. (Option B)



Solution to Question 10:

Transcervical hysteroscopic resection of the septum has the best obstetric outcome.

The procedure involves the resection of the septum under hysteroscopic guidance. Hence, it doesn't scar the uterus and avoids the necessity of cesarean sections for future pregnancies improving obstetric outcomes.

Metroplasty is the surgical resection of the intrauterine septum. This is done by approaching the uterus transabdominally via 3 methods:

- Jones metroplasty
- Strassmann metroplasty
- Tompkins metroplasty

All of these methods are not preferred nowadays as they cause uterine scarring.

Solution to Question 11:

The sequence of pubertal changes in girls is thelarche → pubarche → menarche.

Puberty marks the normal physiologic transition from childhood to sexual and reproductive maturity. The order of changes in puberty are:

- Thelarche - It refers to breast budding, and is the first visible sign. It indicates the competency of the hypothalamo-pituitary-ovarian axis. It appears between the age of 9-11 years, and is described using Tanner staging.
- Adrenarche - It refers to the generation of androgens (estrogen and testosterone) from the adrenal gland.
- Pubarche - It refers to the development of pubic hair, and occurs usually 6 months after thelarche. It is also described using Tanner staging.
- Menarche - It refers to the first occurrence of menstruation.

Growth spurt usually occurs between ages 10.5 to 13.5 years.

Solution to Question 12:

Pubarche or the development of axillary and pubic hair is not mediated by estrogen. It occurs due to testosterone in both sexes. Around 50% of testosterone comes from the ovaries and the rest from the adrenal gland.

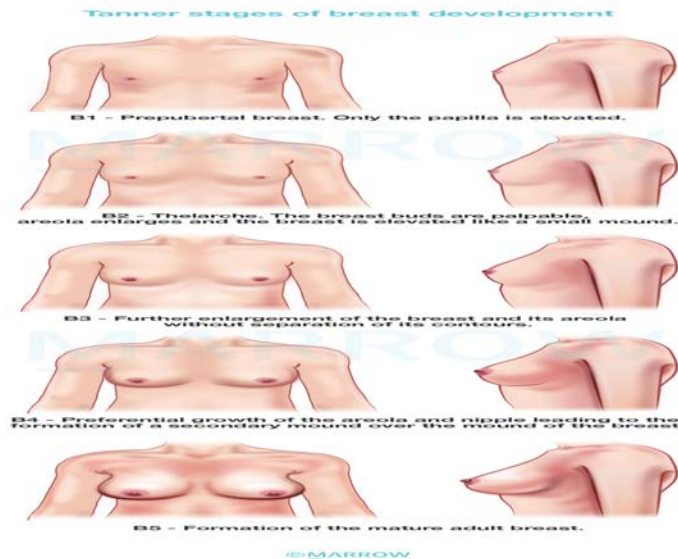
Puberty in girls involves breast development (thelarche), an increase in adrenal androgen secretion (adrenarche), and the beginning of menstrual cycles (menarche).

Estrogen has the following actions during puberty:

- Breast development
- Development of the reproductive tract
- Starting of menstruation (option D)
- Increased fat in the hips
- Regulation of pubertal growth spurt
- Vaginal cornification (option C)
- Cervical mucus production (option B)
- Epiphyseal closure
- Maintenance of bone formation - acts against parathyroid hormone.

Solution to Question 13:

The Tanner staging of the breast, in case of mature adult breast, is denoted as B5.



Solution to Question 14:

The breast elevated as a small mound and pigmented, slightly curved pubic hair is seen in Tanner stage 2.

Tanner staging	Breast changes	Pubic hair changes
Stage 1	B1 - Denotes the prepubertal breast. At this infantile stage, only the papilla is elevated.	P1 - No pubic hair
Stage 2	B2 - Denotes thelarche. The breast buds are palpable, areola enlarges and the breast is elevated like a small mound.	P2 - The appearance of pigmented hair sparsely over the labia, either long or slightly curved.
Stage 3	B3 - There is further enlargement of the breast and its areola without separation of its contours.	P3 - Darker, coarser and curly hair
Stage 4	B4 - Preferential growth of the areola and nipple leads to formation of a secondary mound over the mound of the breast.	P4 - Preadult stage cover short inverted triangular distribution.
Stage 5	B5 - Formation of the mature adult breast.	P5 - Adult inverted triangular distribution.

Solution to Question 15:

The growth rate of boys and girls is similar till the age of 10.5 years.

The growth spurt in girls overtakes that of boys for 1-2 years and reaches a plateau. In boys, the growth spurt starts later and continues for a longer duration. As a result, boys grow taller than girls in terms of final height.

Solution to Question 16:

The given case scenario is suggestive of puberty menorrhagia. Menstruation tends to be irregular and often prolonged.

Since the hypothalamo-pituitary-ovarian axis is maturing, initial cycles are anovulatory. Therefore, the pattern of bleeding is irregular. Also, as the endometrium is under the prolonged influence of estrogen, it gets thickened more than normal. Shedding of this thickened endometrium leads to prolonged bleeding.

Solution to Question 17:

Puberty would be considered to be delayed in any girl who has not started menstruating by the age of 15 years.

Delayed puberty in girls can be declared on the basis of 2 grounds. These are:

- No secondary sexual characters are present by 13 years of age.
- Secondary sexual characters are present, but no menses have occurred by 15 years of age.

If any one of these is present, it's considered to be a case of delayed puberty.

Precocious puberty is considered if breast development or onset of puberty occurs in children less than 8 years of age.

Solution to Question 18:

The given case scenario is suggestive of delayed puberty. The most common cause of this condition is constitutional.

Puberty is considered to be delayed if:

- Breast development doesn't occur by 13 years in girls.
- Testicular enlargement doesn't occur by 14 years in boys.

Causes of delayed puberty:

- Constitutional or physiological delay (most common)
- Chronic anovulation - PCOS
- Anatomical:

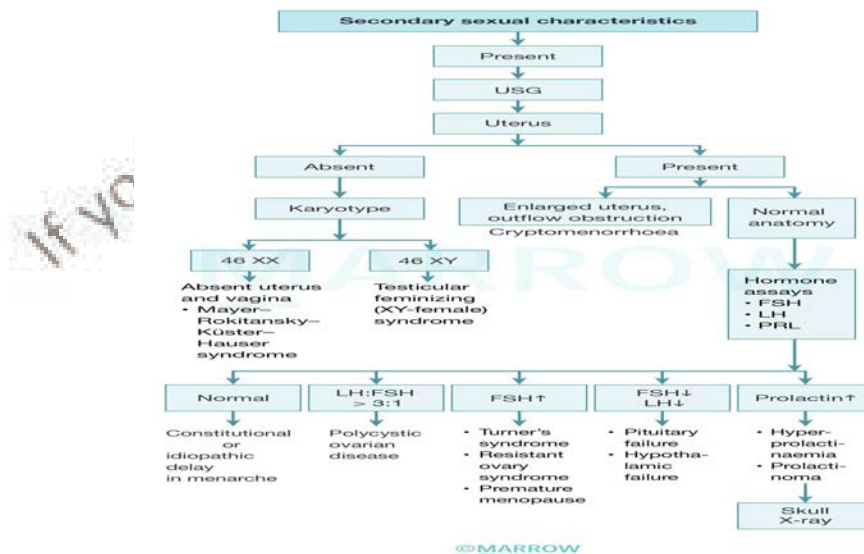
- Outlet obstruction
- Gonadal agenesis
- Androgen insensitivity syndrome
- Hypergonadotropic hypogonadism - E.g., Premature ovarian failure
- Hypogonadotropic hypogonadism - e.g., Kallman syndrome

Precocious puberty is considered if signs of puberty are seen in children less than 8 years of age.

Causes of precocious puberty:

- Idiopathic (most common)
- Central
- CNS tumors
- Infections
- Tumors producing
- Estrogen
- Testosterone
- hCG
- Congenital adrenal hyperplasia

The protocol followed for evaluation of delayed puberty in girls is given below.



Solution to Question 19:

The scenario of a 16-year-old girl with absent secondary sexual characters, normal stature, with XX karyotype and high FSH and LH is suggestive of pure gonadal dysgenesis.

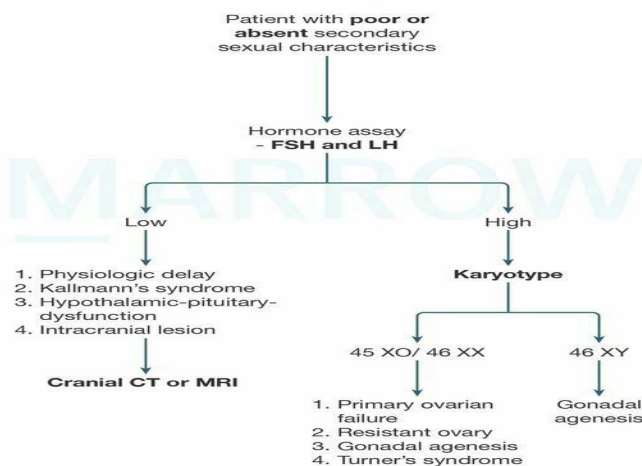
46 XX and 46 XY gonadal dysgenesis without the phenotype of Turner's syndrome (short stature, webbed neck) is described as pure gonadal dysgenesis. It is characterized by the presence of primary amenorrhea with or without normal secondary sexual characteristics.

Option A: Testicular feminizing syndrome occurs in karyotype XY. Secondary sexual characters are present and the uterus is absent.

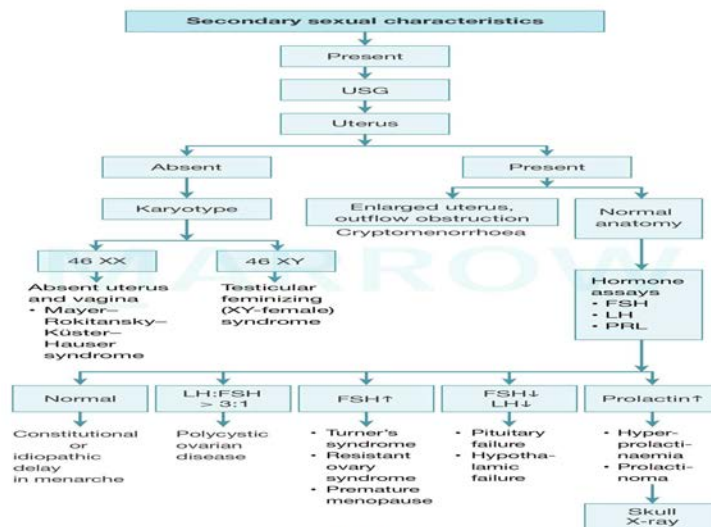
Option C: Kallmann syndrome is a form of hypogonadotropic hypogonadism. GnRH synthesis is reduced, therefore FSH and LH are low, and secondary sexual characters are absent.

Option D: Turner syndrome is associated with short stature and karyotype Xo. Secondary sexual characters and the uterus is present, but the ovaries are streaked.

Evaluation of primary amenorrhea



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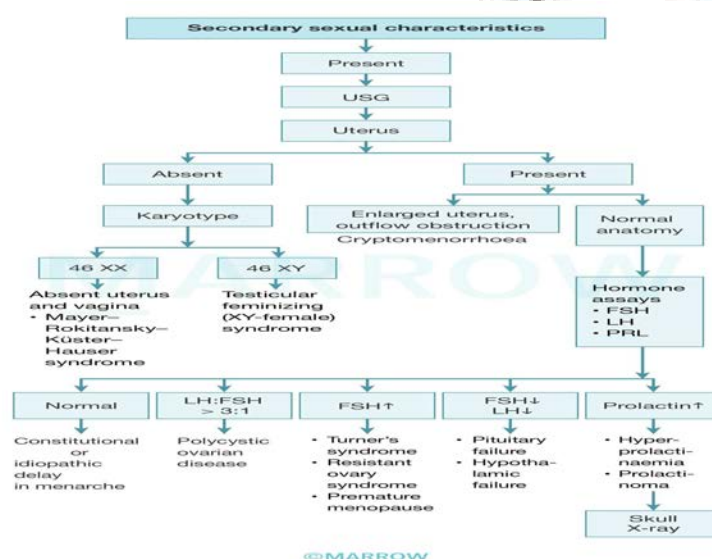
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Solution to Question 20:

Secondary sexual characters appear normal in the following:

- Imperforate hymen
- Transverse vaginal septum
- Absent vagina and functioning uterus
- Absent vagina and nonfunctioning uterus - Mayer–Rokitansky–Küster–Hauser syndrome [MRKH]
- Resistant ovary syndrome

In androgen insensitivity syndrome, secondary sexual characteristics may or may not develop normally. The genotype is XY, however a female phenotype develops. These patients have well-developed breasts, but they have scanty body hair. This is because growth of sexual hair is only mediated via androgens. Thus, secondary sexual characters cannot be considered normal.



Solution to Question 21:

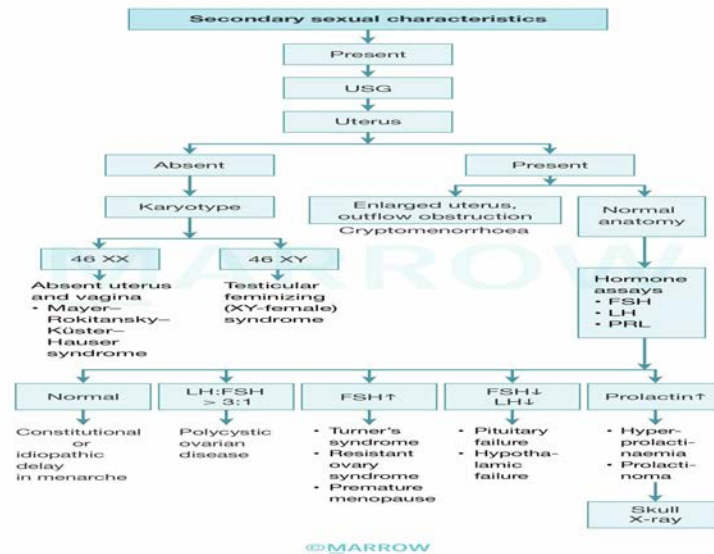
The given case scenario is suggestive of Mullerian agenesis. It is differentiated from androgen insensitivity syndrome as follows:

Option C: Kallman syndrome is hypogonadotropic hypogonadism. There is a lack of GnRH synthesis, and sexual development is delayed. Uterus is present.

Option D: Swyer syndrome is pure gonadal dysgenesis. The karyotype is XY and the uterus is underdeveloped.

Presentation	Mullerian agenesis	Androgen insensitivity /Testicular feminization syndrome
Inheritance pattern	Sporadic	X-linked recessive
Karyotype	46, XX	46, XY

Presentation	Mullerian agenesis	Androgen insensitivity / Testicular feminization syndrome
Breast development	Present	Present
Axillary and pubic hair	Present	Absent
Uterus	Absent	Absent
Gonad	Ovary	Testis (Intraabdominal)
Testosterone	Female levels	Male levels
Associated anomaly	Present	Absent



Solution to Question 22:

The given case scenario is suggestive of an imperforate hymen, which has led to cryptomenorrhoea. This is a condition where menstruation occurs but is not visible due to an obstruction of the outflow tract.

The menstrual fluid gets collected in the reproductive tract. When this collection occurs in the vagina, it is known as hematocolpos. If the collection extends to the uterus, it is known as hematometra. The collection of blood in the fallopian tubes is called hematosalpinx.

The most common causes of cryptomenorrhea are:

- Intact/imperforate hymen - a bluish convex bulge is seen.
- Transverse vaginal septum - a pinkish concave covering is seen

Ultrasound pelvis confirms the diagnosis.

Management:

Imperforate hymen - A simple cruciate incision of the hymen allows the free drainage of the collected menstrual blood, and leads to normal reproductive function.

Transverse vaginal septum - Extensive dissection and vaginoplasty is required.

Option A: Bartholin's cyst appears as a swelling on the inner side of the labia majora. Larger cysts can bulge across the introitus. Small cysts can remain asymptomatic.

Option D: Endometriomas are endometriotic lesions seen on the ovary. They present as adnexal masses. Endometriosis is a complication in cryptomenorrhea, due to retrograde menstruation.

Solution to Question 23:

The given clinical scenario is suggestive of gonadal dysgenesis. Hence, the patient will have high LH and FSH levels as it is a condition of hypergonadotrophic hypogonadism.

Gonadal dysgenesis is the most frequent cause of premature ovarian failure. In this disorder, a normal complement of germ cells is present in the early fetal ovary. However, oocytes undergo accelerated atresia, and the ovary is replaced by a fibrous streak. This is called a streak gonad.

There are two types of gonadal dysgenesis:

1. Pure gonadal dysgenesis:

- Karyotypes seen:
- 46 XX is seen in pure gonadal dysgenesis due to FSH gene mutation.
- 46 XY - Swyer syndrome
- Associated with primary amenorrhoea
- They are phenotypically female due to:
- Absence of testosterone
- Absence of Anti-Müllerian hormone secretion by testes
- The underdeveloped gonad ultimately fails, which is indicated by elevated gonadotropin (LH, FSH) levels
- The presence of a Y chromosome can result in a germ cell tumor in the dysgenetic gonad
- Treatment:
- Hormone replacement therapy - with estrogen and progesterone
- Infertility - Pregnancy can occur with a donated ovum

2. Mixed gonadal dysgenesis:

- Karyotypes seen:
- 45 Xo - e.g. Turner's syndrome
- 46 XX
- 46 XY
- Mosaicism
- Dysgenetic gonad can develop into a germ cell tumor.

Solution to Question 24:

The given case scenario is suggestive of primary amenorrhea. The first line of investigation in primary amenorrhea should be hormonal assay (LH, FSH, estrogen).

Following this investigation, USG pelvis and MRI pelvis can be performed to enable the anatomical assessment.

LH and FSH levels help to assess whether it is a hypogonadotropic or hypergonadotropic condition. The hormone replacement therapy is planned based on LH and FSH levels.

The various types of hypogonadism are as follows:

Note: In a patient with primary amenorrhea with absent sexual characteristics, the first line of investigation is hormonal assay (LH and FSH levels).

Type of Hypogonadism	LH/FSH	Estrogen	Primary defect
Hypergonadotropic	High	Low	Ovary
Hypogonadotropic	Low	Low	Hypothalamus/Pituitary
Eugonadotropic	Normal	Normal	Varied

Solution to Question 25:

The given clinical scenario is suggestive of premature ovarian failure. The next step in management would be to check serum estradiol levels to evaluate amenorrhea.

Categories of amenorrhea are based on gonadotrophin and estrogen levels as follows:

- Hypergonadotropic hypogonadism - also known as premature ovarian failure.
- Hypogonadotropic hypogonadism
- Eugonadotropic hypogonadism

Hypergonadotropic hypogonadism/premature ovarian failure is present when:

- Ovarian function is decreased/absent - Serum estrogen levels are low. Serum prolactin is high.
- LH and FSH (gonadotropins) levels are high, as sex-steroid feedback is absent.

Differential diagnosis of premature ovarian failure:

- Gonadal dysgenesis
- Specific gene defects
- Fragile X (FMR1 premutation)
- Galactosemia

- Iatrogenic
- Ovarian surgery
- Gonadal radiation
- Systemic chemotherapy
- Autoimmune disease
- Toxins
- Viruses
- Miscellaneous

Note: Since FSH, LH, and TFT are already estimated, it suggests that all preliminary tests (including UPT) are done. Moreover, serum FSH and LH values are quite low during pregnancy. This is due to the inhibitory effects of elevated estrogen and progesterone.

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Physiological Changes During Pregnancy

Question 1:

Which of the following is true about vaginal changes in a normal pregnancy?

- a) Decreased number of Lactobacilli
- b) Increased glycogen metabolism in the epithelium
- c) pH $\geq$ 6.8
- d) Epithelial thinning

Question 2:

In pregnancy, vaginal cytology shows a predominance of _____.

- a) Intermediate cells
- b) Superficial cells
- c) Basal cells
- d) Parabasal cells

Question 3:

A 28-year-old woman has come for pre-conceptional counselling. Her BMI is 30 kg/m² and she is worried about weight gain during pregnancy. How much weight should she gain during pregnancy?

- a) 5-9 kg
- b) 8-11 kg
- c) 10-13 kg
- d) 14-16 kg

Question 4:

In pregnancy, the diaphragm is elevated by _____.

- a) 3 cm
- b) 2 cm

- c) 4 cm
- d) 5 cm

Question 5:

A pregnant woman at her 30th week of pregnancy undergoes a pulmonary function test. Provided there is no respiratory abnormality, which of the following would be decreased?

- a) Respiratory rate
- b) Vital capacity
- c) Functional residual capacity
- d) Inspiratory capacity

Question 6:

Which of the following statements is true about prolactin levels in pregnancy?

- a) Lowest in pregnancy and increases after delivery
- b) Highest at term and reduces after delivery
- c) Increases during pregnancy and plateaus during lactation
- d) Pulsatile release during pregnancy

Question 7:

Which of the following produce estrogen and progesterone in the first two months of pregnancy?

- a) Corpus albicans
- b) Adrenal gland
- c) Placenta
- d) Corpus luteum

Question 8:

A pregnant woman presents to an antenatal clinic for the first time during the second trimester. In view of a family history of thyroid disorders, a thyroid function test is sent for. Altered levels of which of the following would not be a cause for concern?

- a) Free T₃ level
- b) Free T₄ level
- c) Total T₃ level
- d) TSH level

Question 9:

Which of the following do not synthesise relaxin during pregnancy?

- a) Placenta
- b) Fetal adrenal gland
- c) Corpus luteum
- d) Decidua

Question 10:

Which among the given features is not a major physiological change in pregnancy?

- a) Insulin resistance
- b) Increase in BMR
- c) Hypothyroidism
- d) Increase in blood volume

Question 11:

A pregnant woman with diabetes mellitus is advised to increase her insulin dose as pregnancy is characterised by insulin resistance. Which of the following is not implicated in this phenomenon?

- a) Human placental lactogen
- b) Progesterone
- c) hCG
- d) Estrogen

Question 12:

Which of the following changes are not seen in the renal system during pregnancy?

- a) Increase in kidney size
- b) Increase in serum creatinine
- c) Increased GFR
- d) Increased renal blood flow

Question 13:

Which of the following is increased during pregnancy?

- a) Serum potassium
- b) Hemoglobin concentration
- c) Total oxygen carrying capacity of blood
- d) Serum iron concentration

Question 14:

Which of the following is not a physiological change seen in pregnancy?

- a) Increase in intravascular volume
- b) Increase in cardiac output
- c) Decrease in systolic BP
- d) Increase in peripheral vascular resistance

Question 15:

A primigravida presents in the 2nd trimester complaining that her slippers do not fit anymore. On examination, her pulse is 88/min and blood pressure is 130/84mm Hg. You note bilateral pitting edema up to the level of ankles. What would you advise her?

- a) Oral furosemide
- b) Oral labetalol
- c) Restrict salt intake
- d) Bed rest

Question 16:

Which of the following is not a physiological change seen during pregnancy?

- a) Pulmonary blood flow is increased by 2,500 mL/min
- b) Blood flow to the skin decreases to 300 mL/min
- c) Blood flow to the uterus increases to 750 mL/min
- d) Renal blood flow increases by 400 mL/min

Question 17:

A 23-year-old primigravida presents to the antenatal clinic at 30 weeks of gestation. During examination, which of the following cardiovascular findings would not be a cause for alarm?

- a) S₃
- b) S₄
- c) Fixed split S₂
- d) Pericardial knock

Question 18:

Which of the following coagulation factors are increased in pregnancy?

- a) 2, 3, 4
- b) 2, 3, 4, 5
- c) 1, 2, 3, 4, 5
- d) 1, 2, 3

Question 19:

A 24-year-old primigravida at 38 weeks of gestation presents to the antenatal clinic with complaints of dizziness on lying down for the past one week. She feels better on sitting up or turning to the side. What is the cause of this condition?

- a) Increased abdominal pressure
- b) Inferior vena caval compression
- c) Aortic compression
- d) Increased uteroplacental blood flow

Question 20:

Which of the following is true regarding the sensitivity of uterine musculature?

- a) Enhanced by progesterone
- b) Inhibited by estrogen
- c) Enhanced by both estrogen and progesterone
- d) Enhanced by estrogen and inhibited by progesterone

Question 21:

A pregnant woman has cramps in her legs. What will you suggest her to relieve the pain?

- a) Extend knee and dorsiflex foot
- b) Flex knee and dorsiflex foot
- c) Extend knee and plantar flex foot
- d) Flex knee and plantar flex foot

Question 22:

Theca lutein cysts are not associated with which of the following conditions?

- a) Anti-D allo-immunization
- b) Fetal hydrops
- c) Normal pregnancy
- d) Hyperprolactinemia

Question 23:

In which trimester does acute fatty liver manifest most commonly during pregnancy?

- a) First trimester
- b) Second trimester
- c) Third trimester
- d) Both (a) and (b)

Question 24:

What does the term 'placental sign' denote?

- a) Alteration of fetal heart rate on pressing the head into the pelvis
- b) Spotting on the expected date of period in early months of pregnancy
- c) Permanent lengthening of the cord in 3rd stage of labor
- d) Slight gush of bleeding in 3rd stage of labor

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	c
5	c
6	b
7	d
8	c
9	b
10	c
11	c
12	b
13	c
14	d
15	d
16	b
17	a
18	d
19	b
20	d
21	a
22	d
23	c
24	b

Detailed Explanations

Solution to Question 1:

There is an increase in the metabolism of glycogen in the vaginal epithelium during normal pregnancy.

The changes in the vagina during pregnancy include:

- Increased vascularity and hyperemia resulting in a violet/purplish color characteristic of Chadwick's sign
- Vaginal epithelium undergoes marked proliferation (option D) with the extension of the squamocolumnar junction downwards
- Vaginal pH decreases and becomes more acidic (option C) and varies from 3.5-6.0.
- Increase in the number of Lactobacillus acidophilus (option A). It metabolizes the glycogen stores and causes an increased production of lactic acid. Therefore, the pH becomes acidic.
- Increased volume of cervical secretion during pregnancy results in the production of thick white vaginal discharge.

Solution to Question 2:

During pregnancy, vaginal cytology shows a predominance of intermediate cells.

Solution to Question 3:

The patient in the given question is obese. The recommended gestational weight gain in such a patient is 5-9 kg.

Ideally, weight gain should depend on the pre-pregnancy BMI (body mass index) level. The gestational weight gain recommended for each BMI category by the Institute of Medicine (IOM) are as follows:

Category	BMI (in kg/m ²)	Recommended weight gain
Underweight	<18.5	12.5–18 kg
Normal	18.5–24.9	11.5–14 kg
Overweight	25–29.9	7–11.5 kg
Obese	>30	5–9.1 kg

Solution to Question 4:

The diaphragm is elevated by 4 cm in pregnancy.

The anatomical changes in the respiratory system during pregnancy are:

- Elevation of diaphragm by 4 cm
- Transverse diameter of chest increases by 2 cm
- Chest circumference increases by 5-7 cm
- Subcostal angle increases from 68 degrees to 103 degrees

Solution to Question 5:

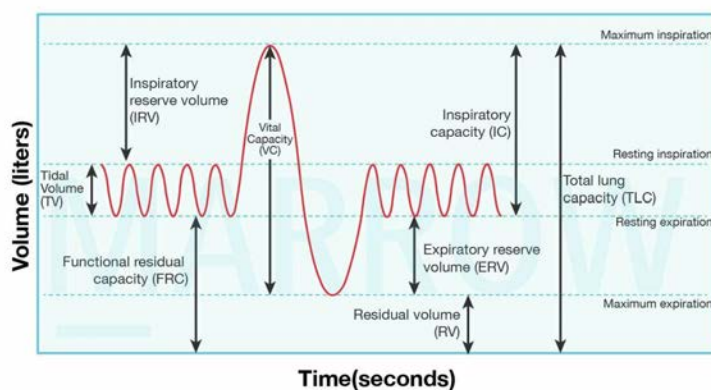
Functional residual capacity decreases in pregnancy.

The changes in the respiratory system during pregnancy are:

(Mnemonic: TIMM increases in pregnancy)

Increase	Decrease	Unaffected
Tidal volume Inspiratory capacity Minute O ₂ uptake Minute ventilation	Functional residual capacity Expiratory reserve volume Residual volume Total lung capacity	Respiratory rate Vital capacity Inspiratory reserve volume

Lung volumes and Capacities



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Solution to Question 6:

The maternal plasma prolactin level is highest at term and reduces after delivery. Despite this high level, milk secretion does not occur during pregnancy due to the action of estrogen and

progesterone.

Prolactin secretion returns to the nonpregnant level a few weeks after delivery. The levels of estrogen and progesterone fall and hence, prolactin can promote milk secretion even at a lesser concentration.

Solution to Question 7:

In the first two months of pregnancy, estrogen and progesterone are produced by the corpus luteum.

It functions maximally during the second month of pregnancy and thereafter contributes relatively little. Estrogen is synthesized in the placenta from the 7th week and progesterone from the 8th week.

Solution to Question 8:

Total T3 levels are altered throughout pregnancy.

Total T3 and total T4 increase due to stimulation of thyroid tissue by hCG during the early period of gestation. However, there is no change in free T3 and T4 due to an increase in thyroid-binding globulin (TBG).

This leads to a transient fall in TSH during the early weeks of gestation. The level of TSH becomes normal during the second and third trimester.

Hormone secretion during pregnancy:

- Increased
- GH
- Insulin
- Prolactin
- Aldosterone
- Testosterone
- Total T3, T4
- Decreased
- LH
- FSH
- DHEAS (dihydroepiandrosterone)
- Unchanged
- TSH
- ADH (vasopressin)

- Free T₃, T₄

Solution to Question 9:

Relaxin is not synthesised by the fetal adrenal gland in a pregnant woman. It is synthesised by corpus luteum, decidua and placenta.

Relaxin is a placental protein hormone. Plasma levels peak between 8 and 12 weeks of gestation and decline thereafter to lower levels that persist until term.

Functions of relaxin are:

- Relaxation of myometrium and quiescence of early pregnancy
- Enhancement of the glomerular filtration rate
- Endometrial vascular remodelling by inducing secretion of VEGF

Solution to Question 10:

Among the given features hypothyroidism is not a major physiological change in pregnancy.

The α -subunits of hCG and TSH are identical. An increase in hCG during pregnancy stimulates thyroid and leads to a decrease in TSH levels. The thyroid undergoes enlargement and boosts the production of hormones. This is accompanied by increased production of thyroid-binding globulin(TBG). All these changes increase total T₄ and T₃ in the first trimester. But the physiologically important free T₄ and T₃ levels remain unchanged. Hence, physiologically pregnant women remain euthyroid.

Option A: Normal pregnancy is characterized by insulin resistance and hyperinsulinemia. Human placental lactogen(hPL) secreted during pregnancy, helps in adaptation to fetal energy requirements. Insulin resistance causes prolonged hyperglycemia and glucose flow to the fetus. It also favours protein synthesis. To counterbalance the increased insulin resistance and prevent maternal hyperglycemia, maternal insulin levels are increased.

Option B: By the 3rd trimester, there is an increase in the maternal basal metabolic rate by 10 to 20% in comparison to a nonpregnant woman.

Option D: During normal pregnancy, mean arterial pressure and vascular resistance decrease, while blood volume and cardiac output rate increase. This helps in meeting metabolic demands of the enlarged uterus, provide nutrients to rapidly growing placenta and fetus, safeguard mother against the effects of parturition-associated blood loss.

Solution to Question 11:

hCG is not implicated in insulin resistance during pregnancy.

As the fetus is dependant on maternal glucose for sustenance, the blood glucose levels are maintained at a high level in the mother. In response to this, the insulin level in the mother increases, which reduces the maternal blood glucose. To counteract this, the placenta secretes certain hormones which increase insulin resistance in the mother. This resistance is maximal between 24 to 28 weeks.

Hormones increasing insulin resistance include:

- Human placental lactogen - Increases lipolysis and liberation of free fatty acids
- Progesterone and estrogen
- Placental growth hormone - Major determinant of maternal insulin resistance after mid-pregnancy
- Cortisol
- Prolactin

Solution to Question 12:

Serum creatinine levels decrease during normal pregnancy from a mean of 0.7 mg/dL to 0.5 mg/dL.

Physiological changes in the renal system during pregnancy include:

- Increase in:
 - Renal blood flow(+80%)
 - GFR(+50%)
 - Creatinine clearance
 - Glucosuria
- Decrease in:
 - Plasma osmolality
 - Creatinine
 - Uric acid
 - Na/K/Cl

Anatomical changes in the renal system during pregnancy include:

- Both kidneys enlarge by 1 cm
- Hydroureter and hydronephrosis - More common on the right side due to dextrorotation of the uterus
- Congestion of the bladder causing decreased bladder capacity
- Increased bladder pressure and intraurethral pressure

Solution to Question 13:

The total oxygen-carrying capacity of the blood increases during pregnancy.

During pregnancy, there is an increase in total haemoglobin (Hb) resulting in increased total oxygen-carrying capacity of the blood. However, there is a relative decrease in Hb concentration (hematocrit) due to a greater increase in plasma volume. This decreases oxygen-carrying capacity per unit volume of blood.

Maternal iron stores are not sufficient to meet the increased demand during pregnancy and hence the serum iron and ferritin concentration falls.

The plasma volume expansion also lowers serum potassium.

Solution to Question 14:

Decrease in peripheral vascular resistance is a physiological change seen in pregnancy.

Cardiovascular parameters that increase in pregnancy are:

- Cardiac output
- Stroke volume
- Heart rate

Cardiovascular parameters that decrease in pregnancy are:

- Peripheral vascular resistance (due to smooth muscle relaxation by progesterone)
- Both diastolic BP and systolic BP (decrease in DBP > SBP)
- Mean arterial BP
- Artery venous O₂ gradient

Solution to Question 15:

The clinical features are suggestive of physiological edema during pregnancy for which the most appropriate treatment is bed rest.

It occurs due to increased venous pressure of the inferior extremities by the gravid uterus pressing on the common iliac veins.

Features of physiological edema in pregnancy are:

- Slight degree of ankle edema
- Usually confined to one leg, more commonly the right leg
- Not associated with any other features of preeclampsia (like BP > 140/90 mm Hg, proteinuria)
- Disappears with rest
- Other cardiac, renal and hematological pathologies are absent

Solution to Question 16:

The blood flow to the skin and mucous membranes increases during pregnancy to a maximum of 500 mL/min by the 36th week. Symptoms experienced during pregnancy such as increased heat sensation, sweating or having a stuffy nose can be explained by the increased blood flow.

Regional distribution of blood flow during pregnancy:

- Uterine blood flow - increased from 50 mL/min in the non-pregnant state to about 750 mL near term.
- Pulmonary blood flow - increased by 2,500 mL/min.
- Renal blood flow - increases by 400 mL/min at 16th week and remains at this level till term.

Solution to Question 17:

S₃ is normally heard during pregnancy.

The third heart sound (S₃) occurs due to rapid diastolic filling in hypervolemic states such as pregnancy, anemia or thyrotoxicosis. Hypervolemia is a normal physiological change during pregnancy.

The auscultation of a fixed split S₂, fourth heart sound (S₄) or a pericardial knock are suggestive of cardiac pathology.

Solution to Question 18:

All clotting factors increase in pregnancy except factors XI and XIII.

Pregnancy is a hypercoagulable state with physiological changes such as:

- All clotting factors are increased (except factors XI and XIII)
- Fibrinogen levels are also increased
- The viscosity of blood decreases due to hemodilution
- Clotting time and bleeding time is unaffected

The most common cause of decreased platelet count in pregnancy is benign gestational thrombocytopenia. This is due to hemodilution and increased platelet consumption and the women are generally asymptomatic.

Solution to Question 19:

The clinical features are suggestive of supine hypotension syndrome due to inferior vena caval compression.

During late pregnancy, the gravid uterus produces a compression effect on the inferior vena cava (IVC) when the patient is in the supine position. To compensate, there will be an opening up of the collaterals of paravertebral and azygous veins.

When collaterals fail to open, venous return of the heart is reduced leading to hypotension, tachycardia, and syncope. The normal blood pressure is quickly restored by turning the patient to lateral position.

Solution to Question 20:

The sensitivity of uterine musculature is enhanced by estrogen and inhibited by progesterone.

Estrogen stimulates the synthesis of myometrial contractile proteins through cAMP. This increases the excitability of uterine muscles and the frequency of action potentials.

Progesterone decreases the frequency and intensity of uterine contractions. This helps to prevent the expulsion of the implanted ovum. However, it increases the amplitude of contraction.

Solution to Question 21:

Extension at the knee joint and dorsiflexion at the ankle joint helps in stretching the calf muscles which relieves the pain.

Leg cramps are minor ailments during pregnancy mostly due to a deficiency of diffusible serum calcium or elevation of serum phosphorus. Management includes massaging the leg, stretching exercises, application of local heat, and supplementation of vitamin B1 and calcium.

Solution to Question 22:

Theca lutein cysts are not seen in hyperprolactinemia.

Theca-lutein cysts are benign ovarian lesions resulting from exaggerated physiological follicle stimulation. They are usually associated with markedly elevated serum levels of hCG.

Theca lutein cysts are found in association with:

- Large placenta
- Diabetes
- Anti-D alloimmunization
- Multifetal gestations
- Fetal hydrops
- Gestational trophoblastic disease - Molar pregnancy

- Reduced hCG clearance - Chronic renal failure

They may also be found in normal pregnancy due to an exaggerated response of the ovaries to normal circulating levels of hCG.

Solution to Question 23:

Acute fatty liver of pregnancy (AFLP) occurs most commonly in the third trimester or early postpartum period.

Risk factors include Pre-eclampsia/HELLP syndrome, male fetus, previous history of AFLP, and multiple gestations.

AFLP (acute yellow atrophy or acute fatty metamorphosis) is the most common cause of acute liver failure in pregnancy. It usually occurs in the later stages of pregnancy, particularly in the third trimester. It occurs due to fetal deficiency of long-chain acyl-coenzyme A dehydrogenase (LCHAD) or poor placental mitochondrial function.

The patients present with persistent nausea and vomiting, progressive jaundice, and epigastric pain. Features like acute hemolysis, hypoglycemia, hypertension, proteinuria, thrombocytopenia, and edema can also be present. Symptoms may progress to fulminant hepatic failure with encephalopathy, disseminated intravascular coagulation, and death.

Liver function tests such as AST, ALT, and serum bilirubin levels are raised. Hypofibrinogenemia, hypocholesterolemia, and prolonged clotting time can be seen. An elevated d-dimer is seen. Microscopy reveals a characteristic microvesicular fatty infiltration.

The condition worsens till delivery after which the hepatic dysfunction resolves. A trial of labor by induction or delivery by Cesarean is done. Intensive supportive measures including blood transfusion should be provided.

The clinical and laboratory findings of acute liver diseases in pregnancy are as follows:

The condition worsens till delivery after which the hepatic dysfunction resolves. Appropriate management is immediate termination of pregnancy irrespective of gestational age by induction or delivery by Cesarean (for obstetric reasons) is done. Intensive supportive measures including blood transfusion should be provided.

Disorders	HELLP syndrome	Obstetric cholestasis	Acute fatty liver of pregnancy	Viral hepatitis
Onset in pregnancy	Mid to late	Late	Late	Variable
Differentiating features	Epigastric pain, hypertension, elevated indirect bilirubin	Pruritus, elevated direct bilirubin	Signs of liver failure like hypoglycemia	Fever
AST (U/L)	≤300	≤200	200-800	2000+
Bilirubin (mg/dL)	1-4	1-5	4-10	5-20
Platelets	↓↓	Normal	↓↓	↓
Fibrinogen	Normal	Normal	↓↓↓	Normal

Disorders	HELLP syndrome	Obstetric cholestasis	Acute fatty liver of pregnancy	Viral hepatitis
D-dimer	↑	Normal	↑	Normal
Prothrombin time	Normal	Normal	↑↑	↑
Hemolysis	↑↑	No	↑↑↑	No

Solution to Question 24:

Spotting on the expected date of menstruation in the early months of pregnancy is known as placental sign.

Cyclic bleeding may occur up to 12 weeks of pregnancy until the decidual space is obliterated by the fusion of decidua parietalis with decidua capsularis. The bleeding is usually scanty and lasts for a shorter duration than in a regular menstrual cycle.

Option A: Slowing of the fetal heart rate (FHR) on pressing the head down into the pelvis is suggestive of placenta previa (especially of posterior type) and is called Stallworthy's sign. The fetal heart rate recovers as soon as the pressure is released.

Options C and D: Permanent lengthening of the cord and slight gush of bleeding in 3rd stage of labor are suggestive of separation of the placenta.

Diagnosis of Pregnancy and Antenatal Care

Question 1:

What is the duration of the 4th trimester in pregnancy?

- a) 6 weeks postpartum
- b) 12 weeks postpartum
- c) 4 weeks postpartum
- d) 38 to 40 weeks of pregnancy

Question 2:

What is the ideal number of antenatal visits?

- a) 12-14
- b) 6-8
- c) 7-9
- d) 10-11

Question 3:

Which of the following statements is not true about parity?

- a) Parity refers to the number of previous pregnancies reaching viability
- b) If a woman has twins, the parity remains one
- c) If a woman has a stillborn child, her parity decreases
- d) Multipara is a woman who has completed 2 or more pregnancies past the period of viability

Question 4:

A 30-year-old woman came to the OPD with a twin pregnancy at 10 weeks of gestation. She has a 3-year-old female child who was born at term and had 2 first trimester abortions previously. Which of the following is the correct representation of her obstetric score.

- a) G4 P2 L1 A2
- b) G4 P1 A2 L1

- c) G5 Po L1 A2
- d) G5 P1 A2 L1

Question 5:

A lady who is 38 weeks pregnant comes to the OPD for a routine checkup. She has a history of normal twin delivery at term 4 years ago. What is her gravida and para score?

- a) G2P2
- b) G2P1
- c) G3P2
- d) G3P1

Question 6:

A woman whose obstetric score is G2 P1 L1 had an LMP of Jan 9th 2017. Calculate the EDD according to Naegele's rule.

- a) September 19
- b) October 19
- c) October 16
- d) September 16

Question 7:

The recommended calorie intake for a pregnant woman in the second trimester, who is a sedentary worker would increase by _____.

- a) 600 kcal/day
- b) 520 kcal/day
- c) 450 kcal/day
- d) 350 kcal/day

Question 8:

A pregnant lady of 49 kg is in her first trimester. What are her additional calorie requirements in this trimester?

- a) 500
- b) 300
- c) Continue same diet
- d) 600

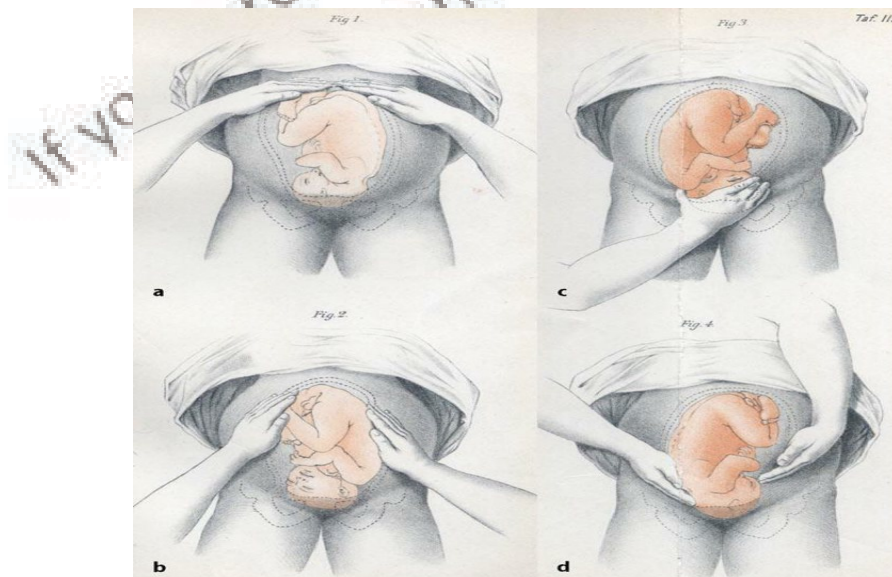
Question 9:

A 22-year-old pregnant woman at 13 weeks of gestation informs you about accidental exposure to radiation from a chest X-ray at 7 weeks of gestation. What should be the next step advised to her?

- a) Continue pregnancy
- b) Terminate pregnancy
- c) Pre-invasive diagnostic test
- d) Chromosome screening

Question 10:

You notice an obstetrics resident performing the following manoeuvres on a 37-week primigravida. Choose the correctly matched set.



- a) 1-a, 2-c, 3-b, 4-d
- b) 1-a, 2-b, 3-c, 4-d
- c) 1-b, 2-a, 3-d, 4-c

d) 1-c, 2-b, 3-a, 4-d

Question 11:

A dusky hue of the vestibule and anterior vaginal wall visible at about 8 weeks of pregnancy is called _____.

- a) Hegar's sign**
- b) Chadwick's sign**
- c) Oslander's sign**
- d) Palmer's sign**

Question 12:

A primigravida comes for a routine ANC visit at 22 weeks of gestation. Which of the following signs would you not expect?

- a) Chloasma**
- b) Quickening**
- c) Lightening**
- d) Braxton Hick's contraction**

Question 13:

Which of the following statements is not true regarding uterine souffle?

- a) It is a soft blowing systolic murmur heard on the sides of pregnant uterus**
- b) The sound is synchronous with the maternal pulse**
- c) It is due to increased blood flow through the umbilical arteries**
- d) It can also be heard in a large fibroid**

Question 14:

A 32-year-old woman nearing her due date was admitted following regular contractions. You note that the head is at +1 station. Where is the exact position of the head?

- a) At the perineum**
- b) Just below ischial spine**

- c) Just above ischial spine
- d) High up in the false pelvis

Question 15:

While examining a pregnant woman, you measure her fundal height to be 32 cm and the station of the fetal head is at -2. What could be the probable weight of the fetus?

- a) 2.3Kg
- b) 3.5Kg
- c) 3.1Kg
- d) 4.0Kg

Question 16:

All of the following are components of Manning score except _____.

- a) Non stress test (NST)
- b) Oxytocin challenge test
- c) Fetal body movement
- d) Respiratory activity of fetus

Question 17:

A 28-year-old G2P1 woman presents to the antenatal clinic at 24 weeks for a routine check-up. USG shows a normal for gestational age fetus in frank breech position, with no other abnormalities. What is the next best step in management?

- a) 75gm oral glucose tolerance test
- b) Non-stress test
- c) External cephalic version
- d) Immediate lower segment Caesarean section

Question 18:

Which of the following vaccines is absolutely contraindicated in pregnancy?

- a) Hepatitis B

- b) Cholera
- c) Rabies
- d) MMR

Question 19:

A newly married couple planning for pregnancy has come to your clinic for preconception counselling. Which of the following drugs will you prescribe to the patient to decrease the chances of having neural tube defects in the baby?

- a) Vitamin B9
- b) Vitamin B12
- c) Vitamin C
- d) Vitamin B6

Question 20:

Which of the following conditions would not cause raised beta-hCG levels in pregnancy?

- a) Down's syndrome
- b) Multifetal pregnancy
- c) Spontaneous abortion
- d) Rh-incompatibility

Question 21:

Which of the following is not considered as a factor for a high-risk pregnancy?

- a) Obesity
- b) Polyhydramnios
- c) Anaemia
- d) Previous manual removal of placenta

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	b
5	b
6	c
7	d
8	c
9	a
10	b
11	b
12	c
13	c
14	b
15	c
16	b
17	a
18	d
19	a
20	c
21	d

Detailed Explanations

Solution to Question 1:

The duration of the fourth trimester is 12 weeks postpartum.

The concept of fourth trimester was introduced to address common concerns raised by women in puerperium such as post-partum pain, breastfeeding related problems, stress and postpartum depression, and lack of awareness of newborn care. The components of postpartum care for the 4th trimester include:

- Dedicated care team
- Lactation support
- Infant feeding plan
- Reproductive life plan

- Contraception
- Addressing complications of pregnancy
- Assessment of cardiovascular risk
- Mental health assessment
- Chronic health conditions

ACOG recommendation for postpartum visits:

- Initial visit: 3 weeks postpartum
- Final visit: 12 weeks postpartum

Government of India recommendation for postpartum visits.

- First visit: 1st day after delivery
- Second visit: 3rd day after delivery
- Third visit: 7th day after delivery
- Fourth visit: 6 weeks after delivery

Solution to Question 2:

The ideal number of antenatal visits is 12-14.

The antenatal visits should ideally be: monthly once up to 28 weeks, once in two weeks between 28-36 weeks, and once weekly visits from 36 weeks onwards.

This translates to a total of 12-15 visits.

The WHO increased the minimum number of antenatal visits from 4 to 8 visits to reduce perinatal mortality.

In the Indian scenario, the earlier minimum of 4 visits were implemented as follows:

- 1st visit: within 12 weeks (preferably as soon as pregnancy is suspected)
- 2nd visit: between 14-26 weeks
- 3rd visit: between 28-34 weeks
- 4th visit: between 36 weeks and term

A booked case is one that has at least 3 antenatal visits with two visits in the third trimester.

Solution to Question 3:

If a female has stillbirth, her parity does not decrease.

Parity is determined by the number of previous pregnancies reaching a period of viability, which is 28 weeks in India.

Multifetal pregnancies, such as twins or triplets, count as one for the purpose of this notation. The current pregnancy is not included in this count.

Multipara refers to a woman who has completed two or more pregnancies past the period of viability. When the pregnancy is preceded by five or more previous viable pregnancies, the woman is called a grand multipara.

Note: Gravida indicates the number of times the mother has been pregnant, regardless of whether these pregnancies were carried to term. The current pregnancy is included in this count.

Solution to Question 4:

The obstetric score is G4 P1 A2 L1.

The obstetric score is used to summarise the past obstetric history as follows:

- G - Gravidity, denoting the total number of pregnancies, including the present one
- P - Parity, denoting the total number of pregnancies crossing the period of viability
- This includes both live births and stillbirths
- The current pregnancy is not included
- A - Abortus or the number of pregnancies that were lost for any reason, including induced abortions or miscarriages.
- L - Living children

Multifetal pregnancies count as one for both gravida and para. A pregnant woman who delivers twins will be expressed as primipara.

The GTPAL system includes the following:

- Gravida
- Term (>37 weeks)
- Preterm (20w - 36w 6d)
- Abortions (<20 weeks)
- Number of living children

Note: The period of viability is when the fetus is capable of independent existence. The WHO defines it to be 24 weeks while the Indian definition is 28 weeks.

Solution to Question 5:

The woman has had two pregnancies in total - the previous twin pregnancy and the current pregnancy, hence the gravida score is 2. The first pregnancy resulted in delivery at term (crossed viable period), hence her para score is 1 (parity doesn't include current pregnancy). Hence, her obstetric score would be G2P1 (Gravida - 2, Para - 1).

Multifetal pregnancies count as one for both gravida and para. A pregnant woman who delivers twins for the first time is expressed as primipara.

The obstetric score is used to summarise the past obstetric history as follows:

- G - Gravidity, denoting the total number of pregnancies, including the present one
- P - Parity, denoting the total number of pregnancies crossing the period of viability. This includes both live births and stillbirths. The current pregnancy is not included.
- A - Abortion is the termination of pregnancy, either spontaneous or induced, before fetal viability.
- L - Living children.

The GTPAL system includes the following:

- Gravida
- Term (≥ 37 weeks)
- Preterm (20w - 36w 6d)
- Abortions (≤ 20 weeks)
- Number of living children

Note: The period of viability is when the fetus is capable of independent existence. The WHO defines it to be 24 weeks while the Indian definition is 28 weeks.

Solution to Question 6:

As per Naegele's rule, the EDD (expected date of delivery) of this woman is October 16, 2017.

Naegele's rule states that EDD is obtained by adding 7 days and subtracting 3 months from the LMP (the 1st day of the last menstrual period). Another method is adding 9 months and 7 days to LMP. Less than 5% of births follow Naegele's rule.

It is based on a 28-day regular cycle. If the cycle is shorter/longer than 28 days, corrected EDD is calculated as follows:

- For a 35-day cycle, add another 7 ($35-28=7$) days to the calculated EDD to get the corrected EDD.
- For a 25-day cycle, subtract 3 ($28-25=3$) days from the calculated EDD to get the corrected EDD.

Naegele's rule cannot be followed in IVF pregnancies. Considering the human embryo takes 266 days to develop, EDD is calculated in IVF pregnancies by the following method:

- Same day as fertilization (day 0): $EDD = \text{date of transfer} + 266 \text{ days}$
- 3 days after fertilization (day 3): $EDD = \text{date of transfer} + 263 \text{ days}$
- 5 days after fertilization (day 5): $EDD = \text{date of transfer} + 261 \text{ days}$

Solution to Question 7:

The recommended calorie intake for a pregnant woman in the second trimester would increase by ~ 350 kcal/day.

Solution to Question 8:

The pregnant lady can continue the same diet since there are no additional calorie requirements in the first trimester.

Pregnancy requires an additional 80,000 kcal, mostly during the last 20 weeks. To meet this demand, a caloric increase of 100 to 300 kcal/d is recommended during pregnancy. This greater intake, however, should not be divided equally during the course of pregnancy. The Institute of Medicine (2006) recommends adding 0, 340, and 452 kcal/d to the estimated nonpregnant energy requirements in the first, second, and third trimesters respectively.

Whenever caloric intake is inadequate, protein is metabolized rather than being spared for its vital role in fetal growth and development.

Solution to Question 9:

In the given clinical case, the patient should be advised to continue the pregnancy.

The radiation dose to the fetus during the declared term of pregnancy should be less than 1 mSv. In chest X-ray PA view, the absorbed dose of radiation by the fetus is less than 0.1 mSv. Hence, pregnancy can be continued.

An X-ray has a very limited role in obstetric imaging. There are very few indications where an X-ray will be useful:

- Fetal: congenital malformation of the fetus and neonates, skeletal malformations, birth injuries like fractures or dislocations.
- Maternal: Patients having cardiopulmonary disease may require an X-ray chest during pregnancy and that should be done beyond 12 weeks.

Radiation can cause fetal morbidity (FGR, genetic mutations, neurologic abnormalities, mental retardation, childhood leukemia) and mortality. The risks are high with radiation after the first 2 weeks postconception and within the first trimester (period of organogenesis).

Radiation during pregnancy can damage the early phase of the embryo but the risk can be minimized by using the '10-day rule'. This rule states that no woman should be exposed to an X-ray for a nonurgent indication outside 10 days from her last period during the reproductive period unless the pregnancy is excluded.

The given table below shows the absorbed radiation by the fetus in different diagnostic radiation procedures:

$$1 \text{ cGy} = 10 \text{ mSv}$$

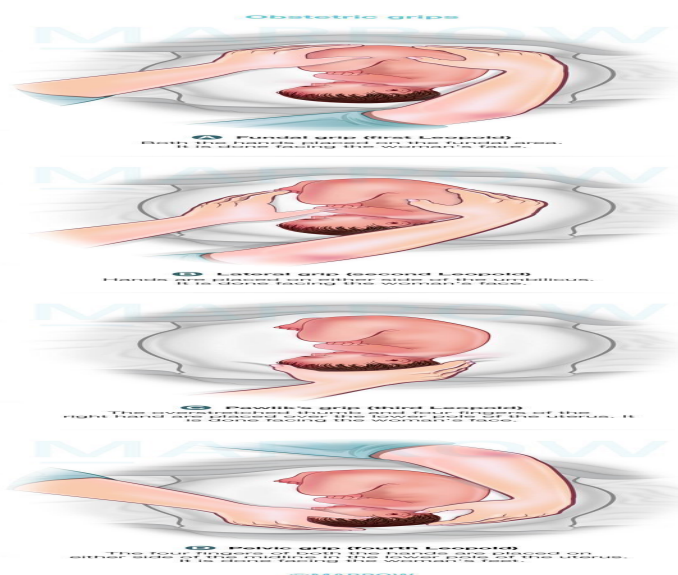
Procedure	Dose (c Gy)
Abdominal X-ray	0.1-0.3
Pelvic X-ray	0.5-1.1
Chest X-ray	<0.001
Abdominal CT	0.4-0.8
Chest CT	0.002-0.02
Abdominopelvic CT	2.5-3.5
Ventilation scan	0.007-0.05
PET scan	1-1.5

Solution to Question 10:

Fetal pole, lie, presenting part, engagement, and attitude of fetal head are assessed by Leopold's manoeuvres:

- Fundal grip (first Leopold):
- Assesses the uterine fundus.
- Helps in identification of fetal lie and determination of fetal pole at the fundus.
- The breech gives the sensation of a large, irregular mass.
- The head feels hard and round and is more mobile.
- Lateral grip (second Leopold):
- Assesses the fetal orientation.
- The palms are placed on either side of the maternal abdomen, and gentle but deep pressure is exerted.
- The back is felt on one side as a hard, resistant structure.
- The fetal extremities are felt on the other side as numerous small, irregular, mobile parts are felt.
- Fetal orientation can be determined by noting whether the back is directed anteriorly, transversely, or posteriorly.
- The position of the anterior shoulder is mainly assessed by this grip.
- Pawlik's grip / 1st pelvic grip (third Leopold):
- Assesses the presenting part in respect to the true pelvis.
- The thumb and fingers of one hand grasp the lower portion of the maternal abdomen just above the symphysis pubis.

- If the presenting part is not engaged, a movable mass will be felt, usually the head.
- Deep pelvic grip / 2nd pelvic grip (fourth Leopold):
- Assesses the engagement, attitude of the fetal head and determine the degree of descent.
- The examiner faces the mother's feet, and the fingertips of both hands are positioned on either side of the presenting part.
- They exert inward pressure and then slide along the axis of the pelvic inlet.
- If the head has already descended into the pelvis, the anterior shoulder or the space created by the neck may be differentiated readily from the head.
- It is the only grip performed facing the feet.



Solution to Question 11:

A dusky hue of the vestibule and anterior vaginal wall visible at about 8th week of pregnancy is called Chadwick's sign or Jacquemier's sign. The discoloration is due to local vascular congestion.

Solution to Question 12:

Lightening is felt in the 3rd trimester (38th week) in a primigravida. It is a sense of relief from the pressure symptoms due to engagement of the presenting part.

The 2nd trimester of pregnancy is characterised by the following:

Sign	Feature	Seen at
Chloasma	Pigmentation over forehead and cheeks	24 weeks

Sign	Feature	Seen at
Breast changes	Appearance of secondary areola.	20 weeks
Uterine changes- size and Braxton Hick contractions	Irregular, infrequent spasmodic and painless contractions without any effect on dilatation of cervix.	Begin in early pregnancy and continue till term
Palpation of fetal parts and active fetal movements		20 weeks
Ballotment of uterus	Feeling of floating object inside uterus on bimanual examination. Not exclusive to pregnancy	16 to 20 weeks
Auscultation of fetal heart sounds	Most conclusive sign of pregnancy	18 to 20 weeks by stethoscope and 10 weeks by doppler
Quickening	Feeling of life inside womb	16 to 20 weeks

Solution to Question 13:

Uterine souffle is due to an increase in blood flow through the dilated uterine vessels, not umbilical vessels.

It is a soft blowing systolic murmur heard low down at the sides of the uterus, better heard on the left side.

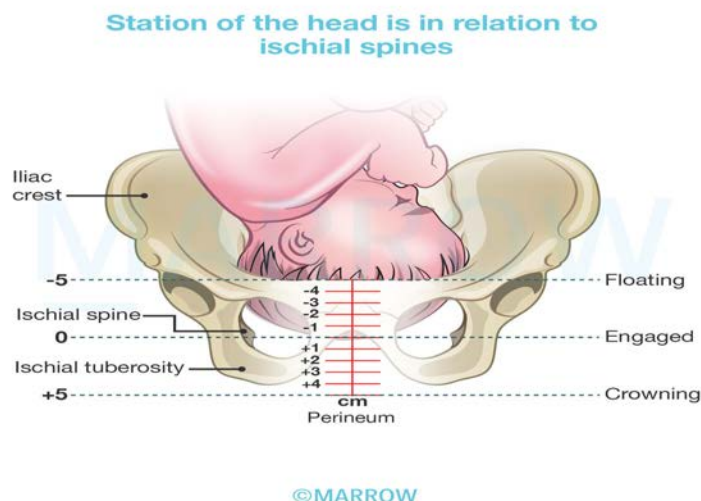
Uterine Souffle	Fetal/Funic Souffle
Soft, blowing sound	Sharp, whistling sound
Synchronous with maternal pulse	Synchronous with fetal pulse.
Caused by the passage of blood through dilated uterine vessels	Caused by the rush of blood through the umbilical arteries
It can also be heard in a large uterine fibroid	When present it is diagnostic of pregnancy

Solution to Question 14:

+1 station of the fetal head refers to it being just below the ischial spine.

The station of the head is expressed in relation to ischial spines as they are the most prominent bony projections felt on internal examination. The level of ischial spines is halfway between the pelvic inlet and outlet. This level is known as station zero (0).

The station is stated in minus figures if it is above the spines (–1 cm, –2 cm, –3 cm, –4 cm and –5 cm) and in plus figures if it is below the spines (+1 cm, +2 cm, +3 cm, +4 cm and +5 cm).



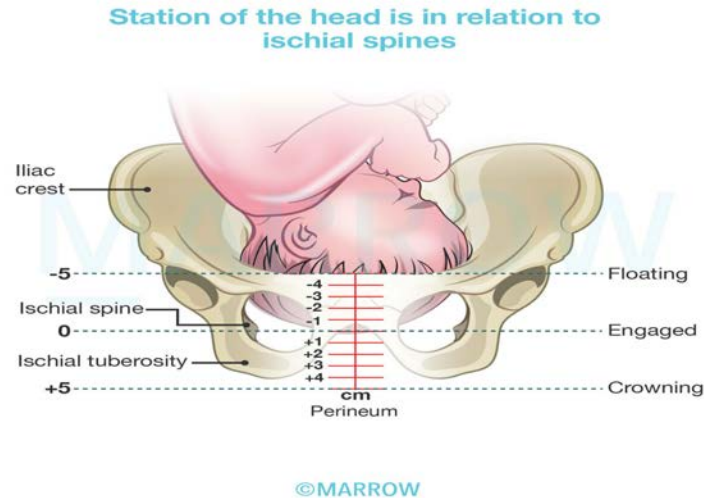
Solution to Question 15:

The fetal weight in the above case is 3.1Kg.

In the above case, fetal weight can be estimated using Johnson's formula. Height of uterus is 32cm, the station is –2 implying the head is above the ischial spines. Therefore, it is calculated as follows: $[32-12] \times 155 = 3100 \text{ grams} = 3.1\text{Kg}$

Estimation of fetal weight:

- Johnson's formula: This is only applicable in vertex presentation.
- If station of head is below the level of the ischial spine: $[\text{Height of uterus above pubic symphysis (in cm)} - 11] \times 155 \text{ in grams}$
- If station of head is above the level of the ischial spine: $[\text{Height of uterus above pubic symphysis (in cm)} - 12] \times 155 \text{ in grams}$
- Sonography: Fetal weight has been estimated by combining a number of biometric data, e.g. biparietal diameter (BPD), head circumference (HC), abdominal circumference (AC) and femur length (FL), using the following formulae:
- Shepard's formula
- Hadlock's formula.



Solution to Question 16:

The oxytocin challenge test is not a component of Manning score.

Earlier, oxytocin challenge test/contraction stress test was done after a non-reactive NST to confirm the diagnosis of hypoxia. However, it is associated with the risk of initiating labor and is hence not done.

Biophysical profile (BPP)/Manning score components: (Mnemonic: BATMaN)

- Breathing movements of the fetus
- Amniotic fluid volume
- Tone of fetus
- Movements of fetus
- Non-stress Test (NST)

Normal score: 8-10; Abnormal: ≤ 5

Modified Biophysical score:

- VAST (Vibroacoustic stimulation test)/NST
- Amniotic fluid index

A score ≤ 4 indicates the need for urgent delivery.

Solution to Question 17:

75gm oral glucose tolerance test (OGTT) is the most appropriate next step of management. The screening strategy for detecting gestational diabetes mellitus (GDM) recommends performing blood glucose testing at 24-28 weeks in case of average risk.

Breech presentation at 24 weeks is not a cause for concern because it can spontaneously rotate and becomes cephalic at term. Hence, external cephalic version (ECV) and immediate caesarean section are not indicated at this time.

Solution to Question 18:

MMR is absolutely contraindicated in pregnancy.

Killed vaccines are safe in pregnancy while live vaccines (MMR, BCG, varicella) are contraindicated in pregnancy.

Even though yellow fever vaccine is a live vaccine, it is recommended in endemic areas or during outbreaks. Pregnant women and nursing mothers should be counselled on the potential benefits and risks of vaccination.

Solution to Question 19:

Preconceptional intake of vitamin B₉ (folic acid) results in decreased incidence of neural tube defects like spina bifida, encephalocele, and anencephaly.

All women of reproductive age should be given 400 µg of folic acid daily. Women, who had fetal neural tube defects in the previous pregnancy, should be given 4 mg of folic acid daily beginning 1 month before conception to about 12 weeks of pregnancy.

Solution to Question 20:

Beta-hCG levels are reduced in the case of spontaneous abortion.

Human chorionic gonadotropin (hCG) is detectable in the plasma of pregnant women as early as 7-8 days after conception by radioimmunoassay. It doubles every 2 days and reaches a maximum level at 9 weeks. Thereafter, it declines to reach the lower level at 20 weeks and disappears from circulation 2 weeks postpartum.

Increased levels of beta-hCG are found in:

- Multiple pregnancies
- Rh isoimmunization
- Down's syndrome
- Choriocarcinoma
- Hydatidiform mole

Decreased levels of beta-hCG are found in:

- Ectopic pregnancies
- Impending spontaneous abortion

- Missed abortion
- Edward syndrome.

Solution to Question 21:

Previous manual removal of the placenta does not indicate a high-risk pregnancy.

The high-risk pregnancies according to the Pregnancy Risk Assessment Monitoring System by the CDC are:

Current pregnancy complications	Obstetric history	Other disorders
Preterm labor (<37 Wks) and Post-term (> 42 wk)	Intrauterine growth restriction	Cardiovascular factors like chronic hypertension, severe heart failure, and renal disorders
Preterm premature rupture of membranes	Stillbirth	Metabolic disorders like Obesity, Insulin-dependent diabetes, Previous endocrine ablation, Thyroid disorders
Moderate to severe preeclampsia	Late abortion (16–20 wk)	Hematological problems like sickle cell disease, severe anemia Thrombophilia
Polyhydramnios (hydramnios) or oligohydramnios	Fetal exchange transfusion (Rh incompatibility)	Infections (malaria, HIV) and Autoimmune disorders
Uterine rupture	Multifetal pregnancy	Epilepsy
Antepartum hemorrhage (Placenta previa, and Abruptio placentae)	Previous brachial plexus injury	Anatomic abnormalities like uterine malformation and cervical incompetence
Fetal tachycardia or bradycardia for >30 min	Abnormal fetal position	Exposure to teratogens like alcohol, drugs, smoking
Fetal weight < 2.5 kg or >4kg	Previous h/o myomectomy, repair of complete perineal tear, vesicovaginal fistula, and stress incontinence	Family history of diabetes, hypertension or multiple pregnancy and congenital malformation
Chorioamnionitis		Low socioeconomic status
Prolapsed cord		Abnormal cervical cytologic findings
Fetal acidosis pH ≤7		

Antenatal Investigations

Question 1:

Which of the following statements is false about non-invasive prenatal testing?

- a) It uses cell-free circulating fetal DNA in the maternal serum
- b) It is used for the diagnosis of trisomy 13, 18 and 21
- c) It can be performed after 10 weeks of gestation
- d) It is done after the routine first trimester ultrasound assessment

Question 2:

A pregnant woman has a fetus with Down's syndrome. Triple test done at 15-16 weeks will have _____.

- a) Low MSAFP, high hCG, low UE₃
- b) High MSAFP, low hCG, high UE₃
- c) Low MSAFP, high hCG, high UE₃
- d) High MSAFP, low hCG, low UE₃

Question 3:

Which of the following is not included in the quadruple screening test?

- a) Inhibin B
- b) Unconjugated Estriol (E₃)
- c) AFP
- d) hCG

Question 4:

Quadruple testing is done during:

- a) 9-11 weeks
- b) 12-14 weeks

- c) 15-20 weeks
- d) 22-28 weeks

Question 5:

Results of the quadruple screening test are tabulated below for three congenital anomalies. Choose the correctly matched set.

- a) 1- Trisomy 21; 2- Trisomy 18; 3- Neural tube defect
- b) 1- Trisomy 18; 2- Trisomy 21; 3- Neural tube defect
- c) 1- Trisomy 21; 2- Neural tube defect; 3- Trisomy 18
- d) 1- Trisomy 18; 2- Trisomy 13; 3- Neural tube defect

Question 6:

In which of the following conditions is an increased level of alpha fetoprotein seen?

- a) Gestational trophoblastic diseases
- b) Molar pregnancy
- c) Under estimated gestational age
- d) Spontaneous abortion

Question 7:

Increased maternal serum alpha-fetoprotein is detected in a primigravida. What is the next step in management?

- a) Repeat measurement of MSAFP at a later date
- b) Targeted sonography
- c) Amniocentesis
- d) Termination of pregnancy

Question 8:

Which of the following statements is true regarding alpha fetoprotein?

- a) MSAFP is detected at 16-18 weeks of gestation

- b) Diabetic patients have increased AFP levels
- c) MSAFP is unrelated to period of gestation
- d) Lowest fetal AFP is seen around 13 weeks of gestation

Question 9:

During genetic counseling, a pregnant woman states that both she and her husband have sickle cell trait. You offer the option of chorionic villus sampling for prenatal diagnosis. What is the best time to perform this procedure?

- a) 6-8 weeks
- b) 7-9 weeks
- c) 9-10 weeks
- d) 11-13 weeks

Question 10:

Which of the following is not true about amniocentesis?

- a) Traditionally done between 16-20 weeks
- b) Limb reduction defect is a known complication
- c) Down's syndrome can be diagnosed
- d) Approximately 20 to 30 mL is collected for fetal karyotyping

Question 11:

A 28-year-old woman presents with 6 weeks of amenorrhea and a positive urine pregnancy test. She states her brother had cystic fibrosis and asks if her child might have the same. Which of the following methods would not be useful for prenatal diagnosis of this condition?

- a) Amniocentesis
- b) Cordocentesis
- c) Chorionic villous sampling
- d) Fetal skin biopsy

Question 12:

A G2P1 woman at 35 weeks of pregnancy presents with complaints of decreased fetal movement. What is the next step in management?

- a) Observation
- b) Non-stress test
- c) Biophysical profile
- d) Induce labour

Question 13:

The CTG shown in the image below may be seen in which of the following conditions?



- a) Head compression
- b) Cord compression
- c) Fetal hypoxia
- d) Normal

Question 14:

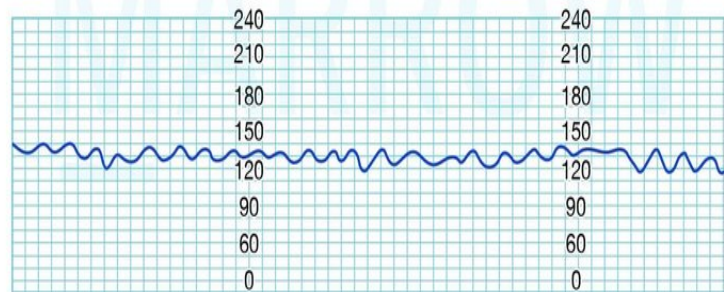
A 35-year-old G2P1 at 38 weeks of gestation came to the labor room with complaints of decreased fetal movements. Non-stress test was done and late decelerations were noted. What does this indicate?

- a) Head compression
- b) Cord compression

- c) Fetal hypoxia
- d) Breech presentation

Question 15:

A pregnant woman brings her son to the hospital for routine immunization. While awaiting his turn, she realizes that her seat is stained with blood. She is rushed to the nearby antenatal clinic. Pulse - 96/min, BP - 110/80 mmHg. Vaginal bleeding is noted and her NST is shown below. What is the likely diagnosis?



©MARROW

- a) Placenta previa
- b) Vasa previa
- c) Battledore placenta
- d) Abruptio placentae

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	c

5	a
6	c
7	b
8	a
9	d
10	b
11	d
12	b
13	a
14	c
15	b

Detailed Explanations

Solution to Question 1:

Non-invasive prenatal testing (NIPT) is used for the screening of trisomy 13, 18 and 21 and Turner's syndrome. It is not a diagnostic test.

It is a novel genetic test that uses cell-free circulating fetal DNA in the maternal serum for screening of chromosomal anomalies.

DNA from the fetus is found circulating in the maternal blood in two forms - either inside the intact fetal cells or as cell-free DNA from the breakdown of placental fetal cells. In NIPT, the cell-free chromosome fragments are measured and the quantitative differences from normal fetuses are noted.

NIPT can be performed after 10 weeks of gestation and is recommended after the routine first-trimester ultrasound assessment in high-risk pregnancies or if either 1st or 2nd-trimester screening becomes positive. Its disadvantage is that it is very expensive and it can be used for screening of only a few chromosomal anomalies such as trisomy 13, 18, and 21 and it doesn't analyse the entire karyotype.

Solution to Question 2:

The triple test findings in pregnancies with fetal Down's syndrome are characterized by:

- Low MSAFP (maternal serum alpha-fetoprotein) levels, approximately 0.7 MoM (multiples of median).
- Low unconjugated estriol levels, approximately 0.8 MoM.
- High hCG levels, approximately 2.0 MoM.

The triple test is done between 15-18 weeks for the diagnosis of Down's syndrome.

Various antenatal screening tests for aneuploidy are as follows:

First-trimester screening with biochemical analytes (↓) PAPP-A and (↑) hCG and USG measurement of NT (↑) can improve the detection rate (87%) of Down's syndrome. For confirmation, a prenatal genetic study (CVS, amniocentesis or cordocentesis) has to be performed.

Second-trimester screening (quad screening) at 15–18 weeks: (↓) MSAFP, (↓) uE3, (↑) inhibin A and (↑) hCG can detect trisomy 21 in 85% of cases with a false-positive rate of 0.9%. When it is combined with first-trimester screening, the detection rate is increased to 95%.

Dual test (1sttrimester)	Triple test (2ndtrimester)	Quadruple test (2ndtrimester)
Beta HCG	Beta HCG	Beta HCG
PAPPA	Unconjugated estriol	Unconjugated estriol
This can be combined with nuchal translucency by USG	MSAFP	MSAFP
	Seruminhibin A	

Solution to Question 3:

Inhibin B is not a component assayed under the quad-screening test. Dimeric inhibin-A is evaluated in this screening test.

Quadruple testing is done in the second trimester between 15-21 weeks of gestation. It comprises:

- Maternal serum AFP
- Human chorionic gonadotropin
- Unconjugated estriol (E3)
- Dimeric inhibin A

The following table summarizes the findings of the quad-screening test in various conditions.

Various antenatal screening tests for aneuploidy are as follows:

Parameter	Trisomy 21	Trisomy 18	Neural tube defect or abdominal wall defect
MSAFP	Low	Low	High
β hCG	High	Low	Normal
Unconjugated estriol	Low	Low	Normal
Inhibin A	High	Normal/Low	Normal

Dual test (1st trimester)	Triple test (2nd trimester)	Quadruple test (2nd trimester)
1. Beta HCG 2. PAPP (Pregnancy-associated plasma protein A) This can be combined with nuchal translucency by USSG	1. Beta HCG 2. Unconjugated estriol 3. MSAFP	1. Beta HCG 2. Unconjugated estriol 3. MSAFP 4. Serum inhibin

Solution to Question 4:

The quadruple screening test is ideally done in the second trimester, during 15-20 weeks of gestation. The most accurate values are seen during 16-18 weeks of gestation.

It is the measurement of maternal serum alpha-fetoprotein (MSAFP), unconjugated estriol, hCG, and inhibin-A.

In Down's syndrome (trisomy 21) or Edward's syndrome (trisomy 18), quadruple-marker screenings do not provide any added benefit over first-trimester screenings. It is generally used as a stand-alone test if screening is done for the first time in the second trimester or if first-trimester screening is not available.

	Trisomy 21	Trisomy 18	Neural tube defect or abdominal wall defect	Trisomy 13/ Patau syndrome
MSAFP	Low	Low	High	Normal
β hCG	High	Low	Normal	Normal
Unconjugated estriol	Low	Low	Normal	Normal
Inhibin A	High	Normal/Low	Normal	High

Solution to Question 5:

Quadruple testing is done in the second trimester between 15-21 weeks of gestation and involves the following parameters:

- Maternal serum alpha fetal protein (MSAFP)
- β hCG
- Unconjugated estriol
- Inhibin A

Parameter	Trisomy 21/ Down syndrome	Trisomy 18/ Edward syndrome	Neural tube defect or abdominal wall defect
MSAFP	Low	Low	High

Parameter	Trisomy 21/ Down's syndrome	Trisomy 18/ Edward syndrome	Neural tube defect or abdominal wall defect
β hCG	High	Low	Normal
Unconjugated estriol	Low	Low	Normal
Inhibin A	High	Normal/Low	Normal

Solution to Question 6:

In underestimated gestational age, there is an increased level of maternal serum alpha fetoprotein (MSAFP).

Elevated levels of MSAFP are seen in:

- Underestimated gestational age
- Severe oligohydramnios
- Multifetal pregnancy
- Neural tube defects
- Abdominal wall defects- Omphalocele, Gastroschisis
- Hepatoma
- Endodermal sinus tumour

Reduced levels of MSAFP are seen in:

- Overestimated gestational age
- Overweight mother
- Gestational trophoblastic diseases
- Spontaneous abortion
- Trisomy 18 and 21

Solution to Question 7:

The next best step is targeted sonography. Maternal serum alpha-fetoprotein (MSAFP) levels are only used as a screening test and must be followed by a diagnostic test.

Amniocentesis is a diagnostic test for amniotic fluid AFP measurement. It was once considered the standard for the diagnosis of open neural tube disorders but has been replaced in most centers by targeted sonography.

Defects in fetal integuments, such as ventral wall and neural tube defects, permit fetal AFP to leak into the amniotic fluid. This results in increased MSAFP levels. This is detected on screening performed from 15 to 20 weeks of gestation.

Conditions with elevated MSAFP:

- Wrong gestational age
- Neural tube defects
- Multiple pregnancies
- Renal anomalies

Conditions with decreased MSAFP:

- Trisomies (Down's syndrome)
- Gestational trophoblastic disease.

Solution to Question 8:

The maternal serum AFP screening is generally performed from 15-20 weeks of gestation. Hence, it is detected in this period.

Alpha-fetoprotein (AFP) is the major serum protein in the embryo and fetus. It is produced by the yolk sac and fetal liver.

Option B: Diabetic patients have low levels of AFP.

Option C: Gestational age influences the MSAFP levels. The maternal serum concentration increases by approximately 15% per week during the second trimester.

Option D: At about 13 weeks of gestation, the AFP concentration in the fetal serum and amniotic fluid are highest, after which it declines. Conversely, AFP is found in steadily increasing quantities in maternal serum after 12 weeks.

Solution to Question 9:

The best time to perform chorionic villous sampling is 11-13 weeks.

Chorionic villus sampling:	
Study material	Chorionic villi from trophoblastic cells
Time	10 to 13 weeks(transcervically) 10 weeks to term (transabdominally)
Indications	To detect chromosomal abnormalities such as Down syndrome. To detect inborn errors of metabolism like phenylketonuria. For DNA analysis in hemoglobinopathies, thalassemia, and sickle cell anemia.

Chorionic villus sampling:	
Complications	Abortion (CVS> Amniocentesis) Limb reduction defects and oro-mandibular defects (when done before 10 weeks)
Drawback	Not suitable for the detection of neural tube defects.

Solution to Question 10:

Limb reduction defect is a complication of chorionic villous sampling.

Amniocentesis	
Study material	Amniotic fluid with amniocytes
Time	16-20 weeks
Early amniocentesis	11-14 weeks (associated with increased fetal loss, foot deformities)
Diagnostic indications	Sex-linked disorders, karyotyping, inborn errors of metabolism, neural tube defects
Therapeutic indications	Abortion induction, decompression of uterus in acute hydrops, amnioinfusion in oligohydramnios
Procedure	Under direct sonographic guidance, a 20–22 gauge spinal needle is directed into a clear pocket of amniotic fluid while avoiding the fetus and umbilical cord. Approximately 20 to 30 mL of fluid is collected.
Complication	Chorioamnionitis, premature Rupture Of Membranes (PROM), fetal trauma/loss, fetomaternal hemorrhage.

Solution to Question 11:

Fetal skin biopsy is not a recommended method of prenatal diagnosis of cystic fibrosis. It is done to detect inherited skin conditions such as:

- Ichthyosis

- Epidermolysis bullosa
- Oculocutaneous albinism

Invasive procedures for prenatal diagnosis of CFTR gene mutations in cystic fibrosis include:

- Chorionic villus sampling (CVS)
- Amniocentesis
- Cordocentesis or percutaneous umbilical blood sampling (PUBS)

Solution to Question 12:

In case of reduced fetal movements, the first step of management is to do a non-stress test (NST).

Interpretation of NST:

- Reactive
- ≥ 2 accelerations of ≥ 15 beats/minute above the baseline and longer than 15 seconds in duration are present in a 20-minute observation
- Reassure the patient and send back home
- Non-reactive - biophysical profile (BPP) has to be assessed.

Solution to Question 13:

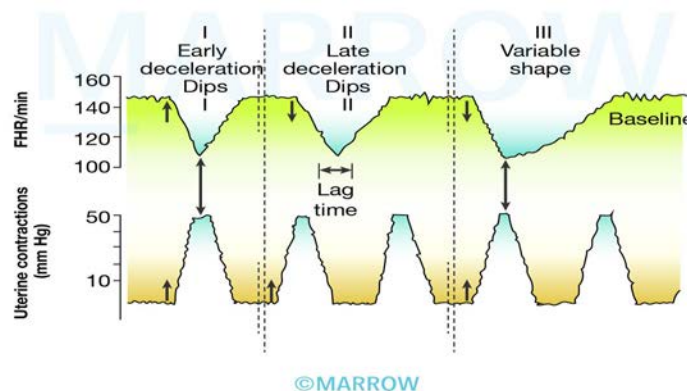
The given CTG shows early deceleration which is seen in cases of head compression.

Solution to Question 14:

The given scenario is suggestive of fetal hypoxia. Late decelerations are seen on an NST in this condition.

Deceleration is a decrease in fetal heart rate below the baseline by 15 bpm for more than 15 seconds. Absent variability of FHR with late decelerations is an indication for immediate Cesarean section.

Graphic representation showing various types of decelerations in relation to uterine contractions



Solution to Question 15:

The given clinical scenario along with the NST showing a sinusoidal pattern point towards a diagnosis of vasa previa.

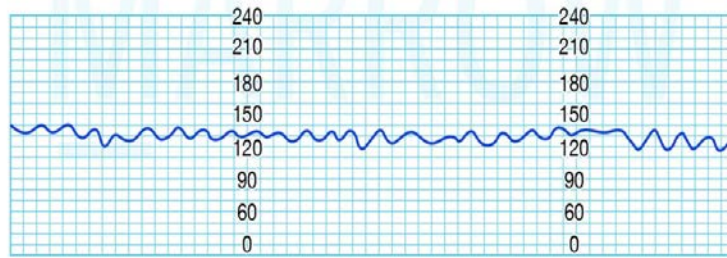
Vasa previa is a condition where fetal vessels travel within the membranes and overlie the cervical os. Hence, there can be bleeding during cervical dilatation or membrane rupture.

Fetal mortality is high due to fetal exsanguination. When bleeding is present in vasa previa, an emergency cesarean section is indicated.

Causes of sinusoidal pattern on NST include:

- Severe fetal anemia - Rh isoimmunisation, vasa previa, fetomaternal hemorrhage, or twin to twin transfusion
- Opioids - meperidine, morphine, alphaprodine, butorphanol
- Amnionitis
- Fetal distress
- Umbilical cord occlusion.

Sinusoidal pattern showing absence of baseline variability



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Obstetrical Imaging

Question 1:

A 26-year-old woman presents with 2 months of amenorrhea and a positive UPT. On ultrasound examination, all of the following are assessed except _____.

- a) Fetal number
- b) Fetal cardiac activity
- c) Amniotic fluid volume
- d) Crown rump length

Question 2:

A primigravida presents to the antenatal clinic for the first time at 18 weeks of amenorrhea. She does not remember her LMP and no ultrasound scans were done in the first trimester. In this situation, gestational age can be most accurately estimated by which of the following parameters?

- a) Crown rump length
- b) Biparietal diameter
- c) Femur length
- d) Abdominal circumference

Question 3:

A 25-year-old woman presented with complaints of 7 weeks of amenorrhea and a positive UPT. Her ultrasound scan is shown below. What does the red arrowhead represent?



- a) Yolk sac
- b) Decidua parietalis
- c) Decidua capsularis
- d) Chorion

Question 4:

Head circumference is measured from which of the following ultrasonographic views?

- a) Transthalamic view
- b) Bitemporal view
- c) Transcranial view
- d) Bimastoid view

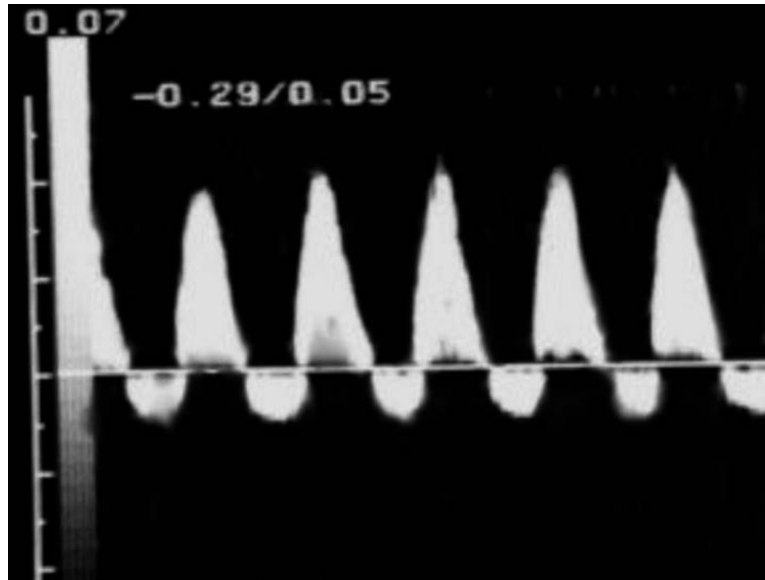
Question 5:

What is the normal range of femur length to abdominal circumference ratio?

- a) 44 - 50
- b) 20 - 24
- c) 50 - 40
- d) 30 - 40

Question 6:

A woman presents at 33-weeks of gestation and the umbilical artery Doppler imaging shows the following finding. What is the next step of management?



- a) Continue pregnancy till term
- b) Immediate termination of pregnancy
- c) Terminate the pregnancy at 34 weeks of gestation
- d) Continue steroids until term

Question 7:

Which of the following statements is true regarding the first-trimester ultrasound?

- a) A gestational sac can first be seen 4 weeks after LMP
- b) Accuracy of USG dating decreases after the 1st trimester
- c) Yolk sac is the first sign of pregnancy on USG
- d) A normal gestational sac implants in the midline of the endometrial cavity

Question 8:

What is the ideal time for nuchal translucency evaluation?

- a) 6-8 weeks
- b) 11-14 weeks

- c) 14-16 weeks
- d) 16-20 weeks

Question 9:

Which of the following are sonographic markers for Down's syndrome?

- a) 1 and 2
- b) 1, 4 and 5
- c) 1, 2, 3 and 4
- d) 1 and 3

Question 10:

The anomaly scan of a G3A2 woman is given below. Identify the sign.



- a) Banana sign
- b) Lemon sign
- c) Mickey Mouse sign
- d) Humming bird sign

Question 11:

A doctor tells a 31-year-old woman, who is 20 weeks pregnant, that her baby has a developmental defect. She has come to you for a second opinion. The following USG finding is seen. Identify the defect.



- a) Anencephaly
- b) Omphalocele
- c) Cystic hygroma
- d) Encephalocele

Question 12:

What is Robert's sign?

- a) Gas shadow in bowel
- b) Gas shadow in portal vein
- c) Gas shadow in heart
- d) Gas shadow in cranium

Question 13:

What is the frequency of a transabdominal transducer?

- a) 20000 hertz
- b) 4-6 megahertz
- c) 5-10 megahertz

- d) 10-15 megahertz

Question 14:

When is fetal cardiac activity first detected on transabdominal USG?

- a) 5th week
- b) 4th week
- c) 6th week
- d) 9th week

Question 15:

A pregnant woman has her first antenatal visit in the 8th month of gestation. On examination, the uterus corresponds to 30 weeks and the fetal heart rate is 136/min. What is the best parameter for the estimation of fetal age by USG in this patient?

- a) Femur length
- b) Biparietal diameter
- c) Abdominal circumference
- d) Crown rump length

Question 16:

Which of the following is false regarding sonography in pregnancy?

- a) Must follow ALARA principle
- b) All machines must display thermal index
- c) Tissue heating is higher with shorter examination time
- d) Risks are greater during organogenesis than later in pregnancy

Question 17:

A pregnant diabetic woman presents for a routine antenatal visit during the third trimester. During the evaluation, you perform obstetrical Doppler imaging. Which among the following would indicate fetal compromise?

- a) Increased pulsatility index in umbilical artery

- b) Decreased systolic/diastolic ratio
- c) Increased cerebral artery flow
- d) Absent end diastolic flow

Answer Key

Question No.	Correct Option
1	c
2	b
3	a
4	a
5	b
6	b
7	b
8	b
9	c
10	a
11	a
12	c
13	b
14	c
15	a
16	c
17	d

Detailed Explanations

Solution to Question 1:

Amniotic fluid volume is assessed on ultrasound in the second and third trimesters.

Ultrasonographic evaluation during the first trimester includes:

- Gestational sac size, location, and number
- Embryo and/or yolk sac identification
- Crown-rump length

- Fetal number, including amnionicity and chorionicity of multifetal gestations
- Embryonic/fetal cardiac activity
- Maternal uterus, adnexa, and cul-de-sac
- Fetal nuchal translucency.

Solution to Question 2:

In the second trimester, the parameter to assess gestational age most accurately is the biparietal diameter (BPD). Accuracy varies from 7 to 10 days.

BPD is measured in the transthalamic view, at the level of the thalamus and cavum septum pellucidum (CSP). It is measured from the outer edge of the skull in the near field to the inner edge of the skull in the far field.

Trimester	Parameter
First	Crown rump length (CRL)
Second	Biparietal diameter (BPD)
Third	Femur length (FL)

Solution to Question 3:

The red arrowhead represents the yolk sac. The yellow arrowhead shows the double ring - outer decidua parietalis and inner decidua capsularis.

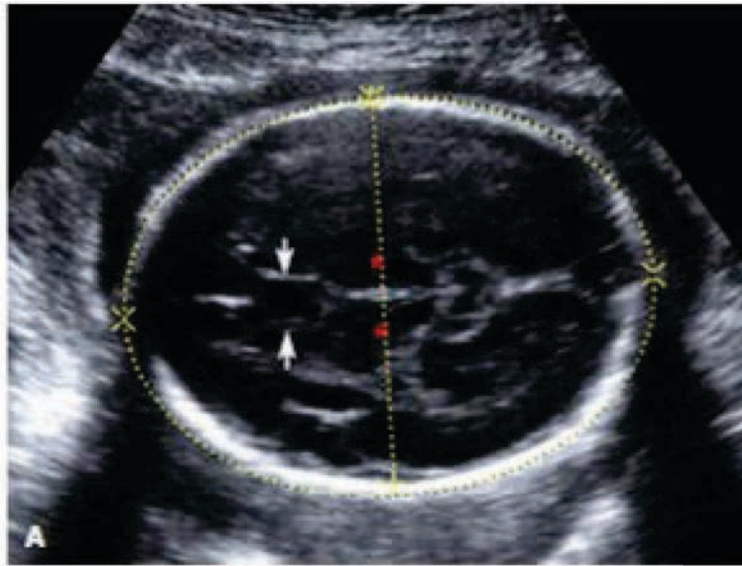
It is referred to as the double decidua sign and is seen in the first trimester transvaginal sonogram. It is useful to confirm an early intrauterine location for the pregnancy.



Solution to Question 4:

Head circumference (HC) is measured from the transthalamic view by placing an ellipse around the outer edge of the skull

HC can also be calculated from the biparietal diameter (BPD) and occipital-frontal diameter (OFD).



Solution to Question 5:

The normal range for the femoral length (FL) to abdominal circumference (AC) ratio is 20-24 percent.

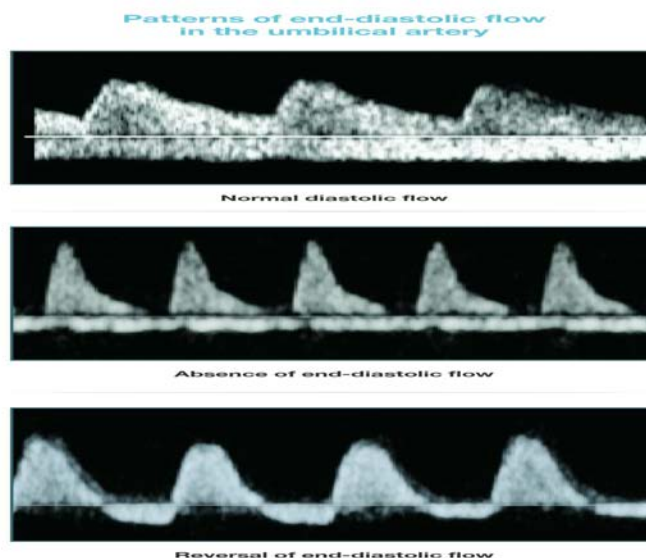
Low FL/AC (< 16) ratio suggests a lethal skeletal dysplasia, particularly if other skeletal abnormalities are present.

Solution to Question 6:

The umbilical artery Doppler depicts reversal of end-diastolic flow or REDF (waves appearing below the baseline) which is seen in intrauterine growth restriction. As the period of gestation has exceeded 32 weeks, the pregnancy can be terminated.

In case of absent end-diastolic flow (AEDF), the waves just touch the baseline, but don't appear below it. This suggests a fetal compromise, hence the pregnancy is terminated by C-section if > 34 weeks.

Reversed end-diastolic flow (REDF) suggests impending fetal death, hence the pregnancy is terminated by C-section if ≥ 32 weeks.



Solution to Question 7:

The accuracy of determining gestational age using ultrasound is the highest in the first trimester and decreases afterwards.

Option A: Gestational sac is reliably seen at 5 weeks of gestation on transvaginal sonography(TVS).

Option C: The first sign of pregnancy seen on sonography is the gestational sac. Visualisation of the yolk sac within the gestational sac confirms an intrauterine location for pregnancy.

Option D: A normal gestational sac implants eccentrically in the endometrium, whereas a pseudosac is seen in the midline of the endometrial cavity.

Solution to Question 8:

Nuchal translucency evaluation, a component of first-trimester aneuploidy screening, is done between 11-14 weeks.

It represents the maximum thickness of the subcutaneous translucent area between the skin and soft tissue, overlying the fetal spine, at the back of the neck. When the nuchal translucency is increased, the risk of fetal aneuploidy and structural anomalies such as heart defects is significantly elevated.

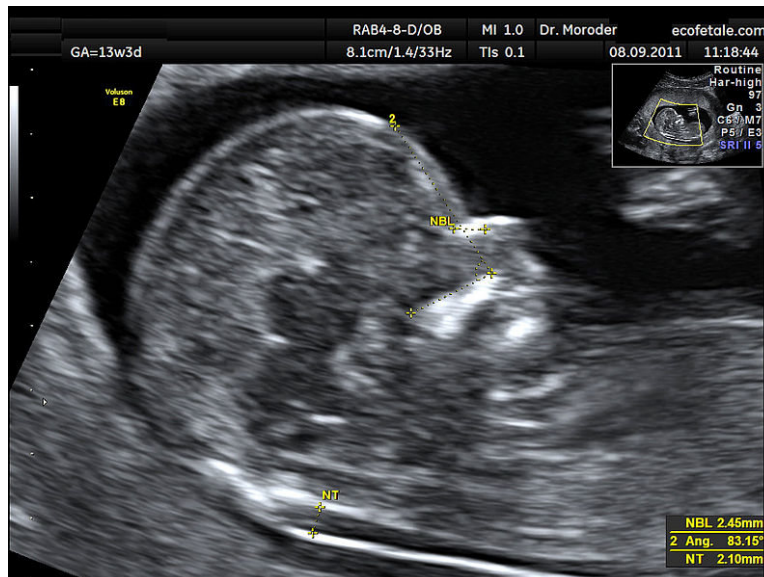
First trimester aneuploidy screening combines two maternal serum analytes and a sonographic measurement:

- Human chorionic gonadotropin (hCG)

- Pregnancy-associated plasma protein A (PAPP-A)
- Nuchal translucency (NT)

It is performed between 11 and 14 weeks of gestation. In cases of fetal Down syndrome, the first-trimester serum-free β -hCG level is higher and the PAPP-A level is lower. With trisomy 18 and trisomy 13, levels of both analytes are lower.

Note: Nuchal fold is a normal fold of skin seen at the back of the fetal neck during the second trimester of pregnancy. Increased nuchal fold thickness is associated with fetal aneuploidy.



Solution to Question 9:

Ultrasound markers for Down's syndrome:

- First trimester
- Nuchal translucency >3 mm
- Second trimester
- Echogenic intracardiac focus
- Ventriculomegaly
- Nuchal thickness >6 mm
- Echogenic bowel
- Hypoplastic/absent nasal bone
- Shortened humerus, femur
- Mild pylectasis
- Aberrant subclavian artery.

Solution to Question 10:

The given image shows banana sign.

An abnormal anterior curvature of cerebellum along with effacement of cisterna magna produces this appearance. It is seen in spina bifida associated with Arnold-Chiari malformation.



Lemon sign: Axial USG image of the fetal head at the level of the biparietal diameter shows inward concavity of both frontal bones instead of the normally seen convexity. This gives the head the appearance of a lemon. It has a strong association with spina bifida and is very useful for detecting this condition.



Solution to Question 11:

The given ultrasound image shows an open defect in the calvaria and skin leading to the exposure of the cranial neural tube. This is consistent with the diagnosis of anencephaly.

Anencephaly results from incomplete closure of the neural tube. It is one of the earliest detectable congenital defects that can be detected as early as 10 to 11 weeks of gestation. Screening can also be done by serum estimation of maternal serum alpha-fetoprotein (MSAFP). The following are the radiological signs seen in anencephaly.

- Frog eye sign - Absence of cranial calvaria, prominent orbits thereby resembling the face of a frog.
- Shower cap sign - The absence of the cranium could lead to the protrusion of a disorganized mass of brain tissue. This resembles a shower cap giving rise to a characteristic triangular facial appearance.

Neural-tube defects can be prevented with pre-conceptional supplementation of folic acid. A dose of 4 mg daily is recommended in subsequent pregnancies.

The frog eye and showercap signs are shown below:

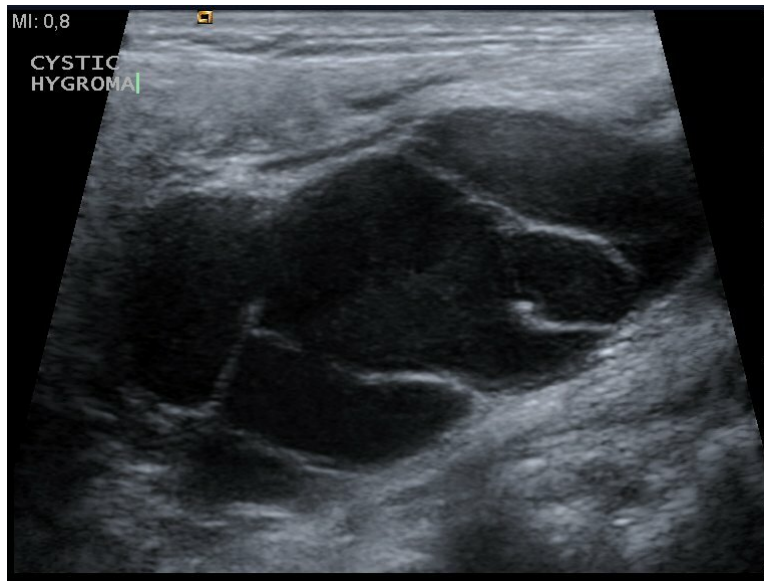


Option B: Omphalocele is a midline anterior abdominal wall defect through which the solid abdominal viscera and/or bowel may herniate. The extruded abdominal contents are covered in a sac. Given below is an ultrasound image of the fetus suggestive of an omphalocele (asterix).



Cystic hygroma is a congenital multiloculated lymphatic lesion that is classically found in the left posterior triangle of the neck. It occurs as a result of sequestration or obstruction of developing lymph vessels. This defect can be picked up in a first-trimester ultrasound. The images below show the MRI and USG appearance of cystic hygroma, respectively:





Encephalocele is caused due to defects in the skull through which the meninges and/or neural tissue herniate. The squamous part of the occipital bone is the most frequently affected bone. The image shows a newborn with encephalocele.



Solution to Question 12:

Robert's sign is a gas shadow in the fetal heart. It is seen in fetal demise in utero and may appear as early as 12 hours.

Other sonographic signs seen in fetal demise in utero are:

- Absent fetal heartbeat
- Absent fetal movements

- Occasional findings
- Overlapping of skull bones - Spalding's sign
- Gross distortion of fetal anatomy (maceration)
- Soft tissue edema - skin >5 mm
- Echogenic amniotic fluid (fetal demise fragments)
- Uncommon findings
- Thrombus in fetal heart
- Gas shadow in the fetal heart - Robert's sign.

Solution to Question 13:

The frequency of trans abdominal transducer is 4-6 megahertz (MHz) since a lower frequency penetrates better.

On the other hand, a vaginal transducer has a frequency of 5-10 megahertz since a higher frequency yields a better image with minimal penetration.

Solution to Question 14:

The cardiac activity of the fetus is first detected on transabdominal USG (TAS) at the 6th week of intrauterine life.

Fetal structure	TVS(weeks)	TAS(weeks)
Gestational sac	4.5	5.5
Yolk sac	5	6
Cardiac activity	6 - 6.5	6 - 7

Solution to Question 15:

Femur length is the best parameter for the estimation of fetal age by USG in the 3rd trimester.

Best USG parameters for estimation of fetal age:

Trimester	Parameter
First	Crown-rump length (CRL)

Trimester	Parameter
Second	Biparietal diameter (BPD)
Third	Femur length (FL)

Solution to Question 16:

Tissue heating is higher with longer examination time.

Sonography should be performed in compliance with the ALARA principle which is As Low As Reasonably Achievable. The theoretical risks are greater during organogenesis (8-15 weeks) than later in pregnancy.

All sonography machines must display 2 indices - the thermal index and the mechanical index.

Solution to Question 17:

Absent end diastolic flow indicates fetal compromise in Doppler.

Umbilical artery vessels:

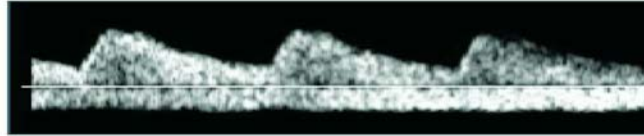
- They are characterized by forward low-resistance flow.
- As gestation increases, there is a gradual fall in all the resistance indices so the amount of forward diastolic flow increases.
- S/D ratio decreases as the period of gestation increases.

Abnormal flow patterns in umbilical artery:

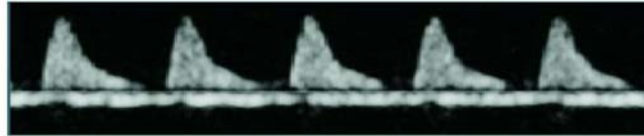
- Reduced end-diastolic flow - Uteroplacental insufficiency (systolic/diastolic ratio increases)
- Absent end-diastolic flow - Fetal compromise (hence pregnancy is terminated by cesarean section if ≥ 34 weeks)
- Reverse end-diastolic flow - Impending death (pregnancy is terminated irrespective of gestational age)

Doppler of the middle cerebral artery helps in detecting fetal anemia. Increased fetal diastolic flow in the middle cerebral artery with absent diastolic flow in the aorta implies fetal acidemia.

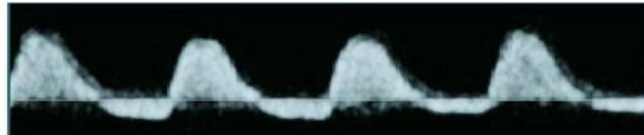
Patterns of end-diastolic flow
in the umbilical artery



Normal diastolic flow



Absence of end-diastolic flow



Reversal of end-diastolic flow

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Normal Labour

Question 1:

Which of the following acts as the primary trigger for onset of labour in pregnant women?

- a) ACTH in mother
- b) ACTH in fetus
- c) Prostaglandin
- d) Oxytocin

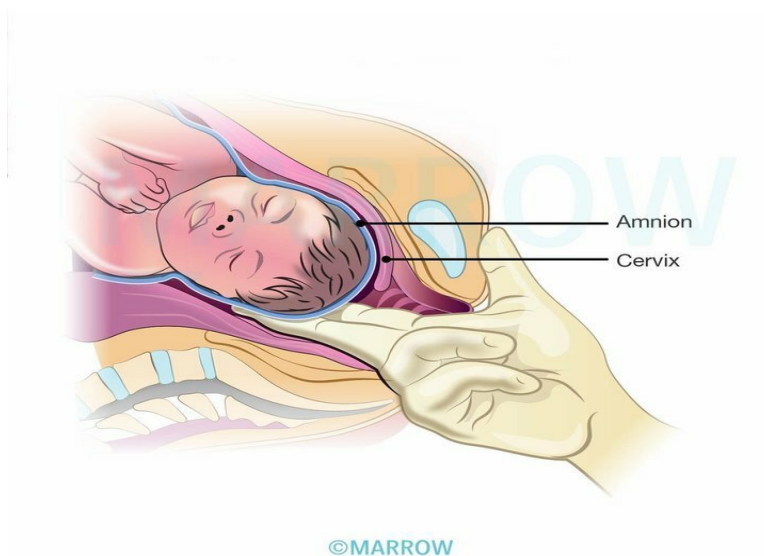
Question 2:

All of the following changes are seen during cervical ripening in a patient except _____.

- a) Collagen fibril diameter is increased
- b) Collagen fibrils are tightly packed
- c) Increase in glycosaminoglycans
- d) Stromal invasion with inflammatory cells

Question 3:

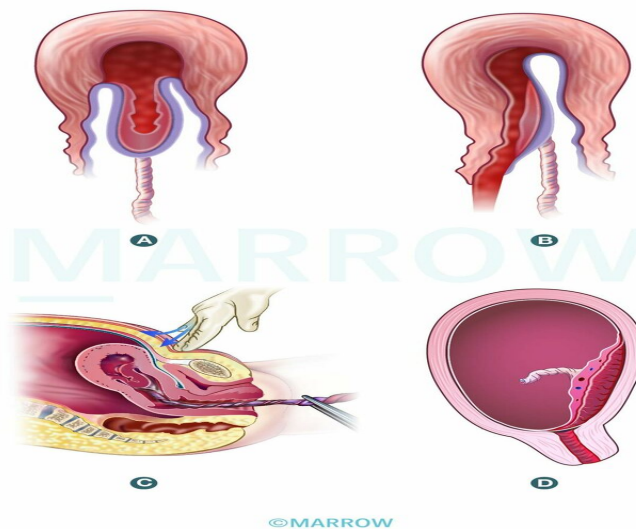
An obstetrician performs the following procedure for labor induction in a patient. What is the reflex elicited in the process?



- a) Andrews reflex
- b) Johnsons reflex
- c) Gails reflex
- d) Ferguson reflex

Question 4:

Which of the following image shows the Duncan method?



- a) A
- b) B
- c) C
- d) D

Question 5:

In which stage of labour, is the following expelled?



- a) Second
- b) Third
- c) Fourth
- d) First

Question 6:

All of the following are included in the first stage of labour for a pregnant woman except:

- a) Effacement of cervix
- b) Dilatation of cervix
- c) Crowning of fetal head
- d) Fetal descent beginning

Question 7:

What does fetal position refer to?

- a) Part of presentation which lies over the internal os
- b) Relation of different parts of fetus to one another
- c) Relation of the denominator to different quadrants of the pelvis
- d) Part of fetus lying in the lower segment of uterus

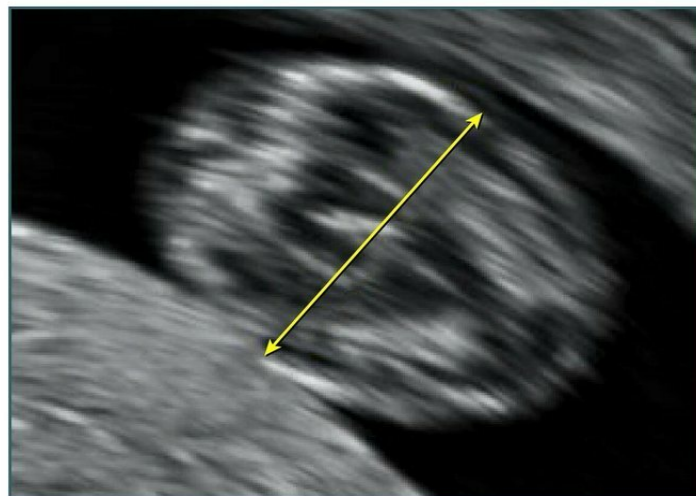
Question 8:

A senior resident examines a primigravida in labour and informs that the fetus is cephalic in attitude of flexion. What is the most common presenting part of this fetus?

- a) Vertex
- b) Sinciput
- c) Occiput
- d) Face

Question 9:

The ultrasonogram shows the diameter of the fetal head which is taken as a reference for engagement. Which diameter does this refer to?



- a) Biparietal diameter
- b) Bitemporal diameter
- c) Occipitofrontal diameter
- d) Suboccipitofrontal diameter

Question 10:

Which of the following is not a part of a partograph?

- a) Fetal heart rate

- b) Maternal heart rate
- c) Maternal temperature
- d) Fetal temperature

Question 11:

What are the cardinal movements of labour?

- a) Engagement—descent—flexion—internal rotation—extension—external rotation—expulsion
- b) Engagement—flexion—descent—internal rotation—extension—expulsion
- c) Engagement—flexion—descent—external rotation—expulsion
- d) Engagement—extension—internal rotation—external rotation—expulsion

Question 12:

A multigravida with gestational diabetes mellitus was admitted in labour at term and progressed to full dilatation. The fetal vertex failed to descend beyond -3 station. The descent of the fetus is facilitated by all except:

- a) Extension and straightening of the fetal body
- b) Pressure of amniotic fluid
- c) Uterine contractions and retractions
- d) Well-flexed head

Question 13:

Which of the following is a prerequisite for internal rotation of the fetal head for a woman in labor?

- a) Straightening of fetal body
- b) Tone of transversus abdominis
- c) Well-extended head
- d) Tone of levator ani muscles

Question 14:

Which of the following statements is true regarding the internal rotation of the fetal head?

- a) Occiput turns towards the transverse axis
- b) Rotation is not essential when the fetus is unusually small
- c) Internal rotation is followed by external rotation
- d) Rotation is earlier in primigravida than in multigravida

Question 15:

What is the minimum rate of cervical dilatation in primigravida during active labour?

- a) 1.5 cm/hr
- b) 1.2 cm/hr
- c) 1.7 cm/hr
- d) 2 cm/hr

Question 16:

Which of the following is part of the WHO guidelines for the management of second stage of labor?

- a) Do a routine episiotomy
- b) Manual support to perineum to maintain flexion of head
- c) Never give warm compress to the perineum
- d) Perform labor in lithotomy position

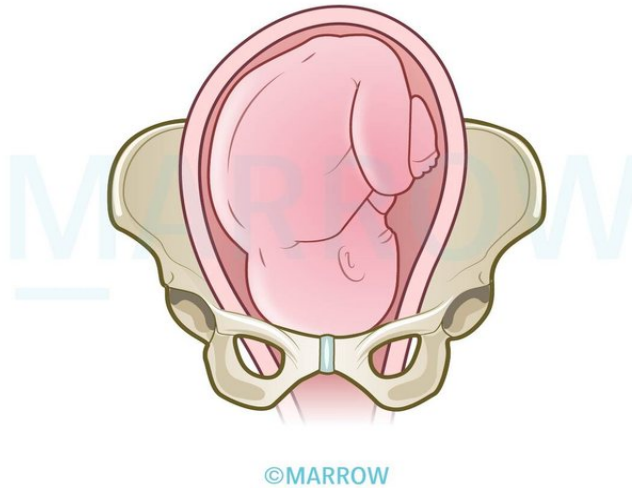
Question 17:

Modified Robson's classification class 1 includes:

- a) Nulliparous, singleton, cephalic, induced labour
- b) Nulliparous, singleton, cephalic, spontaneous labour
- c) Multiparous without previous CS, spontaneous labour
- d) Multiparous without previous CS, induced/CS before labour

Question 18:

Identify the fetal position in the given image.



- a) Right occipito-posterior
- b) Left occipito-anterior
- c) Left occipito-posterior
- d) Right occipito-anterior

Question 19:

Uterine contractions are palpable when its intensity is more than _____.

- a) 10 mm Hg
- b) 15 mmHg
- c) 20 mmHg
- d) 40 mmHg

Question 20:

A primigravida presents to the labor room at 39 weeks of gestation. On examination, you notice the fetal head has appeared at the vulva. Which of the following maneuvers can be used to assist the delivery of the head?

- a) Prague maneuver
- b) Lovset's maneuver
- c) Pinard's maneuver
- d) Ritgen's maneuver

Question 21:

Which dermatome is pain in the early stage of labor limited to?

- a) T10-L1
- b) S1-S3
- c) L4-L5
- d) L2-L3

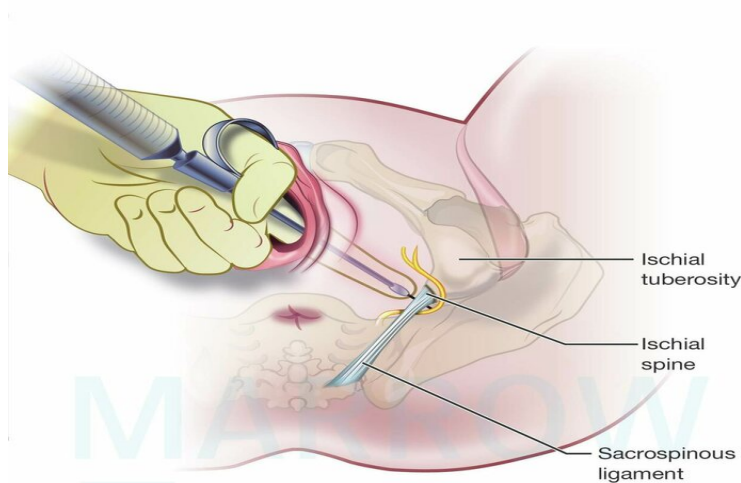
Question 22:

A young primigravida at 39 weeks of gestation presented to the casualty crying of intense abdominal pain. She was diagnosed to be in labour. Which of the following is not a characteristic of this pain?

- a) Painful uterine contraction
- b) Relieved by enema and sedatives
- c) Formation of bag of waters
- d) Progressive descent of presenting part

Question 23:

You notice the gynecology resident administer the nerve block seen in the image before performing an episiotomy. Which nerve roots are blocked?



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- a) L1,2,3
- b) L2,3
- c) S2,3,4
- d) L5,S1,S2

Question 24:

Paracervical block relieves pain from all except:

- a) Cervix
- b) Uterus
- c) Lower third of vagina
- d) Upper third of vagina

Question 25:

All of the following are indications for early clamping of cord in a new born baby except:

- a) Rh isoimmunization
- b) Postdated pregnancy
- c) Birth asphyxia
- d) Maternal diabetes

Question 26:

Which of the following is not a sign of placental separation in stage 3 of parturition of a pregnant woman?

- a) Schroeder's sign
- b) Shortening of cord
- c) Suprapubic bulge
- d) Kustner's sign

Question 27:

Which of the following is false regarding the modified WHO partogram?

- a) Right side of alert line indicates immediate referral to First Referral Unit (FRU)
- b) Alert and action line are separated by a difference of 4 hours
- c) Each small square in partogram is equal to 1 hour
- d) Partograph recording should be started once the cervical dilatation reaches 4 cm

Question 28:

Identify the following instrument commonly used in obstetrics.



- a) Mayo scissors
- b) Curved scissors
- c) Suture remover
- d) Episiotomy scissors

Question 29:

A nursing assistant posted in the obstetric care unit was asked to monitor the progress of labour. After how much dilatation of the cervix, should she start the documentation?

- a) 2cm
- b) 5cm
- c) 4cm
- d) 6cm

Answer Key

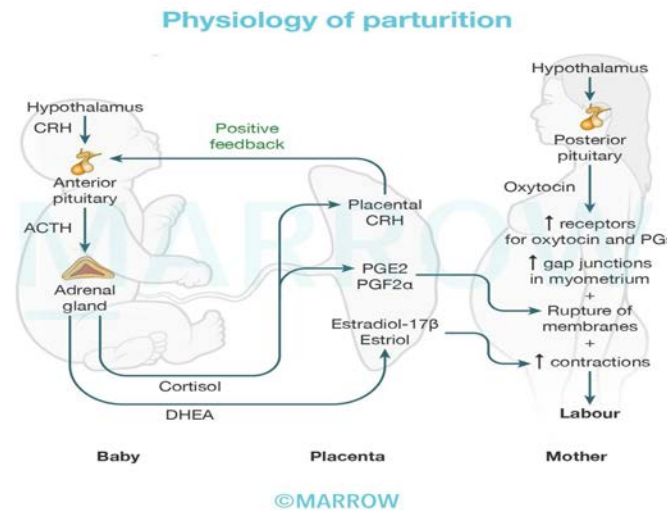
Question No.	Correct Option
1	b
2	b
3	d
4	b
5	b
6	c
7	c
8	a
9	a
10	d
11	a
12	d
13	d
14	b
15	b
16	b
17	b
18	d
19	a
20	d
21	a
22	b
23	c
24	c
25	b
26	b
27	c
28	d
29	b

Detailed Explanations

Solution to Question 1:

The fetal ACTH causes an increase in estrogen acting as the onset of labour in pregnant women. It results in the growth of the fetal adrenal gland which has the capacity for steroid biosynthesis i.e. estrogen.

Daily steroid production of fetal adrenal glands near term is more than in adults.



Solution to Question 2:

Collagen fibrils are loosely packed during cervical ripening.

Changes in cervical ripening:

- Collagen fibril diameter is increased, and there is increased spacing between fibrils leading to increased tissue compliance
- Increase in the amount of proteoglycan and glycosaminoglycan within the matrix
- Stromal invasion with inflammatory cells

Solution to Question 3:

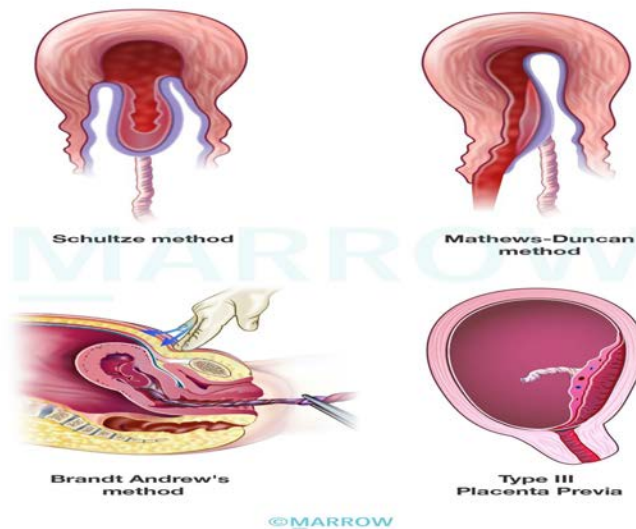
Stripping of membranes is depicted in the image, which elicits the Ferguson reflex.

Fetal ejection reflex or Ferguson reflex is a phenomenon in which mechanical stretching of the internal cervix enhances uterine activity. Release of oxytocin from the maternal pituitary has been suggested but not proven.

It is seen in stripping/sweeping and artificial rupture of membranes. Both of these are methods of induction of labor. It is associated with an increase in blood levels of prostaglandin F₂^o metabolite, also known as dinoprost/carboprost.

Solution to Question 4:

Image B shows the Duncan method of placental separation.



Comparison between two types of placental separation methods:

Brandt Andrew's method or controlled cord traction is a manoeuvre of expressing the placenta from the uterus in the third stage of labour.

One hand grasps the umbilical cord while the other is placed on the mother's abdomen with the fingers over the anterior surface of the uterus. While the hand on the abdomen is pressed backwards and slightly upward, the other applies gentle traction on the cord.

Type III placenta previa is a type of incomplete or partial central placenta previa. Here, the placenta covers the internal os partially i.e. it covers the internal os when closed but does not entirely do so when fully dilated.

Schultze method	Duncan method
Separation begins from the center of the placenta to the periphery.	Separation begins from the periphery to the center.
Blood does not escape externally, as the blood pours into the membrane sac.	Blood escapes from the vagina as the blood collects between the membranes and the uterine wall.

Solution to Question 5:

The placenta and membranes are expelled in the third stage of labour.

The clinical stages of labour may be summarized as follows:

Stages of labour	Definition
First	True labour pain tofull dilatation of cervix
Second	Full dilatation of cervix toexpulsion of fetus
Third	Expulsion of fetus toexpulsion of placenta
Fourth	One hour observationafter the expulsion of placenta

Solution to Question 6:

The encirclement of the largest head diameter by the vulval ring is called crowning. This occurs in the second stage of labour.

First stage of labour is divided into 2 phases:

Latent phase	Active phase
The onset of true labour pain up to 6cm cervical dilatation	From ≥ 6cm cervical dilatation with regular uterine contractions
Duration:10-12 hours- nullipara6-8 hours- multipara	Rate of cervical dilatation:1.2 cm/hr- nullipara1.5 cm/hr- multipara
Mainly cervicaleffacementoccurs	Mainly cervicaldilatationoccurs

Solution to Question 7:

Fetal position refers to the bony fixed point of reference on presenting part with respect to different quadrants of the pelvis.

Definition of various terminologies used for fetus in-utero:

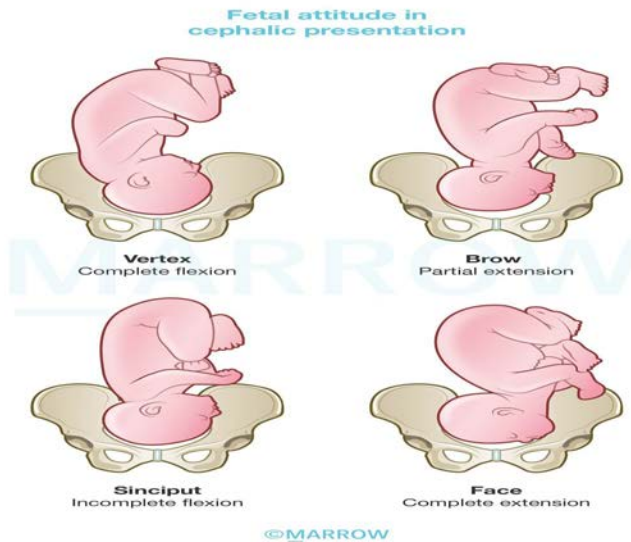
Term:	Definition:
Lie	Relationship of the long axis of fetus to the long axis of the maternal spine

Term:	Definition:
Presentation	Part of the fetus occupying the lower pole of the pelvis
Presenting part	Part of the presentation which overlies the internal os
Attitude	The relation of the different parts of the fetus to one another
Denominator	An arbitrary bony fixed point on the presenting part which comes in relation with the various quadrants of the maternal pelvis
Position	Relation of the denominator to the different quadrants of the pelvis
Engagement	When the biparietal diameter has passed the plane of the pelvic brim, the head is said to be engaged.

Solution to Question 8:

The most common presenting part when the fetus is in attitude of flexion is vertex.

Attitude	Presenting part
Complete flexion	Vertex (occipitoanterior)
Incomplete flexion	Vertex (sinciput)
Deflexion	Vertex (occipitoposterior)
Partial extension	Brow
Complete extension	Face

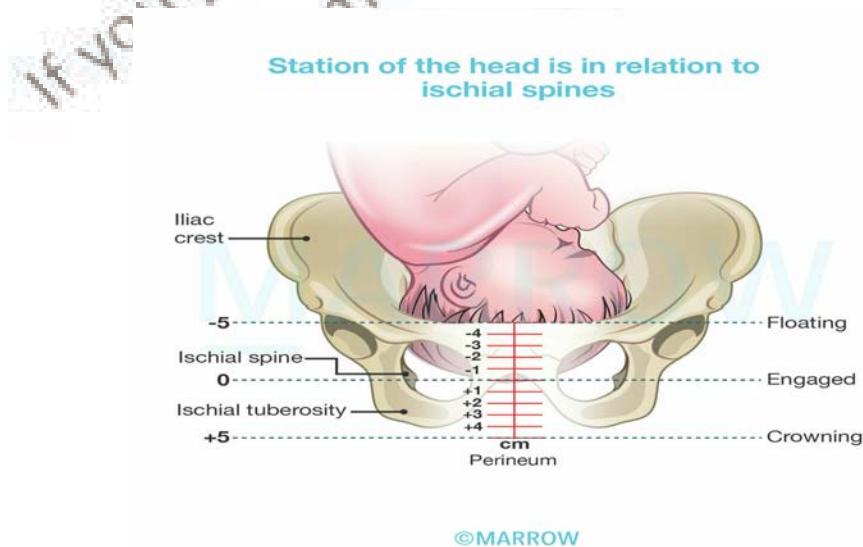


Solution to Question 9:

The fetal head is said to be engaged when the biparietal diameter (shown in the image) passes through the pelvic brim. Engagement of fetal head is with reference to this biparietal diameter.

It is the greatest transverse diameter in an occiput presentation. It also reflects the gestational age in the second trimester, with a variation of 7-10 days.

BPD is measured in the transthalamic view, at the level of the thalamus and cavum septum pellucidum (CSP). It is measured from the outer edge of the skull in the near field to the inner edge of the skull in the far field.



Solution to Question 10:

Fetal temperature is not a part of a partograph.

Assessment of progress of labor is best done by partograph. Components of partograph:

- Patient identification
- Fetal heart rate recorded every 30 min
- Pulse recorded every 30 min
- Blood pressure and temperature recorded every 4 h
- Uterine contractions recorded every 30 min
- Oxytocin-dose and concentration
- Urine analysis
- Vaginal examination done every 4 hours
- State of membranes and colour of liquor

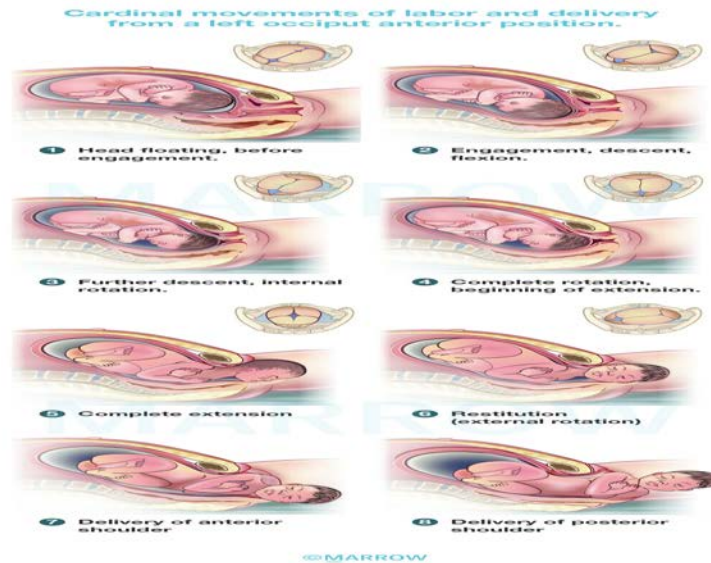
Solution to Question 11:

The cardinal movements of labour are engagement, descent, flexion, internal rotation, extension, external rotation, expulsion.

(Mnemonic: Every Day FIRE Raises Extremely)

- Every: Engagement
- Day: Descent
- F: Flexion
- IR: Internal rotation
- E: Extension
- Raises: external Rotation
- Extremely: Expulsion

The most common presentation is cephalic and presenting part is vertex. Head enters the pelvis through transverse diameter majorly and left occipito transverse is the commonest fetal position. Internal rotation of fetal head occurs at the level of ischial spine. In a vertex delivery, the baby's head is delivered by extension and in a breech delivery, the baby's head is delivered by flexion. The best time to give episiotomy is after crowning has occurred.



Solution to Question 12:

A well-flexed head facilitates internal rotation, not the descent of the presenting part.

The descent of the fetus is brought about by:

- Pressure of amniotic fluid
- Direct pressure of the fundus upon the breech with contractions
- Bearing-down efforts of maternal abdominal muscles
- Extension and straightening of the fetal body

Solution to Question 13:

The tone of levator ani muscles is a prerequisite for internal rotation of the head.

Prerequisites of internal rotation of the head are as follows:

- Well-flexed head
- Efficient uterine contraction
- Favorable shape at mid-pelvic plane
- Tone of the levator ani muscles

Solution to Question 14:

Internal rotation is not essential in the completion of labor when the fetus is unusually small.

Internal rotation begins with the occiput turning away from the transverse axis. In majority, internal rotation is completed by the time the head reaches the pelvic floor. Internal rotation is followed by extension of the head, which is then followed by external rotation.

Rotation occurs earlier in multigravida than primigravida whereas engagement occurs earlier in primigravida than multigravida.

Solution to Question 15:

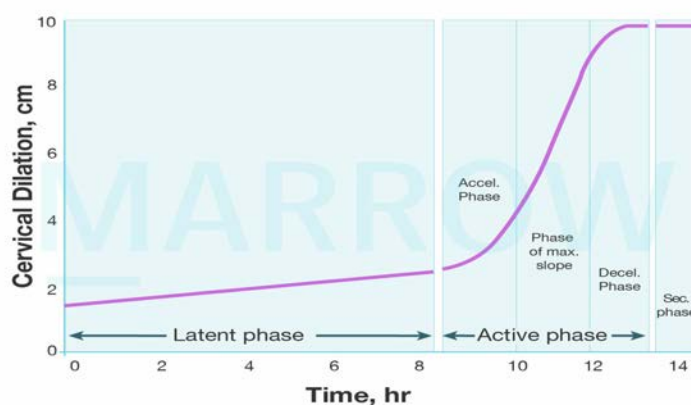
During active labor, the minimal rate of cervical dilatation in primigravida is 1.2 cm/hr.

First stage of labour is divided into 2 phases:

The cervical dilatation ranges from a minimal rate of 1.2 cm - 6.8 cm/hr in nullipara. Also, the multiparas progress faster in the active phase of labor with a minimal dilatation rate of 1.5 cm/hr.

Friedman curve is the sigmoid curve plotting the pattern of cervical dilatation during the latent and active phase of normal labour.

Latent phase	Active phase
The onset of true labor pain up to 6 cm cervical dilatation	Starts from ≥ 6 cm cervical dilatation with regular uterine contractions
Duration: 12 h (nullipara) 8 h (multipara)	The minimum rate of cervical dilatation: 1.2 cm/h (nullipara) 1.5 cm/h (multipara)
Concerned with cervical effacement	Concerned with cervical dilatation



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Solution to Question 16:

The practice of guarding the perineum during delivery of the fetal head is part of the WHO management guidelines for the second stage of labor.

The fingers of one hand support the perineum, while the second hand applies pressure to the fetal head to control the speed of crowning. The advantage of this is to prevent perineal tears.

The second stage of labor begins with full dilatation of the cervix and ends with the delivery of the fetus.

The main events include the descent of the fetus (bearing down by the mother, uterine contractions), cardinal movements of labor, and delivery of the fetus.

Management of the second stage of labor:

- The most common position used is the dorsal position with the hips flexed, knees flexed and abduction of the thighs. WHO recommends any position of choice including the upright position.
- WHO recommendations for the prevention of perineal tears include warm compresses to the perineum, perineal massage, guarding the perineum, and modified Ritgen's maneuver.
- Modified Ritgen's maneuver: A sterile towel or pad is held in one hand and placed over the perineum; the other hand applies gentle downward pressure on the occiput to promote flexion and controlled delivery of the head (ensuring the smallest diameter of the head passes through the outlet). Once the occiput is born—with gentle pressure, the head is extended—forehead, nose, and mouth are then delivered.
- WHO doesn't recommend fundal pressure and routine episiotomy.

Other options:

Option A: A liberal or routine episiotomy is discouraged. Episiotomy is indicated in cases of:

- Shoulder dystocia
- Breech delivery
- Fetal macrosomia
- Operative vaginal deliveries
- Persistent occiput posterior positions

Option C: Manual techniques, such as massage or application of warm compresses, are recommended for healthy pregnant women requesting pain relief during labor.

Option D: The position that is most comfortable to the mother i.e. upright/squatting position is encouraged. This position gives:

- less discomfort and difficulty in bearing down
- less labor pain
- less perineal/vaginal trauma and wound infections

Second stage	Nulliparous	Multiparous
Mean/average duration	50 minutes	30 minutes
Maximum duration	2 hours (+1 with epidural)	1 hour (+1 with epidural)

Second stage	Nulliparous	Multiparous
WHO maximum duration of 2nd stage/diagnosis of arrest	3 hours	2 hours

Solution to Question 17:

Class 1 of modified Robson's classification includes nullipara, equal to or $\geq$ 37 weeks, single, cephalic, spontaneous labor.

It is the most widely accepted classification system available for analyzing characteristics of delivering women worldwide. It consists of 10 patient population categories. The basis for these categories include:

- Parity
- Onset of labor and mode of delivery
- Gestational age
- Fetal presentation
- No. of fetuses

The modified Robson system classification is as follows:

Class 1	Nullipara, equal to or $\geq$ 37 weeks, single, cephalic, spontaneous labor
Class 2	Nullipara, equal to or $\geq$ 37 weeks, single, cephalic2a:induced labor2b:CS before labor
Class 3	Multipara, equal to or $\geq$ 37 weeks, single, cephalic, spontaneous labor (excludes previous CS)
Class 4	Multipara, equal to or $\geq$ 37 weeks, single, cephalic4a:induced labor4b:CS before labor
Class 5	Multipara, previous CS, equal to or $\geq$ 37 weeks, single, cephalic
Class 6	Nullipara, single, breech
Class 7	Multipara, single, breech (including previous CS)
Class 8	Multiple pregnancies (with or without previous CS)

Class 1	Nullipara, equal to or >37 weeks, single, cephalic, spontaneous labor
Class 9	Singleton pregnancy, oblique/transverse lie (with or without previous CS)
Class 10	Single, cephalic <37 weeks (including previous CS)

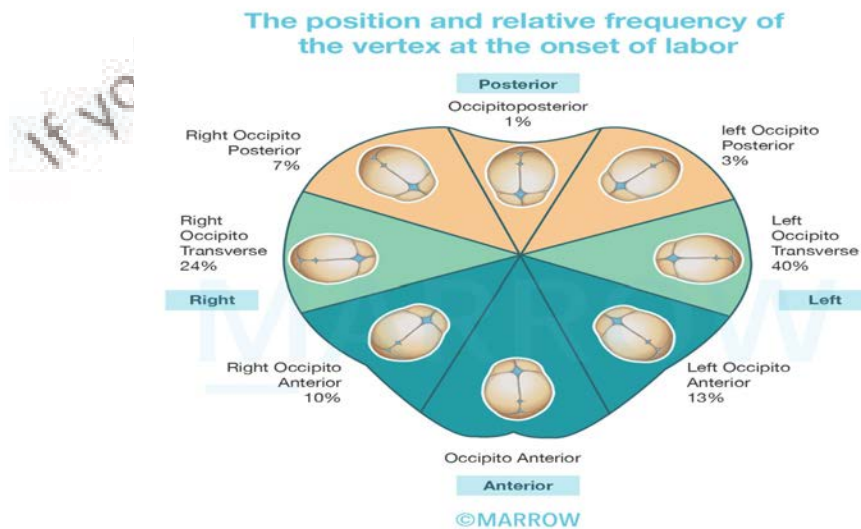
Solution to Question 18:

In the given vertex presentation, the fetal position is right occipitoanterior.

Fetal position refers to the relation of the denominator (arbitrary bony fixed point on the presenting part), with respect to different quadrants of the maternal pelvis. In vertex presentation, the denominator is taken as the occiput (which is close to the triangular posterior fontanelle).

To determine the fetal position the following two steps are followed:

- Determine the direction in which the occiput is facing:
 - Towards the pubic symphysis - occipito-anterior
 - Towards the sacral promontory - occipito-posterior
 - Midway between pubic symphysis and sacral promontory - occipito-transverse
- Determine if the occiput is facing the left or the right in relation to the maternal pelvis



Solution to Question 19:

Uterine contractions are palpable when the intensity is ≥ 10 mmHg.

Uterine contractions cause cervical dilatation at 15 mmHg and are painful beyond 15 mmHg.

Units for measuring the intensity of uterine contractions are mmHg and Montevideo units (MV).

1 MV = intensity of uterine contractions x number of contractions in 10 min.

During labour, normal uterine contractions are between 190 and 300 MV and at the time of delivery, it is 300 MV.

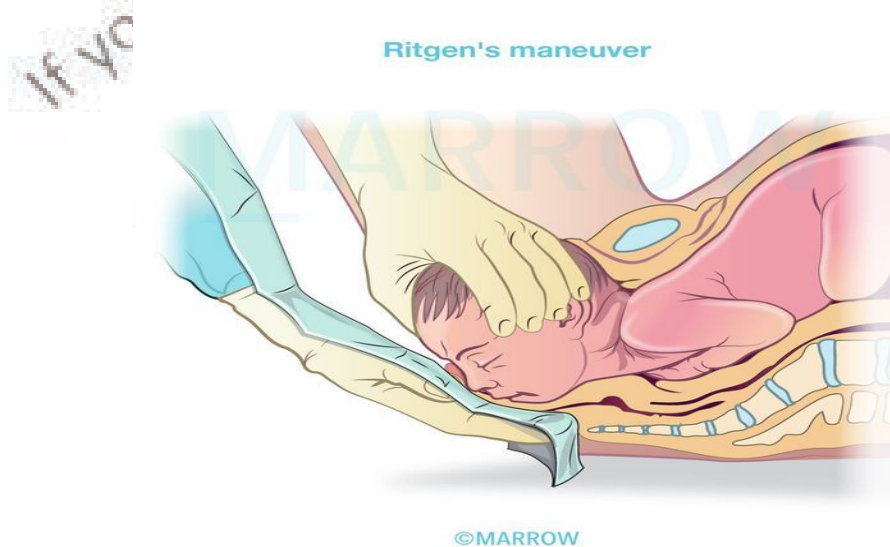
Stage of labour	Pressure
1st	40–50 mmHg
2nd	100–120 mmHg
3rd	100–120 mmHg

Solution to Question 20:

Ritgen's maneuver is done for delivery of head in normal labor.

This maneuver is done when the head distends the vulva and perineum enough to open the vaginal introitus to a diameter of ≥ 5 cm. A towel is draped and a gloved hand may be used to exert forward pressure on the chin of the fetus through the perineum. The other hand exerts pressure superiorly against the occiput.

This maneuver allows controlled delivery of the head.



Solution to Question 21:

Pain in the early stage of labor is limited to dermatomes T10-L1. This is of uterine origin.

Pain in the late stage of labour is due to dilatation of cervix and perineal stretching. The nerves carrying the pain include S2,3,4.

Solution to Question 22:

True labor pain is not relieved by enema and sedatives.

It is characterized by the following:

- Painful uterine contractions at regular intervals
- The frequency of contractions increases gradually
- The intensity and duration of contractions increase progressively
- Associated with 'show'- extrusion of the mucus plug that had previously filled the cervical canal during pregnancy
- Progressive effacement and dilatation of the cervix
- Progressive descent of the presenting part
- Formation of the 'bag of forewaters' i.e., bulging fluid-filled amniotic membrane presenting in front of the fetal head
- Not relieved by enema or sedatives

False labour pain is seen more in primigravidae than in parous women. It usually appears prior to the onset of true labor pain by 1 or 2 weeks in primigravidae and by a few days in multiparae.

Such pains are probably due to stretching of the cervix and lower uterine segment with consequent irritation of the neighboring ganglia.

Features	True labour pains	False labour pains
Cervical dilatation and effacement	Present	Absent
Pain	Lower abdomen and back, radiating to thighs	Lower abdomen only
Show	Present	Absent
Bag of water	Formed	Absent
Relief with enema	No	Yes

Solution to Question 23:

The nerve roots blocked in pudendal nerve block are S2,3,4. It is blocked just above the tip of the ischial spine by piercing the sacrospinous ligament.

Pudendal nerve block can be used for:

- Analgesia in the second stage of labor
- Episiotomy repair
- Pelvic floor relaxation for outlet instrument delivery
- Minor procedures of the lower vagina and perineum
- Perineal analgesia prior to application of vacuum/forceps

Pudendal block does not impact the cervix or upper vagina. It is ineffective for pain relief associated with uterine manipulation such as manual extraction of the placenta.

Solution to Question 24:

Pain from the lower third of the vagina is relieved by pudendal block, not paracervical block.

Paracervical block relieves the pain of uterine contractions and cervical dilatation. It also relieves pain from upper third of the vagina. It is not useful at the time of episiotomy as the lower part of the vagina is not anesthetized.

Note: Paracervical block is no longer considered safe as it causes fetal bradycardia. Epidural analgesia is preferred when complete relief of pain is needed throughout labour.

Solution to Question 25:

Postdated pregnancy is not an indication for early cord clamping.

Early cord clamping is indicated in:

- Birth asphyxia
- Rh isoimmunization
- Diabetic mother

The benefits of delayed cord clamping for up to 60 seconds is an increase in total body iron stores, expand blood volume and a decrease in anemia incidence in the neonate.

On the other hand, delayed cord clamping can lead to a higher hemoglobin concentration, which increases risks for hyperbilirubinemia and extended hospitalization for neonatal phototherapy.

Delayed cord clamping can also hinder timely neonatal resuscitation.

According to AIIMS Protocols in Neonatology, in uncomplicated preterm birth, delayed cord clamping should be attempted. This is done for a minimum time of 60 seconds after delivery for preterm infants not requiring immediate resuscitation after birth. It states that DCC must be done even if the mother has an HIV infection. But as per the NACO guidelines, early cord clamping is advised.

Solution to Question 26:

Lengthening of the umbilical cord is seen in placental separation.

Signs of placental separation:

- The uterus becomes globular, firm, and ballottable (earliest sign to appear)
- Schroeder's sign - Slight rise of fundal height as the separated placenta comes down in the lower segment and uterus rests over it
- Slight suprapubic bulging due to separated placenta distending the lower segment.
- Kustner's sign - On pushing, the uterus cephalad with a hand on the abdomen, the cord no longer recedes
- A sudden gush of blood
- Permanent lengthening of cord

Note: Schroeder's ring is a constriction ring/contraction ring formed due to incoordinate uterine action.

Solution to Question 27:

Each small square in a modified WHO partogram is equal to 30 minutes. Each small square in a composite WHO partogram is equal to 1 hour.

Solution to Question 28:

The instrument given in the picture is episiotomy scissors. The angulation in the scissor is to prevent the extension of pelvic tears into the anal margins and obstetric anal sphincter injuries / complete perineal tears.

Complications of episiotomy:

They can be immediate or remote.

Immediate complications include an extension of the incision into the rectum, vulval hematoma, wound dehiscence and injury to anal sphincter.

Remote complications include dyspareunia, chances of perineal lacerations in subsequent labor and scar endometriosis (rare).

Mayo scissors are long and stout scissors. They can be blunt-tipped/pointed tip/straight or curved. Straight mayo scissors are designed for cutting body tissues near the surface of a wound. They are also used to cut sutures and are also known as suture-cutting scissors.

Curved mayo scissors allow much deeper penetration into the wound and are used to cut thick tissues such as those found in the uterus, muscles, breast, and foot. Mayo scissors used for dissection are placed in tissue with the tips closed. The scissors are then opened so that the tips open and spread out the tissue during the dissection process.

Episiotomy scissor



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Normal Mayo scissors



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Solution to Question 29:

The WHO Labour Care Guide begins the documentation of labour progression at 5 cm of cervical dilatation.

The WHO Labour Care Guide is a tool designed to help health personnel involved in labour care. It aims to

- Guide the monitoring and help in the documentation of the progress of labour.
- Guide the health professionals to offer supportive care throughout labour.
- Assist doctors in promptly identifying and addressing emerging labour complications.
- Prevent the use of unnecessary interventions in labour.
- Support audits and quality improvement.

The differences between the WHO-modified partograph and the latest WHO Labour Care Guide are tabulated below:

Note: Previously the answer to the question was 4 cm (according to the modified WHO partograph). However, this information is outdated, therefore the answer to this question has been revised per the latest WHO guidelines.

WHO modified partograph	WHO Labour Care Guide
Documentation of labour progression begins at 4 cm of cervical dilatation.	Documentation of labour progression begins at 5 cm of cervical dilatation.
Fixed 1 cm/hour 'alert' and 'action' lines	Time-based cut-offs for each centimetre of cervical dilatation to guide the obstetrician to take action. For example, at 5 cm of cervical dilatation, if there is no progress >6 hours, action needs to be taken.
Monitoring of labour is restricted to the first stage only.	Monitoring of the second stage of labour is included
No section on supportive care interventions	Addition of a section on supportive care of the mother- labour companionship, pain relief, oral fluid intake and posture
Documentation of strength, duration, and frequency of uterine contractions	Documentation of only the duration and frequency of uterine contractions

WHO LABOUR CARE GUIDE

Name: _____ Parity: _____ Labour onset: _____ Active labour diagnosis (Date): _____

Impaired membranes (Date): _____ Time: _____ Risk factors: _____

	1	2	3	4	5	6	7	8	9	10	11	12
SPONTANEOUS LABOUR	CONTRADICTIONS Contraction: Pain: Oral fluid: Posture: STAGE Stage: Amniotic fluid: Perineal position: Clap: Anal dilation:											
WOMAN	Pulse: Temperature: Respiratory rate: Uterine: Contraction:											
LABOUR PROGRESS	Contraction: Contraction: Contraction: Contraction:											
DELIVERY	Contraction: Contraction: Contraction: Contraction:											
ASSESSMENT	Contraction: Contraction: Contraction: Contraction:											
PLAN	Contraction: Contraction: Contraction: Contraction:											

INSTRUCTIONS: CHECK ANY OBSERVATION MEETING THE CRITERIA IN THE "ALERT" COLUMN. ALERT THE MIDWIFE IMMEDIATELY ON DOCTOR AND RECORD THE ALERT/DELAY AND ACTION TAKEN. LABOUR DETECTION BEYOND 12H. PLEASE CONTINUE ONLY WITH LABOUR CARE. MONITORING: 1. 10-15 min. 2. 15-30 min. 3. 30-45 min. 4. 45-60 min. 5. 60-75 min. 6. 75-90 min. 7. 90-105 min. 8. 105-120 min. 9. 120-135 min. 10. 135-150 min. 11. 150-165 min. 12. 165-180 min. 13. 180-195 min. 14. 195-210 min. 15. 210-225 min. 16. 225-240 min. 17. 240-255 min. 18. 255-270 min. 19. 270-285 min. 20. 285-300 min. 21. 300-315 min. 22. 315-330 min. 23. 330-345 min. 24. 345-360 min. 25. 360-375 min. 26. 375-390 min. 27. 390-405 min. 28. 405-420 min. 29. 420-435 min. 30. 435-450 min. 31. 450-465 min. 32. 465-480 min. 33. 480-495 min. 34. 495-510 min. 35. 510-525 min. 36. 525-540 min. 37. 540-555 min. 38. 555-570 min. 39. 570-585 min. 40. 585-600 min. 41. 600-615 min. 42. 615-630 min. 43. 630-645 min. 44. 645-660 min. 45. 660-675 min. 46. 675-690 min. 47. 690-705 min. 48. 705-720 min. 49. 720-735 min. 50. 735-750 min. 51. 750-765 min. 52. 765-780 min. 53. 780-795 min. 54. 795-810 min. 55. 810-825 min. 56. 825-840 min. 57. 840-855 min. 58. 855-870 min. 59. 870-885 min. 60. 885-900 min. 61. 900-915 min. 62. 915-930 min. 63. 930-945 min. 64. 945-960 min. 65. 960-975 min. 66. 975-990 min. 67. 990-1005 min. 68. 1005-1020 min. 69. 1020-1035 min. 70. 1035-1050 min. 71. 1050-1065 min. 72. 1065-1080 min. 73. 1080-1095 min. 74. 1095-1110 min. 75. 1110-1125 min. 76. 1125-1140 min. 77. 1140-1155 min. 78. 1155-1170 min. 79. 1170-1185 min. 80. 1185-1200 min. 81. 1200-1215 min. 82. 1215-1230 min. 83. 1230-1245 min. 84. 1245-1260 min. 85. 1260-1275 min. 86. 1275-1290 min. 87. 1290-1305 min. 88. 1305-1320 min. 89. 1320-1335 min. 90. 1335-1350 min. 91. 1350-1365 min. 92. 1365-1380 min. 93. 1380-1395 min. 94. 1395-1410 min. 95. 1410-1425 min. 96. 1425-1440 min. 97. 1440-1455 min. 98. 1455-1470 min. 99. 1470-1485 min. 100. 1485-1500 min. 101. 1500-1515 min. 102. 1515-1530 min. 103. 1530-1545 min. 104. 1545-1560 min. 105. 1560-1575 min. 106. 1575-1590 min. 107. 1590-1605 min. 108. 1605-1620 min. 109. 1620-1635 min. 110. 1635-1650 min. 111. 1650-1665 min. 112. 1665-1680 min. 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163. 2430-2445 min. 164. 2445-2460 min. 165. 2460-2475 min. 166. 2475-2490 min. 167. 2490-2505 min. 168. 2505-2520 min. 169. 2520-2535 min. 170. 2535-2550 min. 171. 2550-2565 min. 172. 2565-2580 min. 173. 2580-2595 min. 174. 2595-2610 min. 175. 2610-2625 min. 176. 2625-2640 min. 177. 2640-2655 min. 178. 2655-2670 min. 179. 2670-2685 min. 180. 2685-2700 min. 181. 2700-2715 min. 182. 2715-2730 min. 183. 2730-2745 min. 184. 2745-2760 min. 185. 2760-2775 min. 186. 2775-2790 min. 187. 2790-2805 min. 188. 2805-2820 min. 189. 2820-2835 min. 190. 2835-2850 min. 191. 2850-2865 min. 192. 2865-2880 min. 193. 2880-2895 min. 194. 2895-2910 min. 195. 2910-2925 min. 196. 2925-2940 min. 197. 2940-2955 min. 198. 2955-2970 min. 199. 2970-2985 min. 200. 2985-3000 min. 201. 3000-3015 min. 202. 3015-3030 min. 203. 3030-3045 min. 204. 3045-3060 min. 205. 3060-3075 min. 206. 3075-3090 min. 207. 3090-3105 min. 208. 3105-3120 min. 209. 3120-3135 min. 210. 3135-3150 min. 211. 3150-3165 min. 212. 3165-3180 min. 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Abnormal Labour

Question 1:

A young woman delivered her baby on the way to the hospital. While examining her, you notice a perineal tear. This injury could have been prevented by all except:

- a) Maintaining flexion of head
- b) Timely episiotomy as a routine
- c) Slow delivery of the head in between contractions
- d) Effective perineal guard

Question 2:

What is the most common cause of true cephalopelvic disproportion in India?

- a) Malpresentations
- b) Rickets
- c) Trauma
- d) Occipito posterior position

Question 3:

You are monitoring uterine contractions in a term primigravida in labor. You find that the contractions are uncoordinated and the frequency is about 8 contractions every 10 minutes. Which of the following is incorrect regarding your finding?

- a) Commonly associated with occipito posterior
- b) It is common in nulliparous labour
- c) Oxytocin administration is beneficial
- d) Management includes wait and watch policy

Question 4:

A 37-week primigravida has mild labor pains for 20 hours and the cervix is persistently 1 cm dilated but non-effaced. What would be the appropriate management?

- a) Sedation and wait
- b) Augmentation with oxytocin
- c) Cesarean section
- d) Amniotomy

Question 5:

A 34-year-old G3P2L2 woman presents to the ER with hypotension and severe abdominal pain. On examination, fetal parts are palpable over the abdomen. USG abdomen revealed a large uterine wall defect. Which of the following is not a feature of this condition?

- a) Maternal bradycardia
- b) Fetal bradycardia
- c) Vaginal bleeding
- d) Hematuria

Question 6:

A young primigravida at 36 weeks of gestation came with complaints of decreased perception of fetal movements for the past 2 days. NST is non-reactive and on ultrasound fetal cardiac activity is absent. What will be the best management in this patient?

- a) Forceps extraction
- b) Wait for spontaneous expulsion
- c) Cesarean section
- d) Artificial rupture of membranes

Question 7:

A G2P1L0 woman presents to OPD at 36 weeks of gestation for a routine visit. She mentions that she has a history of prior stillbirth at 37 weeks. What is the best time for delivery in this patient?

- a) Immediately
- b) 37 weeks
- c) 38 weeks
- d) 39 weeks

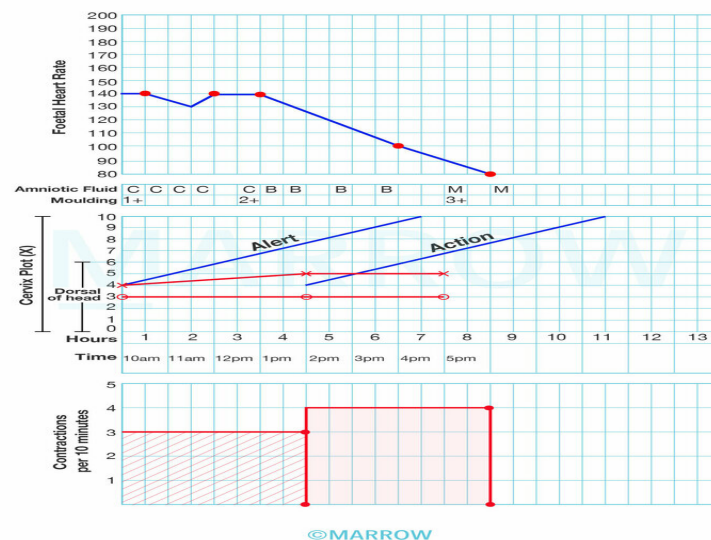
Question 8:

A pregnant woman presents with obstructed labor and is grossly dehydrated. Investigations reveal fetal demise. What will be the management?

- a) Craniotomy
- b) Decapitation
- c) Cesarean section
- d) Forceps extraction

Question 9:

The partogram of a 28 year old G2P1 is shown below. What is the next step in management?



- a) Wait and watch
- b) Oxytocin infusion
- c) Forceps delivery
- d) Cesarean section

Question 10:

After successful delivery of the baby, the obstetrician witnesses that the episiotomy performed has extended backwards beyond the perineal body. Which among the following structures is likely to be injured in this case?

- a) External anal sphincter
- b) Ischiocavernosus
- c) Bulbospongiosis
- d) Urethral sphincter

Question 11:

A midwife is examining a pregnant woman in labour when she noticed that the cord is prolapsed. What should be the next immediate step of management?

- a) Gently replace the cord inside the vagina
- b) Inform the doctor and shift to OT
- c) Put her in Trendelenberg position
- d) Do vaginal examination and find the fetal presentation

Question 12:

Abdominal examination of a 28-year-old primigravida, who is in the second stage of labour, reveals retraction ring between the active upper segment and the distended lower segment. Which would best describe this ring?

- a) Physiological retraction ring
- b) Pathological retraction ring
- c) Physiological contraction band
- d) Pathological contraction band

Question 13:

A 26-year-old term primigravida presents to the clinic for a regular check-up. On examination, uterus corresponds to term and head of the fetus is non-engaged. What is the most common cause for this finding?

- a) Cephalopelvic disproportion
- b) Deflexed head
- c) Transverse lie
- d) Breech

Question 14:

Which of the following is not a cardinal movement occurring during labor?

- Flexion
- Extension
- Internal rotation
- Asynclitism

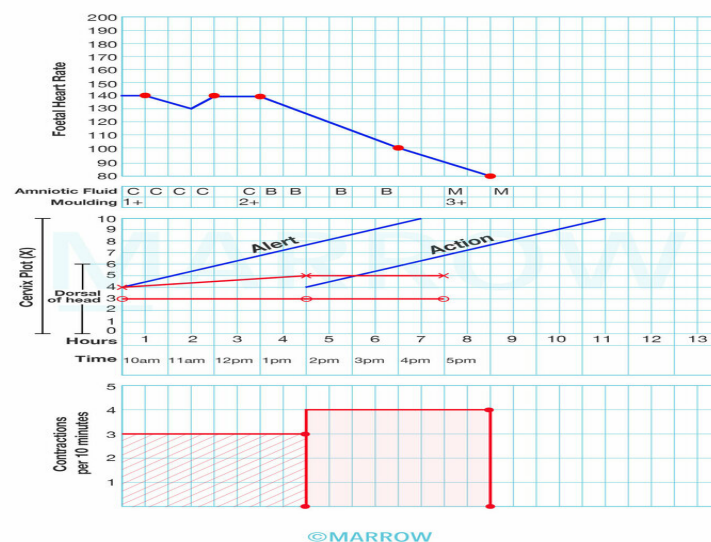
Question 15:

A 28-year-old primigravida is brought to the labour room in the latent phase of labor. How long will you wait till you declare her latent phase to be prolonged?

- a) 12 hours
b) 14 hours
c) 18 hours
d) 20 hours

Question 16:

All of the following findings are seen in the partogram of a patient except:



- a) Arrest of dilatation**
b) Arrest of descent

- c) Fetal distress
- d) Uterine tachysystole

Question 17:

A 26-year-old primigravida presents with labour pains. At 39 weeks of gestation, she was diagnosed with prolonged phase of labour. Which of the following is false regarding this condition?

- a) Greater than 14 hrs in nullipara
- b) Oxytocin stimulation given
- c) Excessive sedation can prolong this phase
- d) Morphine is used in management

Question 18:

Protracted active phase dilatation in a multiparous woman is when cervical dilatation is _____.

- a) ≤ 1 cm/hour
- b) ≤ 1.2 cm/hour
- c) ≤ 1.5 cm/hour
- d) ≤ 2 cm/hour

Question 19:

Which among the following is false with respect to active phase of labor?

- a) The minimum rate of cervical dilatation is 1.2cm/hr in nullipara
- b) Arrest of dilatation is cessation of dilatation for ≥ 2 hours
- c) Arrest of descent is cessation of descent for ≥ 1 hour
- d) Poor cervical dilatation causes prolonged active phase

Question 20:

Protracted descent in labor in a nullipara is _____

- a) ≤ 1 cm /hour

- b) <1.2 cm /hour
- c) 1.5 cm /hour
- d) <2 cm /hour

Question 21:

A multigravida is posted for C-section due to arrest of descent in labor. It is defined as no descent for _____

- a) > 0.5 hr
- b) >1 hr
- c) >1.5 hr
- d) >2 hr

Answer Key

Question No.	Correct Option
1	b
2	b
3	c
4	a
5	a
6	b
7	d
8	c
9	d
10	a
11	c
12	b
13	b
14	d
15	d
16	d
17	a
18	c

19	d
20	a
21	b

Detailed Explanations

Solution to Question 1:

Perineal injury cannot be prevented in normal labor by routine timely episiotomy. Routine episiotomy is associated with an increased incidence of disruption of anal sphincters and rectal tears.

Indications of episiotomy:

- Shoulder dystocia
- Breech delivery
- Fetal macrosomia
- Operative vaginal deliveries
- Persistent occiput posterior positions

Methods of prevention of perineal laceration:

- Avoid delivery by early extension
- Deliver head in between contractions
- Perform timely episiotomy when indicated (not routine).

Solution to Question 2:

The most common cause of true cephalopelvic disproportion (CPD) in India is rickets.

The most common cause of true CPD in developing countries is vitamin D deficiency and rickets. In developed countries, it is trauma to the pelvis. It occurs due to pelvic dystocia (larger fetus and smaller pelvis).

Relative CPD is the most common type of CPD. It is caused due to fetal malposition (occiput posterior position/brow presentation).

Solution to Question 3:

The given clinical finding is suggestive of uterine tachysystole. Oxytocin administration is not beneficial here as it leads to hyperstimulation of the uterus.

Tachysystole (hypertonic uterine dysfunction):

- The uterine contractions are uncoordinated and increased in frequency (>5 contractions in 10 minutes)
- Contractions occur with an increase in the baseline tone without rest in between them
- It is associated with late active phase dysfunction i.e., when the cervix is almost 7 cm dilated
- The most common cause is occipito-posterior position. Hence, a wait and watch policy is adopted
- It is more common in nulliparous females
- Oxytocin administration leads to hyperstimulation of the uterus and is not recommended.

Hypotonic uterine dysfunction:

- Uterine contractions are inadequate
- It is associated with early active phase
- It is more common in nulliparous females.

Solution to Question 4:

The given clinical scenario is suggestive of either false labor pain or prolonged latent phase. The appropriate management is to sedate the patient and wait.

False labor can be differentiated from the latent phase of labor by therapeutic rest i.e., the patient is sedated with morphine.

Augmentation with oxytocin is done only in those patients who even after therapeutic rest continue to be in prolonged latent phase.

There is no indication of cesarean section unless cephalopelvic disproportion is present. Amniotomy is not useful and should be avoided in patients with prolonged latent phase.

Solution to Question 5:

The given clinical scenario is suggestive of uterine rupture. Maternal tachycardia is seen in this condition.

Clinical features of uterine rupture:

- Maternal tachycardia
- Fetal bradycardia/absent fetal heart sounds
- Hypotension
- Abdominal tenderness
- Palpable fetal parts on per abdomen examination
- Bandl's ring
- Bleeding per vagina

- Hematuria

Common causes of uterine rupture are previous caesarean delivery, myomectomy, silent rupture (in previous pregnancy), excess use of oxytocin, grand multipara, uterine overdistension, obstructed labour, difficult forceps delivery, placenta percreta

Solution to Question 6:

The given clinical picture is suggestive of intrauterine fetal death (IUFD). The best management for such cases at 36 weeks is waiting for spontaneous expulsion.

Forceps extraction is only necessary if spontaneous expulsion doesn't take place.

Artificial rupture of membranes (ARM) is not done due to the risk of infections. Cesarean section is avoided as far as possible in case of fetal death.

Waiting till term is not recommended in cases of uterine infections, disseminated intravascular coagulation, and psychological upset.

Solution to Question 7:

The best time for elective delivery in patients with a history of previous stillbirth is 39 weeks, either by induction or cesarean section.

The recommendations for the management of subsequent pregnancy after stillbirth is as follows:

- Ultrasonographic screening for fetal growth restriction after 28 weeks
- Kick counts starting from 28 weeks of gestation
- Antepartum fetal surveillance starting at 32 weeks or 1-2 weeks earlier than prior stillbirth
- Support and reassurance

Labor induction is suitable in such cases. Cesarean delivery is done for those with a contraindication to induction.

Solution to Question 8:

The best management in the given clinical scenario is cesarean section.

Obstructive labor management involves administering intravenous fluids to correct dehydration, providing antibiotics to prevent infection, and promptly performing a cesarean section to relieve the obstruction. If the fetus is alive, an immediate cesarean section is typically performed. Management of obstructed labor never involves a wait-and-watch approach or augmenting with oxytocin.

Clinical features of obstructed labor include maternal exhaustion, dehydration, tachycardia, tachypnea, genital sepsis, postpartum hemorrhage, and shock.

On P/A examination the upper uterine segment is tender and tonically contracted. The lower uterine segment is thin and stretched. A groove/depression is felt between the upper uterine segment and the lower uterine segment. This is called a Bandl's ring.

Complications of obstructed labor include rupture of the uterus and vesicovaginal fistula.

Solution to Question 9:

The above partogram shows a case of obstructed labour and management is caesarean section.

There is a secondary arrest of dilatation for >2 hours and arrest of descent for >1 hour. The patient has adequate uterine contractions as shown. The amniotic fluid is stained with meconium and there is fall in fetal heart rate which indicates that the baby is in distress.

There is no place for wait and watch and use of oxytocin to stimulate uterine contractions in obstructed labor.

Forceps can be applied only when the cervix is fully dilated and the head is low down.

Solution to Question 10:

External anal sphincter is the structure injured in an episiotomy extended posteriorly beyond the perineal body.

Perineal tear have been divided into four degrees:

- First degree- laceration is restricted to the skin of the fourchette
- Second degree- muscles of the perineal body are torn through
- Third degree- the tear extends partially backward through the external sphincter of the anus
- Fourth degree- the external anal sphincter is torn and anal mucosa is also involved

A central tear of the perineum is when the head penetrates first through the posterior vaginal wall, then through the perineal body and appears through the skin of the perineum. It usually occurs in patients with a contracted outlet.

Solution to Question 11:

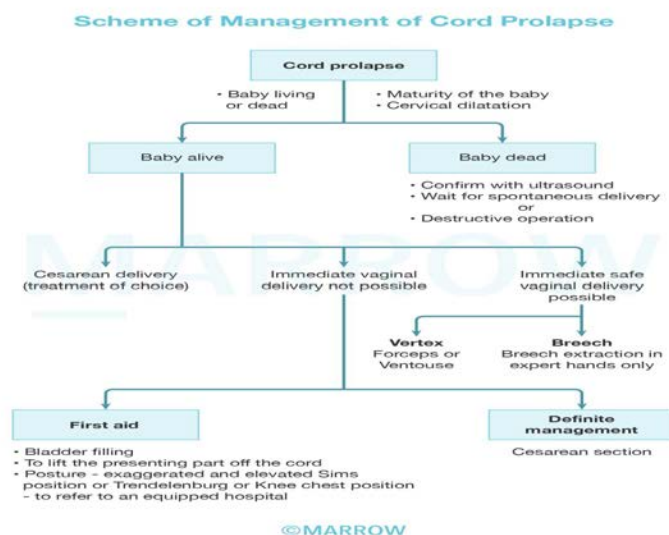
In this scenario the midwife should put the pregnant women in a Trendelenburg position or in an exaggerated and elevated Sims' position with a pillow or wedge under the hip or thigh.

Caesarean section is the recommended treatment of choice for cord prolapse. Prior to performing the procedure, several measures can be taken to alleviate cord compression. These measures involve:

- Manually elevating the fetal presenting part using two fingers or the entire hand through the vagina

- Positioning the patient in steep Trendelenburg, exaggerated Sim's position, or knee-chest position
- Filling the bladder with 500–700 cc or more of saline (known as the Vagos method)

After the diagnosis of cord prolapse is made, it is crucial not to attempt cord replacement. Such attempts are not only ineffective but also result in the inevitable rupture of the membranes, leading to cord prolapse.



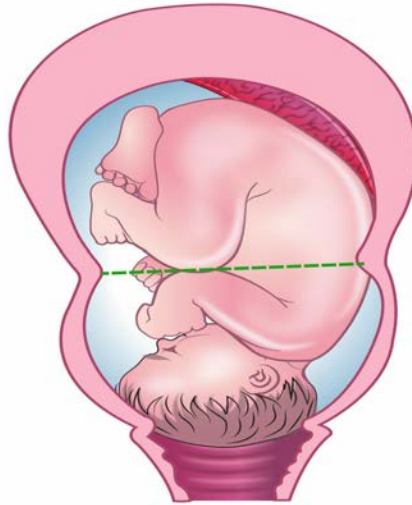
Solution to Question 12:

Bandl's ring refers to pathological retraction ring. It is formed in cases of obstructed labour.

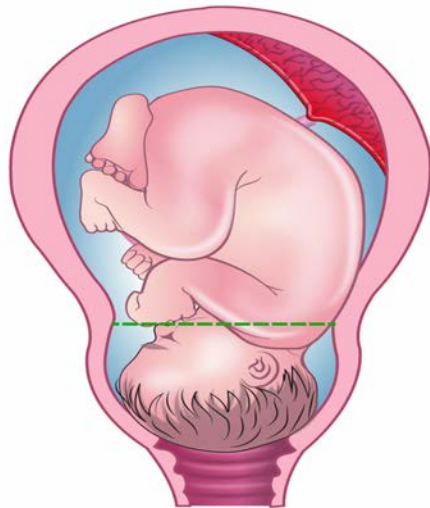
The upper segment of the uterus contracts and retracts vigorously in an attempt to overcome the obstruction while the lower segment dilates.

With each contraction there is myometrial shortening so that the actively contracting upper segment becomes progressively thicker and shorter. The passive lower segment is progressively stretched and becomes thinner.

The junction between the two segments stands out prominently as a pathological retraction ring.



Schroeder's constriction ring is formed due to incoordinate uterine action when there is localized spastic contraction of a ring of circular muscle fibres of the uterus. It is formed due to injudicious use of oxytocin and premature rupture of membranes (PROM).



Solution to Question 13:

The most common cause of non-engagement at term in a primigravida is deflexed head followed by cephalopelvic disproportion.

Non-engagement at the onset of active phase of labour is a risk factor for caesarean section.

The problems of prolonged labour are that the woman is exposed to high risk of infection, ketosis and obstructed labour while the fetus faces the danger of asphyxia and infection.

Solution to Question 14:

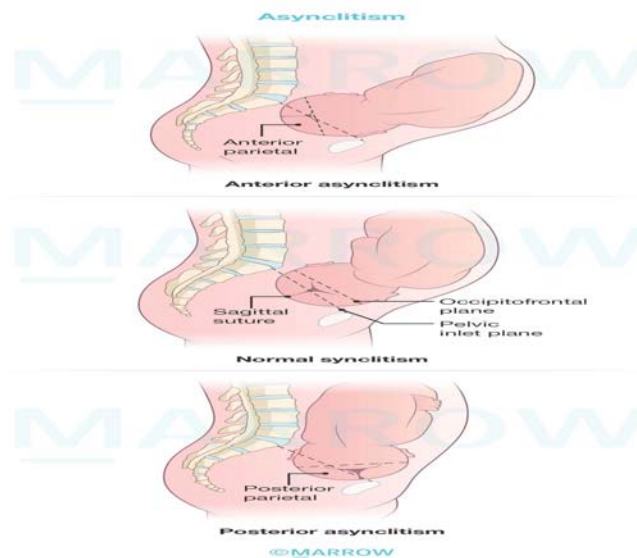
Asynclitism is not a cardinal movement during labor.

Asynclitism refers to the position of a baby in the uterus such that the head of the baby is presenting first and is tilted to the shoulder. This causes the fetal head to no longer be in line with the birth canal.

Moderate degrees of asynclitism are the rule in normal labor. However, if severe, this condition is a common reason for cephalopelvic disproportion even with an otherwise normal sized pelvis.

Asynclitism can be of 2 types:

- When the sagittal suture lies anteriorly, the posterior parietal bone becomes the leading presenting part and is called posterior asynclitism.
- When the sagittal suture lies posteriorly, the anterior parietal bone becomes the leading presenting part and is called anterior asynclitism.



Solution to Question 15:

Latent phase is said to be prolonged in a primigravida if it lasts for more than 20 hours.

Friedman defined prolonged latent phase as:

- A nullipara who has not entered the active phase by 20 hours after the onset of the latent phase
- A multipara who has not entered the active phase by 14 hours after the onset of the latent phase

Solution to Question 16:

The above partogram shows secondary arrest of dilatation for >2 hour, arrest of descent for >1 hour and a fall in fetal heart rate suggestive of fetal distress.

These are characteristic findings of obstructed labour and management is caesarean section.

Solution to Question 17:

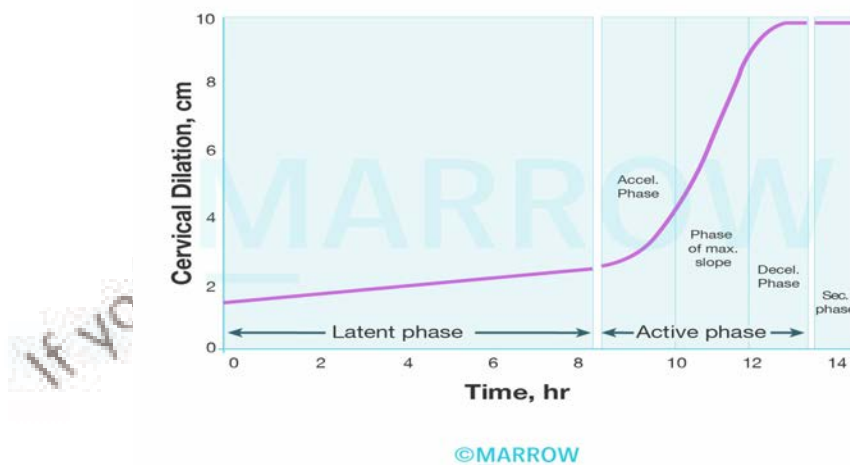
The given clinical features suggest that the patient is in latent phase of labor. Latent phase is said to be prolonged if it is:

- >20 h in nullipara
- >14 h in multipara

Causes of prolonged latent phase:

- Excessive sedation/epidural analgesia
- Poor cervical conditions (thick, uneffaced)
- False labour (most common cause in multipara)

Friedman's curve is used in the diagnosis of prolonged labour. It is the sigmoid curve plotting the pattern of cervical dilatation during the latent and active phase of normal labour.



Management includes morphine administration, rest and oxytocin stimulation.

Solution to Question 18:

Protracted active phase dilatation in a multiparous woman is when cervical dilatation is less than 1.5 cm/hour.

Solution to Question 19:

Poor cervical dilatation causes a prolonged latent phase.

Arrest of cervical dilatation is defined as the lack of cervical dilatation for ≥ 2 hours during the active phase of labour.

According to the Obstetric Care Consensus Committee, when cervical dilatation has reached ≥ 6 cm with ruptured membranes, and there is no cervical change despite 4 hours of adequate uterine contractions or 6 hours of inadequate contractions, a cesarean delivery is warranted.

Solution to Question 20:

The protracted descent in labor in a nullipara is less than 1 cm/hour.

Solution to Question 21:

Arrest of descent in labor in a multigravida is no descent for more than 1 hour.

Sold by @itachibot
If you purchased this from someone else,
you may have been scammed.

Induction and Augmentation of Labour

Question 1:

What is the technique of stimulating uterine contractions in a pregnant patient prior to the onset of spontaneous labour known as?

- a) Augmentation
- b) Induction
- c) Stimulation
- d) Termination

Question 2:

Which of the following is not a contraindication for the induction of labor?

- a) Pelvic tumor
- b) Herpes infection
- c) Heart disease
- d) History of lower transverse cesarean section in last three pregnancies

Question 3:

A 30-year-old G3P2 woman was admitted to the ward at 40 weeks of gestation for induction of labor. You were asked to calculate her Bishop's score. What should be the minimum score before performing this procedure?

- a) 3
- b) 5
- c) 6
- d) 8

Question 4:

During rounds in the postnatal ward, you notice a patient's case sheet mentioning she had uterine tachysystole following labor induction. What is the criteria for this diagnosis?

- a) More than 5 contractions in a 5-min period
- b) More than 5 contractions in a 10-min period
- c) More than 10 contractions in a 5-min period
- d) More than 10 contractions in a 10-min period

Question 5:

A primigravida was admitted to the labor room at 41 weeks of gestation for the induction of labor. What is the dose of misoprostol used for this procedure?

- a) 25 mg
- b) 25 mcg
- c) 50 mg
- d) 50 mcg

Question 6:

Oxytocin is being given for augmentation of labour in a multigravida admitted to the labour room. Which of the following is not a mechanism of action of this drug?

- a) Receptor mediation through G-protein coupled receptors
- b) Activation of voltage mediated Ca channel
- c) Production of prostaglandins
- d) Competitive blocking of acetylcholine receptors

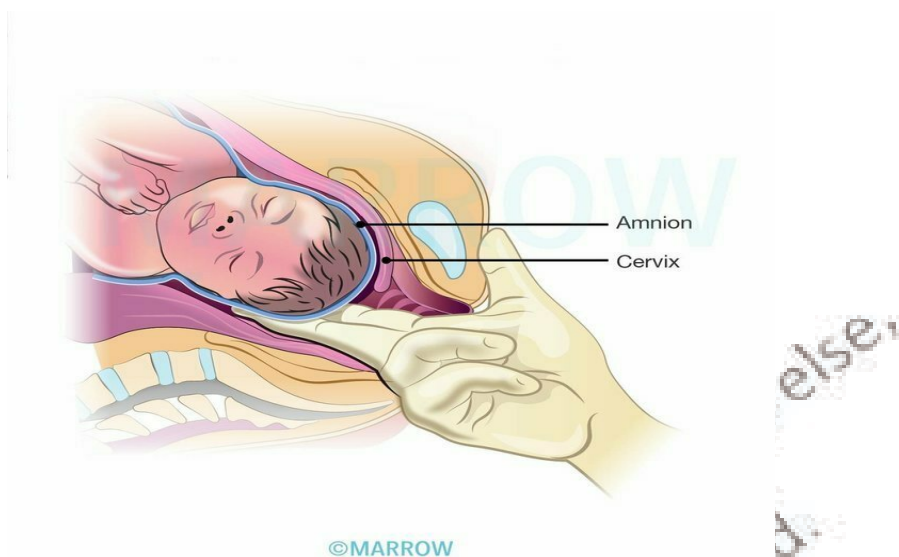
Question 7:

In which of the following patients would you perform a controlled artificial rupture of membranes?

- a) G3P2 lady with absent fetal cardiac activity at 34 weeks of gestation
- b) Multigravida with HIV at 36 weeks of gestation
- c) G2P1 lady with AFI of 28 cm at 38 weeks of gestation
- d) Term primigravida with red, tender, vesicular lesions on her vulva

Question 8:

An obstetrician performs the following procedure for labor induction in a patient. Which of the following is false about the reflex elicited here?



- a) Occurs due to pressure at the external os of cervix
- b) It is associated with increased level of PGF_2°
- c) Artificial rupture of membranes initiates this reflex
- d) It is an example of positive feedback mechanism

Question 9:

All of the following are true about laminaria tents except:

- a) Hygroscopic osmotic cervical dilator
- b) Leads to the release of endogenous prostaglandins
- c) Ripening the cervix
- d) No effect on Bishop's score

Question 10:

A nursing assistant posted in the obstetric care unit was asked to plot a partograph to monitor the progress of labour. Where should she start marking the cervical dilatation?

- a) Right of alert line
- b) Left of alert line
- c) Right of action line

- d) On the action line

Question 11:

You are monitoring the number, duration, and intensity of uterine contractions in a 34-year-old G2P1 woman with a term pregnancy. How long should they last for you to classify them as moderate?

- a) 20 – 40 seconds
- b) < 10 seconds
- c) 10-20 seconds
- d) >40 seconds

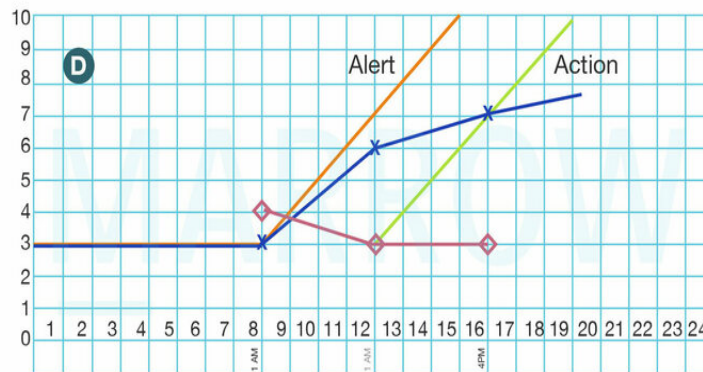
Question 12:

A G2P1 woman with polyhydramnios, presents to the clinic with respiratory distress at 38 weeks of gestation. The obstetrician plans for induction of labor. What is a safe method of induction in this patient?

- a) High rupture of membranes
- b) Prostaglandins
- c) Abdominal amniocentesis followed by oxytocin drip
- d) Low rupture of membranes

Question 13:

Given below is the partograph of a primigravida in labor. What should be the next step in managing her condition?



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- a) Wait and watch
- b) Induction of labor by prostaglandins
- c) Caesarean section
- d) Induction of labor by mechanical methods

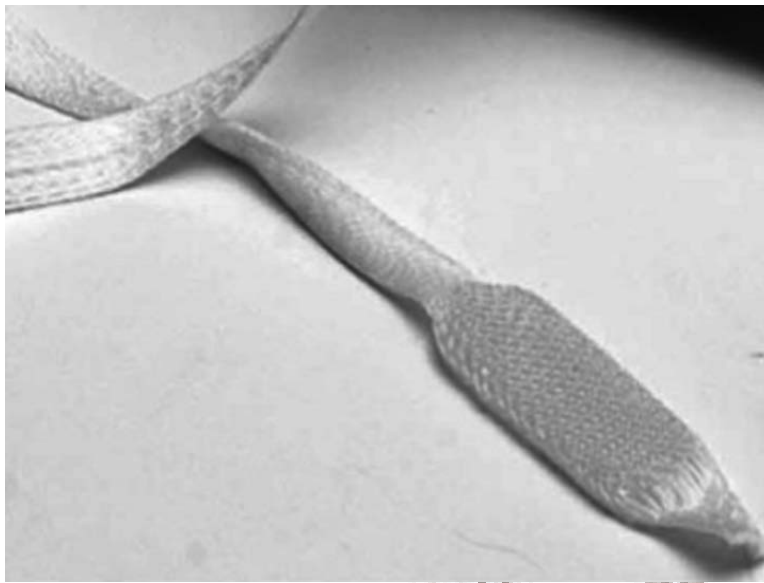
Question 14:

What is the half-life of oxytocin?

- a) 1-3 min
- b) 3-5 min
- c) 1-2 min
- d) 6-8 min

Question 15:

During your OBG rotations, one of the senior residents asks you to induce labor in a primigravida admitted at 40 weeks of gestation. Identify the following device that you will use for the same.



- a) Lippes loop
- b) Cervidil vaginal insert
- c) Misoprostol vaginal insert
- d) Mirena

Question 16:

According to the WHO Labour Care Guidelines, documentation of labour progress begins at a cervical dilatation of -

- a) 3 cm
- b) 4 cm
- c) 6 cm
- d) 5 cm

Question 17:

Which of the following modifications have been made in the WHO Labour care guidelines as compared to the modified partograph

- a) 1, 2, 4
- b) 1, 4, 5
- c) 2, 4, 5
- d) 1, 3, 4

Answer Key

Question No.	Correct Option
1	b
2	c
3	c
4	b
5	b
6	d
7	c
8	a
9	d
10	b
11	a
12	c
13	c
14	b
15	b
16	d
17	b

Detailed Explanations

Solution to Question 1:

The technique of stimulating uterine contractions prior to the onset of spontaneous labor is known as induction of labor.

It can be done irrespective of the rupture of membranes.

Augmentation is the enhancement of spontaneous contractions that are considered inadequate because of failed cervical dilation and fetal descent.

Solution to Question 2:

Heart disease is currently not considered a contraindication for the induction of labor. Vaginal delivery is preferred in such patients.

Solution to Question 3:

The minimum Bishop Score required before proceeding with induction of labour is 6.

A Bishop score ≥ 8 conveys a high likelihood of a successful induction. The latest guidelines state that a cut-off score of 6 can be used to choose between pharmacological and mechanical methods of induction.

This score is calculated based on the:

- station of the fetal head
- dilation of cervix
- cervical length
- position of cervix
- consistency of the cervix.

This score is an important determinant of a successful induction.

Solution to Question 4:

Uterine tachysystole is defined as ≥ 5 contractions in a 10 min period and may occur in spontaneous or stimulated labor. It should always be qualified by the presence or absence of fetal heart rate abnormalities. Tachysystole interrupts maternal blood flow to the placenta during contractions and may lead to a reduction in fetal oxygenation. It is also associated with placental abruption.

It is caused by pharmacological agents used for cervical ripening and induction such as misoprostol (both oral and vaginal) and dinoprostone. Management includes stopping oxytocin infusion and terbutaline 0.25 mg SC.

Solution to Question 5:

Misoprostol (PGE₁) is given as 25mcg 3-6 hourly vaginally or 25mcg 2 hourly orally for the induction of labor.

It is associated with the following side effects:

- Uterine tachysystole
- Meconium stained amniotic fluid
- Uterine rupture

Solution to Question 6:

Competitive blocking of acetylcholine is not a mechanism of action of oxytocin.

Mechanism of action of oxytocin is as follows:

- Receptor mediation through G protein-coupled receptors
- Activation of voltage-gated Ca channel leading to calcium entry
- Prostaglandin production by a positive feedback mechanism. This induces production of more prostaglandins and stronger contractions.

Solution to Question 7:

Controlled artificial rupture of membranes (ARM) is performed in a patient with amniotic fluid index (AFI) ≥ 25 cm. This points towards a diagnosis of polyhydramnios.

This is because if ARM is not done under controlled conditions in these patients, it can cause sudden decompression of the fetus. This can lead to abruption of placenta.

Contraindications to ARM:

- Intrauterine fetal death
- Active genital herpes infection
- Maternal HIV-AIDS

Solution to Question 8:

Stripping of membranes is being depicted in the image, which elicits the Ferguson reflex. It is also known as the fetal ejection reflex and is seen due to the pressure at the internal os of the cervix.

It is associated with an increase in blood levels of a prostaglandin $F2^\circ$ metabolite, also known as dinoprost. This is released due to mechanical stretching of the internal os. This enhances the uterine activity and causes release of more prostaglandins. It is an example of a positive feedback mechanism.

Solution to Question 9:

The Bishop's score is rapidly improved by the use of laminaria tents.

It is a mechanical cervical dilator and acts by hygroscopic osmotic action, leading to slow dilatation of the cervix. The gentle dilatation by laminaria tents induces an endogenous synthesis of prostaglandins causing a softening of the cervix. This is useful in cervical ripening before induction of labour.

Solution to Question 10:

In a partograph, the first marking of cervical dilatation should be made on the left of the alert line. In a normal partograph, the plotting remains on the left of the alert line.



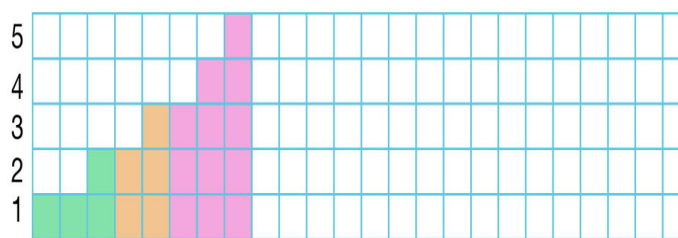
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Solution to Question 11:

Moderate uterine contractions last for 20–40 s.

Weak contractions last for <20 s and strong contractions are >40 s.

During active labour, uterine contractions are 5 min apart for 1 hour, i.e., ≥ 12 contractions in 1 hour. This is accompanied by cervical dilatation of ≥ 3 –5 cm.



Uterine contractions per 10 mins

Weak (<20 secs.) Moderate (20-40 secs.) Strong (>40 Secs.)

Solution to Question 12:

Abdominal amniocentesis followed by oxytocin drip is a safe method for induction of labor in polyhydramnios.

Abdominal amniocentesis drains a good amount of liquor (amnioreduction), after which the oxytocin infusion can be started. This will minimize the sudden decompression and reduce the risk of placental separation, cord prolapse and change in lie of the fetus.

Although polyhydramnios does not constitute an absolute contraindication for the application of prostaglandins, these substances should be administered with caution. There is a marked increase in the incidence of postpartum hemorrhage and amniotic fluid embolism with their use. Hence, amnioreduction is a preferred mode of treatment in polyhydramnios.

Solution to Question 13:

As the plotting on the above partograph has crossed the action line, the next step of management is Caesarean section.

Solution to Question 14:

The half-life of oxytocin is approximately 3–5 min. Therefore, once oxytocin infusion is stopped, its concentration in plasma rapidly falls.

Solution to Question 15:

This is an image of a cervidil vaginal insert. It is used for cervical ripening and improves the Bishop's score.

It contains 10 mg of dinoprostone designed to release at a rate of approximately 0.3 mg/hr over 10-hours.

Solution to Question 16:

According to the WHO Labour Care Guidelines, documentation of labour progress begins at a cervical dilatation of 5cm.

WHO Labour Care Guidelines:

- Time axis - examination findings are plotted every one hour

- Alert column- has values which if reached indicates the need for urgent medical intervention
- Active first stage - findings are plotted every hour upto 12 hours. If the first stage continues, a new sheet is used.
- Second stage - plotted every hour upto 3 hours

1) Section 1: Preliminary information

Name, parity, labour onset, active labour diagnosis (date), ruptured membranes (date&time), risk factors

2) Section 2: Supportive care

- Companionship
- Pain relief
- Oral fluid
- Posture

3) Section 3: Baby

- Baseline FHR
- FHR deceleration
- Amniotic fluid
- Fetal position
- Caput
- Moulding

4) Section 4: Vitals

Pulse, SBP, DBP, temperature, and urine for proteins are recorded every 4 hours

5) Section 5: Labour progress

- Count the number of uterine contractions over a 10-minute period.
- Assess the duration of contractions
- Cervical dilatation: starts at 5 cm and is plotted as X; it is assessed every hour. For example - if the cervical dilatation remains at 5cm for ≥ 6 hrs, it indicates the need for urgent intervention. Similarly, the cut-offs for - 6cm is ≥ 5 hrs; 7cm is ≥ 3 hrs; 8cm is ≥ 2.5 hrs; 9cm is ≥ 2 hrs
- Descent - plotted as O

6) Section 6: Medication

- Oxytocin
- Medicine
- IV fluids

7) Section 7: Shared decision making

- Assessment
- Plan

*If birth is not completed by 3 hours from the start of the active second stage in nulliparous and 2 hours in multiparous women, intervention is needed

WHO LABOUR CARE GUIDE													
Name	Date	Time	Parity	Labour onset	Active labour diagnosis (Date)								
Ruptured membranes (Date)		Time		Risk factors									
SPONTANEOUS	Comprimen	M											
	Pain relief	M											
	Anal block	M											
	Puncture	LP											
PAIN	Duration	min											
	Character	L											
	Location	M + + + L											
	Fetal position	RT											
	Caput	+++											
WATER	Membranes	+++											
	Pelvic	<60, n=120											
	Synthetic BP	<90, n=140											
	Spontaneous BP	<90											
	Uterine	F + + , J + + +											
LABOUR PROGRESS	Contraction	10											
	Interval	2											
	Duration	30											
	Intensity	5											
	Frequency	1											
RECEIVED	Oxytocin (10L, 10mg/ml)	M											
	Medication	M											
	Per Mead	M											
	Assessment	M											
	Plan	M											
INITIALS													

Solution to Question 17:

WHO modified partograph	WHO Labour Care Guide
Documentation of labour progression begins at 4 cm of cervical dilatation.	Documentation of labour progression begins at 5 cm of cervical dilatation.
Fixed 1 cm/hour 'alert' and 'action' lines	Time-based cut-offs for each centimeter of cervical dilatation to guide the obstetrician to take action. For example, at 5 cm of cervical dilatation, if there is no progress > 6 hours, action needs to be taken.
Monitoring of labour is restricted to the first stage only.	Monitoring of second stage of labour is included
No section on supportive care interventions	Addition of a section on supportive care of the mother- labour companionship, pain relief, oral fluid intake and posture
Documentation of strength, duration, and frequency of uterine contractions	Documentation of only the duration and frequency of uterine contractions

The figure consists of three vertically stacked graphs sharing a common x-axis representing time from 10am to 1pm.

- Top Graph: Fetal Heart Rate**
 - Y-axis:** Fetal Heart Rate (bpm), ranging from 80 to 200.
 - X-axis:** Time (10am to 1pm).
 - Legend:** C = Cervical Moulding, B = Breathing, M = Molding.
 - Data Points:**
 - 10am: 140 bpm (C)
 - 10:30am: 140 bpm (C)
 - 11am: 130 bpm (C)
 - 11:30am: 140 bpm (C)
 - 12pm: 140 bpm (C)
 - 12:30pm: 100 bpm (B)
 - 1pm: 80 bpm (M)
- Middle Graph: Cervical Dilation**
 - Y-axis:** Dilation of Cervix (cm), ranging from 0 to 10.
 - X-axis:** Time (10am to 1pm).
 - Legend:** Alert (blue line), Action (red line).
 - Data Points:**
 - 10am: 4 cm (Alert), 3 cm (Action)
 - 11am: 5 cm (Alert), 3 cm (Action)
 - 12pm: 6 cm (Alert), 3 cm (Action)
 - 1pm: 7 cm (Alert), 3 cm (Action)
- Bottom Graph: Contractions**
 - Y-axis:** Contractions per 10 minutes, ranging from 0 to 5.
 - X-axis:** Time (10am to 1pm).
 - Data:**
 - 10am to 12pm: 3 contractions per 10 minutes (shaded area).
 - 12pm to 1pm: 4 contractions per 10 minutes (shaded area).

else,

[illegible]

THIS INSTRUMENT, ONCE ANY OBSERVATION MEETING THE CRITERIA IN THE "ALERT" COLUMN, ALERTS THE SENIOR MEMBERS OF DOCTOR AND RECORD THE ASSESSMENT AND ACTION TAKEN. IF LABOUR EXTENDS BEYOND 120, PLEASE CONTINUE OUR NEW LABOUR CASE GIVE ALTERNATIVE Y = Yes, N = No, A = Not Sure, D = Disagree, S = Unknown, SP = Support, M = Mislead, L = Liable, V = Varies, I = Irrelevant, C = Check, M = Mismatch, B = Block, A = Advice, P = Pointless, T = Transition, P = Pledge, A = Action

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IF NO

Malpresentations

Question 1:

A woman with gestational diabetes mellitus is admitted to the labour room. The delivery of the baby's shoulder is delayed after the delivery of the head. When is shoulder dystocia considered?

- a) Head to shoulder delivery time ≥ 30 sec
- b) Head to shoulder delivery time ≥ 45 sec
- c) Head to shoulder delivery time ≥ 60 sec
- d) Head to shoulder delivery time ≥ 120 sec

Question 2:

A 25-year-old primigravida with a history of gestational diabetes mellitus is in active labour. After delivery of the head, turtle sign was noticed. Which of the following can be used for the management of this patient?

- a) B, C, E, and F
- b) A, D, E, and F
- c) A, B, C, and D
- d) A, B, D, and F

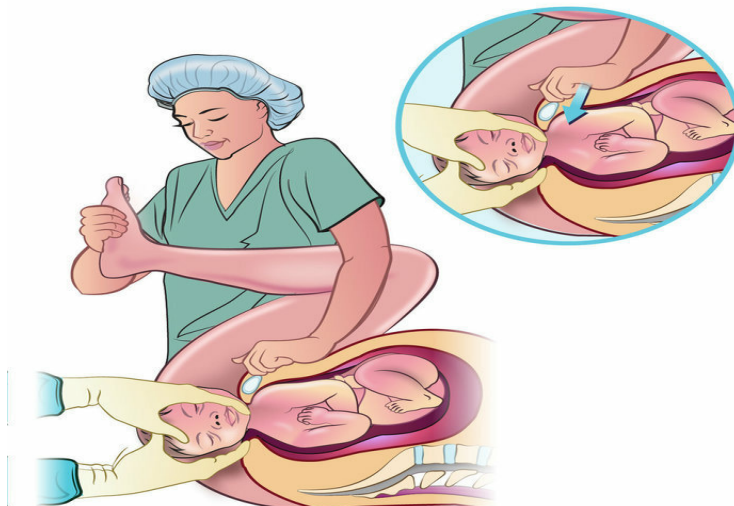
Question 3:

The first manoeuvre for shoulder dystocia consists of:

- a) Forceful adduction by extension of legs on abdomen
- b) Forceful adduction by flexion of legs on abdomen
- c) Forceful abduction of legs by extending them on abdomen
- d) Forceful abduction of legs by flexing them on abdomen

Question 4:

A G2P1 obese woman was diagnosed with arrest of descent during labour. Identify the following manoeuvre performed for this patient.



- a) Rubin manoeuvre
- b) McRoberts manoeuvre
- c) Zavanelli manoeuvre
- d) Wood corkscrew manoeuvre

Question 5:

Which of these steps would you undertake for the management of a patient with shoulder dystocia along with performing the McRoberts manoeuvre?

- a) Apply fundal pressure
- b) Rotate posterior shoulder to 90 degrees
- c) Flex the legs over the abdomen
- d) Apply suprapubic pressure

Question 6:

What is the most common fetal malpresentation?

- a) Occipitoposterior
- b) Face presentation
- c) Breech presentation

d) Brow presentation

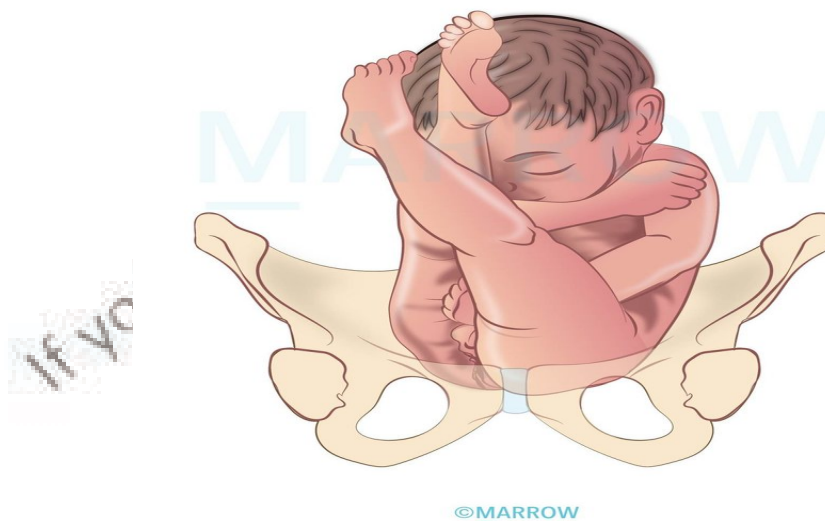
Question 7:

In which of the following conditions is Vagos method used?

- a) Cord prolapse
- b) Shoulder presentation
- c) Impacted breech
- d) Compound presentation

Question 8:

A pregnant woman presented to the clinic with the following fetal presentation. What is the likely diagnosis?



- a) Stargazer breech
- b) Complete breech
- c) Flexed breech
- d) Frank breech

Question 9:

A multigravida presented to the clinic at 36 weeks of gestation. She was found to have a breech presentation clinically and on sonography. Which of the following is not an indication for cesarean section in this type of presentation?

- a) Hyperextended fetus
- b) Flexed breech
- c) Footling breech
- d) Severe IUGR

Question 10:

A footling breech comes under _____ type of breech presentation.

- a) Frank
- b) Complete
- c) Flexed
- d) Incomplete

Question 11:

A neonate is noted to have Klumpke's paralysis of the left arm following delivery. Which of the following is not a risk factor for the associated fetal presentation?

- a) Preeclampsia
- b) Hydrocephaly
- c) Twins
- d) Placenta previa

Question 12:

What is the most common cause of the following presentation?



- a) Prematurity
- b) Hydrocephalus
- c) Placenta previa
- d) Polyhydramnios

Question 13:

The following procedure is planned for a pregnant woman at 37 weeks of gestation. What is an absolute contraindication for this?



- a) Placenta previa

- b) Severe pre-eclampsia
- c) Previous LSCS
- d) Multifetal gestation

Question 14:

A multiparous woman presents in labour at 38 weeks of gestation. On examination, the uterus appears asymmetrical and fundal height is less than the period of amenorrhea. On fundal grip, the fetal pole is not palpable and in pelvic grip, the lower pole of the uterus is empty. What is the likely fetal presentation?

- a) Vertex
- b) Shoulder
- c) Brow
- d) Breech

Question 15:

An obstetrician is performing a breech delivery. Which of the following manoeuvres can be used for chin to pubis delivery?

- a) Pinards manoeuvre
- b) Zavanelli manoeuvre
- c) Prague manoeuvre
- d) Lovset manoeuvre

Question 16:

Which of the following methods is not used for delivery of the aftercoming head in breech?

- a) Burns-Marshall method
- b) Mauriceau-Smellie-Veit method
- c) Piper's forceps
- d) Keilland's forceps

Question 17:

Which of the following is not used in the management of entrapment of aftercoming head?

- a) Dührssen's incision
- b) IV nitroglycerin
- c) Zavanelli manoeuvre
- d) Kristellar manoeuvre

Question 18:

A woman in labour presented to the ER. A vaginal breech delivery was planned for her. What statement regarding the management is false?

- a) Breech identified for the first time during delivery is a contraindication for vaginal delivery
- b) Breech delivery is done in an hospital where a facility for emergency cesarean delivery is available
- c) Continuous FHR monitoring is required for breech delivery during labour
- d) Should be done by a skilled obstetrician

Question 19:

On doing PV examination of a woman in labour, mouth with hard alveolar margins and malar eminence is palpable. Which of the following is not a cause of this fetal presentation?

- a) Anencephaly
- b) Tumour of neck of fetus
- c) Multiparity
- d) Gynecoid pelvis

Question 20:

A multiparous woman was admitted to the labour room at term with complaints of draining per vaginum. On examination, the uterus corresponds to term size, presentation is cephalic and the presenting part is face. Which diameter is engaged in this type of presentation?

- a) Mentovertebral
- b) Submentovertebral
- c) Occipitofrontal
- d) Suboccipitofrontal

Question 21:

A G2P1 woman presented to the clinic at 40 weeks of gestation. On examination, the uterus corresponds to term size and the fetal head is mobile. What is the most common cause to be suspected here?

- a) True cephalopelvic disproportion
- b) Occipitoposterior
- c) Placenta previa
- d) Full bladder of mother

Question 22:

A 34-year-old G2P1 woman in labour was admitted to the hospital at 38 weeks of gestation. On examination, the fetus was found to be in occipitoposterior position. What is the most appropriate management?

- a) Emergency caesarean section
- b) Wait and watch for progress of labour
- c) Prepare for forceps delivery
- d) Prepare for ventouse delivery

Question 23:

What is the most common complication seen in face to pubis delivery?

- a) Intracranial injury
- b) Cephalhematoma
- c) Paraurethral tears
- d) Complete perineal tears

Answer Key

Question No.	Correct Option
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1	c
2	d
3	d
4	b
5	d
6	c
7	a
8	d
9	b
10	d
11	a
12	a
13	a
14	b
15	c
16	d
17	d
18	a
19	d
20	b
21	b
22	b
23	d

Detailed Explanations

Solution to Question 1:

Shoulder dystocia is defined as head to shoulder delivery time of more than 60 seconds. It occurs when either the anterior or the posterior fetal shoulder gets impacted on the maternal symphysis or on the sacral promontory respectively.

Prelabour risk factors:

- Diabetes Mellitus
- Obesity
- Postdatism and prior history
- Excessive weight gain during pregnancy

Intrapartum risk factors:

- Prolonged 1st and 2nd stage of labour
- Instrumental delivery
- Augmentation of labour with oxytocin

Solution to Question 2:

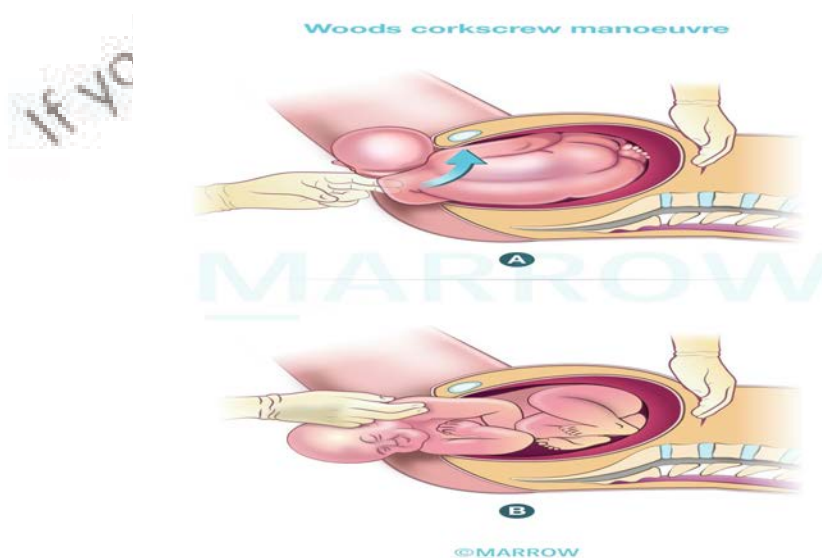
A patient with gestational diabetes and the presence of turtle sign is suggestive of shoulder dystocia. Mc Robert's manoeuvre, Wood's corkscrew manoeuvre, Gaskin's manoeuvre, and Rubin's manoeuvre are all used in its management.

Shoulder dystocia occurs when either the anterior or the posterior fetal shoulder gets impacted on the maternal pelvis. Turtle sign may be seen which refers to the recoil of the fetal head against the perineum.

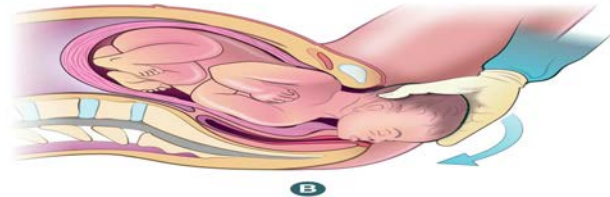
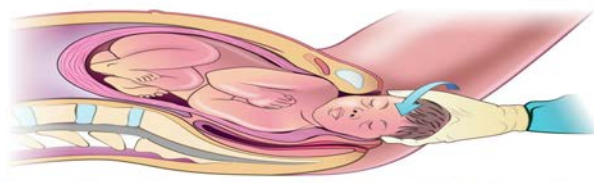
Management of labour in case of shoulder dystocia:

- 1st line of management - Mc Roberts manoeuvre (flexion and abduction of thigh)
- 2nd line of management - Rubin's (rotation of the easily accessible shoulder by 30-45°), Woods corkscrew manoeuvre (rotate the posterior shoulder to 180° in a cork screw fashion) ; OR Gaskin (all four's position)
- Last line of management - Zavanelli manoeuvre (head is rotated to its pre-restitution position and caesarian section is performed); OR procedures such as cleidotomy (fracture of the clavicle of the baby); symphysiotomy (divide pubic symphysis of mother)

In a case of bilateral shoulder dystocia, directly proceed to perform cesarean section after doing Zavanelli manoeuvre.

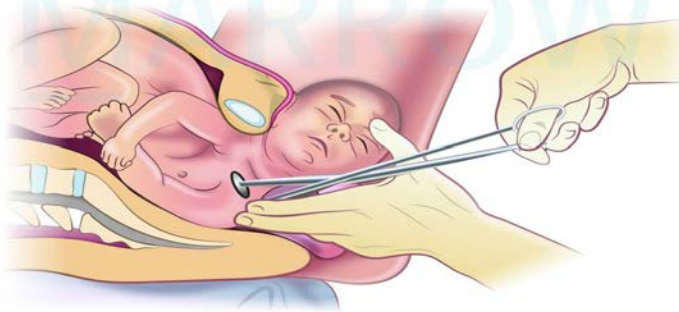


Zavanelli manoeuvre



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Cleidotomy



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Solution to Question 3:

The given clinical scenario is suggestive of shoulder dystocia. Mc Roberts manoeuvre is the first manoeuvre to be performed in shoulder dystocia. It involves flexion and abduction of the maternal hips, positioning the maternal thighs on her abdomen.

Solution to Question 4:

The given clinical scenario is suggestive of shoulder dystocia and the manoeuvre shown above is the McRoberts manoeuvre. This is the most effective intervention and should be the first manoeuvre to be performed in shoulder dystocia.

McRoberts manoeuvre:

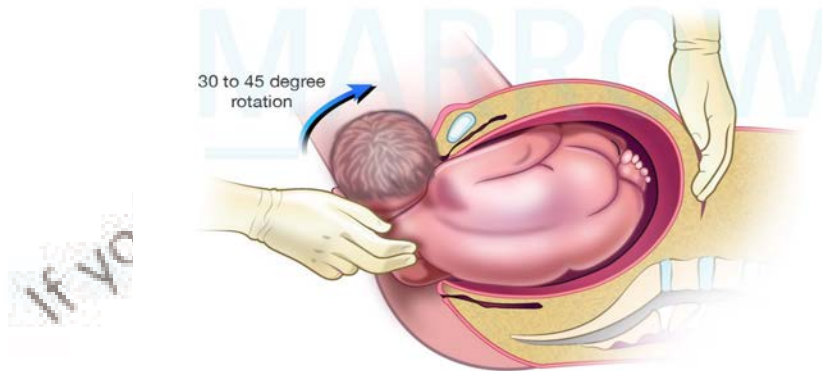
- Flexion and abduction of the maternal hips (positioning the maternal thighs on her abdomen).
- The pelvic rotation which occurs due to the straightening of the sacrum tends to free the impacted anterior shoulder of the baby.
- Suprapubic pressure may be used simultaneously.
- The lateral cutaneous nerve of the thigh is the most common nerve to get injured during this manoeuvre.

If the first line of management fails, proceed with internal rotational manoeuvres such as Rubin's and Wood corkscrew in McRoberts position.

Rubin's manoeuvre:

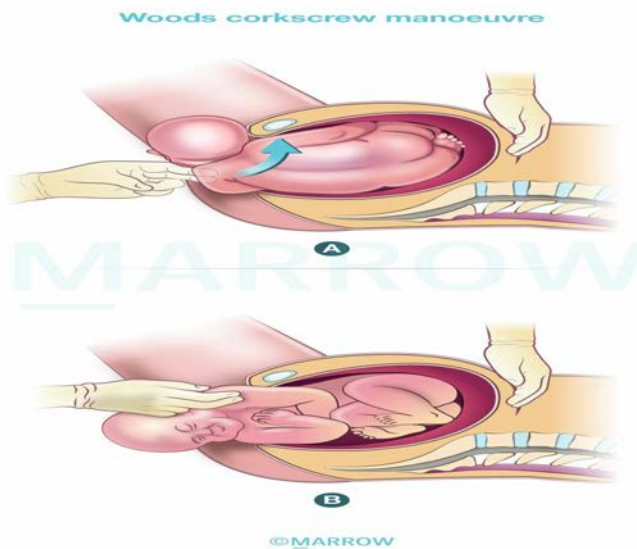
- Internal rotational manoeuvre used to deliver the baby in case of shoulder dystocia.
- It involves dislodging the easily accessible shoulder towards the anterior aspect of the chest from under the pubic symphysis by 30-45° while staying in the McRoberts' position.
- Pressure is applied on the scapular aspect (posteriorly) of the shoulder.

Rubin's manoeuvre



Wood corkscrew manoeuvre:

- Internal rotational manoeuvre used to deliver the baby in case of shoulder dystocia.
- It involves dislodging the posterior shoulder by 180° in a corkscrew fashion to lie beneath the pubic symphysis while staying in McRobert's position.
- Pressure is applied on the clavicular aspect (anteriorly) of the shoulder.



Solution to Question 5:

Applying a gentle suprapubic pressure is the first step in the management of shoulder dystocia along with McRoberts manoeuvre. Fundal pressure is not a method of delivery in such cases.

Hernandez and Wendel suggested the use of a shoulder dystocia drill to better organize emergency management:

- Call for help and mobilize labour assistants, anaesthesia and pediatric doctors.
- An attempt at traction is tried followed by episiotomy to afford room posteriorly.
- Apply pressure suprapubically, not fundally, with the palm or fist, directing the pressure on the anterior shoulder.
- This should be both downward (below the pubic bone) and laterally (toward the baby's face or sternum) in conjunction with the McRoberts manoeuvre.

These manoeuvres will resolve most cases of shoulder dystocia. If the above-listed steps fail, the following steps may be attempted, and any of the manoeuvres may be repeated:

- Rubin's manoeuvre
- Woods corkscrew manoeuvre
- Gaskin (all four's position)
- Zavanelli manoeuvre
- Cleidotomy (fracture of the clavicle of baby)
- Symphysiotomy (divide pubic symphysis of mother).

Solution to Question 6:

The most common fetal malpresentation is the breech presentation.

The most common breech presentation in primigravida is frank breech and in multigravida, it is flexed/complete breech. It is most commonly seen with prematurity.

In breech presentation:

- The most common position is left sacroanterior
- The engaging diameter is bitrochanteric.
- Cord prolapse is common in footling presentation and rare in frank breech.

However, among all malpresentations, cord prolapse is most common in transverse lie.

Solution to Question 7:

Vagos method is used in cord prolapse.

Vagos method is also known as the bladder filling method. Here, the bladder is distended with saline to push the presenting part higher up, thereby preventing compression in case of cord prolapse. The bladder is filled with 400–750 mL of normal saline with a Foley's catheter. Then, the balloon is inflated and the catheter is clamped.

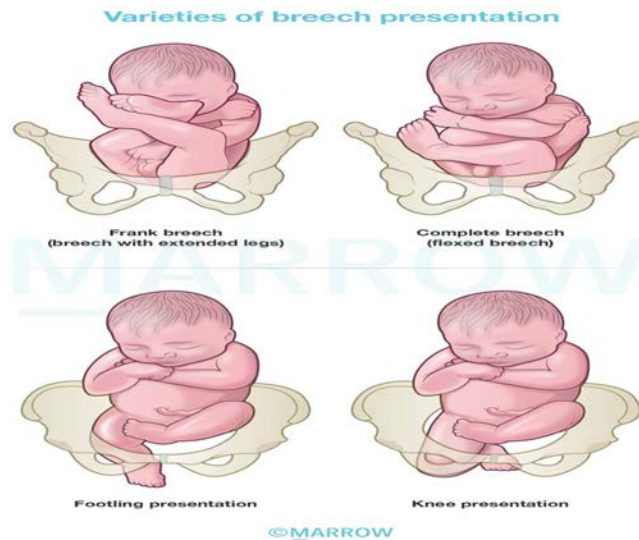
Solution to Question 8:

The given clinical scenario is suggestive of frank breech.

Breech presentation can be divided into the following:

- Incomplete: Breech with extended legs, footling, knee presentation
- Complete: Breech with flexed legs

Breech type	Description	Presenting part
Frank breech(with extended legs)	Thighs flexed on the trunk, legs extended at the knee joint.	Buttocks and genitalia
Footling	Both thighs and legs are partially extended	Foot
Knee	Thighs extended, Knee flexed	Knee
Complete/flexed	Both thighs and legs are flexed.	Buttocks, genitalia, foot



Solution to Question 9:

Flexed breech is not an indication for cesarean section in breech presentation. This baby can be delivered vaginally.

Factors favouring cesarean delivery of the breech fetus:

- Large fetus (> 3800 to 4000 g)
- Severe IUGR
- Footling presentation (due to risk of cord prolapse)
- Hyperextended head
- Contracted pelvis
- Previous cesarean section
- Preterm breech
- Primi with breech
- Twins, with the first baby in the breech
- Lack of surgeon experience
- Patient requests for cesarean delivery.

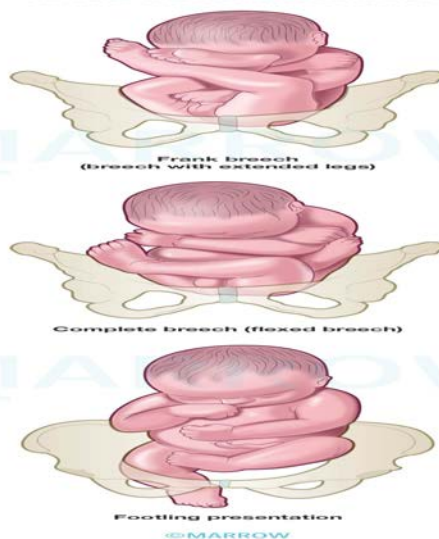
Solution to Question 10:

A footling breech is a type of incomplete breech presentation with one or more feet below the breech.

Breech presentation can be divided into the following:

- Incomplete: Breech with extended legs, footling or knee presentation
- Complete: Breech with flexed legs

Breech type	Description	Presenting part
Frank breech(with extended legs)	Thighs flexed on trunk, legs extended at knee joint.	Buttocks and genitalia
Footling	Both thighs and legs are partially extended	Foot
Knee	Thighs extended, knee flexed	Knee
Complete/flexed	Both thighs and legs are flexed.	Buttocks, genitalia, foot



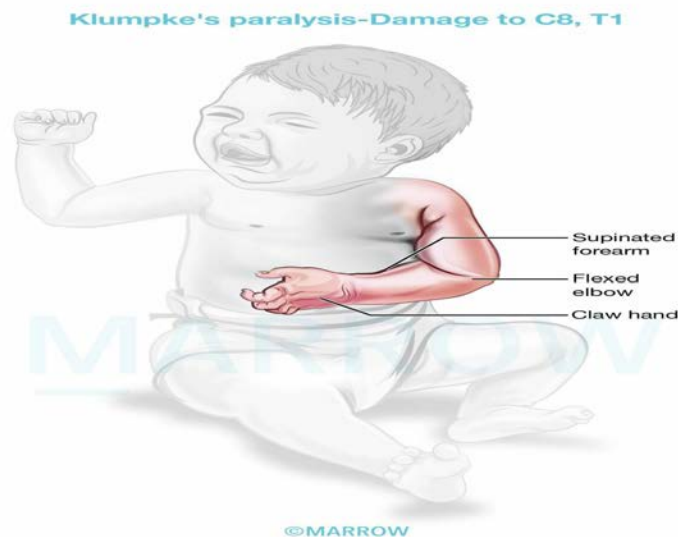
Solution to Question 11:

Klumpke's paralysis is associated with difficult breech delivery, which is due to hyperabduction of the arm during traction and delivery of the fetus. Preeclampsia is not a risk factor for breech presentation.

Risk factors for breech presentation:

- Early gestational age
- Abnormal amniotic fluid volume
- Multifetal gestation
- Hydrocephalus
- Anencephaly
- Uterine anomalies

- Placenta previa
- Fundal placental implantation
- High parity with uterine relaxation
- Prior breech delivery
- Prior cesarean delivery
- Smoking.



Solution to Question 12:

The presentation depicted in the image is breech presentation. The most common cause of this presentation is prematurity.

The incidence of breech at term is 3%. At 28 weeks of pregnancy, 20% of women have a breech presentation. The fetus undergoes spontaneous version usually between 30 - 34 weeks thereby correcting its position by itself.

Persistent breech occurs when this spontaneous version is prevented by any of the following causes:

- Intrauterine death of the fetus
- Anomalies such as trisomy 13, 18, 21
- Anomalies of uterus (septate/bicornuate)
- Oligohydramnios and polyhydramnios
- Twins
- Short cord
- Hydrocephalus
- Placenta previa

- Contracted pelvis

Recurrent/habitual breech is when breech recurs in 3 or more pregnancies. It is usually due to uterine anomalies and cornuofundal attachment of the placenta.

Solution to Question 13:

Placenta previa is an absolute contraindication for external cephalic version (ECV).

Relative contraindications for ECV:

- early labor
- oligohydramnios
- ruptured membranes
- known nuchal cord
- structural uterine abnormalities
- fetal-growth restriction
- multifetal gestation
- prior abruption or its risk factors
- severe pre-eclampsia and hypertension
- previous LSCS

Note: In twin gestation, with a cephalic-noncephalic presentation, after the delivery of the first twin, an internal podalic version or intrapartum external cephalic version can be attempted for the second non-cephalic twin. However, ECV in this case, can lead to an increased incidence of fetal distress when compared with the internal podalic version. Hence, ECV is only a relative contraindication in twin pregnancy.

Previous LSCS is a relative contraindication for ECV. However, it is not attempted often and needs to be performed with caution.

Solution to Question 14:

The given examination findings are suggestive of transverse lie. The likely fetal presentation is shoulder presentation.

The most common cause of transverse lie is prematurity. In term pregnancy, it is commonly seen in patients with placenta previa. Although the most common cause of cord prolapse is transverse lie, hand prolapse is also frequently seen in such conditions. The transverse lie is managed by the caesarean section.

In a twin pregnancy, management of 1st twin with a transverse lie is caesarean section whereas the 2nd twin is managed by internal podalic version.

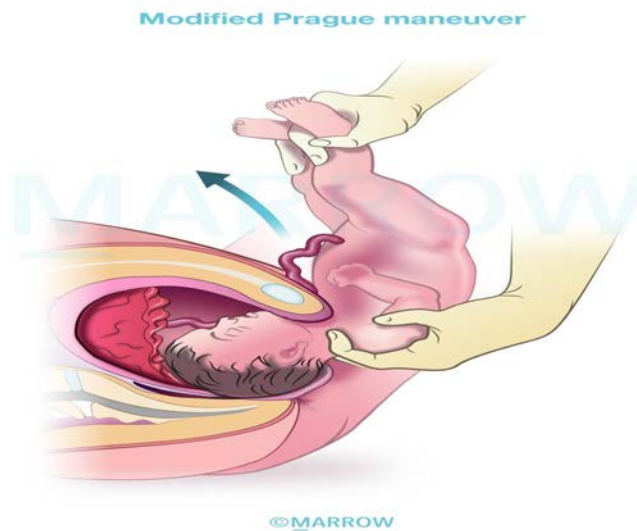
Solution to Question 15:

In a breech presentation, chin to pubis delivery can be done by Prague manoeuvre.

Rarely, the back of the fetus fails to rotate to the anterior. In such cases, rotation of the posteriorly oriented back of the fetus to the anterior may be achieved by using stronger traction on the fetal legs or bony pelvis.

If this still does not succeed, the fetus still may be delivered using the modified Prague manoeuvre. This consists of two fingers of one hand grasping the shoulders of the back-down fetus from below while the other hand draws the feet up and over the maternal abdomen.

If the back is oriented anteriorly, extraction may be accomplished using the Mauriceau manoeuvre and delivering the fetus back down.

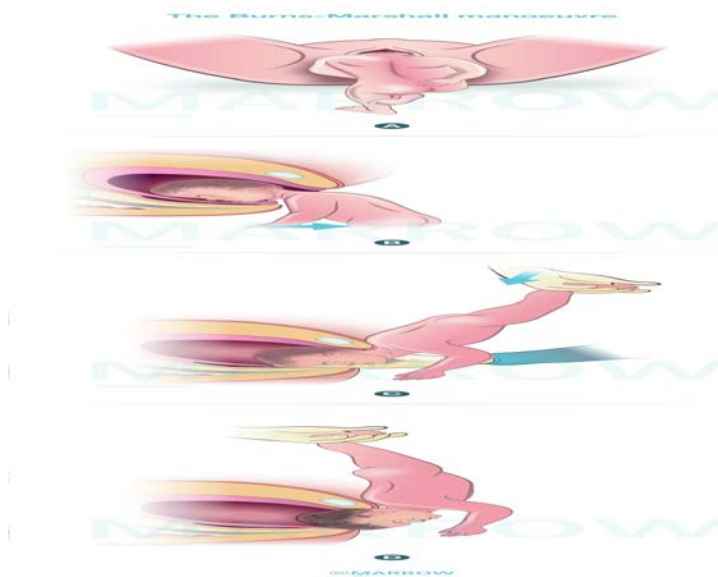


Solution to Question 16:

Delivery of the aftercoming head in breech is not done with the help of Keilland's forceps. These forceps are used for the rotation of the fetal head. It is not used anymore.

The methods used for delivery of the aftercoming head include the following:

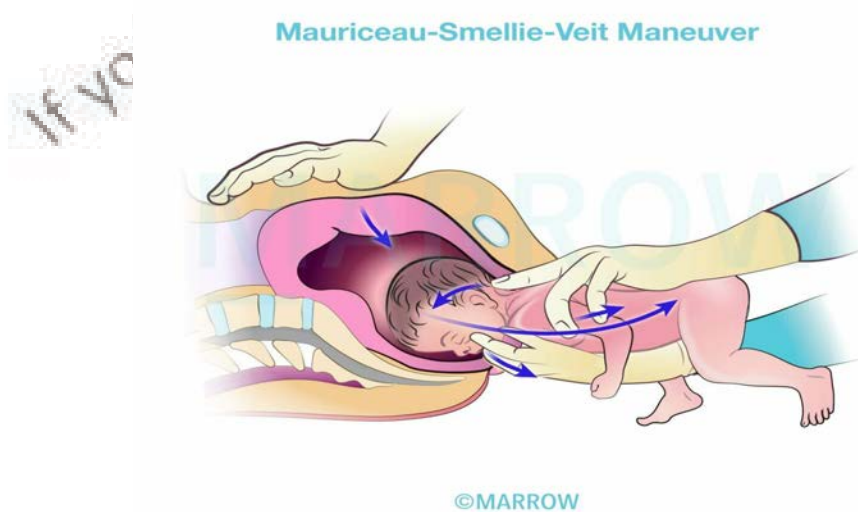
Burns-Marshall manoeuvre: When the occiput is seen under the pubic symphysis, steady traction is given on the feet. The baby is swung in an arc towards the mother's abdomen. Simultaneous suprapubic pressure is given and the head is born by flexion.



Malar flexion and shoulder traction (modified Mauriceau-Smellie-Veit technique):

The baby is placed on the supinated left forearm (preferred) with the limbs hanging on either side. The middle and the index fingers of the left hand are placed over the malar bones. The ring and little fingers of the pronated right hand are placed on the child's right shoulder. The index finger is placed on the left shoulder and the middle finger is placed on the suboccipital region.

Traction is now given in a downward and backward direction till the nape of the neck is visible under the pubic arch. The assistant gives suprapubic pressure during the period to maintain flexion. The fetus is carried in an upward and forward direction toward the mother's abdomen releasing the face and brow. Lastly, the trunk is depressed to release the occiput and vertex.

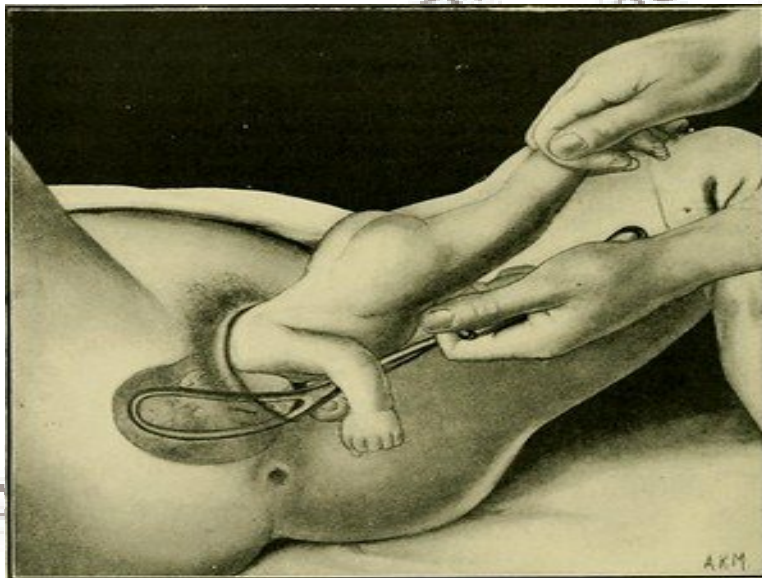


The image below shows Piper's forceps, identified by its distinct perineal curve.

Piper forceps



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Solution to Question 17:

The management of entrapment of aftercoming head does not involve Kristellar manoeuvre.

Kristeller manoeuvre is the application of uterine fundal pressure during the second stage of labour to shorten the second stage of labour. This should be used sparingly as it leads to avulsion of the levator ani.

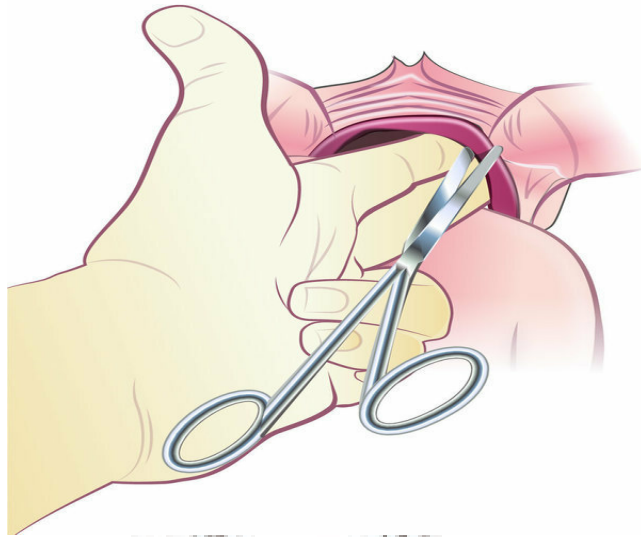
An incompletely dilated cervix will constrict around the neck and impede the delivery of the aftercoming head in breech.

Management options:

With gentle traction on the fetal body, the cervix may be manually slipped over the occiput.

If this is not successful, then Dührssen incisions may be necessary. These are a set of three incisions in the cervix:

- Can be made at the 2 and 10 o'clock position on the cervix. An additional incision at the 6 o'clock position is sometimes needed.
- Done for delivery of aftercoming head in breech or delivery of head through an incompletely dilated cervix.
- There is a significant risk of haemorrhage; the cervical incision may extend upwards within the broad ligament causing broad ligament haematoma.



Other alternatives include intravenous nitroglycerin, typically 100 µg, to provide cervical relaxation. General anaesthesia with halogenated agents is another option.

As a last resort, the following methods are used:

- Zavanelli manoeuvre: It involves the replacement of the fetal head into the pelvis followed by cesarean delivery. This is performed for the protruding head and intractable shoulder dystocia.
- Symphysiotomy: The intervening symphyseal cartilage and ligamentous supports are cut to widen the symphysis pubis. It is also used to aid the delivery of an entrapped aftercoming head.

Solution to Question 18:

Breech identified for the first time during delivery is not a contraindication for normal delivery.

Vaginal breech delivery is considered when:

- Fetal weight is between 1.5 kg and 3.5 kg
- There is an adequate pelvis
- The presentation is frank breech
- The fetal head is flexed

- There are no medical or obstetric complications
 - Facilities for emergency cesarean section are available (anesthetists, neonatologists) (option B)
 - Facilities for continuous labor monitoring are available (option C)
 - A skilled obstetrician is present (option D)
 - Detailed informed consent has been obtained, explaining the risks of vaginal breech delivery
- Cesarean delivery for breech presentation is considered in the following cases:
- Lack of operator experience
 - Patient's request for cesarean delivery
 - Prior perinatal death or neonatal birth trauma
 - Large fetus: > 3800 to 4000 g
 - Severe IUGR, term weight < 2.5 kg to 2.8 kg
 - Incomplete breech presentation
 - Oligohydramnios
 - Fetal anomaly incompatible with vaginal delivery
 - Hyperextended head
 - Contracted pelvis
 - Prior cesarean delivery
 - Preterm breech
 - Twins, with the first baby in the breech position

Solution to Question 19:

The given clinical scenario is suggestive of face presentation. Gynecoid pelvis is not a cause of this fetal presentation.

Causes of face presentation:

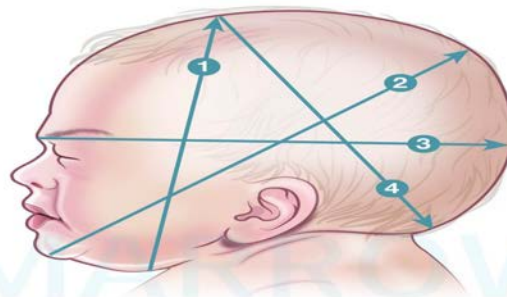
- Anencephaly (most common cause)
- Intrauterine death
- Multiparity
- Prematurity
- Twist of cord around the neck
- Anthropoid pelvis
- Platypelloid pelvis
- Tumours of neck of the fetus.

Solution to Question 20:

Engaging diameter in face presentation is submentovertical and submentobregmatic.

Presentation	Engaging diameter	Measurement
Vertex (Complete flexion)	Suboccipitobregmatic	9.5 cm
Vertex (Incomplete flexion)	Suboccipitofrontal	10 cm
Vertex (Marked Deflexion)	Occipitofrontal	11.5 cm
Face (Incomplete extension)	Submentovertical	11.5 cm
Face	Submentobregmatic	9.5 cm
Brow	Mentovertical	14 cm

Fetal skull dimensions



- 1 Submento-bregmatic 9.5 cm (Face)
- 2 Mento-vertical 14 cm (Brow)
- 3 Occipito-frontal 11.5 cm (Deflexed vertex)
- 4 Suboccipito-bregmatic 9.5 cm (Flexed, vertex)

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Solution to Question 21:

Occipitoposterior position (OP) is the most common cause of a mobile head at term.

It is the most common fetal malposition. It is an abnormal position seen in the vertex presentation, not an abnormal presentation.

- Occipitoposterior position is commonly seen in android pelvis/deflexed head.
- The engaging diameter of this position is sub-occipitofrontal/occipitofrontal.
- The most common type in occipitoposterior position is right OP.
- A persistent occipitoposterior position results from a failure of internal rotation prior to birth.

Cephalopelvic disproportion (CPD) is a disproportion in the size of the fetus relative to the maternal pelvis. This can result in the failure of progress of labour in the second stage. This usually occurs due to fetal malposition e.g., extended or asynclitic fetal head, occipitoposterior or transverse position. It is also seen in malpresentations like mentum posterior, brow.

The diagnosis of CPD is subjective. However, true CPD may occur if the fetus has a large anomaly e.g., teratoma or conjoined twin, or in cases of a large fetus. It can also occur when the maternal pelvis is small or deformed like after pelvic trauma.

Solution to Question 22:

The most appropriate management for pregnant women in labour whose fetus is in occipitoposterior position is to wait and watch for progress of labour. In a majority of cases, the occiput rotates through 3/8th of the circle and the baby is born as occipitoanterior.

Hence, when the diagnosis of occipitoposterior is made, labour is allowed to proceed normally when there are good uterine contractions and favourable pelvis.

Indications of caesarean section in occipitoposterior position:

- Arrest of labour
- Inadequate uterine contraction
- Fetal distress
- Other obstetrical indications.

Solution to Question 23:

The most common complication seen in face to pubis delivery is complete perineal tears.

In face to pubis delivery, the occiput is posterior, and therefore, the largest biparietal diameter (9.4 cm) distends the perineum rather than the smaller bitemporal diameter (8 cm). A liberal episiotomy helps to avoid perineal tears.

Operative Vaginal Delivery

Question 1:

A forceps delivery is being planned for a patient with prolonged second stage of labour. Which of the following is not a prerequisite for this procedure?

- a) Bladder should be empty
- b) No suspicion of cephalopelvic disproportion
- c) Well engaged head
- d) Partially dilated cervix

Question 2:

All of the following are criteria for the use of the following instrument except:



- a) Fetal skull has reached pelvic floor
- b) Fetal head is on the perineum
- c) Scalp is visible at the introitus without separating the labia
- d) Rotation of fetal head is more than 45 degrees

Question 3:

Which of the following is true about instrumental vaginal delivery?

- a) Full cervical dilatation is the only prerequisite
- b) Forceps can be used for delivery of the after-coming head of breech
- c) Ventouse cannot be used in rotational occipitotransverse/ posterior delivery
- d) Engaged head is not a prerequisite

Question 4:

A 25-year-old woman at 35 weeks of gestation is wheeled into the labour room following the onset of labor. A forceps delivery is planned for her. Which of the following is not a contraindication for this?

- a) Fetal haemophilia
- b) Mentoposterior face presentation
- c) Incompletely dilated cervix
- d) Preterm labor

Question 5:

What is the given device used for?



- a) Vaginal delivery
- b) Suction and evacuation
- c) Continuous fetal monitoring

- d) Fetal scalp pH monitoring

Question 6:

A 28-year-old G2P1 woman with a history of mitral valve prolapse was planned for ventouse delivery. Which statement is true regarding the mode of delivery planned for this patient?

- a) Minor scalp abrasions and subgaleal hematomas in newborns are frequently seen
- b) It can be applied when fetal head is above the level of ischial spine
- c) Maternal trauma is seen more frequently when compared to forceps
- d) It cannot be used when fetal head is not fully rotated

Question 7:

Contraindications of ventouse include all except:

- a) Preterm delivery
- b) Deep transverse arrest
- c) Face presentation
- d) Fetal coagulopathy

Question 8:

You are an intern posted in the labor room assisting the delivery of a primigravida in labor. The obstetrician planned for a ventouse delivery because of maternal exhaustion. She asked you to help place the ventouse cup. Where would you place it?

- a) On the sagittal suture 3 cm in front of posterior fontanelle
- b) On the sagittal suture 3 cm behind of anterior fontanelle
- c) On the posterior fontanelle
- d) On the coronal suture 6 cm away from anterior fontanelle

Question 9:

A 24-year-old term primigravida was brought to the hospital in labor. Prophylactic ventouse was used due to ineffective expulsive efforts by the mother. What is the maximum negative pressure used in this procedure?

- a) 1 kg/ sq. cm
- b) 0.9 kg/ sq. cm
- c) 0.8 kg/ sq. cm
- d) 0.6 kg/ sq. cm

Question 10:

A prophylactic ventouse was planned for an anaemic pregnant woman. Delivery through this mode should occur within_____.

- a) 4 pulls over 30 min
- b) 4 pulls over 15-20 min
- c) 5 pulls over 15- 20 min
- d) 3 pulls over 15-20 min

Question 11:

A 24-year-old primigravida delivered a full-term male baby by assisted forceps delivery. What is not an advantage of this mode of delivery?

- a) No maternal effort is needed
- b) Quick in case of fetal distress
- c) Can be used in preterm
- d) Promotes autorotation

Question 12:

What is the advantage of ventouse over forceps?

- a) Less expertise required
- b) Delivery failure is less
- c) Can be used in preterm head
- d) No maternal effort is needed

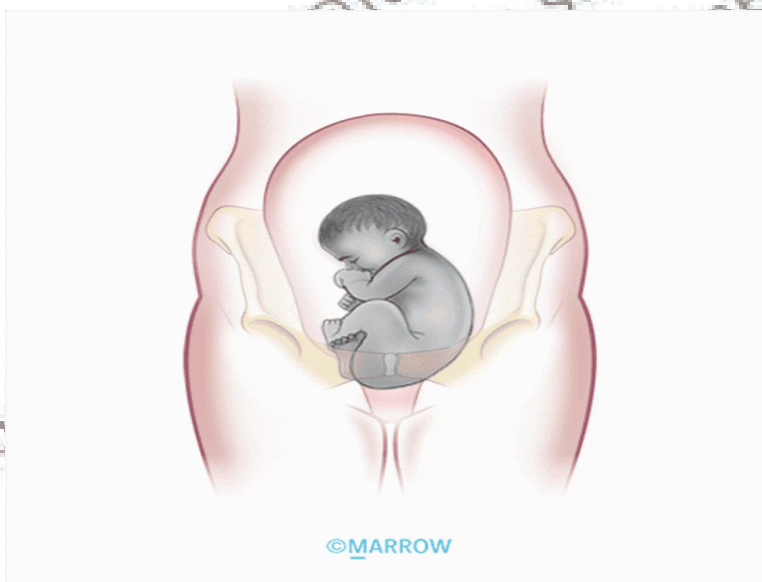
Question 13:

During a routine check-up of a multigravida at 32 weeks of gestation, the fetus was found to be in breech presentation. The obstetrician planned to do an external cephalic version. What is the ideal time to do this procedure?

- a) In labour
- b) At 39 weeks
- c) At 37 weeks
- d) At 34 weeks

Question 14:

A 25-year-old multigravida at 36 weeks presents for routine ANC. On examination, a hard, ballotable structure is felt at the fundus. NST is reactive. The obstetrician performs the following procedure on her. When will this procedure be contraindicated?



- a) Intra-uterine growth restriction
- b) Flexed breech
- c) Multiparity
- d) Unengaged fetus

Question 15:

At which angle would you perform a mediolateral episiotomy in a primigravida?

- a) 15 degrees

- b) 45 degrees
- c) 60 degrees
- d) 30 degrees

Answer Key

Question No.	Correct Option
1	d
2	d
3	b
4	d
5	a
6	a
7	b
8	a
9	c
10	d
11	d
12	a
13	c
14	a
15	c

Detailed Explanations

Solution to Question 1:

Completely dilated cervix is a prerequisite for forceps delivery.

Prerequisites for forceps delivery:

- Experienced operator
- Engaged head
- Ruptured membranes
- Vertex presentation
- Completely dilated cervix

- Precisely assessed fetal head position
- Cephalopelvic disproportion not suspected
- Absent fetal coagulopathy or bone demineralization disorder

Before operative vaginal delivery, the bladder should be emptied to provide additional pelvic space and minimize bladder trauma.

Solution to Question 2:

The image shows Wrigley's forceps, which is a type of outlet forceps. Rotation of the fetal head should not exceed 45 degrees for the application of outlet forceps. This is because it increases the risk of vaginal wall lacerations.

Solution to Question 3:

Forceps can be used for delivery of the aftercoming head of breech. The advantages are as follows:

- Delivery can be controlled by giving pull directly on the head and the force is not transmitted through the neck.
- Flexion is better maintained.
- Mucus can be sucked out from the mouth more effectively.

Prerequisites for instrumental vaginal delivery:

- Ruptured membranes vertex presentation
- Cephalopelvic disproportion not suspected
- No fetal coagulopathy or bone demineralization disorder
- Completely dilated cervix
- Precisely assessed fetal head position
- Engaged head
- Experienced operator

Note: Ventouse can be used for occipitotransverse or posterior position.

Solution to Question 4:

Forceps can be used to deliver a preterm baby where it is believed to form a protective cage. Vacuum is contraindicated in these cases.

Contraindications of forceps delivery:

- Unengaged head

- Cephalopelvic disproportion
- Malpresentations like brow and face (mentoposterior position)
- Cervix not completely dilated
- Inability to define position
- Fetal coagulopathy or osteogenesis imperfecta

Note: Forceps delivery is only reserved for mentoanterior position.

Solution to Question 5:

The image shown is a vacuum delivery system. It is used for vacuum-assisted vaginal delivery of the fetus.

Theoretical advantages of ventouse (vacuum extraction) compared to forceps include:

- Simpler requirements for positioning of the fetal head
- Avoidance of space occupying blades within the vagina
- Lower maternal trauma rates

Solution to Question 6:

Minor scalp abrasions and subgaleal hematomas in newborns are more frequently seen with ventouse delivery than with forceps delivery.

Maternal morbidity is seen more with forceps deliveries and perinatal morbidity is seen more with vacuum deliveries.

Maternal morbidity seen in operative vaginal deliveries include:

- Increased third and fourth-degree vaginal lacerations
- Pelvic floor disorders
- Postpartum hemorrhage

Perinatal morbidity seen in operative vaginal deliveries include:

- Vacuum-assisted perinatal injuries: Cephalohematoma, subgaleal hemorrhage, retinal hemorrhage, neonatal jaundice secondary to these hemorrhages, shoulder dystocia, clavicular fracture, intracranial hemorrhage, and scalp lacerations.
- Forceps-assisted perinatal injuries: Higher rates of facial nerve injury, brachial plexus injury, depressed skull fracture, and corneal abrasion.

In cardiac patients, vaginal delivery is preferred. It is conducted with the laboring woman in an upright or left decubitus position. Prophylactic use of vacuum is done to cut short the second stage of labor. This is to minimize aortocaval compression. Caesarean section is not preferred since it is associated with more blood loss and higher thromboembolic and infection risk.

Solution to Question 7:

Deep transverse arrest is an indication for ventouse delivery.

In ventouse delivery technique, suction is created within a cup placed on the fetal scalp such that traction on the cup aids fetal expulsion. The important step in ventouse extraction is proper cup placement over the flexion point.

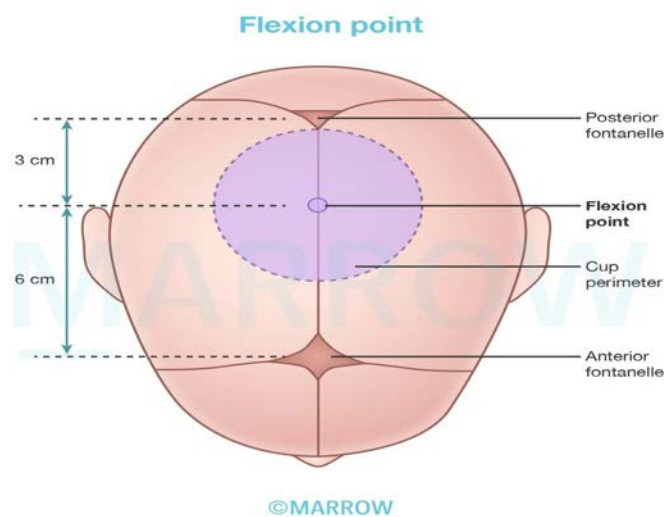
The vacuum is gradually created by increasing the suction in increments of 0.2 kg/cm^2 every 2 minutes until a total negative pressure of 0.8 kg/cm^2 is reached. Progressive descent should accompany each traction attempt.

Solution to Question 8:

An important step in vacuum extraction is proper cup placement over the flexion point. It is a point on the sagittal suture that lies 3 cm in front of the posterior fontanelle.

Advantages of correct positioning of cup in vacuum extraction:

- Maximum traction
- Minimum cup detachment
- Twisting of fetal head is avoided
- Smallest head diameter is delivered through the pelvic outlet
- Success rate is improved
- Fetal scalp injury rates are lowered
- Perineal trauma is reduced because the smallest fetal head diameter distends the vulva



Solution to Question 9:

The maximum negative pressure used in ventouse is 0.8 kg/cm²

Gradual vacuum creation is recommended. This is generated by increasing the suction in increments of 0.2 kg/cm² every 2 minutes. This is done until a total negative pressure of 0.8 kg/cm² is reached.

This slow incremental approach would allow for a more firm attachment of the vacuum cup to the fetal head and thereby a lower failure rate.

Solution to Question 10:

Delivery with ventouse should occur within 3 pulls over 15-20 min.

Solution to Question 11:

Autorotation is seen with vacuum extraction and not assisted forceps delivery.

Advantages of ventouse over forceps:

- No need of anaesthesia
- Less expertise required
- Less maternal injuries
- Promotes autorotation
- Does not occupy the sides
- Less traction force needed

Advantages of forceps over ventouse:

- No maternal effort is needed
- Quick in case of fetal distress
- Can be used in face and breech
- Can be used in preterm
- Less cephalohematoma
- Delivery failure is less

Solution to Question 12:

The advantage of ventouse over forceps is that less expertise is required in this technique.

Advantages of ventouse over forceps:

- No need of anaesthesia
- Less expertise required
- Less maternal injuries
- Promotes autorotation
- Does not occupy the sides
- Less traction force needed

Advantages of forceps over ventouse:

- No maternal effort is needed
- Quick in case of fetal distress
- Can be used in face and breech
- Can be used in preterm
- Less cephalohematoma
- Delivery failure is less

Solution to Question 13:

The ideal time to do external cephalic version is 37 weeks. This procedure is attempted in a woman who has completed 36 weeks of gestation and has a breech fetal presentation.

Before 37 weeks, a breech presentation has a high likelihood of correcting spontaneously. Moreover, if performed too early, there is time to return back to breech.

After 37 weeks, the success rates of ECV are very low. Although it can be performed until the early stages of labour, it is not recommended.

Solution to Question 14:

The procedure shown here is external cephalic version. A relative contraindication for this procedure is intra-uterine growth restriction.

Solution to Question 15:

The mediolateral episiotomy is directed at an angle of 60 degrees.

The mediolateral episiotomy incision starts in the midline at the fourchette and is directed at an angle 60 degrees from the midline. This 60-degree angle takes into consideration a distortion of perineal anatomy during crowning and ultimately yields an incision that is 45 degrees from the midline following delivery.

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Caesarean Section and Vaginal Birth After Caesarean (VBAC)

Question 1:

When should an elective caesarean section ideally be performed?

- a) After 37 weeks
- b) After 38 weeks
- c) After 40 weeks
- d) After 39 weeks

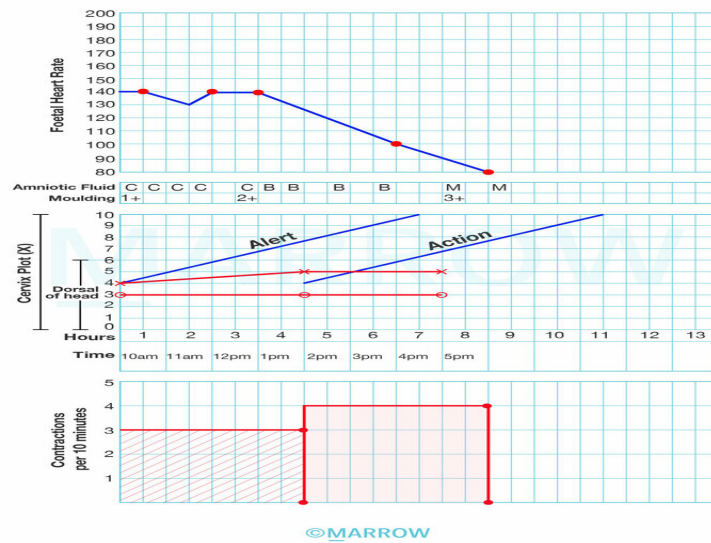
Question 2:

A 28-year-old pregnant woman is set to undergo an elective LSCS 2 days later. What is the ideal time to administer antibiotics to this patient?

- a) 24 hours before surgery
- b) 6 hours before surgery
- c) After cord clamping
- d) 30-60 min before surgery

Question 3:

The partogram of a 26-year-old G2P1 woman is shown below. What is the prophylactic antibiotic of choice for this patient before definitive management is done?



- a) Azithromycin
- b) Cefazolin
- c) Doxycycline
- d) Ciprofloxacin

Question 4:

You are asked to administer antibiotic prophylaxis against infective endocarditis to a patient posted for elective LSCS. What condition does the patient likely have?

- a) Atrial septal defect
- b) Prosthetic valves
- c) Ventricular septal defect
- d) Patent ductus arteriosus

Question 5:

A 25-year-old obese pregnant woman is scheduled for caesarean section. While going through her file, you find that she is allergic to penicillin. What is/are the antibiotic(s) you would prescribe for her?

- a) Clindamycin 600 mg alone
- b) Clindamycin 900 mg plus an aminoglycoside
- c) Clindamycin 900 mg alone

- d) Clindamycin 600 mg plus an aminoglycoside

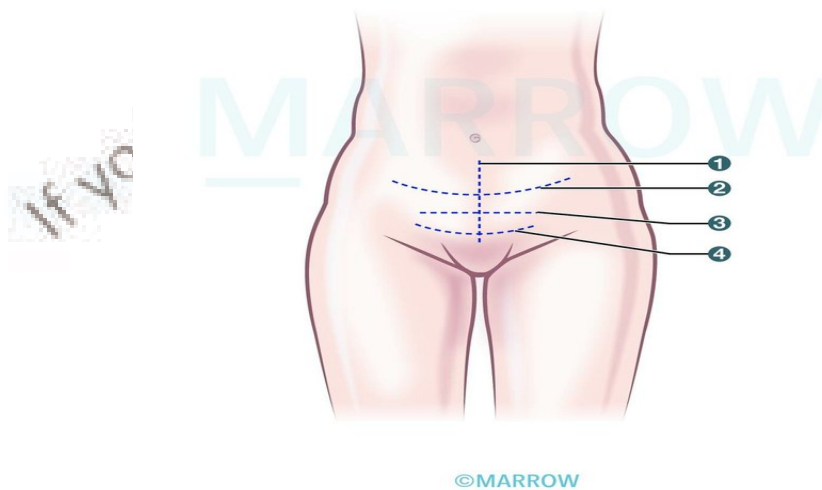
Question 6:

A woman with preeclampsia presents with haemorrhage at 38 weeks of gestation. You decide to perform a caesarean section and terminate the pregnancy. What is the anaesthesia of choice?

- a) Combined spinal epidural anaesthesia
- b) Single shot spinal anaesthesia
- c) General anaesthesia
- d) Epidural anaesthesia

Question 7:

An elective cesarean section is being performed in a primigravida due to suspected cephalopelvic disproportion. In which of the following abdominal incisions are the recti muscles transected?



- a) 4
- b) 3
- c) 1
- d) 2

Question 8:

All of the following are benefits of using a transverse abdominal incision for caesarean delivery in a patient except_____.

- a) Wide operating space
- b) Low fascial wound dehiscence
- c) Decreased postoperative pain
- d) Low risk of incisional hernia

Question 9:

An elective C-section is being performed in a primigravida due to suspected contracted pelvis. After opening the abdomen through a modified Pfannenstiel incision, the obstetrician makes a low transverse uterine incision. How should this incision be extended?

- a) By applying lateral and slightly upward pressure with each index finger
- b) By applying lateral and slightly downward pressure with each index finger
- c) By applying lateral and slightly upward pressure with each thumb
- d) By applying lateral and slightly downward pressure with each thumb

Question 10:

The highest risk of uterine rupture is seen with_____.

- a) Multiple low-transverse incisions
- b) T-shaped incision
- c) Low-vertical incision
- d) Single low-transverse incision

Question 11:

An obstetrician performing a C-section exteriorises the uterus before performing uterine repair. All of the following are true about this procedure except_____.

- a) Incision and bleeding points are easily visualized and repaired
- b) Relaxed/atonic uterus can be recognized quickly and massage applied
- c) Vomiting and discomfort is caused by traction due to exteriorization

d) Adnexal exposure is low so tubal sterilization is difficult

Question 12:

Which of the following is not an indication for peripartum hysterectomy?

- a) Uterine atony
- b) Cervical cancer
- c) Uterine rupture
- d) Cervical laceration

Question 13:

What is the most common fetal injury associated with caesarean section?

- a) Skull fracture
- b) Cephalohematoma
- c) Skin laceration
- d) Clavicular fracture

Question 14:

An elective caesarean section is planned for a pregnant woman with a cardiac ailment. Which of the following complications is she not at an increased risk for?

- a) Thromboembolism
- b) Chorioamnionitis
- c) Mendelson syndrome
- d) Haemorrhage

Question 15:

Which of the following statements is true regarding the instrument shown below?



- a) A and B
- b) B and C
- c) A and C
- d) Only C

Question 16:

The risk factors for venous thromboembolism after caesarean section include all except_____.

- a) Preeclampsia
- b) Obesity
- c) Parity ≥ 3
- d) Age ≤ 35 years

Question 17:

A 28-year-old multigravida with a history of C-section is being considered for vaginal delivery. The presence of which of the following features would categorise this procedure as low risk?

- a) Transverse incision in prior caesarean
- b) Breech presentation
- c) Preeclampsia
- d) Classical incision in prior caesarean

Question 18:

A trial of scar is performed on a G2P1L1 woman. The risk of which of the following complications is increased in this baby when compared to their elder sibling?

- a) Birth trauma
- b) Hypoxic ischemic encephalopathy
- c) Transient tachypnea of new born
- d) Neonatal sepsis

Answer Key

Question No.	Correct Option
1	d
2	d
3	b
4	b
5	b
6	c
7	d
8	a
9	a
10	b
11	d
12	b
13	c
14	b
15	c
16	d
17	a
18	b

Detailed Explanations

Solution to Question 1:

An elective caesarean section should ideally be performed after 39 weeks.

It is not done before 39 weeks of gestation as there is an increased risk of:

- Complications related to prematurity including respiratory problems
- Neonatal adaptation problems such as hypothermia and hypoglycemia

Solution to Question 2:

According to ACOG guidelines, the ideal time to administer antibiotics is within a time period of 60 mins prior to the planned caesarean section.

Antimicrobial administration before surgical incision has been shown to lower postoperative infection rates without adverse neonatal effects, as compared with drug administration after umbilical cord clamping. A 1 g dose of cefazolin is an efficacious and cost-effective choice.

For emergency delivery, antibiotic prophylaxis should be given as soon as possible.

Solution to Question 3:

The above partogram is suggestive of obstructed labour and the definitive management in this case is caesarean section. Cefazolin is an efficacious and cost-effective choice for antibiotic prophylaxis before caesarean section.

According to the partogram, there is a secondary arrest of dilatation for >2 hours and arrest of descent for >1 hour. The amniotic fluid is stained with meconium and there is fall in fetal heart rate indicating fetal distress. Hence, the management plan is to perform a caesarean section.

The recommended prophylactic drug is a single intravenous dose of a β -lactam antimicrobial. This could either be a cephalosporin or an extended-spectrum penicillin derivative. A 1 g dose of cefazolin is usually given. For obese women, a 2 g dose is better to provide an adequate coverage.

Solution to Question 4:

Antibiotic prophylaxis against infective endocarditis before surgical procedures is recommended in patients with prosthetic valves, prior history of endocarditis, and unrepaired cyanotic heart diseases.

Pregnant women at the highest risk for endocarditis are those with cyanotic cardiac disease, prosthetic valves, or both. However, endocarditis prophylaxis is not required if the patient is already receiving intrapartum prophylaxis with broad-spectrum antibiotics to prevent endometritis.

Antibiotic prophylaxis is not recommended for atrial septal defect (ASD). The risk of endocarditis with ASD is negligible. Bacterial endocarditis is associated with unrepaired ventricular septal

defects where antibiotic prophylaxis is recommended. Similarly, prophylaxis for bacterial endocarditis is indicated at delivery for unrepaired patent ductus arteriosus.

Solution to Question 5:

In an obese pregnant woman who is allergic to penicillin, a dose of 900 mg clindamycin with an aminoglycoside is used for antibiotic prophylaxis.

In women with significant penicillin or cephalosporin allergy, a single 600 mg intravenous dose of clindamycin combined with a weight-based dose of aminoglycoside is recommended. For obese women, a 900 mg dose is preferred.

Solution to Question 6:

General anaesthesia is the anaesthesia of choice here since the patient presents with haemorrhage.

General anaesthesia may be indicated for emergency cesarean delivery in patients with a reassuring airway examination. It is also recommended for patients with severe coagulopathy where neuraxial anesthesia is contraindicated.

Spinal anesthesia is not preferred when the operation is expected to result in blood loss or in a previously haemorrhaging patient. It is because in such cases the development of severe hypovolemia is likely which is a contraindication for this type of anaesthesia.

Solution to Question 7:

The recti muscles are transected in the Maylard incision for caesarean delivery which is marked as '2' in the given image.

It is a variation of the Pfannenstiel incision in which the rectus abdominis muscles are sectioned transversely to permit wider access to the pelvis. It is also known as the Mackenrodt incision.

The various incisions mentioned in the image are:

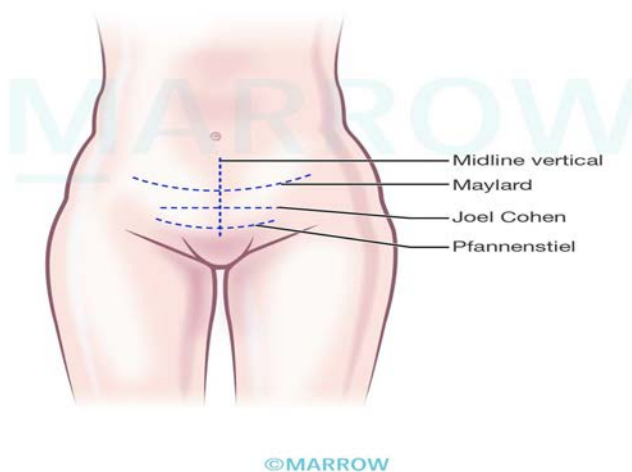
Incision '1' - Midline vertical incision is made in the middle extending from just below the umbilicus to just above the symphysis pubis. This is used less commonly compared to the transverse abdominal incision.

Incision '3' - Joel Cohen is a straight incision higher than the Pfannenstiel incision. The fascia was incised transversely in the midline then extended laterally with blunt finger dissection; finger dissection was used to separate the rectus muscles vertically and laterally.

Incision '4' - Pfannenstiel incision is the most common incision made in caesarean delivery. It follows Langer lines of skin tension. A transverse and slightly curvilinear incision is made in the skin and subcutaneous tissue 3 cm above the superior border of the pubic symphysis.

This is stretched laterally for 12-15 cm to accommodate the delivery of the fetus. Following this, the rectus sheath is incised transversely and rectus muscles are separated in the middle to gain entry to the peritoneum.

Abdominal incisions for caesarean section



Solution to Question 8:

Transverse abdominal incisions offer a narrow operating space.

Solution to Question 9:

In a C-section, the incision on the uterus can be extended by using lateral and slightly upward pressure applied with each index finger.

If the lower uterine segment is thick, then cutting laterally and then slightly upward with bandage scissors will extend the incision.

Note: When scissors are used, the index and middle fingers of the non-dominant hand should be insinuated between the myometrium and the fetal parts to prevent fetal laceration.

Solution to Question 10:

A T-shaped incision has the highest risk of uterine rupture among the given options. Among all the incisions, a previous classical cesarean incision has the highest risk of rupture but it is not an option here.

Overall, the maximum risk of uterine rupture is in patients who have had a previous upper segment uterine rupture. Women with single prior low-transverse hysterotomy have the lowest risk of symptomatic scar separation.

Solution to Question 11:

Adnexal exposure is high and hence tubal sterilization is easy.

After delivery of the placenta, the uterus is lifted through the incision onto the draped abdominal wall. The fundus is covered with a moistened laparotomy sponge and this is known as exteriorisation of uterus.

Benefits of exteriorization:

- Relaxed, atonic uterus can be recognized quickly and massage applied.
- The incision and bleeding points are more easily visualized and repaired.
- Adnexal exposure is superior, and thus, tubal sterilization is easier.

Drawbacks of exteriorization:

- Intraoperative nausea and vomiting
- Need for additional uterotonics
- Longer time duration to the first recognized bowel movement

The below image shows the exteriorisation of the uterus.



Solution to Question 12:

Cervical cancer is not an indication for peripartum hysterectomy.

Indications for peripartum hysterectomy include:

- Uterine atony - when every other method fails to control PPH
- Uterine extension

- Uterine rupture
- Postpartum uterine infection
- Cervical laceration
- Leiomyoma - large leiomyomas in a woman with completed family
- Ovarian neoplasia

Solution to Question 13:

Skin laceration is the most common fetal injury associated with caesarean section. Other injuries seen include:

- Cephalohematoma
- Brachial plexus injury
- Clavicular fracture
- Skull fracture
- Facial nerve palsy

Solution to Question 14:

Elective caesarean section decreases the risk of chorioamnionitis.

Maternal complications increase with caesarean section when compared with vaginal deliveries. Some of these complications include:

- Post-operative infection
- Haemorrhage
- Thromboembolism
- Anaesthesia-related complications
- Mendelson syndrome/ aspiration pneumonitis

Solution to Question 15:

The given image shows a pneumatic compression device. It is used in women who are not on any thromboprophylaxis and planned for cesarean section (statement A) and it is discontinued once the patient starts ambulating post-surgery (statement C).

The device is used because pregnancy and surgery increase the risk of thromboembolism.

According to recent guidelines, patients with no additional risk factors and not placed on any pharmacological thromboprophylaxis are placed on pneumatic compression devices before

cesarean delivery. Patients undergoing cesarean delivery with additional risk factors are put on both pneumatic compression devices and pharmacological therapy (UFH or LMWH).
UFH-unfractionated heparin, LMWH-lower molecular weight heparin.

Indications of pharmacological thromboprophylaxis:

- Inherited coagulation disorder
- Overweight or obese patient
- Heart failure
- Family history of deep vein thrombosis or pulmonary embolism

Solution to Question 16:

Age ≥ 35 years is a risk factor for venous thromboembolism after caesarean section. The other risk factors include:

- Emergency cesarean
- Obesity
- Parity ≥ 3
- Caesarean hysterectomy
- Concurrent infection
- Preeclampsia
- Prior deep-vein thrombosis or thrombophilia

Solution to Question 17:

Transverse incision in prior caesarean section is a low risk factor in vaginal birth after cesarean (VBAC).

Risk factors for VBAC:

High risk -

- Classical or T-incision in prior cesarean
- Prior rupture
- Obstetrical contraindication such as placenta previa
- Inadequate facilities

Low risk -

- Transverse incision in prior cesarean
- Prior vaginal delivery
- Sufficient personnel and equipment

Solution to Question 18:

Hypoxic ischemic encephalopathy (HIE) in neonates has a higher incidence with trial of scar or VBAC as compared to elective cesarean.

Uterine rupture is a dangerous complication of VBAC. This leads to hemorrhage which can cause the baby to be hypoxic eventually leading to permanent brain damage from hypoxic-ischemic encephalopathy. This results in developmental delay and cerebral palsy.

Birth trauma, transient tachypnea of newborn (TTN) and neonatal sepsis are more commonly seen with elective caesarean in comparison to VBAC.

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Puerperium

Question 1:

When does puerperium begin?

- a) Soon after delivery of fetus
- b) Soon after expulsion of placenta
- c) After the fourth stage of labour
- d) 24 hours after child birth

Question 2:

How long does early puerperium last?

- a) Up to 24 hrs after delivery
- b) Up to 48 hrs after delivery
- c) Up to 72 hrs after delivery
- d) Up to 1 week after delivery

Question 3:

What is the weight of the uterus immediately after delivery?

- a) 300 g
- b) 100 g
- c) 1000 g
- d) 500 g

Question 4:

Which of the following is true regarding puerperium?

- a) The number of muscle fibres decrease
- b) The myometrial cell size decreases
- c) The uterus weighs 1000g by 1 week postpartum

- d) There is decreased activity of uterine collagenase

Question 5:

After how many weeks of postpartum will the uterus becomes a pelvic organ?

- a) 1 week
- b) 2 weeks
- c) 4 weeks
- d) 6 weeks

Question 6:

When does the uterus return to pregravid size sonographically?

- a) 2 weeks
- b) 4 weeks
- c) 6 weeks
- d) 8 weeks

Question 7:

External os reverts back to the pre-pregnant state by how many weeks?

- a) 4 weeks
- b) 6 weeks
- c) 12 weeks
- d) Never returns

Question 8:

Endometrial regeneration following delivery starts by which day?

- a) 5th day
- b) 6th day
- c) 7th day
- d) 10th day

Question 9:

A 34-year-old woman has just delivered a boy by spontaneous vaginal delivery. How often should her BP be recorded for the next 2 hours?

- a) Every 5 minutes
- b) Every 15 minutes
- c) Every 30 minutes
- d) Every 1 hour

Question 10:

A woman on post-natal day 1 presented with abdominal pain which worsens when she is breastfeeding. What is this known as?

- a) After-labour
- b) Involutional pain
- c) Uterine colic
- d) Afterpains

Question 11:

What would be the average duration of lochial discharge in a puerperal woman?

- a) 5-15 days
- b) 10-20 days
- c) 24-36 days
- d) 36-42 days

Question 12:

What is the contribution of postpartum diuresis to weight loss?

- a) 1-2 kg
- b) 2-3 kg
- c) 3-4 kg

d) 4-5 kg

Question 13:

What is the single most important galactopoietic hormone in post-natal women?

- a) Oxytocin
- b) Prolactin
- c) Estrogen
- d) Progesterone

Question 14:

A woman presents with numbness and paraesthesia along the anterolateral thigh two days after vaginal delivery. Which nerve is likely involved in this condition?

- a) Femoral nerve
- b) Iliohypogastric nerve
- c) Ilioinguinal nerve
- d) Lateral femoral cutaneous nerve

Question 15:

Which among the following is the most common presentation of postpartum nerve injury?

- a) Facial palsy
- b) Wrist drop
- c) Foot drop
- d) Claw hand

Question 16:

Puerperal fever is defined as a temperature of _____.

- a) $\geq 37.2^{\circ}\text{C}$
- b) $\geq 37.5^{\circ}\text{C}$
- c) $\geq 38^{\circ}\text{C}$

d) $\geq 38.5^{\circ}\text{C}$

Question 17:

A 30-year-old woman presents with high-grade fever and foul-smelling lochia 2 days after vaginal delivery. Lower abdominal tenderness is elicited on examination. Her WBC count is 25,000/ microlitre. Which is the commonest route for the spread of her likely condition?

- a) Lymphatic
- b) Direct invasion
- c) Hematogenous
- d) Aerosol

Question 18:

What is the first line of treatment of mastitis in a lactating mother?

- a) Dicloxacillin
- b) Cefazolin
- c) Ceftriaxone
- d) Ampicillin

Question 19:

A nursing mother in her third week postpartum presents with breast pain and difficulty in breastfeeding. On examination, warmth and tenderness over the left breast are noted. Her temperature is 38.3°C and her WBC count is 13,000/mm³. Which of the following is false regarding this condition?



- a)** Most common etiological agent is streptococcus
- b)** Immediate source of infection is the infant's nose and throat
- c)** Complication includes toxic shock syndrome
- d)** Incidence is less than 1 %

Question 20:

A mother started to feel sad and tearful over the first two weeks following childbirth. She felt overwhelmed by the responsibilities of nursing her baby and changing diapers in the middle of the night. She felt that she was a failure as a mother. What is false regarding this condition?

- a)** It is observed 4-5 days after delivery
- b)** Lowered tryptophan levels have been noticed
- c)** Nearly 50% of women suffer from this problem
- d)** Antidepressants are the treatment of choice

Question 21:

A lady was discharged after a normal vaginal delivery. On the third day, she came back with fever, tachycardia, and seizures. Fundus showed papilledema with no focal deficits. What is the most probable diagnosis?

- a)** Subarachnoid haemorrhage
- b)** Meningitis

- c) Acute migraine
- d) Cortical vein thrombosis

Question 22:

A mother, previously on OCPs, comes with her 2-week-old baby to the OPD. She informs you that she is currently breastfeeding her child and enquires about re-starting contraception. When would you recommend she start using contraceptives?

- a) 3rd postpartum week
- b) 3rd postpartum month
- c) 4th postpartum month
- d) 6th postpartum month

Question 23:

A woman presents to the OPD with complaints of amenorrhoea for the past year. She has a history of postpartum hemorrhage and lactational failure. What is the probable diagnosis?

- a) Sheehan's syndrome
- b) Rotor syndrome
- c) Prolactinoma
- d) Acromegaly

Question 24:

A woman presents with fever on the 3rd postpartum day and is put on antibiotics. Fever is still not brought under control. On further investigation, she is diagnosed to have septic pelvic thrombophlebitis. Which among the following is the next course of treatment for her?

- a) Stop antibiotics and start heparin
- b) Antibiotics+Heparin
- c) Surgical embolectomy
- d) Hysterectomy

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	b
5	b
6	d
7	d
8	c
9	b
10	d
11	c
12	b
13	b
14	d
15	c
16	c
17	b
18	a
19	a
20	d
21	d
22	b
23	a
24	b

Detailed Explanations

Solution to Question 1:

Puerperium begins as soon as the placenta is expelled and lasts for approximately 6 weeks.

It is the period following childbirth during which body tissues revert back approximately to the pre-pregnant state, both anatomically and physiologically.

Puerperium can be divided as follows:

- Immediate puerperium: within 24 hours

- Early puerperium: up to 7 days
- Remote/late puerperium: up to 6 weeks or 42 days.

Solution to Question 2:

Early puerperium is considered to last up to 1 week after delivery.

Puerperium can be divided as follows:

- Immediate puerperium: within 24 hours
- Early puerperium: up to 7 days
- Remote/late puerperium: up to 6 weeks or 42 days

Solution to Question 3:

Immediately following delivery, the uterus weighs 1000 g.

Involution of the uterus occurs due to a decrease in the size of muscle fibers.

Time following delivery	Features	Weight of the uterus
Immediately following delivery	Uterus corresponds to 20 weeks gestational age Fundus lies just below the umbilicus	1000 g
48 hours	Myometrial involution starts	-
1 week	Rapid involution of uterus	500 g
2 weeks	The uterus becomes a pelvic organ	300 g
4 weeks	Clinically, involution is complete	100 g
6-8 weeks	Uterus returns to normal (pre-pregnant) size Sonographically, involution is complete at 8 weeks	60 g

Solution to Question 4:

During puerperium, the number of muscle fibers is not decreased, but there is a substantial reduction in the myometrial cell size.

There is an increased activity of uterine collagenases and release of proteolytic enzymes.

Within the puerperal uterus, larger blood vessels become obliterated by hyaline changes. These vessels are gradually resorbed and replaced by smaller ones.

Time following delivery	Features	Weight of the uterus
Immediately following delivery	Uterus corresponds to 20 weeks gestational age Fundus lies just below the umbilicus	1000 g
48 hours	Myometrial involution starts	-
1 week	Rapid involution of uterus	500 g
2 weeks	The uterus becomes a pelvic organ	300 g
4 weeks	Clinically, involution is complete	100 g
6-8 weeks	Uterus returns to normal (pre-pregnant) size Sonographically, involution is complete at 8 weeks	60 g

Solution to Question 5:

The uterus becomes a pelvic organ by 2 weeks of postpartum.

Involution of the uterus occurs due to a decrease in the size of muscle fibers.

Time following delivery	Features	Weight of the uterus
Immediately following delivery	Uterus corresponds to 20 weeks gestational age Fundus lies just below the umbilicus	1000 g
48 hours	Myometrial involution starts	-
1 week	Rapid involution of uterus	500 g
2 weeks	The uterus becomes a pelvic organ	300 g
4 weeks	Clinically, involution is complete	100 g
6-8 weeks	Uterus returns to normal (pre-pregnant) size Sonographically, involution is complete at 8 weeks	60 g

Solution to Question 6:

The uterus and endometrium return to pregravid size by 8 weeks when assessed by sonography. On clinical examination, the involution of the uterus is complete by 4 weeks.

Time following delivery	Features	Weight of the uterus
Immediately following delivery	Uterus corresponds to 20 weeks gestational ageFundus lies just below the umbilicus	1000 g
48 hours	Myometrial involution starts	-
1 week	Rapid involution of uterus	500 g
2 weeks	The uterus becomes a pelvic organ	300 g
4 weeks	Clinically, involution is complete	100 g
6-8 weeks	Uterus returns to normal (pre-pregnant) sizeSonographically, involution is complete at 8 weeks	60 g

Solution to Question 7:

External os never returns back to normal whereas the contour of the cervix takes 6 weeks to regain normalcy.



External os before pregnancy



External os after pregnancy

©Marrow

Solution to Question 8:

Endometrial regeneration starts by 7th day and is completed by the 10th day. Regeneration of the epithelium at the placental site takes about 6 weeks.

Endometrium regenerates from zona basalis which remains intact and is the source of new endometrium. This arises from proliferation of the endometrial glandular remnants and the stroma of the interglandular connective tissue.

Solution to Question 9:

For the first 2 hours following delivery, blood pressure and pulse should be recorded every 15 minutes.

Temperature is assessed every 4 hours for the first 8 hours after delivery.

Solution to Question 10:

The given clinical scenario is suggestive of afterpains which are seen during puerperium.

In primiparous women, the uterus tends to remain tonically contracted following delivery. In multiparous women, it often contracts vigorously at intervals and gives rise to afterpains, which are similar to but milder than labor contractions.

These are more pronounced as parity increases and worsen when the infant suckles, likely because of oxytocin release. Afterpains gradually decrease in intensity and become mild by the third day.

Solution to Question 11:

The average duration of lochial discharge in a postpartum woman is about 24-36 days. The average amount of discharge for the first week is 250 mL.

The lochial discharge appears as follows:

Name	Colour	Timing after delivery	Content
Lochia rubra	Red	First few days	Blood, decidua, fetal membranes, vernix, lanugo, meconium
Lochia serosa	Yellowish/Pink	After 3-4 days	WBC, cervical mucus, microorganisms
Lochia alba	Pale White	After 10th day	Decidua, cells, WBC, mucin, cholesterol crystals, and fat cells

Solution to Question 12:

Postpartum diuresis results in a relatively rapid weight loss of 2 to 3 kg.

Decreased estrogen levels are associated with the diuresis of excess extracellular fluid accumulated during pregnancy.

Solution to Question 13:

Prolactin is the single most important galactopoietic hormone in postnatal women.

Function	Definition	Hormone
Mammogenesis	Breast growth and development	Estrogen (ductal growth) Progesterone (alveolar growth)
Lactogenesis	Initiation of milk secretion	Prolactin
Galactokinesis	Ejection of milk	Oxytocin
Galactopoiesis	Maintenance of lactation	Prolactin

Solution to Question 14:

The given clinical features are suggestive of meralgia paraesthetica which is caused by injury to the lateral femoral cutaneous nerve.

The lateral femoral cutaneous nerve is at risk of injury during prolonged pushing with hip flexion as the nerve is compressed under the inguinal ligament.

It is the most common obstetrical neuropathy after vaginal delivery, followed by femoral nerve involvement.

Note: The nerves most commonly injured during cesarean section are the iliohypogastric and ilioinguinal nerves.

Solution to Question 15:

The most common postpartum nerve injury is an injury to the peroneal nerve causing foot drop.

The lateral femoral cutaneous nerve is the most commonly injured nerve after vaginal delivery followed by femoral neuropathies.

Obstetrical neuropathies are due to the pressure on branches of the lumbosacral nerve plexus during labor. They can manifest as complaints of intense neuralgia or cramp-like pains extending down one or both legs as soon as the head descends into the pelvis. Footdrop can be secondary to injury at the level of the lumbosacral plexus, sciatic nerve, or common fibular (peroneal) nerve.

Injury of the iliohypogastric and ilioinguinal nerves may occur with cesarean delivery, which may manifest as an inguinal hernia and loss of sensation at the pubic region respectively.

Note: Since there are no options relating to intense neuralgia in the lower limb, foot drop becomes the best possible answer.

Solution to Question 16:

Puerperal fever is defined as the temperature of 38 °C or higher (100.4 °F) recorded on 2 separate occasions 24 hours apart within the first 10 days following delivery. This excludes the first 24 hours.

In the first 24 hours, a slight rise of temperature is not uncommon and hence the first 24 hours are excluded while defining puerperal pyrexia.

Note: Generally, fever in an adult is defined as an oral temperature of ≥ 37.5 °C.

Solution to Question 17:

The given clinical scenario of high-grade fever, foul-smelling lochia, and pelvic tenderness is suggestive of puerperal sepsis. The most common route of spread of this condition is direct spread.

Streptococcus sp. is the most important causative organism of puerperal sepsis. The commonest manifestation is endometritis.

The route of delivery is an important risk factor for the development of puerperal sepsis (cesarean > vaginal). The most common site of puerperal infection is the placental site in vaginal delivery and uterine incision in cesarean section.

The only proven way of decreasing uterine infection following cesarean section is administering a single dose of ampicillin or 1st generation cephalosporin at the time of caesarean delivery.

Solution to Question 18:

Dicloxacillin is the first line of treatment for mastitis in lactating mothers and should be taken for 10-14 days.

Mastitis refers to the infection of the breast parenchymal tissue. The most common organism causing breast infection is Staphylococcus aureus. It is common in new mothers at the beginning of lactation who are inexperienced with breastfeeding or during weaning when nipple trauma occurs due to the infant's growing teeth.

Clinical features of mastitis include:

- Fever and malaise
- Localized pain and tenderness

- Purulent nipple discharge
- Redness and swelling of the affected breast
- Positive fluctuation test

A culture from nipple discharge or test-needle aspiration of the cellulitic area is done to confirm the organism and its antibiotic sensitivity. An ultrasonogram of the involved breast is done to exclude abscess. They are treated with antibiotics and adequate breast support. Choice of antibiotics in lactational mastitis:

- Dicloxacillin 500 mg 4 times a day is to be started empirically.
- Erythromycin is given to women who are penicillin-sensitive.
- If the infection is caused by resistant, penicillinase-producing *Staphylococcus* species then vancomycin, clindamycin, or trimethoprim-sulfamethoxazole is given for at least 10 days.

Solution to Question 19:

The clinical vignette of breast pain, fever, and leucocytosis with the image showing a red and swollen breast suggests a diagnosis of mastitis. The most common etiological agent is *Staphylococcus aureus*.

Mastitis refers to infection of the breast parenchymal tissue. Symptoms usually appear in the 3rd or 4th week. The infection is usually unilateral, with the source of infection being the infant's nose and throat. Predisposing factors are fissures, abrasions, and nipple cracks.

Management:

- Breastfeeding should be continued. If the infant is unable to nurse on the inflamed breast, the mother may resort to breast pumping. When nursing bilaterally, suckling is best started on the uninvolved breast.
- Dicloxacillin is the first line of treatment of mastitis. Erythromycin is given to women who are penicillin-sensitive.
- If an abscess forms, surgical treatment is done.

Solution to Question 20:

The given clinical scenario is suggestive of puerperal blues. It is also known as 3 day blues/baby blues/maternity blues. It is treated by reassurance and psychological support.

Features	Puerperal blues	Post-partum depression
Onset	4-5 days after delivery	4-6 months after delivery/abortion
Cause	Sudden decrease in hormone levels	Multifactorial

Features	Puerperal blues	Post-partum depression
Duration	Lasts for a few days to 2-3 weeks	-
Incidence in postpartum mothers	50%	10-20%
Treatment	Reassurance and psychological support	Antidepressants (Fluoxetine/ Paroxetine)
Recurrence rate	25%	50-100%

Solution to Question 21:

The given clinical situation likely suggests cortical vein thrombosis.

Risk factors:

- Pregnancy and puerperium
- Oral contraceptives
- Prothrombotic conditions

Most common clinical presentation is headache with or without vomiting, papilledema, or focal deficits.

Investigations:

- Brain MRI with MR venography - most sensitive
- CT - normal in 30 % of cases.

Option A: Subarachnoid hemorrhage - As there are no focal deficits, this diagnosis is unlikely.

Option B: Meningitis - The patient would have presented with neck rigidity or photophobia. Moreover, the presence of early papilledema goes against acute meningitis.

Option C: Acute migraine - It does not present with seizures.

Solution to Question 22:

Lactating women should start using contraceptives in the 3rd postpartum month.

Progesterone-only methods are the hormonal contraception of choice in a lactating mother.

Efficacy of contraceptives used in postpartum period: Levonorgestrel IUD/ Implant > DMPA > Progestin-only pill.

Contraception should be ideally started after the resumption of ovulation. However, there are no contraindications for the use of contraceptives as early as 4-6 weeks in lactating women and immediately in non-lactating women.

Lactating mother	Non-lactating mother
Ovulation occurs after 10 weeks	Ovulation occurs after 4 weeks
Use contraceptive in 3rd postpartum month	Use contraceptive in 3rd postpartum week.

Solution to Question 23:

The clinical scenario points to a diagnosis of Sheehan's syndrome.

Sheehan's syndrome (SS) is postpartum hypopituitarism secondary to the necrosis of the pituitary gland. It usually results from severe hypotension or shock caused by massive hemorrhage during or after delivery. Patients with SS have varying degrees of anterior pituitary hormone deficiency. The hyperplastic enlargement of the pituitary, which occurs normally during pregnancy, increases the risk of infarction.

Affected women may have persistent hypotension, tachycardia, hypoglycemia, and lactation failure. The most prominent symptoms are lactation failure (due to a deficiency of prolactin) and postpartum amenorrhea (due to a deficiency of gonadotropins). Because deficiencies of some or all pituitary-responsive hormones may develop after the initial insult, Sheehan syndrome can be heterogeneous and may not be identified for years.

It can present acutely as pituitary apoplexy, an endocrine emergency, characterized by a sudden onset of headache, nausea, visual deficits, and in severe cases, severe hypoglycemia, hypotension and shock, CNS hemorrhage, and death.

Also, it can present late with the following features:

- Loss of pubic and axillary hair
- Hypothyroidism due to decreased TSH secretion
- Adrenal insufficiency due to low ACTH

Treatment is by hormone replacement - first glucocorticoids, later - thyroid hormone, sex steroids, vasopressin and growth hormones are given. If there is evidence of progressive visual loss, cranial nerve involvement or loss of consciousness, immediate surgical decompression must be carried out. Fertility can be achieved in these women with hMG and hCG. Clomiphene is ineffective.

Option B: Rotor syndrome is a type of hereditary hyperbilirubinemia with autosomal recessive inheritance. Affected individuals have mild jaundice and hyperbilirubinemia.

Option C: Prolactinoma or pituitary adenoids is the condition of hyperprolactinemia diagnosed by a prolactin level ≥ 200 $\mu\text{g/L}$. It usually presents with infertility, amenorrhea, and galactorrhea.

Option D: Acromegaly is characterized by the signs and symptoms pertaining to excessive growth hormone secretion.

Solution to Question 24:

Antibiotics plus heparin is the next course of treatment for septic pelvic thrombophlebitis.

Septic pelvic thrombophlebitis refers to the presence of an endovascular thrombus in the setting of associated bacterial or fungal infection.

The treatment of septic pelvic thrombophlebitis is broad-spectrum antibiotics plus anticoagulation. Anticoagulation is thought to prevent further thrombosis and reduce the spread of septic emboli.

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Multifetal Pregnancy

Question 1:

What is the most common type of multifetal pregnancy?

- a) Monozygotic dichorionic diamniotic
- b) Dizygotic dichorionic diamniotic
- c) Monozygotic monochorionic diamniotic
- d) Monozygotic monochorionic monoamniotic

Question 2:

What is the time of division of the inner cell mass in a diamniotic monochorionic pregnancy?

- a) 1st - 2nd day
- b) 4th – 8th day
- c) 8th - 12th day
- d) 10th - 14th day

Question 3:

Which of the following statements is false regarding multifetal pregnancies?

- a) Number of embryos implanted affects the risk of twinning
- b) Isograft is done between monozygotic twins
- c) Twins of opposite sex are usually dizygotic
- d) Same sex twins having same blood group are always monozygotic

Question 4:

The examination findings of a pregnant woman are suggestive of her having conjoined twins. Which is the most common type of such twins?

- a) Craniopagus
- b) Thoracopagus

- c) Pyopagus
- d) Cephalopagus

Question 5:

Which of the following can be used to calculate the incidence of multifetal gestation?

- a) Tei index
- b) Hellin's rule
- c) CHOP score
- d) Kleihauer-Betke test

Question 6:

In which of the following instances is there an increased chance of twinning?

- a) 1, 4, 5
- b) 3, 4, 5
- c) 2, 3, 4
- d) 1, 2, 3

Question 7:

In superfecundation, which of the following is seen?

- a) Fertilisation of 2 ova released at the same time by sperms released at single intercourse
- b) Fertilisation of 2 ova released in the same cycle by sperms released at intercourse on 2 different occasion
- c) Both of the above
- d) None of the above

Question 8:

Which of the following statements is false regarding the condition given in the image?



- a) It is more common in monochorionic pregnancies
- b) It cannot be detected by sonography
- c) It is diagnosed usually after delivery
- d) Morbidity in the co-twin is due to hypotension

Question 9:

A woman is diagnosed to have twins during her dating scan. What is the most common presentation in such pregnancies?

- a) Both vertex
- b) 1st vertex and 2nd transverse
- c) 1st breech and 2nd vertex
- d) 1st vertex and 2nd breech

Question 10:

The best time to identify the type of twin pregnancy on ultrasound is:

- a) 18-22 weeks
- b) 24-28 weeks
- c) 11-14 weeks
- d) 14-18 weeks

Question 11:

A 35-year-old woman conceived after being treated for infertility. Her USG scan is given below. What is her diagnosis?



- a) Dichorionic diamniotic twins
- b) Monochorionic diamniotic twins
- c) Monochorionic monoamniotic twins
- d) Singleton pregnancy with uterine septum

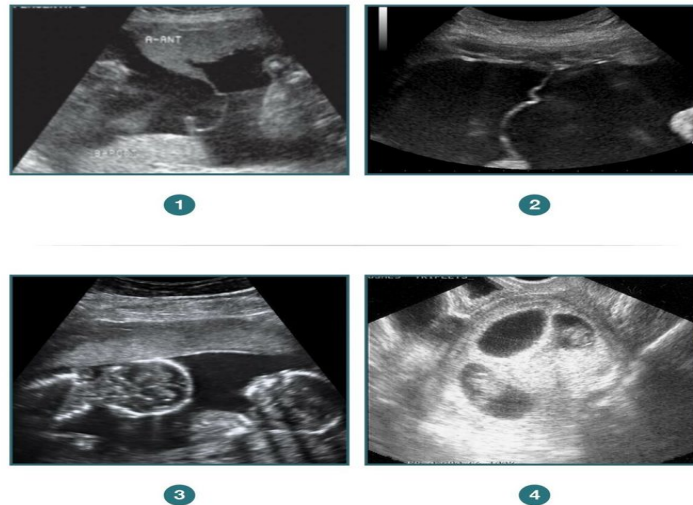
Question 12:

A primigravida presents to the ANC clinic for the first time at 14 weeks of gestation. She complains of excessive nausea and vomiting. On examination, fundal height is midway between pubic symphysis and umbilicus. The T sign is seen on USG. What is her diagnosis?

- a) Dichorionic diamniotic twins
- b) Monochorionic diamniotic twins
- c) Monochorionic monoamniotic twins
- d) Dizygotic twins

Question 13:

Which of the following images depicts monochorionic diamniotic pregnancy?



- a) 3
- b) 4
- c) 1
- d) 2

Question 14:

A mother with twin pregnancy underwent routine USG. Estimated fetal weights were calculated for both twins. What is the criteria to diagnose pathological discordant growth in this pregnancy?

- a) Difference of $\geq 25\%$ with larger twin as index
- b) Difference of $\leq 50\%$ with larger twin as index
- c) Difference of $\geq 50\%$ with smaller twin as index
- d) Difference of $\geq 20\%$ with larger twin as index

Question 15:

In a woman receiving artificial reproductive therapy, the inner cell mass divides on the 7th day. Which of the following complications is not seen in the resulting multifetal pregnancy?

- a) Twin twin transfusion syndrome
- b) Discordant growth
- c) Cord entanglement

d) Acardiac twin

Question 16:

What is the most common type of placental vascular anastomoses seen in monochorionic twin pregnancies?

- a) Artery – vein**
- b) Vein – vein**
- c) Artery – artery**
- d) Both A and B**

Question 17:

A 26-year-old woman, pregnant with twins, is diagnosed to have twin-twin transfusion syndrome. Which of the following statements is true regarding the recipient twin in this condition?

- a) Oliguria is seen**
- b) It is called the stuck twin**
- c) Pulmonary hypoplasia is seen**
- d) Congestive cardiac failure develops**

Question 18:

The ultrasound findings in a monochorionic diamniotic twin pregnancy are indicative of poly-oli syndrome. What is the pathology seen in this condition?

- a) Artery - artery anastomosis**
- b) Vein – vein anastomosis**
- c) Deep artery – vein anastomosis**
- d) Superficial artery - vein anastomosis**

Question 19:

Ultrasound findings in a twin pregnancy are given below:

- a) Stage I**

- b) Stage II
- c) Stage III
- d) Stage IV

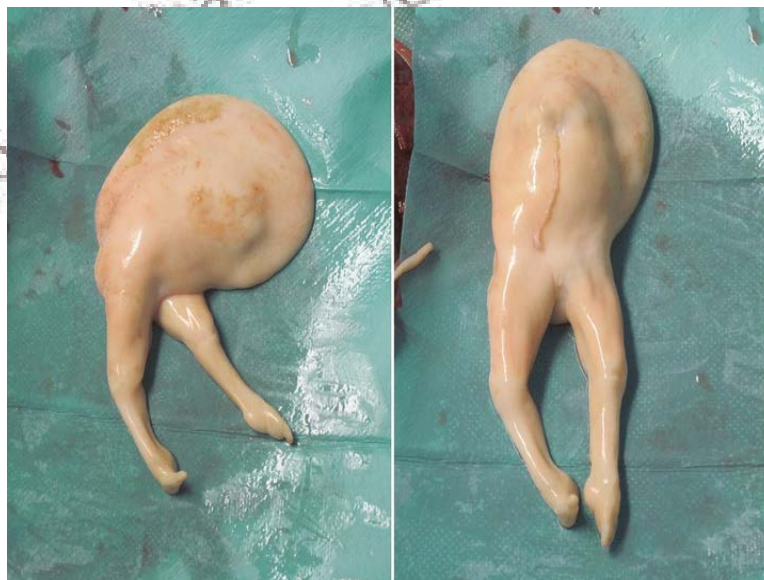
Question 20:

An anxious mother pregnant with twins was recently diagnosed with twin-twin transfusion syndrome. She wants to know the possible fetal outcome of the pregnancy. Which of the following investigations can be used to predict this?

- a) Deepest vertical pocket - USG
- b) Fetal hemoglobin estimation
- c) MPI - ventricle Doppler
- d) Peak systolic velocity - MCA Doppler

Question 21:

What is/are the placental vascular abnormality that can be seen in the condition depicted in the image given below?



- a) Artery – artery anastomosis
- b) Vein – vein anastomosis
- c) Artery – vein anastomosis
- d) Both A and B

Question 22:

A 25-year-old woman, pregnant with twins, goes into labour. Following the vaginal delivery of the first twin, she was taken to the OT for internal version. What could have been the possible indication for this procedure?

- a) 2nd twin breech presentation
- b) 2nd twin knee presentation
- c) 2nd twin transverse lie
- d) 2nd twin shoulder presentation

Question 23:

Identify this condition.



- a) Twin to Twin Transfusion Syndrome
- b) Monochorionic diamniotic pregnancy
- c) Monochorionic monoamniotic pregnancy
- d) Dichorionic diamniotic pregnancy

Question 24:

A pregnant woman was diagnosed with triplets on ultrasound. The obstetrician has planned to do a selective fetal reduction of one of the fetuses to improve the outcome of the pregnancy. What is the ideal timing and the method for the procedure?

- a) 10-13 weeks with intracardiac KCl
- b) 7-9 weeks with intracardiac KCl
- c) 14-16 weeks with intraumbilical KCl
- d) 9-12 weeks with intraumbilical KCl

Question 25:

A 35-year-old primigravida who conceived after in-vitro fertilization therapy came for an antenatal checkup at 38 weeks of gestation. Her obstetric history revealed that she was pregnant with DCDA twins. On examination, the first twin was found to be in a breech position and the maternal blood pressure was found to be greater than 140/90 mm Hg on two occasions with grade 1+ proteinuria. What should be done next?

- a) Monitor BP and terminate pregnancy if BP rises
- b) Immediate Caesarean section
- c) Terminate pregnancy at 40 weeks of gestation
- d) Induce labour with PGE₂ gel

Question 26:

A 26-year-old primigravida with dichorionic diamniotic twins presents to the clinic in active labor. The first twin is delivered vaginally but the second has a transverse lie. A trial of ECV was unsuccessful. What is not a complication of the procedure performed next for the delivery of the second twin?

- a) Face presentation
- b) Uterine atony
- c) Uterine rupture
- d) Vaginal laceration

Answer Key

Question No.	Correct Option
1	b
2	b
3	d
4	b
5	b
6	a
7	b
8	b
9	a
10	c
11	a
12	b
13	d
14	d
15	c
16	a
17	d
18	c
19	c
20	c
21	d
22	c
23	c
24	a
25	b
26	a

Detailed Explanations

Solution to Question 1:

The most common type of multifetal pregnancy is dizygotic twinning, which is always dichorionic diamniotic.

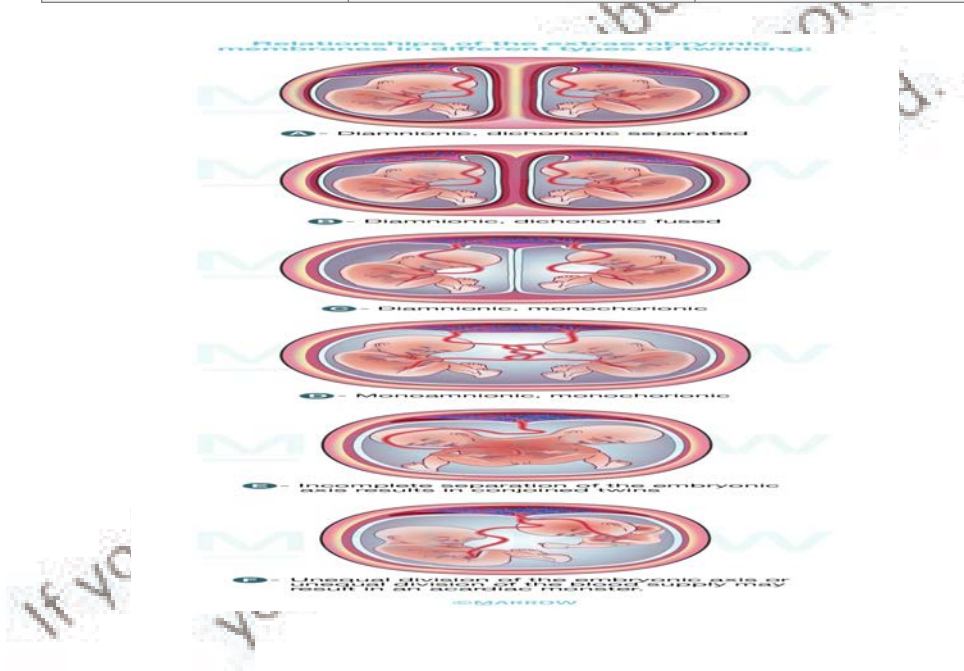
The incidence of dizygotic twins is 80% among twins. Of the monozygotic twins, monochorionic diamniotic is the most common (13-14%).

Zygosity refers to the type of conception while chorionicity refers to the type of placentation.

Types of twin pregnancies:

- Dizygotic: 2 ova are fertilized by 2 sperms to form 2 separate zygotes.
- Monozygotic: Single zygote (1 ovum fertilized by 1 sperm) divides into 2.

Feature	Monozygotic	Dizygotic
Phenotype	Identical	Non-identical
Genotype	Identical	Non-identical
Incidence	1:250 (constant worldwide)	Varies with demography
Incidence after ART	Increases	Increases
Perinatal complications	High	Low



Solution to Question 2:

Diamniotic monochorionic pregnancy occurs if the division of inner cell mass takes place between 4th - 8th day after fertilization.

The timing of the division of zygote for each type of twinning is given below.

Monozygotic twin types	Time of zygote division
Dichorionic diamniotic	<72 hours
Monochorionic diamniotic (more common)	4-8 days
Monochorionic monoamniotic	8-12 days

Monozygotic twin types	Time of zygote division
Conjoined twins	>12 days

Solution to Question 3:

Same sex twins of the same blood group are not always monozygotic but can also be dizygotic twins as well. The genetic relationship between dizygotic twins is the same as any siblings.

Option A: More the number of embryos transferred in IVF, more is the risk of twinning. Hence, single embryo transfer is being recommended by the American Society for Reproductive Medicine (2017) to prevent twin gestation.

Option B: Isograft is done between monozygotic twins as they have HLA homology.

Option C: Twins of the opposite sex are usually dizygotic. Rarely, somatic mutations or chromosomal aberrations can change the sex of 1 or both of the monozygotic twins.

Solution to Question 4:

Thoracopagus is the most common type of conjoined twins.

Division of the inner cell mass after the 12th day of fertilization results in the development of conjoined or Siamese twins. Based on the parts of the body joined, they are of the following types:

- Ventral
- Rostral
- Omphalopagus (abdomen)
- Thoracopagus (upper chest)
- Cephalopagus (head)
- Caudal
- Ischiopagus (conjoined spines at 180° angle)
- Lateral
- Paraphagus (side by side fusion with a shared pelvis)
- Dorsal
- Craniopagus (skull)
- Rachipagus (vertebral arches)
- Pyopagus (pelvis)

Sharing of organs is common among conjoined twins. MRI imaging is done later in pregnancy for the evaluation of organ sharing between conjoined twins who are being planned for surgical separation.



Solution to Question 5:

The incidence of multiple pregnancies is calculated by Hellin's rule.

According to this rule, incidence = $1 : 80(n-1)$, where n is the number of fetuses.

Factors affecting the incidence of twin pregnancies are:

- Monozygotic
- ART
- Dizygotic
- Race
- Heredity
- Maternal age
- Parity
- Family history of twinning
- Good nutritional status
- Infertility therapy

Option A: Tei index or myocardial performance index (MPI) is calculated using ventricular doppler imaging. It is used to assess cardiac function in TTTS.

Option C: CHOP score is used for cardiac evaluation of recipient twin in TTTS.

Option D: Kleihauer-Betke test is used to evaluate the approximate volume of fetomaternal hemorrhage in cases of Rh-D alloimmunization.

Solution to Question 6:

Increasing maternal age, family history of twinning, and conception after stopping OCPs increase the risk of twinning.

Taller, well-built women have an increased risk of twinning compared to shorter undernourished women. IVF increases the risk of twinning but single embryo transfer reduces the risk of multifetal pregnancies.

Factors affecting the incidence of twin pregnancies:

- Monozygotic
- ART
- Dizygotic
- Race
- Heredity
- Maternal age
- Parity
- Family history of twinning
- Good nutritional status
- Infertility therapy

Note- A family history of twinning (maternal and paternal) causes an increased chance of twinning. It is seen that maternal history of twinning has a higher chance of twinning as compared to a paternal history of twinning.

Solution to Question 7:

Superfecundation is the fertilization of two different ova released in the same cycle, by two separate acts of coitus within a short period of time.

Superfecundation is the fertilization of two different ova in the same menstrual cycle (as suggested by the same gestational age) by separate acts of coitus. It can occur even after artificial reproductive therapy. Hence, after embryo transfer, intercourse must be avoided. Heteropaternal superfecundation is used to refer to instances of two different males fathering fraternal twins. This leads to the possibility of twins also being half-siblings. This may give rise to the doubt of adultery and infidelity.

Superfetation is the fertilization of two ova released in different menstrual cycles. This is not seen in humans. The nidation and development of one fetus over another fetus is theoretically possible until the decidual space is obliterated by 14-16 weeks of pregnancy.

Solution to Question 8:

The above image shows fetus compressus which appears as a tan ovoid mass compressed against the fetal membranes. It can be detected early by sonography.

In multifetal pregnancies, if one of the fetuses dies early in pregnancy, the surviving fetus continues to develop normally. It can lead to the following outcomes:

Factors affecting the prognosis of the surviving twin:

- Gestational age at fetal death: 2nd and 3rd trimester deaths increase the risk of morbidity in the surviving twin
- Chorionicity: Monochorionic twins with early fetal death have an increased risk of morbidity in the surviving twin. This is due to neurological abnormality and vascular anastomoses.
- Duration between fetal demise and delivery of surviving twin
- Surviving twin may be born preterm leading to an increased risk of fetal demise

Death in the surviving twin occurs mostly due to vascular anastomoses between the twins. The surviving twin can develop sudden hypotension due to these anastomoses after fetal demise.

Outcome	Type
Disappears	Vanishing twin
Gets compressed	Fetus compressus
Flattened through desiccation	Fetus papyraceus

Solution to Question 9:

The most common presentation in twins is both vertex/ cephalo-cephalic presentation (60%). The other presentations include:

- Cephalo-Breech (20%)
- Breech-Cephalic (10%)
- Breech-Breech (6-10%)
- Transverse-Transverse (Rarest)

Mode of delivery depends on the presentation of the 1st twin as:

Presentation of the 1st twin	Mode of delivery
Vertex	Vaginal
Breech	Caesarean section
Transverse	Caesarean section

Solution to Question 10:

The best time to identify the type of twin pregnancy on ultrasound is 11-14 weeks.

Twin pregnancies should ideally be dated when the crown–rump length (CRL) measurement is between 45 and 84 mm (i.e. 11 + 0 to 13 + 6 weeks of gestation).

Chorionicity should be ideally determined between 11 + 0 and 13 + 6 weeks of gestation (the earliest time to identify chorionicity is 7 weeks). This is done using the membrane thickness at the site of insertion of the amniotic membrane into the placenta. The T sign (in monochorionic twins) or lambda sign (in dichorionic twins) can be seen here. As the determination of chorionicity is most accurate when the amnion and chorion have not yet been fused, the first trimester scan in twin pregnancy is highly important.

Markers of dichorionicity include 4 layers of membranes between the twins, membrane thickness ≥ 2 mm, different sex of twins, 2 different placentas, and positive twin peak sign.

Women with an uncomplicated dichorionic twin pregnancy should have a first trimester scan, a detailed second trimester scan, and scans every 4 weeks thereafter.

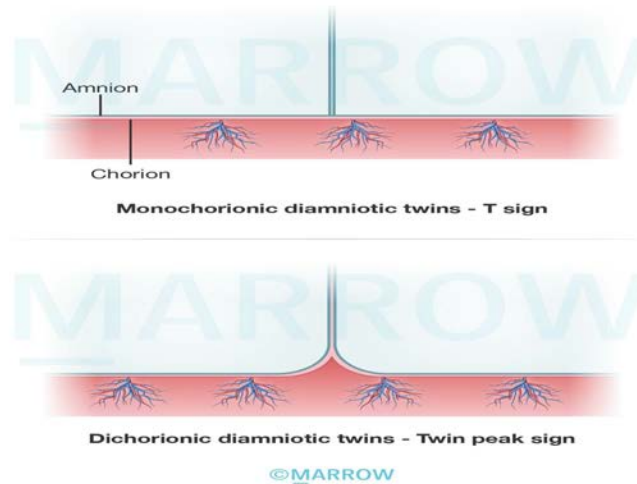
Women with uncomplicated monochorionic twins should have a first trimester scan and then scans every 2 weeks from 16 weeks onwards. This is done in order to detect TTTS and TAPS in a timely manner.

Solution to Question 11:

The given image shows the twin peak sign which is seen in dichorionic diamniotic twins. This is also known as the lambda sign or delta sign.

In dichorionic twins, the intervening membrane between the twins has 4 layers i.e. 2 chorions and 2 amnions. The placenta protrudes from the base to the space between the 2 amnion layers. This gives rise to the twin peak sign i.e. triangle with the base at placenta and apex in the intervening membrane.

USG findings in Twin Pregnancy



Sonographic assessment of dichorionicity is done by observing the following:

- Number of placental masses - 2
- Presence of intervening membrane - twin peak or lambda or delta sign
- Thickness of intervening membranes $\geq 2\text{mm}$

The best time of delivery of dichorionic diamniotic twins is 38 weeks.

Solution to Question 12:

The given clinical scenario and the T sign seen on USG points towards a diagnosis of monochorionic diamniotic twins.

The T sign is formed at the point at which the amnions meet the placenta. In monochorionic diamniotic twins, the separating membrane is made up only by the two amniotic membranes with no intervening chorion. This can be seen in the latter part of the first trimester or the beginning of the second trimester.

Symptoms of multifetal pregnancy:

- Increased nausea and vomiting in early months
- Palpitations and shortness of breath
- Increased swelling of legs, varicose veins and haemorrhoids
- Unusual rate of uterine growth
- Excessive fetal movements

Signs of multifetal pregnancy:

- Increased weight gain
- Hypertension

- Barrel shaped abdomen
- Increased fundal height and abdominal girth
- Multiple fetal parts
- Palpation of 2 fetal heads
- FHS at different spots with FHR difference of at least 10bpm

Monochorionic diamniotic pregnancy is terminated between 34-37 weeks+6 days.

The below image shows the T sign.

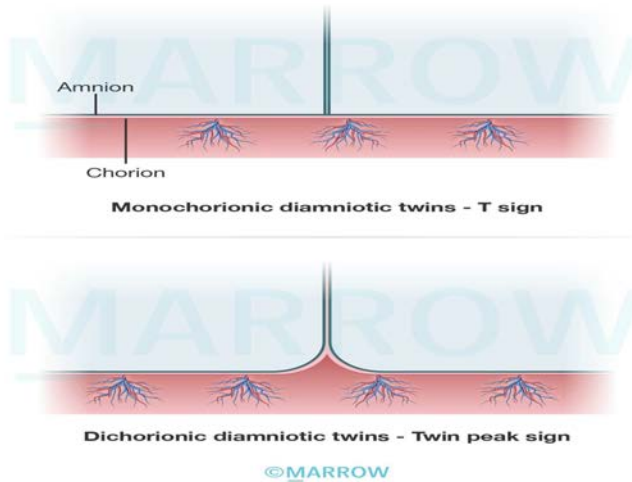


Solution to Question 13:

The USG in image 2 depicts monochorionic diamniotic pregnancy. It is identified by the T sign.

In monochorionic diamniotic twins, the intervening membrane has only 2 layers (i.e) 2 amnions. This thin membrane is seen ending at the common placenta giving rise to the T sign.

USG findings in Twin Pregnancy



Sonographic assessment of monochorionicity is done by observing the following:

- Single placental mass
- Presence of intervening membrane - T sign
- Thin intervening membrane $< 2\text{mm}$

The best time of delivery of monochorionic diamniotic twins is 34 - 37 weeks + 6 days.

Image 1: Dichorionic twins with twin peak or lambda sign

Image 3: Monochorionic monoamniotic twins with no intervening membranes between them

Image 4: Triplets with 3 separate gestational sacs

Solution to Question 14:

A difference in estimated fetal weight of $> 20\%$ with larger twin as index is one of the criteria for diagnosing discordant growth.

Discordant growth is the unequal size/ weight among the fetuses in a multifetal gestation. It can be diagnosed by either of the following:

- Estimated fetal weight difference $> 20\%$
- Discordance = $(\text{weight of larger twin} - \text{weight of smaller twin} / \text{weight of larger twin}) \times 100$

Discordant growth in monochorionic twins is due to placental vascular anastomoses. In dichorionic twins, it occurs due to differences in genetic growth potential and suboptimal implantation site for one of the fetuses.

Solution to Question 15:

The division of the inner cell mass on the 7th day will result in monochorionic diamniotic type of twin pregnancy. Cord entanglement is seen in monoamniotic twins and not in diamniotic twins.

The timing of the division of zygote for each type of twinning is given below.

Complications in monochorionic diamniotic twins occur due to placental vascular anastomoses. They include:

- Twin-twin transfusion syndrome (TTTS)
- Twin reversed arterial perfusion (TRAP) or acardiac twin
- Twin anaemia polycythemia syndrome (TAPS)

Monochorionic monoamniotic twins have a high fetal death rate. This is due to the following complications:

- Preterm birth
- Congenital anomalies
- Low birth weight
- Cord entanglement

Monozygotic types	Time of zygote division
Dichorionic diamniotic	<72 hours
Monochorionic diamniotic (more common)	4-8 days
Monochorionic monoamniotic	8-12 days
Conjoined twins	>12 days

Solution to Question 16:

Artery-vein anastomoses are most commonly seen in monochorionic pregnancies.

All monochorionic placentas have anastomoses that can be:

- Superficial artery-vein
- Deep artery-vein
- Artery-artery
- Vein-vein
- One vessel with connections to multiple arteries and veins

These anastomoses are not functional as long as the fetuses are hemodynamically balanced. When a pressure gradient develops between the fetuses, complications can occur like:

- TTTS - Twin-twin transfusion syndrome
- TAPS - Twin anemia polycythemia sequence

- TRAP - Twin reversed arterial perfusion

The most common complication in monochorionic twins is twin-twin transfusion syndrome (TTTS). Deep artery-vein anastomosis is the pathology seen in this condition.

Solution to Question 17:

Congestive heart failure is seen in the recipient twin.

Twin-twin transfusion syndrome is a deep arteriovenous anastomotic complication seen in monochorionic diamniotic twins. Deoxygenated blood from the donor enters through the anastomosis into a shared cotyledon for oxygen exchange. Oxygenated blood is taken up by the recipient twin causing imbalance in the blood volumes.

Diagnosis of TTTS depends on 2 USG criteria:

- Monochorionic diamniotic gestation
- Hydramnios (DVP > 8cm) in one twin and oligohydramnios (DVP < 2cm) in the other twin

Treatment modalities for TTTS:

- Serial amnioreduction
- Laser ablation of vascular placental anastomoses
- Selective fetal termination

Note: Donor twin is called the stuck twin as the oligohydramnios limits fetal movement.

Donor twin characteristics	Recipient twin characteristics
Oliguria	Polyuria
Oligohydramnios	Polyhydramnios
Hypotension	Hypertension
Anaemia	Polycythemia
Activation of renin-angiotensin-aldosterone system (RAAS)	Activation of atrial natriuretic peptide (ANP)
Renal failure, growth restriction, pulmonary hypoplasia, contractures	Congestive cardiac failure, premature rupture of membranes

Solution to Question 18:

In poly - oli or polyhydramnios - oligohydramnios syndrome, the basic pathology is the presence of a deep artery-vein anastomosis.

TTTS can present as poly-oli syndrome or twin oligohydramnios polyhydramnios syndrome (TOPS). This is usually seen in mid-pregnancy. In this syndrome, there is decreased donor renal perfusion, causing oliguria and oligohydramnios. Because of the anastomosis, the recipient twin

has increased renal perfusion and increased urine production causing polyhydramnios.

Donor twin characteristics	Recipient twin characteristics
Oliguria	Polyuria
Oligohydramnios	Polyhydramnios
Hypotension	Hypertension
Anemia	Polycythemia
Renal failure	Congestive cardiac failure
Growth restriction	Premature rupture of membranes
Pulmonary hypoplasia	Hyperbilirubinemia
Contractures	

Solution to Question 19:

The given USG findings are indicative of twin-twin transfusion syndrome (TTTS) with twin A being the recipient and twin B being the donor twin. Oligohydramnios (AFI<5cm), absent urine and abnormal Doppler flow findings in the donor twin indicate stage III of twin-twin transfusion syndrome according to Quintero staging.

Quintero staging system for TTTS classification

Stage	USG findings
I	Discordant amniotic fluid volumes + urine visible in donor's bladder
II	Discordant amniotic fluid volumes + urine not visible in donor's bladder
III	Stage II + abnormal Doppler studies of umbilical artery/vein or ductus venosus
IV	Ascites or frank hydrops in either twin
V	Death of either twin

Solution to Question 20:

In twin-twin transfusion syndrome (TTTS), the fetal outcome depends on the cardiac function of the recipient twin. Tei index or myocardial performance index (MPI) is used to assess the

ventricular function using doppler.

Option A: Deepest vertical pocket on USG is used to diagnose polyhydramnios-oligohydramnios in TTTS.

Options B and D: Fetal haemoglobin and MCA peak systolic velocity are used to diagnose TAPS (Twin anaemia polycythemia syndrome).

Solution to Question 21:

The given image shows an acardius acephalus fetus. This refers to the acardiac twin seen in twin reversed arterial perfusion (TRAP). Both artery-artery and vein-vein anastomosis can be seen in TRAP.

TRAP is a placental vascular anastomotic complication seen in monochorionic twins. In this disorder, deoxygenated blood from the donor twin enters the recipient twin through the anastomosis and supplies the lower half of the recipient's body. Hence, the heart and upper body of the recipient is poorly developed.

The below image depicts the circulation pattern seen in TRAP.



Based on the supply of deoxygenated blood, the recipient twin might have the following fates:

The donor twin develops high output cardiac failure.

The treatment for this condition is radiofrequency ablation of the acardiac or recipient twin.

Fate	Type
Lower body is perfused with deoxygenated blood, leading to failed head growth	Acardius acephalus
Partially developed head with identifiable limbs	Acardius myelecephalus

Fate	Type
No recognizable structure	Acardius amorphous

Solution to Question 22:

The second twin being in transverse lie is the only indication for internal version in modern obstetrics.

In a twin pregnancy, if the first twin has vertex presentation, vaginal delivery can be performed. Following the vaginal delivery of the first twin, if the second twin is in transverse lie, it is corrected by external version (preferably cephalic) into a longitudinal lie. If external version fails and the membranes are intact, internal version is done under general anaesthesia, followed by breech extraction.

If the membranes are ruptured, caesarean section is done immediately.

Mode of delivery depends on the presentation of the 1st twin.

Presentation of the 1st twin	Mode of delivery
Vertex	Vaginal
Breech	Caesarean section
Transverse	Caesarean section

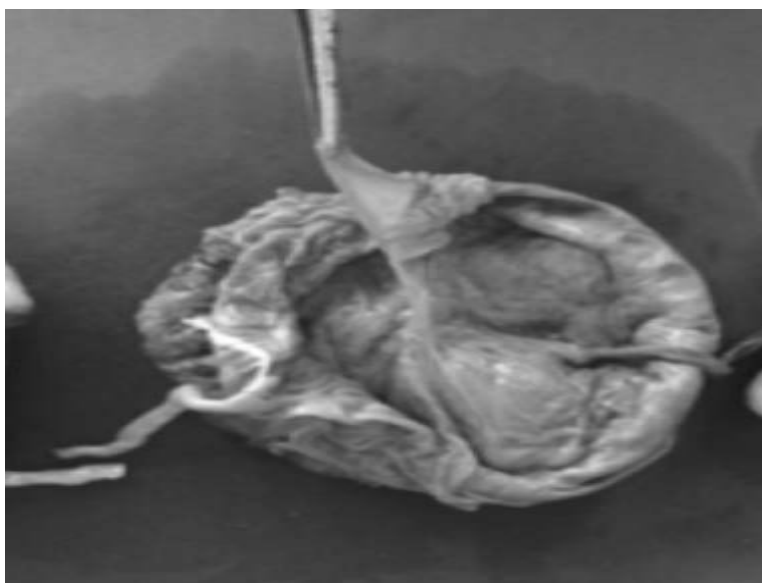
Solution to Question 23:

The given specimen depicts a monochorionic monoamniotic pregnancy as:

- 2 umbilical cords are arising from 1 placenta (monochorionic)
- No intervening amniotic membrane is attached to the placenta. This means that both fetuses must have been inside a single amniotic cavity (monoamniotic).

Option B: In Monochorionic diamniotic pregnancy, on gross examination, there will be 1 placenta and intervening 2 layers of amnion attached to the placenta. In this type of pregnancy, each fetus will have its own amniotic cavity (diamniotic) with a shared placenta (monochorionic). This is seen if the fertilized zygote divides into 2 between 4-8 days.

The below given gross image of monochorionic diamniotic pregnancy shows two umbilical cords arising separately from the placenta with the intervening amniotic membrane held by the forceps.

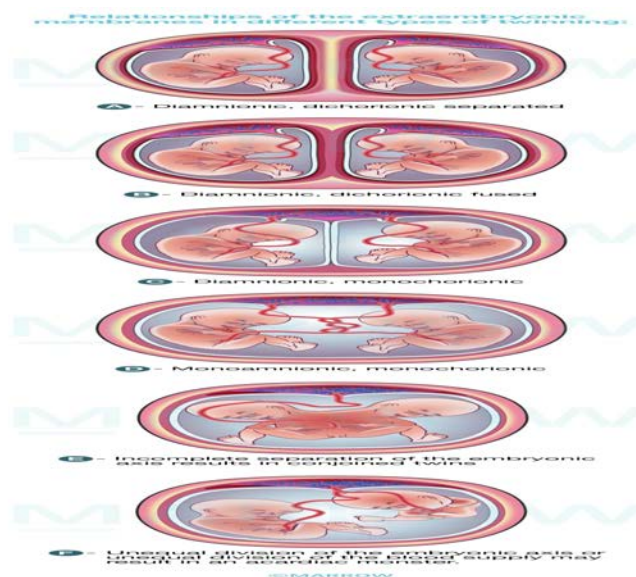


Option D: In dichorionic diamniotic pregnancy each fetus has it's own amnion & chorion. The placentas here may be fused or separate. This is seen in dizygotic twins as well as monozygotic twins that divide within 0-4 days.

Option A: Twin to twin transfusion syndrome (TTTS) most commonly complicates monochorionic (single placenta) twin pregnancy. In TTTS blood is transfused from the donor twin to the recipient twin through the abnormal vascular connections within the placenta leaving the donor anemic while the recipient becomes polycythemic.

The box below describes the type of amnion and chorion seen in different types of twins depending on the time of division.

Dizygotic twins	Dichorionic diamniotic
Monozygotic twins- division occurs within 0-4 days 4-8 days 8-12 days >13 days	Dichorionic diamniotic Monochorionic diamniotic Monochorionic monoamniotic Monochorionic monoamniotic conjoined twins



Solution to Question 24:

Selective fetal reduction is done between 10-13 weeks via intracardiac injection of KCl.

In multi-chorionic multifetal gestations, selective reduction of the smallest or anomalous fetus/fetuses is done to increase the chance of survival of the other fetuses.

Laser ablation or bipolar coagulation of the umbilical cord of the targeted fetus is done. This protects the surviving fetuses from the adverse effects of the drug.

Solution to Question 25:

Immediate cesarean section is done as the first twin is in breech presentation. Uncomplicated dichorionic diamniotic twin pregnancies are terminated at 38 weeks of gestation.

Timing of delivery depends on the type of twin pregnancy:

The mode of delivery depends on the 1st twin:

After delivery of 1st twin, if the 2nd twin is in breech, assisted breech delivery is done. If the 2nd twin is in a transverse lie, the internal podalic version is done under GA in the OT followed by breech extraction.

Complications in attempting vaginal delivery in non-vertex presentation (i.e) transverse and breech:

- Interlocking of twins
- Fetal head trapping above the cervix
- Umbilical cord prolapse

According to ACOG criteria (2019), preeclampsia requires proteinuria of grade 2+ or more. In the case of pre-eclampsia, termination of pregnancy must be done at 37 weeks. According to the updated criteria, this patient does not have pre-eclampsia. Hence, a Caesarean section can be done at 38 weeks as for any uncomplicated DCDA twin pregnancy.

Type of twin pregnancy	Timing of delivery
Dichorionic diamniotic	38 weeks
Monochorionic diamniotic	34 - 37 weeks + 6 days
Monochorionic monoamniotic	32 - 34 weeks by Caesarean section

1st twin presentation	Mode of delivery
Vertex	Vaginal
Breech	Cesarean section
Transverse	Cesarean section

Solution to Question 26:

The given clinical procedure is suggestive of internal podalic version for the second twin. Shoulder presentation, and not face presentation, is a complication of this procedure.

The complications seen with internal podalic version are as follows:

- Maternal-
 - Failure of the procedure
 - Uterine rupture
 - Vaginal and perineal trauma
 - Placental abruption
 - Postpartum haemorrhage
 - Ascending infection
 - Anaesthetic risks
- Fetal-
 - Cord prolapse
 - Shoulder presentation with prolapsed arm
 - Skeletal injury (femur, hips, humerus, skull)
 - Visceral injury (kidney, liver, spleen)

- Neural injury (facial nerve)
- Hypoxic brain injury
- Death

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Ectopic Pregnancy

Question 1:

Arrange the following sites of ectopic pregnancy from the most common to the least common.

- a) 3>2>4>1
- b) 1>2>3>4
- c) 2>4>1>3
- d) 3>1>4>2

Question 2:

Which of the following factors maximally increases the risk of tubal pregnancy?

- a) Intrauterine device
- b) Pelvic Inflammatory disease
- c) Prior tubal surgery
- d) Artificial reproductive technique

Question 3:

The fall in which of the following hormone/s leads to endometrial sloughing in ectopic gestation?

- a) hCG
- b) Estrogen
- c) Estrogen and Progesterone
- d) Progesterone and hCG

Question 4:

Which type of tubal pregnancy ruptures at a later stage?

- a) Ampullary
- b) Interstitial

- c) Fimbrial
- d) Isthmic

Question 5:

Which of the following clinical features is not seen in a patient with ruptured ectopic pregnancy?

- a) Shoulder tip pain
- b) Decidual cast
- c) Vertigo and syncope
- d) Shock index of 0.7

Question 6:

A woman presented with complaints of amenorrhoea for 6 weeks and a history of prior ectopic pregnancy. Serial serum beta hCG levels were estimated in her. Which of the following findings will increase the suspicion of ectopic gestation in her?

- a) Low levels with prolonged doubling time
- b) High levels with decreased doubling time
- c) Undetectable
- d) Double the expected value

Question 7:

A patient presented with 6 weeks of amenorrhoea and a positive UPT. TVS revealed a tubal ectopic pregnancy. Which of the following findings can be usually picked up on colour Doppler?

- a) Enhanced blood flow pattern in the uterine artery
- b) Arias stella reaction
- c) Ring of fire pattern
- d) Whirlpool sign

Question 8:

A pregnant woman came for routine antenatal USG. Which of the following features is suggestive of a pseudogestational sac?

- a) Presence of yolk sac
- b) Intradecidual sac sign
- c) Central location
- d) Double decidual sign

Question 9:

A patient with suspected tubal pregnancy underwent transvaginal ultrasound which was inconclusive. What is the next step of management?

- a) Laparotomy
- b) Diagnostic laparoscopy
- c) Serum β hCG levels
- d) Methotrexate therapy

Question 10:

A pregnant woman is diagnosed with unruptured ectopic gestation. Which of the following is not an indication for expectant management in this patient?

- a) Initial serum beta hCG level of 200 mIU/mL
- b) Gestational sac size of 5 cm
- c) Asymptomatic
- d) Absence of fetal cardiac activity

Question 11:

A primigravida in her first trimester came for a routine ANC. USG findings revealed an empty uterine cavity. Serum beta hCG was measured and was found to be 1700 IU/mL. All of the following drugs can be used in managing this patient except:

- a) Potassium chloride
- b) Methotrexate
- c) Actinomycin
- d) Oxytocin

Question 12:

A patient with unruptured ectopic pregnancy is planned for medical management with methotrexate. What is the multi-dose protocol for this drug?

- a) 1 mg/kg on days 1, 2, 5, and 8
- b) 0.1 mg/kg on days 1, 3, 5, and 7
- c) 0.1 mg/kg on days 1, 4, 6, and 8
- d) 1 mg/kg on days 1, 3, 5, and 7

Question 13:

Which of the following must be avoided by a patient with tubal pregnancy on methotrexate?

- a) 1, 3, 5
- b) 1, 4, 5
- c) 2, 3, 5
- d) 1, 2, 4

Question 14:

A 30-year-old woman was brought to the casualty in a drowsy state with complaints of bleeding PV associated with abdominal pain and a history of amenorrhoea for 8 weeks. On examination, BP was 90/60 mmHg, PR was 140 bpm, and UPT was positive. What is the next step of management?

- a) Admit to the ICU
- b) Resuscitation and emergency laparotomy
- c) Suction and evacuation
- d) FAST followed by laparoscopic salpingectomy

Question 15:

Match the following criteria used for the diagnosis of ectopic pregnancy:

- a) 1-A, 2-C, 3-B
- b) 2-B, 3-A, 4-C

- c) 1-B, 2-C, 4-A
- d) 1-C, 2-A, 3-B

Question 16:

An O-ve primigravida underwent surgery for ectopic pregnancy at 13 weeks of gestation. What will be the dose of anti-D immunoglobulin to be given to her?

- a) 300 µg
- b) 75 µg
- c) 100 µg
- d) 150 µg

Question 17:

A 30-year-old female patient (G2PoA1) with a history of FSH therapy and in vitro fertilization presented with lower abdominal pain, vomiting, and vaginal discharge. On examination, she was hemodynamically stable. Transvaginal ultrasound showed a viable intrauterine pregnancy of 8 weeks associated with a heterogeneous right adnexal mass showing a positive bagel sign. What is the most appropriate management?

- a) Methotrexate
- b) MTP
- c) Surgery
- d) Expectant management

Question 18:

Methotrexate was administered to a patient with unruptured ectopic pregnancy. At what serum beta hCG levels will this drug be most effective?

- a) <7000 IU/L
- b) <5000 IU/L
- c) <6000 IU/L
- d) <10000 IU/L

Question 19:

Studdiford's criteria is used for the diagnosis of _____

- a) Ovarian pregnancy
- b) Cervical pregnancy
- c) Tubal pregnancy
- d) Abdominal pregnancy

Answer Key

Question No.	Correct Option
1	a
2	c
3	d
4	b
5	d
6	a
7	c
8	c
9	c
10	b
11	d
12	d
13	d
14	b
15	c
16	a
17	c
18	b
19	d

Detailed Explanations

Solution to Question 1:

The order of the most common site to the least common site in the fallopian tube for ectopic pregnancy is ampulla>isthmus>fimbriae>interstitium.

An ectopic pregnancy is any pregnancy implanted outside of the endometrial cavity.

Extrauterine sites: The most common site of ectopic pregnancy is the extrauterine fallopian tube.

- Tubal - 97% (most common)
- Ampulla - 55%
- Isthmus
- Infundibulum
- Interstitium
- Ovarian
- Abdominal

Intrauterine ectopic sites:

- Cervical
- Caesarean scar
- Rudimentary horn of a bicornuate uterus
- Heterotopic pregnancy

Solution to Question 2:

Prior tubal surgery for ectopic pregnancy has the highest risk of tubal pregnancy among the given options.

Risk factors for ectopic pregnancy:

- Prior ectopic pregnancy
- Prior tubal surgery for ectopic, recanalization of tube post-sterilization
- Pelvic inflammatory disease - most common cause for ectopic pregnancy
- Infertility
- Artificial reproductive therapy
- Endometriosis
- Smoking
- Maternal age >40 years
- Contraceptive methods - intrauterine device, progesterone-only pills

Solution to Question 3:

Endometrial sloughing in ectopic gestation is due to the fall in the levels of progesterone and hCG. During normal pregnancy, endometrium is maintained by the high levels of hCG and progesterone. In ectopic pregnancy, a fall in the levels of these hormones will cause endometrial sloughing and uterine bleeding. If the decidua is expelled as a single piece through the cervix, it is called a decidual cast.

Solution to Question 4:

Interstitial tubal pregnancy ruptures at a later stage i.e. at 12-16 weeks. This is because pregnancy at this site is closer to the uterine myometrium which provides support as pregnancy progresses.

Isthmic and ampullary pregnancies terminate by tubal rupture. However, isthmic pregnancy ruptures earlier. Ampullary and infundibular pregnancy can terminate by tubal abortion as well.

Solution to Question 5:

A shock index of >0.85 is seen in a case of ruptured ectopic pregnancy. There is massive blood loss (SBP <110 mmHg) sometimes leading to shock.

The shock index is calculated as HR/systolic BP.

Option A: Sub-diaphragmatic blood irritates the phrenic nerve. This causes a referred pain at the shoulder tip. This is called as Danforth sign.

Option B: The expulsion of decidua in a single piece through the cervix is called a decidual cast. This is due to the fall in the levels of β hCG and progesterone.

Option C: Vertigo and syncope are due to reflex vasomotor disturbances due to peritoneal irritation from hemoperitoneum.

Condition	Range of shock index
Normal value of shock index (non-pregnant)	0.5 - 0.7
Normal value of shock index in pregnancy (i.e) obstetrical shock index	0.7 - 0.8
Significant or massive blood loss	0.9 - 1.1

Solution to Question 6:

Low levels of β hCG and prolonged doubling time increase the suspicion of ectopic pregnancy.

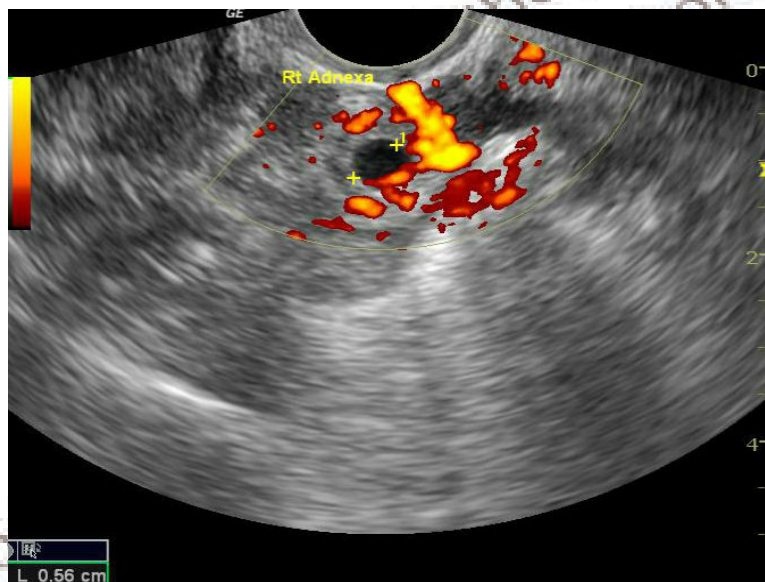
Urine pregnancy test is very sensitive and found positive in most ectopic pregnancies. Hence, serial serum β hCG levels are estimated to detect ectopic pregnancy. In early normal pregnancy, the doubling time of β hCG concentrations in plasma is 1.4 to 2 days.

Solution to Question 7:

Ring of fire pattern is seen on color doppler in a tubal ectopic pregnancy. This is due to placental blood flow in the tubal pregnancy.

This finding may also be seen in corpus luteum cysts but the wall is less echogenic compared to tubal ectopic.

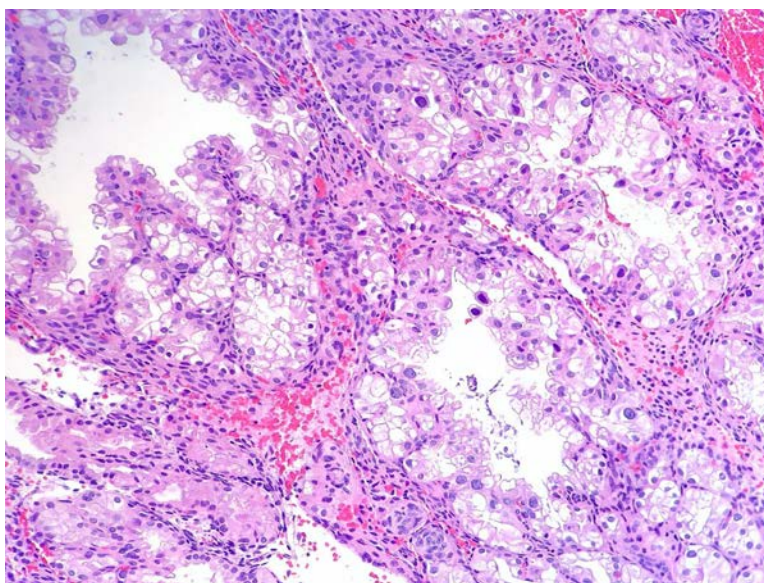
Option A: There is an enhanced blood flow pattern outside the uterine cavity in case of ectopic pregnancy.



Option B: Arias-Stella reaction/phenomenon is a histological finding. It is an atypical endometrial change due to the action of progesterone on the endometrial tissue seen in normal, ectopic, and molar pregnancies.

On histology, there are:

- large cells with abundant eosinophilic or vacuolated cytoplasm
- nuclear enlargement
- hyperchromasia



Option D: Whirlpool sign refers to the twisted ovarian pedicle seen on USG of ovarian torsion.

Solution to Question 8:

Central location is a feature of the pseudogestational sac seen in ectopic pregnancies.

Pseudogestational sac is a collection of fluid between the endometrial layers, surrounded by a thick decidua. Its presence increases the suspicion of ectopic pregnancy.

Differences between pseudogestational sac and true gestational sac:

Feature	Pseudogestational sac	True gestational sac
Location within the endometrial cavity	Central	Eccentric
Shape	Usually ovoid	Usually round
Fetal and yolk sac poles	Absent	Present
Double ring sign	Absent	Seen
Intradecidual sign	Absent	Seen

Solution to Question 9:

In the given scenario, the next step would be to confirm the tubal ectopic by measuring serum β hCG. This can be used to differentiate tubal ectopic pregnancy from missed abortion and early intrauterine pregnancy.

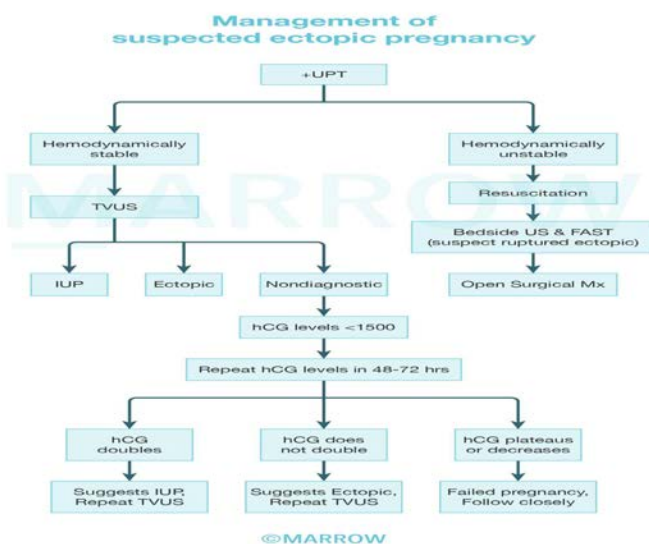
If the level is ≥ 2000 IU/L, ectopic pregnancy is suspected. If the level is < 1500 IU/L, a repeat serum β hCG is taken after 48 hours. If the levels are:

- Doubled - early intrauterine pregnancy
- Decreased - abortion
- Increased but less than doubled value - ectopic pregnancy

Investigations for ectopic pregnancy:

- Serum β hCG - lower levels compared to normal pregnancy, prolonged doubling time
- Transvaginal ultrasound - empty uterine cavity, fluid in pouch of Douglas, adnexal mass
- Colour Doppler - ring of fire pattern
- Laparoscopy
- Dilatation and curettage - decidua identified without villi
- Serum progesterone
- Culdocentesis

Diagnostic laparoscopy may be done when the diagnosis is still uncertain after serial β -hCG monitoring and repeat ultrasonography. In high-risk situations, such as a history of previous ectopic pregnancy, earlier diagnostic laparoscopy is considered.



Solution to Question 10:

A gestational sac size of < 4 cm is an indication of expectant management.

Expectant management is done only in the case of unruptured tubal pregnancy. In this, observation is done till there is spontaneous resolution. It can be done but is not preferred due to prolonged follow-up and uncertainty in resolution.

The indications are:

- Initial serum β hCG <1500 IU/L
- Falling serum β hCG levels
- Gestational sac size <4 cm
- No fetal cardiac activity on TVS
- No evidence of rupture or bleeding on TVS

Solution to Question 11:

Oxytocin is not used in the management of unruptured ectopic pregnancy.

Indications for medical management of unruptured tubal ectopic pregnancy:

- Hemodynamically stable patient
- Serum β hCG <5000 IU/L
- Tubal diameter <4 cm
- No fetal cardiac activity
- No intra-abdominal haemorrhage

Medical management of unruptured tubal ectopic:

- Direct local therapy (laparoscopy or USG guidance)
- Methotrexate
- Potassium chloride
- Actinomycin
- Hyperosmolar glucose
- PGF₂^o
- Systemic therapy - single dose of methotrexate 50 mg/m² IM

Follow-up after medical management is done by measuring the decline in serial serum β hCG.

Solution to Question 12:

In the multi-dose protocol, methotrexate is given at a dose of 1 mg/kg as an IM injection on alternate days 1, 3, 5, and 7.

This regimen is continued on alternating days until the hCG level drops by 15% over two days. Up to 4 doses may be given. Along with this, leucovorin rescue (folinic acid) is given at 0.1 mg/kg after 24 hrs (on days 2, 4, 6, and 8) to buffer the side effects of methotrexate.

Note: If the decline is $\geq 15\%$, the patient is followed up weekly till the levels fall to <10 IU/L.

MTX Protocol	Multi-dose regimen	Single-dose regimen
Dose	1 mg/kg(+ 0.1 mg/kg LEU)	50 mg/m ² (Body surface area)
Regimen	Alternate daily doses	Day 0
hCG measurement	Days 0, 1, 3, 5, 7	Days 0, 4, 7
Treatment success	hCG declines >15%from the previous value	hCG declines >15%between Day 4 and 7
When to administer additional dose	2nd, 3rd or 4th dose to be given if hCG does not decline 15 % from the previous value. Maximum 4 doses	Second dose on Day 7 if hCG does not decline 15%

Solution to Question 13:

Sunlight exposure, sexual intercourse and NSAIDs must be avoided when patients are on methotrexate therapy.

The following should be avoided when methotrexate therapy has been initiated for the treatment of ectopic gestation:

- Nonsteroidal anti-inflammatory drugs - can cause reduced renal blood flow and delay drug excretion
- Alcohol - can predispose to concurrent hepatic enzyme elevation
- Sunlight - may provoke methotrexate-related dermatitis
- Sexual intercourse - can rupture the ectopic pregnancy
- Phenytoin, tetracyclines, salicylates, and sulfonamides - can increase serum methotrexate levels

Solution to Question 14:

The given clinical scenario is suggestive of ruptured ectopic pregnancy which must be immediately managed with simultaneous resuscitation and laparotomy. On laparotomy, salpingectomy is the definitive surgery.

Diagnosis of ruptured ectopic pregnancy is made based on clinical features:

- Amenorrhoea for 6-8 weeks followed by bleeding PV
- Abdominal pain
- Hemodynamically unstable
- Urine pregnancy test - positive

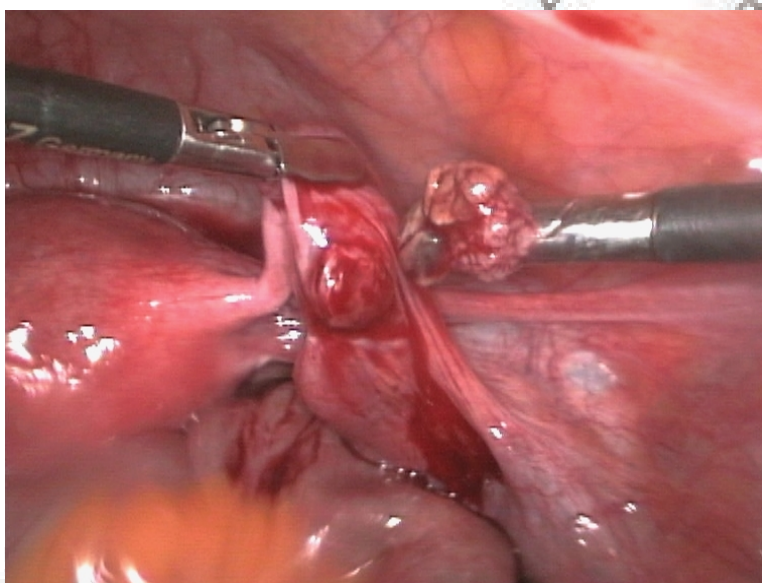
Surgical treatment of ectopic pregnancy includes salpingectomy or salpingostomy, performed either through laparoscopy or laparotomy.

Management of suspected ruptured ectopic pregnancy consists of initial resuscitation and stabilization, followed by point-of-care ultrasound (FAST) to identify intraperitoneal free fluid and surgical exploration.

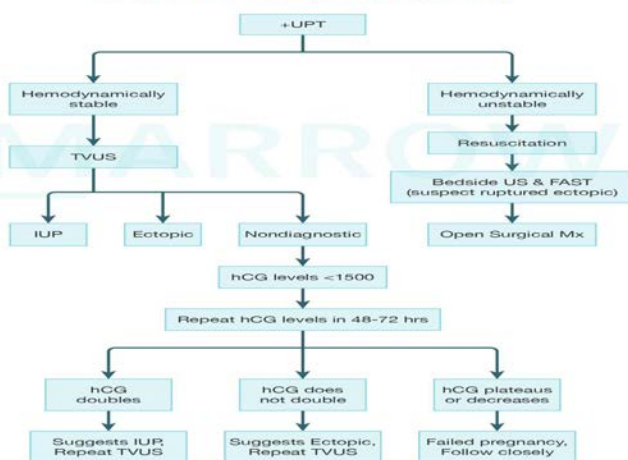
Laparotomy is done for hemodynamically unstable patients with ruptured ectopic pregnancies. Salpingectomy is the treatment of choice and is done with preservation of the ipsilateral ovary and its vascular supply.

For patients wishing to preserve future fertility, if the contralateral fallopian tube is damaged, salpingostomy is preferred. However, salpingostomy may result in inadequate evacuation of the products of conception and a recurrence of symptoms. Therefore, after a patient undergoes salpingostomy, it is important to check the β -hCG level on a weekly basis to ensure that it reaches zero.

The image given below shows a laparoscopic visualization of a tubal pregnancy.



Management of suspected ectopic pregnancy



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Solution to Question 15:

Spiegelberg's criteria used for the diagnosis of ovarian pregnancy include intact ipsilateral tube, pregnancy occupying ovary connected by the utero-ovarian ligament.

Studdiford's criteria used for the diagnosis of primary abdominal pregnancy include the presence of normal bilateral tubes and pregnancy related to the peritoneal surface.

Rubin's criteria used for the diagnosis of cervical pregnancy includes presence of soft, enlarged cervix and closed internal os and partially opened external os.

Ectopic pregnancy	Criteria
Abdominal	Studdiford
Cervical	Rubin
Ovarian	Spiegelberg

Solution to Question 16:

Anti-D is given at a dose of 300 µg intramuscularly for alloimmunization after surgical management of ectopic pregnancy.

The dose of anti-D in a surgically treated case of ectopic is 50 µg if gestation <12 weeks. If the gestation is ≥12 weeks, 300 µg of anti-D is given intramuscularly.

Anti-D is not given when the ectopic is treated with medical methods.

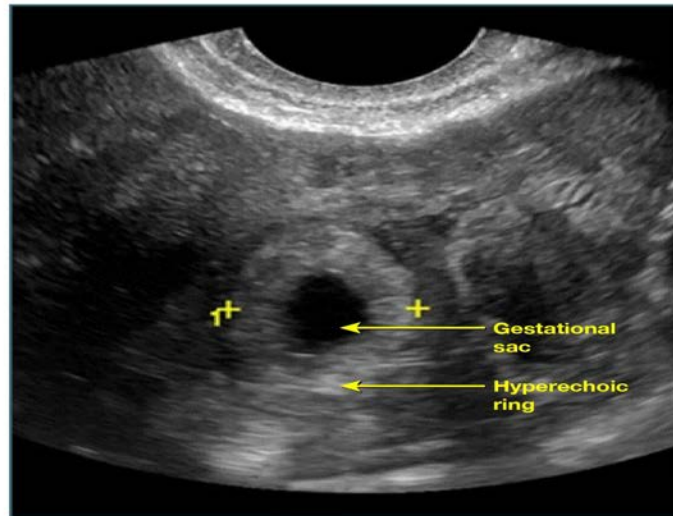
Solution to Question 17:

The given clinical scenario is suggestive of heterotopic pregnancy. Surgery is the ideal treatment in heterotopic pregnancy.

A normal uterine pregnancy co-existing with an extrauterine (ectopic) pregnancy is termed a heterotopic pregnancy. Although heterotopic pregnancy can occur spontaneously, it is more likely after ART and ovulation induction.

Tubal ring sign also referred to as bagel sign is one of the ultrasound signs of a tubal ectopic. It comprises an echogenic ring that surrounds an unruptured ectopic pregnancy.

USG image showing bagel sign or tubal ring sign:



Methotrexate is not to be given as it harms even the normal intrauterine fetus. Although potassium chloride may be injected into the sac of the ectopic, the safer method would be surgery to remove the ectopic.

Solution to Question 18:

Methotrexate is most effective when given at β hCG levels are <5000 IU/L.

Methotrexate is given at a dose of 50 mg/m^2 of body surface area. Monitoring is done by measuring β hCG levels on days 4 and 7. If the decline is $\geq 15\%$ of the initial value, the patient is followed up weekly till the levels fall to <10 IU/L. If the decline is $<15\%$, a 2nd dose of methotrexate is given on day 7.

Solution to Question 19:

Studdiford's criteria is used for the diagnosis of primary abdominal pregnancy.

Primary abdominal pregnancy is the rarest type of ectopic gestation. The conceptus implants on the peritoneal surface of the abdomen.

Studdiford's criteria:

- The presence of normal bilateral tubes and ovaries with no evidence of recent or past pregnancy.
- No evidence of a uteroperitoneal fistula.
- The presence of pregnancy, related exclusively to the peritoneal surface, early enough to eliminate the possibility of secondary implantation

Criteria used in the diagnosis of ectopic pregnancy are:

- Cervical pregnancy: Paalman and McElin (clinical criteria); Rubin's criteria (histological criteria)

- Ovarian pregnancy: Spiegelberg's criteria

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Abortion and Medical Termination of Pregnancy

Question 1:

Abortion is the expulsion of fetus weighing _____.

- a) <250 g
- b) <500 g
- c) <750 g
- d) <1000 g

Question 2:

A patient presented with vaginal bleeding following 6 weeks of amenorrhoea. Her UPT was positive. What is the most common cause of this condition?

- a) Luteal phase defect
- b) Infections
- c) Chromosomal abnormality
- d) Cervical incompetence

Question 3:

A patient presented with a spontaneous abortion at 7 weeks of gestation. Which of the following is the least likely chromosomal abnormality that can lead to this condition?

- a) Trisomy 16
- b) Trisomy 1
- c) Trisomy 21
- d) Trisomy 22

Question 4:

A woman with 6 weeks of amenorrhea had a positive urine pregnancy test. The following finding was seen on transabdominal USG. What is the most probable diagnosis?



- a) Singleton pregnancy
- b) Missed abortion
- c) Anencephaly
- d) Molar pregnancy

Question 5:

You are examining a primigravida with complete abortion. Which of the following findings would you not expect?

- a) Uterus smaller than period of amenorrhea
- b) Open cervical os
- c) Uterus firm on palpation
- d) Trace amounts of vaginal bleeding

Question 6:

Recurrent miscarriage is the loss of:

- a) ≥ 3 consecutive pregnancies before 20 weeks
- b) ≥ 2 consecutive pregnancies before 20 weeks
- c) ≥ 3 consecutive pregnancies after 24 weeks
- d) ≥ 2 consecutive pregnancies before 28 weeks

Question 7:

Which of the following conditions does not usually cause recurrent miscarriages?

- a) Asherman syndrome
- b) Polycystic ovarian syndrome
- c) Bicornuate uterus
- d) Infections

Question 8:

A primigravida had a spontaneous abortion at 18 weeks. On examination, the fetus was morphologically normal. Further investigations were performed upon the patient's insistence and the results are given below. What is the likely diagnosis?

- a) Syphilis
- b) Anti-phospholipid antibody syndrome
- c) Hemophilia B
- d) Factor V Leiden mutation

Question 9:

A woman comes to the casualty at 18 weeks of gestation with complaints of vaginal bleeding. She has a history of two early pregnancy losses. USG reveals the absence of fetal cardiac activity. Which of the following investigations need not be done in this patient?

- a) Extensive infection work-up
- b) Serum prolactin level
- c) Antiphospholipid antibodies
- d) Parental karyotyping

Question 10:

A pregnant lady with a history of recurrent abortions is now diagnosed to have antiphospholipid syndrome. What would be the best mode of treatment for her?

- a) Aspirin only
- b) Aspirin+heparin

- c) Aspirin+heparin+steroids
- d) Aspirin+steroids

Question 11:

A G3A3 patient was admitted to the hospital with catastrophic APLS following a spontaneous abortion. Which of the following is false about this condition?

- a) It is a rapidly progressive thromboembolic disorder
- b) At least 2 organ systems should be involved to make a diagnosis
- c) Infections can act as a trigger
- d) IVIG can be used for treatment

Question 12:

Which of the following features can be seen in a patient with cervical incompetence?

- a) Painful vaginal bleeding at <20 weeks
- b) Repeated mid-trimester painless cervical dilatation
- c) Repeated third trimester painless cervical dilatation
- d) Bleeding per vaginum in the first trimester

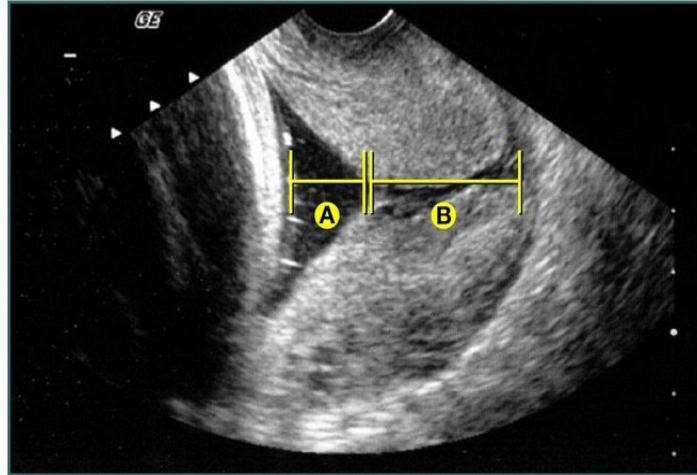
Question 13:

Which of the following increase the risk of developing cervical incompetence?

- a) 2, 3, 4
- b) 1, 2, 5
- c) 1, 3, 4
- d) 2, 4, 5

Question 14:

The transvaginal ultrasound of a pregnant woman is given below. What are the cutoff values of the parameters marked A and B for a diagnosis of cervical incompetence?



- a) A $>1\text{cm}$ and B $<25\text{mm}$
- b) A $<25\text{ mm}$ and B $>1\text{cm}$
- c) A $>2\text{ cm}$ and B $<35\text{mm}$
- d) A $>2.5\text{cm}$ and B $<25\text{mm}$

Question 15:

A 26-year-old woman presents with 6 weeks of amenorrhea and a positive UPT. Her past obstetric history was significant for 2 mid-trimester abortions. On USG, cervical length was 20 mm and no bulging membranes were seen. What is the next step in her management?

- a) Expectant management
- b) Immediate rescue cerclage
- c) Prophylactic cerclage at 12 weeks
- d) Immediate termination of pregnancy

Question 16:

A patient with cervical insufficiency is posted for McDonald's procedure. Which of the following is not a contraindication for this surgery?

- a) Placental abruption
- b) Premature membrane rupture
- c) Cervical dilation $>4\text{ cm}$

d) >20 weeks of gestation

Question 17:

A woman with cervical incompetence is posted for prophylactic McDonald's cerclage. Which of the following statements is false about this procedure?

- a) Removal of stitch must be done by 37th week
- b) A mersilene tape is tied around the cervix
- c) Done at the level of internal os
- d) Chorioamnionitis is a post-operative complication

Question 18:

A 20-year-old female patient came for medical termination of pregnancy. From whom will you obtain consent for the procedure?

- a) Patient and her male partner
- b) Only the patient
- c) Patient's guardian
- d) Patient's caregiver

Question 19:

A patient with contraceptive failure requires MTP at 5 weeks of gestation. You plan to use a medical method of abortion in her case. Which of the following is not a contraindication for this?

- a) Intrauterine device in situ
- b) Long term corticosteroids therapy
- c) Uncontrolled hypertension
- d) Chronic diabetes mellitus

Question 20:

A 23-year-old pregnant rape victim was posted for a suction and evacuation procedure. This procedure can be done up to _____ of gestation.

- a) 6 weeks
- b) 12 weeks
- c) 18 weeks
- d) 20 weeks

Question 21:

You have performed MTP on a multigravida at 10 weeks of gestation. As a part of post-op advice, you must prescribe her a prophylactic antibiotic. Which of the following drugs would you prefer?

- a) Doxycycline 200 mg orally
- b) Azithromycin 500 mg orally
- c) Amoxicillin 250 mg orally
- d) Cefotaxime 1 g iv

Question 22:

You are about to administer misoprostol to a woman who requires MTP in the 8th week of her pregnancy. What is the dose of this drug when given by the mucosal route, according to WHO guidelines?

- a) 200 mcg
- b) 600 mcg
- c) 400 mcg
- d) 800 mcg

Question 23:

Which of the following statements is false regarding the methods of MTP?

- a) Pain is more in medical methods of abortion
- b) Incomplete abortion is more common in medical methods
- c) Bleeding is prolonged in surgical methods
- d) Failure rate is low in surgical methods of abortion

Question 24:

A 21-year-old woman is brought in a drowsy state from a PHC with complaints of fever, severe lower abdominal pain, and purulent vaginal discharge. Her mother revealed that she had had an unsafe abortion a week ago. On examination, her abdomen was distended, BP was 80/60 mmHg, PR was 140 bpm, and RR was 36 breaths/ minute. Which of the following is not a causative agent of her present condition?

- a) Streptococcus pyogenes
- b) Clostridium perfringens
- c) Clostridium sordellii
- d) Escherichia coli

Question 25:

Which of the following is not a risk factor for stillbirth?

- a) Maternal diabetes and hypertension
- b) Illicit drug abuse by the mother
- c) Untreated seizure disorder in pregnancy
- d) Pregnancy following assisted reproductive technology

Question 26:

A young female has history of a spontaneous abortion. She has had amenorrhea since then. Her FSH now is 6 mIU/ml. Which of the following can be a cause for her amenorrhea?

- a) Pituitary dysfunction
- b) Ovarian dysfunction
- c) First pregnancy
- d) Uterine synechiae

Answer Key

Question No.	Correct Option
1	b
2	c
3	b

4	b
5	b
6	a
7	d
8	b
9	a
10	b
11	b
12	b
13	b
14	a
15	c
16	d
17	b
18	b
19	d
20	b
21	a
22	d
23	c
24	d
25	c
26	d

Detailed Explanations

Solution to Question 1:

According to WHO, abortion is defined as pregnancy termination or loss before 20 weeks of gestation or with a fetus weighing ≤ 500 g.

Some common causes of spontaneous abortion are:

- Chromosomal abnormalities
- Luteal phase defects
- Diabetes mellitus and thyroid disorders
- Cervical incompetence
- Antiphospholipid antibody syndrome

- Maternal infections
- Alcohol and cigarette smoking

Solution to Question 2:

The given clinical scenario is suggestive of first-trimester abortion which is most commonly caused by chromosomal abnormalities.

Autosomal trisomies are the most common (50%) genetic abnormality overall. Of these, the most common trisomy involved in abortions is trisomy 16. Turner syndrome (monosomy X) is the single most common chromosomal abnormality in miscarriages.

Causes of spontaneous abortion are:

- Genetic -
- Numerical chromosomal abnormalities
- Chromosomal rearrangements
- Mendelian etiology
- Endocrine -
- Luteal phase defects
- Thyroid abnormalities
- Diabetes mellitus
- Anatomical -
- Cervical incompetence
- Congenital uterine malformations
- Uterine fibroid
- Uterine synechiae
- Infectious (TORCH infections)
- Immunological -
- Antiphospholipid antibody syndrome
- Autoimmunity
- Maternal illness -
- Cyanotic heart diseases
- Hemoglobinopathies
- Environmental -
- Radiation exposure
- Alcohol
- Cigarette smoking

Solution to Question 3:

The least likely chromosomal abnormality involved in first-trimester abortions is trisomy 1.

The most common chromosomal abnormality in first trimester abortions is trisomy. The common trisomies are:

- Edwards syndrome (Trisomy 18)
- Down syndrome (Trisomy 21)
- Patau syndrome (Trisomy 13)
- Trisomies 16 and 22

Turner syndrome (Monosomy X) is the single most frequent chromosomal abnormality. It usually results in abortion.

Solution to Question 4:

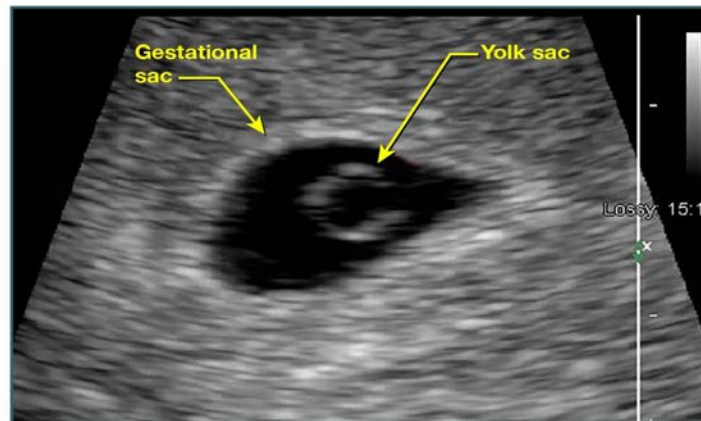
The given USG image shows an empty gestational sac which is seen in missed abortion before 12 weeks. At this gestational age, the entire ovum is covered by the chorionic villi giving rise to the empty gestational sac.

If this condition occurs beyond 12 weeks of gestation, a dead fetus is seen on USG. This is not preceded by any warning signs and the fetus is not expelled outside spontaneously.

Missed abortion



Normal pregnancy



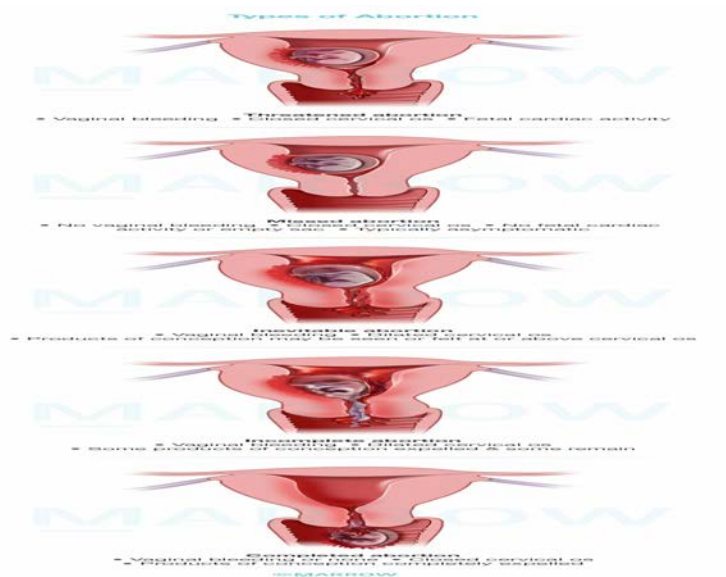
Solution to Question 5:

The cervical os is closed in complete abortion.

Complete Spontaneous abortion



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Solution to Question 6:

Recurrent miscarriage is defined as ≥ 3 consecutive pregnancy losses at ≤ 20 weeks or with a fetal weight < 500 g.

It can be of 2 types:

- Primary recurrent miscarriage where there are multiple losses in a woman who has never delivered a liveborn
- Secondary recurrent miscarriage where multiple pregnancy losses occur in a patient who has had a prior live birth

Solution to Question 7:

Infections do not usually cause recurrent miscarriages.

Common causes of recurrent miscarriage:

- Parental chromosomal abnormalities
- Anatomical factors -
- Uterine synechiae or Asherman syndrome
- Uterine leiomyomas
- Congenital genital tract anomalies
- Immunological factors -
- Antiphospholipid antibody syndrome
- Endocrine factors -

- Luteal phase defect
- Polycystic ovarian syndrome

Solution to Question 8:

The given lab findings are suggestive of anti-phospholipid antibody syndrome (APLS).

Criteria for diagnosis of antiphospholipid antibody syndrome:

Clinical criteria -

- Obstetric
- Three or more unexplained consecutive spontaneous abortions before 10 weeks
- One or more deaths of a morphologically normal fetus at or beyond 10 weeks
- Severe pre-eclampsia necessitating delivery before 34 weeks
- Vascular
- One or more episodes of arterial, venous, or small vessel thrombosis in any tissue or organ

Laboratory criteria -

- Presence of lupus anticoagulant
- Presence of IgG or IgM anti-cardiolipin antibodies
- Presence of IgG or IgM anti-beta-2 glycoprotein I antibodies

At least one clinical and one laboratory criteria is needed for the diagnosis.

Non-specific tests for lupus anticoagulant are increased aPTT, Kaolin clotting time, and dilute Russell viper venom time.

Solution to Question 9:

The given clinical scenario is suggestive of recurrent abortions, and extensive infection workup is not done.

Even though recurrent abortions are defined as ≥ 3 pregnancy losses, investigations must be done after 2 consecutive losses.

Diagnostic evaluation of recurrent pregnancy loss based on etiology is given as follows:

Etiology	Diagnostic evaluation
Genetic	Parental karyotyping
Anatomic	Hysterosalpingography Ultrasound Saline-infusion sonography

Etiology	Diagnostic evaluation
Endocrine	TSH Testing for insulin resistance Serum prolactin level Ovarian reserve testing Antithyroid antibodies
Infectious	No evaluation is recommended unless the patient has evidence of chronic endometritis/cervicitis on examination, or is immunocompromised
Autoimmune	Anticardiolipin antibody levels (IgG and IgM) Lupus anticoagulant Homocysteine Factor V Leiden Prothrombin promoter mutation Activated protein C resistance

Solution to Question 10:

The best mode of treatment for a pregnant woman with antiphospholipid syndrome (APS) is aspirin and heparin.

APS is a prothrombotic disorder (autoantibody-mediated thrombophilia) affecting both arterial and venous systems.

Patients present with recurrent pregnancy loss due to thrombosis of uteroplacental vessels, poor placental perfusion, and prevention of normal decidual changes after zygote implantation. They may also present with preterm delivery, pre-eclampsia, intrauterine growth restriction (IUGR), and fetal death.

Criteria for APS include:

- at least one otherwise unexplained fetal death at or beyond 10 weeks;
- at least one preterm birth before 34 weeks gestation because of eclampsia, severe preeclampsia, placental insufficiency; or
- at least three unexplained consecutive spontaneous abortions before 10 weeks.

Aspirin (60-80mg daily) with heparin (5000-10,000 U subcutaneously every 12h) is the most effective therapy for APS. Treating APS with only aspirin (option A) is not recommended as heparin is crucial in preventing thrombus formation at the decidual-thrombotic interface, Corticosteroids (option C, D) should not be used in primary APS (without an associated connective tissue disorder) but are indicated in patients with pre-existing systemic lupus erythematosus (SLE) and APS (secondary APS).

Solution to Question 11:

Catastrophic antiphospholipid antibody syndrome (CAPS) is a complication of APLA syndrome, in which ≥ 3 organ systems are involved. It is also known as Asherson syndrome.

It is seen in both primary and secondary APLA syndrome. Triggering factors include infections, trauma, and obstetric causes. IVIG can be used in CAPS when the first-line therapies with aspirin and heparin have failed.

Note: Asherman syndrome is an acquired condition characterised by formation of intrauterine adhesions or synechiae.

Solution to Question 12:

Cervical incompetence or insufficiency is characterized by repeated mid-trimester painless cervical dilatation. Escape of amniotic fluid followed by painless expulsion of products of conception can also be seen.

Risk factors for cervical incompetence:

- Congenital uterine anomalies
- Acquired -
 - Dilatation and curettage
 - Dilatation and evacuation
 - Vaginal operative delivery through an undilated cervix
 - Amputation of cervix
 - Multiple gestations
 - Prior preterm birth

Solution to Question 13:

In-utero exposure to DES, twin pregnancy, and bicornuate uterus increase the risk of cervical incompetence.

Offsprings born to pregnant mothers who were treated with diethylstilbestrol may develop cervical hypoplasia and malformation. They may also present with cervical incompetence during their 2nd trimester.

Cervical incompetence can be seen following MTP with surgical methods and not with medical methods.

The retentive power of the internal os may be compromised by anatomical abnormalities like bicornuate uterus and overdistended uterus due to twin gestations.

Risk factors for cervical incompetence:

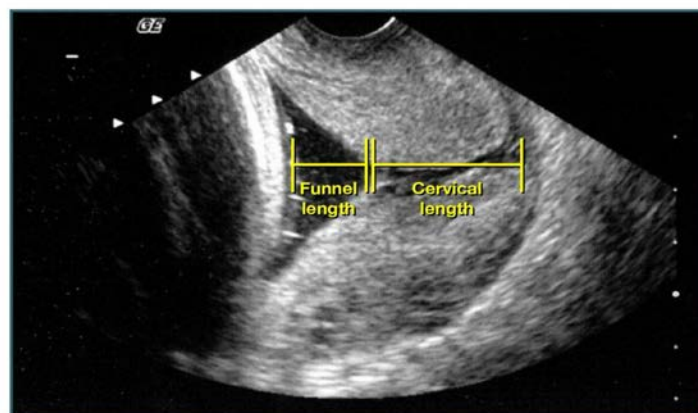
- Congenital uterine anomalies
- Acquired -

- Dilatation and curettage
- Induced abortion by dilatation and evacuation
- Vaginal operative delivery through an undilated cervix
- Amputation of cervix
- Multiple gestations
- Prior preterm birth

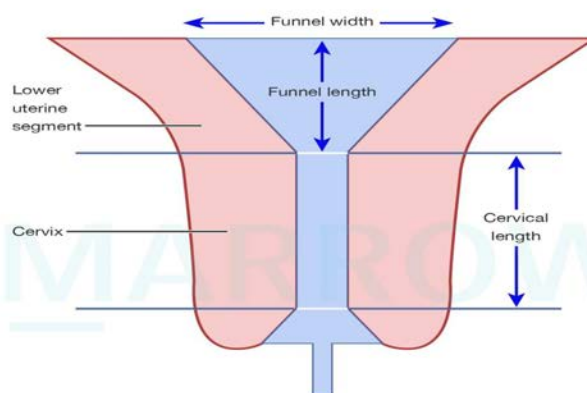
Solution to Question 14:

In the given image, the parameter marked A is funnelling of the internal os and the parameter marked B is cervical length. Funnelling of internal os and cervical length < 25 mm are diagnostic of cervical incompetence on transvaginal ultrasound.

Transvaginal ultrasound showing funnelling of the internal os



Diagrammatic representation of the cervical length and funnelling of internal os



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Solution to Question 15:

The given clinical scenario is suggestive of cervical incompetence. It can be managed with prophylactic cervical cerclage done between 12 and 14 weeks of gestation.

Rescue or emergency cerclage is done anytime within 24 weeks of gestation. It should only be done when there is bulging of membranes through the dilated cervix.

Investigations for cervical incompetence:

- In interconceptional period -
- Passage of number 6 - 8 Hegar's dilator beyond the internal os without resistance or pain
- Funnel-shaped shadow in premenstrual hystercervicography
- During pregnancy -
- Per speculum examination showing cervical dilatation with herniation of membranes
- USG revealing cervical length ≤ 2.5 cm and funnelling of internal os ≥ 1 cm

Solution to Question 16:

McDonald's procedure is a type of cervical cerclage. It can be done up to 24 weeks of gestation.

Contraindications to cervical cerclage are:

- Absolute
- ≥ 24 weeks of gestation
-

Active labour

-

Active vaginal bleeding

-

Abruptio placenta

-

Premature rupture of membranes

-

Chorioamnionitis

- Severe uterine irritability
- Fetal death or defect
- Relative

- Prolapsed membranes
- Vaginal spotting

Solution to Question 17:

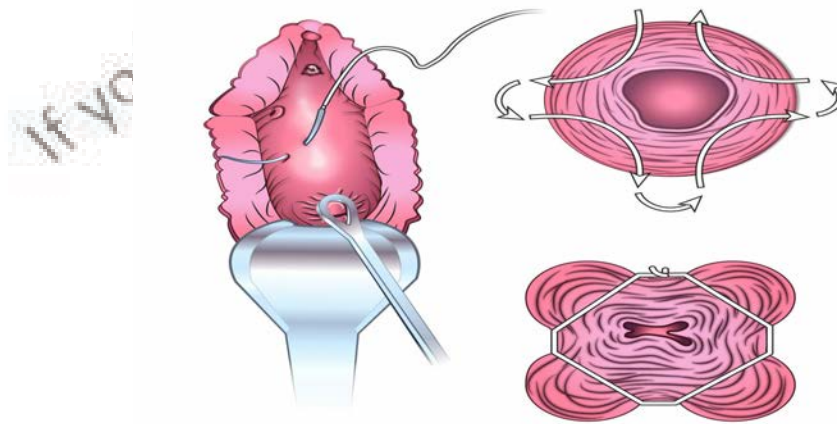
In McDonald's cerclage, a nonabsorbable suture, such as Mersilene, is placed as a purse-string suture as high as possible at the level of the internal os, positioned at the junction where the rugose vaginal epithelium transitions to the smooth vaginal portion of the cervix, below the bladder.

Another cerclage procedure is Modified Shirodker's technique. In this, a non-absorbable (mersilene) tape is passed subcutaneously through a transverse anterior incision and is tied posteriorly.

Removal of cerclage should be done by 37 weeks of gestation. It can be removed earlier in case of preterm labour.

Complications of cervical cerclage:

- Slipping of suture
- Chorioamnionitis
- Rupture of membranes
- Abortion or preterm labour
- Cervical laceration or scarring



McDonald stitch

Solution to Question 18:

Only the patient's consent is required for medical termination of pregnancy (MTP) and her male partner's consent is not required.

Consent of parent/guardian is needed if the age of the patient is <18 yrs or if the patient is mentally ill.

Solution to Question 19:

Chronic diabetes mellitus is not a contraindication to medical termination of pregnancy (MTP) with medical methods.

Contraindications to medical methods of abortion are:

- Women with in situ intrauterine device
- Severe anemia, coagulopathy, or anticoagulant use
- Long term corticosteroid therapy
- Severe liver, renal, pulmonary, or cardiovascular disease
- Inherited porphyria
- Uncontrolled hypertension

Contraindications to surgical methods of abortion are:

- Suction and evacuation - uterine size >10 weeks
- Intrauterine instillation of hypertonic solution - cardiovascular or renal lesion and severe anaemia

Solution to Question 20:

Suction and evacuation can be done up to 12 weeks of gestation. It is a method of termination of pregnancy in the first-trimester.

The various methods of termination of pregnancy are:

Timing	Medical methods	Surgical methods
1st trimester(up to 12 weeks)	Mifepristone Mifepristone and Misoprostol Methotrexate and Misoprostol Tamoxifen and Misoprostol	Menstrual regulation Vacuum aspiration (MVA/EVA) Suction evacuation and/or curettage Dilatation and evacuation - Rapid method Slow method
2nd trimester(13 - 20 weeks)	Misoprostol Dinoprostone Carboprost	Dilatation and evacuation Intrauterine instillation of hyperosmotic solutions Oxytocin infusion Hysterotomy

Solution to Question 21:

Doxycycline is the preferred drug for post-abortion antibiotic prophylaxis in this case. Oral doxycycline 100 mg is given 1 hour before surgical evacuation. 200 mg is given after the procedure is complete.

The best method of MTP at 7-12 weeks is by suction and evacuation. Following this procedure, antibiotic prophylaxis is given with doxycycline.

Solution to Question 22:

Misoprostol is given at a dose of 800 mcg by mucosal route for MTP.

Mucosal misoprostol administration has increased efficacy and minimizes gastrointestinal side effects. Hence, it is preferred over oral administration.

WHO guidelines for medical management in induced abortion for <12 weeks of gestation:

- Combined regimen (recommended): 200 mg of oral Mifepristone followed 1-2 days later by 800 mcg of buccal, vaginal, or sublingual misoprostol
- Misoprostol only regimen (alternative): 800 µg misoprostol administered vaginally, sublingually, or buccally

Solution to Question 23:

Bleeding is prolonged and unpredictable in medical methods of abortion.

Comparison of drawbacks of medical and surgical methods of MTP:

Factor	Medical methods	Surgical methods
Invasiveness	Usually not invasive	Invasive
Pain	More	Less
Vaginal bleeding	Prolonged, unpredictable	Light, predictable
Incomplete abortion	More common	Uncommon
Failure rate	2-5%	1%
Severe bleeding	0.1%	0.1%
Infection rate	Low	Low
Anesthesia	Usually not given	Always given
Time involved	Multiple visits for follow-up	Usually one visit No follow-ups required

Solution to Question 24:

The given clinical scenario is suggestive of toxic shock syndrome (TSS) following septic abortion. *Escherichia coli* can be detected in septic abortion, but it is not a causative agent for TSS.

Organisms causing toxic shock syndrome in septic abortion:

- *Streptococcus pyogenes*
- *Clostridium perfringens*
- *Clostridium sordellii*

Clinical features of septic abortion:

- Fever with chills and rigors
- Abdominal or chest pain
- Diarrhoea and vomiting
- Renal angle tenderness
- Offensive purulent vaginal discharge.

Clinical grading of septic abortion:

- Grade I - infection localized in the uterus
- Grade II - infection spreads to parametrium, tubes, ovaries, or pelvic peritoneum
- Grade III - generalized peritonitis, endotoxic shock, jaundice, or acute renal failure

Empirical antibiotic therapy in septic abortion is started for the broadest microbial coverage. Piperacillin-tazobactam or carbapenem and clindamycin are given intravenously. Antibiotics can be changed once culture sensitivity reports are available.

Solution to Question 25:

An untreated seizure disorder in pregnancy is not a risk factor for stillbirth. However, the use of certain anti-epileptic drugs like valproate could lead to the development of neural tube defects in the baby.

Recent studies haven't consistently shown that untreated women with epilepsy face a higher risk, though it's hard to rule out entirely because those who manage without medication usually have milder symptoms. Recent research found that birth defects were more common in children of women who used anticonvulsants.

Risk factors for stillbirth are:

- Advanced maternal age
- African-American race
- Smoking
- Illicit drug use

- Maternal medical diseases - overt diabetes or chronic hypertension
- Assisted reproductive technology
- Nulliparity
- Obesity
- Previous adverse pregnancy outcomes - prior preterm birth or IUGR

Note: Stillbirth rates appear to be slightly increased in pregnancies conceived via assisted reproductive technology.

Solution to Question 26:

The development of uterine synechiae is the most likely cause of amenorrhoea in the given clinical scenario.

Certain types of spontaneous abortions such as inevitable, incomplete and missed abortions might require medical assistance for complete removal of products of conception including procedures such as dilatation and curettage.

Vigorous uterine curettage in early pregnancy damages the basal layer of endometrium leading to scarring and adhesions in the uterus that interferes with normal endometrial growth and shedding. This causes the clinical feature of secondary amenorrhea as seen in this patient.

These intrauterine adhesions are known as uterine synechiae or Asherman syndrome.

As there is no defect in the hypothalamo-pituitary-ovarian axis in the Asherman syndrome, the levels of FSH remain within the normal range of 4 - 25 IU/L as seen in this patient.

In the case of Pituitary dysfunction (option A) the FSH levels are usually decreased.

In ovarian dysfunction (option B) the negative feedback of ovarian hormones (estrogen) on pituitary and hypothalamus is lost. Hence, the levels of FSH are increased (≥ 25 IU).

Asherman syndrome is characterised by a spectrum of intra uterine scarring including thin adhesions, dense bands, or complete obliteration of the uterine cavity. It occurs due to endometrial damage following vigorous curettage, usually in miscarriage, or elective abortion. Damage may also occur from other uterine surgeries such as myomectomy or cesarean delivery or from an infection related to an intrauterine device.

Depending on the degree of scarring, patients may present with amenorrhea, hypomenorrhea or recurrent pregnancy loss due to inadequate placentation.

Asherman syndrome shows multiple filling defects on hysterosalpingography (HSG) or saline-infusion sonography. Definitive diagnosis requires hysteroscopy. Hysteroscopic lysis of adhesions is the preferred surgical treatment for the syndrome.

Antepartum Hemorrhage

Question 1:

Antepartum hemorrhage is defined as bleeding from the genital tract after ____.

- a) 22nd week of pregnancy
- b) 24th week of pregnancy
- c) 26th week of pregnancy
- d) 28th week of pregnancy

Question 2:

What is the most common cause of antepartum haemorrhage?

- a) Abruptio placenta
- b) Placenta previa
- c) Local trauma
- d) Cervical varicosities

Question 3:

Transvaginal ultrasound findings in a multiparous woman revealed a low lying placenta. What does this mean?

- a) Placental edge within 2 cm from the internal os
- b) Placental edge within 3 cm from the internal os
- c) Placental edge within 4 cm from the internal os
- d) Placenta overlying the internal os

Question 4:

Which of the following types are included in the major degrees of placenta previa?

- a) 5 only
- b) 4 and 5

- c) 1, 2, 3 and 4
- d) 3, 4 and 5

Question 5:

A 30-year-old multiparous woman presented at 32 weeks of gestation with complaints of recurrent painless vaginal bleeding. On examination, fundal height corresponded to gestational age and fetal heart rate was 140 bpm. Which of the following is not a risk factor for this condition?

- a) Pre eclampsia
- b) Multifetal gestation
- c) Cigarette smoking
- d) Artificial reproductive technology

Question 6:

Per speculum examination findings in a pregnant woman with vaginal bleeding showed placenta protruding through the cervical os. What is the type of placenta previa in this case?

- a) Type 1
- b) Type 2
- c) Type 3
- d) Type 4

Question 7:

A patient at 34 weeks of gestation presented with placenta previa. Which of the following is false about this condition?

- a) Painful recurrent vaginal bleeding is seen
- b) The size of the uterus corresponds to gestational age
- c) The uterus is relaxed
- d) Floating head can be palpated

Question 8:

Abdominal examination during a routine antenatal checkup gave a positive Stallworthy sign. What could be the most probable cause?

- a) Concealed abruptio placenta
- b) Vasa previa
- c) Posterior placenta previa
- d) Cord compression

Question 9:

A G2P1 woman presents to the casualty at 30 weeks with a history of self-limiting painless bleeding PV 3 hours ago. On examination, BP - 110/70 mmHg and PR - 80 bpm. What is the next best step in management?

- a) Transvaginal ultrasound to confirm the diagnosis
- b) Start injection corticosteroids for fetal lung maturity
- c) Immediate caesarean section
- d) Expectant management till 37 weeks

Question 10:

A G2P1L1 woman at 30 weeks of gestation was rushed to the casualty with complaints of painless vaginal bleeding. She had woken up to find herself in a pool of blood. On examination, her vitals were unstable. Active bleeding per vaginum was present. Which of the following complications would not be anticipated in her?

- a) Birth asphyxia
- b) Preterm labour
- c) Precipitate labour
- d) Postpartum haemorrhage

Question 11:

Transvaginal ultrasound in a multiparous woman at 32 weeks confirmed placenta previa. Her vitals are stable and there is no active bleeding PV. What is the next step in management?

- a) Immediate caesarean delivery
- b) Immediate induction and vaginal delivery

- c) Manage as per McAfee and Johnson method
- d) Manage as per Sher and Page grading

Question 12:

Suitable cases for expectant management in case of placenta previa include all except:

- a) Assured fetal well-being
- b) Duration of pregnancy ≥ 37 weeks
- c) Mother in good health
- d) No active vaginal bleeding

Question 13:

What is the most common cause of consumptive coagulopathy in obstetrics?

- a) Intrauterine death
- b) Amniotic fluid embolism
- c) Sepsis
- d) Placental abruption

Question 14:

Which of the following does not increase the risk of placental abruption?

- a) Pre-eclampsia
- b) Cigarette smoking
- c) Post term pregnancy
- d) Twin pregnancy

Question 15:

Which of the following clinical features can be seen in a patient presenting with accidental haemorrhage?

- a) Painless bleeding PV
- b) Uterus is tender

- c) Uterine size is disproportionately small
- d) Positive Stallworthy sign

Question 16:

A multiparous woman at 34 weeks of gestation presented with bleeding per vaginum and was diagnosed to have grade 2 abruptio placenta. Which of the following features may be seen in her?

- a) Hypovolemic shock
- b) Anuria
- c) Coagulation defect
- d) Fetal distress

Question 17:

A 23-year-old primigravida at 32 weeks of gestation presents to the labour room with lower abdominal pain. Her vitals are BP - 90/60 mmHg and PR - 120 bpm. Fetal distress was noted and she was taken up for a caesarean section. The uterus appears as shown below. Which of the following is true regarding this condition?



- a) It is seen in revealed abruptio placenta
- b) It is an indication for hysterectomy
- c) It is a case of uteroplacental apoplexy
- d) Broad ligament is spared

Question 18:

Colour doppler studies in a patient with velamentous cord insertion revealed fetal vessels which were overlying the cervical os. Which of the following is false regarding her condition?

- a) Maternal mortality is high
- b) Bleeding is triggered by membrane rupture
- c) Bleeding is managed by category I caesarian section
- d) Apt test is diagnostic

Question 19:

Sample from a patient with bleeding PV at 30 weeks showed a positive Apt test. What is the most probable cause?

- a) Cervical polyp
- b) Vasa previa
- c) Abruptio placenta
- d) Placenta previa

Question 20:

Which of the following investigations can be used in the antenatal diagnosis of vasa previa in a patient with no history of bleeding PV?

- a) Singer's alkali denaturation test
- b) Transabdominal ultrasound
- c) Colour doppler
- d) MRI

Question 21:

A multiparous woman at term presents to the labour room with bleeding per vaginum and ruptured membranes. NST shows fetal distress and sinusoidal heart rate pattern. How will you manage this patient?

- a) Rapid induction and delivery by vacuum

- b) Rapid induction and delivery by forceps
- c) Immediate LSCS
- d) Adopt Vagos method and rapid vaginal delivery

Question 22:

The classical triad of amniotic fluid embolism includes all except _____.

- a) Metabolic acidosis
- b) Hypotension
- c) Hypoxia
- d) DIC

Question 23:

A G2P1L1 woman presented with placental abruption and was delivered by normal vaginal delivery. The next day, she develops petechiae, cold extremities, and her BP drops to 90/60 mmHg. Her blood reports show increased FDP, D-dimer, and decreased platelets. Which of the following is not used in the management of her condition?

- a) Epsilon-aminocaproic acid
- b) Packed RBCs transfusion
- c) Intravenous fluids
- d) Cryoprecipitate

Answer Key

Question No.	Correct Option
1	d
2	a
3	a
4	d
5	a
6	d
7	a

8	c
9	a
10	c
11	c
12	b
13	d
14	c
15	b
16	d
17	c
18	a
19	b
20	c
21	c
22	a
23	a

Detailed Explanations

Solution to Question 1:

Antepartum haemorrhage (APH) is defined as bleeding from or into the genital tract after 28 weeks of gestation up to the delivery of the baby. The lower limit is the period of viability and the upper limit includes the 1st and 2nd stages of labour.

Causes of APH:

- Placental causes:
 - Abruptio placenta (most common)
 - Placenta previa (2nd most common)
 - Circumvallate placenta
 - Circummarginate placenta
- Fetal causes:
 - Vasa previa
- Extraplacental causes (local cervicovaginal lesions):
 - Local trauma
 - Varicose veins
 - Cervical polyp

- Carcinoma cervix.

Note: According to RCOG criteria, the lower limit for APH is 24 weeks. However, in India, the period of viability is 28 weeks, so it is taken as the lower limit for APH.

Solution to Question 2:

Abruptio placenta is the most common cause of antepartum haemorrhage (APH).

Placental abruption or abruptio placenta is the premature separation of a normally situated placenta. The bleeding can be revealed, concealed or mixed type.

Causes of APH:

- Placental causes:
 - Abruptio placenta (most common)
 - Placenta previa (2nd most common)
 - Circumvallate placenta
 - Circummarginate placenta
- Fetal causes:
 - Vasa previa
- Extraplacental causes (local cervicovaginal lesions):
 - Local trauma
 - Varicose veins
 - Cervical polyp
 - Carcinoma cervix.

Solution to Question 3:

In low lying placenta, the edge of the placenta lies within 2 cm from the internal os but does not cover the internal os.

Placenta previa is a condition where the placenta is implanted over the lower uterine segment (partially or completely).

ACOG classification of placenta previa:

- True placenta previa - placenta covers the internal os partially or completely
- Low lying placenta - placenta does not cover the internal os but the edge is within 2 cm of the internal os

Solution to Question 4:

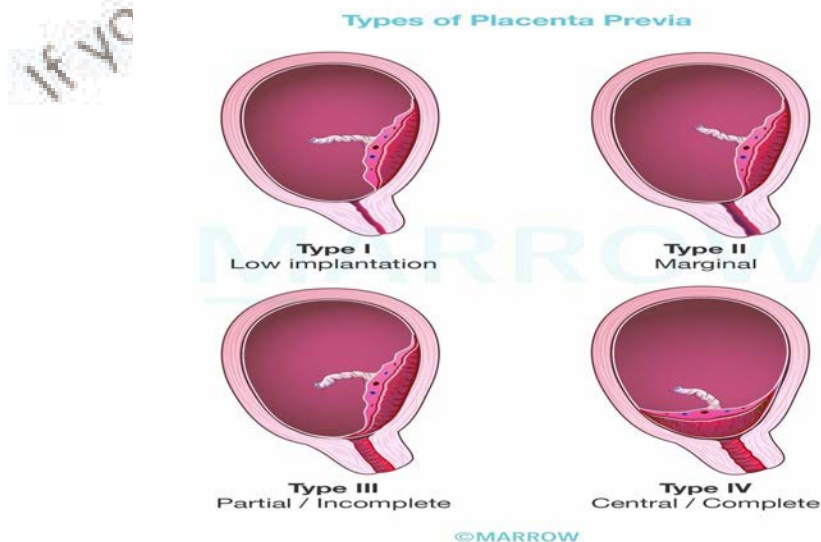
Major degree of placenta previa includes posterior marginal, partial and complete central.

Clinical classification of placenta previa:

- Mild degree
- Type I or lateral or low lying
- Type II anterior or anterior marginal
- Major degree
- Type II posterior or posterior marginal or dangerous placenta previa
- Type III or partial central
- Type IV or complete

Older (Browne's) classification of placenta previa:

Type	Position	Definition
Type I	Lateral	The placenta dips into the lower segment but does not reach up to the os
Type II	Marginal	The placental edge reaches the os
Type III	Incomplete or partial central	The placenta partially covers the internal os
Type IV	Complete or central	The placenta completely covers the internal os



Solution to Question 5:

The given clinical scenario is suggestive of placenta previa. Pre-eclampsia is a risk factor for abruptio placenta, not placenta previa.

Painless bleeding PV, fundal height proportional to gestational age and presence of FHS are the differentiating features of placenta previa from abruptio placenta.

Risk factors for placenta previa are:

- Demographic factors
- Maternal age $\geq$ 35 years
- Multiparity
- Cigarette smoking
- Clinical factors
- Prior caesarean section
- Past history of placenta previa
- Artificial reproductive technology
- Multifetal pregnancy
- Succenturiate lobe.

Solution to Question 6:

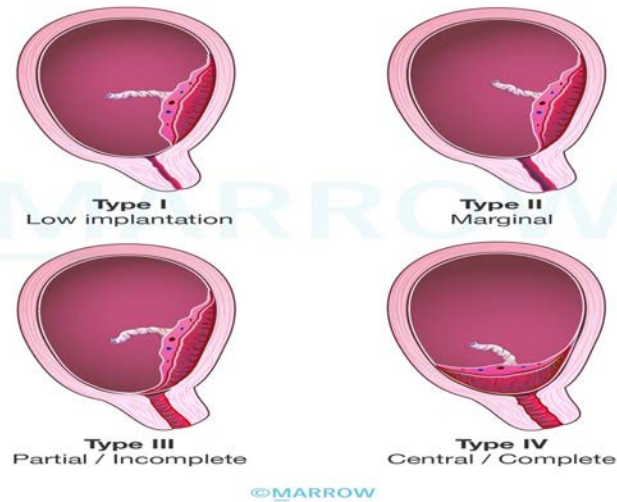
Placenta protruding through the internal os on per speculum examination indicates complete placenta previa or type IV placenta previa.

Older (Browne's) classification of placenta previa:

Cervical digital examination must only be done in suspected cases of placenta previa in an operation theater with provisions for cesarean section kept ready.

Type	Position	Definition
Type I	Lateral	The placenta dips into the lower segment but does not reach up to the os
Type II	Marginal	The placental edge reaches the os
Type III	Incomplete or partial central	The placenta partially covers the internal os
Type IV	Complete or central	The placenta completely covers the internal os

Types of Placenta Previa



Solution to Question 7:

Painless recurrent vaginal bleeding is seen in placenta previa.

Clinical features of placenta previa:

- Bleeding per vaginum - sudden onset, painless, recurrent, bright red in colour, sometimes a history of warning haemorrhage is present
- Uterus - relaxed, soft, no tenderness, size proportional to the period of gestation
- Floating head i.e., persistent displacement of fetal head as it cannot be pushed down into the pelvis
- Fetal heart sound - present
- Stallworthy sign - seen in low lying placenta, especially the posterior type. Slowing of fetal heart rate (FHR) on pressing the head down into the pelvis, which recovers promptly as pressure is relieved, is called Stallworthy sign.

Solution to Question 8:

Stallworthy sign is observed in posterior placenta previa.

The slowing of fetal heart rate on pressing the head down into the pelvis, which recovers promptly when the pressure is relieved, is called Stallworthy sign.

Note: This sign is not always significant as this could also occur in normal pregnancies due to fetal head compression.

Solution to Question 9:

The given clinical scenario is suggestive of antepartum haemorrhage (APH) due to placenta previa (painless bleeding PV). The next step in the management is to confirm the diagnosis with transvaginal ultrasound (TVS) as the patient is stable.

Transabdominal ultrasound (TAS) is used to detect the cause of APH.

If the placenta is in the lower uterine segment on TAS, confirmation of placenta previa is done by TVS. TVS provides a clearer image of the relationship between the edge of the placenta and the internal cervical os than TAS.

Approach to placenta previa after confirmation with TVS:

- Minor placenta previa - repeat USG at 36 weeks
- Major placenta previa or accreta - USG at 32 weeks followed by elective caesarean delivery
- Bleeding present - individual management

On TAS, if the placenta is in the upper uterine segment with evidence of retroplacental blood clot, abruptio placenta is suspected.

Per vaginal examination in placenta previa must not be done as it may further separate the placenta and cause excessive bleeding. However, transvaginal ultrasound is safe as long as the probe is 2-3 cm away from the cervix.

Solution to Question 10:

Then given clinical scenario is suggestive of placenta previa and precipitate labour is not a complication. Precipitate labour is seen in abruptio placenta and uterine tachysystole.

Vaginal bleeding in placenta previa is sudden in onset, recurrent, painless, unrelated to activity, and often occurs during sleep.

Complications of placenta previa:

Maternal complications:

Fetal complications:

- Low birth weight
- Asphyxia
- Intrauterine death
- Birth injuries
- Congenital malformations.

Period	Complication
Antepartum	Antepartum haemorrhage Malpresentation Preterm labour

Period	Complication
Intrapartum	Early rupture of membranes low dilatation of cervix
Postpartum	Postpartum haemorrhage
Puerperium	Sepsis Subinvolution

Solution to Question 11:

In the given clinical scenario, the next best step of management is conservative management till 37 weeks i.e., McAfee and Johnson's regimen.

The aim of conservative management is to continue the pregnancy for fetal maturity without compromising maternal health.

Prerequisites:

- Blood for transfusion
- Facilities for cesarean section

Selection of cases for expectant management for placenta previa:

- Mother is hemodynamically stable (Hb $\geq$ 10 g% and/or hematocrit $\geq$ 30%)
- Duration of pregnancy $\geq$ 37 weeks
- No active vaginal bleeding
- Fetus reactive on cardiotocography and no congenital anomalies on USG

Conduct of expectant treatment:

- Corticosteroids for fetal lung maturity
- Bed rest or pelvic rest
- Periodic monitoring of mother for bleeding PV
- Fetal surveillance with USG
- Tocolytics if contractions are present
- Anti-D to all Rh-negative unsensitized women

Option A: Immediate caesarean section after resuscitation is done if the patient is unstable.

Option B: Vaginal delivery is considered when the edge of the placenta is $\geq$ 2 cm away from the internal os.

Option D: Sher and Page grading is for abruptio placenta.

Solution to Question 12:

The duration of pregnancy must be less than 37 weeks for expectant management which was advocated by McAfee and Johnson.

The aim of conservative management is to continue the pregnancy for fetal maturity without compromising maternal health.

Prerequisites:

- Availability of blood for transfusion
- Facilities for cesarean section available

Selection of cases for expectant management for placenta previa:

- Mother is hemodynamically stable (Hb $\geq$ 10 g% and/or hematocrit $\geq$ 30%)
- Duration of pregnancy $\geq$ 37 weeks
- No active vaginal bleeding
- Fetus normal on USG and reactive on CTG

Conduct of expectant treatment:

- Corticosteroids for fetal lung maturity
- Bed rest or pelvic rest
- Periodic monitoring of mother for bleeding
- Fetal surveillance with USG
- Tocolytics if contractions are present
- Anti-D to all Rh-negative (unsensitized) women.

Solution to Question 13:

Placental abruption is the most common cause of severe consumptive coagulopathy in obstetrics.

Consumptive coagulopathy, also known as disseminated intravascular coagulation (DIC), is a pathologic disruption of hemostasis. It is characterized by systemic activation of blood coagulation, which results in thrombosis and fibrin deposition in the small blood vessels throughout the body.

Other causes of DIC in pregnancy:

- Amniotic fluid embolism
- Sepsis
- Preeclampsia and HELLP syndrome
- Acute fatty liver of pregnancy
- Postpartum haemorrhage.

Solution to Question 14:

Post term pregnancy does not increase the risk of abruptio placenta.

Risk factors for placental abruption:

- Prior abruption
- Increased age and parity
- Pre-eclampsia and chronic hypertension
- Chorioamnionitis
- Preterm ruptured membranes
- Multifetal gestation
- Cocaine use
- Cigarette smoking
- Uterine leiomyomas
- Single umbilical artery.

Solution to Question 15:

Abruptio placenta is also known as accidental haemorrhage or premature separation of placenta. Tender uterus is seen in abruptio placenta.

Abruptio placenta is defined as premature separation of a normally situated placenta. It is of the following types:

- Concealed
- Revealed
- Mixed

Clinical features of abruptio placenta:

In mixed type, features of concealed variety will be predominant.

Features	Revealed	Concealed
General condition	Proportionate to visible blood loss	Shock may be pronounced which may be out of proportion to visible bleeding.
Symptoms	Abdominal discomfort or pain followed by bleeding	Abdominal acute intense pain No or minimal bleeding
Uterine height	Proportionate to the period of gestation	Maybe disproportionately enlarged
FHS	Usually present	Usually absent

Solution to Question 16:

Grade 2 abruptio placenta is characterized by fetal distress.

Sher and Page clinical classification of placental abruption:

Grade 0 of abruptio placenta is seen in an asymptomatic patient, where a retroplacental clot is observed only after delivery of placenta.

Feature	Grade 1	Grade 2	Grade 3
Vaginal bleeding	Slight	Mild to moderate	Moderate to severe or concealed type
Uterine tenderness	Minimal	Present	Marked
Maternal BP and pulse rate	Unaffected	BP maintained Pulse rate increased	Hypovolemic shock Anuria
Fetal condition	FHS good	Fetal distress	Fetal death
Maternal fibrinogen	Unaffected	Decreased	Coagulation defect

Solution to Question 17:

The given image shows a couvelaire uterus or uteroplacental apoplexy which is seen in severe forms of concealed abruptio placenta.

There is massive extravasation of blood into the uterine musculature up to the serosa. Thus, the uterus has a port wine colour, which can be patchy or diffuse. They can also extend to the broad ligament, ovaries and peritoneal cavity.

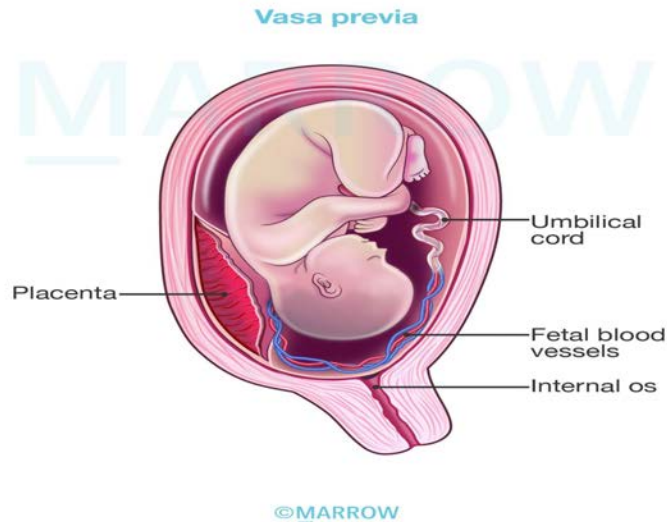
It is usually observed during caesarean section. It is not an indication of hysterectomy.

Solution to Question 18:

The doppler findings are suggestive of vasa previa. Maternal mortality is not seen in vasa previa, but fetal mortality is seen.

Vasa previa is a condition where fetal vessels travel within the membranes and overlies the cervical os. Hence, there can be bleeding during cervical dilatation or membrane rupture.

Fetal mortality is high and is due to fetal exsanguination.



Investigations of vasa previa:

- Colour flow Doppler (TVS)
- Singer's alkali denaturation test - to detect nucleated red blood cells
- Apt test (alkali denaturation test) - to detect fetal haemoglobin

Management of vasa previa:

- No bleeding
- 28-32 weeks - antenatal corticosteroids
- 30-34 weeks - admission for surveillance
- ≥ 37 weeks - elective caesarean section
- Bleeding present
- Category I emergency caesarean section is done (immediate threat to the life of mother or fetus and decision delivery interval ≤ 30 mins)

Solution to Question 19:

Apt test is an alkali denaturation test to detect fetal haemoglobin (HbF). It is positive in vasa previa and Rh incompatibility.

Apt test is based on the principle that HbF is resistant to alkali denaturation, while HbA is not. It is done by taking the blood sample in a test tube containing water and adding 10% NaOH. Maternal blood turns yellow-brown, while fetal blood remains pink in the test tube.

Bleeding due to vasa previa must be managed immediately with an emergency caesarean section. Hence, Apt test is not practically done in all cases.

Other options: Bleeding due to cervical polyp, placenta previa, and abruptio placenta is of maternal origin. So, Apt test will be negative in these cases.

Solution to Question 20:

The antenatal diagnosis of vasa previa can be done by colour doppler.

Vasa previa is a condition where fetal vessels travel within the membranes and overlie the cervical os. Hence, there can be bleeding during cervical dilatation or membrane rupture.

Fetal mortality is high due to fetal exsanguination.

Investigations of vasa previa:

- Colour flow Doppler (TVS)
- Singer's alkali denaturation test - to detect nucleated red blood cells
- Apt test - to detect fetal haemoglobin

Management of vasa previa:

- No bleeding
- 28-32 weeks - antenatal corticosteroids
- 30-34 weeks - admission for surveillance
- ≥ 37 weeks - elective caesarean section
- Bleeding present
- Emergency caesarean section done by category I

Options A and B: Colour Doppler is superior to MRI and TAS in the antenatal diagnosis of vasa previa.

Option D: Singer's alkali denaturation test to detect fetal RBCs can be used in intrapartum diagnosis of vasa previa (as there might be bleeding).

Solution to Question 21:

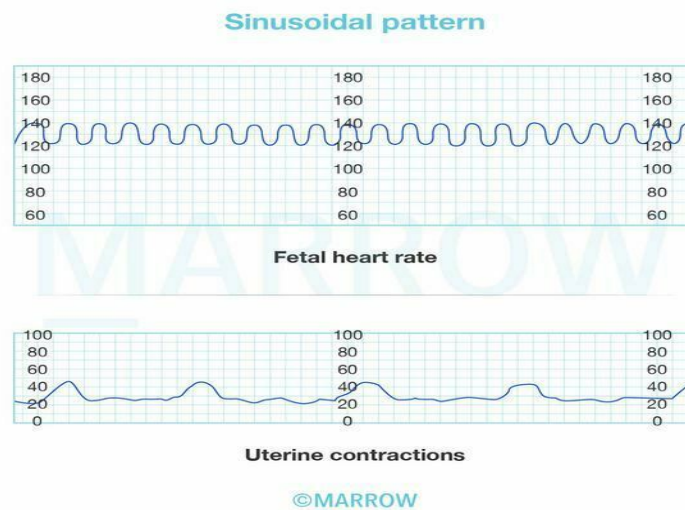
The given clinical scenario is suggestive of vasa previa since bleeding is associated with fetal distress and sinusoidal pattern in NST. Actively bleeding vasa previa in a term gestation is managed by immediate LSCS.

Sinusoidal pattern in CTG is a regular, smooth, undulating signal resembling a sine wave. It can be seen in conditions associated with severe fetal anemia like:

- Anti-D allo-immunisation
- Fetal-maternal haemorrhage
- Twin-to-twin transfusion syndrome
- Ruptured vasa previa
- Acute fetal hypoxia
- Infection

- Cardiac malformations
- Hydrocephalus
- Gastroschisis

Sinusoidal pattern on CTG:



Management of vasa previa:

- No bleeding
- 28-32 weeks - antenatal corticosteroids
- 30-34 weeks - admission for surveillance
- ≥ 37 weeks - elective caesarean section
- Bleeding present
- Emergency caesarean section

Note: Vagos method is used in the management of umbilical cord prolapse.

Solution to Question 22:

The classical triad of amniotic fluid embolism does not include metabolic acidosis.

Classical triad of amniotic fluid embolism:

- Hypotension
- Hypoxia
- DIC

Conditions predisposing to the development of amniotic fluid embolism:

- Rapid labour

- Meconium-stained amniotic fluid
- Tears in uterine and other large pelvic veins
- Older maternal age
- Post-term pregnancy
- Labour induction or augmentation
- Eclampsia
- Cesarean/forceps/vacuum delivery
- Placental abruption/placenta previa
- Polyhydramnios.

Solution to Question 23:

The given clinical scenario is suggestive of disseminated intravascular coagulation (DIC). Epsilon-aminocaproic acid and tranexamic acid are not used in the treatment of DIC.

Evaluation and management of DIC:

- Vigorous restoration and maintenance of circulation to treat hypotension
- Crystalloids and colloids to treat hypotension
- Whole blood and packed cells
- Fresh frozen plasma - contains fibrinogen, antithrombin III, factors V, XI, XII
- Cryoprecipitate - contains fibrinogen, factor VIII, von Willebrand's factor, factor XIII
- Platelet replacement if low
- Identification of the cause:
 - Abruptio placenta
 - PPH
 - HELLP syndrome
 - Amniotic fluid embolism
 - Sepsis
 - Acute fatty liver
- Assess the fetal condition and plan for termination of pregnancy.

Postpartum Haemorrhage

Question 1:

In a patient who has completed the 3rd stage of labor following vaginal delivery, what is the minimum amount of blood loss required for her to be diagnosed with postpartum hemorrhage?

- a) 500 mL
- b) 750 mL
- c) 1000 mL
- d) 1500 mL

Question 2:

A grand multipara had continuous, moderate vaginal bleeding immediately after delivery. What could be the most probable cause of bleeding in this patient?

- a) Uterine atony
- b) Retained tissues
- c) Genital tract trauma
- d) Uterine inversion

Question 3:

Which of the following is the most probable cause of bleeding per vaginum occurring 24 hours after delivery?

- a) Episiotomy wound
- b) Retained cotyledon
- c) Placenta accreta
- d) Uterine atony

Question 4:

Which of the following is not a cause of secondary PPH?

- a) Retained cotyledon
- b) Endometritis
- c) Placental polyp
- d) Placenta previa

Question 5:

Which of the following complications during pregnancy does not increase the risk of PPH?

- a) Oligohydramnios
- b) Fibroid uterus
- c) Pre eclampsia
- d) Macrosomia

Question 6:

In a patient with GDM complicating pregnancy, there was severe bleeding following the delivery of the placenta. On abdominal palpation, the fundus felt soft and later became hard on massaging. Which of the following is not a risk factor for this condition?

- a) Obesity
- b) High parity
- c) Prolonged labour
- d) Uterine rupture

Question 7:

Which of the following is not included in the active management of the third stage of labor to prevent postpartum hemorrhage?

- a) Uterotonic within 1 minute of delivery
- b) Delayed clamping, cutting and ligation of cord
- c) Gentle massage of uterus
- d) Controlled cord traction

Question 8:

A patient with pre-eclampsia presents with profuse bleeding after delivery. Which of the following drugs can be used in the medical management of this patient?

- a) Methergine
- b) Carboprost
- c) Misoprostol
- d) Dinoprost

Question 9:

In a case of atonic PPH, a methyl PGF₂ alpha derivative was given every 15 mins as a uterotonic. What is the maximum number of times the drug can be administered?

- a) 5 times
- b) 8 times
- c) 15 times
- d) 32 times

Question 10:

As per 2017 WHO guidelines for the treatment of PPH, which of the following statements is true?

- a) Uterine massage is not recommended in PPH when oxytocin is administered
- b) Intravenous colloids are the recommended fluids for initial resuscitation
- c) If the PPH is not responsive to uterotonics, uterine artery embolisation is the next step
- d) Tranexamic acid is administered early in all cases of PPH

Question 11:

Following normal vaginal delivery, a patient developed PPH and lost about 700mL of blood. Her PR was 100 bpm and her BP was 100/60mmHg. What is the value and corresponding interpretation of her shock index?

- a) 0.7 - normal
- b) 0.7 - significant
- c) 1 - significant
- d) 1 - normal

Question 12:

A patient was referred to the casualty from a district hospital with atonic PPH. Which of the following measures is least likely to be done?

- a) Carry out uterine massage
- b) Use non-pneumatic anti-shock garment to stabilize the patient
- c) Continuous iv infusion of oxytocin
- d) Volume resuscitation with iv crystalloids

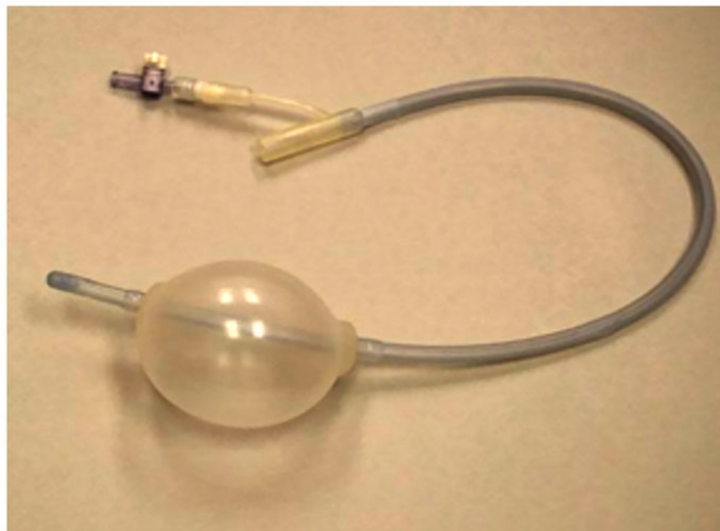
Question 13:

A 35-year-old multipara with atonic PPH continues to bleed profusely despite medical management and tamponade. Which of the following methods cannot be used to reduce bleeding in this patient?

- a) Hysterectomy
- b) Thermal endometrial ablation
- c) Internal iliac artery ligation
- d) Angiographic selective artery embolisation

Question 14:

Identify the instrument in the image given below.



- a) Bakri balloon
- b) Foley's catheter
- c) Sengstaken blakemore tube
- d) Condom catheter

Question 15:

What is the capacity of balloon tamponade to arrest PPH?

- a) 100 - 250 mL
- b) 300 - 500 mL
- c) 500 - 750 mL
- d) 700 - 900 mL

Question 16:

A P3 woman bleeds profusely after delivery of the placenta. You initiate oxytocin infusion. The patient continues to bleed and the uterus fails to contract despite bimanual compression. She becomes tachycardic and hypotensive. What is the next best step in her management?

- a) Tight intrauterine packing with gauze
- b) Balloon tamponade
- c) Ligation of uterine arteries
- d) Hysterectomy

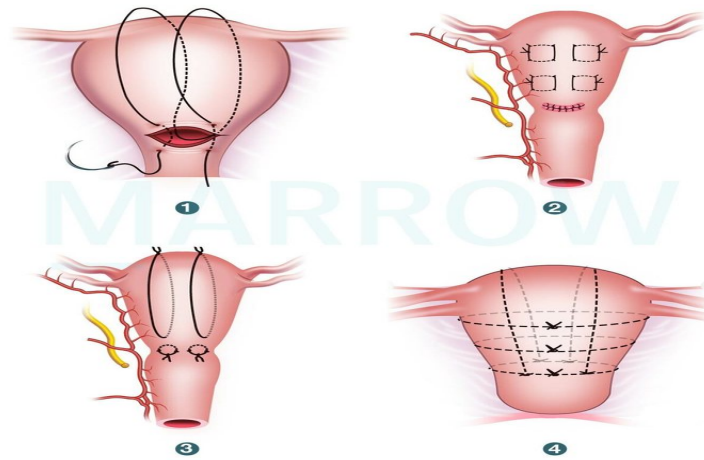
Question 17:

A pregnant lady delivered her baby unattended while she was on the way to the hospital. While examining her, you note a perineal tear in which the anal sphincter is disrupted. This tear is classified as ____.

- a) 1st degree
- b) 2nd degree
- c) 3rd degree
- d) 4th degree

Question 18:

Identify B lynch suture from the image given below.



- a) 1
- b) 2
- c) 3
- d) 4

Question 19:

A patient with severe PPH is shifted to the emergency OT for B lynch compression sutures. Which of the following statements is false about this procedure?

- a) The suture loops over the fundus
- b) Not used in atonic PPH
- c) Uterine cavity is compressed antero-posteriorly
- d) Uterine ischaemic necrosis is a complication

Question 20:

Which of the following statements are true?

- a) 1, 2, 3
- b) 1, 3, 5

c) 1, 4, 5

d) 1, 2, 4

Question 21:

In a patient with hydramnios complicating pregnancy, the third stage of labour was managed by applying fundal pressure and controlled traction of the cord. On palpation, the uterus was still soft and boggy. Within a few minutes, there was moderate bleeding per vaginum following which a reddish-purple mass was expelled. The patient then developed hypotension, bradycardia and went into shock. What is the most probable diagnosis?

a) Amniotic fluid embolism

b) Atonic postpartum haemorrhage

c) Uterine inversion

d) Placenta accreta

Question 22:

Which among the following is the most potent drug to control PPH?

a) Oxytocin

b) Methergin

c) Carboprost

d) Misoprostol

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	d
5	a
6	d
7	c

8	c
9	b
10	d
11	c
12	b
13	b
14	a
15	b
16	b
17	c
18	a
19	b
20	a
21	c
22	c

Detailed Explanations

Solution to Question 1:

Postpartum haemorrhage (PPH) in vaginal delivery is defined as a blood loss of more than 500 ml after completion of the third stage of labour.

In Caesarean section, blood loss of more than 1000 mL from the genital tract is called PPH.

Depending upon the amount of blood loss, PPH can be:

- Minor (<1L)
- Major (>1L)
- Severe (>2L)

In most cases, the magnitude of haemorrhage cannot be assessed accurately as visual estimates are unreliable.

Note: as per the latest ACOG guidelines, PPH is defined as cumulative blood loss ≥ 1000 mL or bleeding associated with hypovolaemia in the first 24 hours after birth, irrespective of the route of delivery. It is added that despite the new definition, blood loss ≥ 500 mL in a vaginal delivery should raise suspicion and mandates further evaluation.

Solution to Question 2:

Uterine atony is the most common cause of postpartum haemorrhage in a grand multipara.

Uterine atony is the failure of the uterus to contract sufficiently after delivery. Contraction helps to arrest bleeding from vessels at the placental implantation site.

Types of PPH:

- Primary/early: Haemorrhage occurring within 24 hours of childbirth. Uterine atony is the most common cause of primary PPH.
- Secondary/late: Hemorrhage occurring after 24 hours and up to 12 weeks postpartum. Retained placenta is the most common cause.

The risk factors for atonic uterus include:

- Primiparity or grand multiparity
- Previous PPH
- Overdistended uterus as in multiple pregnancies, hydramnios, macrosomia
- Precipitate labour
- Prolonged labour
- Uterine malformations
- Uterine fibroid
- Induction of labour with prostaglandins or oxytocin
- Placental abnormalities like placenta previa, accreta, increta, percreta, or abruptio placenta
- Obesity
- Age >40 years

Note: Uterine inversion occurs rarely and causes secondary PPH.

Solution to Question 3:

Retained cotyledon is the most probable cause of late or secondary PPH i.e., bleeding that occurs after 24 hours of delivery.

Other options are causes of early PPH.

Types of PPH include:

Primary/early: Haemorrhage occurring within 24 hours of childbirth. The causes are: (mnemonic 4T)

- Tone: Atonic PPH (80%)
- Trauma: Traumatic PPH
- Thrombin: Coagulopathy
- Tissue: retained products of conception

Risk Factors for primary PPH:

Atonic PPH:

- Primiparity or grand multiparity
- Previous PPH
- Overdistended uterus as in multiple pregnancy, hydramnios, macrosomia
- Precipitate labour
- Prolonged labour
- Uterine malformations
- Uterine fibroid
- Induction of labour with prostaglandins or oxytocin
- Placental abnormalities like placenta previa/accreta/increta/percreta/abruptio placenta
- Obesity
- Age >40 years

Traumatic PPH:

- Instrumental delivery
- Episiotomy
- Precipitate labour
- Macrosomia

Coagulopathy:

- Sepsis
- IUD
- HELLP syndrome

Secondary/ late PPH: Hemorrhage occurring after 24 hours and up to 12 weeks postpartum.

Causes include:

- Retained parts of cotyledon or membranes
- Endometritis
- Abnormal involution of placental site
- Secondary haemorrhage from caesarean section wound
- Placental polyp
- Infected fibroid
- Uterine AV fistula
- Puerperal uterine inversion
- Carcinoma cervix.

Solution to Question 4:

Placenta previa does not cause secondary postpartum hemorrhage. It causes primary postpartum hemorrhage.

Types of PPH:

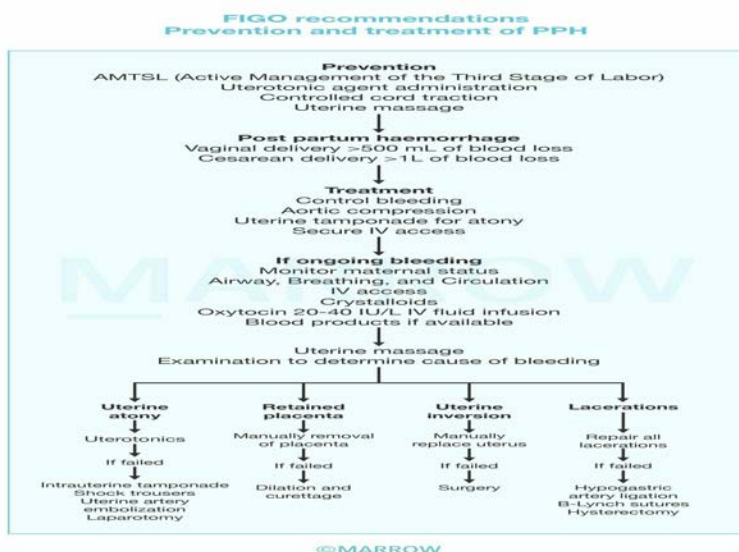
Primary/early: Hemorrhage occurring within 24 hours of childbirth.

Secondary/late: Hemorrhage occurring after 24 hours and up to 12 weeks postpartum.

Causes and risk factors for PPH:

The image below shows the prevention and management of PPH:

Primary PPH	Secondary PPH		
Atonic PPH	Traumatic PPH	Coagulopathy	
Primiparity or grand multiparity	Instrumental delivery	Sepsis	Retained bits of cotyledon or membranes
Placental abnormalities like placenta previa/accreta/increta/percreta/abruptio placenta	Precipitate labour	HELLP syndrome	Secondary haemorrhage from Caesarean section wound
Overdistended uterus as in multi-gestational pregnancy, hydramnios, macrosomia	Macrosomia	IUD	Puerperal uterine inversion
Induction of labour with prostaglandins or oxytocin	Episiotomy		Abnormal involution of placental site
Prolonged labour			Infected fibroid
Precipitate labour			Endometritis
Previous h/o PPH			Placental polyp
Obesity			Carcinoma of cervix
Age >40 years			Uterine av fistula



Solution to Question 5:

Oligohydramnios does not increase the risk of PPH. However, polyhydramnios can increase the risk by causing uterine atony, due to an overdistended uterus.

Risk factors for primary PPH:

Atonic PPH:

- Primiparity or grand multiparity
- Previous PPH
- Overdistended uterus as in multi-gestational pregnancy, hydramnios, macrosomia
- Precipitate labour
- Prolonged labour
- Uterine malformations
- Uterine fibroid
- Induction of labour with prostaglandins or oxytocin
- Placental abnormalities like placenta previa/accreta/increta/percreta/ abruptio placenta
- Obesity
- Age >40 years

Traumatic PPH:

- Instrumental delivery
- Episiotomy
- Precipitate labour
- Macrosomia

Coagulopathy:

- Sepsis
- IUD
- HELLP syndrome

Causes of secondary PPH:

- Retained bits of cotyledon or membranes
- Endometritis
- Abnormal involution of placental site
- Secondary haemorrhage from Caesarean section wound
- Placental polyp
- Infected fibroid

- Uterine av fistula
- Puerperal uterine inversion
- Carcinoma cervix.

Solution to Question 6:

In the above scenario, a soft uterus that becomes hard on massaging indicates an atonic uterus. Uterine rupture is a risk factor of traumatic PPH, not atonic PPH.

Uterine atony is the failure of the uterus to contract sufficiently after delivery. Thus, bleeding from the placental implantation site is not arrested.

GDM complicating pregnancy can cause macrosomia, which is a risk factor of this condition.

Risk factors for atonic PPH:

- Primiparity or grand multiparity
- Previous PPH
- Overdistended uterus as in multi-gestational pregnancy, hydramnios, macrosomia
- Precipitate labour
- Prolonged labour
- Uterine malformations
- Uterine fibroid
- Induction of labour with prostaglandins or oxytocin
- Placental abnormalities like placenta previa/accreta/increta/percreta/ abruptio placenta
- Obesity
- Age >40 years

Solution to Question 7:

As per the latest guidelines, uterine massage is no longer a core component of Active Management of the Third Stage of Labor (AMTSL) that is performed for the prevention of postpartum hemorrhage (PPH).

Uterine massage or fundal massage is not recommended as an intervention to prevent PPH in women who have already received prophylactic uterotonics such as oxytocin (core component of AMTSL). Hence, it is now an optional component of AMTSL that is done in patients with uterine atony as a part of the management of PPH.

AMTSL is recommended by the WHO to prevent postpartum hemorrhage (PPH). Uterotonics are the most important components in AMTSL to do so.

The currently recommended components of AMTSL are as follows:

Note: In the event that uterine atony is diagnosed during the assessment of uterine tone, uterine massage can be performed to control PPH.

1.	Injecting uterotonics within 1 minute of delivery of the baby or immediately after delivery of the anterior shoulder	Most important step Oxytocin 10 IU im/iv infusion is the preferred uterotonic. (Not recommended by WHO-carboprost)
2	Delayed cord clamping	Done within 1-3 mins of delivery 80 ml of additional blood goes to the fetus Prevents fetal anemia
3	Delivery of placenta by controlled cord traction: Modified Brandt Andrews technique	The uterus is pushed upwards and backward The traction of cord is downwards and forwards
4	Intermittent assessment of uterine tone	Done every 15 mins for 2 hours

Solution to Question 8:

PPH in a patient with pre-eclampsia can be treated with misoprostol. It is a PGE₁ analogue used as a second line uterotonic in the treatment of PPH.

The drug of choice for PPH in normal patients is oxytocin.

Option A: Methergine and other ergot derivatives cause dangerous hypertension in pre-eclampsia.

Options B and D: Carboprost and Dinoprost cause severe hypertension in pre-eclampsia.

Medical Management of PPH:

Drug	Dose	Side effects	Contraindications
Oxytocin	10-40 U in 1L crystalloid (never given as iv bolus)	Water intoxication	-
Methergine	0.2 mg	Hypertension	Pre-eclampsia
Carboprost	0.25 mg (8 doses max)	Diarrhoea Hypertension Vomiting	Bronchial asthma Cardiac, renal or hepatic diseases
Misoprostol	600-1000 mcg	Fever Tachycardia	-

Solution to Question 9:

Carboprost is a methyl PGF₂ alpha derivative. The normal dosage is 0.25mg. The maximum dose is 2mg.

So, $2/0.25 = 8$ times.

Carboprost is a 15-methyl PGF₂ alpha derivative used as a 2nd line drug in the treatment of PPH that is unresponsive to oxytocin.

Side effects:

- Diarrhoea
- Hypertension
- Nausea
- Vomiting

Contraindications:

- Bronchial asthma
- Suspected amniotic fluid embolism
- Active cardiac, renal or hepatic diseases.

Solution to Question 10:

Early administration of tranexamic acid is done in all cases of PPH.

Tranexamic acid for the treatment of PPH is used as follows:

- Tranexamic acid should be given in all cases of PPH, regardless of the cause.
- Early use of iv tranexamic acid (i.e) within 3 hours of birth
- Fixed-dose of 1 g in 10 mL (100 mg/mL) iv at 1 mL per minute
- If bleeding continues after 30 minutes, the second dose of 1 g iv is given

Option A: Uterine massage must be done in addition to oxytocin in all cases of PPH. This will help in the contraction of an atonic uterus. In normal patients, uterine massage is not compulsory as per AMTSL.

Option B: IV crystalloids are the recommended fluids for resuscitation.

Option C: Balloon tamponade is the next step after uterotonics.

Solution to Question 11:

Shock index = 1 and it indicates significant blood loss.

The shock index is calculated as HR/systolic BP.

Here, PR=100 and systolic BP=100.

Shock Index = $100/100 = 1$

If the shock index is 0.9 – 1.1, it indicates massive blood loss and the need for intensive resuscitation.

Description	Range of shock index
Normal value of shock index (non-pregnant)	0.5 - 0.7
Normal value of shock index in pregnant (i.e) Obstetrical shock index	0.7 - 0.8
Significant or massive blood loss	0.9 - 1.1

Solution to Question 12:

Using a non-pneumatic anti-shock garment (NASG) is not one of the first measures for the management of atonic PPH. NASG is used to reduce bleeding and stabilize the patient for shifting to tertiary centers.

The below image shows a non-pneumatic anti-shock garment.



Non-pneumatic anti-shock garment (NASG) works by applying circumferential counterpressure. It decreases blood flow to the abdomen, pelvis, and lower extremities. It increases the blood flow to the heart, lungs, and brain.

First step measures in the management of PPH must be done immediately and simultaneously. This includes:

- Continuous iv oxytocin infusion
- Fundal massage and bimanual uterine compression
- Volume resuscitation with crystalloids
- Insertion of Foley's catheter to monitor urine output

- Alert the anesthesia team to begin a manual exploration of the uterus under GA to check for:
- Retained placental fragments
- Uterine abnormalities
- Lacerations
- Rupture
- Blood transfusion should be done if vitals are still unstable

If bleeding still continues, further steps are taken depending on the patient profile:

- Balloon tamponade
- Surgical methods:
- B lynch compression sutures
- Uterine artery ligation
- Internal iliac artery ligation
- Angiographic embolization
- Hysterectomy.

Solution to Question 13:

Thermal endometrial ablation is a treatment modality for Abnormal Uterine Bleeding (AUB). It is not used in the treatment of PPH.

In patients who are unresponsive to initial resuscitation and medical management, the following methods can be used:

- Balloon tamponade
- Surgical methods:
- B lynch compression sutures
- Uterine artery ligation
- Internal iliac artery ligation
- Angiographic embolization
- Hysterectomy.

Solution to Question 14:

The image shows a Bakri balloon. It is used in uterine tamponade in patients with atonic PPH who are unresponsive to medical management.

The Bakri balloon has a three-way valve and a separate tubing to drain the blood from the uterine cavity.

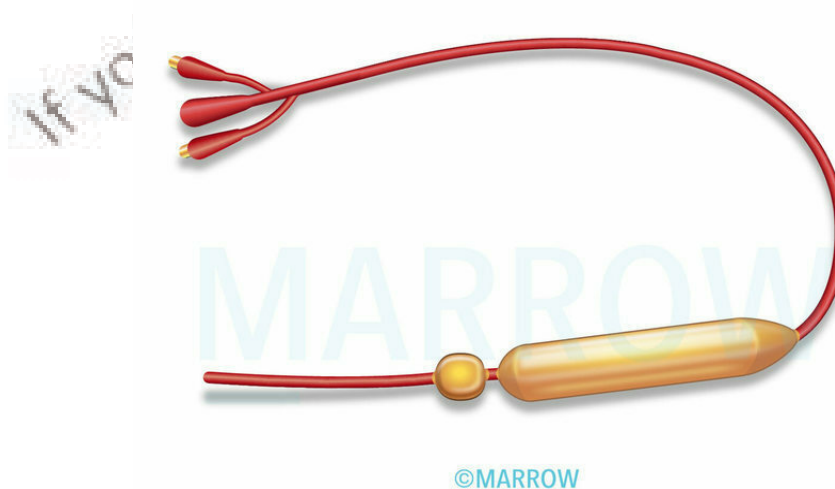


It acts by stimulating uterine contraction and direct hemostatic pressure to the open uterine sinuses (tamponade effect).

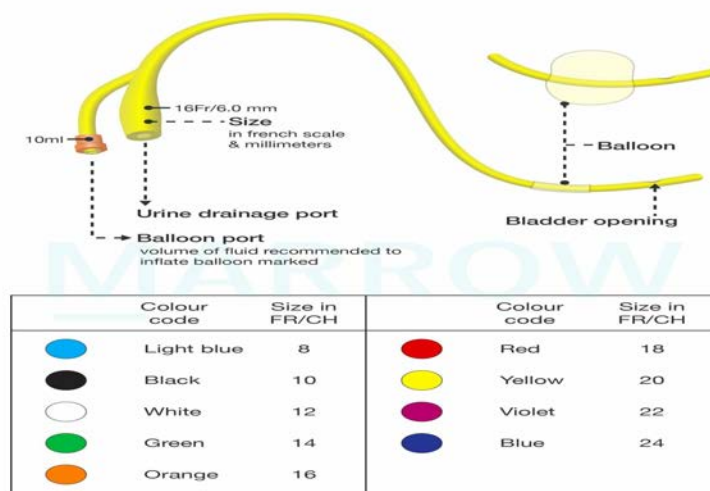
The maximum capacity of the balloon is 500mL. It is inflated with normal saline (200-500mL) and can be retained for up to 12 hours. Success rates are high and it can be used to avoid hysterectomy.

Other options: Sengstaken Blakemore tube, condom catheter, and 34F Foley with 60mL balloon are other instruments that can be used for uterine tamponade.

The image below depicts Sengstaken Blakemore tube.



The image below depicts Foley's catheter.



Solution to Question 15:

The capacity of balloon tamponade to arrest PPH is 300 to 500 mL.

Bakri balloon is used in uterine tamponade in cases of atonic PPH unresponsive to medical management.

Insertion of Bakri balloon requires three team members.

- One to perform abdominal sonography during the procedure.
- Another member places the deflated balloon into the uterus and stabilizes it.
- The third member instills fluid to inflate the balloon.

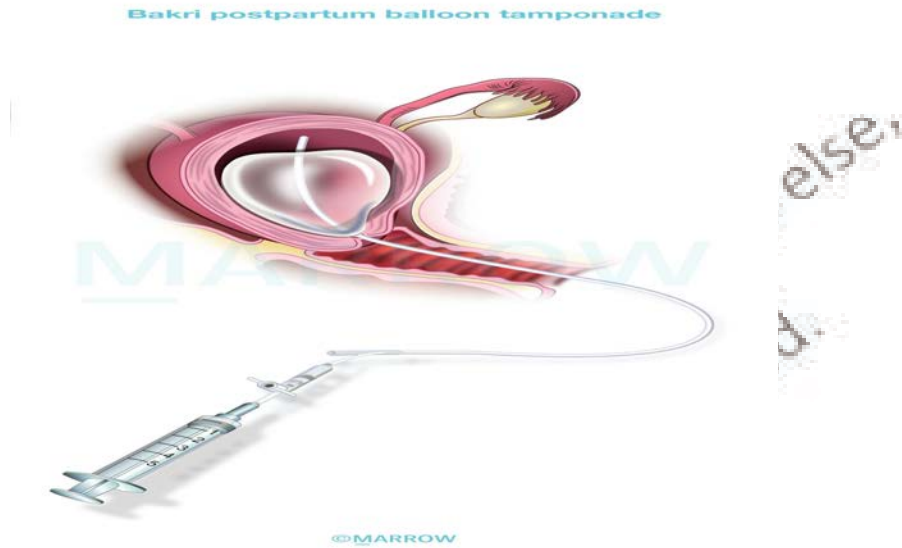
Initially, rapid infusion of at least 150 mL of normal saline is done. It is followed by further infusion over a few minutes up to 300 to 500 mL.

Bakri postpartum balloon tamponade



Solution to Question 16:

Balloon tamponade is the next step in the management of atonic PPH unresponsive to medical management and bimanual uterine massage.



Option A: Tight intrauterine packing has become unpopular due to the dangers of concealed bleeding and infections. It can be used in low-resource settings while shifting the patient to tertiary care centers.

Option C: Uterine artery ligation is done when tamponade fails (Bakri balloon and compression sutures).

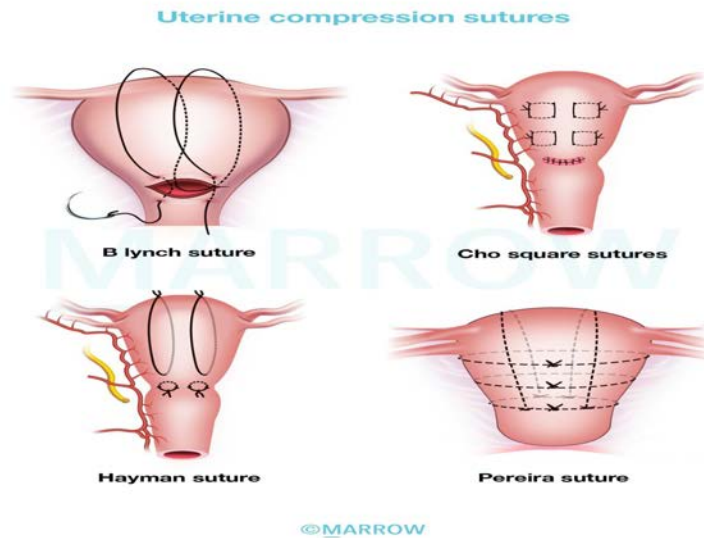
Option D: Hysterectomy is the last option when all the other treatment modalities fail. Usually, the uterus contracts and bleeding stops with ligation or embolization. However, hysterectomy is done sooner rather than later in cases of placenta accreta and uterine rupture.

Solution to Question 17:

The perineal tears in which anal sphincter is disrupted are classified as third-degree perineal tear.

Solution to Question 18:

Image 1 depicts B lynch suture.

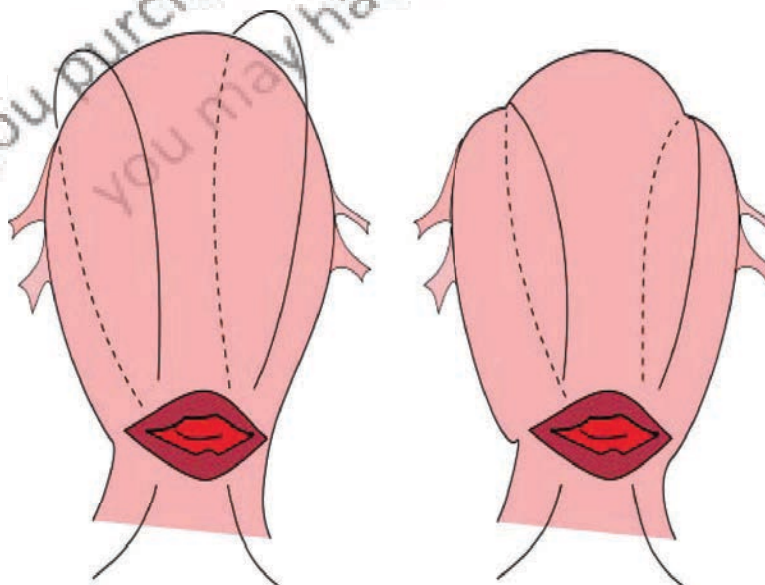


All these stitches are types of uterine compression sutures. They used to surgically oppose the walls of the uterus to arrest bleeding in case of atonic PPH.

Solution to Question 19:

The procedure is B-lynch compression sutures and it is used in atonic PPH.

B lynch suture is a technique to arrest hemorrhage in severe postpartum atony.



The suture bites are taken over the lower uterine segment anteriorly. Then it is looped over the fundus to go posterior. Another bite is taken posteriorly. Then it is looped back over the fundus and taken anteriorly. The final bite is taken over the lower uterine segment and the free ends are tied in a surgical knot.

The uterus is compressed antero-posteriorly.

The success rate is high.

Ischaemic uterine necrosis is a complication of this procedure.

Solution to Question 20:

Statements 1, 2 and 3 are true.

Statement 4: Hematomas <5 cm can be managed conservatively with cold compress. Larger hematomas need to be explored surgically.

Statement 5: Lower uterine segment rupture causes supra-levator hematoma.

Types of pelvic hematomas:

Feature	Infralevator hematoma	Supralevator hematoma
Incidence	Common	Rare
Location	Below the levator ani	Above the levator ani
Causes	Improper hemostasis during repair of vaginal/ perineal tears or episiotomy wound	Extension of cervical laceration, lower uterine segment rupture
Presentation	Severe pain with rectal tenesmus	Unexplained shock with features of internal haemorrhage
Treatment	Small hematomas can be conservatively managed with cold compress	Laparotomy

Solution to Question 21:

The most probable diagnosis is uterine inversion.

Hydramnios is a risk factor for uterine inversion due to uterine over-distension. Fundal pressure while the uterus is atonic (soft, relaxed, and boggy) causes uterine inversion. It results in haemorrhage followed by complete uterine inversion protruding out of the vulva. Neurogenic type of shock develops due to stretching of the infundibulopelvic ligament and the ovaries.

Option A: Amniotic fluid embolism causes cardiac arrest, acute hypoxia and severe haemorrhage.

Option B: There was only moderate bleeding per vaginum and soft uterus. But since there is protrusion of the uterus per vaginum, the most probable diagnosis is uterine inversion.

Option D: Placenta accreta rarely causes uterine inversion.

Uterine inversion is an extremely rare but life-threatening complication in the 3rd stage of labor, in which the uterus is turned inside out partially or completely.

Causes:

- Most common cause is mismanagement of 3rd stage of labour

- Spontaneous
- Placenta accreta (rarely)

Features:

- Shock immediately after delivery
- The degree of shock is out of proportion to the amount of bleeding
- Shock is either neurogenic or hemorrhagic
- Per abdomen examination - fundus of the uterus is not palpable.

Management of uterine inversion:

Vaginal operations	Abdominal operations	Hydrostatic method
Kustner	Huntingtons	O Sullivan
Spinelli	Haultain	Oguch

Solution to Question 22:

Carboprost is the most potent drug to control PPH.

Though oxytocin is the most commonly used drug for the prevention and treatment of PPH, Carboprost is highly efficacious than oxytocin in treating PPH. If the question would have been "Drug of choice to treat PPH" then the answer will be Oxytocin.

Initially for PPH, oxytocin or methergin (ergometrine) is given. If it fails to control the bleeding then Misoprostol is given.

Drug	Use in PPH
Oxytocin	DOC to prevent and treat PPH
Methergin	Drug to prevent and treat PPH
Carboprost(PGF ₂)	Best drug to control PPH
Misoprostol(PGE ₁)	2nd line drug to treat PPH

Preterm Labor and Postterm Pregnancy

Question 1:

Preterm labour is defined as when the labour starts before _____

- a) 35 weeks
- b) 36 weeks
- c) 37 weeks
- d) 38 weeks

Question 2:

Which of the following is not a risk factor for preterm labour?

- a) Pregnancy following artificial reproductive technique
- b) Asymptomatic bacteriuria
- c) Periodontal disease
- d) Progesterone intake

Question 3:

Late preterm births include births between _____.

- a) 30-32 weeks
- b) 32-34 weeks
- c) 34-36 weeks
- d) 36-38 weeks

Question 4:

A 31-year-old G3P2 woman presents at 29 weeks of gestation with a blood-tinged vaginal discharge. She is suspected to have preterm labour. Which of the following findings will not help confirm her diagnosis?

- a) Bishop score of 5
- b) Cervical dilatation of 24 mm

- c) Cervical length of 32 mm
- d) Fetal fibronectin $> 50\text{ ng/ml}$

Question 5:

Which of the following cannot be used to predict preterm labour in a patient having irregular uterine contractions at 30 weeks of gestation?

- a) IL-6
- b) TNF alpha
- c) Fetal fibronectin
- d) IL-2

Question 6:

A 30-year-old G2P1L0 woman came for a routine antenatal checkup at 20 weeks of gestation. She is worried about preterm labour as her previous baby was born preterm. What will be the timing and cut-off limit for fetal fibronectin (fFN) assay to predict this condition?

- a) > 34 weeks, fFN $> 15\text{ ng/mL}$
- b) < 37 weeks, fFN $> 1.5\text{ ng/mL}$
- c) 20-24 weeks, fFN $> 5\text{ ng/mL}$
- d) 24-34 weeks, fFN $> 50\text{ ng/mL}$

Question 7:

A pregnant woman presented with blood-tinged vaginal discharge at 35 weeks of gestation. A nitrazine paper test was performed on her. What is this test used to detect?

- a) Fetal hemoglobin in vasa previa
- b) Fetal blood cells in Rh isoimmunisation
- c) Premature rupture of membranes
- d) Abruptio placenta

Question 8:

A sample of the vaginal fluid from the posterior fornix of a patient gave a positive nitrazine paper test. Which of the following conditions does not give a false positive result?

- a) Abruptio placenta
- b) Recent sexual intercourse
- c) Bacterial vaginosis
- d) Oligohydramnios

Question 9:

Transvaginal sonography in a case of clinically suspected preterm labour revealed a short cervix and funnelling of the internal os. What is the shape of the cervix that is likely to be seen here?

- a) U
- b) T
- c) C
- d) O

Question 10:

A multigravida presents with preterm premature rupture of membranes. According to RCOG guidelines, which of the following antibiotics is preferred for her?

- a) Ampicillin
- b) Amoxicillin
- c) Erythromycin
- d) Azithromycin

Question 11:

A primigravida at 30 weeks of gestation presents with labour pains but her membranes were intact. You decide to start her on corticosteroids and a tocolytic. Which of the following drugs will not be useful in this case?

- a) Magnesium sulphate
- b) Nifedipine
- c) Indomethacin

d) Progesterone

Question 12:

A primigravida at 32 weeks of gestation was admitted for management of preterm labour. Which of the following tocolytics would be preferred for this?

- a) Nifedipine**
- b) Glyceryl trinitrate**
- c) Ritodrine**
- d) Magnesium sulphate**

Question 13:

Which of the following is not a contraindication for tocolysis in a patient with preterm labour?

- a) Chorioamnionitis**
- b) Fetal distress**
- c) Anencephaly**
- d) Placenta previa**

Question 14:

A G3P1A1 woman presents with preterm labour at 27 weeks of gestation. She tells you that her previous child who was born preterm developed cerebral palsy. Which of the following drugs can you use to reduce the incidence of this condition?

- a) Terbutaline**
- b) Atosiban**
- c) Magnesium sulphate**
- d) Nifedipine**

Question 15:

A primigravida was administered tocolytics and corticosteroids for preterm labour. The following day, she suddenly started coughing and gasping for air. On examination, her BP

was 160/100 mmHg and SpO₂ was 80% in room air. Which of the following drugs could have caused this?

- a) Atosiban
- b) Dexamethasone
- c) Indomethacin
- d) Ritodrine

Question 16:

Which of the following tocolytics can cause multiple fractures in the fetus if given for a prolonged duration?

- a) Atosiban
- b) Magnesium sulphate
- c) Nifedipine
- d) Nitric oxide

Question 17:

Post-term pregnancy refers to pregnancy continuing beyond _____.

- a) 41 weeks of gestation
- b) 42 weeks of gestation
- c) 44 weeks of gestation
- d) 40 weeks of gestation

Question 18:

A woman presented for an antenatal checkup 2 weeks past her EDD. Which of the following is not an etiological factor for her condition?

- a) Anencephaly
- b) Sedentary habit
- c) Primiparity
- d) Maternal stress

Question 19:

You have performed an artificial rupture of membranes in a primigravida for augmentation of labour. You note the amniotic fluid to be saffron-coloured. What is the likely diagnosis?

- a) Postmaturity
- b) Rh incompatibility
- c) Intrauterine death
- d) Concealed placental abruption

Question 20:

A newborn baby was found to have wrinkled, patchy skin, a long thin body and long nails. Which of the following could be a cause for his condition?

- a) Placental aromatase deficiency
- b) Placental progesterone deficiency
- c) Placental pregnenolone deficiency
- d) Placental sulphatase deficiency

Question 21:

A G2P1L1 woman came for a routine ANC at 41 weeks of gestation. On examination, her Bishop score was 2. She was admitted for fetal monitoring and induction. Which of the following complications would you not anticipate in the fetus even if her pregnancy continues further?

- a) Shoulder dystocia
- b) Polycythemia
- c) Cord compression
- d) Hyperglycemia

Question 22:

A primigravida was admitted at 42 weeks for induction of labour. Which of the following conditions is she not at an increased risk for?

- a) Postpartum hemorrhage
- b) Meconium stained liquor

- c) Shoulder dystocia
- d) Oligohydramnios

Question 23:

The term small for gestational age implies birth weight _____

- a) <10th percentile
- b) <15th percentile
- c) <2.5 kg
- d) <2 kg

Question 24:

What is the normal pH of amniotic fluid?

- a) 8.1 - 8.7
- b) 7.1 - 7.3
- c) 6.1 - 6.3
- d) 6.5 - 6.9

Answer Key

Question No.	Correct Option
1	c
2	d
3	c
4	c
5	d
6	d
7	c
8	d
9	a
10	c
11	d

12	a
13	d
14	c
15	d
16	b
17	b
18	d
19	a
20	d
21	d
22	a
23	a
24	b

Detailed Explanations

Solution to Question 1:

Preterm labour is defined as when labour starts before 37 completed weeks (<259 days) counting from the first day of the last menstrual period.

Timing of delivery	Terminology
<33 weeks	Early preterm
34 to 36 weeks	Late preterm
37 to 38 weeks	Early term
39 to 40 weeks	Full term

Solution to Question 2:

Progesterone intake is not a risk factor for preterm labour. Progesterone is used for prophylaxis of preterm labour.

Risk factors for preterm labour:

- Maternal history -
- Previous history of spontaneous abortion or preterm delivery

- Artificial reproductive technique
- Asymptomatic bacteriuria or recurrent UTI
- Smoking
- Maternal stress
- Pregnancy complications -
- Pre-eclampsia
- Antepartum haemorrhage
- PROM
- Overdistended uterus - polyhydramnios, multiple pregnancy
- Cervical incompetence
- Uterine malformations
- Intrauterine death
- Maternal chronic diseases -
- Hypertension
- Nephritis
- Diabetes
- Severe anaemia
- Periodontal diseases

Option A: Multifetal pregnancies are common following artificial reproductive techniques and can be associated with preterm labour.

Option B: Asymptomatic bacteriuria or bacterial vaginosis can cause spontaneous preterm labour.

Option C: Chronic diseases like periodontal disease increase the risk of preterm labour.

Solution to Question 3:

Late preterm births include births between 34 to 36 weeks of gestational age.

Timing of delivery	Terminology
<33 weeks	Early preterm
34 to 36 weeks	Late preterm
37 to 38 weeks	Early term
39 to 40 weeks	Full term

Solution to Question 4:

A cervical length of 32 mm (3 cm) does not help in confirming preterm labour as it is highly unlikely when the cervical length is ≥ 3 cm. The length of the cervix in preterm labour, as measured in transvaginal ultrasound is ≤ 2.5 cm.

Preterm labor is the onset of labor before 37 completed weeks of pregnancy (less than 259 days of gestation).

Features of preterm labor:

- Uterus - regular contractions at least one in every 10 minutes, with/without pain
- Cervix - dilatation ≥ 2 cm and effacement $\geq 80\%$
- TVS - length of cervix ≤ 2.5 cm and funneling of the internal os ≥ 1 cm
- Pelvic pressure, backache, vaginal discharge or bleeding PV

Predictors of preterm labor:

- Clinical predictors:
 - History of prior preterm birth
 - Multiple pregnancies
 - Presence of genital tract infection
 - Symptoms of preterm labour
- Biophysical predictors:
 - Uterine contractions ≥ 4 /hour
 - Bishop score ≥ 4
 - Cervical length (TVS) ≤ 25 mm
- Biochemical predictors:
 - Fetal fibronectin (fFN) in cervical/vaginal discharge ≥ 50 ng/mL
 - Increased levels of IL-6, IL-8, TNF-alpha

The best predictors of preterm labor in a woman with a previous history of preterm birth are fetal fibronectin ≥ 50 ng/ml and cervical length ≤ 2.5 cm on TVS.

Solution to Question 5:

IL-2 is not a biochemical predictor of preterm labour.

Biochemical predictors of preterm labor:

- Fetal fibronectin
- IL 6

- IL 8
- TNF α

Investigations of preterm labour:

- Fetal fibronectin >50 ng/ml
- TVS - length of cervix ≤ 2.5 cm
- Urine culture
- High vaginal swab
- Maternal salivary estriol
- Uterine activity monitoring

Solution to Question 6:

Fetal fibronectin value of >50 ng/mL in the cervicovaginal discharge between 24 and 34 weeks is predictive of preterm labor.

Fibronectin is a glycoprotein that binds the fetal membranes to the decidua. It is normally found in the cervicovaginal discharge before 22 weeks and after 37 weeks of gestation.

It is measured using an enzyme-linked immunosorbent assay. When the test is negative, it reassures that the delivery will not occur within the next 7 days.

Solution to Question 7:

The nitrazine paper test is used to detect premature rupture of membranes.

The nitrazine paper test detects the pH of the secretions from the vaginal pool. It turns blue at pH >6 . Amniotic fluid is slightly alkaline with a pH of 7.1 - 7.3 and vaginal secretions have a pH of 4.5 - 6.0.

False-positive nitrazine test can be seen in contamination of the discharge with blood, semen, and bacterial infections.

The below image shows the nitrazine paper.



Option A: Fetal hemoglobin in vasa previa is detected by Apt test.

Option B: Estimation of the volume of fetal blood entering the maternal circulation in Rh isoimmunisation is done by Kleihauer-Betke test.

Option D: In abruptio placenta, the diagnosis is mainly based on clinical features.

Solution to Question 8:

Oligohydramnios shows a false negative nitrazine paper test.

Nitrazine paper test detects the pH of the secretions from the vaginal pool. It turns blue at pH ≥ 6 . Amniotic fluid is slightly alkaline with a pH of 7.1 - 7.3 and vaginal secretions have a pH of 4.5 - 6.0.

False-positive nitrazine test can be seen in contamination of the discharge with blood, semen, and bacterial infections. False-negative nitrazine test can be seen in the case of scanty amniotic fluid.

Solution to Question 9:

U-shaped cervix is seen on tranvaginal sonography (TVS) in preterm labour.

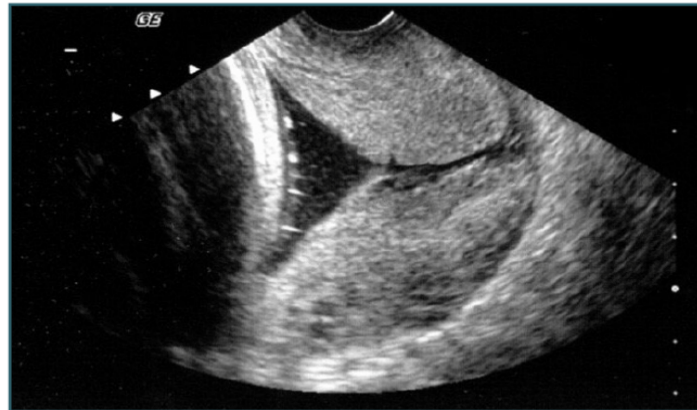
On TVS, the cervix should be assessed to determine whether funnelling (internal os width ≥ 5 mm) is present.

The following is a TVS image showing a V-shaped cervix:

Shape of cervix	Inference
T shaped	Normal

Shape of cervix	Inference
Y shaped	Suspicion of preterm labour
V shaped	Funneling of internal os
U shaped	Incompetent osPreterm labour

Transvaginal ultrasound showing V-shaped cervix - Preterm labour



Solution to Question 10:

According to RCOG guidelines, erythromycin is the preferred antibiotic in the case of PPRM (preterm premature rupture of membranes).

Spontaneous rupture of membranes before the onset of labour between 28 to 37 weeks is called preterm premature rupture of membranes.

If PPRM occurs between 28 to 34 weeks, prophylactic antibiotics and corticosteroids are given to the patient.

Solution to Question 11:

The given clinical scenario is suggestive of preterm labour with intact membranes and progesterone is not used as a tocolytic in these cases.

Progesterone can be used for the prophylaxis of preterm labour in women with a previous history of spontaneous preterm labour.

Glucocorticoid therapy is given in preterm labour when the period of gestation is ≤ 34 weeks to hasten fetal lung maturity. Short term therapy with tocolytics is given in conjunction with steroids for 1-3 days to delay labour for at least 48 hours or to transfer the patient to a tertiary centre (with

NICU).

Tocolytics that can be given as short term therapy include:

- Terbutaline
- Magnesium sulphate
- Indomethacin
- Nifedipine
- Atosiban

Solution to Question 12:

Nifedipine is the preferred tocolytic as it has minimum maternal side effects.

According to ACOG guidelines, the use of tocolytics in preterm labour does not prolong gestation but may delay delivery upto 48 hours. Hence, short-term therapy with tocolytics is given for 1-3 days to delay labour for the action of corticosteroid therapy or to transfer the patient to a tertiary centre (with NICU).

Tocolytics used in preterm labour and their side effects:

Tocolytics	Maternal side effects	Fetal/ neonatal side effects
Beta-adrenergic agonists: Ritonavir, Terbutaline, Isoxsuprine	Headache, palpitations, tachycardia, hypotension, pulmonary edema, cardiac failure, ARDS	Fetal heart failure, intrauterine death, Neonatal hypoglycemia, intraventricular haemorrhage
Calcium channel blockers: Nifedipine, Nicardipine, Verapamil	Hypotension, headache, flushing, nausea	-
Oxytocin antagonists: Atosiban (not approved for use in India)	Nausea, vomiting, chest pain	-
Magnesium sulphate	Flushing, perspiration, headache, muscle weakness, respiratory paralysis	Neonatal cardiac arrest, lethargy, hypotonia, respiratory depression
COX inhibitors: Indomethacin, Sulindac	Heartburn, asthma, GI bleeding, thrombocytopenia, renal injury, platelet dysfunction, oligohydramnios	Constriction of ductus arteriosus, pulmonary hypertension, intrauterine growth restriction

Solution to Question 13:

Placenta previa is not a contraindication for tocolysis. In placenta previa, nifedipine and magnesium sulphate are used as tocolytics.

Contraindications to tocolysis:

- Chorioamnionitis
- Preeclampsia/Eclampsia
- Advanced labour
- Fetal distress
- Intra-uterine death
- Congenital anomalies not compatible with life
- Pregnancy ≥ 34 weeks
- Abruptio placenta

Solution to Question 14:

Magnesium sulphate has a neuroprotective effect and hence decreases the risk of developing cerebral palsy.

Magnesium sulphate can be used for neuroprotection in preterm labour when the pregnancy is ≤ 32 weeks. The dose is 4 g IV over 20 mins followed by an infusion of 1 g/hr.

Solution to Question 15:

The given clinical scenario is suggestive of acute pulmonary edema which is a side effect of the tocolytic ritodrine.

Acute pulmonary edema in pregnancy can be due to:

- Non-cardiogenic causes -
- Pre-eclampsia
- Acute haemorrhage
- Sepsis
- Tocolysis with ritodrine and magnesium sulphate
- Vigorous IV fluid therapy
- Cardiogenic causes

Tocolytics used in preterm labour and their side effects:

Tocolytics	Maternal side effects	Fetal/ neonatal side effects
Beta-adrenergic agonists: Ritodrine, Terbutaline, Isoxsuprine	Headache, palpitations, tachycardia, hypotension, pulmonary edema, cardiac failure, ARDS	Fetal heart failure, intrauterine death, Neonatal hypoglycemia, intraventricular haemorrhage
Calcium channel blockers: Nifedipine, Nicardipine, Verapamil	Hypotension, headache, flushing, nausea	-

Tocolytics	Maternal side effects	Fetal/ neonatal side effects
Oxytocin antagonists: Atosiban (not approved for use in India)	Nausea, vomiting, chest pain	-
Magnesium sulphate	Flushing, perspiration, headache, muscle weakness, respiratory paralysis	Neonatal cardiac arrest, lethargy, hypotonia, respiratory depression
COX inhibitors: Indomethacin, Sulindac	Heartburn, asthma, GI bleeding, thrombocytopenia, renal injury, platelet dysfunction, oligohydramnios	Constriction of ductus arteriosus, pulmonary hypertension, intrauterine growth retardation

Solution to Question 16:

Magnesium sulphate can cause multiple fractures and hypocalcemia in a preterm baby.

The FDA warned against prolonged use of magnesium sulphate to arrest preterm labour. The antagonistic effect of magnesium lowers the calcium levels in the fetus. This causes bone thinning and fractures in fetuses exposed to the drug for more than 5 to 7 days.

Solution to Question 17:

ACOG defines post-term pregnancy as a pregnancy continuing beyond 42 completed weeks of gestation.

For clinical purposes, a pregnancy continuing beyond 2 weeks of the expected date of delivery (> 294 days) is called a post-term pregnancy.

Inaccurate LMP is the most common cause of misdiagnosis of post-term pregnancy. Hence, early ultrasound dating is important to determine fetal gestational age.

Solution to Question 18:

Maternal stress is associated with preterm labour, not post-term pregnancy.

Etiological factors of post-term pregnancy are:

- Inaccurate LMP
- Heredity
- Primiparity
- Previous prolonged pregnancy
- Sedentary habits
- Elderly multipara

- Anencephaly
- Placental sulphatase deficiency

Solution to Question 19:

Saffron-coloured amniotic fluid is seen in postmaturity.

Solution to Question 20:

The given clinical scenario is suggestive of postmaturity syndrome which can occur due to placental sulphatase deficiency.

Etiology of post-term pregnancy:

- Wrong dates - inaccurate LMP (most common)
- Heredity
- Previous prolonged pregnancy
- Anencephaly
- Placental sulphatase deficiency (leads to decreased estrogen synthesis)
- Primipara/ elderly multipara

Features of postmaturity syndrome in a newborn:

- Wrinkled, patchy, peeling skin, particularly prominent on the palms and soles
- Long, thin body suggesting wasting
- Long nails
- Eyes open and unusually alert
- Looks old and worried

Note: Placental aromatase deficiency causes disorder of sexual development of fetus.

Solution to Question 21:

In the given scenario, the pregnancy continuing further would cross 42 weeks of gestation and thus turn into a post-term pregnancy. Hypoglycemia is seen in a post-term neonate.

Complications of post-term pregnancy:

- During labour:
- Fetal hypoxia
- Labour dysfunction

- Macrosomia
- Shoulder dystocia
- Oligohydramnios
- Cord compression
- Birth trauma
- Meconium aspiration
- Intracranial haemorrhage
- Following birth:
- Meconium aspiration syndrome
- Hypoxia
- Hypoglycemia
- Polycythemia
- Neonatal convulsions

Solution to Question 22:

The risk of post-partum haemorrhage does not increase in post-term pregnancy.

Complications of post-term pregnancy are due to placental insufficiency and placental ageing. It manifests as placental infarcts and calcifications. During post-term pregnancy, diminished placental function can cause oligohydramnios.

Complications of post-term pregnancy are:

- During labour:
- Fetal hypoxia
- Labour dysfunction
- Macrosomia
- Shoulder dystocia
- Oligohydramnios
- Cord compression
- Birth trauma
- Meconium aspiration
- Following birth:
- Chemical pneumonitis, atelectasis, pulmonary hypertension (due to meconium aspiration)
- Hypoxia
- Hypoglycemia

- Polycythemia
- Neonatal convulsions

Solution to Question 23:

Small for gestational age implies birthweight is <10th percentile for gestational age.

With respect to the size, a newborn can be appropriate, small or large for gestational age.

Birth weight	Terminology
<10th percentile	Small for gestational age
>90th percentile	Large for gestational age
Between 10th and 90th percentile	Appropriate for gestational age

Solution to Question 24:

The normal pH of the amniotic fluid is 7.1 - 7.3

The normal pH of the vaginal secretions is 4.5 - 5.5. In premature rupture of membranes (PROM), the pH of the vaginal secretions becomes 6 - 6.2 as the amniotic fluid mixes with it. This forms the basis for nitrazine paper test.

Gestational Trophoblastic Diseases

Question 1:

What is the chromosome number of a partial hydatidiform mole?

- a) 46 XX
- b) 45 XY
- c) 69 XY
- d) 69 XXY

Question 2:

Which of the following statements is false regarding hydatidiform moles?

- a) Complete moles show diffuse trophoblastic hyperplasia
- b) Fetal tissue is seen in partial mole
- c) Early USG is sensitive for partial moles
- d) Complete moles have higher malignant potential

Question 3:

Which of the following is not associated with the development of a hydatidiform mole?

- a) Maternal age ≥ 40 years
- b) Previous history of GTD
- c) AB blood group
- d) Vitamin C deficiency

Question 4:

A 20-year-old primigravida with bleeding PV was diagnosed to have a complete hydatidiform mole. Which of the following is not true in this condition?

- a) Serum β -hCG levels are highly elevated
- b) Urine pregnancy test is negative

- c) Theca lutein cysts are rare
- d) Serum free T4 levels normalise after uterine evacuation

Question 5:

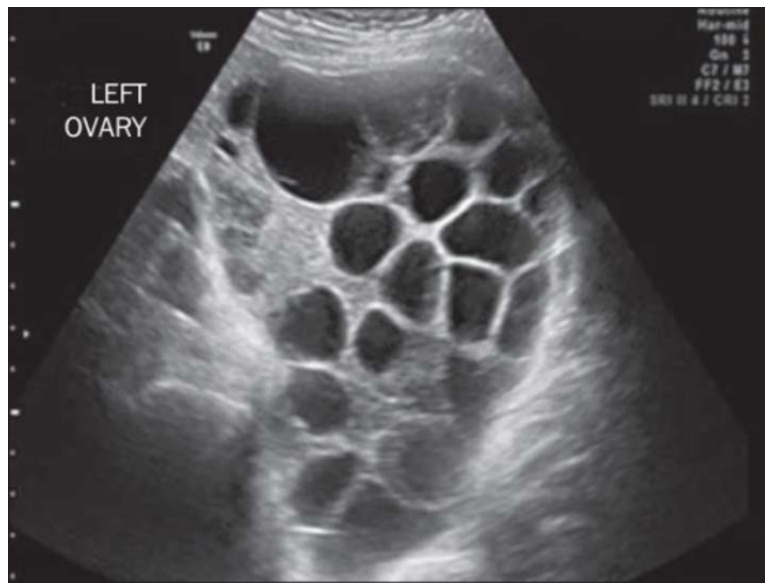
A pregnant woman at 12 weeks of gestation presents with spotting PV, hyperemesis, and lower abdominal pain for the past 2 days. USG done is shown below. What is the most probable diagnosis?



- a) Missed abortion
- b) Complete mole
- c) Partial mole
- d) Choriocarcinoma

Question 6:

A primigravida at 12 weeks of gestation presented with complaints of the expulsion of grape-like vesicles PV for the past day. A diagnosis of molar pregnancy was made on USG examination. Incidentally, the following adnexal lesion was also noted. What is the lesion?



- a) Mucinous cystadenoma
- b) Multilocular endometrial cysts
- c) Theca lutein cyst
- d) Polycystic ovarian disease

Question 7:

A patient who was treated for molar pregnancy came for follow-up after 4 months. Her USG revealed a large, twisted theca lutein cyst. How will you manage this patient?

- a) Untwisting of the cyst
- b) Laparoscopic oophorectomy
- c) Wait and watch
- d) Transabdominal aspiration under USG guidance

Question 8:

A pregnant woman at 14 weeks of gestation complains of excessive vomiting and mild vaginal bleeding for 2 weeks. On examination, fundal height corresponds to 16 weeks and fetal heart sounds are absent. USG shows a snowstorm appearance and no fetus. What is the best step of management?

- a) Chemotherapy
- b) Hysterectomy
- c) Suction and evacuation

d) Serum beta hCG

Question 9:

A 36-year-old woman underwent suction evacuation for a hydatidiform mole. Which of the following statements is true regarding her follow-up with β -hCG estimation?

- a) Start estimation 72 hours following evacuation**
- b) Twice weekly estimation for a month following evacuation**
- c) Monthly follow-up for 6 months after three consecutive normal values**
- d) After 6 months of follow-up, regular yearly estimation should be done**

Question 10:

Which of the following are complications that can be seen in a patient with molar pregnancy?

- a) 1, 2, and 5**
- b) 1, 2, 3 and 5**
- c) 2, 3 and 4**
- d) 1, 3, 4 and 5**

Question 11:

Gestational trophoblastic neoplasm does not include

- a) Partial mole**
- b) Choriocarcinoma**
- c) Placental site trophoblastic tumor**
- d) Invasive mole**

Question 12:

All of the following are non-molar trophoblastic neoplasms except _____.

- a) Choriocarcinoma**
- b) Placental site trophoblastic tumor**
- c) Invasive mole**

d) Epithelioid trophoblastic tumor

Question 13:

What is the most common non-metastatic gestational trophoblastic neoplasia that develops following hydatidiform mole?

- a) Choriocarcinoma
- b) Placental site trophoblastic tumour
- c) Choriodenoma destruens
- d) Epithelioid trophoblastic tumour

Question 14:

Which of the following are risk factors for malignant change in a patient with molar pregnancy?

- a) 1, 2, 5
- b) 2, 3, 4
- c) 1, 2, 4
- d) 1, 3, 4

Question 15:

All of the following can be indicators of choriocarcinoma except:

- a) Increased number of theca lutein cysts
- b) Persistently enlarged uterus
- c) Plateau of high serum hCG
- d) Suburethral nodule

Question 16:

A patient with a history of molar pregnancy was treated with suction and evacuation. Which method of contraception would you not recommend for her?

- a) Barrier methods
- b) Intrauterine contraceptive device

- c) Oral contraceptive pills
- d) Depot medroxyprogesterone acetate

Question 17:

What will be the WHO score in a patient with low-risk gestational trophoblastic neoplasia?

- a) 0-3
- b) 0-5
- c) 0-6
- d) 0-7

Question 18:

A patient was diagnosed to have choriocarcinoma two years post-evacuation of molar pregnancy. Which of the following is the most common site of metastasis in this patient?

- a) Vagina
- b) Kidney
- c) Liver
- d) Ovaries

Question 19:

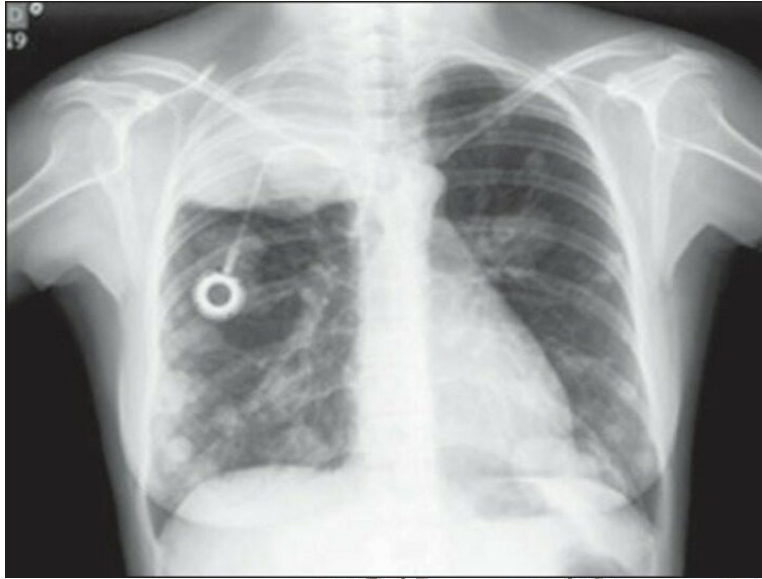
A case of gestational trophoblastic neoplasm is detected to have metastasis in the lungs. It is considered as stage _____.

- a) I
- b) II
- c) III
- d) IV

Question 20:

A 43-year-old woman presents with complaints of hemoptysis, vaginal bleeding, and abdominal pain for 7 days. Chest X-ray is shown below. Transabdominal USG revealed a solid mass occupying the uterus which was found to be choriocarcinoma with a WHO score of 8.

How would you manage this patient?



- a) Hysterectomy
- b) R-CHOP
- c) EMA-CO
- d) MVAC

Question 21:

Placental-site trophoblastic tumor arises from _____.

- a) Cytotrophoblast
- b) Syncytiotrophoblast
- c) Intermediate trophoblast
- d) Chorionic villi

Question 22:

A 26-year-old woman with a history of early miscarriage presented with amenorrhoea. Her serum beta hCG levels were persistently low and USG revealed a single solid mass in the uterine fundus. Histopathological examination of the biopsy showed abundant intermediate trophoblastic cells. What is the next step of management?

- a) Methotrexate
- b) Actinomycin D

- c) EMA-CO regimen
- d) Hysterectomy

Answer Key

Question No.	Correct Option
1	d
2	c
3	d
4	c
5	b
6	c
7	a
8	c
9	c
10	d
11	a
12	c
13	c
14	a
15	a
16	b
17	c
18	a
19	c
20	c
21	c
22	d

Detailed Explanations

Solution to Question 1:

Partial hydatidiform mole has a 69XXY or 69XYY karyotype.

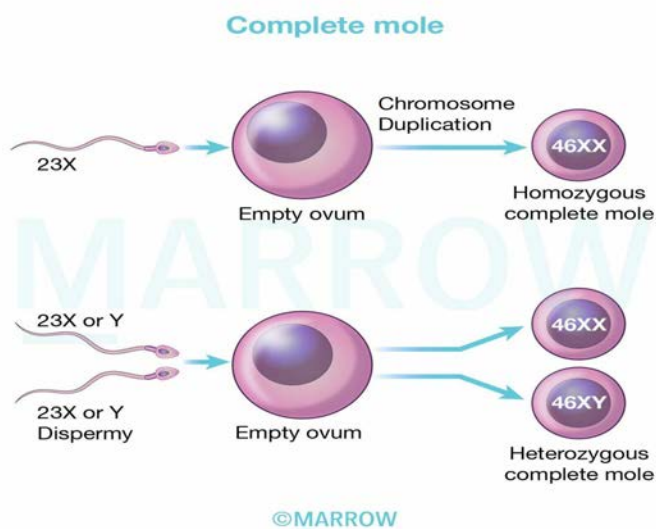
Hydatiform mole belongs to the benign group of gestational trophoblastic diseases (GTD). It is an abnormality in chorionic villi making them partly proliferative and degenerative, thereby resulting in the formation of multiple small cysts. Based on the portion of the chorionic villi involved, they could be complete (diffused) or partial (focal)

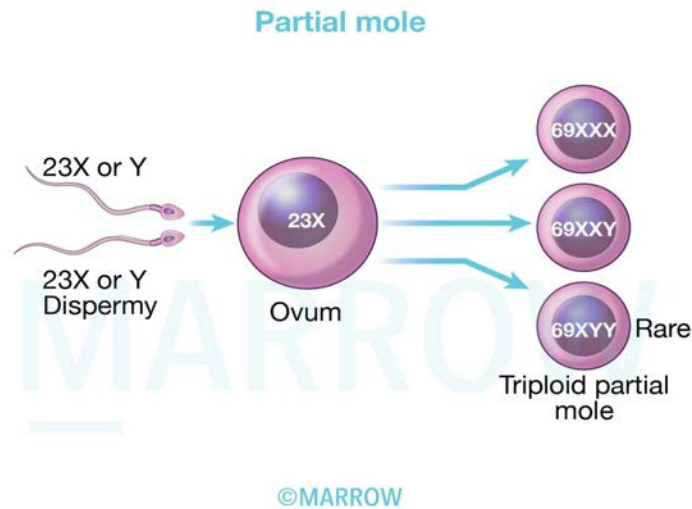
Solution to Question 2:

Early ultrasonography is not highly sensitive in a patient with partial molar pregnancy. However, USG features like thickened, multicystic placenta along with fetal tissue or even a fetus may be seen.

Hydatidiform mole or molar pregnancy is an abnormal condition of the placenta. It is characterized by trophoblastic proliferation and stromal edema. They are classified into complete and partial moles based on the absence or presence of embryonic elements.

Pathogenesis of complete and partial moles are as follows:





Solution to Question 3:

Vitamin A deficiency, not Vitamin C deficiency is a risk factor for hydatidiform mole.

Risk factors for hydatidiform mole are:

- Maternal age - pregnancy in teenage and women >35 years
- Previous history of gestational trophoblastic disease
- Dietary deficiency - vitamin A deficiency, inadequate protein and animal fat intake
- AB blood group.

Solution to Question 4:

Theca lutein cysts are rare in partial molar pregnancy, but may be seen in 25-30% of patients with complete molar pregnancy.

Serum β -hCG levels are elevated more than expected for gestational age in a complete mole.

The sensitivity of most hCG tests is set to the pregnancy hCG range. An excessive β -hCG hormone causes an oversaturation of the test assay giving a false negative UPT. This is called the hook effect.

The thyrotropin-like effects of hCG frequently cause serum free-T₄ levels to be elevated which normalize after uterine evacuation.

Solution to Question 5:

The given clinical scenario and USG showing snowstorm appearance is suggestive of a complete mole.

The snowstorm appearance is due to an echogenic uterine mass that has numerous anechoic cystic spaces. Fetus and amniotic sac are absent.

Investigations for molar pregnancy are:

- USG - shows snowstorm appearance
- Serum β hCG - elevated to ≥ 2 MoM (multiples of median) for corresponding gestational age
- Histological examination of products of conception

Snowstorms in radiology:

- Complete hydatiform mole (ultrasound)
- Extracapsular breast implant rupture (ultrasound)
- Carcinoma thyroid - pulmonary metastases (chest X-ray).

Solution to Question 6:

The given USG image shows theca lutein cysts in the ovary. These cysts are usually seen in patients with complete molar pregnancies. They are usually bilateral and are also seen in multiple pregnancies.

Theca lutein cysts result from the overstimulation of lutein elements by massive amounts of β hCG. The cystic fluid is rich in chorionic gonadotropin, estrogen, and progesterone.

Solution to Question 7:

Untwisting of the cyst is the only management in an asymptomatic patient with a twisted theca lutein cyst.

Laparoscopic oophorectomy is done only when there is associated extensive necrosis or hemorrhage.

Theca-lutein cysts regress within 2 months after pregnancy evacuation, and hence expectant management is preferred in uncomplicated cases. If the cysts are symptomatic, they can be aspirated trans-abdominally under ultrasound guidance.

Solution to Question 8:

The given clinical scenario is suggestive of molar pregnancy which is managed by suction and evacuation, regardless of uterine size.

If the cervix is unfavorable, slow dilatation is done with vaginal PGE₁. Volume resuscitation with IV fluids and parenteral antibiotics can be given as supportive therapy. Blood transfusion is done

if the patient is anaemic.

Intraoperative sonography is recommended to help ensure that the uterine cavity has been emptied.

Indication for hysterectomy in molar pregnancy:

- Women who have completed their family
- Age $\geq$ 40 years
- Uncontrolled hemorrhage
- Perforation during surgical evacuation.

Solution to Question 9:

In the given scenario, the patient is followed up with serum β hCG every month for 6 months.

Follow-up after the termination of molar pregnancy is done with serum β hCG levels. During the follow-up period, pregnancy must be avoided as it might interfere with the β hCG levels.

The initial β -hCG level is obtained within 48 hours after evacuation and used as a baseline.

It is measured weekly till it becomes negative, and three consecutive normal values are obtained. This normally happens in 4-8 weeks.

After this, β hCG levels are measured monthly for 6 months. Women who undergo chemotherapy must be followed up for 1 year.

Solution to Question 10:

Thyrotoxicosis is seen as a complication of molar pregnancy, not hypothyroidism.

Complications in molar pregnancy include:

- Immediate:
- Early-onset pre-eclampsia - molar placentas produce more antiangiogenic proteins than normal placentas.
- Thyrotoxicosis - as α component of hCG mimics TSH and causes thyroid storm
- Acute pulmonary insufficiency - due to embolization of trophoblastic cells
- Electrolyte imbalance i.e., hypokalemia, hyponatremia - due to hyperemesis caused by β hCG
- Shock - hemorrhage due to the separation of vesicles and intra-peritoneal haemorrhage
- Disseminated intravascular coagulation
- Late: Choriocarcinoma.

Solution to Question 11:

A partial mole is not a gestational trophoblastic neoplasm (GTN).

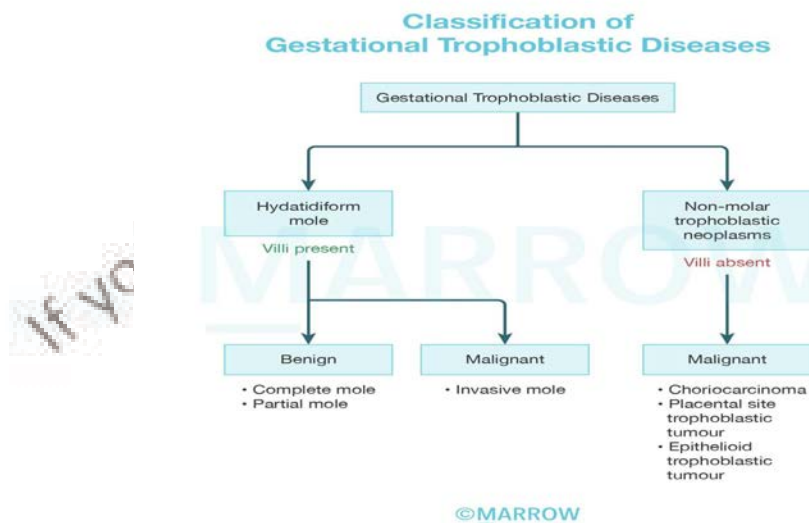
Gestational trophoblastic disease (GTD) is the term used to describe a group of tumors due to abnormal trophoblast proliferation. Malignant forms of GTD are termed gestational trophoblastic neoplasia (GTN).

Modified WHO Classification of GTD

Molar Pregnancies	Trophoblastic tumors
Hydatidiform mole (benign) - Partial and Complete Invasive mole (malignant)	Choriocarcinoma (malignant) Placental site trophoblastic tumor (malignant) Epithelioid trophoblastic tumor (malignant)

Solution to Question 12:

Invasive mole is a molar trophoblastic neoplasm.

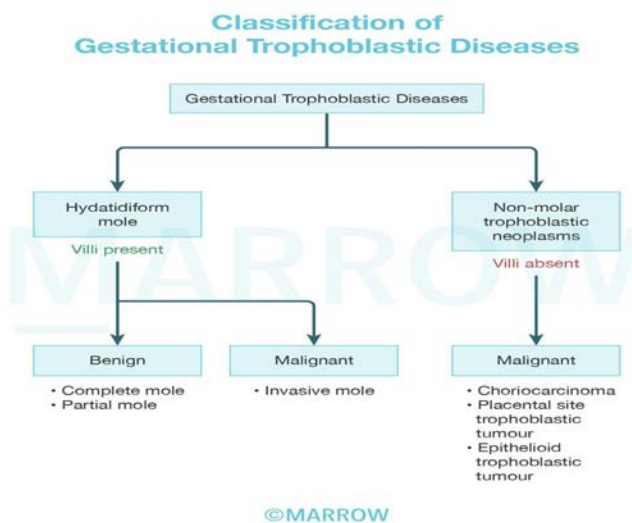


Solution to Question 13:

Chorioadenoma destruens or invasive mole is the most common non-metastatic GTN that follows hydatidiform moles. Almost all invasive moles arise from partial or complete moles.

Invasive mole is characterized by extensive tissue invasion by trophoblast and whole villi. They can penetrate deep into the myometrium, peritoneum, parametrium, or vaginal vault. They are locally aggressive and do not usually metastasize.

The most common GTN following term pregnancy or miscarriage is choriocarcinoma.



Solution to Question 14:

Maternal age ≥ 40 years, β -hCG levels $\geq 100,000$ mIU/mL and grand multiparity are risk factors for malignant change in a patient with molar pregnancy.

Risk factors for malignant change are:

- Patient's age ≥ 40 years or ≤ 20 years
- Parity ≥ 3
- Uterine size ≥ 20 weeks
- Previous history of molar pregnancy
- Complete moles have 15-20% risk of malignant transformation, partial moles have 1-5% risk
- Serum β hCG $\geq 1,00,000$ mIU/mL
- Large theca lutein cysts ≥ 6 cm in diameter

Post-evacuation uterine sonographic surveillance can be used as a predictor of subsequent neoplasia. Findings indicative of malignant transformation are:

- Suburethral nodules
- Myometrial nodules
- Hypervascularity.

Solution to Question 15:

Increased number of theca lutein cysts is not an indicator of choriocarcinoma. Increased size or persistence of theca lutein cysts indicates choriocarcinoma.

Plateau of high serum β -hCG level for four measurements for ≥ 3 weeks is an indicator of choriocarcinoma.

Risk factors for malignant change are:

- Molar pregnancy - complete moles have 15-20% risk, partial moles have 1-5% risk
- Patient's age ≥ 40 years or < 20 years
- Parity ≥ 3
- Serum β hCG $> 1,00,000$ IU/mL
- Uterine size > 20 weeks
- Previous history of molar pregnancy
- Large theca lutein cysts > 6 cm in diameter

Post-evacuation uterine sonographic surveillance can be used as a predictor of subsequent neoplasia. Findings indicative of malignant transformation are:

- Suburethral nodules
- Myometrial nodules
- Hypervascularity.

Solution to Question 16:

IUCD (intrauterine contraceptive device) is contraindicated in this patient as the uterus is friable and can cause further bleeding.

Accepted methods of contraception after suction evacuation in molar pregnancy include:

- Oral contraceptive pills
- Depot medroxyprogesterone acetate
- Barrier methods.

Solution to Question 17:

Low-risk GTN has a WHO score of 0 to 6. High-risk GTN has a WHO score of ≥ 7 .

Features	Low risk	High risk
Total score	0-6	≥ 7
Age	≤ 39 years	≥ 40 years
hCG	≤ 105 IU/L	≥ 105 IU/L

Features	Low risk	High risk
Type of antecedent pregnancy	Molar pregnancy	Full term pregnancy
Duration of antecedent pregnancy	<4 months	>12 months
Metastasis site	Lungs, vagina	Brain, liver
Number of metastases	<4	>8
Size of the tumor	<3 cm	>5 cm
Antecedent chemotherapy	Single-agent chemotherapy	Multiple-agent chemotherapy

Solution to Question 18:

The common sites of metastases of choriocarcinoma are in the following order:

- Lungs - most common
- Vagina - second most common
- Vulva
- Kidneys
- Liver
- Ovaries
- Brain
- Bowel

Lung metastases present as cannon-ball or snowstorm appearance on chest X-ray. Vaginal metastases present as purplish sub-urethral nodules.

Solution to Question 19:

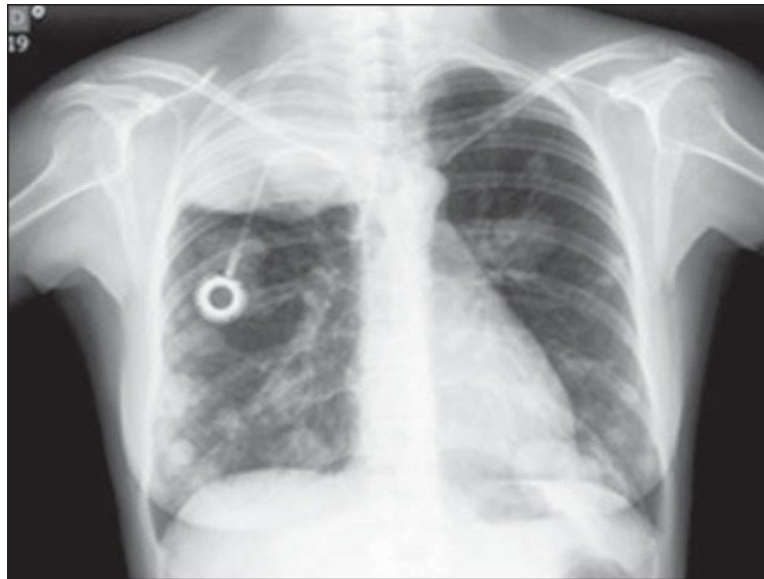
Metastases in the lungs are considered stage III GTN (gestational trophoblastic neoplasia) in FIGO classification.

International Federation of Gynecology and Obstetrics (FIGO) staging for GTN is as follows:

The chest X-ray below shows lung metastasis - stage III GTN.

Stage	Description
I	GTN confined to the uterus

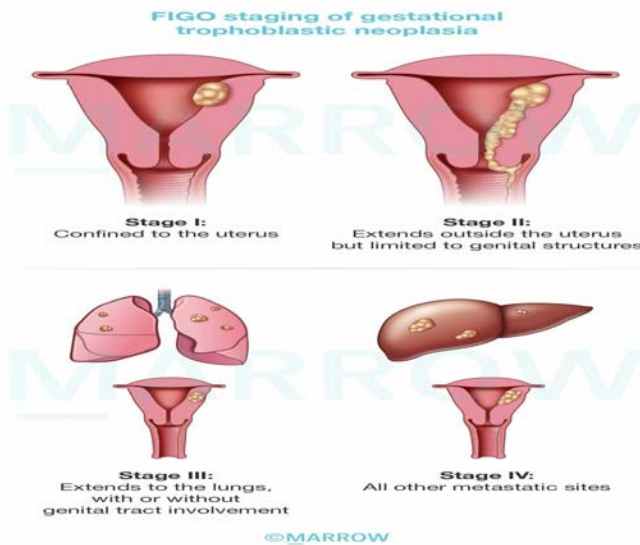
Stage	Description
II	GTN extends outside of the uterus but is limited to the genital structures (adnexa, vagina, broad ligament)
III	GTN extends to the lungs, with or without known genital tract involvement
IV	Allothermetastatic sites i.e. kidney, liver, brain, bowel, etc.



Gestational trophoblastic neoplasia (GTN) refers to the aggressive invasion of the endometrium and myometrium by trophoblast cells. It includes invasive mole, gestational choriocarcinoma, placental site trophoblastic tumors, and epithelioid trophoblastic tumors. GTN is diagnosed based on elevated β -hCG levels. FIGO criteria for diagnosis of GTN:

- Persistence of β -hCG level plateau in 4 measurements for 3 weeks or longer
- Rise in β -hCG levels in 3 or more consecutive measurements done weekly, over a period of 2 weeks or longer
- Elevated β -hCG level for 6 months or more
- Histologic diagnosis of choriocarcinoma

Treatment for low-risk disease is chemotherapy with methotrexate, while those with high-risk disease are treated with etoposide, methotrexate, and dactinomycin alternating with cyclophosphamide and vincristine (EMA/CO) chemotherapy.



Solution to Question 20:

The given clinical scenario is suggestive of high-risk (WHO) stage III choriocarcinoma which is treated with EMA-CO regimen. The given X-ray shows numerous, dense, bilateral metastatic nodules, of varying sizes indicative of lung metastases.

The components of EMA-CO are:

- Etoposide
- Methotrexate
- Actinomycin D
- Cyclophosphamide
- Oncovin i.e. vincristine

Option A: PSTT (placental-site trophoblastic tumor) and ETT (epithelioid trophoblastic tumors) are treated by hysterectomy only, as they are resistant to chemotherapy.

Option B: R-CHOP regimen is used for B-cell non-Hodgkin's lymphoma.

Option D: MVAC regimen is used for advanced bladder carcinoma.

Risk	Criteria	Treatment
Low	FIGO stage I GTN FIGO Stage II or III GTN with a WHO risk score < 6	Single agent chemotherapy with Methotrexate or Actinomycin D.
High	Stages II and III with WHO risk score > 7 Stage IV disease	EMA-CO regimen

Solution to Question 21:

Placental-site trophoblastic tumor arises from implantation site intermediate trophoblast. As syncytiotrophoblastic cells are generally absent in this site, it is composed mainly of cytotrophoblastic cells.

Patients with placental site trophoblastic tumors present with vaginal bleeding and modestly elevated serum β -hCG levels. Local invasion into the myometrium and lymphatics occurs. Hysterectomy is the preferred treatment.

Solution to Question 22:

The given clinical scenario is suggestive of placental-site trophoblastic tumor (PSTT). Hysterectomy is the treatment of choice because they are locally invasive tumors and resistant to chemotherapy.

PSTT is a rare tumor arising from intermediate trophoblastic of the placental bed. Patients with placental site trophoblastic tumors present with vaginal bleeding and modestly elevated serum β -hCG levels. They produce variant forms of hCG.

Anemia in Pregnancy

Question 1:

A pregnant woman presented to the clinic with complaints of dizziness. On examination, pallor was present. She will be considered severely anemic if her hemoglobin is less than _____.

- a) 4 g/dl
- b) 5 g/dl
- c) 7 g/dl
- d) 7.9 g/dl

Question 2:

Which of the following is the second most common pathological cause of anemia in pregnancy?

- a) Anemia associated with chronic disease
- b) Sickle cell anemia
- c) Iron-deficiency anaemia
- d) Megaloblastic anemia

Question 3:

What is the average maternal requirement of iron during a singleton pregnancy?

- a) 300 mg
- b) 1000 mg
- c) 400 mg
- d) 800 mg

Question 4:

You initiate iron and folic acid supplements for a woman at 13 weeks of gestation. According to the ministry of health & family welfare (MoHFW), what is the minimum duration for

which she should take these supplements?

- a) 150 days
- b) 180 days
- c) 120 days
- d) 90 days

Question 5:

Identify the correctly matched dose for folic acid supplementation that should be given in pregnant women with the following conditions?

- a) Sickle cell anaemia - 5 mg
- b) Previous child with neural tube defect - 1 mg
- c) Thalassemia - 500 mcg
- d) Epilepsy - 8 mg

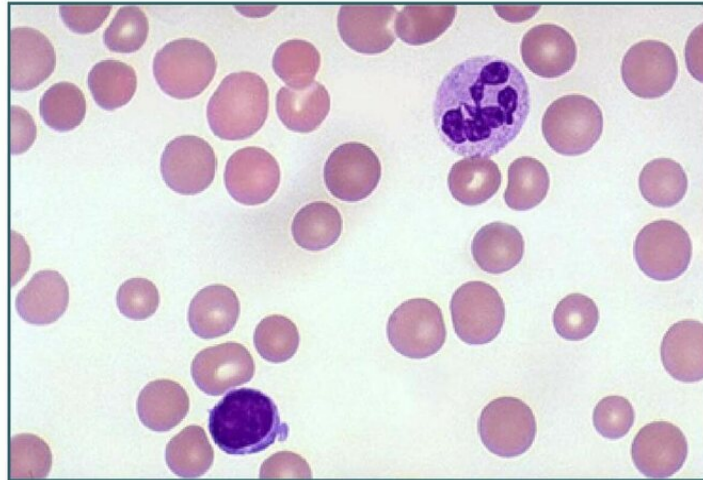
Question 6:

A pregnant woman at 10 weeks of gestation presented with insomnia due to a compulsive urge to keep moving her legs. Blood tests revealed a Hb value of 10.1 g/dl. Which of the following is the most sensitive indicator for her underlying condition?

- a) Serum iron levels
- b) Serum ferritin
- c) Serum transferrin
- d) Serum erythropoietin

Question 7:

A pregnant teacher presented with complaints of difficulty concentrating at work. Investigations revealed normal MCV with blood picture as shown below. What is the most likely diagnosis?



- a) Iron deficiency anaemia
- b) Dimorphic anemia
- c) Sickle cell anaemia
- d) Folic acid deficiency

Question 8:

A woman at 36 weeks of gestation was brought to the clinic by her concerned husband. He mentioned that over the last few weeks he noticed her chewing on cardboard pieces and ice cream sticks on multiple occasions. On examination, pallor was present and Hb was 8.6 g/dl. What is the most appropriate treatment regimen?

- a) Ferrous sulfate
- b) Red cell transfusion
- c) Ferrous sucrose
- d) Ferrous ascorbate

Question 9:

Which of the following is the most effective parenteral iron preparation prescribed for postpartum women with iron deficiency anemia?

- a) Ferrous sucrose
- b) Ferrous biglycinate
- c) Ferric carboxymaltose

d) Iron dextran

Question 10:

You have been asked to start parenteral iron infusion for a woman at 38 weeks of gestation. What formula would you use to calculate the required dosage?

- a) $2.4 \times (\text{target Hb} - \text{actual Hb}) \times \text{body weight (kg)}$**
- b) $2.4 \times (\text{target Hb} - \text{actual Hb}) \times \text{body weight (kg)} + 500 \text{ mg}$**
- c) $2.4 \times (\text{target Hb} - \text{actual Hb}) \times \text{body weight (kg)} + 500 \text{ mcg}$**
- d) $2.4 \times (\text{target Hb} - \text{actual Hb}) \times \text{body weight (kg)} + 50 \text{ mg/kg}$**

Question 11:

Which of the following is true regarding parenteral iron therapy for iron deficiency anemia in pregnant women?

- a) Can be given in first trimester**
- b) Avoided in chronic liver disease**
- c) Sensitivity test is mandatory before administration**
- d) Can be given alongside oral iron therapy**

Question 12:

A G3P2 female patient who is an agricultural laborer presents with complaints of fatigue at 36 weeks of gestation. On examination, pallor and mild pedal edema are present. Her hemoglobin is 6.7 g/dl. How will you manage this patient?

- a) Iron sucrose infusion**
- b) Ferric carboxymaltose infusion**
- c) Packed cell transfusion**
- d) Iron dextran infusion**

Question 13:

A pregnant woman was brought by her husband after fainting at home. He gave a history of her being clumsy and forgetful over the last few weeks. Investigations revealed high MCV

with hypersegmented neutrophils. How will you manage this patient?

- a) 100 µg of IM cyanocobalamin
- b) 1000 µg of IM cyanocobalamin + folic acid
- c) 400mcg of folic acid
- d) 500 µg of Oral cyanocobalamin

Question 14:

An anxious primigravida with sickle cell trait comes to you for her first antenatal checkup. Which of the following is she at a higher risk of developing during the course of this pregnancy?

- a) 4 & 5
- b) 1, 3, & 5
- c) Only 3
- d) 2 & 4

Question 15:

A sexually active woman with sickle cell anemia comes to the clinic for advice regarding contraception. Which of the following is the ideal contraceptive for this patient?

- a) Progesterone implant
- b) Oral contraceptive pills
- c) Copper IUD
- d) Barrier methods

Question 16:

A pregnant woman with thalassemia major presents to the clinic at 24 weeks of gestation with fatigue and shortness of breath. Which of the following is not true regarding the management of this condition?

- a) Parenteral iron therapy is the treatment of choice
- b) Oral iron therapy is contraindicated
- c) Breast feeding is safe after delivery
- d) Repeated transfusions are necessary to maintain Hb levels above 10 g/dl

Answer Key

Question No.	Correct Option
1	c
2	a
3	b
4	b
5	a
6	b
7	b
8	c
9	c
10	b
11	b
12	c
13	b
14	c
15	a
16	a

Detailed Explanations

Solution to Question 1:

Severe anemia is defined as Hb $\leq$ 7 g/dL.

Anemia in pregnancy is said to occur when Hb level is $\leq$ 11 g/dL. It is classified as shown below:

Hemoglobin values (g/dL)	WHO classification of anemia in pregnancy	ICMR classification of anemia in pregnancy
≥ 11	Normal In the second trimester, Hb ≥ 10.5 g/dL is considered normal	Normal
10-10.9	Mild	Mild
7-9.9	Moderate	Moderate
< 7	Severe	Severe

Hemoglobin values (g/dL)	WHO classification of anemia in pregnancy	ICMR classification of anemia in pregnancy
<4	-	Very severe

Solution to Question 2:

The second most common pathological cause of anemia during pregnancy is anemia associated with chronic disease. The most common pathological cause of anemia during pregnancy is iron deficiency anemia.

Women with chronic disorders may develop anemia for the first time during pregnancy. Causes include:

- Chronic renal insufficiency - the most common cause. Recombinant erythropoietin is used for treating anemia due to chronic renal failure.
- Inflammatory bowel disease
- Connective tissue disorders
- Granulomatous infections
- Malignant neoplasms
- Rheumatoid arthritis
- Chronic suppurative conditions.

Note: The most common cause of anemia in pregnancy is physiological. It is normally seen during pregnancy. This occurs due to the expansion of plasma volume by 30–40%. This increase is much more than the increase in red cell mass (20-30%) during pregnancy. This leads to hemodilution and an apparent decrease in hemoglobin levels.

Solution to Question 3:

Maternal requirement of iron during a singleton gestation averages close to 1000 mg.

The average iron stores in a non-pregnant female is around 300 mg. This falls short of the requirement of iron during pregnancy which is 1000 mg. Hence iron supplementation is necessary to prevent iron deficiency anemia during pregnancy. This also forms the basis for prophylactic iron supplementation during pregnancy.

Note: The Federation of Obstetrics and Gynaecology (FOGSI) states that the iron requirement in pregnancy is 840 mg, as about 160 mg of iron is saved due to amenorrhea during pregnancy.

Iron requirement during pregnancy	
Fetus and placenta	300 mg

Iron requirement during pregnancy	
Expansion of maternal hemoglobin mass	500 mg
Losses through normal excretion routes	200 mg
Total	1000 mg

Solution to Question 4:

The guideline recommends 60 mg of elemental iron and 500 µg of folic acid daily for at least 180 days, starting from the 4th month of pregnancy, and to be continued throughout pregnancy. Ideally, this must be continued until 180 days postpartum.

Solution to Question 5:

The amount of folic acid to be given to pregnant women suffering from sickle cell anemia is 5 mg. This is to reduce the risk of neural tube defects and to compensate for the increased demand for folate during pregnancy.

Normally in all sickle cell patients (irrespective of age or sex), 1mg of folic acid is given to prevent folate deficiency.

Note: William's Obstetrics 26th Edition mentions folic acid 4 mg daily for pregnant women with sickle cell disease. However various other sources and RCOG guidelines mention it as 5 mg daily. Hence this is the best answer.

Solution to Question 6:

In the given scenario, history suggestive of restless leg syndrome and Hb of 10.1g/dL points to a diagnosis of iron deficiency anemia. Serum ferritin is the most sensitive indicator of iron deficiency anemia in pregnant women.

This is because its concentrations are not affected by the recent intake of iron-rich foods. Serum ferritin concentration $<15 \mu\text{g/L}$ indicates iron deficiency anemia.

However, treatment needs to be initiated when the concentration falls $<30 \mu\text{g/L}$, as this indicates early iron depletion.

Restless leg syndrome also called Willis-Ekbom Disease, causes unpleasant or uncomfortable sensations in the legs along with an irresistible urge to move them. It may be associated with iron deficiency anemia.

Solution to Question 7:

In the given clinical scenario presence of microcytic hypochromic RBCs indicates iron deficiency anemia, and hyper segmented neutrophils indicate anemia due to folic acid or vitamin B12 deficiency. This is also known as dimorphic anemia.

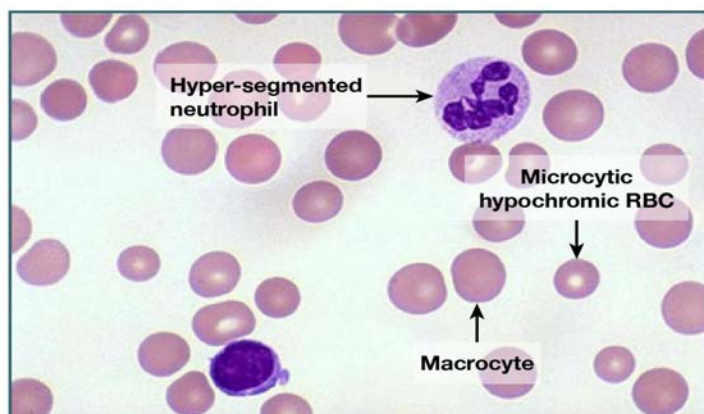
Blood picture in iron deficiency anemia:

- Microcytic hypochromic RBCs
- Typical pencil cells
- Presence of poikilocytes (pre-erythrocytes)

Blood picture in folic acid/vitamin B12 deficiency anemia:

- Hyper segmented neutrophils - earliest change
- Macrocytic RBCs

Dimorphic anaemia - iron deficiency and folic acid deficiency



Solution to Question 8:

The given scenario is suggestive of pica associated with moderate iron-deficiency anemia. Since the anemia was diagnosed late in pregnancy, the most appropriate treatment is parenteral iron therapy. Ferrous sucrose is a form of parenteral iron given as an IV infusion.

If anemia is diagnosed:

- In the first trimester - Oral iron therapy is preferred
- In the second trimester - Oral iron therapy should be switched over to parenteral iron if Hb falls $< 8\text{g/dl}$ or if serum ferritin is $< 15\text{ng/ml}$
- In the third trimester - Parenteral iron therapy is preferred

Ferrous sucrose is the most commonly used preparation for IV infusion. It is safe and does not require a test dose.

Other iron preparations include:

- Oral preparations- Ferrous sulfate, ferrous gluconate, ferrous ascorbate
- IM preparations- Iron sorbitol, Iron polymaltose
- IV preparations- Ferric carboxymaltose
- Both IM and IV- Iron dextran.

Solution to Question 9:

Ferric carboxymaltose is considered superior to all other parenteral iron supplementation especially for iron deficiency anemia in postpartum women.

Ferric carboxymaltose (FCM) is a dextran-free IV iron preparation. It allows rapid administration of high doses of iron (up to 1000 mg iron in 15 min).

It is safe and effective in improving the mean Hb level. It avoids the need for blood transfusions.

Solution to Question 10:

The formula for calculating dose of parenteral iron is:

Total dose in mg = $2.4 \times (\text{Target Hb} - \text{Actual Hb}) \times \text{Body Wt.} + 500 \text{ mg}$

- 2.4 - conversion factor
- Target Hb - Hb you want to achieve
- Actual Hb - level of current Hb
- 500 mg - additional amount to replenish the iron stores

Iron sucrose must be diluted only using normal saline at a concentration of 1-2 mg of iron/mL of saline. It is given over 20-30 minutes intravenously.

Solution to Question 11:

Parenteral iron therapy for iron deficiency anemia in pregnancy is avoided if the patient has a history of chronic liver disease.

It is used as an alternative therapy in people who cannot tolerate oral iron.

Contraindications include:

- A history of prior anaphylactic reaction to parenteral iron therapy

- First trimester of pregnancy - there is no evidence on the use of IV iron in the first trimester of pregnancy
- Chronic liver disease
- Active infection (acute or chronic)

A sensitivity test prior to the infusion is not mandatory. IV iron should be given with caution even if previous administrations have been well tolerated.

Oral iron should be stopped at least 24 hours prior to therapy to avoid iron toxicity.

Solution to Question 12:

The treatment of choice for severe anemia ($< 7\text{g/dL}$) in the third trimester is packed cell transfusion. It is preferred over whole blood as it avoids volume overload.

Indications of blood transfusion in pregnancy include:

- Pregnancy < 34 weeks
- Hb $< 5\text{ g/dL}$ with or without signs of cardiac failure or hypoxia
- Hb $5-7\text{ g/dL}$ – in presence of impending heart failure
- Pregnancy > 34 weeks - Hb $< 7\text{ g/dL}$

Solution to Question 13:

The given scenario points to vitamin B12 deficiency. It is treated by vitamin B12 injections with folic acid supplementation. $1000\text{ }\mu\text{g}$ of IM cyanocobalamin should be given initially every week for 6 weeks, following which it should be given once every month.

Megaloblastic anemia caused by Vit B12 deficiency should be identified and treated accordingly because folic acid supplementation alone will not help.

Causes of Vit B12 deficiency leading to anemia in pregnancy:

- Gastric resection
- Addisonian pernicious anemia
- Crohn's disease
- Ileal resection
- Bacterial overgrowth in the small bowel.

Solution to Question 14:

Among the given options, patients with sickle cell trait are at a higher risk of developing urinary tract infections.

Sickle cell disease, on the other hand, is associated with increased risk of :

- Miscarriages
- Fetal growth restriction
- Preterm labor
- Venous thromboembolism
- Pre-eclampsia
- Placental abruption
- Fetal distress in labor

Note: Sickle cell trait is a benign condition in which the affected person has only one abnormal hemoglobin beta gene allele whereas, in sickle cell disease, the affected person has two abnormal hemoglobin beta gene alleles and is hence severely symptomatic. Unlike individuals with sickle cell disease, those with sickle cell trait do not have symptoms related to sickling and mostly lead a normal life.

Solution to Question 15:

Low-dose oral progestins or progesterone injections or implants are ideal contraceptives in a patient with sickle cell anemia. Progesterone has been known to prevent painful sickle cell crises.

It is considered very safe in pregnancy. It is rated level 1 indicating that there is no restriction in their use.

Contraceptive methods and their safety profile in sickle cell anemia:

Level 1 contraceptives (no restriction):

- Progestogen-only injection (DMPA)
- Levonorgestrel intrauterine system (Mirena)
- Emergency contraception containing progestins

Level 2 contraceptives (advantages outweigh disadvantages):

- Combined oral contraceptive
- Copper intrauterine device.

Solution to Question 16:

In thalassemia major, both oral and parenteral iron preparations are avoided. This is because these patients are already in a state of iron overload due to multiple blood transfusions.

Pregnancy in patients with thalassemia major is very rare. In case pregnancy occurs the following strategies are adopted:

- Iron chelating agents should be stopped

- 5mg folic acid supplementation should be given daily
- Pneumococcal vaccination and boosters are advised prior to pregnancy
- Repeated blood transfusions are given to maintain Hb levels above 10 g/dL
- Breastfeeding is safe and iron chelation therapy can be resumed post-delivery.

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If you purchased this from someone else,
you may have been scammed.

Hypertensive Disorders in Pregnancy

Question 1:

Gestational hypertension is defined as:

- a) BP $\geq$ 130/80 after 22 weeks of gestation
- b) BP $\geq$ 130/80 after 20 weeks of gestation
- c) BP $\geq$ 140/90 after 22 weeks of gestation
- d) BP $\geq$ 140/90 after 20 weeks of gestation

Question 2:

What is the minimum cut-off for defining proteinuria in a random urine sample?

- a) 0.3 g/L
- b) 0.2 g/L
- c) 0.1 g/L
- d) 0.4 g/L

Question 3:

In a normal pregnancy, when does the remodeling of spiral arteries begin?

- a) 6 weeks
- b) 16 weeks
- c) 14 weeks
- d) 12 weeks

Question 4:

A 28-year-old G2P1 patient at 17 weeks of gestation comes for a routine antenatal visit. Her blood pressure is 142/94 mm of Hg. A repeat blood pressure after 4 hours is measured to be 148/92 mm Hg. What is the most likely diagnosis?

- a) Preeclampsia
- b) Preeclampsia superimposed on chronic hypertension

- c) Chronic hypertension
- d) Gestational hypertension

Question 5:

Which of the following does not put a patient at an increased risk for preeclampsia?

- a) Twin pregnancy
- b) Family history of hypertension
- c) New paternity
- d) Smoking

Question 6:

Which of the following does not occur in patients developing pre-eclampsia?

- a) Abnormal trophoblastic invasion of uterine vessels
- b) Maladaptive immunological tolerance to fetal antigens
- c) Maladaptation to cardiovascular changes of normal pregnancy
- d) Increased placental perfusion due to inflammation

Question 7:

Which of the following statements is false regarding the management of patients with an increased risk of pre-eclampsia?

- a) Low dose aspirin may reduce severity of disease
- b) Calcium supplementation above RDA reduces the incidence
- c) Treatment with magnesium lowers endothelin-1 concentrates
- d) Megadoses of vitamin C has no proven benefit

Question 8:

Which of the following statements is false regarding the pathogenesis of pre-eclampsia?

- a) Vascular sensitivity to angiotensin 2 is increased
- b) Angiotensinase activity is depressed

- c) There is an increase in sFlt -1 levels
- d) A decrease in the level of soluble endoglin is seen

Question 9:

A 24-year-old primigravida at 22 weeks of gestation came for a regular antenatal checkup. Her BP is 152/96mmHg. Urine analysis shows proteinuria of 3+. All of the following would be consistent with her probable diagnosis, except:

- a) Thrombocytopenia
- b) Hemolysis
- c) Increased Hemoglobin
- d) Low fibronectin

Question 10:

Which finding is most likely to be seen in a patient with pre-eclampsia?

- a) Low uric acid level
- b) Decreased GFR
- c) Low serum creatinine level
- d) High urinary calcium

Question 11:

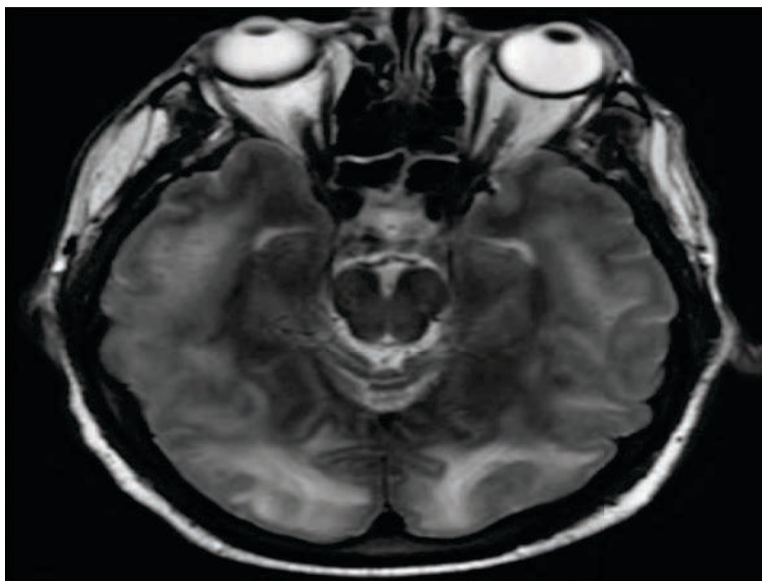
Ophthalmic manifestations in patients with preeclampsia syndrome include all, except:

- a) Purtscher's retinopathy causing reversible blindness
- b) Central vision loss due to temporary scotoma
- c) Permanent visual loss due to retinal artery occlusion
- d) Permanent blindness due to serous retinal detachment

Question 12:

A patient, diagnosed with preeclampsia was brought to the emergency room with three episodes of generalized tonic-clonic seizures, urinary incontinence, and altered sensorium for 3 hours. After stabilization, an MRI revealed the following findings. Which of the following is

true regarding this condition?



- a) Anterior circulation is most commonly involved
- b) These lesions may be seen in most pre-eclampsia patients
- c) The condition is reversible
- d) Basal ganglia are always spared

Question 13:

Which of the following are indicators of severe disease in a patient with preeclampsia?

- a) 1, 3, 2, 5
- b) 1, 4, 5
- c) 2, 3, 4, 5
- d) 1, 3, 4

Question 14:

A patient at 32 weeks of gestation with pre-eclampsia is admitted to your ward. All of the following symptoms would warn you of impending eclampsia, except:

- a) Ankle clonus
- b) Oliguria
- c) Scotoma
- d) Decreased deep tendon reflex

Question 15:

A G2P1 woman with a history of pre-eclampsia presents at 24 weeks of gestation for an antenatal check-up. You decide to evaluate her risk of developing pre-eclampsia in the present pregnancy. Which of the following results is not an indicator?

- a) Absence of diastolic notch on Doppler
- b) High resistance index in uterine artery Doppler in the second trimester
- c) Average MAP $\geq$ 90 mmHg
- d) Presence of free fetal DNA in maternal serum

Question 16:

A 30-year-old G3P2 at 34 weeks of gestation was admitted as her BP was 150/90mmHg and urine dipstick was 2+ on antenatal checkup. Which of the following drugs is not suitable to control BP in this patient?

- a) Methyl dopa
- b) Labetalol
- c) Hydralazine
- d) Captopril

Question 17:

Which is the drug of choice for controlling BP in a patient with severe pre-eclampsia?

- a) Labetalol
- b) Hydralazine
- c) Nifedipine
- d) Methyldopa

Question 18:

A G3P1A1, preeclamptic patient at 33 weeks of gestation came for a regular antenatal check-up. At present, her BP is 130/90 and an NST is reactive. What is the plan of management for this patient?

- a) Wait till 37 weeks for delivery
- b) Deliver at term
- c) Induce at 34 weeks
- d) Perform LSCS at 34 weeks

Question 19:

A 33-week-pregnant lady with pre-eclampsia presents with features suggestive of HELLP syndrome. Which of the following is false about her condition?

- a) Corticosteroids aid in increasing platelet count
- b) It is characterised by microangiopathic hemolysis
- c) MgSO₄ can be used in the management of HELLP syndrome
- d) Termination of pregnancy is the definitive management

Question 20:

Which among the following is false regarding HELLP syndrome?

- a) PT, PTT, fibrinogen levels are normal
- b) All patients with HELLP syndrome have abnormal BP
- c) Tennessee criteria is used for diagnosis
- d) It usually manifests at 27-37 weeks

Question 21:

A G2P1 patient at 25 weeks of gestation is brought to the emergency with right upper quadrant pain and one episode of seizure. CBC reveals a Hb of 7.6gm/dl and a platelet count of 70,000/microL. What is the definitive management?

- a) Termination of pregnancy
- b) Administration of steroids
- c) MgSO₄ regimen
- d) Platelet transfusion

Question 22:

A pre-eclamptic patient at 30 weeks of gestation is rushed to the hospital as she suddenly lost her vision 1 hour ago. During admission, she has a seizure. Which of the given statements is correct?

- a) Convulsions within 7 days of delivery are called postpartum eclampsia
- b) Thrombocytosis is likely to be found
- c) Blindness due to this condition is usually reversible
- d) The seizure is usually focal

Question 23:

A G3P2 patient with preeclampsia develops seizures and is started on the Pritchard regimen. What level of serum magnesium should be maintained in this patient in order to prevent seizures?

- a) 4-7 meq/L
- b) 7-10 meq/L
- c) 1-3 meq/L
- d) 6-8 meq/L

Question 24:

The mechanism of action of MgSO_4 includes all, except:

- a) It is a membrane stabiliser
- b) It is a neuroprotector
- c) It reduces motor endplate sensitivity to Ach
- d) It blocks neuronal sodium influx

Question 25:

Which of the following is an indication to skip the next dose in a patient with severe pre-eclampsia taking MgSO_4 prophylactically?

- a) Knee jerks are sluggish
- b) Urine output exceeds 30 ml/hr
- c) Respiratory rate more than 14 /min
- d) Heart rate more than 100/min

Question 26:

A pregnant woman with pre-eclampsia is started on prophylactic MgSO_4 in view of signs of impending eclampsia. An hour later, examination reveals RR - 16 breaths/minute and absent patellar reflexes. What is her plasma concentration of MgSO_4 most likely to be?

- a) 6-8 mEq/L
- b) 14-16 mEq/L
- c) 16-30 mEq/L
- d) 10-12 mEq/L

Question 27:

Magnesium sulphate has no role in:

- a) Prevention of seizures in severe pre-eclampsia
- b) Prevention of recurrent seizures in eclampsia
- c) Prevention of RDS in premature baby
- d) Reducing contractility of uterus

Question 28:

A primigravida at 33 weeks of gestation was rushed to the hospital with recurrent seizures. Her husband says that prior to the seizures, she had complained of a headache and visual disturbances. She was on antihypertensives from the first trimester and there is no history of neurological disease. Which of the following is true regarding her management?

- a) First step of management is intravenous MgSO_4
- b) Diazepam is recommended as an anticonvulsant if MgSO_4 fails
- c) Treatment may be based on recommendations from the Magpie trial
- d) Maternal alkalosis commonly occurs and needs correction

Question 29:

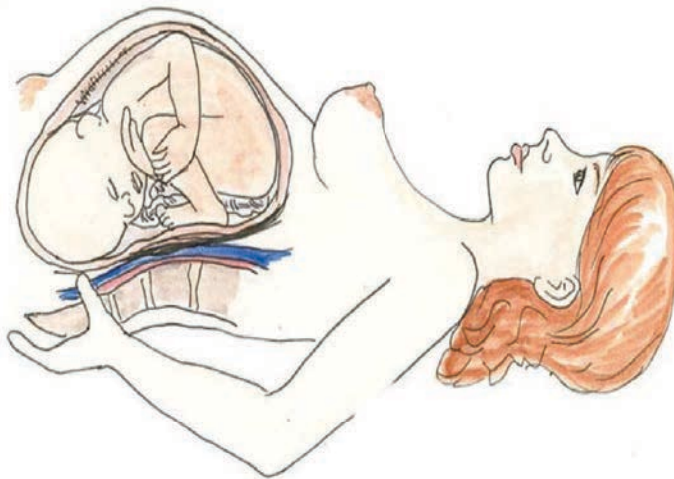
Which of the following is true regarding eclampsia?

- a) 1, 3

- b) 3, 5
- c) 2, 4
- d) 1, 4

Question 30:

A pregnant lady with PIH was being monitored with bed rest in supine position. Which of the following is a complication of the below representation?



- a) Ascending aorta syndrome
- b) Superior vena caval syndrome
- c) Abdominal aorta syndrome
- d) Supine venacava syndrome

Answer Key

Question No.	Correct Option
1	d
2	a
3	d
4	c
5	d

6	d
7	b
8	d
9	d
10	b
11	d
12	c
13	b
14	d
15	a
16	d
17	a
18	a
19	a
20	b
21	a
22	c
23	a
24	d
25	a
26	d
27	c
28	c
29	d
30	d

Detailed Explanations

Solution to Question 1:

Gestational hypertension is defined as BP $\geq 140/90$ after 20 weeks of gestation.

Hypertension in pregnancy is defined as BP of $\geq 140/90$ mmHg measured 2 times with at least a 4-hour interval.

Proteinuria is not an essential criterion to define pre-eclampsia.

Pre-eclampsia is hypertension with any of the following features:

- Proteinuria
- Thrombocytopenia

- Renal insufficiency
- Liver involvement
- Cerebral symptoms
- Pulmonary edema.

Note: A normotensive pregnant woman may have an acute increase in BP from baseline levels and develop eclamptic seizures/HELLP syndrome, despite being normotensive. This condition is known as delta hypertension.

	Gestational hypertension	Pre-eclampsia	Chronic hypertension in pregnancy
Rise in BP	After 20 weeks of pregnancy	After 20 weeks of pregnancy	Before 20 weeks of gestation
Proteinuria	Absent	May be present: >300 mg in 24 hrs urine >30 mg/dl in a random urine sample Urine dipstick 2+	Absent
Fall in BP to normal levels	Within 12 weeks of delivery	Within 12 weeks of delivery	Does not return to normal

Solution to Question 2:

The cut-off for defining proteinuria is ≥ 30 mg/dl (0.3 g/L) in a random urine sample.

Proteinuria in preeclampsia is also defined by:

- ≥ 0.3 g protein in a 24-hour urine specimen.
- Random urine protein: creatinine ratio (PCR) ≥ 0.3 mg protein/mg creatinine
- Protein $\geq 1+$ on urine dipstick

A urine dipstick is used only if the above quantitative methods are not available.

Note: Proteinuria is not an essential criterion to define pre-eclampsia.

Solution to Question 3:

In a normal pregnancy, the remodeling of spiral arteries begins at 12 weeks.

Remodeling of arteries includes invasion of arteries by trophoblasts that convert them from low-flow, high-resistance to high-flow, low-resistance vessels. These modified vessels are more capable of meeting the demands of the developing fetus.

Uteroplacental vessel development takes place in two stages:

The first stage occurs before 12 weeks post-fertilization. During this stage, invasion of spiral arteries up to myometrium occurs.

The second stage occurs between 12-16 weeks. It involves invasion of the endovascular trophoblasts into the intra-myometrial segments of spiral arteries. Interstitial trophoblasts surround the arteries.

Incomplete trophoblastic invasion leads to pregnancy-induced hypertension (PIH) where the vessels remain in a state of high resistance leading to placental ischemia and release of inflammatory mediators and vasoactive substances leading to endothelial injury.

As the second stage of invasion occurs by 18 weeks, PIH develops after 20 weeks of pregnancy.

Solution to Question 4:

In the above scenario of hypertension in a patient with gestational age <20 weeks, the diagnosis is chronic hypertension.

Chronic hypertension is diagnosed when the patient is a known hypertensive (or) BP > 140/90 mmHg before 20 weeks of gestation.

Gestational hypertension and preeclampsia are only diagnosed after 20 weeks of gestation.

Solution to Question 5:

Smoking is not a risk factor for pre-eclampsia rather it is protective.

Although smoking during pregnancy causes various adverse pregnancy outcomes, it is associated with a reduced risk for hypertension in pregnancy.

Risk factors for pre-eclampsia are:

- Previous history of pre-eclampsia
- Primigravida: young or elderly
- Family history: Hypertension, pre-eclampsia.
- Placental abnormalities:
- Hyperplacentalosis: excessive exposure to chorionic villi (molar pregnancy, twins, diabetes)
- Placental ischemia
- Thrombophilias: APLA, Protein C, Protein S deficiency, Factor V Leiden mutation, homocysteinemia
- Twin pregnancy
- Molar pregnancy
- Rh-negative pregnancy
- New paternity - A new partner for subsequent pregnancies

Solution to Question 6:

Placental perfusion is reduced in patients developing pre-eclampsia.

The development of pre-eclampsia may be explained by:

- Abnormal trophoblastic invasion of uterine vessels, leading to decreased placental perfusion.
- Maternal inflammatory response directed against foreign fetal antigens leading to immunological maladaptation.
- Maternal maladaptation to cardiovascular and inflammatory changes of normal pregnancy.
- Genetic factors and epigenetic influences.

Solution to Question 7:

Calcium supplementation above the RDA does not reduce the incidence of pre-eclampsia.

However, women who have a calcium-deficient state may benefit from calcium supplementation to reduce their risk of preeclampsia.

Prophylactic measures for the prevention of pre-eclampsia include low-dose aspirin administration (50–150 mg/day). It is started after the 1st trimester, continued throughout pregnancy, and is to be stopped 7 days before labor.

Magnesium administration may be beneficial as it lowers endothelin-1 concentrates.

There is no role of a low salt diet, anti-oxidants, or vitamin C.

Solution to Question 8:

In pre-eclampsia, there is an increase in the level of soluble endoglin.

Excessive amounts of antiangiogenic factors, increased sensitivity to angiotensin, and lowered angiotensinase levels worsen hypoxia at the uteroplacental interface.

Trophoblast of women who develop preeclampsia overproduces at least two antiangiogenic peptides that enter the maternal circulation. They increase months before clinical preeclampsia develops.

- Soluble endoglin (sEng) is derived from the placenta and blocks endothelial nitric oxide-dependent vasodilatation. It also inhibits various TGF β isoforms from binding to endothelial receptors.
- Soluble Fms-like tyrosine kinase 1 (sFlt-1) is a variant of the Flt-1 receptor for placental growth factor (PlGF) and for VEGF. Increased maternal sFlt-1 levels inactivate and decrease circulating free PlGF and VEGF concentrations, leading to endothelial dysfunction.

Solution to Question 9:

The given clinical picture is suggestive of pre-eclampsia and low fibronectin is not a feature of pre-eclampsia.

Features associated with pre-eclampsia:

- Fibronectin is a glycoprotein associated with vascular endothelial cell basement membrane and is elevated in women with preeclampsia.
- Thrombocytopenia (HELLP syndrome)
- Hemolysis (HELLP syndrome)
- Increased hemoglobin, caused by a decrease in plasma volume.

Hemoconcentration is a hallmark of eclampsia. In preeclampsia, hemoconcentration is present but usually not marked.

Solution to Question 10:

Decreased GFR is seen in pre-eclampsia.

During normal pregnancy, renal blood flow and GFR are increased. However, in pre-eclampsia, renal blood flow and GFR decrease due to increased renal afferent arteriolar resistance. The hallmark renal lesion is glomerular endotheliosis. Diminished filtration causes serum creatinine levels and plasma uric acid concentration levels to rise. The elevation is also due to enhanced tubular reabsorption.

The most common system involved in pre-eclampsia is the renal system.

Several studies have demonstrated that pre-eclampsia is associated with low urinary calcium. This is often used as a predictive test.

Solution to Question 11:

Serous retinal detachment does not cause permanent visual loss.

The most common visual symptom of preeclampsia is scotoma.

Blindness can be due to occipital cortex edema, Purtscher's retinopathy (retinal ischemia), or retinal detachment. The blindness is mostly reversible and corrects automatically after delivery.

Retinal artery occlusion sometimes causes permanent vision loss.

Solution to Question 12:

The above clinical scenario and radiological image are suggestive of posterior reversible leukoencephalopathy syndrome (PRES). This condition is usually reversible.

Endothelial dysfunction and cerebral autoregulation failure are implicated as etiologies of PRES. PRES lesions are almost universal in women with eclampsia but less frequent in

pre-eclampsia.

PRES is characterized by the following:

- Severe headaches
- Altered sensorium
- Seizures
- Visual disturbances.

Radiological findings in PRES:

- Hyperintense T2 lesions in the subcortical and cortical regions of the parietal and occipital lobes.
- The basal ganglia, brainstem, and cerebellum are commonly involved.
- Posterior circulation is commonly involved, and anterior circulation involvement is seen in severe cases.

Neurological manifestations of preeclampsia include:

- Headaches
- Convulsions
- Cerebral edema
- Blindness
- Cerebral hemorrhage.

Solution to Question 13:

Among the given statements, the presence of pulmonary edema, scotoma, or epigastric pain is indicative of severe pre-eclampsia.

Criteria for diagnosing severe preeclampsia include:

Signs and symptoms of significant end-organ dysfunction are:

- New-onset cerebral or visual disturbance i.e., photopsia, scotoma, altered mental status, and severe headache
- Severe, persistent right upper quadrant or epigastric pain, unresponsive to medication and not accounted for by an alternative diagnosis (OR) Serum transaminase concentration ≥ 2 times upper limit of normal.
- Thrombocytopenia with platelets $< 100,000/\mu\text{L}$.
- Progressive renal insufficiency (serum creatinine $> 1.1 \text{ mg/dL}$)
- Pulmonary edema.

Systolic BP	Diastolic BP	Proteinuria	Signs of end-organ damage
≥160 mm Hg	≥110 mm Hg	+	+/-
≥140 mm Hg	≥90 mmHg	+/-	+

Solution to Question 14:

Decreased deep tendon reflex is not a symptom of impending eclampsia. There will be increased deep tendon reflex i.e., hyperreflexia.

Symptoms and signs of impending eclampsia include:

- Severe occipital or frontal headache
- Oliguria - diminished urine output <400 ml/day
- Epigastric pain
- Visual symptoms - blurring, scotomas, blindness
- Disturbed sleep
- Ankle clonus
- Severe elevation in blood pressure
- Pulmonary edema.

Solution to Question 15:

In this above scenario, the absence of a diastolic notch at 24 weeks on Doppler is not a predictor of pre-eclampsia.

In normal pregnant women, a diastolic notch in the uterine artery disappears by 24 weeks. Thus, a persistent diastolic notch present in uterine artery flow between 22-24 weeks is predictive of preeclampsia.

Predictors of preeclampsia:

- Free fetal DNA in maternal plasma
- Diastolic notch present in uterine artery flow between 22-24 weeks (high resistance index in the second trimester)
- Average MAP >90 mmHg

Predictors no longer used include:

- Hemoconcentration
- Elevated fibronectin

- Increase in uric acid levels
- Hypocalciuria - calcium in urine ≤ 12 mg/dl in 24 h
- Increased homocysteine levels
- Roll-over test - screening test is done between 28-32 weeks.

Solution to Question 16:

Captopril is an ACE inhibitor and it is contraindicated in pregnancy.

High BP and proteinuria are diagnostic of preeclampsia. The following drugs are used for the management of pre-eclampsia.

ACE inhibitors inhibit the conversion of angiotensin I to the potent vasoconstrictor angiotensin-II. When given in the second and third trimesters, they can cause severe fetal malformations such as:

- Bony malformations
- Persistent PDA
- Respiratory distress syndrome
- Limb contracture
- Pulmonary hypoplasia.

Anti-hypertensives to be avoided in pregnancy include:

- ACE inhibitors
- Angiotensin receptor blockers
- Diuretics (contraindicated as anti-hypertensive but can be given in CCF)
- Reserpine
- Sodium nitroprusside (used only as a drug of last resort in refractory severe hypertension)
- Diazoxide.

Drug	Dose
Labetalol	100 mg TID or QID
Methyl-dopa	250–500 mg TID or QID
Nifedipine	10–20 mg BD
Hydralazine	10–25 mg BD

Solution to Question 17:

Intravenous labetalol is the drug of choice for controlling BP in severe pre-eclampsia.

It is the first-line agent for urgent control of BP in pregnancy. It is effective, has a rapid onset of action, and has a good safety profile.

Labetalol is a nonselective β blocker and vascular α blocker. It should be avoided in patients with asthma and congestive heart failure due to bronchoconstriction.

Option B: Hydralazine causes maternal hypotension and fetal distress.

Option C: Nifedipine causes reflex tachycardia and headache in the mother.

Option D: In methyldopa, a change in BP is an indirect action. So it is less effective in controlling BP.

Note: Experimental antihypertensive drugs for pre-eclampsia treatment are calcitonin gene-related peptide (CGRP) and anti-digoxin antibody Fab (DIF). DIF is directed against endogenous digitalis-like factors, also called cardiotoxic steroids.

Choice of anti-hypertensives in pregnancy	
PIH /pre-eclampsia / severe pre-eclampsia	Labetalol
Acute hypertension/ Hypertensive crisis	IV labetalol, nifedipine, hydralazine
Chronic hypertension in pregnancy	Labetalol > methyl dopa.

Solution to Question 18:

The scenario is suggestive of mild pre-eclampsia as the patient has a BP of 130/90 mmHg and no fetal compromise. Hence, the plan of management would be to wait till 37 weeks for termination.

In all cases of hypertension during pregnancy, the preferred mode of delivery is vaginal. Cesarean is indicated only for any co-existing obstetrical reasons.

Hypertensive condition	Time of delivery
Gestational hypertension	38–39 weeks
Mild pre-eclampsia	≥ 37 weeks
Severe pre-eclampsia	≥ 34 weeks
Eclampsia, HELLP syndrome	Immediate termination irrespective of gestational age

Solution to Question 19:

Corticosteroid treatment for thrombocytopenia is not recommended in patients with HELLP syndrome.

HELLP syndrome is a serious complication in pregnancy characterized by HEmolysis, ELevated liver enzymes, and Low Platelet count. It occurs in 0.5-0.9% of all pregnancies and in 10–20% of patients with severe preeclampsia.

Management of HELLP syndrome is supportive, including:

- Antiseizure prophylaxis with MgSO₄
- Platelet transfusion if platelet count $\leq 50,000/\text{mm}^3$, FFP transfusion
- Plasmapheresis
- Immediate termination of pregnancy in the interest of the mother.

The most common cause of maternal death in HELLP syndrome is abruptio placenta and DIC.

Solution to Question 20:

15% of pregnant women with HELLP syndrome have normal blood pressure.

HELLP syndrome occurs at 27-37 weeks. PT, PTT, and fibrinogen levels are normal. This differs from DIC where PT and PTT are elevated while the plasma fibrinogen level is decreased.

Tennessee criteria is used for diagnosis and Mississippi classification is used to grade HELLP syndrome.

- Tennessee criteria for diagnosis of true/complete HELLP
- Platelet count ≤ 1 lakh/cu mm
- AST ≥ 70 IU/L,
- LDH ≥ 600 IU/L
- Mississippi classification divides HELLP into 3 grades based on the platelet count.

Solution to Question 21:

The clinical scenario is suggestive of eclampsia and HELLP syndrome. The definitive management is the termination of pregnancy.

In cases of pre-eclampsia, indications for termination of pregnancy, irrespective of gestational age, are as follows:

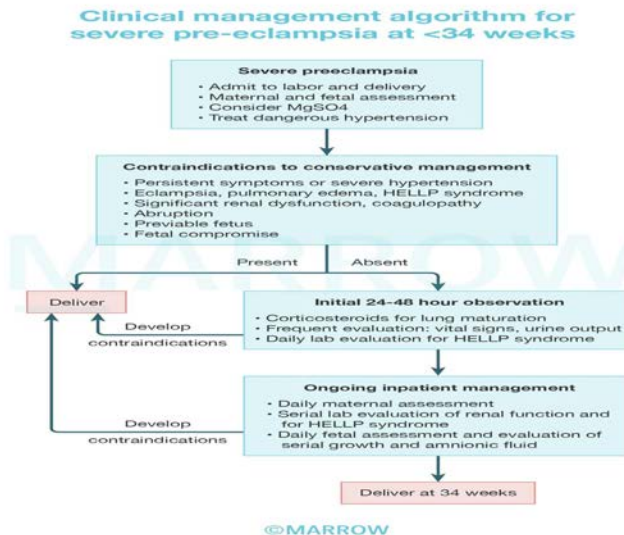
- Eclampsia (give MgSO₄ followed by induction of labor)
- HELLP syndrome
- Persistent symptoms of severe pre-eclampsia or severe hypertension
- Pulmonary edema
- Significant renal dysfunction, coagulopathy

- Abruption of placenta
- Fetal compromise

In mild pre-eclampsia, steroids are given and the patient is delivered at 37 weeks.

In severe pre-eclampsia:

- If pregnancy is ≥ 34 weeks - MgSO₄ is given for prophylaxis of eclampsia and the pregnancy is terminated.
- If pregnancy is at 27-34 weeks - in the absence of contraindications, expectant management is followed and the baby is delivered at 34 weeks, as shown below in the flowchart.



Solution to Question 22:

The clinical scenario is suggestive of eclampsia. Blindness due to eclampsia is usually reversible.

Blindness due to eclampsia can be due to a central or peripheral cause. It is irreversible only if the etiology is a combination of retinal infarction and bilateral lateral geniculate nuclei lesions.

Option A: Convulsions within 48 hours of delivery are called postpartum eclampsia.

Option B: The most common hematological finding in pregnancy-induced hypertension is thrombocytopenia.

Option D: The seizures are generalized seizures.

Note: The most common cause of death in patients with eclampsia is intracerebral hemorrhage.

Solution to Question 23:

The clinical picture is suggestive of eclampsia and to control fits, optimum serum magnesium levels of 4-7 mEq/L have to be maintained.

MgSO₄ regimens for management of eclampsia and severe pre-eclampsia:

Regimen		Loading Dose	Maintenance Dose
Intramuscular (Pritchard)		4 g (20% solution) IV over 3–5 min followed by 10 g (50%), deep IM (5 g in each buttock)	5 g (50%) IM 4 hourly in alternate buttocks
Intravenous (Zuspan or Sibai)		4g IV slow over 15–20 minutes	1–2 g/hr IV infusion

Solution to Question 24:

MgSO₄ blocks calcium influx, not sodium.

Mechanism of action of MgSO₄:

- It is a membrane stabilizer and acts as a neuroprotector by reducing motor endplate sensitivity to Ach.
- It blocks neuronal calcium influx.
- It reduces the presynaptic release of the neurotransmitter glutamate.
- It blocks glutaminergic NMDA receptors and potentiates adenosine action.

Solution to Question 25:

Repeat injections of MgSO₄ are given only if knee jerks are present.

The next dose of MgSO₄ is only given when all signs of magnesium toxicity are absent.

Signs of magnesium toxicity:

- Loss of deep tendon reflexes
- Decreased respiratory rate (<12 per min)
- Urine output <30 ml/hr
- Chest pain, heart block

Management of magnesium toxicity:

- Stop magnesium therapy
- Estimation of serum magnesium and creatinine levels
- Injection of calcium gluconate 10 ml (10% solution) IV

Side effects of MgSO₄:

- Anuria
- Pulmonary edema.

Solution to Question 26:

Patellar reflexes disappear when plasma magnesium concentration reaches 10 mEq/L. As there is no respiratory depression, the plasma concentration is likely to be 10-12 mEq/L.

The disappearance of the patellar reflex is the first sign of impending magnesium toxicity. MgSO₄ is prophylactically used in impending eclampsia, severe pre-eclampsia, and HELLP syndrome.

Signs of magnesium toxicity	Plasma concentrations of magnesium (mEq/L)
Loss of patellar reflex	10–12
Respiratory depression	12–13
Cardiac arrest	>30

Solution to Question 27:

Magnesium sulfate has no role in the prevention of respiratory distress syndrome (RDS) in a premature baby, where there is a deficit of surfactant (which results in alveolar collapse).

Magnesium in the MgSO₄ competes with calcium on voltage-gated calcium channels, thereby altering its uptake and release. Functions of MgSO₄ include :

- Membrane protector and neuro stabilizer. It is used in the prevention of cerebral palsy in very low birth weight neonates.
- Reduction of motor endplate sensitivity to acetylcholine, and thus is used as an adjunct in the management of acute severe asthma in children and adults
- Blocking of neuronal calcium influx, and is thus used in the conservative management of pre-term labor (tocolytic, option D)
- Inducing cerebral vasodilatation, dilating the uterine arteries, increasing the production of endothelial prostacyclin, and inhibiting platelet activation. Hence it is used in the management of pre-eclampsia (option A) and to prevent progression to eclampsia (option A).

Solution to Question 28:

The clinical scenario is suggestive of eclampsia and management may be based on recommendations from the Magpie trial.

The Magpie trial showed that prophylactic use of magnesium sulfate lowers the risk of eclampsia.

The first step in the management of eclampsia is maintaining the airway, ensuring oxygen delivery to both mother and fetus. Then, MgSO₄ is administered to control seizures.

Thiopentone & phenobarbitone may be used to control seizures. No attempt should be made to shorten the convulsion using diazepam as it often leads to thrombophlebitis and respiratory depression.

Maternal acidosis usually occurs and is corrected when present.

Solution to Question 29:

Antepartum eclampsia has the worst prognosis and the most common MRI finding of eclampsia is subcortical white matter edema.

Antepartum eclampsia is the most common type of eclampsia. It has the highest rate of maternal mortality.

The outcome of eclampsia depends on the duration of hypertension and not severity. The most common cause of death in eclampsia and severe pre-eclampsia is intracranial bleeding.

Solution to Question 30:

Supine vena cava syndrome is the complication of the given representation. The image shows compression of the inferior vena cava by the gravid uterus which can sometimes lead to supine hypotension.

During late pregnancy, the gravid uterus compresses the great vessels against the hard spine, when the patient is in the supine position. Of the two vessels, the vena cava is compressed, but the aorta, owing to its structure is relatively resistant.

Normally, the vena cava compression results in opening up collateral circulation by means of paravertebral and azygous veins but in some cases, when the collateral circulation fails to open up, the venous return of the heart may be seriously curtailed resulting in hypotension, tachycardia, and syncope and is called supine vena cava syndrome.

The normal blood pressure is quickly restored by turning the patient to the lateral position.

Diabetes in Pregnancy

Question 1:

Which of the following are not risk factors for developing gestational diabetes?

- a) Family history of diabetes
- b) Gestational diabetes in previous pregnancy
- c) Persistent glucosuria
- d) Obesity in first-degree relatives

Question 2:

You discover that there is evidence of cardiac malformation in a fetus during a routine USG scan at 16 weeks. What is the minimum level of fasting blood sugar at which you can suspect overt diabetes?

- a) 106 mg/dL
- b) 126 mg/dL
- c) 116 mg/dL
- d) 130 mg/dL

Question 3:

A patient underwent a one-step procedure for gestational diabetes screening. Which of the following statements regarding this procedure is false?

- a) Fasting blood glucose ≥ 92 mg/dL is diagnostic
- b) 100 grams of glucose is given after overnight fast
- c) 1-hour blood glucose ≥ 180 mg/dL is diagnostic
- d) 2-hour blood glucose ≥ 153 mg/dL is diagnostic

Question 4:

A woman at 28 weeks of gestation is started on insulin therapy after being diagnosed with gestational diabetes mellitus. What should be the fasting blood glucose target in her?

- a) <85 mg/dl
- b) <100 mg/dl
- c) <90 mg/dL
- d) <95 mg/dL

Question 5:

According to Priscilla White's classification for diabetes in pregnancy, all of the following classes should be managed with insulin except _____

- a) B
- b) H
- c) A1
- d) F

Question 6:

Which of the following statements is false regarding the management of hyperglycemia during pregnancy?

- a) Metformin crosses the placenta and has teratogenic effects on the fetus
- b) Diet control is the first line management
- c) HbA1c monitored at the end of the first trimester and then every 3 months thereafter
- d) Insulin is the first line pharmacological agent

Question 7:

A primigravida on screening was found to have a fasting blood glucose of more than 100 mg/dl. Which of the following medications will you not prescribe to her?

- a) Glyburide
- b) Metformin
- c) Insulin
- d) Tolbutamide

Question 8:

A primigravida at 30 weeks of gestation is admitted with preterm labour contractions. She was diagnosed with gestational diabetes, on screening. Which of the following tocolytic drugs should be avoided in her?

- a) Magnesium Sulfate
- b) Nifedipine
- c) Terbutaline
- d) Indomethacin

Question 9:

According to the American college of obstetricians, macrosomia is defined as the fetal birth weight of _____.

- a) ≥ 3 kg
- b) ≥ 3.5 kg
- c) ≥ 4.5 kg
- d) ≥ 5 kg

Question 10:

In gestational diabetes, which of the following is an independent and substantial risk factor for macrosomia?

- a) BMI
- b) Glucose intolerance
- c) Cord serum c peptide levels
- d) HbA_{1c}

Question 11:

According to ACOG, routine cesarean delivery is advocated in a woman with gestational diabetes mellitus if the sonographically estimated fetal weight is _____ g.

- a) >3000
- b) >3500
- c) >4000
- d) >4500

Question 12:

A pregnant woman has gestational diabetes mellitus (GDM) and her blood glucose levels are well controlled on insulin. What will be the appropriate gestational age for her delivery?

- a) 37 weeks
- b) 39 weeks
- c) 38 weeks
- d) 40 weeks

Question 13:

Which of the following complications is unlikely in a patient with gestational diabetes?

- a) Abortion
- b) Preeclampsia
- c) Respiratory distress syndrome
- d) Oligohydramnios

Question 14:

On screening, a primigravida is found to have fasting blood glucose of 110mg/dl. What is the likelihood that she will develop overt diabetes within the next 20 years?

- a) 25%
- b) 40%
- c) 50%
- d) 10%

Question 15:

A 29-year-old woman with overt diabetes comes to you for pre-conception counseling. Which of the following congenital heart defects are least likely to occur if she were to have a child?

- a) Tetralogy of Fallot
- b) Transposition of great arteries
- c) Ventricular septal defect

- d) Double outlet right ventricle

Question 16:

Hypocalcemia in a newborn is defined as serum calcium _____ mg/dl.

- a) < 8
- b) < 8.5
- c) < 9
- d) < 7.5

Question 17:

You are performing capillary blood glucose charting for newborns in the post-natal ward. Below what level is it considered hypoglycemia?

- a) < 40
- b) < 45
- c) < 50
- d) < 55

Question 18:

Identify the maternal condition that can lead to the findings in the baby as shown below.



- a) Pre-gestational diabetes
- b) Gestational diabetes mellitus
- c) Effect of valproate
- d) Hypertensive on ACE inhibitors

Question 19:

Which of the following will be the most effective contraception in a female with hypertension and a history of gestational diabetes mellitus?

- a) Oral contraceptives
- b) Intra uterine device
- c) Barrier methods
- d) DMPA injection

Answer Key

Question No.	Correct Option
1	d
2	b
3	b
4	d
5	c
6	a
7	d
8	c
9	c
10	a
11	d
12	b
13	d
14	c
15	a
16	a
17	b

18	a
19	b

Detailed Explanations

Solution to Question 1:

Obesity in first-degree relatives is not a high-risk factor for gestational diabetes mellitus. However, severe obesity is a risk factor for developing gestational diabetes.

Gestational diabetes mellitus is defined as carbohydrate intolerance of variable severity with onset or first recognition during the present pregnancy.

It usually presents in the second or third trimester. More than 50% of women with GDM ultimately develop overt diabetes within 20 years.

It is recommended to screen for gestational diabetes mellitus in the presence of the following high-risk factors:

- Severe obesity
- Strong family history of type-2 diabetes
- Previous history of gestational diabetes mellitus
- Impaired glucose metabolism or glucosuria.

Solution to Question 2:

Overt diabetes is when fasting blood sugar is ≥ 126 mg/dl. Overt diabetics are women who are known to have diabetes before pregnancy.

A patient is said to have overt diabetes if:

- Fasting blood sugar ≥ 126 mg/dL
- A random blood sugar of ≥ 200 mg/dL
- HbA1c of $\geq 6.5\%$

The maternal complication of diabetes:

- increased Miscarriage and preterm labor.
- increased incidence of preeclampsia
- polyhydramnios
- prolongation of labor due to shoulder dystocia, operative interference, and cephalo-pelvic disproportion.

Fetal and Neonatal complications of diabetes:

- fetal macrosomia (birth weight > 4 kg)

- congenital malformations - Musculoskeletal, urogenital, and CNS anomalies.

Solution to Question 3:

In one-step procedure, the patient is given 75g of glucose after an overnight fast, and blood glucose is measured hourly for 2 hours.

Diagnostic criteria with one-step procedure is as follows:

- Fasting ≥ 92 mg/dL
- 1-hour ≥ 180 mg/dL
- 2-hour ≥ 153 mg/dL

In two-step approach patient is given glucose and subsequent values measured in two separate steps.

- Step 1 - the patient is given 50 grams of glucose and blood glucose is measured after one hour. If it is ≥ 140 mg/dl then the screen is positive and step 2 is done.
- Step 2 - the patient is given 100g of glucose and values measured hourly, for 3 hours. A diagnosis is made if at least two values are high using one of two methods.

DIPSI (Diabetes In Pregnancy Societies of India) recommends a one-step procedure without regard to the time of the last meal. Here the patient is given 75 grams of oral glucose. A blood glucose value of ≥ 140 mg/dL after 2-hours is diagnostic.

Solution to Question 4:

The target fasting blood glucose level in a patient with gestational diabetes mellitus is 95 mg/dl.

In gestational diabetes mellitus, the targets are:

- Fasting ≤ 95 mg/dl
- 1-h postprandial ≤ 140 mg/dl
- 2-h postprandial ≤ 120 mg/dl
- HbA_{1c} $\leq 6.0\%$.

Solution to Question 5:

Class A1 is managed by diet therapy. Other than A1, all the other classes are managed by insulin therapy.

Priscilla White's classification for diabetes in pregnancy is not used anymore. It was modified by the American College of Obstetricians and Gynaecologists in 1986.

Class	Description
Gestational Diabetes Mellitus	
A1	Diet-controlled GDM
A2	Insulin-controlled GDM
Diabetes Mellitus	
B	Onset >20yDuration <10y
C	Onset 10-19yDuration 10-19y
D	Onset <10yDuration >20yBenign retinopathy
F	Nephropathy
R	Proliferative retinopathy
H	Heart disease

Solution to Question 6:

Metformin crosses the placenta. However, no teratogenic effects have been observed as yet.

Insulin is the preferred treatment in gestational diabetes mellitus because it does not cross the placenta (American Diabetes Association).

Management of diabetes in pregnancy:

- Preconceptional counseling:
- To achieve tight control of diabetes before the onset of pregnancy.
- HbA1C level measured to plan pregnancy.
- Appropriate advice about diet and insulin given
- Antenatal care (at monthly intervals up to 20 weeks and thereafter at 2 weeks intervals):
- Diabetic diet—30 Kcal/kg for normal BMI, 24 Kcal/kg for overweight women, and 12 Kcal/kg for morbidly obese women
- Frequent monitoring of blood glucose
- HbA1c determination at the end of the first trimester and then every 3 months thereafter. HbA1c level of $\leq 6\%$ is desirable
- Insulin therapy - initiated when diabetes is not controlled by diet alone
- Oral hypoglycemic drugs- glibenclamide and metformin are safe.

Solution to Question 7:

Tolbutamide is not used for the treatment of gestational diabetes.

First-generation sulfonylureas such as tolbutamide and chlorpropamide can cause teratogenicity, neonatal hypoglycemia, and fetal hyperinsulinemia.

First-line glycemic control agents in pregnancy:

- Insulin
- Glyburide
- Metformin

Solution to Question 8:

Terbutaline should be avoided in pregnant women with diabetes.

Terbutaline (β -mimetic drugs) given for tocolysis can cause hyperglycemia, which can cause diabetic ketoacidosis. Factors precipitating ketoacidosis in pregnancy:

- Hyperemesis gravidarum
- Infection
- Non-compliance
- Insulin pump failure
- β -sympathomimetic drugs
- Corticosteroids
- Poor glycemic control

Solution to Question 9:

According to the American college of obstetricians (ACOG), macrosomia is defined as the fetal birth weight of ≥ 4.5 kg.

Macrosomia refers to growth beyond a specific threshold, regardless of gestational age. In developed countries, the most commonly used threshold is the weight above 4.5 kg, but weight above 4 kg is also commonly used.

The WHO defines the cut-off at 4.0 kg for fetal macrosomia in its guidelines for the management of common maternal conditions. This is followed in India.

Solution to Question 10:

BMI (body mass index) is an independent and substantial risk factor for macrosomia in both diabetic and non-diabetic mothers.

BMI levels are associated with increased birth weight, regardless of glucose levels. Gestational diabetes prevalence increases by approximately 1 percent for every 1 kg/m² increase in BMI.

BMI cutoffs for pregnancy:

Underweight - $< 18 \text{ kg/m}^2$

Normal - $18.5 \text{ to } 24.9 \text{ kg/m}^2$

Overweight - $25 \text{ to } 29.9 \text{ kg/m}^2$

Obese - $\geq 30 \text{ kg/m}^2$

Weight distribution also seems to play a role because the risk of gestational diabetes is increased with truncal obesity. HbA1c levels are associated with fetal malformations in pre-gestational diabetes.

Solution to Question 11:

According to the American College of Obstetricians and Gynecologists (ACOG), routine cesarean delivery is advocated in a woman with gestational diabetes mellitus if the sonographically estimated fetal weight is $\geq 4500 \text{ g}$.

Solution to Question 12:

The most appropriate time for the delivery in this patient would be 39 weeks as she has good glycemic control with insulin.

The timing for delivery in a pregnant woman with GDM would vary depending on the control of glucose levels & mode of treatment.

Appropriate timing for delivery in gestational diabetes	
Well-controlled with diet	39 weeks to 40 weeks + 6 days
Well-controlled with insulin/oral agent	39 weeks to 39 weeks + 6 days
Co-morbidity or suboptimal control	Induction before 39 weeks
With poor glycemic control	Reasonable to deliver at 37 to 38 weeks + 6 days of gestation In extreme cases where all efforts for glycemic control have failed induction prior to 37 weeks (no clear cut guideline)

Solution to Question 13:

Polyhydramnios is a complication of diabetes, not oligohydramnios.

Gestational diabetes mellitus (GDM) is commonly defined as glucose intolerance first recognized during pregnancy.

Complications of gestational diabetes:

- Spontaneous abortion
- Preterm labor
- Unexplained fetal demise
- Hypoglycemia
- Malformations
- Altered fetal growth
- Hypocalcemia
- Hypertrophic cardiomyopathy
- Pre-eclampsia
- Respiratory distress syndrome
- Polyhydramnios

Solution to Question 14:

50 % of women with gestational diabetes develop overt diabetes within 20 years. Hence, they are recommended to undergo screening and reevaluation from time to time.

Fasting blood glucose of ≥ 92 mg/dL indicates gestational diabetes.

Solution to Question 15:

Tetralogy of Fallot (TOF) is usually not seen in infants of diabetic mothers.

Ventricular septal defect and transposition of great arteries are the most common cardiac anomalies in infants of diabetic mothers.

Cardiac defects that present more often in infants of diabetic mothers are:

- Transposition of the great arteries (TGA)
- Double outlet right ventricle (DORV)
- Ventricular septal defect (VSD)
- Truncus arteriosus
- Tricuspid atresia
- Patent ductus arteriosus (PDA)

The most common cardiac anomaly in an infant of a diabetic mother is a controversial topic. Most resources mention VSD and TGA are the most common and do not specify which among them is more common.

Solution to Question 16:

Hypocalcemia in a newborn is defined as serum calcium < 8 mg/dL.

Hypocalcemia is one of the potential metabolic derangements in neonates of diabetic mothers.

There is hypomagnesemia in the mother and the infant due to increased maternal urinary excretion of magnesium due to diabetes. There is maternal hyperparathyroidism due to hypomagnesemia which leads to increased maternal calcium.

This maternal calcium passes to the infant through the placenta and suppresses fetal PTH synthesis and impairs PTH response to postpartum hypocalcemia. PTH suppression, in this case, can be severe and lead to convulsions in the early neonatal period and may persist for months.

Solution to Question 17:

Neonatal hypoglycemia is defined as blood glucose levels < 45 mg/dL.

Maternal hyperglycemia stimulates the fetal pancreas to secrete excess insulin, which causes increased neonatal fat deposition, abdominal girth, and macrosomia (Pederson's hypothesis). It also leads to a rapid drop in plasma glucose concentration after delivery.

Solution to Question 18:

The image showing a poorly developed lower limb suggests the diagnosis of caudal regression sequence/sacral agenesis seen in pregestational diabetes. It is the most specific anomaly seen in diabetes.

The caudal regression sequence is a rare malformation, frequently associated with maternal pre-gestational diabetes. It consists of a spectrum of structural defects of the caudal region, including incomplete development of the sacrum and the lumbar vertebrae.

Elevated maternal HbA1c levels correlate with fetal malformations. It occurs approximately 200 times more frequently in infants of diabetic mothers than in other infants.

Overt diabetes (pregestational diabetes) is associated with higher malformation rates than the general population. Neural tube defects (encephalocele, meningocele, anencephaly), transposition of great vessels, ventricular septal defect, atrial septal defect, hypoplastic left heart, aortic stenosis and coarctation of the aorta are commonly seen.

Small left colon syndrome is a rare anomaly that develops in the second and third trimesters.

Option B: Malformations are not associated with gestational diabetes mellitus (GDM).

Malformations occur due to the disruption of organogenesis, usually occurring in the first

trimester. The abnormal glycemic status in GDM manifests only in the second trimester.

Option C: Valproate leads to neural tube defects and not sacral agenesis.

Option D: ACE inhibitors can cause oligohydramnios.

Solution to Question 19:

Intrauterine devices are the most effective contraceptive choice in patients with gestational diabetes mellitus and hypertension.

It is suitable in patients with diabetes and comorbidities like obesity, hypertension, or dyslipidemia. It is a method without potential cardiovascular consequences, unlike hormonal contraceptives.

Low-dose hormonal contraceptives can be used safely in women with gestational diabetes if there are no associated comorbidities.

Even though the barrier method will not lead to complications in such patients, the failure rate is high due to incorrect use and non-compliance. Hence it is not preferred.

Option D: DMPA (depot-medroxyprogesterone acetate) is an injectable hormonal contraceptive that is contraindicated in diabetic pregnant women with co-morbidities.

Cardiovascular Conditions in Pregnancy

Question 1:

The most common valvular heart disease in pregnancy is _____

- a) Aortic stenosis
- b) Mitral stenosis
- c) Mitral regurgitation
- d) Mitral valve prolapse

Question 2:

What is the commonest congenital valvular lesion observed in pregnancy?

- a) Mitral stenosis
- b) Mitral regurgitation
- c) Mitral valve prolapse
- d) Aortic stenosis

Question 3:

A 24-year-old lady with rheumatic mitral stenosis becomes pregnant. At what point during her pregnancy would you be least concerned about her developing cardiac failure?

- a) 24 weeks of gestation
- b) 30 weeks of gestation
- c) During labor
- d) Immediate post-partum

Question 4:

An intern posted in the antenatal OPD auscultates a patient who came for her routine checkup. Presence of which of the following features would not suggest a physiological change of pregnancy?

- a) Loud S1 with splitting

- b) Ejection systolic murmur at left sternal border
- c) Continuous murmur at 2nd to 4th intercostal space
- d) Late systolic murmur

Question 5:

Which of the following is not an absolute indication for therapeutic termination of pregnancy?

- a) Primary pulmonary hypertension
- b) Eisenmenger's syndrome
- c) Pulmonary venoocclusive disease
- d) Prosthetic valve replacement

Question 6:

A pregnant lady with heart failure planned for valve replacement surgery is posted for cesarean section. Which of the following is not a cardiac indication for cesarean delivery?

- a) Coarctation of aorta
- b) Aortic dissection
- c) Recent myocardial infarction
- d) Primary pulmonary hypertension

Question 7:

You are performing CPR on a woman who is 38-weeks pregnant. She was admitted to the emergency after a massive pulmonary embolism led to cardio-pulmonary arrest. Despite the initial management of airway, oxygenation and vasopressor support, there is no improvement. What is the cut off time for perimortem cesarean delivery in this patient?

- a) 1 minute
- b) 2 minutes
- c) 5 minutes
- d) 3 minutes

Question 8:

A G2P1 patient with mitral valve prolapse goes into labor. Which of the following should not be done in the management of this patient?

- a) Cut short second stage of labour using ventouse
- b) Fluid restriction
- c) Antibiotic prophylaxis
- d) Trendelenburg position during first stage of labor

Question 9:

A pregnant woman with cardiac disease is on warfarin therapy. When will the final switch to heparin be advised?

- a) At the onset of labour
- b) 34 weeks
- c) 36 weeks
- d) 40 weeks

Question 10:

A 28-year-old G2P1 was admitted to the labor room at 38 weeks of gestation. She is a known case of mitral valve prolapse. Which of the following is the best for managing the second stage of labor in this patient ?

- a) Unassisted vaginal delivery
- b) Using ventouse delivery
- c) Zavanelli maneuver
- d) Emergency cesarean section

Question 11:

General anesthesia is the preferable mode of analgesia in pregnant women with _____

- a) Aortic stenosis
- b) Prosthetic heart valves
- c) MVP with MR
- d) Intracardiac shunts

Question 12:

A pregnant lady with mitral stenosis requires anticoagulation. Which of the following statements is false regarding her management?

- a) LMWH or UFH is used until 13 weeks of pregnancy
- b) From 13 weeks till close to delivery substitute UFH with warfarin
- c) Following vaginal delivery, anticoagulation with LMWH or UFH is restarted after 6 hours
- d) Warfarin is contraindicated during lactation

Question 13:

Which of the following statements is false regarding hypertrophic cardiomyopathy in pregnancy?

- a) Most affected patients are asymptomatic during pregnancy
- b) Diuretics are generally not used in this patient
- c) She may develop chest pain, palpitations, and syncope
- d) The most common cause for death is cardiac failure

Question 14:

All are true regarding peripartum cardiomyopathy except _____

- a) Development of cardiac failure within 5 months of delivery
- b) Left ventricular diastolic dysfunction
- c) Absence of identifiable cause
- d) Dilated left ventricle

Question 15:

Which of the following agents has been proposed to be responsible for peripartum cardiomyopathy?

- a) Estrogen
- b) Progesterone
- c) Beta hCG
- d) Prolactin

Question 16:

A P1L1 lady presents with complaints of fatigue and paroxysmal nocturnal dyspnea 3 months after her delivery. On examination, she has pedal edema and echo shows left ventricular systolic dysfunction. Which of the following drugs has shown to be effective for the treatment of her condition?

- a) Ribavirin
- b) Misoprostol
- c) Danazol
- d) Bromocriptine

Question 17:

A 34-year-old lady came to the antenatal OPD with 6 weeks of amenorrhea. Her UPT is positive. She is a known case of mitral stenosis for 5 years. Which is the best time to do surgery for mitral stenosis in this patient?

- a) 8–10 weeks
- b) 10–14 weeks
- c) 14–18 weeks
- d) 24–26 weeks

Question 18:

A lady with ventricular septal defect becomes pregnant. Which of the following is not a common complication that is expected to occur in this patient?

- a) Congestive cardiac failure
- b) Arrhythmia
- c) Hypotension
- d) Left ventricular dysfunction

Question 19:

What is the most common arrhythmia encountered in pregnancy?

- a) Atrial fibrillation
- b) Atrial flutter
- c) PSVT
- d) Bradyarrhythmia

Question 20:

Women with the highest risk of endocarditis in pregnancy are those with _____

- a) 1, 4 and 5
- b) 3 and 5
- c) 1 and 2
- d) 2, 3 and 4

Question 21:

Which of the following is associated with the highest risk of maternal mortality during pregnancy?

- a) Coarctation of aorta
- b) Mitral stenosis
- c) Eisenmenger syndrome
- d) Aortic stenosis

Question 22:

A 32-year-old primigravida delivered a full-term male baby by vaginal delivery. She is a known case of rheumatic mitral stenosis for 4 years. After the delivery of the placenta, she started to bleed profusely. Which among the following should not be used to control bleeding?

- a) Methyl ergometrine
- b) Carboprost
- c) Misoprostol
- d) Oxytocin

Answer Key

Question No.	Correct Option
1	b
2	c
3	a
4	d
5	d
6	d
7	c
8	d
9	c
10	b
11	d
12	d
13	d
14	b
15	d
16	d
17	c
18	c
19	c
20	c
21	c
22	a

Detailed Explanations

Solution to Question 1:

Mitral stenosis is the most common valvular heart disease in pregnancy.

The commonest cardiac lesions are of rheumatic origin followed by congenital conditions. The incidence of cardiac lesions is less than 1% amongst hospital deliveries. Rheumatic valvular lesions predominantly include mitral stenosis (80%).

Solution to Question 2:

Mitral valve prolapse (MVP) is the commonest congenital valvular lesion observed in pregnancy. Mitral stenosis is the commonest acquired heart lesion during pregnancy.

Condition in pregnancy	Most common lesion
Congenital heart disease	Atrial septal defect
Congenital valvular lesion	Mitral valve prolapse
Cyanotic Congenital Heart disease	Tetralogy of Fallot
Acquired valvular heart disease	Mitral stenosis > Mitral regurgitation > Aortic stenosis

Solution to Question 3:

Cardiac failure in pregnancy is least likely to occur during 24 weeks, considering the given options.

The order of risk of cardiac failure in pregnancy is immediate postpartum > 2nd stage of labor > 30 weeks > 22 weeks.

Maternal cardiac risk predictors: [Mnemonic: NOPE]

N-NYHA class >2

O-Obstructive lesions of left heart (Mitral valve area <2 cm² or aortic valve area <1.5 cm², peak LV outflow tract gradient >30 mmHg)

P-Prior cardiac events before pregnancy

E-Ejection fraction <40%

Scoring systems used to calculate risk are Cardiac Disease in Pregnancy (CARPREG) and ZAHARA scores.

Solution to Question 4:

Late systolic murmur is pathological in pregnancy.

CVS physiological changes in pregnancy:

- Pulse rate: Increases
- Diastolic BP: Decreases
- S1: Loud and split
- S2: Normal
- S3: prominent (occasional)
- Jugular venous distension and venous hum

- Murmurs
- Ejection systolic murmur at left sternal border.
- Continuous murmur at 2nd-4th intercostal space (known as mammary souffle)
- Heart is displaced upwards and to the left
- ECG changes
- Left axis deviation (due to elevation of diaphragm)
- T wave inversion, biatrial enlargement
- Mild ST changes in the inferior leads
- Premature atrial and ventricular contractions are relatively frequent.

Solution to Question 5:

Prosthetic valve replacement is not an indication for therapeutic termination as it comes under the WHO category risk 3. Only conditions included under WHO category risk 4 are considered as indications for termination of pregnancy.

WHO class 4 comprises of diseases with very high risk of maternal mortality or severe morbidity, such as:

- Pulmonary arterial hypertension
- Severe systemic ventricular dysfunction (NYHA III-IV or LVEF 30-40%)
- Previous peripartum cardiomyopathy with any residual impairment of left ventricular function
- Severe left heart obstruction
- Marfan syndrome with aorta dilated ≥ 40 mm

Termination of pregnancy is also advised in patients with Eisenmenger syndrome.

Solution to Question 6:

Primary pulmonary hypertension is not a cardiac indication for caesarean delivery.

In primary pulmonary hypertension, maternal mortality is about 50% and the majority die (75%) postpartum.

Simpson's criteria for cesarean delivery for cardiac disease in pregnancy:

- Dilated aortic root ≥ 4 cm or aortic aneurysm
- Acute severe congestive heart failure
- Recent myocardial infarction
- Severe symptomatic aortic stenosis
- Warfarin administration within 2 weeks of delivery

- Need for emergency valve replacement immediately after delivery

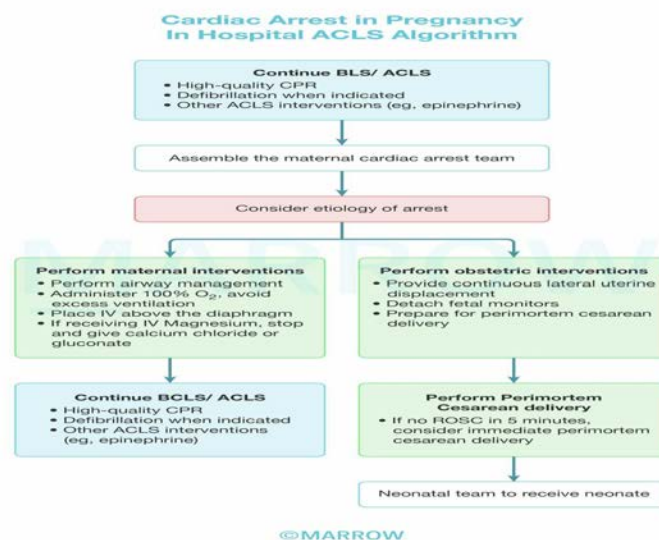
Coarctation of the aorta and aortic dissection are other indications for cesarean delivery.

In general, vaginal delivery is preferred, and labor induction is usually safe. Cesarean delivery is limited to obstetric indications, and considerations are given for the specific cardiac lesions only. Epidural anesthesia is given during LSCS in heart disease. The major problem with epidural/spinal analgesia is maternal hypotension. This is especially dangerous in women with intracardiac shunts in whom flow may be reversed (blood passes from right to left and bypasses the lungs). So in case of a shunt lesion, only general anesthesia should be given.

Solution to Question 7:

Perimortem cesarean delivery should be considered if there is no return of spontaneous circulation in 5 minutes, according to the AHA 2020 Guidelines.

The goal of perimortem cesarean section (PMCS) should be primarily aimed at saving the life of the mother. PMCS improves maternal survival by decreasing compression of the IVC by the gravid uterus. This improves venous return and diaphragmatic displacement, which in turn improves respiratory dynamics.



Solution to Question 8:

Lateral recumbent position is advised during the first stage of labor in this patient.

Management of labor in patients with heart disease:

First stage:

- Lateral recumbent or propped up position
- Oxygen administration (5 to 6 L/min)

- Analgesia by epidural
- Fluids should not be infused more than 75 ml/h
- Prophylactic antibiotics for bacterial endocarditis during labour and 48 h after delivery
- Central venous pressure monitoring

Second stage:

- No maternal pushing
- Cut short second stage by forceps or ventouse under pudendal and/or perineal block anaesthesia
- The benefit of avoiding haemodynamic fluctuations resulting from pushing must be weighed against an increased risk of perineal trauma, haemorrhage and fetal head injury with forceps or vacuum delivery.

Third stage:

- Oxytocin by infusion
- IV frusemide if needed

Solution to Question 9:

The final switch to heparin in a pregnant woman is advised at 36 weeks of pregnancy and is continued till 7 days postpartum.

This switch is made from warfarin to heparin at 36 weeks, as warfarin is a strong anticoagulant which can cause post partum hemorrhage in the mother at the time of labour and can also cross the placenta into the fetal blood and can increase the risk of intracranial hemorrhage in the fetus

The below given algorithm is followed for administration of anticoagulants in a pregnant female:

LMWH- low molecular weight heparin

UFH- Unfractionated heparin

Period of pregnancy	Anticoagulation of choice
First trimester	If dose of warfarin < 5mg/day –continue warfarin till 36 weeks If dose of warfarin > 5mg/day or warfarin embryopathy present –replace it with heparin / LMWH
From 12 weeks to 36 weeks	warfarin
At 36 weeks	Switch to LMWH
1 week before EDD / 3 days before EDD	Switch to UFH (as LMWH have longer T _{1/2} – increased chances of PPH)
On the day of delivery	Stop all anticoagulant

Period of pregnancy	Anticoagulation of choice
6 hrs after vaginal delivery / 6-12 hrs after cesarean	Restart UFH and Warfarin. When INR is 2.5 to 3 withdraw UFH & continue warfarin

Solution to Question 10:

The best technique for managing the second stage of labor in a patient with heart disease is using ventouse.

For most cardiac patients, vaginal delivery is preferred and cesarean section is reserved for obstetric indications since the cesarean section is associated with more blood loss and higher thromboembolic and infection risk. If vaginal delivery is selected as the mode of choice, it will be conducted with the laboring woman in an upright position, or in the left decubitus position. This minimizes aortocaval compression.

Active pushing during the second stage may also be limited, depending on the degree of cardiac compromise. In moderate to severe cardiac disease, it may be prudent to completely avoid an active second stage. Regional anesthesia can be used to allow the fetal head to descend to the perineum without active pushing. Delivery can be further facilitated by forceps or vacuum extraction.

The benefit of avoiding hemodynamic fluctuations resulting from pushing must be weighed against an increased risk of perineal trauma, hemorrhage, and fetal head injury with forceps or vacuum delivery.

Some journals report that ventouse is preferable to forceps, as it can be applied without putting the patient in lithotomy position (raising the legs increases the cardiac load) however it is controversial.

Solution to Question 11:

In pregnant ladies with intracardiac shunts, general anesthesia is the preferable mode of analgesia.

The major problem with epidural/spinal analgesia is maternal hypotension. This is especially dangerous in women with intracardiac shunts in whom flow may be reversed (blood passes from right to left and bypasses the lungs). So in case of a shunt lesion, only general anesthesia should be given.

Note: Aortic stenosis will also be affected by the hypotension caused by epidural/spinal analgesia. However, in aortic stenosis, slow epidural anesthesia is adequate and doesn't compromise ventricular preload. Only a few severe cases of aortic stenosis warrant GA. Hence, in the context of this MCQ, 'intracardiac shunts' would be the most appropriate answer.

Solution to Question 12:

Warfarin is not contraindicated during lactation.

Warfarin, low-molecular-weight heparin (LMWH), unfractionated heparin (UFH) do not accumulate in breast milk. They do not induce an anticoagulant effect in the infant.

The choice of anticoagulants is the main problem in pregnancy due to the following reasons:

- Warfarin: Although safe for the mother, can lead to warfarin embryopathy, miscarriage, IUGR, and stillbirths.
- Heparin: It is safe for the fetus as it does not cross the placenta but less effective than warfarin.
- LMWH: Inadequate and ACOG does not recommend its use in pregnant women with prosthetic heart valves.

If the patient delivers when the anticoagulant is still effective and bleeds excessively, then protamine sulfate is given intravenously.

Period of gestation	Anticoagulant
Till 12 completed weeks	LMWH or UFH
13 to 36 weeks	Warfarin
36 weeks to 6 hours before delivery	LMWH or UFH
6 hours after vaginal and 24 hours after cesarian section	LMWH or UFH (ACOG recommendation)

Solution to Question 13:

The most common cause of death in HOCM in pregnant women is ventricular tachyarrhythmias.

Most affected women are asymptomatic, but some patients may develop:

- Dyspnea
- Anginal or atypical chest pain
- Syncope
- Arrhythmias.

Strenuous exercise is prohibited during pregnancy. Abrupt positional changes are avoided to prevent reflex vasodilation and decrease of preload. Diuretics and vasodilators are generally not used as they decrease preload. If angina develops, beta-blockers or calcium channel blockers are used.

The mode of delivery depends upon obstetric indications and if cesarean section is done, general anesthesia is considered the safest.

Solution to Question 14:

Left ventricular systolic dysfunction is a feature of peripartum cardiomyopathy (also known as postpartum cardiomyopathy) and not diastolic dysfunction.

Peripartum cardiomyopathy (PPCM) is a type of dilated cardiomyopathy. It develops during the last month of pregnancy or within the first 5 months following delivery.

The predisposing factors include Cardiac gene mutations involving several genes like *TTNC1*, *TTN*, and *STAT3*, Twin pregnancy/ multifetal gestation, Smoking, and Preeclampsia. Other proposed causes include viral myocarditis, abnormal immune response to pregnancy, malnutrition, inflammation, etc.

PPCM is associated with increased secretion of prolactin by the pituitary and antiangiogenic molecule sFlt-1 (soluble fms-like tyrosine kinase) by the placenta at term gestation.

Pearson's diagnostic criteria for PPCM:

- Development of cardiac failure in the last month of pregnancy or within 5 months after delivery
- Absence of an identifiable cause for the cardiac failure
- Absence of recognizable heart disease prior to the last month of pregnancy, and
- Left ventricular systolic dysfunction demonstrated by echocardiographic criteria, such as ejection fraction $\leq 45\%$ or fractional shortening $\leq 30\%$ along with a dilated left ventricle.

Management involves treating the features of heart failure. Bromocriptine has been found useful in this condition. Many women recover within 6 months of delivery, while some continue to have residual effects.

Solution to Question 15:

Prolactin has been proposed to be responsible for peripartum cardiomyopathy.

Oxidative stress during late pregnancy leads to the proteolytic cleavage of prolactin. The resulting 16-kDa prolactin fragment has been found to be cardiotoxic and can impair the metabolism and contractility of cardiomyocytes.

The etiology of peripartum cardiomyopathy remains unknown. Many potential causes have been proposed but are not proven.

- Viral myocarditis
- Abnormal immune response to pregnancy
- Abnormal response to the increased hemodynamic burden of pregnancy
- Malnutrition
- Inflammation
- Apoptosis
- Prolactin

Solution to Question 16:

The given scenario is suggestive of peripartum cardiomyopathy and bromocriptine is effective for its treatment.

Oxidative stress during late pregnancy leads to the proteolytic cleavage of prolactin. The resulting 16-kDa prolactin fragment has been found to be cardiotoxic and can impair the metabolism and contractility of cardiomyocytes. Based on this proposed mechanism, bromocriptine therapy has been suggested because it inhibits prolactin secretion.

Solution to Question 17:

The best time of surgery for mitral stenosis in pregnancy is between 14–18 weeks.

Balloon valvuloplasty is the procedure of choice for mitral stenosis in pregnancy. Any cardiac surgery in pregnancy should usually be postponed until after delivery, but if required, it can be performed safely in the 2nd trimester.

Solution to Question 18:

Hypertension is a common maternal complication of congenital cardiac disease in pregnancy.

The common maternal complications in patients with congenital cardiac disease include:

- Congestive cardiac failure
- Pulmonary edema
- Arrhythmia
- Hypertension

Solution to Question 19:

The most common arrhythmia seen in reproductive-aged women is paroxysmal supraventricular tachycardia.

Solution to Question 20:

Women at the highest risk for endocarditis are those with cyanotic cardiac disease and prosthetic valves.

The American heart association recommends prophylaxis for dental procedures with:

- Prosthetic valves
- Unrepaired cyanotic heart defects

- Valvulopathy after heart transplantation

American College of Obstetricians and Gynecologists do not recommend endocarditis prophylaxis for either vaginal or cesarean delivery in the absence of pelvic infection.

Solution to Question 21:

Eisenmenger syndrome is the heart disease with the worst prognosis, having a maternal mortality rate of 50%.

The most common cause of death in Eisenmenger's syndrome is right ventricular failure with cardiogenic shock. If antenatal diagnosis of Eisenmenger syndrome is made in the first trimester, termination of pregnancy is advised. Eisenmenger syndrome has the highest risk of maternal mortality. However, since mitral stenosis is the most common cardiac condition in pregnancy, it is the most common cause of maternal mortality overall.

Mitral regurgitation, mitral valve prolapse and pulmonary stenosis are well tolerated during pregnancy. Severe aortic stenosis can be life-threatening. However, women with mild to moderate aortic valve stenosis may have a relatively low-risk pregnancy.

Solution to Question 22:

The given scenario indicates postpartum hemorrhage (PPH) in a primigravida with rheumatic mitral stenosis. Methyl ergometrine is contraindicated in patients with organic heart disease.

Contraindications of methyl ergometrine are:

- Twin pregnancy - Second baby gets compromised by tetanic contractions of uterus if given after delivery of first baby
- Organic cardiac disease - Causes overloading of right heart by squeezing blood from uterine circulation.
- Severe pre-eclampsia and eclampsia
- Rh negative mother - Increases chance of fetomaternal transfusion.

Rhesus Isoimmunization

Question 1:

The Rhesus blood group antigen complex is located on which chromosome?

- a) 16
- b) 14
- c) 4
- d) 1

Question 2:

Antigens of which of the following blood group systems do not cause alloimmunization in a newborn/fetus?

- a) Kidd
- b) MNS
- c) Kell
- d) Lewis

Question 3:

A pregnant Rh-negative woman presents for her regular antenatal visit. Her Indirect Coombs test is negative. When would you give her prophylactic Anti-D?

- a) 28 weeks
- b) 24 weeks
- c) 28 weeks and then monthly thereafter
- d) 24 weeks and then monthly thereafter

Question 4:

Which of the following events in Rh-negative patients does not require Anti-D prophylaxis?

- a) Miscarriage
- b) Chorionic villi sampling

- c) Amniocentesis
- d) Postpartum hemorrhage

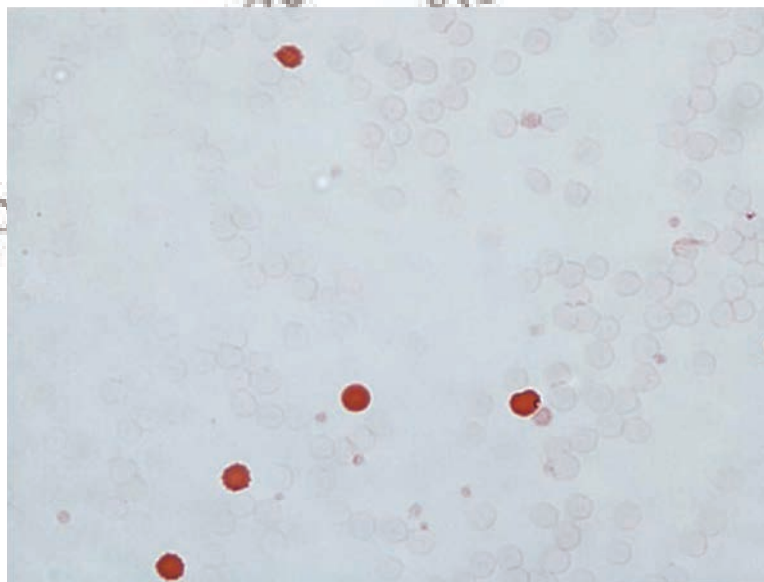
Question 5:

Chorionic villus sampling was performed on a G2P1 Rh-negative woman at 10 weeks of gestation. What is the dose of Anti-D immunoglobulin that should be administered?

- a) 50 microgram
- b) 100 microgram
- c) 150 microgram
- d) 300 microgram

Question 6:

An Rh-negative G2P1 mother delivered her Rh-positive baby by cesarean section. Kleihauer Betke test, as shown in the image, was done. Which of the following statements is true regarding the test?



- a) Fetal erythrocytes are the ghost cells
- b) The red cells contain HbA
- c) The reagent used is citric acid
- d) It is a qualitative test

Question 7:

If the volume of fetomaternal hemorrhage is greater than 30 mL whole blood, the dose of Rh-immunoglobulin calculated is:

- a) 10 µg for every 0.5 mL of fetal RBCs
- b) 10 µg for every 5 mL of fetal RBCs
- c) 0.5 µg for every 1 mL of fetal whole blood
- d) 5 µg for every 10 mL of fetal whole blood

Question 8:

A 25-year-old Rh-negative primigravida presents to the antenatal clinic at 12 weeks of pregnancy. Her husband is found to be Rh-positive. An Indirect Coombs test is also positive. At what critical titer of the test, would the fetus require monitoring for signs of anemia?

- a) 1:4
- b) 1:8
- c) 1:16
- d) 1:18

Question 9:

In modern obstetrics, which of the following investigations is used for monitoring the fetus of an Rh-negative sensitized mother?

- a) Biophysical profile
- b) Fetal blood sampling
- c) Amniospectrometry
- d) MCA doppler peak systolic velocity

Question 10:

In modified Liley's (Queenan) chart, which zone represents a severely affected fetus?

- a) Zone 1
- b) Zone 2
- c) Zone 3

d) Zone 4

Question 11:

You are performing a Doppler assessment of a fetal middle cerebral artery. What is the threshold value of its peak systolic velocity that would prompt you to consider fetal blood sampling?

- a) 0.5
- b) 1
- c) 1.5
- d) 2.5

Question 12:

A 32-year-old G4P3 Rh-negative patient with an antibody titer of 1:32 presents to the clinic at 30 weeks. A USG reveals features of fetal edema, ascites, and polyhydramnios. Cordocentesis was done for fetal blood sampling. At what hematocrit value is intrauterine blood transfusion indicated?

- a) $<40\%$
- b) $<25\%$
- c) $<30\%$
- d) $<35\%$

Question 13:

A patient presented for an antenatal checkup at 25 weeks of gestation. She had bilateral pitting pedal edema and a history of mild headaches. While updating her antenatal card, a weight gain of 13 kgs in the past 2 weeks was noted. USG revealed gross fetal hydrops and a hyperechoic placenta. Which of the following is not true about the patient's condition?

- a) It is also known as Ballantyne syndrome
- b) Hydrops could be due to rhesus isoimmunization
- c) It is associated with elevated liver enzymes
- d) It is not a form of pre-eclampsia

Question 14:

An ultrasound was done on a pregnant Rh-negative patient to screen for hydrops fetalis. What is the earliest sign that may be seen on USG?

- a) Scalp edema
- b) Pleural effusion
- c) Pericardial effusion
- d) Ascites

Question 15:

An Rh-negative patient who did not have access to antenatal care delivered a stillborn fetus as shown in the image. Which of the following is not associated with the fetal condition?



- a) Oligohydramnios
- b) Extramedullary hematopoiesis
- c) Cardiac failure
- d) Hepatosplenomegaly

Answer Key

Question No.	Correct Option
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1	d
2	d
3	a
4	d
5	a
6	c
7	a
8	c
9	d
10	c
11	c
12	c
13	d
14	a
15	a

Detailed Explanations

Solution to Question 1:

The genetic locus for the Rh antigen complex is on the short arm of chromosome 1.

RhCcEe and RhD are the two distinct genes located within the Rh locus. The presence of D antigen on one or both chromosomes 1, determines heterozygosity or homozygosity respectively.

Solution to Question 2:

The Lewis antigens do not cause alloimmunization or hemolytic disease of the fetus and newborn.

Alloimmunization/isoimmunisation is defined as an immune response to foreign antigens after exposure to genetically different cells or tissues.

The Kell blood group system is complex and contains many antigens that are highly immunogenic. These antigens are the third most potent, after those of the ABO and Rh blood groups, at triggering an immune reaction. Anti-Kell causes severe fetal anemia by suppressing fetal RBC synthesis.

MNS and Kidd blood types cause alloimmunization. Hemolytic disease of newborn is mild with Kidd whereas MNS causes potentially severe anemia.

Solution to Question 3:

In a non-immunized Rh-negative primigravida, Anti D is given at 28 weeks of gestation. It is also given within 72 hours of delivery.

Routine postpartum administration of anti-D immune globulin to at-risk pregnancies within 72 hours of delivery decreases the alloimmunization rate by 90 percent.

Additionally, the provision of Anti-D immunoglobulin at 28 weeks of gestation reduces the third-trimester alloimmunization rate from approximately 2 percent to 0.1 percent.

Solution to Question 4:

Postpartum hemorrhage is not an indication of Anti-D prophylaxis.

Anti-D prophylaxis is administered to Rh-negative patients in the following circumstances:

- Pregnancy loss due to:
- Ectopic pregnancy
- Spontaneous abortion/hydatiform mole
- Elective abortion
- Fetal death (any trimester)
- Procedures:
- Amniocentesis
- Chorionic villus sampling
- Fetal blood sampling
- External cephalic version
- Delivery
- Trauma
- Placental abruption
- Unexplained vaginal bleeding during pregnancy

Solution to Question 5:

The dose of Anti-D immunoglobulin to be given in the first trimester is 50 µg.

All Rh-negative unsensitized women should receive 50 µg of Rh-immune globulin IM in the first trimester in the following conditions:

- Within 72 hours of induced or spontaneous abortion
- Ectopic/molar pregnancy
- Chorionic villus sampling

Women with a pregnancy beyond 12 weeks should receive a full dose of 300 µg. Generally, a 300 µg dose will protect a woman from fetal hemorrhage of up to 30 mL of fetal whole blood.

The dose in the 3rd trimester and post-delivery is 300 µg or 1500 international units (IU).

Solution to Question 6:

The reagent used for the Kleihauer-Betke test is citric acid.

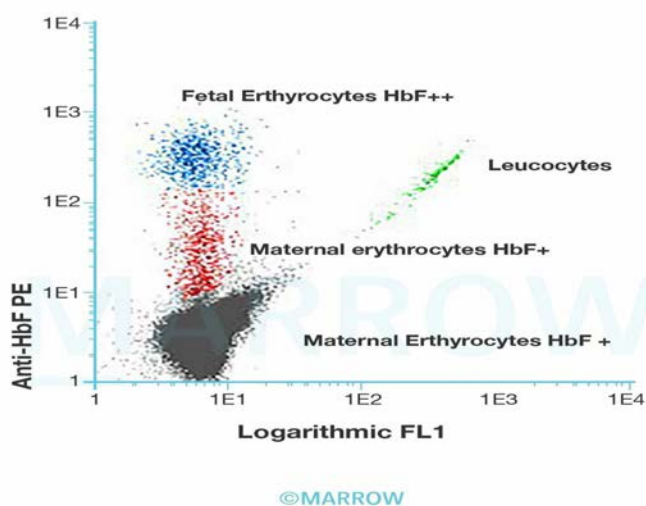
Kleihauer-Betke test is the most commonly used quantitative test used to detect fetomaternal hemorrhage i.e., the approximate volume of fetal blood entering the maternal circulation. It is also known as the acid elution test.

The principle behind it is that fetal erythrocytes contain hemoglobin F, which is more resistant to acid elution than hemoglobin A.

After exposure to acid, only fetal hemoglobin remains, such that after staining, the fetal erythrocytes appear red and adult cells appear as "ghosts." Fetal cells are counted and expressed as a percentage of adult cells.

Flow cytometry is the best test for the quantification of fetomaternal hemorrhage. It is more sensitive and accurate as compared to the Kleihauer-Betke test.

Flow cytometry can analyze a larger number of cells and is not influenced by maternal levels of fetal hemoglobin and fetal levels of maternal hemoglobin. It utilizes monoclonal antibodies to HbF or to the D antigen followed by quantification by fluorescence.



Solution to Question 7:

If the volume of fetomaternal hemorrhage is greater than 30 mL whole blood, the dose of Rh-immune globulin calculated is 10 µg for every 0.5 mL of fetal RBCs.

The calculation is done as follows:

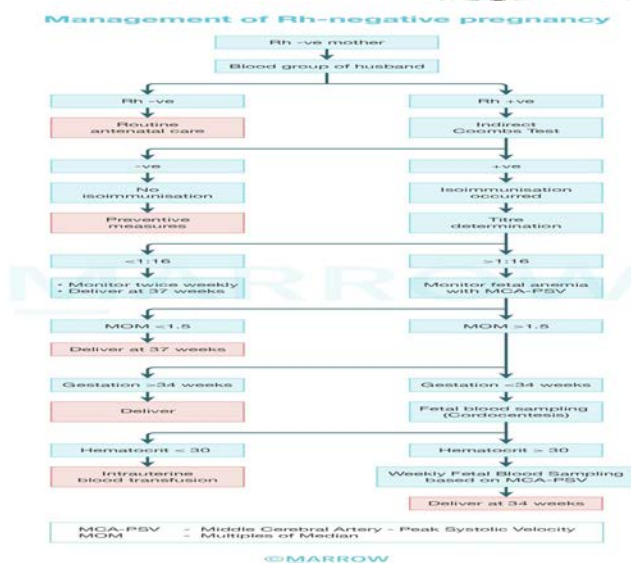
- 300 micrograms of Anti-D can neutralize 15 ml of fetal RBCs or 30ml of fetal whole blood.
- 50 micrograms of Anti-D can neutralize 2.5 ml of fetal RBCs.

The half-life of anti D is 16-24 days. It is administered intramuscularly either in the deltoid or in the anterolateral aspect of the thigh.

Solution to Question 8:

The critical titer for the indirect Coombs test (ICT) is 1:16

Protocol for management Rh isoimmunisation is as follows:



Solution to Question 9:

MCA Doppler peak systolic velocity (PSV) is used for the monitoring of fetuses in Rh-negative sensitized mothers.

The middle cerebral artery is used as other fetal vessels require a larger insonating angle and confers significant measurement error.

MCA PSV is enhanced due to increased cardiac output and decreased viscosity of fetal blood, which permits the reliable, noninvasive detection of fetal anemia in cases of blood-group alloimmunization.

If the MCA PSV exceeds 1.5 MoM (multiples of the median) and the gestational age is less than 34 or 35 weeks, fetal blood sampling should be considered. MCA Doppler PSV has replaced amniocentesis as the preferred test for the detection of anemia.

Solution to Question 10:

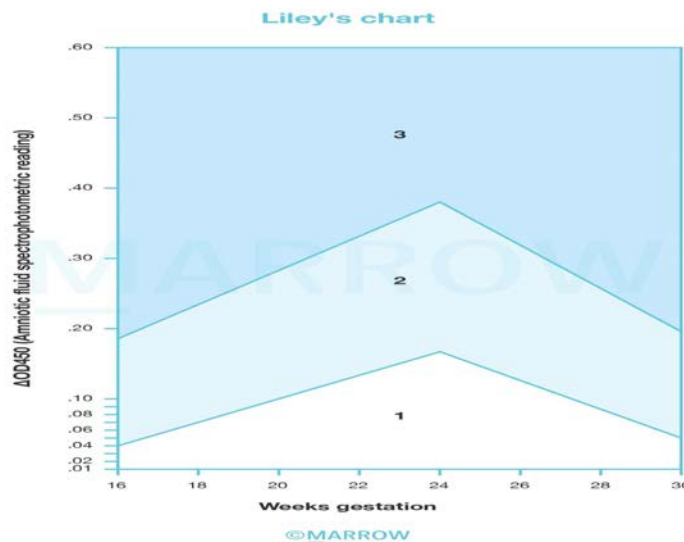
A severely affected fetus is represented by zone 3.

Liley developed a chart of changing amniotic fluid bilirubin levels (delta OD₄₅₀) and weeks of gestation. Amniotic fluid bilirubin is measured by a spectrophotometer and is demonstrable as a change in optical density absorbance at 450 nm (ΔOD_{450}).

However, middle cerebral artery peak systolic velocity (MCA PCV) is non-invasive and more accurate than optical density at 450 nm, hence Liley's chart is no longer used in practice.

Liley's chart has three zones as shown below, delineating the severity of rhesus disease.

- Liley's zone I (low zone) - mild disease.
- Liley's zone II (mid zone) - fetal anemia.
- Liley's zone III (high zone) - severely affected fetus and fetal death is imminent.



Solution to Question 11:

Threshold (critical value) of middle cerebral artery (MCA) peak systolic velocity is 1.5 multiples of the median (MoM).

If the MCA peak systolic velocity exceeds 1.5 MoM, further evaluation by fetal blood sampling is necessary to assess the need for fetal transfusion.

The anemic fetus shunts blood preferentially to the brain to maintain adequate oxygenation. The velocity increases because of increased cardiac output and decreased blood viscosity.

In many centers, serial measurement of the peak systolic velocity of the fetal middle cerebral artery has replaced amniocentesis for the detection of fetal anemia. This method provided 100 percent sensitivity.

Solution to Question 12:

Intrauterine transfusion is done if fetal haematocrit is $\leq 30\%$.

The blood for transfusion is usually O negative and cross-matched with the mother's blood. It is highly packed to achieve a hematocrit of 75 -80%.

Intrauterine transfusions may be done intraperitoneally where blood is absorbed through diaphragmatic lymphatics. It may also be done intravascularly through the umbilical vein. The segments of the umbilical vein used are the intrahepatic portion or at its insertion into the placenta.

Complications of intrauterine transfusion include:

- Fetal demise
- Chorioamnionitis
- Placental abruption
- Pre-term labor
- Bradycardia/tachyarrhythmia
- Umbilical cord hematoma

Solution to Question 13:

The clinical scenario is suggestive of maternal mirror syndrome/Ballantyne syndrome. It may be considered a form of severe pre-eclampsia.

Maternal mirror syndrome is the association between fetal hydrops and the development of maternal edema. The syndrome is named as such because the mother mirrors the fetus. It is characterized by triple edema i.e., the fetus, mother, and placenta become edematous.

The pathogenesis of mirror syndrome, although not well established, mimics trophoblastic damage and maternal vascular endothelial dysfunction, as is also seen in pre-eclampsia. Hence, these two conditions may have a similar clinical presentation, with common features such as:

- Edema
- Hypertension
- Proteinuria
- Liver enzyme elevation
- Headache and visual disturbances.

However, in contrast to preeclampsia, the following features are noted in maternal mirror syndrome:

- Low maternal hematocrit i.e., hemodilution rather than hemoconcentration
- Polyhydramnios rather than oligohydramnios
- The fetus always shows signs of hydrops.

The etiology of fetal hydrops is not related to the development of mirror syndrome. Hydrops could be due to rhesus isoimmunization, twin-twin transfusion syndrome, congenital infections, fetal cystic hygroma, Ebstein anomaly, sacrococcygeal teratoma, vein of Galen aneurysms, etc.

In most cases with mirror syndrome, prompt delivery is indicated, because women with severe features of preeclampsia can deteriorate rapidly. In few cases, identification and treatment of the cause of fetal hydrops resulted in the resolution of symptoms for both the mother and baby for the remainder of the pregnancy.

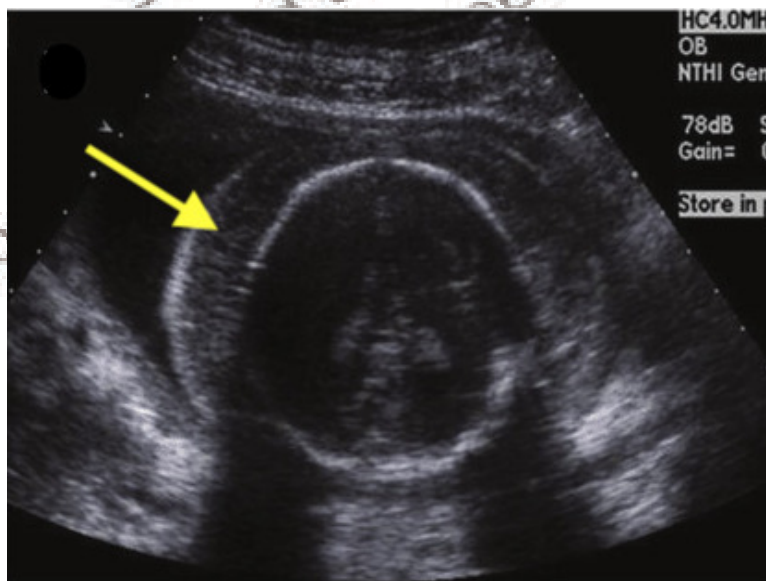
Solution to Question 14:

The earliest sign of hydrops fetalis on a USG is scalp edema. Subcutaneous scalp edema is also known as the Halo sign.

Other USG signs include:

- Ascites
- Pleural effusion
- Pericardial effusion
- Typical position termed as Buddha's position

The image below shows skin thickening at the level of the fetal skull, on ultrasonography:



Solution to Question 15:

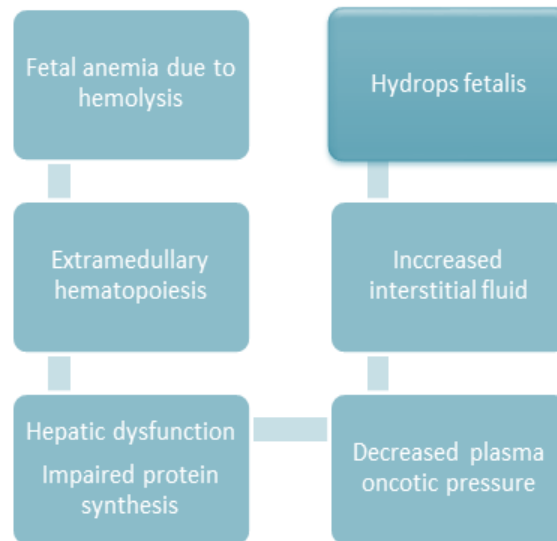
The pale, oedematous fetus in the image is suggestive of fetal hydrops. Fetal hydrops is associated with polyhydramnios, not oligohydramnios.

It often occurs as an effect of iso-immunization in an Rh-negative pregnancy. Other complications that may occur are fetal anemia and hyperbilirubinemia.

Fetal anemia occurs in the initial stage of iso-immunization. It is associated with polyhydramnios and increased placental thickening (hyper-placentosis). An increase in the right atrium size of the fetus is seen.

Fetal hydrops is usually accompanied by severe anemia. It results in stillbirth or fetal demise soon after birth. It is also associated with:

- Congestive cardiac failure
- Ascites
- Pleural effusion
- Pericardial effusion
- Hepatosplenomegaly.



Hepatic Disorders and Infections in Pregnancy

Question 1:

Which of the following is not true regarding obstetric cholestasis?

- a) Pruritus is worse at night
- b) Serum bilirubin >5 mg/dl
- c) Pruritus usually develops in the third trimester
- d) Aspartate transaminase <200 IU /L

Question 2:

A woman at 34 weeks of gestation presents with severe itching over her palms and soles for 2 weeks. She gives a history of similar complaints in her last pregnancy. On examination, she is icteric. Which of the following is the preferred investigation to diagnose this condition?

- a) Serum bile acid
- b) Serum bilirubin
- c) Alanine amino transferase
- d) 5' nucleosidase

Question 3:

A food critic at 32 weeks of gestation presents with nausea and mild right upper abdominal pain. She also gives a history of itching over palms and soles for 2 weeks. On examination, she is icteric. Investigations done are shown below. What is the preferred treatment in this condition?

- a) Antihistamines + nutritional support
- b) Antiviral therapy
- c) Ursodeoxycholic acid
- d) Cholestyramine

Question 4:

Which of the following features are not likely to be seen in a patient with acute fatty liver of pregnancy?

- a) Only 1
- b) 2 & 4
- c) 3, 4 & 5
- d) 1, 3 & 4

Question 5:

A woman at 37 weeks of gestation with icterus, nausea, and yellowish discoloration of urine has abnormal liver function tests. She is suspected to have acute fatty liver of pregnancy. Which of the following is false regarding this condition?

- a) Symptoms improve post delivery
- b) More common in first pregnancy
- c) Associated with female fetus
- d) Occurs due to heterozygous defect for fatty acid oxidation

Question 6:

Which of the following will not increase the risk for urinary tract infection in pregnant women?

- a) Diabetes
- b) Hypertension
- c) Sickle cell anaemia
- d) Vesicoureteric reflux

Question 7:

A G3P2 lady at 13 weeks of gestation presented for a regular antenatal checkup. A urine routine revealed the presence of 2,00,000 gram-negative bacilli/mL. Repeat urinalysis showed the following results. Which of the following is not true about this condition?

- a) Incidence is similar to that in non pregnant women
- b) It is higher in people with sickle cell trait
- c) Antibiotic therapy is required only if symptoms develop

- d) There is risk of developing chronic renal lesions

Question 8:

A woman presented to the clinic with severe lower abdominal pain and dysuria at 28 weeks of gestation. Suprapubic tenderness was noted on palpation. Urine examination revealed pyuria with microscopic hematuria. All of the following can be used to treat this condition except?

- a) Amoxycillin
- b) Cefazolin
- c) Ciprofloxacin
- d) Cefixime

Question 9:

Which stage of pregnancy is most often complicated by acute pyelonephritis?

- a) 1st trimester
- b) 2nd trimester
- c) 3rd trimester
- d) Puerperium

Question 10:

Which of the following congenital infections are associated with the triad of Gregg in a neonate?

- a) Toxoplasma
- b) Parvovirus
- c) Varicella zoster virus
- d) Rubella

Question 11:

Which of the following statements is true about toxoplasmosis in pregnancy?

- a) Maximum transmission and damage in 1st trimester
- b) Maximum transmission in 1st trimester and maximum damage in 3rd trimester

- c) Maximum damage in 1st trimester and maximum transmission in 3rd trimester
- d) Maximum transmission and damage in 3rd trimester

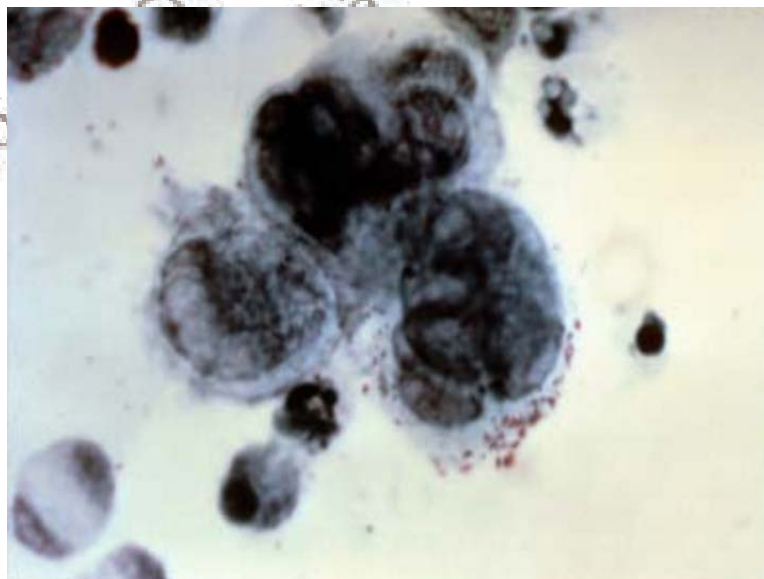
Question 12:

A woman presents for a regular antenatal check-up at 12 weeks of gestation. She works at a rescue center that held an adoption drive for cats a few weeks ago. On examination, mild cervical lymphadenopathy is noted. Sabin-Feldman dye test is positive. Which of the following will be used to treat this condition?

- a) Spiramycin
- b) Pyrimethamine-sulfonamide
- c) Doxycycline
- d) Cotrimoxazole

Question 13:

A per-vaginal examination performed on a lady in labor reveals multiple painful tiny vesicular ulcers over the vulva and vaginal walls. A slide showing the microscopic findings is given below. What is the best step in the management of delivery in this patient?



- a) Vaginal delivery
- b) Cesarean section
- c) Give tocolytics and treat the infection
- d) Local application of antibiotic.

Question 14:

In a pregnant woman with herpes simplex infection, the risk of transmission to the fetus will be maximum during which stage of the pregnancy?

- a) First trimester
- b) Second trimester
- c) Third trimester
- d) Labour

Question 15:

The risk of fetal varicella syndrome is maximum when the mother is infected between ____.

- a) 6 to 10 weeks
- b) 8 to 12 weeks
- c) 13 to 20 weeks
- d) 20 to 28 weeks

Question 16:

A young woman delivered a baby with limb hypoplasia and cicatricial skin lesions. Which of the following infections did the mother most likely have during her pregnancy?

- a) Rubella
- b) Cytomegalovirus
- c) Chicken pox
- d) Toxoplasmosis

Question 17:

In a pregnant woman with syphilis, the infection can get transmitted to the fetus during which stage in pregnancy?

- a) During delivery
- b) 28-32 weeks
- c) Before 6 weeks

d) 10-15 weeks

Question 18:

You are posted in the labor room. While inspecting the placenta to rule out any abnormalities, you notice that it appears pale and enlarged. Which of the following infections does the mother most likely have?

- a) Rubella**
- b) Measles**
- c) Syphilis**
- d) Toxoplasmosis**

Question 19:

A sexually active woman presents with amenorrhea for the past 3 months and a history of a painless genital ulcer. UPT is positive and routine antenatal tests revealed a reactive VDRL report. What is the dose of benzathine penicillin that should be given for this patient?

- a) 3–4 million units intravenously**
- b) 2.4 million units intramuscularly as a single injection**
- c) 2.4 million units intramuscularly daily, plus probenecid 500 mg orally for 14 days**
- d) 2.4 million units intramuscularly weekly for three doses**

Question 20:

A woman at 24 weeks of gestation presented with complaints of fatigue, joint pain, and the following rash for 3 weeks. Urinalysis showed proteinuria and anti-Smith antibodies were positive. Which of the following complications is least likely in this patient?



- a) Thrombocytopenia
- b) Eclampsia
- c) Post term delivery
- d) Thrombophilia

Question 21:

A woman with HIV presents for a routine antenatal checkup at 35 weeks of gestation. Investigations revealed a viral load of >1000 copies/ml. What will be the best mode of delivery in this patient?

- a) Normal vaginal delivery
- b) Cesarean delivery
- c) Forceps assisted vaginal delivery
- d) Vacuum assisted vaginal delivery

Question 22:

A woman is diagnosed to be positive for HIV in her first trimester. What is the appropriate management?

- a) Start ART immediately and continue till 6 weeks after delivery
- b) Start ART after 1st trimester and continue life-long
- c) Start ART immediately and continue life-long

- d) Start ART after 1st trimester and continue till 6 weeks after delivery

Question 23:

A woman with HIV on antiretroviral therapy is in labor. Which of the following will be given to her newborn baby prophylactically?

- a) 6 weeks of nevirapine
- b) 6 weeks of zidovudine and nevirapine
- c) 12 weeks of zidovudine, nevirapine and lamivudine
- d) 12 weeks of zidovudine and lamivudine

Question 24:

The risk of developing a stroke is highest during what period of gestation in a pregnant woman?

- a) 1st trimester
- b) 2nd trimester
- c) Postpartum period
- d) Intrapartum period

Question 25:

Which of the following is false regarding epilepsy in pregnancy?

- a) It increases risk of fetal malformations by 5%
- b) Seizure risk is more at the time of delivery
- c) Seizure frequency increases in 30% of women
- d) Ideal contraceptive in breast feeding mother with seizure disorder is DMPA

Answer Key

Question No.	Correct Option
1	b

2	a
3	c
4	d
5	c
6	b
7	c
8	c
9	b
10	d
11	c
12	a
13	b
14	d
15	c
16	c
17	d
18	c
19	b
20	c
21	b
22	c
23	a
24	c
25	a

Detailed Explanations

Solution to Question 1:

Obstetric cholestasis is a condition in which cholestasis occurs in pregnancy. Serum bilirubin increases very late in the disease and total plasma concentrations rarely exceed 5 mg/dL.

Hyperbilirubinemia results from the retention of conjugated pigment.

It is also referred to as recurrent jaundice of pregnancy, cholestatic hepatitis, and icterus gravidarum. There is a significant genetic influence. It may be more common in multifetal pregnancies.

Patients present with pruritus that develops usually during the third trimester, typically involving the palms and soles. It is worse at night. Icterus may also be present.

Disorder	AST (SGOT) U/L	Bilirubin (mg/d L)
Cholestasis of pregnancy	Up to 200	1–5
Acute fatty liver of pregnancy	200–800	4–10
Hepatitis	2000+	5–20

Solution to Question 2:

The given clinical picture is suggestive of cholestasis in pregnancy. Serum bile acid is the earliest and most consistent parameter to increase in obstetric cholestasis.

Serum cholic acid is preferred compared to serum chenodeoxycholic acid.

Although serum bile acid increases after meals it is still considered an important investigation among women with obstetric cholestasis. This is because serum bile acid levels remain significantly elevated in women with obstetric cholestasis in comparison to pregnant women without it.

Solution to Question 3:

The given scenario is suggestive of obstetric cholestasis. Ursodeoxycholic acid is the drug of choice in obstetric cholestasis.

It relieves pruritus, lowers bile acid and serum enzyme levels (like SGOT, ALP). It may also reduce certain neonatal complications such as preterm birth, fetal distress, and respiratory distress syndrome.

Antihistamines and topical emollients may provide some relief.

Cholestyramine causes decreased absorption of fat-soluble vitamins, which leads to vitamin K deficiency. Fetal coagulopathy may develop, which may lead to intracranial hemorrhage and stillbirth, and therefore it is not preferred.

Disorder	AST (SGOT) U/L	Bilirubin (mg/d L)
Cholestasis of pregnancy	Up to 200	1–5
Acute fatty liver of pregnancy	200–800	4–10
Hepatitis	2000+	5–20

Solution to Question 4:

Hyperglycemia, hypercholesterolemia, and thrombocytosis are not likely to be seen in acute fatty liver of pregnancy (AFLP).

Clinical features of AFLP include:

- Persistent nausea and vomiting
- Progressive jaundice
- Hypoglycemia
- Hypertension
- Proteinuria
- Edema
- Thrombocytopenia
- Moderate to severe liver dysfunction - manifested by:
 - Hypofibrinogenemia
 - Hypoalbuminemia
 - Hypocholesterolemia
 - Prolonged clotting time.

Solution to Question 5:

Acute fatty liver of pregnancy (AFLP) is more commonly associated with male fetuses and not female fetuses.

AFLP is a rare, potentially fatal complication also known as acute fatty metamorphosis or acute yellow atrophy. It is more common in the third trimester. It is due to a defect in the long-chain-3-hydroxy acyl-CoA-dehydrogenase (LCHAD) pathway.

It is also associated with:

- First pregnancy - uncommon in subsequent pregnancies
- Maternal obesity
- Multiple pregnancies

Investigations:

- Liver function tests such as AST, ALT, and serum bilirubin levels are raised.
- Hemolysis with coagulopathy:
- Decreased platelets and fibrinogen
- Elevated d-dimers and prothrombin time
- Biopsy reveals a characteristic microvesicular fatty infiltration.

The definitive management of AFLP is rapid delivery of the fetus and supportive care. Hepatic dysfunction usually normalizes within a week of delivery.

Solution to Question 6:

Hypertension is not a risk factor for urinary infections in pregnancy.

Risk factors for urinary infections in pregnancy include:

- Urinary retention
- Vesicoureteric reflux of pregnancy
- Diabetes in pregnancy
- Sickle cell anemia

Solution to Question 7:

The given clinical scenario is suggestive of asymptomatic bacteriuria. Antibiotic therapy should be given regardless of the symptoms.

Asymptomatic bacteriuria refers to persistent, actively multiplying bacteria within the urinary tract in asymptomatic women.

Organisms that cause urinary infections are those from the normal perineal flora. The most common causative agent is E.coli. Others include Klebsiella, and Proteus.

Incidence is similar to that in non-pregnant women. Prevalence depends on parity (more in higher parity), race, socioeconomic status (more in low SES).

It is more common in patients with sickle cell anemia/trait, diabetes. There is a high risk of developing chronic renal lesions in later life.

It is also associated with:

- Hypertension
- Anemia
- Premature labor
- Intrauterine growth retardation
- Urinary tract abnormalities in recurrent cases.

Diagnostic criteria:

- A positive bacterial count of the same species - over 1,00,000/mL
- Organisms in midstream clean-catch specimen of urine on two occasions
- No symptoms of urinary infection

Treatment:

- Single-dose therapy - Amoxicillin 3g, Nitrofurantoin 200mg
- Regular 10-14 days therapy - Amoxicillin, Nitrofurantoin, Cephalexin, or Amoxiclav

- Recurrent cases need to be treated until delivery with prophylactic doses.

Solution to Question 8:

The given clinical scenario is suggestive of cystitis. Ciprofloxacin, a fluoroquinolone antibiotic, is contraindicated in pregnancy as it causes cartilage and bone defects in the fetus.

Antibiotics are used for treating cystitis in pregnancy.

Antibiotics safe in pregnancy include:

- Penicillin group
- Cephalosporins
- Erythromycin
- Metronidazole (except in the 1st trimester)

Antibiotics unsafe in pregnancy include:

- Tetracycline
- Aminoglycoside
- Chloramphenicol
- Metronidazole (in the first trimester)
- Fluoroquinolones

Solution to Question 9:

Acute pyelonephritis in pregnancy most often complicates 2nd trimester of pregnancy.

Acute pyelonephritis occurs more commonly in those with preexisting renal disease. It is caused by *E. coli*, *Streptococcus* sp., *S. aureus*, *Enterobacter*, *Klebsiella* sp.

Clinical features include:

- Fever with chills
- Loin pain
- Vomiting
- Right-sided renal angle tenderness

Patients may develop acute respiratory distress syndrome due to endotoxin-mediated alveolar injury. Hemolysis can also occur due to endotoxins, leading to anemia.

Treatment includes intravenous fluids with antibiotics.

Solution to Question 10:

Triad of Gregg is seen in a neonate infected congenitally with rubella virus. The features include:

- Patent ductus arteriosus (PDA)
- Cataract
- Sensorineural Hearing loss (SNHL)

Other infections which cause congenital malformations are:

Organism	Effect
Toxoplasma	Hydro/microcephaly, convulsions, chorioretinitis, cerebral calcification
Varicella	Skin scarring, hypoplastic limbs, eye defects, microcephaly
Parvovirus	Non-immune hydrops
Syphilis	Hutchinson's triad: Interstitial keratitis Hutchinson's teeth Eighth nerve deafness

Solution to Question 11:

Toxoplasmosis causes maximal complications in 1st trimester and is maximally transmitted in the 3rd trimester.

This is because the risk of transmission increases and the risk of damage decreases as pregnancy progresses further.

Clinical features of affected neonates include:

- Hydrocephalus or microcephaly
- Chorioretinitis
- Intracranial calcifications
- Convulsions
- Low birthweight
- Hepatosplenomegaly
- Jaundice
- Anemia

Solution to Question 12:

The given scenario points to a diagnosis of maternal toxoplasmosis. It is treated using spiramycin. Pyrimethamine-sulfonamide is used when fetal infections are confirmed. It should be avoided in the first trimester.

Toxoplasmosis is caused by *Toxoplasma gondii*. Cats are definitive hosts. It can be transmitted to humans either by ingestion of raw or inadequately cooked infected meat or by ingestion of oocysts via exposure to cat litter or soil.

Acute infection is diagnosed either by the presence of IgG and IgM antibodies or by a four-fold increase in paired specimens 4 weeks apart.

Solution to Question 13:

Painful genital ulcers and painful inguinal lymphadenopathy are suggestive of herpes virus infection. The cytological image showing multinucleate giant cells, Tzanck cells confirms the diagnosis of HSV. The best step in the management of delivery in this patient is a cesarean section.

Herpes infection in the newborn may occur during birth, particularly if the mother has genital lesions due to HSV-2. In such cases, a cesarean section may prevent neonatal infection.

Tzanck smear



Solution to Question 14:

The maximum risk of HSV transmission from mother to child is during labor. This is due to the shedding of the virus from an active infection in the genital tract during delivery.

HSV is a sexually transmitted infection. There is a higher risk of transmission during primary infections than with recurrences. Reactivations are less severe.

Cesarean delivery reduces the risk of vertical transmission during labor and is indicated in:

- Cases of active genital lesions - both primary and recurrent
- Prodromal symptoms - like vulval itching or burning

Suspicious lesions should be sent for PCR testing or culture. Antiviral treatment with acyclovir should be given if found positive.

Women without active genital lesions, without symptoms, or those with active lesions in non-genital areas can undergo normal labor and delivery.

Management of HSV infection during pregnancy:

- Primary or recurrent infections during pregnancy - Oral acyclovir or valacyclovir
- Severe or disseminated infection - IV acyclovir followed by oral acyclovir
- Pain relief - Oral analgesics or topical anesthetics
- Urinary retention - catheterization.

Solution to Question 15:

The risk of developing fetal varicella syndrome (congenital varicella syndrome) is highest when the mother develops varicella infection between 13 to 20 weeks.

Congenital varicella syndrome occurs in 2% of infants with maternal varicella between 13 and 20 weeks of gestation and in 0.4% of infants before 13 weeks of gestation. It is rare after 20 weeks.

Congenital varicella syndrome is characterized by cicatricial skin scarring in a zoster-like pattern (zig-zag scarring), and limb hypoplasia (shown below). They can also present with neurological abnormalities (microcephaly, cortical atrophy, mental retardation), ophthalmic features (chorioretinitis, microphthalmia, cataract), and renal and autonomic system abnormalities (hydronephrosis, neurogenic bladder). Low birth weight is also common.

Infants born to mothers with varicella in the period from 5 days prior to delivery up to 2 days after delivery are at high risk for severe neonatal varicella. They acquire the infection transplacentally.

Since the mother has not developed a strong antibody response, the infant lacks the protective effect of maternal anti-varicella-zoster virus (VZV) antibodies. Therefore varicella-zoster immune globulin (VZIG) should be administered as soon as possible after birth to newborns whose mothers have varicella 5 days before and up to 2 days after the delivery.

Since perinatally acquired varicella can be potentially life-threatening, the infant should typically be treated with intravenous acyclovir (10 mg/kg every 8 hours) when lesions appear.

For pregnant women with uncomplicated primary varicella or herpes zoster that does not require hospitalization, the recommended treatment is oral acyclovir (800 mg five times daily for seven days).



Solution to Question 16:

The given scenario is suggestive of congenital varicella syndrome. It is seen in babies born to mothers who had chickenpox infections during their pregnancy.

Features of congenital varicella syndrome include;

- Chorioretinitis
- Limb hypoplasia
- Cicatricial skin lesions
- Microphthalmia
- Cerebral cortical atrophy
- Growth restriction
- Hydronephrosis

The image below shows limb hypoplasia in fetal varicella syndrome:



Solution to Question 17:

A pregnant woman with syphilis can transmit *T. pallidum* to the fetus at 10–15 weeks of gestation.

Features of congenital syphilis include:

- Hutchinson's triad - It is seen in late congenital syphilis.
- Hutchinson's teeth - small widely spaced, peg-shaped, notched upper incisors
- Interstitial keratitis
- Sensorineural hearing loss
- Mulberry molars
- Rhagades - deep fissuring of mucocutaneous junctions in the nares, lips, and anus
- Frontal bossing
- Saber shins - anterior tibial bowing
- Clutton's joints - painless, symmetric hydrarthrosis of the knees and elbows

The image below shows Hutchinson incisors.



Note: The risk of fetal infection during untreated early maternal syphilis is ~75–95%, decreasing to ~35% for maternal syphilis of ≥ 2 years duration.

Solution to Question 18:

Enlarged and pale placenta is due to infection by syphilis.

Other clinical manifestations and complications include:

- Spontaneous abortion
- Stillbirth
- Non-immune hydrops
- Intrauterine growth restriction
- Perinatal death.

Solution to Question 19:

The given clinical scenario is suggestive of early syphilis in a pregnant woman. 2.4 million units of benzathine penicillin should be given intramuscularly as a single injection.

Treatment of syphilis includes:

- Early syphilis - Benzathine penicillin G, 2.4 million units intramuscularly as a single injection
- Late syphilis (≥ 1 year duration) - Benzathine penicillin G, 2.4 million units intramuscularly weekly for three doses

Treatment of neurosyphilis:

- Aqueous crystalline penicillin G, 3–4 million units intravenously every 4 hours for 10–14 days

- Procaine penicillin, 2.4 million units intramuscularly daily, plus probenecid 500 mg orally four times daily, both for 10–14 days.

Solution to Question 20:

The above scenario points to a diagnosis of SLE. Preterm labor is a complication in pregnant women with SLE, not post-term delivery.

Complications in a pregnant patient with SLE include:

- Thrombophilia
- Preterm labor
- Hypertension, eclampsia, and preeclampsia
- Anemia
- Thrombocytopenia
- Stroke
- Renal failure

Features of neonatal lupus include:

- Cardiac abnormalities - Complete heart blocks
- Cutaneous lupus
- Hepatobiliary diseases - liver failure, transient conjugated hyperbilirubinemia, transient transaminase elevations
- Hematologic abnormalities - thrombocytopenia, neutropenia, or anemia

Solution to Question 21:

Cesarean delivery is the preferred mode of delivery in a woman with HIV viral load above 1000 copies/ml at 34-36 weeks of gestation.

Cesarean delivery is usually scheduled at 38 weeks of gestation, or two weeks before the due date.

Pregnant women with HIV on ART throughout pregnancy, with a viral load $\leq$ 1000 copies/mL evaluated within 4-6 weeks of delivery, may choose to have a vaginal delivery.

Solution to Question 22:

ART should be initiated immediately and continued lifelong in all pregnant and breastfeeding women with HIV. This is regardless of their WHO clinical stage and CD4 cell count.

*Note: Pearl attached below has been updated according to the NACO ART guidelines 2021. Efavirenz has been replaced by Dolutegravir as per the new guidelines.

Solution to Question 23:

In India, according to the NACO guidelines, nevirapine is given in the syrup form for 6 weeks to an infant of a mother who is on long-term ART.

In infants born to mothers with HIV who are not on ART, syrup nevirapine should be continued for 12 weeks.

According to the NIH guidelines, zidovudine should be given to the neonate for 4 weeks.

Solution to Question 24:

The highest incidence of stroke in pregnancy is 5 days to 2 weeks postpartum. This is followed by the intrapartum period.

There is an increased risk of cerebrovascular thrombosis post-delivery due to a sudden reduction in the blood volume, a fall in the level of circulating antithrombotic factors, and venous stasis.

Risk factors of stroke in pregnancy include:

- Hypertensive disorders
- Gestational diabetes
- Antepartum hemorrhage
- Cesarean delivery.

Solution to Question 25:

Untreated epilepsy has no association with fetal malformations. Anti-epileptic medications such as valproate, phenytoin, etc., are responsible for fetal malformations.

Seizure frequency is unchanged in 50%, increased in 30%, and decreased in 20% women.

Seizure risk increases during labor as pain and hyperventilation decrease the seizure threshold.

The ideal contraceptive in a breastfeeding mother with a seizure disorder is depot medroxyprogesterone acetate (DMPA), as progestins increase the seizure threshold.

Disorders of Menstruation

Question 1:

Which of the following statements is true regarding amenorrhoea?

- a) MRKH syndrome is associated with secondary amenorrhoea
- b) Primary amenorrhoea is seen in Turner syndrome
- c) Primary amenorrhea is absence of menarche by 13 years
- d) Secondary amenorrhea is absence of menses for 6 months

Question 2:

While evaluating a patient with secondary amenorrhoea, you administer her estrogen and progesterone. However, withdrawal bleeding is not seen. What is the likely site of lesion here?

- a) Pituitary
- b) Endometrium
- c) Ovary
- d) Hypothalamus

Question 3:

The risk of Asherman syndrome is highest if D&C is done for ____.

- a) Dysfunctional uterine bleeding
- b) Postpartum haemorrhage
- c) Missed abortion
- d) Medical termination of pregnancy

Question 4:

Which of the following statements is true regarding oligomenorrhea?

- a) Refers to frequent menstruation
- b) Regularly timed episodes of bleeding

- c) Menstruation occurring at an interval of more than 30 days
- d) Can progress to amenorrhea

Question 5:

A patient with uterine fibroids complains of frequent menstruation with an average cycle duration of 19-21 days. What is this menstrual irregularity called?

- a) Hypomenorrhea
- b) Polymenorrhea
- c) Oligomenorrhea
- d) Menorrhagia

Question 6:

Which of the following statements is true regarding menorrhagia?

- a) Heavy and regularly timed episodes of bleeding
- b) Coagulation studies are routinely performed
- c) Amount of bleeding greater than 50 ml
- d) Duration of bleeding greater than 5 days

Question 7:

A 33-year-old female patient comes with complaints of menorrhagia for the past 3 months. The duration and regularity of her menstrual cycles are normal. Non-hormonal management has been tried without success. What is the most appropriate next step?

- a) Uterine artery embolisation
- b) Endometrial ablation
- c) Hormonal therapy
- d) Hysterectomy

Question 8:

Which of the following is not a feature of menometrorrhagia?

- a) Excessive bleeding

- b) Prolonged bleeding
- c) Irregularly timed bleeding
- d) Infrequent bleeding

Question 9:

A patient who had undergone a dilatation and curettage procedure comes to the clinic 6 months later for a follow-up visit. She informs you that since the procedure, she has had scanty blood loss during menses which only lasts for 1-2 days. What is this pattern of bleeding called?

- a) Oligomenorrhea
- b) Polymenorrhea
- c) Epimenorrhea
- d) Hypomenorrhea

Question 10:

A 36-year-old woman presents to the gynecology OPD with complaints of occasional bleeding PV between her usual menstrual cycles. Which of the following is true about her condition?

- a) Bleeding is always pathological
- b) Bleeding is usually scanty
- c) Occurs between 2 abnormal cycles
- d) Occurs in consecutive cycles

Question 11:

A young girl with McCune-Albright syndrome is noted to have precocious menstruation. This condition is defined as the occurrence of menstruation at_____.

- a) <8 years
- b) <7 years
- c) <10 years
- d) <11 years

Question 12:

A 45-year-old woman presents with profuse, painless vaginal bleeding. She has a history of 6-8 weeks of amenorrhea and her UPT is negative. On ultrasound, endometrial thickness is 21mm. What is the likely diagnosis?

- a) Senile endometritis
- b) Halban disease
- c) Metropathia haemorrhagica
- d) Endometriosis

Question 13:

Which does 'I' stand for in the PALM-COEIN classification for abnormal uterine bleeding?

- a) Infectious causes
- b) Iatrogenic causes
- c) Idiopathic bleeding
- d) Inflammatory causes

Question 14:

A 28-year-old P1L1 woman presented with heavy menstrual bleeding and occasional spotting between cycles. What is the first investigation to be done for her?

- a) Ultrasound abdomen
- b) Pap smear
- c) Urine beta-hCG
- d) Transvaginal ultrasound

Question 15:

A woman with AUB-E, who was unresponsive to medical management, underwent a hysteroscopic ablation. Which of the following is false about this procedure?

- a) Safe, effective with less morbidity
- b) Quicker recovery
- c) Fertility is preserved
- d) Method destroys 2-3 mm of myometrium

Question 16:

When is dilatation and curettage contraindicated?

- a) Pelvic inflammatory disease
- b) Ectopic pregnancy
- c) Menstrual irregularities
- d) Endometriosis

Question 17:

A 31-year-old woman presented with acute abdominal pain and spotting. UPT was negative. On examination, she was tachycardic and afebrile. Guarding and abdominal tenderness was present. CT abdomen showed air in the uterine wall and pneumoperitoneum. Which of the following is her condition most likely to be a complication of?

- a) Endometrial biopsy
- b) Dilatation and curettage
- c) Laparoscopy
- d) Hysteroscopy

Question 18:

An 18-year-old girl presents to you with cramping abdominal pain. It starts at the onset of menses and is more pronounced in the first two days. Which condition is she likely suffering from?

- a) Congestive dysmenorrhoea
- b) Spasmodic dysmenorrhoea
- c) Unilateral dysmenorrhoea
- d) Acquired dysmenorrhoea

Question 19:

An increase in which of the following is spasmodic pain during dysmenorrhoea attributed to?

- a) PGE₁
- b) PGE₂

- c) PGD₂
- d) PGF₂^o

Question 20:

A 28-year-old woman presents to the psychiatry OPD with complaints of mood swings and depression 10 days prior to her menses. This is associated with fatigue, breast tenderness, and bloating. Her symptoms resolve once menses begin. What is the treatment of choice in this case?

- a) Evening primrose oil
- b) Leuprolide
- c) Fluoxetine
- d) Drospirenone

Answer Key

Question No.	Correct Option
1	b
2	b
3	b
4	d
5	b
6	a
7	c
8	d
9	d
10	b
11	c
12	c
13	b
14	c
15	c
16	a
17	b

18	b
19	d
20	c

Detailed Explanations

Solution to Question 1:

Primary amenorrhoea is seen in Turner syndrome

Amenorrhoea is a symptom and not a disease entity. It refers to the absence of menstruation that can be physiological or pathological.

Primary amenorrhea is defined as an absence of menstruation by the age of 15 years. At age 13 years, if no menses have occurred and there is a complete absence of secondary sexual characteristics such as breast development, evaluation for primary amenorrhea should begin. Mullerian agenesis (MRKH syndrome), androgen insensitivity syndrome, and Turner syndrome present with primary amenorrhea. In Turner's syndrome, gonadal dysgenesis causes primary ovarian failure, resulting in hypogonadism and amenorrhea.

Secondary amenorrhea is defined as the absence of menses for 3 or more months or fewer than 9 cycles per year after the establishment of regular menstrual cycles. Polycystic ovarian syndrome and hypothyroidism are associated with secondary amenorrhea. Pregnancy is the most common physiological cause of secondary amenorrhea.

Solution to Question 2:

The absence of withdrawal bleeding after the estrogen-progesterone challenge test points towards a local endometrial lesion.

Estrogen-progesterone challenge test is indicated as an investigation in case of secondary amenorrhea. It depends on estrogen-primed endometrium being present in the uterus.

Procedure:

- Ethinyl estradiol 0.02 mg or conjugated equine estrogen 1.25 mg is to be taken daily for 25 days.
- Medroxyprogesterone acetate 10 mg daily is added from day 15–25.
- Alternately, one course of oral contraceptive pill is given and to observe whether withdrawal bleeding occurs or not.

Results:

- If there is no bleeding, a local endometrial lesion is present such as uterine synechiae. This is to be confirmed by HSG or hysteroscopy.
- If withdrawal bleeding occurs, the endometrium is responsive but the endogenous estrogen production is inadequate. Hence, the defect lies in the ovary, hypothalamus, or pituitary.

Solution to Question 3:

The risk of developing Asherman syndrome is the highest if D&C is done for postpartum hemorrhage.

This is because the postpartum uterus is larger and bulkier. Hence, a greater area of the uterine wall is curetted. This increases the risk of developing Asherman syndrome.

Asherman syndrome is the presence of uterine synechiae, due to the destruction of large areas of the endometrium. The treatment of Asherman syndrome is by hysteroscopic adhesiolysis.

Solution to Question 4:

Oligomenorrhoea can progress to develop into amenorrhoea.

Oligomenorrhoea means infrequent, irregularly timed episodes of bleeding usually occurring at intervals of more than 38 days.

Solution to Question 5:

The given case scenario is suggestive of polymenorrhoea, also known as epimenorrhea. It refers to frequent episodes of menstruation, usually occurring at intervals of 21 days or less.

The definitions of various menstrual cycle irregularities are:

Amenorrhoea refers to the absence of menstruation.

Intermenstrual bleeding refers to occasional episodes of bleeding (usually not excessive) that occur between otherwise normal menstrual cycles.

Hypomenorrhea refers to regularly timed cycles, but having scanty episodes of bleeding.

Menorrhagia refers to regularly timed menstrual cycles with excessive bleeding (>80 mL in amount) and/or increased duration of flow (>8 days).

Metrorrhagia refers to normal cyclical bleeding with superimposed irregularly timed episodes of bleeding.

Menometrorrhagia refers to excessive, prolonged bleeding that occurs at irregularly timed and frequent intervals.

Oligomenorrhea refers to infrequent, irregularly timed cycles with intervals of more than 38 days.

Precocious menstruation refers to the occurrence of menstruation before the age of 10 years.

Solution to Question 6:

Menorrhagia is defined as heavy but regular menstrual bleeding.

Menorrhagia denotes regularly timed episodes of bleeding that are excessive in amount (>80 mL) and/or duration of flow (>8 days). Nowadays, menorrhagia has been replaced by the term heavy menstrual bleeding. This is defined as greater than 80 mL per cycle, which will result in anemia if recurrent

Investigations done in case of menorrhagia:

- Complete blood count (CBC) - to rule out anemia and thrombocytopenia.
- Coagulation studies are ordered only in case of clinical suspicion and correlation with CBC
- Sonohysterography/Hysteroscopy - to diagnose endometrial polyps and fibroids
- Endometrial biopsy - in case of suspected endometrial hyperplasia
- Transvaginal ultrasound - supplements biopsy

Solution to Question 7:

Hormonal therapy is the next best step in the given scenario of menorrhagia that is unresponsive to non-hormonal therapy.

Menorrhagia/heavy menstrual bleeding is defined as heavy but regular menstrual bleeding, >80 mL of blood loss, or >8 days of bleeding per menstrual period. It can be acute (<6 months) or chronic (>6 months).

It is managed as follows:

- Medical - first line of treatment.
- NSAIDs
- Tranexamic acid
- OCPs/progestogens - if non-hormonal management fails
- Mirena IUCD
- GnRH agonists
- Blood transfusion, if required
- Surgical - if medical management fails, depending on the diagnosis:
- Endometrial ablation - hysteroscopic, TCRE, RITEA, microwave, NovaSure, etc (option B)
- Uterine artery embolization (option A)
- Uterine tamponade
- Hysterectomy - if all minimally invasive methods fail (option D)

Solution to Question 8:

Menometrorrhagia refers to menstrual bleeding at frequent intervals.

Menometrorrhagia means excessive, prolonged bleeding that occurs at irregularly timed and frequent intervals. It can be caused by cervical cancer and granulosa cell tumors.

Solution to Question 9:

In the given clinical scenario, the pattern of bleeding is called hypomenorrhea. Hypomenorrhea along with a prior history of dilatation and curettage are suggestive of partial Asherman's syndrome.

Hypomenorrhoea refers to regularly timed but scanty episodes of bleeding. It can be seen in:

- Partial Asherman's syndrome - uterine synechiae, causing scarring
- Oral contraceptive use
- Perimenopausal age group

Solution to Question 10:

The given case scenario is suggestive of intermenstrual bleeding, also known as metrorrhagia. Usually, this is scanty, and not excessive.

Intermenstrual bleeding or metrorrhagia refers to irregularly timed episodes of bleeding superimposed on normal cyclical bleeding. It can be caused by:

- Ovulation (physiological)
- Polyps
- Cervical cancer
- Lower genital tract infections etc.

Solution to Question 11:

Precocious menstruation refers to the occurrence of menstruation before the age of 10 years.

Precocious puberty in a girl is the appearance of any of the secondary sexual characteristics before the age of 8 years or the occurrence of menarche before the age of 10 years.

McCune Albright syndrome is associated with cafe-au-lait spots, precocious puberty (and precocious menstruation), and polyostotic fibrous dysplasia leading to recurrent fractures. The image given below shows cafe-au-lait spots.



Solution to Question 12:

The given case scenario is suggestive of metropathia hemorrhagica.

Metropathia hemorrhagica is a form of anovulatory abnormal uterine bleeding. It is characterized by continuous painless vaginal bleeding at the onset of menses or preceded by a 6-8 week period of amenorrhea. It is usually seen in the age group of 40-45 years. A history of menorrhagia may be present.

On examination, the uterus is bulky and uniformly enlarged. Histopathological findings are:

- Myohyperplasia of the uterine wall - up to 2.5cm
- Thick endometrium with cystic glandular hyperplasia appearing as swiss cheese pattern
- Absence of secretory endometrium i.e absence of corkscrew glands
- Thin, slender endometrial polyps project into the uterine cavity
- Corpus luteum is absent
- One or both ovaries may contain a cyst

Option A: Senile endometritis occurs after menopause when the endometrium is thin and atrophic. It may be associated with accumulation of pus in the uterine cavity.

Option B: Halban's disease is irregular shedding and is due to persistent corpus luteum. Regular cycles with prolonged bleeding and normal flow is present.

Option D: Endometriosis usually presents in the reproductive age group, with symptoms of menorrhagia, dysmenorrhea and chronic pelvic pain.

Solution to Question 13:

The 'I' in the FIGO PALM-COEIN classification for abnormal uterine bleeding (AUB) stands for iatrogenic causes.

AUB in the reproductive age group was previously known as dysfunctional uterine bleeding, however, that term is now obsolete. AUB can be acute or chronic:

- Acute AUB - it can start sporadically or be superimposed on chronic AUB.
- Chronic AUB - it lasts for 3-6 months minimum.

PALM-COEIN classification of AUB:

- Polyp (AUB-P):
 - Endometrial polyps
 - Endocervical polyps
- Adenomyosis (AUB-A)
- Leiomyoma (AUB-L). Fibroids are further classified as:
 - Submucosal (SM)
 - Other (O)
- Malignancy and hyperplasia (AUB-M)
 - Endometrial hyperplasia
 - Endometrial carcinoma
- Coagulopathy (AUB-C)
 - von Willebrand's disease
 - Hemophilias
- Ovarian causes (AUB-O)
 - PCOD
 - Hypothyroidism
 - Hyperprolactinemia.
- Endometrial causes (AUB-E)
 - Tubercular or chlamydial endometritis
- Iatrogenic causes (AUB-I)
 - Intrauterine contraceptive devices
 - Anticoagulants
 - Tricyclic antidepressants
- Not yet classified (AUB-N) includes rare causes such as
 - Arteriovenous malformations
 - Myometrial hypertrophy

Solution to Question 14:

The first investigation to be done in a case of abnormal uterine bleeding (AUB) in the reproductive age group is urine β -hCG estimation. This is done to rule out pregnancy.

In the reproductive age group, anovulatory uterine bleeding is encountered less often. Therefore, pregnancy must be excluded. Pregnancy complications such as miscarriage, ectopic pregnancies, and molar pregnancies may present with AUB, hence these need to be ruled out by urine β -hCG levels. This is done for all women in the reproductive age group.

Other investigations to be done for AUB are:

- Pap smear - to rule out cervical cancer and preinvasive lesions (CIN).
- Transabdominal ultrasound - to visualize the pelvic organs and lesions such as endometrial hyperplasia, polyps, fibroids, etc.
- Transvaginal ultrasound - to visualize the pelvic organs. It is done after evaluating β -hCG. This is more comfortable for a patient compared to a transabdominal ultrasound as it doesn't require a full bladder.

Solution to Question 15:

Fertility is not preserved with ablative procedures, because the endometrium and superficial myometrium are destroyed.

In ablative procedures, the uterus is preserved, and hysterectomy is avoided. The endometrium along with 2-3mm of superficial myometrium is destroyed.

Various methods of endometrial ablation include:

- Hysteroscopic endometrial ablation
- Transcervical endometrial resection (TCRE) - now replaced by bipolar electrode called Versapoint
- Radiofrequency induced thermal ablation (RITEA)
- Microwave ablation
- NovaSure - impedance-controlled endometrial ablation
- Bilateral uterine artery embolization
- Uterine tamponade

Solution to Question 16:

Dilatation and curettage (D&C) is contraindicated in pelvic inflammatory disease (PID) as it increases the risk of ascending genital infection.

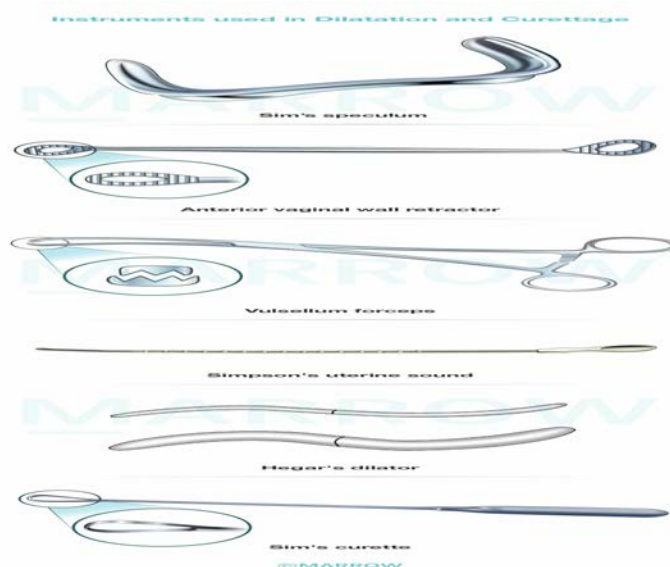
D&C is a minor gynecological procedure of dilating the cervix and curetting the endometrial tissue from the uterine cavity. Diagnostic indications of D&C include AUB, secondary

amenorrhea to detect tubercular endometritis, postmenopausal bleeding to rule out endometrial cancer or hyperplasia, and infertility due to genital TB.

Therapeutic indications of D&C include missed abortion, incomplete abortion, retained products of conception, and endometrial polypectomy.

Absolute contraindication of D&C includes Suspected/ desired pregnancy and lower genital tract infection. However, D&C is performed in some cases with genital infections like septic abortion (where retained pieces of placenta act as a nidus and need to be removed) and pyometra (as there is cervical obstruction with superinfection of menstrual blood and cellular debris, D&C is appropriate to remove the obstruction as well as the source of infection)

Complications of D&C include ascending infection, cervical tear with bleeding, incompetent os, uterine perforation, and ashermann syndrome.



Solution to Question 17:

The given case scenario is suggestive of uterine perforation. In a non-pregnant uterus, uterine perforation is most likely to occur due to dilatation and curettage.

In pregnancy and uterine malignancies, the uterus is soft. Hence, uterine perforation is common during surgeries.

The etiology of uterine perforation is as follows:

- Insertion of intrauterine contraceptive devices (IUCD)- due to the lack of provider experience.
- Spontaneous migration of IUCD within the myometrium
- Dilatation and curettage (D&C)
- Hysteroscopy - diagnostic (e.g, hysteroscopic biopsy) or therapeutic (adhesiolysis)
- Laparoscopy

It can present with fever, acute abdominal pain, nausea and vomiting, and bleeding pv. Imaging reveals air within the uterine wall or pneumoperitoneum.

The management is as follows:

- Conservative management - Fundal injuries are managed conservatively, as the myometrium contracts around the defect.
- Surgical management:
- Small defects can be repaired laparoscopically or hysteroscopically.
- Emergency laparotomy - as abdominal viscera may prolapse through the defect.
- Hysterectomy - if the defect is large, and the patient has completed her family, hysterectomy is done as last resort.

Solution to Question 18:

The given case scenario is suggestive of spasmodic dysmenorrhea. It presents with cramping pain at the onset of menses which is more pronounced in the first two days of menstruation.

Dysmenorrhoea is defined as cramping pain accompanying menstruation. It can be classified as follows:

A. Etiological classification:

- Primary dysmenorrhoea - no identifiable pelvic pathology. Commonly occurs in the age group of 18 to 25 years, with ovulatory cycles.
- Secondary (acquired) dysmenorrhoea - is associated with pelvic pathologies such as fibroids, adenomyosis, PID, and endometriosis.

Unilateral dysmenorrhoea can occur in a rudimentary horn of a bicornuate uterus, IUCD users, and cervical stenosis.

B. Clinical classification:

- Spasmodic dysmenorrhoea:
- Most prevalent
- Cramping pain at the onset of menses, more pronounced on the first 2 days.
- Occurs due to myometrial contractions due to increased PGF2°
- Common in adolescents and young women
- Congestive dysmenorrhoea:
- Pelvic pain and pelvic discomfort occurs, a few days before menses begin
- Occurs due to pelvic venous congestion
- Seen in pelvic endometriosis, PID, fibroids, and IUCD users
- Membranous dysmenorrhoea:
- Least prevalent

- The endometrium is shed as a cast at the time of menstruation
- The passage of the cast is associated with painful uterine cramps

Solution to Question 19:

Spasmodic pain is attributed to myometrial contractions occurring due to increased PGF₂^o. This is secreted under the effect of progesterone.

Progesterone is secreted after ovulation by the corpus luteum. Hence, in anovulatory cycles, menstruation is not accompanied by spasmodic pain.

Solution to Question 20:

The given case scenario is suggestive of premenstrual dysphoric disorder (PMDD). Selective serotonin reuptake inhibitors (SSRIs) such as fluoxetine, sertraline, and paroxetine are the treatment of choice for PMDD.

DSM-5 defines premenstrual dysphoric disorder (PMDD) as the presence of at least five of the associated symptoms most of the time during the premenstrual week. Symptoms improve within a few days after menses and are minimal in the postmenstrual period. These symptoms include:

- Marked emotional lability
- Marked irritability or anger
- Marked depressive mood, self-deprecating thoughts
- Marked anxiety
- Decreased interest in activities
- Difficulty in concentration
- Lethargy
- Appetite changes
- Bloating
- Breast tenderness
- Weight gain

Treatment of PMDD is as follows:

- Lifestyle changes such as - stress reduction, adequate sleep, and exercise, reduction in caffeine intake, smoking cessation
- Selective serotonin reuptake inhibitors (SSRIs) - fluoxetine, paroxetine, sertraline
- Evening primrose oil - to treat breast tenderness
- Ovulation suppression - danazol, leuprolide, drospirenone
- Other less proven methods - cognitive behavioral therapy, calcium supplements, etc.

Note: Premenstrual syndrome has no scientific definition. The constellation of symptoms is now defined under PMDD.

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Uro-gynecology

Question 1:

A woman with urinary incontinence was diagnosed to have a vesicovaginal fistula. What is the most common cause of this condition?

- a) Congenital
- b) Acquired
- c) Idiopathic
- d) Infectious

Question 2:

Which of the following types of genitourinary fistulas has not been reported?

- a) Ureterovaginal
- b) Vesicouterine
- c) Uterourethral
- d) Urethrocervical

Question 3:

Which of the following does not cause genitourinary fistulas?

- a) Hysterectomy
- b) Radiotherapy
- c) Prolonged obstructed labour
- d) Hysteroscopy

Question 4:

A P1L1 woman presents with constant dribbling of urine. She gives a history of unattended vaginal delivery following a difficult and prolonged labor 3 months back. What is the most likely diagnosis?

- a) Ureterovaginal fistula

- b) Vesicouterine fistula
- c) Vesicovaginal fistula
- d) Urethrovaginal fistula

Question 5:

What is the most useful test in vesicovaginal fistula?

- a) Cystoscopy
- b) 3 swab test
- c) MRI
- d) Cystometry

Question 6:

A patient comes with a complaint of dribbling of urine from the vagina during micturition. She has a history of obstructed labor. Methylene blue 3 swab test is done, and the lowest swab gets stained. What is the diagnosis?

- a) Ureterovaginal fistula
- b) Vesicovaginal fistula
- c) Urethrovaginal fistula
- d) Vesicouterine fistula

Question 7:

A 40-year-old woman with a history of colporrhaphy is diagnosed with a vesicovaginal fistula. Which of the following investigations is not preferred in this patient?

- a) Urine culture
- b) Ultrasound abdomen
- c) Cystoscopy
- d) MRI

Question 8:

A 40-year-old woman came for a follow-up visit one week following a vaginal hysterectomy. You note she has developed a large vesicovaginal fistula. What is the best time for fistula repair in this case?

- a) Immediately
- b) After 3 weeks
- c) After 6 weeks
- d) After 3 months

Question 9:

A 34-year-old woman presents to the OPD with complaints of blood in her urine. She informs you that this happens every month for 1 week and stops on its own. She underwent a C-section 1 year ago and has no history of incontinence. During clinical examination, a catheter is inserted into the bladder, and the following finding is seen. What is the likely diagnosis?



- a) Urethrouterine fistula
- b) Vesicouterine fistula
- c) Vesicovaginal fistula
- d) Ureterovaginal fistula

Question 10:

Intravenous pyelography in a patient suspected to have a genitourinary fistula revealed a ureteric fistula. Which of the following is false about this condition?

- a) It is mostly caused by gynecological surgeries
- b) Continuous dribbling of urine occurs, but the patient can also micturate
- c) In 3-swab test, the swab is stained blue
- d) Hydronephrosis is present on the affected side

Question 11:

You need to locate the ureter while performing an abdominal hysterectomy. Which of the following features will help you identify it?

- a) 1, 2, 5
- b) 1, 2, 3
- c) 2, 3, 4
- d) 2, 3, 5

Question 12:

During post-op rounds, you note one of the patients has developed fever and flank pain. The nurses inform you that her abdominal drain outputs have been consistently high. Which of the following procedures has the highest incidence of the complication suspected here?

- a) Vaginal hysterectomy
- b) Wertheim's hysterectomy
- c) Abdominal hysterectomy
- d) Emergency cesarean section

Question 13:

A patient with uterine fibroids is set to undergo a hysterectomy. While obtaining informed consent for the procedure, you counsel her about the potential risk of ureteric injury. Which of the following sites is the injury least likely to occur at?

- a) Infundibulopelvic ligament
- b) Ureteric canal
- c) Uterosacral ligament

d) Ureteropelvic junction

Question 14:

What is the most common cause of rectovaginal fistula?

- a) Improper suturing of perineal tears
- b) Tuberculosis of the anus
- c) Carcinoma cervix
- d) Radiotherapy

Question 15:

Which of the following is not a prerequisite for maintaining female urinary continence?

- a) Normal urethral closure pressure more than 20 cm of water
- b) Urethrovesical angle $> 100^\circ$
- c) Internal urinary sphincter must lie above levator ani
- d) Innervation of internal sphincter by Onuf's nucleus

Question 16:

An elderly woman presented with a mass per vaginum and incontinence. She complains that she has to push the mass inwards to be able to urinate normally. What is the type of incontinence seen here?

- a) Urge incontinence
- b) Stress incontinence
- c) Overflow incontinence
- d) Neurogenic bladder

Question 17:

A gynecologist performs a Marshall and Bonney test in the OPD on an elderly woman. Which of the following conditions is this used to diagnose?

- a) Vesicovaginal fistula
- b) Genuine stress incontinence

- c) Prolapse
- d) Rectovaginal fistula

Question 18:

A wrestler in the post-partum period complains of spurting of urine during weight-lifting practice. Which of the following investigations will help aid your diagnosis?

- a) Q-tip test
- b) Methylene blue 3 swab test
- c) Intravenous pyelography
- d) Transvaginal ultrasound

Question 19:

A young mother presented with dribbling of urine on coughing or sneezing. Marshall and Bonney's test was positive. What is the next step in management?

- a) Conservative management
- b) Venlafaxine therapy
- c) Burch colposuspension
- d) Marshall–Marchetti–Krantz operation

Question 20:

A 24-year-old primiparous woman with Grade 2 genuine stress incontinence did not respond to conservative management. Which of the following surgeries will not be beneficial for her condition?

- a) Pereyra operation
- b) Tension-free vaginal taping
- c) Burch colposuspension
- d) Latzko procedure

Question 21:

A 62-year-old woman presented with complaints of urinary urgency followed by involuntary escape of urine. What is the etiology of the condition she's likely suffering from?

- a) Pelvic mass in pouch of Douglas
- b) Prolapse of anterior vaginal wall
- c) Detrusor instability
- d) Retroverted gravid uterus

Question 22:

Which of the following patients with urinary incontinence will not have detrusor instability?

- a) 1 and 4
- b) 2 and 5
- c) 2 only
- d) 3 only

Question 23:

A 34-year-old G3P2L2 woman presents to the ER with hypotension, tachycardia, and severe abdominal pain. On examination, fetal parts are palpable in the abdomen. USG abdomen revealed a large uterine wall defect. What's the most common cause of this condition?

- a) C-section scar dehiscence
- b) Obstructed labor
- c) Blunt abdominal trauma
- d) Prolonged induction of labor

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	c

5	a
6	c
7	d
8	c
9	b
10	c
11	d
12	b
13	d
14	a
15	d
16	b
17	b
18	a
19	a
20	d
21	c
22	d
23	a

Detailed Explanations

Solution to Question 1:

Vesicovaginal fistulas are mostly acquired.

In developing countries, the most common cause is obstetric trauma. In developed countries, iatrogenic injury during pelvic surgery is responsible for 90% of the vesicovaginal fistulas.

Congenital genitourinary fistulas and fistulas resulting from infectious conditions are rare.

Solution to Question 2:

Uterourethral type of fistula has not been reported.

A genitourinary fistula is defined as an abnormal communication between the urinary system (ureter, bladder, urethra) and the genital systems (vagina, cervix, uterus).

Most genitourinary fistulas are acquired. In developing countries, the most common cause is obstetric trauma. In developed countries, iatrogenic injury during pelvic surgery is responsible

for 90% of the vesicovaginal fistulas.

Congenital genitourinary fistulas are rare. If present, they are commonly associated with other renal or urogenital abnormalities.

Solution to Question 3:

Hysteroscopy is not an etiology for genitourinary fistulae. It can occur following hysterectomy.

Most genitourinary fistulas are acquired. Its causes are enumerated below:

- Prolonged obstructed labor - major cause in developing countries
- Difficult instrumental or manipulative deliveries
- Injury to bladder or ureter during hysterectomy - major cause in developed countries
- Congenital (rare)
- Other causes include:
 - Radiation
 - Malignancy
 - Trauma
 - Infections - PID, tuberculosis etc.

Solution to Question 4:

The given case scenario is suggestive of a vesicovaginal fistula.

Vesicovaginal fistulas usually present with:

- Constant dribbling of urine from the vagina
- Excoriation in the groin region due to constant leakage of urine
- History of pelvic surgery, vaginal delivery or prolonged labor
- Amenorrhea may be present

On examination, the fistula may be visualized by observing urine collection in the vagina. Bimanual examination reveals its fixity and extent.

Note: Ureterovaginal fistulas usually occur following accidental ureteric injury during C-sections in developing countries.

Solution to Question 5:

Cystoscopy is the most useful test and also the investigation of choice for vesicovaginal fistula (VVF).

Cystoscopy or cystourethroscopy helps to visualize the size of the fistulas, the exact location of multiple fistulas and their proximity to the ureteral orifice. It can also assess the bladder mucosa for any injuries.

The 3-swab test (Option B) is done to identify the types of genitourinary fistula.

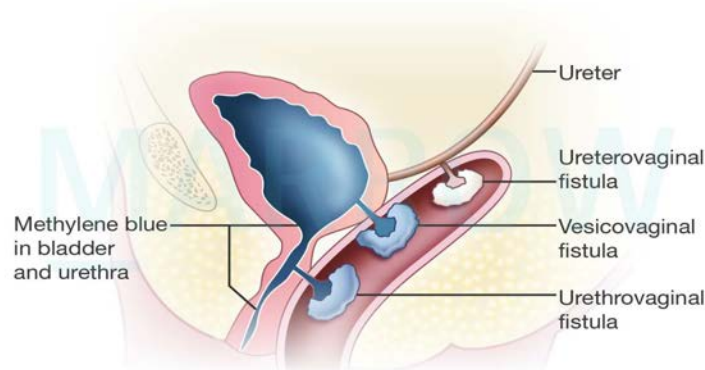
In this test, three sterile pieces of cotton swabs are serially placed in the vaginal cavity, while 50-100ml of methylene blue solution is instilled in the bladder using a catheter. The cotton swabs are placed one at the vault, one at the middle, and one just above the introitus.

Interpretation:

- Uppermost swab gets wet and not stained - the leak is from the ureter, as seen in ureterovaginal fistula
- Middle swab gets stained - the leak is from the bladder, as seen in vesicovaginal fistula
- Lowermost swab gets stained- the leak is from the urethra, as seen in urethrovaginal fistula

The below image shows the 3-swab test.

Three swab test for genitourinary fistula



Option C: MRI is usually not needed for the diagnosis of VVF. They can be used to study vesicouterine fistulas.

Option D: Cystometry is done to measure the capacity of the bladder. It is used to investigate urge incontinence.

Solution to Question 6:

The given clinical scenario is suggestive of a urethrovaginal fistula.

Urethrovaginal fistulas have the following clinical features:

- Continent and dry at rest
- Urinary dribbling during micturition

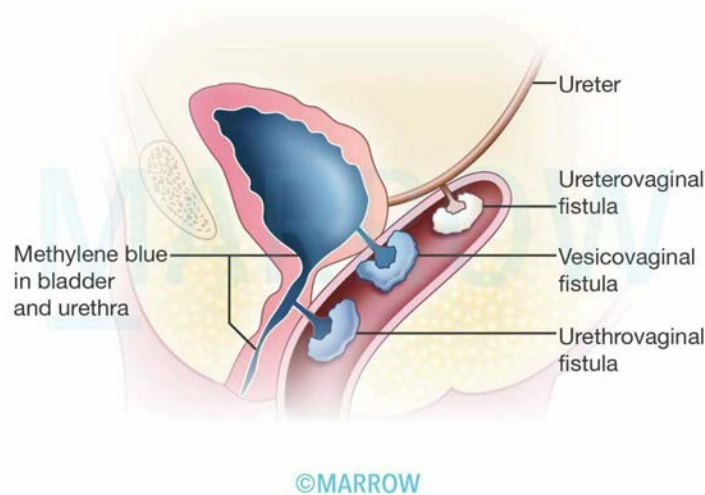
- Fistulous opening is seen on per speculum examination

In the methylene blue 3-swab test, a catheter is introduced into the bladder through the urethra. The vaginal cavity is packed with three sterile cotton swabs. One at the vault, one at the middle, and one just above the introitus. 50–100 mL of dilute methylene blue dye is injected into the bladder using a catheter.

Interpretation:

- The uppermost swab gets wet and not stained - the leak is from the ureter, seen in ureterovaginal fistula
- Middle swab gets stained - the leak is from the bladder, seen in vesicovaginal fistula
- The lowermost swab gets stained- the leak is from the urethra, seen in urethrovaginal fistula

The given image shows the 3-swab test.



Solution to Question 7:

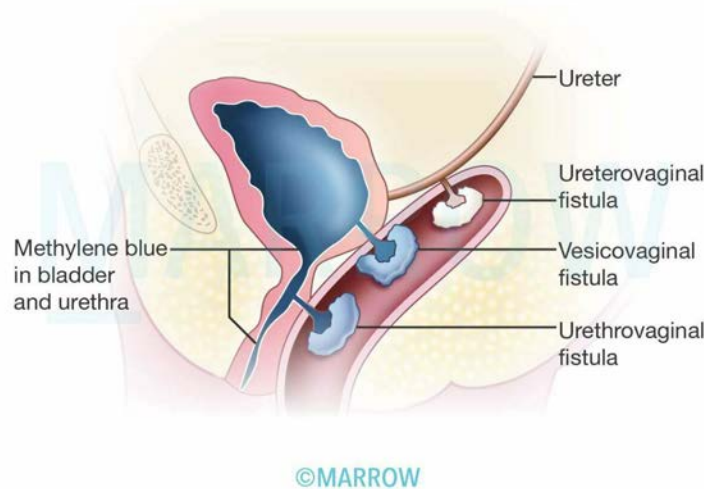
MRI is not preferred for a case of vesicovaginal fistula as other investigations are more specific and useful.

Investigations in VVF include:

- Urine culture - to treat concurrent UTI.
- Ultrasound abdomen - kidney, bladder and ureter should be observed for the presence of a urinoma or hydronephrosis.
- Intravenous pyelography - to look for hydronephrosis
- Cystoscopy - used to visualise the number of fistulae and their location.

Methylene blue 3-swab test in urinary fistulae - A catheter is introduced into the bladder through the urethra and 50-100ml methylene blue is injected. The vaginal cavity is packed with three sterile cotton swabs. It is interpreted as follows:

- The uppermost swab gets wet and not stained - the leak is from the ureter, seen in ureterovaginal fistula.
 - Middle swab gets stained - the leak is from the bladder, seen in vesicovaginal fistula.
 - The lowermost swab gets stained- the leak is from the urethra, seen in urethrovaginal fistula.
- The given image shows the methylene blue 3-swab test.



Solution to Question 8:

In the given scenario, the best time to perform fistula repair is 6 weeks after the inciting surgery.

In uncomplicated fistulas, early surgical intervention is preferred within the first 24 to 48 hours following the inciting surgery. This avoids the brisk postoperative inflammatory response.

However, if extensive and severe inflammation is present, repair of fistula can be delayed to 6 weeks after the surgery. This is usually done in a patient who has undergone major surgery,

During this delay it is ensured that all the tissue inflammation has subsided, tissue vascularisation has improved and local infection has cleared. Estrogen therapy should be instituted in postmenopausal women in this time period.

In case the fistula repair fails, a second attempt is made only after 3 months.

Solution to Question 9:

The given image shows the catheter emerging from the cervical os. This, along with the history of cyclical hematuria, is suggestive of Youssef syndrome seen in a case of vesicouterine fistula.

Vesicouterine fistulas can develop after cesarean section, uterine rupture, and placenta accreta. Most cases present weeks to years later.

Vesicouterine fistulas can present with cyclical hematuria, urinary incontinence or complete continence, menstrual abnormalities, and urinary tract infections. Youssef syndrome is seen in cases of vesicouterine fistulas, and it is associated with a classic triad:

- Amenorrhea
- Cyclical hematuria - also called menouria: Menstrual blood trickles into the bladder through the fistula, causing hematuria
- Urinary continence (there is no vaginal leakage of urine)

Hysteroscopy and cystoscopy are gold-standard investigations.

Vesicouterine fistulas are treated as follows:

- Prolonged bladder catheterization - leads to spontaneous closure
- Surgical repair via abdominal route

Solution to Question 10:

Swabs stained with methylene blue are obtained in a bladder fistula and not in a ureteric fistula.

Ureteric fistulas can present with the following features:

- Constant urinary dribbling
- Flank pain or loin tenderness
- Renal angle tenderness
- Fever
- Hematuria
- Oliguria

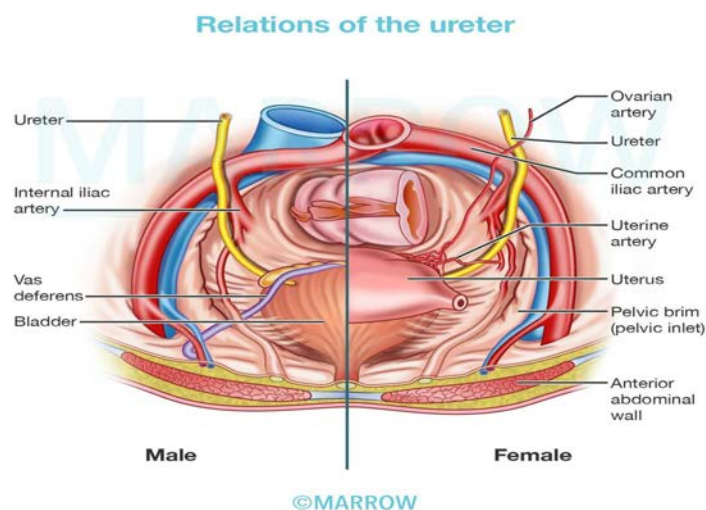
Ureteric fistulas are most commonly caused by iatrogenic trauma during pelvic surgeries, especially gynecological surgeries. An excretory urogram (intravenous pyelography) or a CT urogram are the investigations of choice. Hydronephrosis may be present on the affected side.

It is managed by performing a ureteroneocystostomy and percutaneous nephrostomy.

Solution to Question 11:

The ureter is identified intraoperatively by the following features:

- Ovarian artery crosses anterior to it
- It is cord-like on palpation
- Peristalsis occurs on palpation
- It has a pale, glistening appearance
- Uterine artery crosses it, from lateral to medial, 2 cm lateral to the cervix



Solution to Question 12:

The given case scenario is suggestive of ureteric injury. The highest incidence of this complication is seen with Wertheim's hysterectomy.

Urinary tract injury occurs almost exclusively in major gynecologic surgery that involves surgical dissection in proximity to the ureters or bladder. Vaginal hysterectomy tends to have lower rates of ureteric injuries than open (abdominal, Wertheim) or laparoscopic routes. The highest risk for ureteric injury is associated with the following procedures:

- Surgical debulking of gynecological cancers
- Advanced pelvic reconstructive surgery
- Laparoscopic hysterectomy

Patients with ureteric injuries have a complicated post-operative period. They can present with the following features:

- Fever
- Flank pain
- Abdominal distension
- Prolonged paralytic ileus - bowel sounds are absent
- High drain outputs, in case abdominal drains are placed.

CT urogram and retrograde pyelography are the investigations of choice. They reveal urinary extravasation from the ureter. Ureteric injuries are prevented by careful division of tissue planes during pelvic or genitourinary surgeries. The condition is managed by prompt surgical repair by ureteroureterostomy or percutaneous ureterostomy.

Solution to Question 13:

Iatrogenic injury to the ureter is unlikely to occur at the ureteropelvic junction during gynaecological surgeries like a hysterectomy.

Ureteric injury is most commonly caused by gynecological surgeries such as laparoscopic hysterectomy, abdominal hysterectomy, etc. It can occur at:

- The infundibulopelvic ligament on the lateral pelvic wall
- In the ureteric canal when the uterine vessels are ligated
- Near the internal cervical os
- Near the uterosacral ligament

Solution to Question 14:

Improper suturing of complete perineal tears is the most common cause of rectovaginal fistula.

Rectovaginal fistulas occur most commonly due to obstetric causes such as perineal tears, obstructed labor, and instrumental deliveries. Other causes include:

- Tuberculosis
- Lymphogranuloma venereum
- Bartholinitis - rupture of an abscess into the rectum can cause a fistula
- Diverticulitis
- Carcinoma cervix
- Radiotherapy - can cause ulceration which can develop into a fistula

Solution to Question 15:

The external urinary sphincter is innervated by the Onuf's nucleus, but not the internal one.

The following factors are prerequisites for continence in women:

- Position of internal urinary sphincter lies above levator ani muscles
- Normal urethral closure pressure is >20 cm of water
- Urethrovesical angle $>100^\circ$
- The tone of pelvic floor muscles

Solution to Question 16:

The given case scenario is suggestive of a cystocele in a case of genital prolapse. Stress incontinence is seen in these cases.

In stress incontinence, involuntary leakage of urine occurs when there is increased intraabdominal pressure, such as during laughing, sneezing, or coughing. There is no normal urge to pass urine.

Option A: Urge incontinence is the involuntary voiding of urine when there is an imminent urge to urinate.

Option C: Overflow incontinence is an old term that refers to the inability to empty the bladder, with involuntary, continuous urinary dribbling. Currently, it refers to another presentation of urge incontinence.

Option D: Neurogenic bladder occurs due to defects anywhere in the pathway of the micturition reflex, e.g., spinal cord trauma, diabetes etc. It presents with with urgency or urge incontinence.

Solution to Question 17:

Marshall and Bonney's test is used to diagnose genuine stress incontinence. It also helps in assessing how successful a surgical repair will be.

It is performed as follows:

Test	Marshall test	Bonney's test
Procedure	The vagina in the region of the bladder neck is infiltrated with local anesthetic and it is elevated.	Two fingers are placed in the vagina at the urethrovesical junction and the bladder neck is elevated.
Positive test	Absence of leakage of urine on straining or coughing	Absence of leakage of urine on straining or coughing

Solution to Question 18:

The given case scenario is suggestive of stress incontinence. It can be diagnosed with the help of the Q-tip test.

This test is performed by placing a xylocaine-covered Q-tip cotton swab stick in the patient's urethra. Initially, the stick will be parallel to the floor. Then the patient is asked to strain or cough. Normally the stick will reach an angle of around 10-15°.

In genuine stress incontinence (GSI), this angle increases by 20° or more. A positive test indicates a sufficient degree of bladder neck descent. Unfortunately, all patients with GSI may not have a positive test, and it is not specific.

Tests done for GSI(Genuine stress incontinence) include:

- Stress test
- Cotton swab stick test
- Marshall and Bonney test

- Urethroscopy
- Urodynamic evaluation- cystometry, urethrocystometry, uroflowmetry etc

Solution to Question 19:

The given scenario is suggestive of stress incontinence. In this case, the first line of treatment is conservative management.

Stress incontinence is managed as follows:

Conservative management:

- It is the first line of treatment.
- It includes:
 - Kegel pelvic floor exercises for 4–6 months
 - Electric/magnetic stimulation for nerve damage, magnetic stimulation
 - Artificial urinary sphincter in neurological condition
 - Vaginal cones
 - It is also done in case surgery fails.

Medical management:

- Estrogen cream in menopausal woman
- Venlafaxine 75 mg daily
- Imipramine 10–20 mg BD

Surgical management:

- Kelly's repair - vaginal approach
- Marshall–Marchetti–Krantz operation - abdominal approach. It not done anymore as it causes osteitis of the pubis.
- Burch colposuspension
- Combined vaginal and abdominal suspension - Pereyra operation
- Slings
- Tension-free vaginal taping
- Transobturator taping
- Laparoscopic suspension of the bladder neck
- Periurethral collagen injection

Vaginal sling operations have the highest long term cure rate in cases of GSI (genuine stress incontinence). Their long term cure rate is more than 90%.

Solution to Question 20:

Latzko procedure is not beneficial for a patient with genuine stress incontinence as it is done for vesicovaginal fistula.

The various surgeries done for genuine stress incontinence (GSI) are as follows:

- Vaginal approach
- Kelly repair
- Pacey repair
- Marshall-Marchetti-Krantz operation - abdominal approach. It is not done anymore as it causes osteitis of the pubis.
- Burch colposuspension
- Combined vaginal and abdominal suspension - Pereyra operation
- Slings
- Tension-free vaginal taping
- Transobturator taping
- Laparoscopic suspension of the bladder neck
- Periurethral collagen injection

Solution to Question 21:

The given case scenario is suggestive of urge incontinence. It occurs due to detrusor instability.

During normal micturition, the detrusor muscle contracts after being stimulated by the parasympathetic nerves. In detrusor instability, detrusor muscle contracts spontaneously or on provocation during filling of the bladder. It is often caused by overactivity of parasympathetic nerves. It is usually seen in older women and is caused by:

- Decreased bladder capacity - due to fibrosis
- Decreased sensation
- Central nervous system (CNS) inhibition - due to stroke, multiple sclerosis, spinal cord involvement etc.

It is associated with the following symptoms:

- Incontinence
- Urge to urinate
- Increased frequency
- Nocturia
- Urination during sexual intercourse and with the sound of water, hand-washing.

Solution to Question 22:

In patient 3, detrusor instability is not seen, as this is a case of functional incontinence.

During normal micturition, the detrusor muscle contracts after being stimulated by the parasympathetic nerves. In detrusor instability, detrusor muscle contracts spontaneously or on provocation during filling of the bladder. It is often caused by the overactivity of parasympathetic nerves. It is seen usually in older women and is caused by:

- Decreased bladder capacity - due to fibrosis
- Decreased sensation
- Central nervous system (CNS) inhibition - due to stroke, multiple sclerosis, spinal cord involvement etc.

It is associated with the following symptoms:

- Incontinence
- Urge to urinate
- Increased frequency
- Nocturia
- Urination during sexual intercourse and with the sound of water, hand-washing.

Treatment is initially conservative and include decreasing caffeine intake, smoking cessation, fluid restriction and weight reduction. Anticholinergic drugs and bladder training may also be tried.

Note: Functional incontinence occurs in patients with disabilities, whose bladder movements are normal. However, due to environmental or physical barriers, involuntary leakage of urine occurs.

Solution to Question 23:

The given case scenario is suggestive of uterine rupture. The most common cause is scar dehiscence due to a previous cesarean section. This can occur due to trial of labour after C-section, or due to spontaneous labor in a patient with previous C-section. Other causes of uterine rupture are:

- Prolonged induction or augmentation of labor
- Obstructed labor
- Blunt abdominal trauma during pregnancy
- Rupture of hematometra
- Rupture of pyometra

Uterine rupture is managed as follows:

- Immediate cesarean section
- Depending upon the extent of rupture:
- Subtotal hysterectomy

- Hysterectomy

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Prolapse

Question 1:

Which of the following is an uncommon etiology for prolapse?

- a) Menopause
- b) Atonicity of uterus
- c) Obesity
- d) Congenital

Question 2:

What is the most important support of the pelvic viscera?

- a) Levator ani muscle
- b) Anogenital diaphragm
- c) Uterosacral ligaments
- d) Cardinal ligaments

Question 3:

A woman presented to you with the following finding. Which of the following is not a risk factor for this condition?



- a) Vaginal delivery
- b) Premature bearing down
- c) Episiotomy
- d) Chronic bronchitis

Question 4:

A 64-year-old woman presents to the gynecology clinic with complaints of a mass descending PV. On examination, you note that the external os lies at the level of the introitus. What is the degree of uterine prolapse?

- a) First degree
- b) Second degree
- c) Third degree
- d) Procidentia

Question 5:

Which of the following is correctly matched between classification systems of prolapse and their reference point?

- a) Baden-Walker halfway system - Hymen
- b) POP-Q system - Introitus
- c) POP-Q system - Internal os
- d) Shaws - Hymen

Question 6:

Which of the following is seen in congenital elongation of the cervix?

- a) The vaginal portion of the cervix gets elongated
- b) The supravaginal portion of the cervix gets elongated
- c) Both the vaginal and supravaginal portions of the cervix get elongated
- d) There is no true elongation and it is a misnomer

Question 7:

Which of the following differential diagnosis will you not consider in a case of prolapse?

- a) Cervical fibroid
- b) Urethral diverticula
- c) Chronic uterine inversion
- d) Retroverted uterus

Question 8:

A 57-year-old woman presents to you with complaints of mass descending per vaginum for the past 6 months. On examination, grade 3 uterovaginal prolapse and a decubitus ulcer are observed. What causes this ulcer?

- a) Thrombosis
- b) Venous congestion
- c) Infection
- d) Trauma

Question 9:

A post-menopausal woman presents with a mass protruding from the vagina that increases in size on coughing and sneezing. Which of the following is not a complication of this condition?

- a) Incarceration
- b) Hydronephrosis
- c) Squamous cell carcinoma
- d) Urge incontinence

Question 10:

A multiparous woman notices a bulge at the vulva that diminishes in size on lying down. She also finds it difficult to initiate micturition. What is the likely diagnosis?

- a) Uterine prolapse
- b) Vesicovaginal fistula
- c) Cystocoele

- d) Neurogenic bladder

Question 11:

Which of the following is correct about the POP-Q staging system?

- a) Stage 4 involves complete eversion of the vagina.
- b) In stage 3, the leading point is 1 cm above hymen.
- c) Five reference points are present in the vagina.
- d) The points above the hymen are described as plus.

Question 12:

A 35-year-old woman presents with pelvic pain, and a mass descending PV with increases in size on coughing. She is posted for a Fothergill repair. Which of the following is false about this procedure?

- a) It is also known as Manchester operation
- b) It does not preserve the reproductive functions
- c) Recurrent abortions are a sequelae of this operation
- d) Amputation of cervix is done

Question 13:

A 55-year-old lady presents with third degree uterine prolapse. Which of the following is the best management for her?

- a) Fothergill repair
- b) Sling operation
- c) Vaginal hysterectomy with pelvic floor repair
- d) Shirodkar procedure

Question 14:

A 75-year-old woman diagnosed with procidentia is posted for operative repair. During a pre-anesthetic check-up, she was determined to be medically unfit for major surgery. Which of the following procedures is recommended in her?

- a) Fothergill repair
- b) Shirodkar's procedure
- c) Le Fort's repair
- d) Manchester operation

Question 15:

A 25-year-old nulliparous woman presents to you with primary infertility. On examination, stage 2 uterine prolapse is present. What is the next step in management?

- a) Ring pessary
- b) Fothergill's operation
- c) Sling operation
- d) Colpocleisis

Question 16:

A 67-year-old woman diagnosed with uterine prolapse, now presented with dyspareunia and fecal incontinence. After manually reducing the prolapse, a peristaltic bulge is seen in the posterior vaginal wall. What is the preferred method of management in this case?

- a) Moschcowitz repair
- b) Purandare's operation
- c) McCall culdoplasty
- d) Le Fort's repair

Question 17:

How many degrees of vault prolapse are there?

- a) 2
- b) 3
- c) 4
- d) 5

Question 18:

A 55-year-old woman presented with a mass bulging into the vagina, difficult intercourse, and back pain. She has a history of hysterectomy 2 years back. On examination, you observe that the vaginal vault is protruding outside the introitus. What is the best management in this case?

- a) Sacrospinous colpopexy
- b) Colpocleisis
- c) Kelly's repair
- d) Burch colposuspension

Question 19:

A young woman with uterine prolapse following pelvic trauma underwent a hysterectomy. A year later, she presents with vault prolapse. What would you not recommend for her?

- a) Transabdominal sacropexy
- b) Transvaginal sacrocolpopexy
- c) Posterior intravaginal slingplasty
- d) Ring pessary

Question 20:

The given instrument was used to treat a patient with uterine prolapse. Which of the following statements is false about this instrument?



- a) It should be changed every 3 months.
- b) It can be used in pregnant patients.
- c) It is curative.
- d) It may rarely cause carcinoma of vagina.

Question 21:

Which of the following is the first step after fitting the pessary to check if the pessary is adequately retained?

- a) Asking the patient to void and see if the pessary is in place
- b) Seeing if the patient feels any discomfort, which means that the pessary is retained
- c) Asking the patient to perform the Valsalva manoeuvre and see if the pessary is in place
- d) Being able to admit two fingers between the ring and the pessary.

Question 22:

The deepest part of perineal body injury, which causes cystocoele, enterocoele and urethral descent is ____

- a) Pubococcygeous
- b) Bulbospongiosus
- c) Ischiocavernosus
- d) Superficial transverse perineal

Question 23:

All of the following are correctly matched regarding the management of prolapse except

- a) Nulliparous - Abdominal sling operations
- b) Vaginal pessary is contraindicated in pregnancy
- c) Postnatal - Ring pessary and pelvic floor exercises for 3–6 months
- d) Woman beyond 40 years and multipara - Vaginal hysterectomy and pelvic floor repair

Question 24:

All of the following surgeries are done in vault prolapse except _____

- a) Uterosacral suspension
- b) Colposuspension
- c) Sacral colpopexy
- d) Sacrospinous ligament fixation

Answer Key

Question No.	Correct Option
1	d
2	a
3	c
4	b
5	a
6	a
7	d
8	b
9	d
10	c
11	a
12	b
13	c
14	c
15	c
16	a
17	b
18	a
19	d
20	c
21	c
22	a
23	b
24	b

Detailed Explanations

Solution to Question 1:

Congenital prolapse is an uncommon cause of prolapse.

The most important etiological factors that lead to prolapse are the atonicity of the uterus and asthenia, which occurs due to aging and follows menopause.

The other factors which lead to prolapse are as follows:

- Pregnancy
- Vaginal childbirth
- Menopause
- Aging
- Hypoestrogenism
- Peripheral nerve injury to the pudendal nerve
- Premature bearing down before full dilatation of the cervix
- Chronically increased intra-abdominal pressure
- Chronic bronchitis
- Constipation
- Obesity
- Spina bifida occulta
- Congenital weakness of pelvic floor muscles
- Abdominoperineal excision of rectum
- Surgeries such as Stamey and Pereyra operation for stress incontinence

Solution to Question 2:

The most important support of the pelvic viscera is the levator ani muscle. The levator ani forms a large portion of the pelvic floor. It is composed of three separate paired muscles:

- Pubococcygeus
- Puborectalis
- Iliococcygeus

Supports of the genital organs are given by DeLancey's levels of support. However, the broad ligament is a fold of the peritoneum, which doesn't provide support to pelvic organs.

Solution to Question 3:

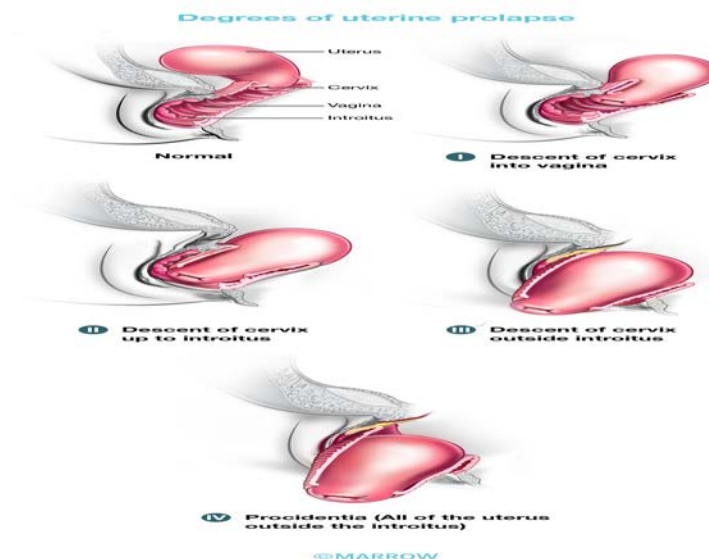
The given image shows uterine prolapse, for which an episiotomy is not a risk factor. Episiotomy shortens the second stage of labor and prevents the atonicity of the uterus, thus reducing the risk of prolapse.

The risk factors for uterine prolapse are:

- Pregnancy
- Vaginal delivery
- Menopause
- Aging
- Hypoestrogenism
- Peripheral nerve injury to the pudendal nerve
- Premature bearing down before full dilatation of the cervix
- Chronically increased intra-abdominal pressure
- Chronic bronchitis
- Constipation
- Obesity
- Spina bifida occulta
- Congenital weakness of pelvic floor muscles
- Abdominoperineal excision of rectum
- Surgeries such as Stamey and Pereyra operation for stress incontinence

Solution to Question 4:

The given case scenario is suggestive of 2nd degree prolapse.



Solution to Question 5:

In Baden-Walker halfway system of classification of prolapse, the reference point is the hymen.

Prolapse classification system	Reference point
Shaws	Introitus
Baden Walker halfway	Hymen
POP-Q	Hymen

Solution to Question 6:

In congenital elongation of the cervix, the vaginal portion of the cervix is elongated. The fornices are hence deep, and there is no accompanying vaginal prolapse.

In 2nd and 3rd degree uterine prolapse, the supravaginal portion of the cervix gets stretched and elongated as it is well supported by the Mackenrodt's ligaments, while the vaginal portion of the cervix prolapses with the vagina. In prolapse, the cervix can elongate to as much as 10cm in length.

However, in procidentia, the entire uterus slides with the vagina, and hence the cervix retains its normal length.

Solution to Question 7:

The given case scenario is suggestive of uterine prolapse. A retroverted uterus is not a differential diagnosis for uterine prolapse.

However, retroversion of the uterus does occur with uterine prolapse.

Differential diagnoses of genital prolapse include:

- Congenital elongation of the cervix - vaginal part of the cervix is elongated. In genital prolapse, supravaginal part is elongated and vaginal part is prolapsed.
- Vulval cyst and Gartner cyst - large cysts may mimic prolapse
- Cyst of the anterior vaginal wall - tense with well-defined margins and cannot be reduced on pressure.
- Urethral diverticula - rare, small, and situated low in the anterior vaginal wall. Urethroscopy helps in the diagnosis.
- Cervical fibroid polyp - cervix remains high up in its anatomical position.

- Chronic inversion of the uterus - Uterine sound is used to confirm the diagnosis. USG and laparoscopy will show fundal depression and the absence of uterine fundus in the pelvis.
- Rectal prolapse

Solution to Question 8:

Decubitus ulcer seen in uterine prolapse is due to venous congestion and local circulatory changes.

Decubitus ulcer refers to the ulceration of prolapsed tissue which occurs due to the chronic dependent position causing venous congestion and stasis. It has a clean edge and usually heals well with prolapse reduction and daily packing with glycerine (hygroscopic agent) and acriflavine (antiseptic). It is a benign and non-infectious disease and does not require further investigation such as biopsy or culture.

Solution to Question 9:

The given case scenario is suggestive of pelvic organ prolapse. Urge incontinence is not a complication of this condition.

Complications of prolapse are:

- Damage to the ureter - can occur during
- Surgical repair
- Kinking due to procidentia
- Urinary tract infection - due to retention of urine in a cystocele
- Hydronephrosis and hydroureter
- Incarceration of the prolapse - the prolapsed organ becomes irreducible
- Decubitus ulcer
- Stress incontinence
- Squamous cell carcinoma over the decubitus ulcer (rare)

Solution to Question 10:

The given case scenario is suggestive of a cystocele.

A cystocele is the herniation of the urinary bladder into the vagina. It is usually associated with uterine prolapse, and presents with the following features:

- Mass descending in the vagina -
- Increases in size on straining and coughing

- Reduces on lying down, or manual reduction
- Difficulty in micturition - difficulty initiating, straining required
- Mass needs to be pushed inwards to urinate - repositioning of cystocele required
- Increased frequency of micturition
- Backache - due to stretching of uterosacral ligaments

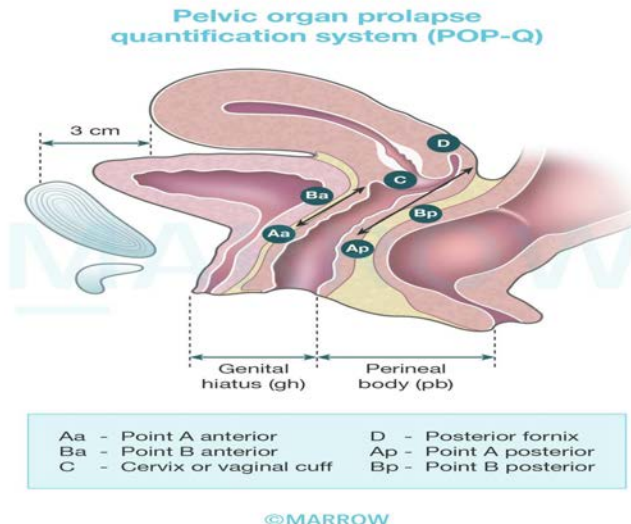
Solution to Question 11:

The Pelvic Organ Prolapse Quantification (POP-Q) system was described by the International Continence Society and has the following features:

- The hymen is taken as a fixed point.
- Six reference points are measured, using a scaled spatula:
- Two on the anterior vaginal wall
- Two on the apical vagina
- Two on the posterior vaginal wall
- Three measurements: The genital hiatus, perineal body, and total vaginal length.
- The points above the hymen are described as minus and points below as plus.

Staging of pelvic organ prolapse according to the POP-Q system:

Stage	Description
Stage 0	No descent of pelvic organs
Stage 1	The leading edge of the prolapse is >1 cm above the level of the hymen.
Stage 2	The leading edge of the prolapse is <1 cm proximal or distal to the hymen.
Stage 3	The leading point of prolapse is >1 cm below the hymen but protrudes no further than 2 cm less than the total length of the vagina.
Stage 4	Complete eversion of the vagina



Solution to Question 12:

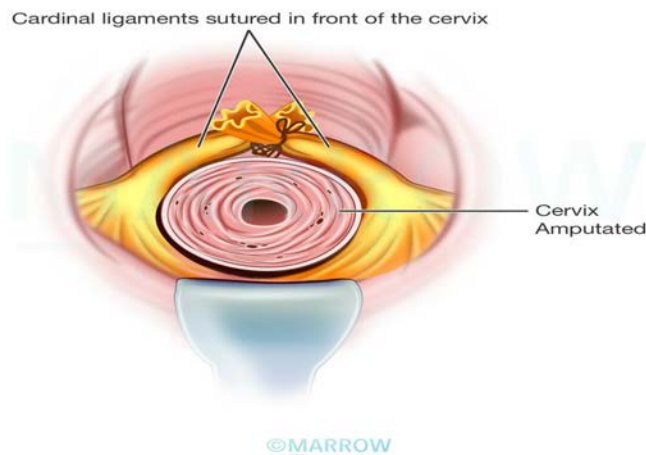
Fothergill's operation preserves menstrual and reproductive functions.

Fothergill's repair is also known as Manchester's operation. It is suitable for women under 40 years who are desirous of retaining their menstrual and reproductive functions. The steps are:

- Preliminary D and C
- Anterior colporrhaphy with amputation of the cervix.
- Raw area covered with the vaginal mucosa.
- Mackenrodt ligaments sutured in the front of the cervix
- Colpoperineorrhaphy

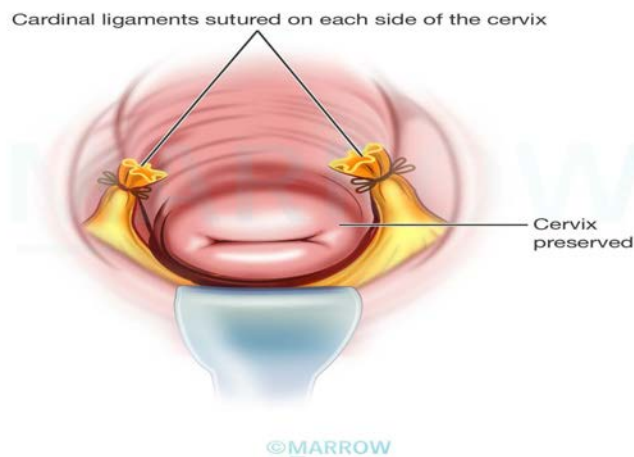
Since the uterus is preserved in this procedure, the patient is able to menstruate and can conceive and carry the pregnancy. However, fertility decreases due to the loss of cervical mucus. Moreover, cervical incompetence occurs, leading to recurrent abortions and preterm deliveries.

Manchester's/Fothergill's procedure



Shirodkar's modification of Fothergill's repair avoids cervical amputation, hence subsequent pregnancy complications are avoided.

Shirodkar's procedure



Solution to Question 13:

In the clinical scenario of a 55-year-old woman with third-degree uterine prolapse, the ideal management is vaginal hysterectomy with pelvic floor repair.

Uterine prolapse is the descent of the uterus from its normal position which usually occurs due to laxity of ligaments that offer support, seen often due to childbirth, aging, or trauma. The uterine descent is clinically categorized into:

- First degree: the external os remains above the vaginal introitus

- Second degree: The external os protrudes below the vaginal introitus but the uterine body remains inside the vagina
- Third degree (Syn: Complete prolapse): The uterine, cervix, body, and fundus descend to lie outside the introitus
- Procidentia involves prolapse of the uterus with the entire vagina everted.

The patient usually presents with a sense of heaviness in the pelvic region with lower back discomfort. Based on the structures involved the patient can additionally have urinary symptoms (like incontinence and urgency), bowel disturbances (constipation and tenesmus), or sexual discomfort during intercourse.

Symptomatic uterine prolapse is managed surgically with different procedures based on the profile of the patient. Conservative management is done for old asymptomatic women with mild degree prolapse.

A vaginal hysterectomy is recommended for patients who satisfy the following criteria:

- >40 years of age
- Have completed their families
- Do not wish to preserve menstrual and child-bearing function

Other options:

Option A: Fothergill repair or Manchester operation is a procedure done to preserve the reproductive function and vaginal prolapse with supravaginal cervix elongation. It initially involves preliminary curettage with low amputation of the cervix. After Mackenrodt's ligament is plicated, anterior colporrhaphy and colpoperineorrhaphy are performed.

Option B: Sling operation (Purandare's operation) is indicated in congenital or nulliparous prolapse without cystocele. The procedure involves the extraperitoneal passage of rectus sheath strips joined to the anterior surface of the cervix by silk sutures.

Option D: Shirodkar procedure is a type of abdominal sling surgery done in young women to retain menstrual and reproductive function. It involves using a sling to fix the cervix to the lumbosacral fascia.

Solution to Question 14:

Le Fort's repair is recommended in this case. This repair is done in patients who satisfy the following criteria:

- Elderly (>60 years)
- Do not wish to retain sexual function
- Medically unfit for major surgeries

Solution to Question 15:

Sling operations are recommended in cases of nulliparous prolapse, who wish to retain menstrual and child-bearing function

Mackenrodt's and uterosacral ligaments of the uterus are strengthened, by using nylon or Dacron tapes as slings. Abdominal sling operations include:

- Abdominal wall cervicopexy - cervical supports are fixed to the abdominal wall
- Shirodkar's sling operation - cervix is fixed to the lumbosacral fascia using Mersilene tape
- Khanna's abdominal sling operation - isthmus of the cervix is fixed to the anterior superior iliac spines. It is not done anymore due to the risk of periosteitis.
- Virkud's sling operation - the tape is fixed from the isthmus and attached to the rectus compartment
- Joshi's sling operation - the uterus is suspended from the pectineal ligaments.

Solution to Question 16:

The given case scenario is suggestive of an enterocele associated with fecal incontinence. It is preferably repaired using an abdominal approach like Moschcowitz repair.

An enterocele is herniation of the pouch of Douglas and small bowel loops into the rectovaginal septum. Enteroceles are most commonly associated with uterine prolapse. Patients present with the following clinical features:

- Fecal incontinence
- Sexual dysfunction
- Constipation/diarrhea
- Mass descending PV
- Pelvic pain or pelvic discomfort
- A bulge in the posterior vaginal wall:
- Upper 1/3rd - enterocele
- Lower 2/3rd - rectocele
- Peristalsis may be observed in the bulge

Enterocele repair is done via the following methods:

- Vaginal approach - it is usually preferred over an abdominal approach. In McCall's culdoplasty, the uterosacral ligaments are used to suspend the vaginal vault. The posterior peritoneum and uterosacral ligaments are approximated.
- Abdominal approach - it is preferred in cases where there is associated fecal incontinence. It includes:
 - Moschcowitz repair - the pouch of Douglas is obliterated by several purse-string sutures.
 - Halban's procedure - the peritoneal sac is obliterated using vertical sutures. This can also be done transvaginally.

Option B: Purandare's sling operation is abdominal wall cervicopexy for uterine prolapse. It is done for young nulliparous women.

Option D: Le Fort's repair is done in elderly patients with uterine prolapse, who are unfit for surgery.

Note: Enteroceles may be present along with uterine prolapse. Failure to identify and repair it during vaginal hysterectomy can lead to vault prolapse.

Solution to Question 17:

There are 3 degrees of vault prolapse.

Vault prolapse is a delayed complication of both abdominal and vaginal hysterectomy. It can also occur due to failure to identify and repair an enterocele during a hysterectomy.

In vault prolapse, level 1 supports become weak and deficient. Risk factors apart from past history of surgery are:

- Advanced age
- Menopause - due to estrogen deficiency
- Parity
- Obesity
- Chronic cough

Solution to Question 18:

The given case scenario is suggestive of vault prolapse. Sacrospinous colpopexy is the gold standard management for vault prolapse.

Vault prolapse presents with pelvic or back pain, the descent of the vault through the vagina, difficulties in intercourse, and difficulty in walking due to the mass. It is commonly associated with a cystocele or a rectocele.

It can be prevented by identifying and repairing an enterocele and by attaching the uterosacral and cardinal ligaments to the vault during a hysterectomy.

Vault prolapse is managed as follows:

- If the patient is fit for abdominal surgery: Transabdominal sacral colpopexy, i.e., uterosacral suspension
- If the patient is not fit for abdominal surgery: Transvaginal sacrospinous ligament fixation/colpopexy - the gold standard
- Laparoscopic colpopexy
- If the patient does not want to retain sexual function:
- Colpocleisis

- Le Fort's repair
- Anterior colporrhaphy in case of cystocele
- Posterior colporrhaphy in case of rectocele
- If the patient is unfit for surgery, ring pessaries can be tried
- Posterior intravaginal slingplasty

Note: Kelly's repair and Burch colposuspension are done to manage stress incontinence.

Solution to Question 19:

In a young patient with vault prolapse, ring pessary is not recommended.

Vault prolapse is managed as follows:

- If the patient is fit for abdominal surgery: Transabdominal sacral colpopexy, i.e., uterosacral suspension - The best method for vault prolapse.
- If the patient is not fit for abdominal surgery: Transvaginal sacrospinous ligament fixation.
- Laparoscopic colpopexy
- If the patient does not want to retain sexual function:
 - Colpocleisis
 - Le Fort's repair
 - Anterior colporrhaphy in case of cystocele
 - Posterior colporrhaphy in case of rectocele
- If the patient is unfit for surgery, ring pessaries can be tried
- Posterior intravaginal slingplasty

Solution to Question 20:

The given image shows a ring pessary, which is only a palliative treatment for genital prolapse.

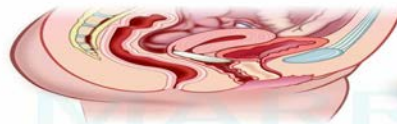
A ring pessary is used for patients who cannot be surgically managed, such as pregnant patients and very elderly women. In these patients, it is used as an adjunct to pelvic floor physiotherapy.

Pessaries provide mechanical support to the pelvic organs. After being placed, they should be changed every 3 months. They may cause vaginitis; dyspareunia; and, rarely, carcinoma if retained for a long time.

The image below shows the types of pessaries:



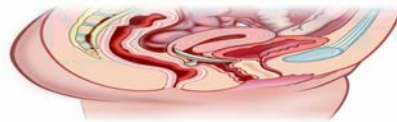
Types of pessaries



Doughnut pessary



Ring pessary with support



Gellhorn pessary



©Marrow

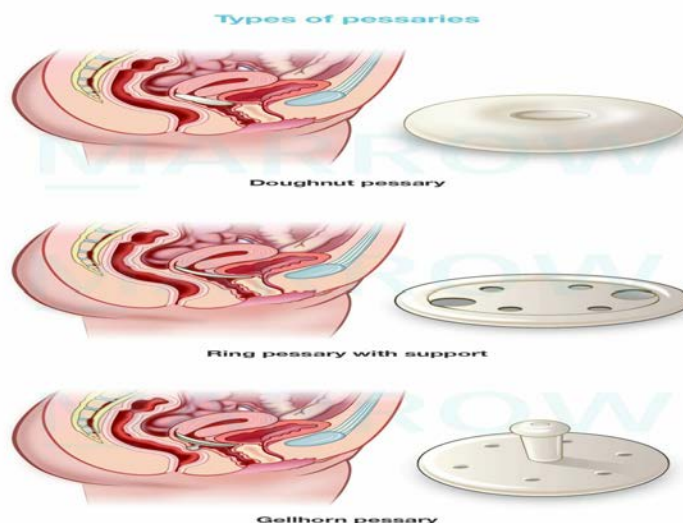
Solution to Question 21:

The first step after fitting the pessary to check if the pessary is adequately retained is to ask the patient to perform Valsalva's maneuver or cough and see if the pessary is in place.

There is some level of descent of the pessary with these maneuvers is to be expected, but the pessary should return to its normal position when these efforts are stopped. If the pessary descends to the introitus with these maneuvers, then a larger size pessary should be tried.

The pessary is well-fitted if you are able to slide the ring of the pessary up and down along the vaginal sidewall. It is well-fitted if you are able to insert one finger between the vaginal sidewall and the edge of the pessary. When the labia are separated, a well-supported pessary is not usually visible.

The patient should be comfortable, and able to void urine with the pessary in place before leaving the clinic.



Solution to Question 22:

Pubococcygeus is the deepest part of the levator ani muscle. Injury to it can cause enterocele, cystocele, and urethral descent.

Levator ani muscles are the most important muscles in the pelvic floor and maintain a constant state of contraction. They provide a stable floor and support the weight of the abdominopelvic contents against intra-abdominal forces. Weakening of levator ani and pelvic fascia from stretching or tearing during childbirth, may decrease support for the vagina, bladder, uterus, or rectum or alter the position of the neck of the bladder and urethra.

The levator ani muscle is a complex unit, which consists of 3 muscle components namely

- Pubococcygeus (anterior fibers)
- Pubovaginalis - elevates the urethra and hence contributes to urinary continence
- Puboperinealis - draws the perineal body towards the pubic symphysis
- Puboanalis - elevates the anus and keeps the urogenital hiatus narrowed
- Puborectalis (intermediate fibers)
- Iliococcygeus (posterior fibers)

Origin: The levator ani muscle has a linear origin from the pelvic surface of the body of the pubis, a tendinous arch of obturator fascia, and the pelvic surface of the ischial spine.

Insertion:

- The anterior fibres form a sling around the prostate (levator prostatae) or vagina (sphincter vaginae) and are inserted into the perineal body (a mass of fibrous tissue) in front of the anal canal.

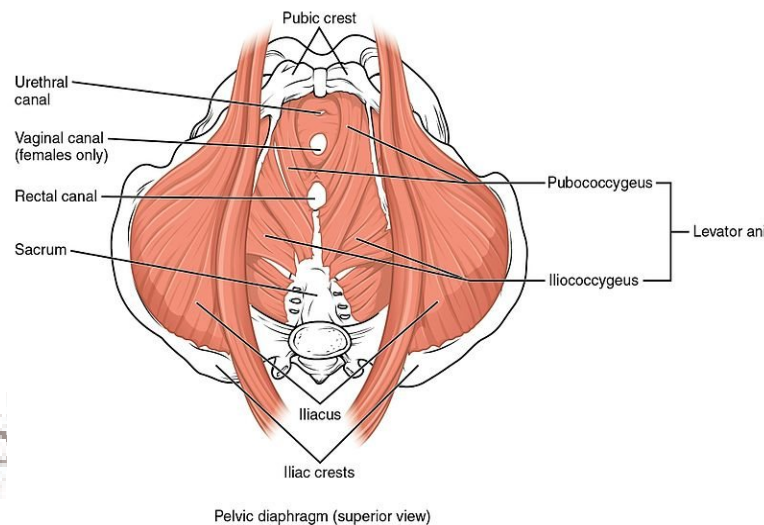
- The intermediate fibres (puborectalis) form a sling around the anorectal junction to be inserted into the anococcygeal raphe (a fibrous raphe which extends from the anorectal junction to the tip of the coccyx).
- The posterior fibres (iliococcygeus) are inserted into the anococcygeal raphe and the coccyx.

Nerve supply:

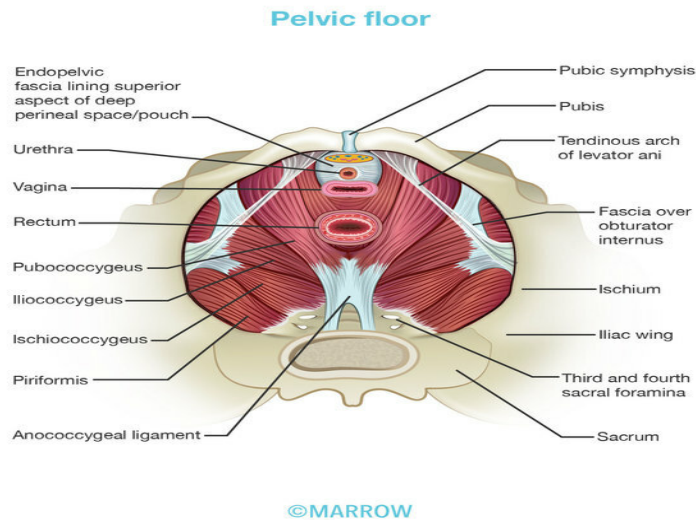
- perineal branch of 4th sacral nerve (S4) from its pelvic surface.
- perineal branch of the pudendal nerve (S2, S3) from its perineal surface.

Actions:

- The levator ani muscles of two sides together support the pelvic viscera and resist the intra-abdominal pressure during straining and expulsive efforts of the anterior abdominal wall muscles.
- They also subserve a sphincteric action on the anorectal junction to maintain the continence of faeces in both sexes and vagina in females.



The given image below shows the pelvic floor:



Solution to Question 23:

The vaginal pessary is not contraindicated in pregnancy.

During pregnancy, depending on the severity and patient preference, prolapse is managed conservatively as follows:

- Intravaginal pessaries - simple, effective, potentially effective in preventing preterm birth
- Pelvic floor exercises
- Pelvic physiotherapy
- Electrical therapy

Solution to Question 24:

Colposuspension is done in stress urinary incontinence and not for vault prolapse.

Management of Vault prolapse:

- If the patient is fit for abdominal surgery: Transabdominal sacral colpopexy, i.e., uterosacral suspension
- The best method for vault prolapse.
- If the patient is not fit for abdominal surgery: Transvaginal sacrospinous ligament fixation.

Fibroid

Question 1:

Which of the following statements is false about fibroids?

- a) More common in Asians than African-Americans
- b) Smoking is a protective factor
- c) More common in nulliparous women
- d) Does not usually increase in size during pregnancy

Question 2:

Which of the following genes is not associated with leiomyomas?

- a) MED12
- b) HMGA2
- c) COL4A5-A6
- d) MLH1

Question 3:

Leiomyomas create a hyperestrogenic state by all of these mechanisms except:

- a) Cells contain a greater density of estrogen receptors
- b) Metabolise androgens to estrogen
- c) Have higher levels of cytochrome P450 aromatase
- d) Maximally convert estradiol to estrone

Question 4:

Initially, all fibroids are ____.

- a) Intramural
- b) Submucosal
- c) Subserous

d) Cervical

Question 5:

A 55-year-old lady comes with abnormal uterine bleeding. A hysterectomy was performed and the following specimen was retrieved. What is the diagnosis?



- a) Leiomyoma
- b) Endometrial carcinoma
- c) Endometrial polyp
- d) Adenomyosis

Question 6:

Which of the following symptoms would you not expect in a patient diagnosed with uterine fibroids?

- a) 1 and 2
- b) 2 and 3
- c) 2, 4 and 5
- d) 1, 3 and 5

Question 7:

A patient presented with intermenstrual bleeding for the past 6 months. Which of the following types of fibroid is usually associated with this symptom?

- a) Interstitial
- b) Subserosal
- c) Submucous
- d) Cervical

Question 8:

A 46-year-old woman presented with heavy menstrual bleeding and dysmenorrhea. You suspect a diagnosis of submucous uterine fibroids. Which of the following investigations will help you confirm this?

- a) 2 and 3
- b) 1, 2 and 3
- c) 4 and 5
- d) 1, 2 and 4

Question 9:

An adnexal mass seen on ultrasound in a 38-year-old woman was diagnosed to be a secondary broad ligament fibroid. Which of the following is true about the same?

- a) It arises from the broad ligament
- b) The uterine vessels always lie medial to it
- c) The ureter always lies lateral to it
- d) It is also known as true broad ligament fibroid

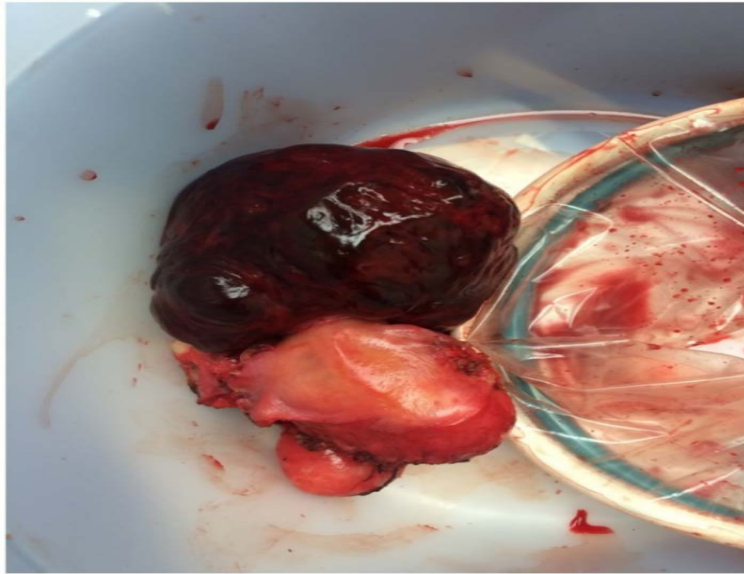
Question 10:

Which of the following is incorrect about fibroids?

- a) Calcification begins at the centre
- b) Hyaline degeneration begins at the centre
- c) Centre is least vascular
- d) Pseudocapsule is present

Question 11:

A patient with a history of miscarriage underwent a myomectomy. The specimen is shown below. Where does the degenerative change shown here begin from?



- a) Centre
- b) Periphery
- c) Subcapsular area
- d) Extracapsular area

Question 12:

A 44-year-old woman presenting with acute abdominal pain had previously been diagnosed with uterine fibroids. Which of the following complications is most unlikely to have occurred here?

- a) Hemorrhage into the fibroid
- b) Hyaline degeneration
- c) Red degeneration
- d) Sarcomatous change

Question 13:

A primigravida presented with severe abdominal pain and fever at 28 weeks of gestation. Ultrasound reveals a healthy fetus in utero and a large submucosal fibroid with multiple,

hyperechoic areas within. Blood investigations reveal leukocytosis and a raised ESR. What is the diagnosis?

- a) Red degeneration of fibroid
- b) Torsion of fibroid
- c) Infection of fibroid
- d) Sarcomatous change

Question 14:

A 34-year old woman with a history of fibroids presented with 1.5 months of amenorrhea. UPT is positive. Which of the following complications will you not expect in this patient?

- a) Uterine inversion
- b) Red degeneration
- c) Postpartum hemorrhage
- d) Fetal malformations

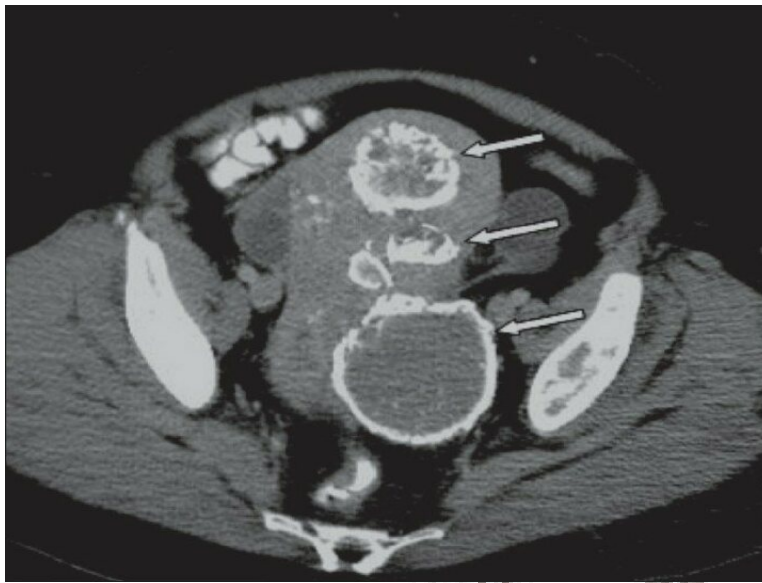
Question 15:

Which of the following changes are not seen in fibroids?

- a) Necrosis
- b) Squamous metaplasia
- c) Atrophy
- d) Sarcomatous changes

Question 16:

CT pelvis of a woman with fibroids is shown below. Which type of degeneration is seen here?



- a) Hyaline degeneration
- b) Cystic degeneration
- c) Calcific degeneration
- d) Hemorrhagic degeneration

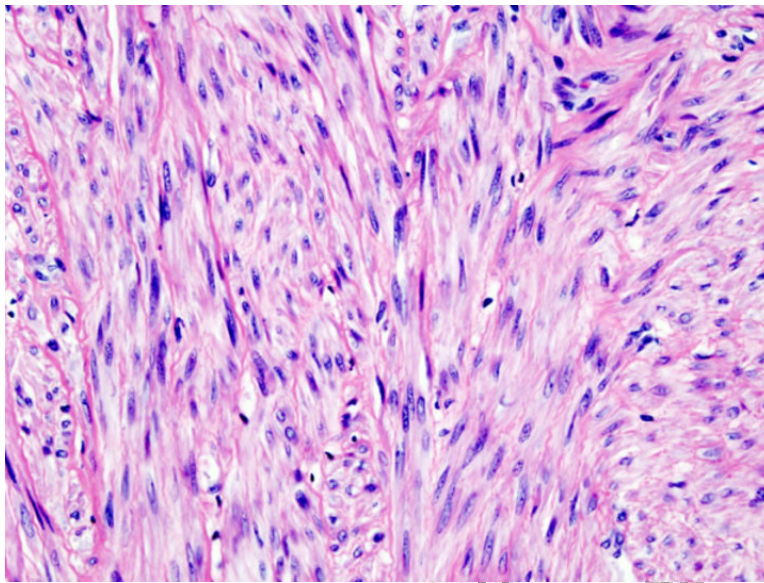
Question 17:

What is the least common secondary change seen in fibroids?

- a) Hemorrhage into fibroid
- b) Sarcomatous change
- c) Atrophy
- d) Necrosis

Question 18:

A 46-year-old woman presents with a lower abdominal mass for the past 2 years. She had undergone uterine artery embolisation for fibroids 5 years ago. Ultrasound showed a homogeneous mass near the uterine fundus. On exploratory laparotomy, the mass was observed to be attached to the greater omentum and it was excised. Histopathology of the mass is shown below. What is your diagnosis?



- a) Pedunculated fibroid
- b) Parasitic fibroid
- c) Leiomyosarcoma
- d) Disseminated intraperitoneal leiomyomatosis

Question 19:

A pregnant woman with a fibroid uterus develops severe acute pain abdomen at 28 weeks of gestation. There is no history of fever and the WBC count is 8000 cells/mL. The most likely diagnosis is:

- a) Red degeneration of fibroid
- b) Preterm labour
- c) Torsion of fibroid
- d) None of the above

Question 20:

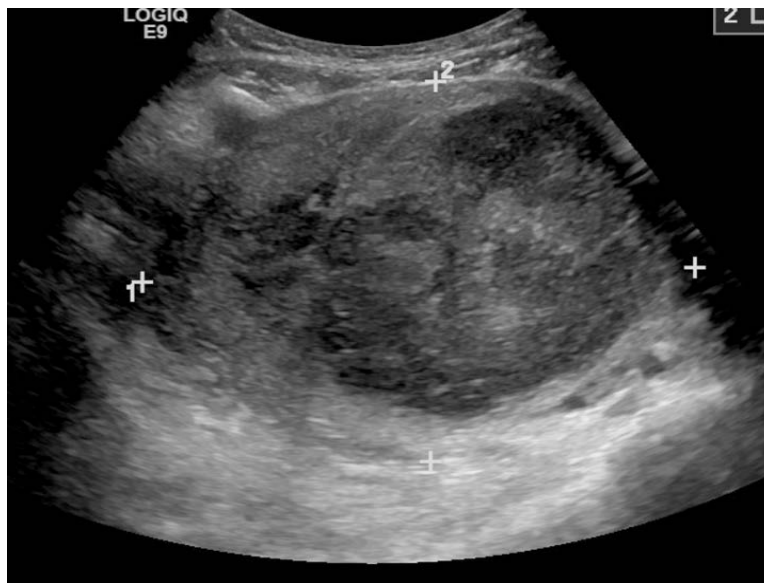
A primigravida at 24 weeks presents to you for a routine antenatal checkup. On ultrasound, you incidentally note the presence of two 3x3cm submucosal fibroids. How will you manage this case?

- a) Emergency laparoscopy
- b) Myomectomy
- c) Cesarean section at 37 weeks

d) Conservative management

Question 21:

A 40-year-old woman presented with menorrhagia and increased frequency of micturition. Ultrasound abdomen shows the following picture. Which of the following drugs is not used in the management of her condition?



- a) Mifepristone
- b) Goserelin
- c) Cetrorelix
- d) Misoprostol

Question 22:

A patient diagnosed with leiomyoma was prescribed an intranasal spray to manage her disease. Which of the following drugs is it unlikely to contain?

- a) Leuprolide
- b) Histrelin
- c) Nafarelin
- d) Buserelin

Question 23:

A pregnant woman was diagnosed with red degeneration of fibroid at 26 weeks of gestation. What is the next step in her management?

- a) Laparotomy
- b) Termination of pregnancy
- c) Fibroid removal during Caesarean section
- d) Analgesics

Question 24:

A 32-year-old woman presented with anemia due to heavy menstrual bleeding in spite of medical management of fibroids. She was posted for a myomectomy as she wanted to have children in the future. Which of the following is not an indication for this procedure?

- a) Recurrent pregnancy loss
- b) Associated infertility
- c) Red degeneration
- d) Pressure symptoms

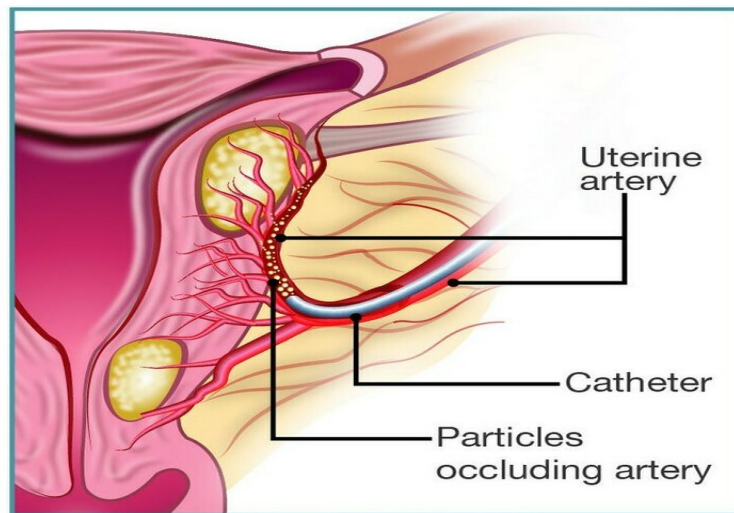
Question 25:

Which of the following modalities is not used for the treatment of leiomyomas?

- a) Uterine artery embolisation
- b) Magnetic resonance guided focused ultrasound
- c) Ultrasound guided transcervical metroplasty
- d) Nd:YAG laser myolysis

Question 26:

The following procedure was planned for a patient with uterine fibroids. Which of the following statements is false about this procedure?



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- a) Can be done in nulliparous women
- b) Ovarian failure is one of the complications
- c) PVA particles are used
- d) Done under radiological guidance

Answer Key

Question No.	Correct Option
1	a
2	d
3	d
4	a
5	a
6	b
7	c
8	d
9	c
10	a
11	a
12	b
13	a
14	d

15	b
16	c
17	b
18	b
19	c
20	d
21	d
22	b
23	d
24	c
25	c
26	a

Detailed Explanations

Solution to Question 1:

Fibroids are more common in African-American women.

Uterine fibroids are benign, smooth muscle tumors present in the uterine musculature. They are most commonly seen during reproductive years. Estrogen and progesterone promote the development of fibroids.

Option B: Hypoestrogenism occurs in smokers, therefore smoking is protective for fibroids.

Option C: Estrogen exposure is increased in nulliparous women, therefore fibroids are more common. Increased parity is protective for fibroids.

Option D: Most fibroids do not increase in size during pregnancy. Pregnancy can have an unpredictable effect on fibroids based on individual genetic and environmental factors.

Solution to Question 2:

MLH1 gene is not associated with leiomyomas. Its mutation leads to endometrial carcinoma.

MED12 and HMGA2 are the most common genes associated with leiomyomas. Other genes involved are COL4A5-A6 and the gene for fumarate hydratase (FH). FH gene mutation leads to hereditary leiomyomatosis and renal cell cancer (HLRCC) syndrome.

Solution to Question 3:

Leiomyomas convert less estradiol to the weaker estrone.

Leiomyomas create a hyperestrogenic environment, which is a requisite for their growth and maintenance. The hyperestrogenic environment is created by the following mechanisms:

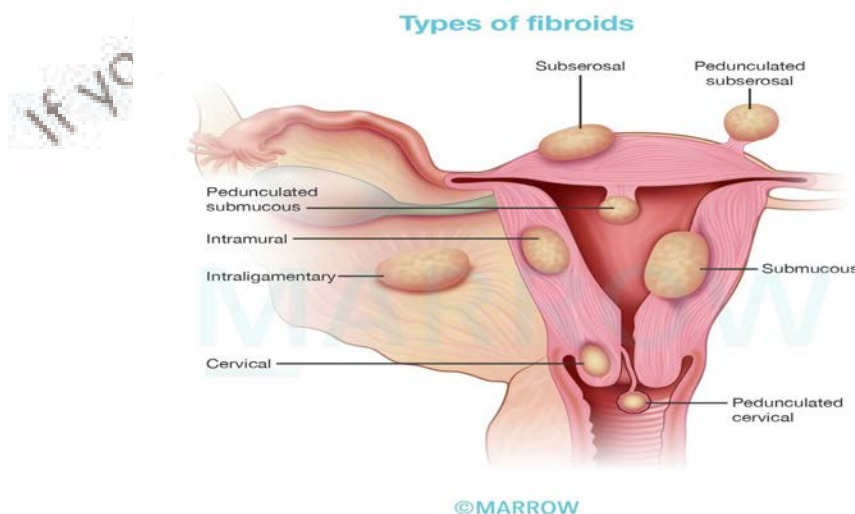
- Compared to normal myometrium, leiomyoma cells contain a greater density of estrogen receptors, which results in greater estradiol binding.
- These tumors convert less estradiol to the weaker estrone.
- There are higher levels of cytochrome P450 aromatase in leiomyomas compared with normal myocytes. This specific enzyme catalyzes the conversion of androgens to estrogen.

Solution to Question 4:

Initially, all fibroids are intramural. Fibroids develop in the myometrium, as intramural/interstitial and then branch out to form the other types.

Classification of fibroids based on anatomical location:

- Body (corporeal) fibroids-
- Interstitial/intramural fibroids: They are the most common (75%).
- Subserosal fibroids: They are the least common (10%). Interstitial fibroids grow outwards to form subserous fibroids and are covered by the peritoneum. These include wandering or parasitic fibroids and broad ligament fibroids.
- Submucous fibroids: Interstitial fibroid grows inwards towards the uterine cavity. It is the most symptomatic type.
- Cervical fibroids



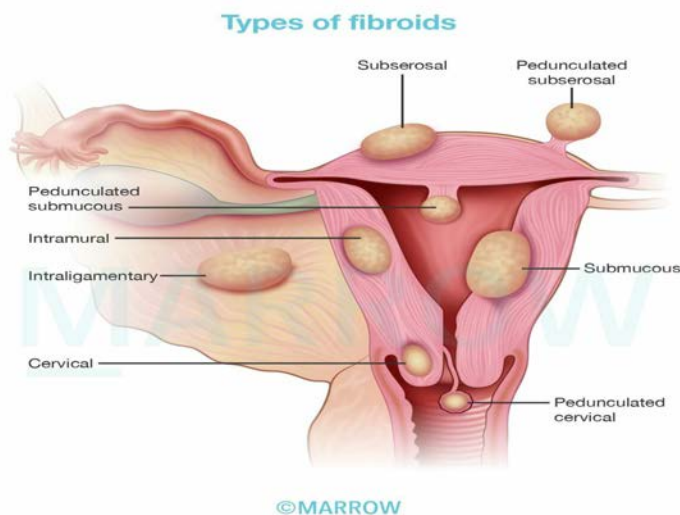
Solution to Question 5:

The given cut section image showing multiple nodules, well-circumscribed, firm, and white-greyish with a whorled appearance is Leiomyoma (Fibroid).

Leiomyoma is the most common benign and the most common pelvic tumor in females and is more common in obese females(hyper-estrogenism). They are mostly asymptomatic and the most common symptom is menorrhagia.

Leiomyomas can also present with bleeding areas, cystic degeneration, and calcifications. The uterine cavity is compressed.

The image below different types of uterine fibroids classified according to their locations:



Sub-mucous fibroids are the most symptomatic type and intra-mural fibroids are the most common type of uterine fibroids.

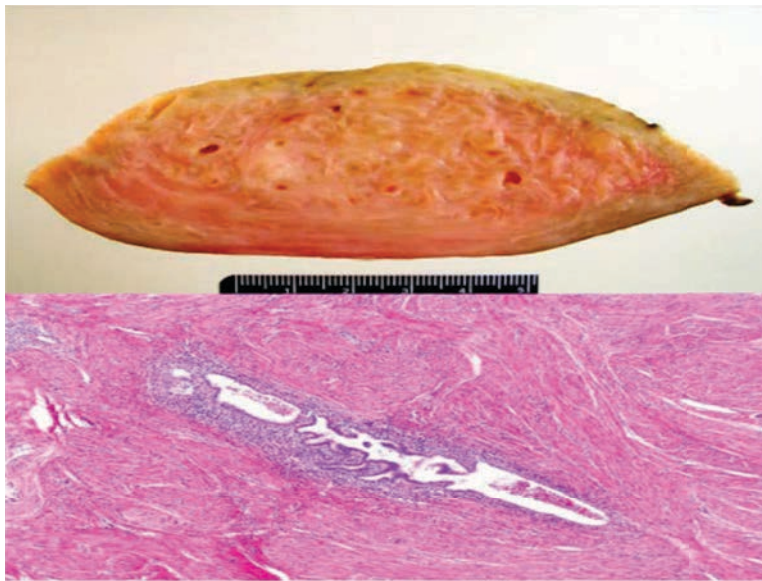
Other options:

Option B: Endometrial cancer presents as a polypoid mass protruding into the uterine cavity or as a diffuse thickening of the endometrium. The cut surface may appear white to yellow and may show areas of necrosis or hemorrhage.

Option C: Endometrial polyps are usually small and may be single or multiple. Grossly, they appear as soft, red to pink, often pedunculated (attached by a stalk) masses protruding into the uterine cavity. It presents with heavy cyclic or intermenstrual bleeding, or without complaints as an incidental finding on imaging. Estrogen and progesterone have been implicated in the growth of endometrial polyps, and higher receptor levels are seen compared with the normal endometrium.

Option D: In Adenomyosis (Endometriosis interna), the uterus may appear bulky and symmetrically enlarged. On the cut section, the myometrium may appear thickened and may show tiny cystic spaces filled with old blood (chocolate cysts). The preferred treatment is hysterectomy. Localised excision or levonorgestrel IUCD is tried in younger women.

The image below shows both the gross and histologic appearances of adenomyosis:



Solution to Question 6:

Menouria and hypomenorrhea are not seen in patients with uterine fibroids. Menouria occurs due to vesicouterine fistulas. Hypomenorrhea is seen in Asherman's syndrome and due to OCP use.

Fibroids present with the following clinical features:

- Menstrual symptoms such as:
 - Menstrual irregularities – heavy menstrual bleeding/menorrhagia which is the most common symptom, polymenorrhoea, and intermenstrual bleeding (metrorrhagia)
 - Dysmenorrhea
 - Postmenopausal bleeding
 - Abdominal pain
 - Abdominal distension, mass abdomen
 - Vaginal discharge
 - Infertility
 - Recurrent abortions
- Pressure symptoms such as:
 - Urinary retention
 - Increased frequency of urination
 - Constipation
- Broad ligament fibroids can cause compression of the ureter leading to hydronephrosis or hydroureter.

Solution to Question 7:

Intermenstrual bleeding is common with submucous fibroids. Metrorrhagia or intermenstrual bleeding refers to bleeding that occurs during the middle of the menstrual cycle.

Metrorrhagia in a woman over 40 years warrants a dilatation and curettage procedure to rule out endometrial cancer. This is associated with fibroids in 3% of cases.

Polymenorrhoea in a patient with fibroids occurs when she has polycystic ovaries or pelvic inflammatory disease (PID).

Progressive menorrhagia is seen in intramural and submucous myoma due to increased vascularity, endometrial hyperplasia, and an enlarged uterine cavity.

Note: Further the distance of the fibroid from the cavity, lesser is the possibility of menorrhagia. For this reason, subserous and pedunculated fibroids do not cause menorrhagia.

Solution to Question 8:

Investigations required to identify submucous fibroids include:

- USG - most commonly done. Transvaginal USG is useful to assess the location, dimension, and associated adnexal pathology.
- MRI gives better soft tissue details but is not routinely done
- Saline infusion sonography and hysteroscopy - help to visualise submucous fibroid more clearly
- Laparoscopy - If uterus <12 weeks size and associated pelvic pain and infertility and to detect pelvic endometriosis and tubal pathology.
- Uterine curettage – A bump is felt in submucous fibroid during endometrial curettage. Dilatation and curettage is done in patients presenting with postmenopausal bleeding.
- Hysterosalpingography – A filling defect is seen.

CT abdomen is not very useful for the identification of submucous fibroids. These cannot be visualised on colposcopy either.

The image below shows the ultrasonographic appearance of a submucous fibroid (asterisk):



The image given below shows hysteroscopic view of a submucous fibroid:



Solution to Question 9:

The ureter and uterine vessels lie lateral to a secondary broad ligament fibroid.

There are 2 types of broad ligament fibroids:

Primary broad ligament fibroid:

- It is also known as true broad ligament fibroid.
- They arise from:
- Uterosacral ligament
- Round ligament

- Tissues in the broad ligament
- Relations:
- The uterine vessels lie medial to it.
- The ureter is usually beneath the fibroid. However, it can be present anywhere in relation to it.

Secondary broad ligament fibroid:

- It is also known as false broad ligament fibroid.
- They arise from the lateral wall of the uterus or cervix, but grow laterally, between the two layers of the broad ligament.
- Relations - the uterine vessels and ureter lie lateral to it.

Solution to Question 10:

Fibroid calcification begins at the periphery and spreads inwards along the vessels.

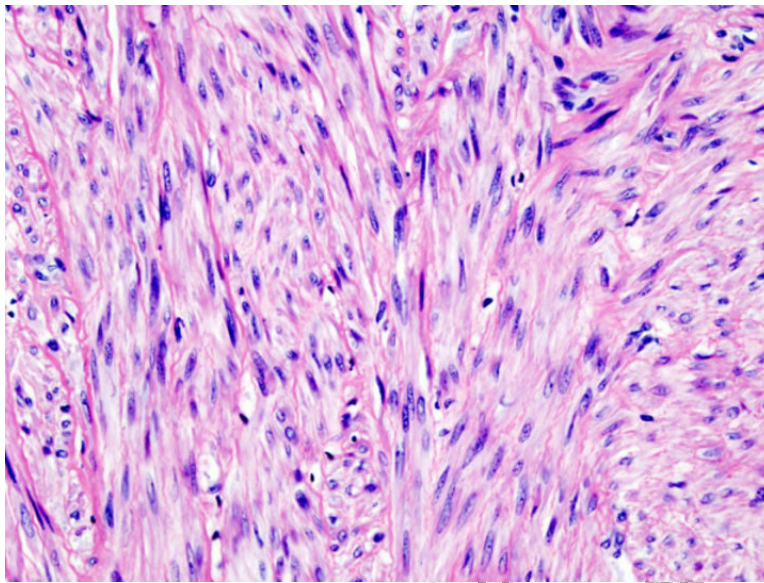
A typical myoma is a well-circumscribed tumor with a pseudo-capsule and firm in consistency. Its cut surface has a whorled appearance.

The vessels that supply blood to the tumor lie in the capsule and send radial branches into the tumor. Therefore, the central portion of the tumour receives the least blood supply, and degeneration is noticeable early in this part of the tumour. However, fibroid calcification begins at the periphery.

The image given below show the gross appearance of fibroids.



The below image shows the histological appearance of fibroids.



Solution to Question 11:

The given specimen shows red degeneration of fibroid. Degeneration is most likely to start from the centre of the fibroid.

In a fibroid, centre of the tumor is the least vascular part and hence degeneration is more common here. The only exception is calcareous degeneration where phosphates and carbonates of lime are deposited in the periphery along the course of the vessels.

Solution to Question 12:

Hyaline degeneration of fibroids is unlikely here as it is usually painless and presents as an abdominal mass causing distension and heaviness.

Acute pain in fibroids is seen in the case of:

- Fibroid torsion
- Hemorrhage into the fibroid
- Red degeneration
- Sarcomatous change - seen in postmenopausal women.

Solution to Question 13:

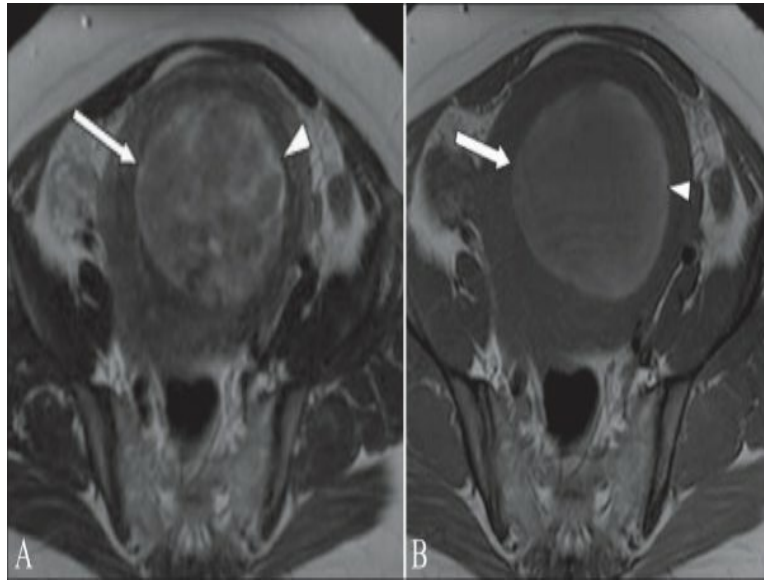
The given case scenario is suggestive of red degeneration of fibroid.

It is most common during the second trimester of pregnancy. The gross appearance is described as follows:

- Fishy odor
- Thrombosis of subcapsular vessels and vessels within the fibroid
- Purplish-red color due to diffusion of blood pigments from the thrombosed vessels.

Red degeneration presents with acute abdominal pain, fever and, vomiting. Investigations reveal leukocytosis and a raised ESR. On radiology, the hemorrhagic areas appear hyperechoic within the fibroid.

The image given below shows the MRI appearance of red degeneration of fibroids



Option B: Torsion of subserous pedunculated fibroid presents with severe pain abdomen, but fever and leukocytosis are absent.

Option C: Infection is more common in puerperium and presents with blood-stained vaginal discharge.

Option D: Sarcomatous changes occur very rarely, and are unlikely in patients <40 years of age. They are more commonly seen in post-menopausal women.

Solution to Question 14:

Fetal malformations have not been recorded to occur due to fibroids.

Pregnancy complications due to fibroids are:

- Increased risk of abortion or miscarriage
- Preterm labor
- Abnormal presentation
- Fibroid degeneration (Red degeneration)
- Accidental hemorrhage
- Dystocic labor

- Postpartum hemorrhage
- Puerperal sepsis
- Uterine inversion

Solution to Question 15:

Squamous metaplasia is not seen in fibroid.

Solution to Question 16:

The given image shows womb-stone appearance on CT, suggestive of calcific degeneration of fibroid.

These are also called calcareous myomas and are seen in old patients with long-standing myomas. This occurs due to the deposition calcium phosphates and calcium carbonates in the periphery of the fibroid, along the course of the vessels. It is easily visualized on radiography.

Solution to Question 17:

Sarcomatous change is the least common secondary change in fibroids. Its incidence is $\leq 0.5\%$.

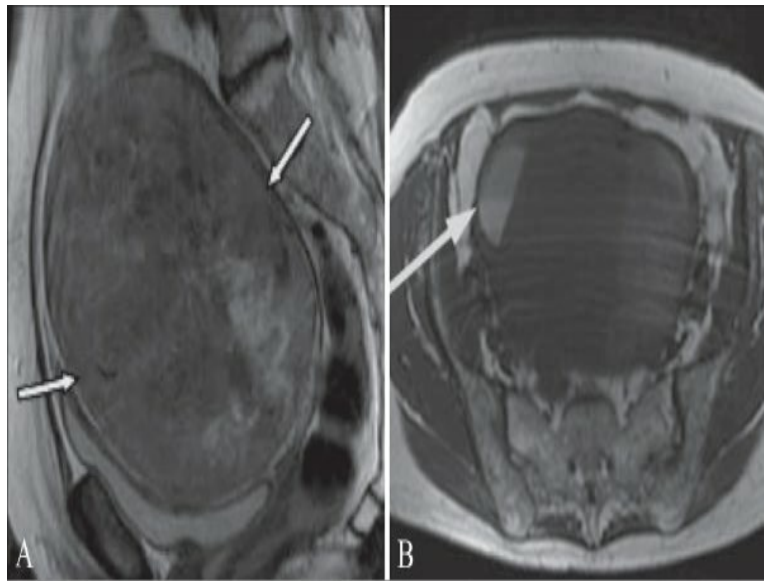
Sarcomatous change leads to the development of malignant leiomyosarcomas. This is seen in postmenopausal women and is rare in women ≤ 40 years of age. Intramural and submucous tumors have a higher potential for sarcomatous change than subserous tumors.

Its gross appearance is described as follows:

- Soft consistency
- Yellowish-grey
- Hemorrhagic spots seen

Radiology shows dark, irregular areas and air levels within the fibroid. These are indicative of sarcomatous change.

The image given below shows the CT appearance of a large fibroid that has undergone sarcomatous change. The image on the right shows a fluid level within the tumor.



Solution to Question 18:

The given case scenario is suggestive of a parasitic fibroid. The given histopathological image shows interlacing bundles of uniform spindle-shaped smooth muscle cells with blunted nuclei that are characteristic of a fibroid.

A parasitic fibroid or a wandering fibroid gets cut off from its uterine origin and derives its blood supply from the vessels of the organ it is attached to. Thus, it can continue to grow in spite of uterine artery embolisation. It is managed by complete resection.

Option A: Pedunculated fibroids have a pedicle with which they're attached to the myometrium.

Option C: Leiomyosarcoma commonly presents in postmenopausal women with sudden increase in size and postmenopausal bleeding. It would show nuclear atypia on histology.

Option D: Disseminated intraperitoneal leiomyomatosis involves large areas of sub peritoneal surfaces. It is seen during pregnancy and while on oral contraceptives.

Solution to Question 19:

The given scenario of severe acute abdominal pain without fever/leucocytosis in a pregnant woman with a fibroid is suggestive of fibroid torsion.

The incidence of fibroid in pregnancy is about 1 in 1,000. Due to increased vascularity in pregnancy, the fibroid increases in size and becomes softer. The torsion is seen in the subserous pedunculated fibroid where the venous return is blocked at its site of attachment in the uterus. This leads to severe acute abdominal pain. Due to the complication being more local than systemic, fever or leucocytosis are generally not seen.

A transvaginal ultrasonogram is immediately done to detect the position and size of the fibroid, which is followed by laparotomic myomectomy as a part of emergency surgical management.

Other options:

Option A: Red degeneration of fibroid is seen commonly in the second half of pregnancy or puerperium. It is due to a hemorrhagic infarction of the tissue and typically presents as acute focal abdominal pain with fever and leucocytosis. Treatment is usually conservative with bed rest, analgesics, and sedatives.

Option B: Preterm labor is where the labor starts before 37 completed weeks. It is characterized by true labor pain. The abdominal pain is slowly increasing in intensity and duration and is also accompanied by show (expulsion of the cervical mucus plug). There are intervals between episodes of contractions where the patient doesn't complain of abdominal pain.

Solution to Question 20:

The given case scenario is suggestive of an uncomplicated fibroid during pregnancy. These are managed conservatively.

Surgery is only indicated in the case of a pedunculated fibroid undergoes torsion.

Solution to Question 21:

The given case scenario and ultrasound image showing a well-defined, heterogeneous hypoechoic mass are suggestive of a uterine fibroid. Misoprostol does not help to reduce the size of the fibroid and so is not recommended for its management.

Medical management of fibroids helps to reduce their size. The drugs used in therapy are as follows:

- Antiprogestosterone – Mifepristone / RU-486
- GnRH analogs
- Agonists – goserelin, leuprorelin, nafarelin, buserelin
- Antagonists – cetrorelix, ganirelix
- Levonorgestrel releasing intrauterine system
- Selective progesterone receptor modulator (SPRM) – Asoprisnil
- Danazol - usually not used due to androgenic side effects.

Solution to Question 22:

The nasal spray is unlikely to contain histrelin, as it is only available as a subcutaneous implant.

GnRH analogues are used to treat leiomyomas. Intranasal preparations are available for leuprolide, nafarelin, and buserelin. However, leuprolide has maximal action as an intramuscular injection.

GnRH analogue therapy in fibroids is given as follows:

- Leuprolide acetate (Lupron) - given intramuscularly. It is also available as an intranasal spray, but isn't commonly used.
- Goserelin (Zoladex) - subcutaneous depot implant
- Triptorelin (Trelstar) IM injection
- Nafarelin and buserelin - given as intranasal sprays bd
- Histrelin - available as subcutaneous implants.

Solution to Question 23:

Red degeneration of fibroid is managed conservatively with bed rest and analgesics.

It is most common during the second trimester of pregnancy. Red degeneration presents with acute abdominal pain, fever and, vomiting. Investigations reveal leukocytosis and a raised ESR. On radiology, the hemorrhagic areas appear hyperechoic within the fibroid.

Solution to Question 24:

Red degeneration is not an indication of myomectomy. It is conservatively managed.

Myomectomy is indicated in the following:

- Symptoms compromising the quality of life such as:
- Heavy menstrual bleeding, causing anemia
- Pelvic pain
- Urinary symptoms and ureteric obstruction.
- The patient is unwilling for hysterectomy
- The patient wishes to preserve childbearing function
- Fibroids causing infertility

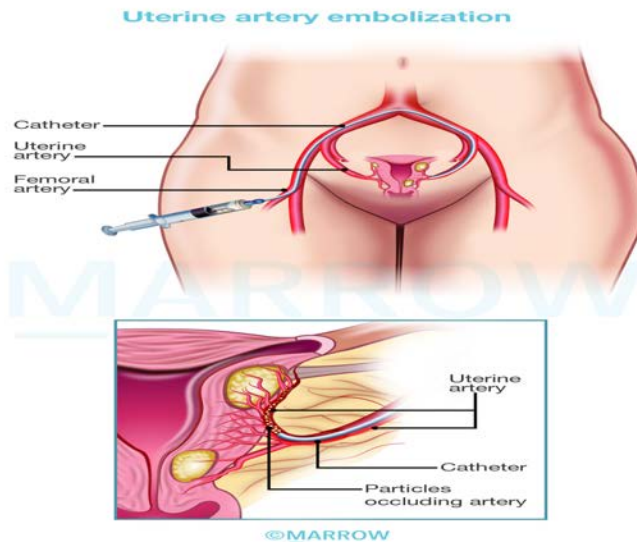
Solution to Question 25:

Ultrasound-guided transcervical metroplasty is not used in the treatment of leiomyomas. It is used for the resection of the uterine septum.

Uterine artery embolization/Uterine fibroid embolization (UAE/UFE) is an angiographic interventional procedure that delivers polyvinyl alcohol (PVA) microspheres or other synthetic particulate emboli into both uterine arteries thus producing ischemia and necrosis. The vessels serving leiomyomas have a larger caliber, hence these microspheres are preferentially directed to the tumors, thus sparing the surrounding myometrium.

Nd:YAG myolysis is laparoscopic myolysis performed with an Nd:YAG laser. It causes coagulation of a subserous fibroid.

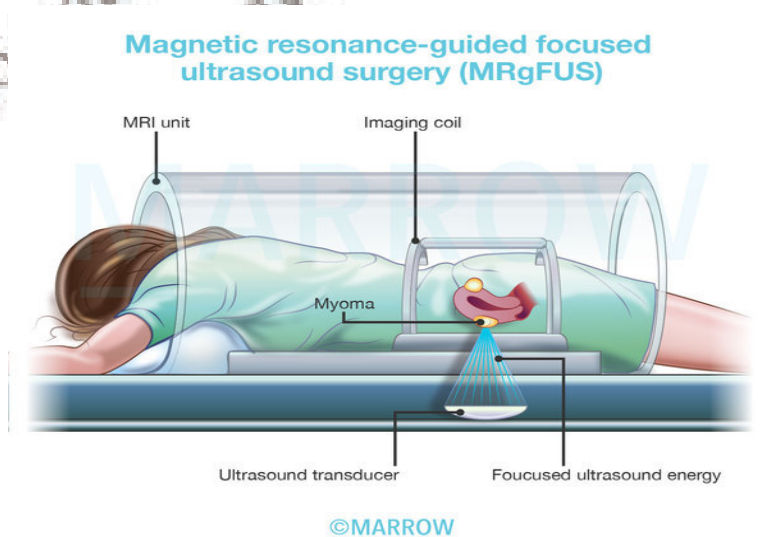
The below image shows uterine artery embolization.



Magnetic Resonance guided Focused Ultrasound (MRgFUS) is also called MR-guided high-intensity focused ultrasound (MR-HIFU).

Ultrasound energy is focused to heat and incite coagulative necrosis in selected myomas. Concurrent MR imaging enables precise targeting and provides real-time tissue temperature feedback to limit surrounding thermal injury.

The given image shows MRgFUS.



Solution to Question 26:

The given image shows uterine artery embolization (UAE). UAE cannot be done in nulliparous women as embolization of ovarian vessels can cause ovarian failure and premature menopause.

Under local sedation, bilateral UAE is approached through percutaneous femoral catheterization. Embolization is done by releasing polyvinyl alcohol (PVA), gel foam particles or metal coils into the uterine artery.

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you may have been scammed.

Endometriosis and Adenomyosis

Question 1:

Which of the following accurately describes Sampson's theory for endometriosis?

- a) Retrograde menstruation
- b) Coelomic metaplasia
- c) Hemogenous metastasis
- d) Induction of stem cells

Question 2:

Which of the following statements is false about endometriosis?

- a) Hormone dependent condition
- b) More common in nulliparous women
- c) In utero exposure to DES increases risk
- d) Most common site is the cervix

Question 3:

Which of the following is the most common symptom associated with endometriosis?

- a) Dysmenorrhea
- b) Dyspareunia
- c) Menorrhagia
- d) Infertility

Question 4:

In a patient diagnosed with endometriosis, which of the following symptoms would you not expect?

- a) Vaginal discharge
- b) Premenstrual spotting

- c) Postmenstrual spotting
- d) Menorrhagia

Question 5:

A 26-year-old lady presented with complaints of severe pain on menstruation and heavy flow. She is subsequently diagnosed with endometriosis. The pain in this condition mainly correlates with _____

- a) Depth of penetration
- b) Multiple sites
- c) CA 125 levels
- d) Stage of the disease

Question 6:

Arrange the following sites of endometriosis in the order of frequency of occurrence.

- a) 4 > 5 > 2 > 1 > 3
- b) 4 > 1 > 3 > 2 > 5
- c) 4 > 5 > 1 > 3 > 2
- d) 5 > 4 > 2 > 3 > 1

Question 7:

A middle-aged woman presented with dysmenorrhea, dyspareunia, and infertility. On examination, you note tender nodularity of the uterosacral ligaments. Which of the following is the most likely diagnosis?

- a) Endometrioma
- b) Endometriosis
- c) Pelvic tuberculosis
- d) Adenomyosis

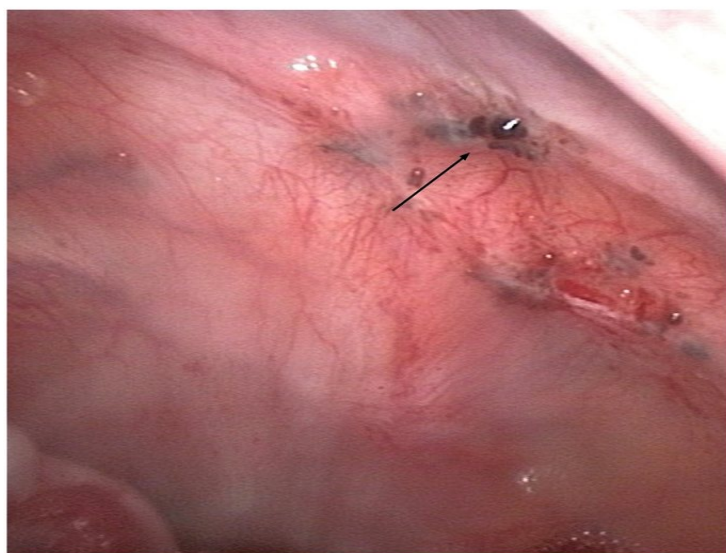
Question 8:

A young woman presented to your clinic with complaints of painful menstruation and postmenstrual spotting. On examination, you note tender nodules in the posterior fornix and a fixed retroverted uterus. Which of the following is the gold standard investigation to confirm the likely diagnosis?

- a) Ultrasound
- b) Laparotomy
- c) Laparoscopy
- d) Hysteroscopy

Question 9:

A diagnostic laparoscopy was performed on a 27-year-old woman with severe dysmenorrhoea and dyspareunia. The following finding was seen on the pelvic peritoneum. What is the most likely diagnosis?



- a) Pelvic Inflammatory Disease
- b) Adenomyosis
- c) Fitz Hugh Curtis syndrome
- d) Endometriosis

Question 10:

In endometriotic lesion, histology represents its:

- a) High cholesterol
- b) High estrogen
- c) High levels of prolactin
- d) High progesterone

Question 11:

Which of the following operative procedures are least likely to be followed by scar endometriosis?

- a) Hysterotomy
- b) Cesarean section
- c) Ventrofixation
- d) Hysterectomy

Question 12:

A 39-year-old nulliparous woman presented with dysmenorrhea and a progressively growing, painless abdominal mass. USG scan revealed a cystic lesion of the left ovary which was excised. The below image shows the excised specimen. Which of the following statements is false about this condition?



- a) It is seen in endometriosis of ovary
- b) It is lined by columnar epithelium
- c) It is usually a bilateral condition

- d) It is a malignant cystic lesion

Question 13:

A patient is being evaluated for chronic abdominal pain. Her lab reports revealed elevated CA-125 levels. Which of the following conditions is the least likely diagnosis?

- a) Endometrioma
- b) Chronic liver disease
- c) Abdominal tuberculosis
- d) Malignant pancreatic tumor

Question 14:

A 30-year-old woman with dyspareunia is diagnosed to have endometriosis. Which of the following drugs is not useful for her treatment?

- a) Estrogen
- b) Progesterone
- c) Danazol
- d) GnRH

Question 15:

A patient with endometriosis requires treatment with GnRH analogues. She is scared of injections so you plan to prescribe a nasal preparation for her. Which of the following GnRH agonists could you prescribe?

- a) Nafarelin
- b) Goserelin
- c) Leuprolide
- d) Triptorelin

Question 16:

Which of the following statements is true regarding adenomyosis?

- a) More common in nulliparous women

- b) Uterus is hard and tender
- c) Menorrhagia, dysmenorrhea are common symptoms
- d) Estrogens are used for treatment

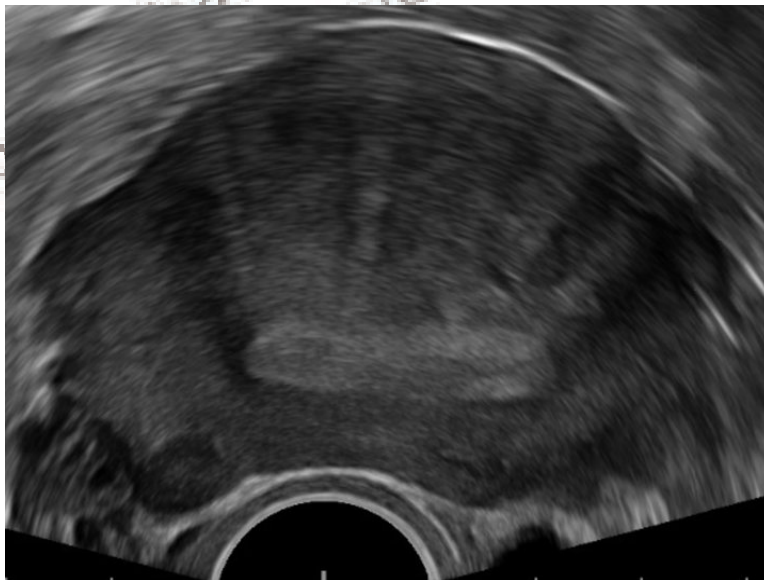
Question 17:

A patient with menorrhagia and backache is diagnosed with adenomyosis. Which of the following is not associated with this condition?

- a) Fibroid uterus
- b) Endometriosis
- c) Sub involution of uterus
- d) Endometrial carcinoma

Question 18:

A 43-year-old multiparous woman presented with menorrhagia and secondary dysmenorrhea. On examination, Halban's sign was positive. The transvaginal USG findings are given below. Which of the following is the definitive treatment for her condition?



- a) Hormonal therapy
- b) Hysterectomy
- c) Laser ablation
- d) Localised excision

Answer Key

Question No.	Correct Option
1	a
2	d
3	a
4	a
5	a
6	b
7	b
8	c
9	d
10	b
11	d
12	d
13	d
14	a
15	a
16	c
17	c
18	b

Detailed Explanations

Solution to Question 1:

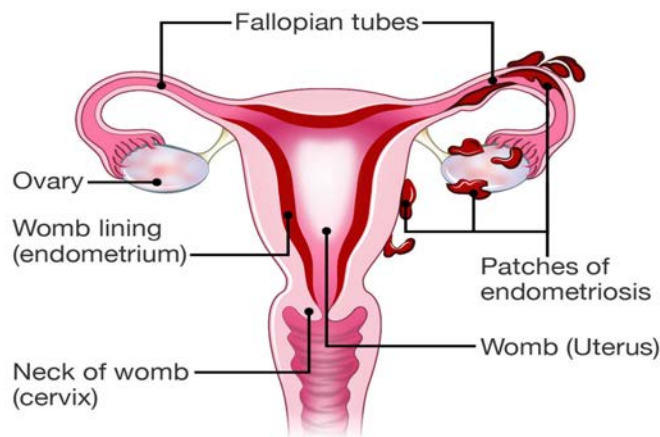
Sampson's theory describes the implantation of endometrial tissue due to retrograde menstruation.

The various theories for the etiology of endometriosis include:

Given below is an illustration showing retrograde menstrual flow that is leading to the development of patches of endometriosis.

Theory	Mechanism
Sampson's implantation theory	Reflux of menstrual endometrium and subsequent implantation in the surrounding structures
Theory of coelomic metaplasia by Meyer and Ivanoff	Metaplastic changes in embryonic mesothelium which respond to hormone stimulation
Metastatic theory by Halban	Embolization of menstrual fragments through vascular and lymphatic channels
Theory of histogenesis by induction	Induction of undifferentiated peritoneal cells to endometrial tissue by an endogenous factor

Reversed Menstrual flow



Solution to Question 2:

The most common site of occurrence of endometriosis is the ovary. Endometriosis of the ovary is known as endometrioma.

By definition, endometriosis is the occurrence of benign endometrial tissue (glands and stroma) outside the cavity of the uterus. Endometriosis can occur anywhere in the body (except spleen) including the abdominal wall, lung, pleura, brain, etc.

It is a proliferative, hormone-dependent disease of the childbearing period. Women between the age groups of 30-40 years are most commonly affected. Risk factors for endometriosis are as follows:

- Family history (7-10x risk with a 1st-degree relative)
- Nulliparity

- In utero exposure to Diethylstilbestrol (DES)
- Estrogen dependent conditions/Estrogen therapy
- Immunological factor - Impaired T cell and NK cell activity and altered immunology in a woman may increase the susceptibility to proliferation and growth.

Solution to Question 3:

The most common symptom of endometriosis is dysmenorrhea.

Pain begins before the onset of menstruation and builds up continuously until the flow starts. Thereafter, the pain gradually declines. This pattern of pain is called triple dysmenorrhea.

Solution to Question 4:

Patients diagnosed with endometriosis are not likely to present with vaginal discharge.

Clinical features of endometriosis include:

- About 30% of the patients are asymptomatic.
- The classic symptom complex includes:
 - Dysmenorrhoea
 - Dyspareunia
 - Menorrhagia
 - Infertility (30-40%)
- Dysmenorrhoea - most common symptom
- Dyspareunia - occurs when the pouch of Douglas and rectovaginal septum are involved.
- Deep-seated pelvic pain
- Premenstrual and postmenstrual spotting

Solution to Question 5:

Pain in endometriosis mainly correlates with the depth of penetration.

Lesions penetrating more than 5mm are responsible for pain, dysmenorrhea and dyspareunia.

The dysmenorrhea associated with endometriosis precedes menses by 24 to 48 hours and is less responsive to NSAIDs and combined OCPs.

Solution to Question 6:

The sites of endometriosis arranged in the order of frequency of occurrence are: Ovary & Pouch of Douglas & Broad ligament & Uterosacral ligament & Fallopian tube.

Endometriotic lesions have been identified on all organs except the spleen.

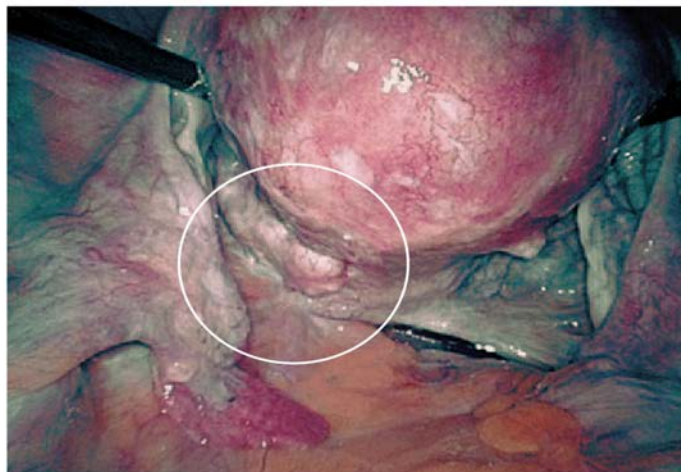
Solution to Question 7:

The given clinical scenario along with the tender nodularity of the uterosacral ligament points towards a diagnosis of endometriosis.

The nodularity of the uterosacral ligament in endometriosis gives rise to the cobblestone appearance seen on laparoscopy. This is referred to as 'sine qua non of endometriosis'.

The below image shows the cobblestone appearance of the uterosacral ligament.

Endometriosis - uterosacral ligament



Note: Endometrioma or chocolate cyst is endometriosis of the ovary. It presents as an adnexal mass.

Solution to Question 8:

The given clinical scenario of dysmenorrhea with tender nodules in the posterior fornix and a retroverted uterus is suggestive of endometriosis. The gold standard investigation for the diagnosis of this condition is a visual inspection by laparoscopy, preferably with histological confirmation. It is also the investigation of choice for the condition.

Role of laparoscopy in endometriosis:

- Detect and diagnose pelvic endometriosis
- Locate the site of endometriosis and staging
- To take biopsy

- To surgically treat endometriosis by ablation

Other investigations for endometriosis include:

- Estimation of CA-125 levels - raised in endometriosis.
- Estimation of MCP-1 (Monocyte chemotactic protein) levels - raised in the peritoneal fluid of women with endometriosis.

Solution to Question 9:

The given clinical history and the image showing characteristic powder burn spots seen over the pelvic peritoneum on laparoscopy point toward a diagnosis of pelvic endometriosis.

On laparoscopy, the typical endometriotic lesions appear like small black, dark brown or bluish dots. These are called powder burns or gunshot appearance of the serosal surfaces of the peritoneum. They are nodules or small cysts containing old haemorrhages surrounded by a variable degree of fibrosis.

Subtle appearances of endometriosis seen on laparoscopy are:

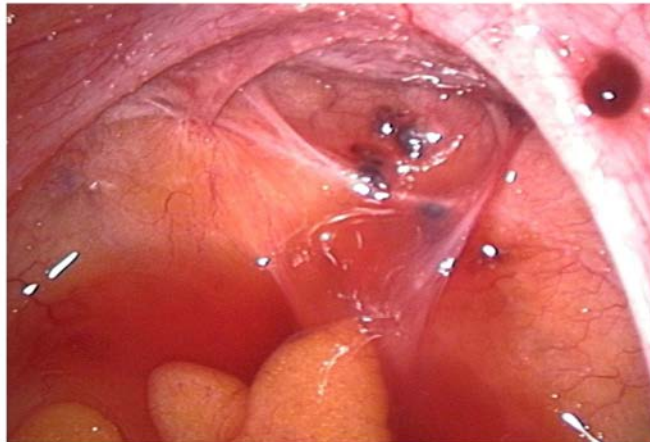
- Red flame-shaped areas
- Red polypoid areas
- Yellow-brown patches
- White peritoneal plaques or scarring
- Peritoneal defects - known as Allen Masters Syndrome
- Subovarian adhesions.

The lesions mentioned above are thought to be more active than the powder burn areas.

Endometriotic lesions in the peritoneal cavity contain yellowish-brown fluid. This fluid contains prostaglandins that are responsible for the pain associated with this condition. Endometriosis of the ovary leads to formation of the chocolate cyst which is a true cyst with columnar epithelium lining.

The image below shows the laparoscopic appearance of endometriotic lesions in the pouch of Douglas and on the right, the sacrouterine ligament. Dyspareunia occurs when the pouch of Douglas and rectovaginal septum are involved.

Endometriosis - pouch of Douglas



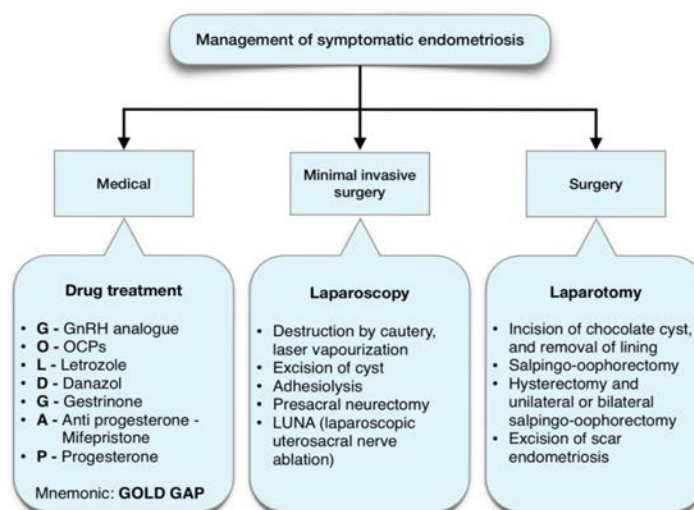
Solution to Question 10:

In endometriotic lesions, histology represents its high estrogen activity.

Endometriosis is defined by the presence of "ectopic" endometrial tissue at a site outside of the uterus. Endometriosis is a proliferative hormone-dependent disease of the childbearing period. It is extremely rare before menarche and disappears after menopause.

The endometriotic implants show certain differences when compared to normal endometrium. There is increased production of estrogen by endometriotic stromal cells due to high levels of the enzyme aromatase. This is absent in normal endometrial stroma.

Management of endometriosis:



Solution to Question 11:

Among the given operative procedures, hysterectomy is least likely to be followed by scar endometriosis.

Operations most likely to be followed by scar endometriosis are:

- Hysterotomy
- Classical Cesarean section
- Myomectomy
- Ventrofixation (fixing uterus to the abdominal wall)
- Episiotomy

Solution to Question 12:

The given clinical history and the image is suggestive of a chocolate cyst of the ovary which is a benign cystic lesion.

Chocolate cysts of the ovary are a manifestation of endometriosis. Both the ovaries are usually affected. On histology, the cyst is lined by columnar epithelium. Pseudoxanthoma cells are present which are large macrophages or scavenger cells that appear brown in colour due to ingested blood pigments such as haemosiderin. They are polyhedral in shape and resemble lutein cells.

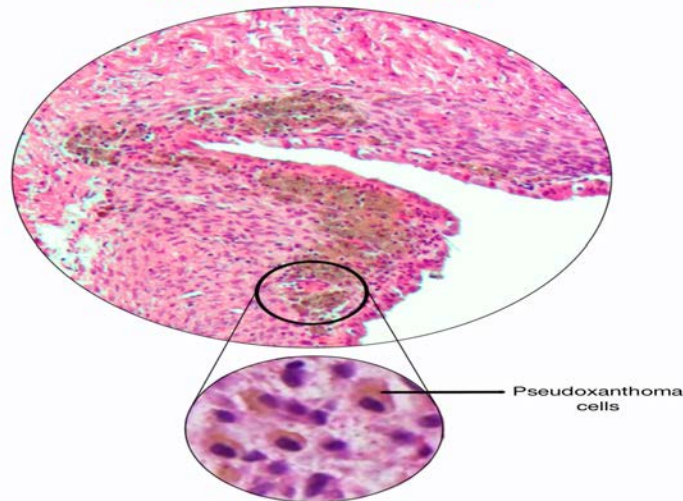
Given below is a gross image of the chocolate cyst of the ovary.

Chocolate cyst - Ovary



Given below is a histopathological section of an endometriotic lesion showing the pseudoxanthoma cells.

Endometriosis



Solution to Question 13:

CA-125 levels are not expected to be elevated in malignant pancreatic tumours.

CA-125 is a glycoprotein and cell surface antigen. Its levels are raised in the following conditions:

- Abdominal or pelvic tuberculosis
- Pelvic inflammatory disease
- Endometriosis
- Malignant epithelial ovarian tumour
- Chronic liver disease
- 2% of normal women, especially during menstruation.

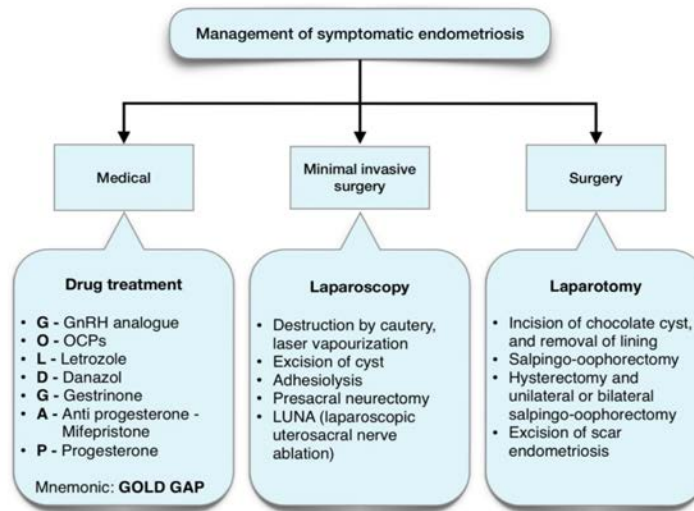
Endometrioma refers to ovarian endometriosis. CA-125 levels are raised to more than 35 U/mL in 80% of cases of endometriosis. The level is directly proportional to the extent of the disease.

Solution to Question 14:

Estrogen is not used in the treatment of endometriosis as the condition is estrogen-dependent.

For minimal disease, the drug of choice is NSAIDs. Oral contraceptive pills are given if associated menstrual complaints are present. OCPs inhibit gonadotropin release and decrease the menstrual flow.

For moderate to severe disease, oral progestogens are the first drugs to be started. They are known to antagonize estrogenic effects on the endometrium causing endometrial atrophy. The drug of choice is GnRH agonists.



Note: The best time to conceive in a patient with endometriosis is immediately after surgery.

Solution to Question 15:

The GnRH agonist available as a nasal preparation for the treatment of endometriosis is nafarelin.

GnRH analogues are used in the management of endometriosis and are given by continuous administration. This suppresses the secretion of pituitary gonadotropins and leads to atrophy of the endometrial tissue in 90% of cases.

Leuprolide and triptorelin are GnRH agonists which are given IM. Goserelin is available as a depot preparation.

Solution to Question 16:

Patients with adenomyosis commonly present with symptoms of menorrhagia and dysmenorrhea.

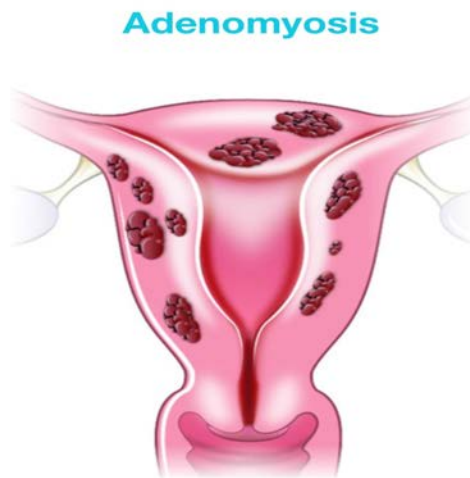
Adenomyosis (endometriosis interna) is the ingrowth of stromal and glandular components of the endometrium directly into the myometrium. It usually occurs following vigorous curettage, repeated childbirth or in conditions with an excess of estrogen. Adenomyosis is more common in multiparous women aged 40 years and above. They commonly present with menorrhagia and dysmenorrhea. Other symptoms include pelvic discomfort, backache and dyspareunia.

On per-vaginal examination, there is a gross symmetrical enlargement of the uterus corresponding to 12-14 weeks size. The uterus is soft and tender on bimanual examination. This is known as Halban's sign.

Diagnosis is mainly based on clinical assessment. A diagnostic hysteroscopy combined with curettage is the initial step in the management because of menorrhagia.

Total hysterectomy is the treatment of choice. Localised excision and levonorgestrel containing IUCD can be tried in younger patients.

The image below is an illustration of adenomyosis.



The image below shows adenomyosis as seen during laparoscopy. The uterus is soft and enlarged; the blue spots represent subserous endometriosis.



Solution to Question 17:

Adenomyosis is not associated with sub involution of the uterus.

Uterine subinvolution is impaired or retarded involution of the uterus. It is associated with uterine fibroid, cesarean section and conditions causing overdistension of the uterus like twin pregnancies or polyhydramnios.

Adenomyosis often coexists with the following conditions:

- Uterine fibromyomas

- Pelvic endometriosis (15%)
- Endometrial carcinoma

Solution to Question 18:

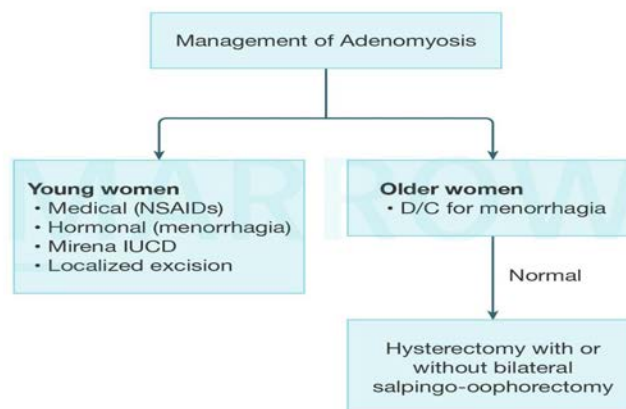
The given clinical vignette and the transvaginal ultrasound(TVS) showing the Venetian blind appearance is suggestive of adenomyosis. The definitive treatment for adenomyosis is hysterectomy.

Adenomyosis (endometriosis interna) is the ingrowth of stromal and glandular components of the endometrium directly into the myometrium. Diagnosis is mainly based on clinical assessment. Painful, symmetrical enlargement of the uterus is suggestive of adenomyosis. On bimanual examination, the uterus is softened and tender. This is known as Halban's sign.

TVS findings include salt and pepper appearance of the myometrium and the presence of subendometrial striations producing the Venetian blind appearance. MRI can also be done for diagnosis. However, the only definitive mode of diagnosis is histopathological examination.

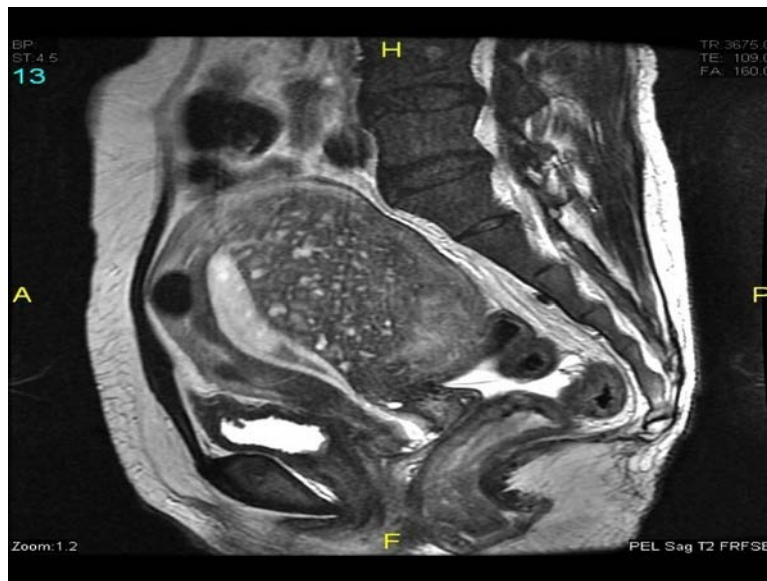
Since most women affected by the condition are past the childbearing age, total hysterectomy is the treatment of choice. Localized excision/Levonorgestrel containing IUCD can be tried in younger women.

Management of Adenomyosis



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The image below is of a sagittal MRI showing a uterus with adenomyosis in the posterior wall. Gross enlargement of the posterior wall is noted, with many foci of hyperintensity.



Sold by @Itachibot
If you purchased this from someone else,
you may have been scammed.

Disorders of Ovary

Question 1:

A young lady who presented with oligomenorrhea and infertility is diagnosed with PCOS. Mutation of which of the following genes can cause this condition?

- a) CYP21
- b) CYP22
- c) CYP23
- d) CYP24

Question 2:

A teenage girl presented with complaints of excessive growth of coarse hair on her jawline and upper lip. Which of the following is the most common cause for this condition?

- a) Antiepileptic medications
- b) Cushing's syndrome
- c) Idiopathic
- d) Polycystic ovarian syndrome (PCOS)

Question 3:

A mother brought her 15-year-old daughter to the OPD with concerns of excessive growth of coarse hair over her face and chest. Which scoring system is used to clinically grade this condition?

- a) Danforth and trotter score
- b) Halter score
- c) Ferriman-Gallwey score
- d) Prader score

Question 4:

Which of the following conditions is the Rotterdam criteria used to diagnose?

- a) Meig's syndrome
- b) Asherman syndrome
- c) Stein-Leventhal syndrome
- d) Mayer-Rokitansky-Küster-Hauser syndrome

Question 5:

An adolescent girl was brought by her mother to the clinic with complaints of irregular periods and acne. A USG scan was performed which showed the following finding. Which of the following is the most likely diagnosis?



- a) Polycystic ovarian disease
- b) Theca lutein cysts
- c) Congenital adrenal hyperplasia
- d) Ovarian endometriosis

Question 6:

A 26-year-old woman who presented with complaints of irregular cycles and weight gain was diagnosed with PCOS. This condition is characterised by multiple ovarian cysts that are less than _____ in size.

- a) 1 cm
- b) 1.5 cm
- c) 2 cm

d) 2.5 cm

Question 7:

A 29-year-old lady presented to your clinics with lesions as shown in the below images. Which of the following is a false statement regarding her condition?



- a) Associated with polycystic ovarian disease
- b) Associated with hypoinsulinemia
- c) More frequent in obese women
- d) Associated with adenocarcinoma stomach

Question 8:

A 25-year-old female patient presented with hirsutism. After further evaluation, a diagnosis of HAIR-AN syndrome was made. Which of the following statements is false regarding this condition?

- a) Severe insulin resistance is seen
- b) Mild hyperandrogenism is seen
- c) Temporal balding is seen
- d) Clitoromegaly is seen

Question 9:

A young lady being evaluated for infertility is diagnosed to have PCOS. Which of the following lab findings would you not expect to see in her?

- a) Raised triglyceride levels
- b) Testosterone levels elevated but < 200 ng/dL
- c) Raised LDL levels
- d) Raised HDL levels

Question 10:

A teenage girl who recently got diagnosed with PCOS presented with skin lesions on her forehead as shown in the image below. On examination, hirsutism was also noted. Which of the following drugs would you prescribe to this patient?



- a) Letrozole
- b) OCPs with cyproterone acetate
- c) Bromocriptine
- d) Clomiphene citrate

Question 11:

A 31-year-old lady has been unable to conceive for the past 2 years in spite of regular intercourse. She has recently been diagnosed with PCOS. Which of the following is the FDA approved medication that you could recommend to her?

- a) Ciproterone acetate
- b) Clomiphene
- c) Dexamethasone
- d) Tamoxifen

Question 12:

In a patient diagnosed with PCOS, which of the following is not an indication for surgical management?

- a) Ovarian hyperstimulation
- b) Infertility
- c) Previous pregnancy losses
- d) Morbid obesity

Question 13:

A woman with a history of infertility is on treatment with human menopausal gonadotropin. She presented with the following finding in the USG. What is the most likely diagnosis?



- a) Ovarian hyperstimulation syndrome
- b) Theca lutein cyst
- c) Polycystic ovarian syndrome
- d) None

Question 14:

A 21-year-old nulliparous woman presented with galactorrhoea and breast engorgement. Which of the following is a false statement regarding this condition?

- a) Prolactin is the most important hormone involved in its pathophysiology
- b) Prolactin levels more than 25 ng/ml cause this condition
- c) All patients with hyperprolactinemia present with galactorrhoea
- d) The drug of choice is Cabergoline

Answer Key

Question No.	Correct Option
1	a
2	d
3	c
4	c
5	a
6	a
7	b
8	b
9	d
10	b
11	b
12	d
13	a
14	c

Detailed Explanations

Solution to Question 1:

Mutation of the CYP21 gene can cause polycystic ovarian syndrome.

The gene for 21-hydroxylase (CYP21) is located in the short arm of chromosome 6. Heterozygosity of the gene has been linked to the development of PCOS. Defects in this gene also lead to congenital adrenal hyperplasia.

Solution to Question 2:

The given clinical history of excessive growth of coarse hair is suggestive of hirsutism. The most common cause for hirsutism in female patients is polycystic ovarian syndrome (PCOS).

Hirsutism is defined as excessive growth of androgen-dependent sexual hair or male distribution of hair in female patients. PCOS accounts for 70-80% of the cases of hirsutism, which typically begins in late adolescence or the early 20s.

Idiopathic hirsutism is the second most frequent cause. The most common rapid onset of hirsutism in young females is a testosterone-producing tumour like granulosa cell or Sertoli-Leydig cell tumours.

Other less common causes of hirsutism include:

- Late-onset congenital adrenal hyperplasia
- Tumours of the ovary and adrenal gland
- Cushing's syndrome
- Drug-induced

Virilisation: Characterized by more extensive androgen-induced changes than hirsutism alone. It has the following features:

- Acne
- Increased oily skin
- Temporal balding
- Clitoromegaly
- Deepening of the voice
- Development of male muscular pattern and body habitus
- Atrophy of breasts

Hypertrichosis: Excessive growth of non-sexual hair.

Solution to Question 3:

The given clinical scenario is suggestive of hirsutism. The Ferriman-Gallwey score is used to clinically assess and quantify the degree of hirsutism.

Within this system, abnormal hair distribution is assessed at nine body areas and each of them scored from 0-4. Increasing numeric scores correspond to greater hair density within a given area. The maximum score is 36. The degree of hirsutism is mild if the score is ≥ 8 . Moderate to

severe hirsutism is present if the score is ≥ 15 .

Note: Ferriman Gallwey score originally had 11 sites which were modified later to exclude 2 sites, viz., forearm, and legs.

Solution to Question 4:

The Rotterdam criteria is used to diagnose Stein-Levinthal syndrome or PCOS.

ESHRE/ASRM or Rotterdam criteria (2003):

Any two out of these three must be present-

- Clinical/ Biochemical hyperandrogenism
- Oligo/anovulation
- Polycystic ovaries

Other criteria used for diagnosing PCOS:

- National Institute of Health (NIH) criteria (1990)
- To include both
- Clinical/Biochemical hyperandrogenism
- Oligo/anovulation
- Androgen excess and PCOS society (AE-PCOS) criteria:
- To include both:
- Clinical/Biochemical hyperandrogenism
- Oligo/anovulation/polycystic ovaries

Option A: Meigs's syndrome refers to ascites and pleural effusions that accompany a benign ovarian tumor.

Option B: Asherman's syndrome refers to the formation of intrauterine synechiae, usually following endometrial destructive procedures like curettage or ablation.

Option C: Mayer-Rokitansky-Küster-Hauser syndrome is characterised by congenital absence of vagina, primary amenorrhoea, rudimentary cornua uteri and morphologically normal ovaries and Fallopian tubes situated on the pelvic sidewall.

Solution to Question 5:

The given clinical history and the necklace appearance on ultrasound are suggestive of polycystic ovarian disease.

The original Rotterdam consensus defined polycystic ovaries when ≥ 12 follicles were seen per ovary on USG. Each follicle should measure between 2 and 9 mm, and/or an ovarian volume should be > 10 ml. Due to improvement in imaging modalities since that time, the threshold for

follicle number has now been raised to 20 per ovary.

Solution to Question 6:

PCOS is characterised by multiple small ovarian cysts, each less than 1 cm in size.

Sonographic findings seen in PCOS are:

- ≥ 12 small cysts of 2 to 9 mm in diameter.
- Increased ovarian volume (> 10 mL) or both.

A single ovary with these findings is sufficient to define PCOS. However, this criteria does not apply to women taking combined oral contraceptive pills.

Note: In settings with improved sonographic imaging resolution, a follicle number ≥ 20 cysts per ovary can be considered significant for PCOS.

Solution to Question 7:

The hyperpigmented areas in the given image are acanthosis nigricans and this is associated with hyperinsulinemia. This skin condition is thought to be a cutaneous marker of insulin resistance.

Acanthosis nigricans is a velvety, darkening of the skin that usually occurs in the intertriginous areas. It has poorly defined borders, usually occurs in skin fold areas, such as the back of the neck, axilla, and groin, and may include thickening of the skin.

The pathogenesis of acanthosis nigricans is linked to growth factor levels and insulin-mediated activation of insulin-like growth factor (IGF) on keratinocytes and increased growth factor levels.

Acanthosis nigricans is usually seen in people younger than 40 years of age and is associated with the following disorders:

- Obesity
- Hypothyroidism
- Acromegaly
- Polycystic ovary disease
- Insulin-resistant diabetes
- Cushing's disease
- Addison's disease
- Gastrointestinal malignancies such as adenocarcinoma of the stomach or pancreas.

The treatment for acanthosis nigricans is directed towards the management of the underlying cause.

Solution to Question 8:

HAIR-AN syndrome is not characterized by mild hyperandrogenism but marked hyperandrogenism (HA). It also includes insulin resistance (IR) and acanthosis nigricans (AN).

HAIR-AN syndrome and ovarian hyperthecosis are exaggerated phenotypes of polycystic ovarian syndrome (PCOS). Women affected by HAIR-AN syndrome often present with symptoms of severe hyperandrogenism (hirsutism, acne) and may occasionally display signs of virilization like clitoromegaly, temporal balding and voice deepening.

Ovarian hyperthecosis is a rare condition characterized by nests of luteinized theca cells distributed throughout the ovarian stroma.

Solution to Question 9:

A depressed HDL level and elevated total cholesterol:HDL ratios are seen in patients with PCOS.

The following are associated with PCOS:

- Insulin resistance
- Impaired glucose tolerance and type 2 diabetes mellitus
- Dyslipidemia: Increased LDL and triglyceride levels, elevated total cholesterol levels:HDL ratios.
- Infertility/subfertility
- A three-fold increased risk of endometrial cancer
- Mild elevations in testosterone levels < 200 ng/dL

Normal range of testosterone in women is 20-70 ng/dL. A mild elevation in testosterone is consistent with the diagnosis of PCOS. However, values > 200 ng/dL may suggest an ovarian tumour and warrant pelvic sonography.

Solution to Question 10:

The lesions shown in the image are acne. The drugs used to treat acne and hirsutism in a patient with PCOS are OCPs with cyproterone acetate.

The first step in the management of PCOS is weight reduction. OCPs are the drugs of choice for regularisation of cycles. They are also the drugs of choice for hirsutism and acne. OCPs used for PCOS should contain a 3rd/4th generation progestagen like cyproterone acetate. OCPs induce regular menstrual cycles, lower androgen levels and causes thinning of the endometrium.

Insulin sensitizers like metformin have been found to improve metabolic and gynecologic issues. It acts by reducing the insulin level and reducing hepatic production of glucose (liver neoglycolysis). Other benefits of metformin include improvement in fertility rate and reduction of the total and free testosterone levels. It also increases levels of sex hormone-binding globulin. Metformin is contraindicated in hepatic and renal disease.

Treatment of infertility associated with PCOS is done using ovulation-inducing drugs like letrozole and clomiphene citrate. If medical management fails, surgical options like laparoscopic ovarian drilling can be considered.

Note:

FOGSI, in 2024, has replaced Letrozole as the first-line drug for induction of ovulation in patients with PCOS. However, the FDA approval for the same is still pending.

Solution to Question 11:

Clomiphene is an FDA approved drug used for the treatment of infertility associated with PCOS.

Clomiphene induces ovulation in 80% of the cases and 40–50% patients conceive. However, abortions are common due to luteal phase defects. Ovarian hyperstimulation occurs in <1% cases. Co-administration with dexamethasone leads to better outcomes. In resistant case, tamoxifen 20–40 mg daily for 5 days should be tried.

Currently, letrozole has proven to be more effective in ovarian infertility in obese women but its use in such cases is pending FDA approval.

Solution to Question 12:

Surgical management is indicated in all the above conditions except for morbid obesity.

Indications of surgery in PCOS include:

- Failed medical therapy
- Ovarian hyperstimulation
- Infertility
- Previous pregnancy losses

Surgery comprises of laparoscopic drilling or puncture of not more than four cysts in each ovary. It is done using a laser or by unipolar electrocautery. It restores the endocrine milieu and improves fertility for a year or so. Thereafter, pelvic adhesions caused by surgery may again reduce the fertility rate. The formation of adhesions can be reduced by hydrofloatation.

The advantages of surgery are as follows:

- Tubal testing with chromotubation can be performed simultaneously
- Other causes of infertility like endometriosis can be looked for
- One-time treatment
- Intense and prolonged monitoring for side effects is not required
- Cost-effective compared to IVF
- Reduces androgen and LH production

- Following surgery, single ovulation occurs with drugs, and hyperstimulation and multiple pregnancies are avoided
- Ovulation occurs in 80–90% and pregnancy in 60–70%

The disadvantages of surgery are as follows:

- It involves giving anaesthesia and performing an invasive procedure like laparoscopy
- Adhesions may form postoperatively
- If cautery is used, premature ovarian failure due to destruction of ovarian tissue may be seen

Solution to Question 13:

The image showing enlarged ovaries with multiple large cysts along with the history of treatment for infertility with human menopausal gonadotropin suggests the diagnosis of ovarian hyperstimulation syndrome (OHSS).

Human menopausal gonadotropin (HMG) is a medication that is composed of FSH and LH. It is used to stimulate the release of ova in women who do not ovulate spontaneously. It can lead to ovarian hyperstimulation syndrome.

Ovarian Hyperstimulation Syndrome (OHSS) is a condition associated with ovarian enlargement due to exogenous gonadotropin therapy. Symptoms include abdominal pain, distention, ascites, gastrointestinal problems, respiratory compromise, oliguria, and thromboembolism. It can occur during ovulation induction or in early pregnancies conceived through exogenous ovarian stimulation.

Predisposing factors for OHSS include multi-follicular ovaries (such as in PCOS), young age, high estradiol levels during ovulation induction, and pregnancy.

Vasoactive substances produced by the corpus luteum, such as vascular endothelial growth factor (VEGF) and angiotensin II, are believed to contribute to enhanced capillary permeability, resulting in the loss of fluid, protein, and electrolytes into the peritoneal cavity and hemoconcentration.

Sonographic examination reveals enlarged ovaries with numerous follicular cysts and ascites.

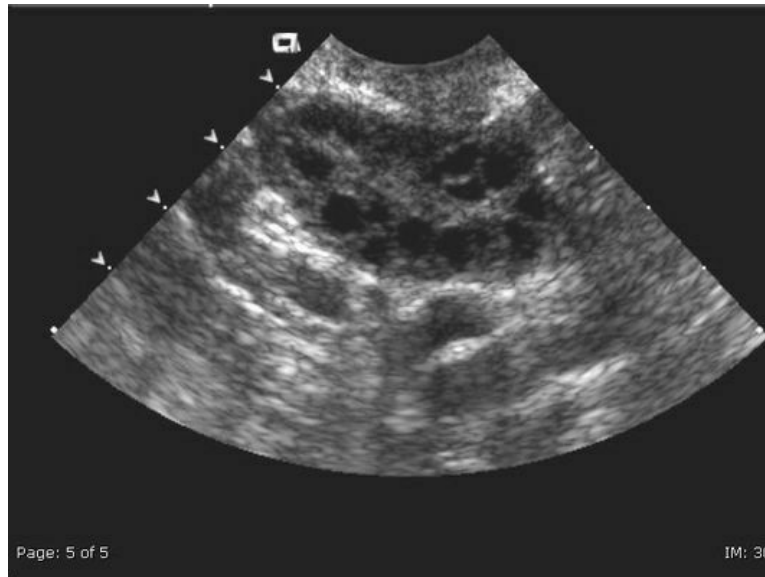
Maintaining fluid balance with isotonic fluids like normal saline is crucial. Strategies to prevent OHSS during exogenous ovulation include reducing follicular stimulation (low FSH dose), coasting (withholding FSH administration one or more days prior to the hCG trigger), prophylactic volume expanders, and substituting hCG for FSH towards the end of ovarian stimulation.

Other options:

Option B: Theca lutein cysts are a rare form of follicular cysts characterized by luteinization and hypertrophy of the theca interna layer. Both ovaries are enlarged and cystic. This condition is commonly associated with gestational trophoblastic disease, multifetal gestation, and placentomegaly.

Option C: Polycystic ovarian syndrome shows 12 or more small follicles each 2-9 mm in size placed peripherally (necklace pattern) (shown below). The patient presents with menstrual

irregularities, hirsutism, acne, etc.



Solution to Question 14:

All patients with hyperprolactinaemia do not present with galactorrhoea.

Galactorrhoea is the secretion of milky fluid from the nipple which is unrelated to childbirth. Prolactin is the most important hormone involved in the pathophysiology of amenorrhoea and galactorrhoea. Majority of women (80%) with high prolactin levels present with galactorrhoea. However, some hyperprolactinemic women may not have this symptom.

Normal prolactin level is 25 ng/mL and levels more than this can cause galactorrhoea. Levels up to 100 ng/mL occur in hyperprolactinaemia but prolactin level over 100 ng/mL is seen in pituitary tumours.

Galactorrhoea is caused by pituitary adenoma and hyperprolactinemia. The condition is associated with amenorrhoea, oligomenorrhoea and infertility. The drug of choice for this condition is cabergoline.

Contraception and Sterilization

Question 1:

Which of the following statements is false about the rhythm method?

- a) Anencephaly is seen on accidental conception
- b) It is a cost effective method
- c) Safe period is determined using temperature or mucous method
- d) Abstinence is required for 7-10 days in a month

Question 2:

What does Billing's method of contraception refer to?

- a) Basal body temperature method
- b) Cervical mucus method
- c) Calendar method
- d) Coitus interruptus method

Question 3:

Identify the contraceptive shown in the given image.



- a) Male condom
- b) Female condom
- c) Diaphragm
- d) Cervical cap

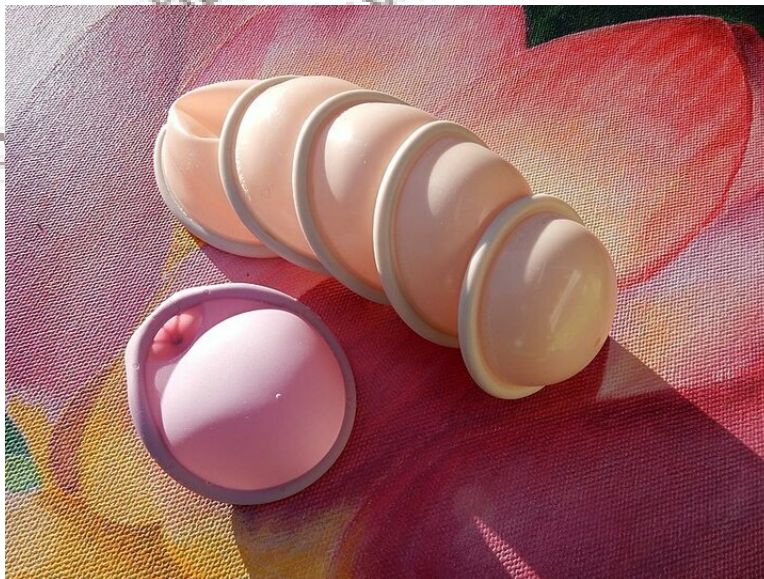
Question 4:

A nulliparous woman has been using Today sponge as a contraceptive. What does this contain?

- a) Nonoxynol 9
- b) Octoxynol 8
- c) Menfegol
- d) Tergitol

Question 5:

Which of the following is not a contraindication for the use of the contraceptive shown below?



- a) Multiple sex partners
- b) Recurrent UTI
- c) Uterine prolapse
- d) Herpes vaginalis

Question 6:

You are prescribing nonoxynol-9 jelly to a woman using a cervical diaphragm for contraception. What is its mechanism of action?

- a) Acrosomal enzyme inhibition
- b) Cervical enzyme alteration
- c) Glucose uptake inhibition by sperms
- d) Disruption of cell membrane

Question 7:

A young woman who is using a cervical diaphragm asks for an alternative contraceptive. You suggest a recently launched contraceptive Phexxi. What is its composition?

- a) Lactic acid, Citric Acid, Potassium bitartrate
- b) Ascorbic acid, Citric acid, Calcium bitartrate
- c) Lactic acid, Ascorbic acid, Potassium bitartrate
- d) Ascorbic acid, Lactic acid, Calcium bitartrate

Question 8:

A newly married woman was prescribed Mala-N at a PHC. What is its composition?

- a) LNG and ethinyl estradiol
- b) Nordestero and ethinyl estradiol
- c) Desogestrel and ethinyl estradiol
- d) Norgestimate and ethinyl estradiol

Question 9:

Which of the following is not an advantage of norgestimate OCPs?

- a) Reduces venous thrombosis
- b) Less metabolic side effects
- c) Reduces acne and hirsutism

- d) Lesser cardiovascular risk

Question 10:

Which of the following is not a contraindication to the use of OCPs?

- a) Ischemic heart disease
- b) Hepatoma
- c) Hypothyroidism
- d) Breast cancer

Question 11:

Which of the following drugs do not lead to failure of OCPs?

- a) Rifampicin
- b) Phenobarbitone
- c) Valproate
- d) Phenytoin

Question 12:

Which of the following is not the mechanism of action of progesterone only pills?

- a) Thicken the cervical mucus
- b) Prevents ovulation
- c) Kills the sperms in the cervical canal
- d) Inhibits penetration of sperm

Question 13:

A patient has come to your clinic for contraceptive advice. As she has a family history of breast cancer, you suggest a non-hormonal oral contraceptive to her. What is the initial dose of this drug?

- a) 30 mg on day 1 of cycle
- b) 60 mg on day 1 of cycle

- c) 60 mg on day 14 of cycle
- d) 30 mg on day 14 of cycle

Question 14:

Which of the following is the most effective reversible method of contraception?

- a) Combined pill
- b) Norplant
- c) Mini pill
- d) Mirena

Question 15:

Which of the following is not a mechanism of action of the device given in the image below?



- a) Chronic endometrial inflammation
- b) Increase the motility of tubes
- c) Enzymatic changes to prevent implantation
- d) Formation of cervical mucus plug

Question 16:

IUCD insertion is being considered in a multiparous woman who wants long-term contraception. This method of contraception is absolutely contraindicated in all of the following except _____.

- a) Uterine leiomyomas
- b) Suspected pregnancy
- c) HIV infection
- d) Pelvic inflammatory disease

Question 17:

You are inserting Nova-T in a patient for spacing of births. This device contains_____.

- a) Copper and silver
- b) Copper and aluminium
- c) Only copper
- d) Copper and selenium

Question 18:

A nursing mother was prescribed progestasert for contraception. How much progesterone is released by this device?

- a) 0.65 µg/day
- b) 65 µg/day
- c) 6.5 µg/day
- d) 6.5 mg/day

Question 19:

Which of the following increases the risk of pelvic inflammatory disease?

- a) OCPs
- b) Mirena
- c) DMPA injectables
- d) Copper IUCD

Question 20:

What is the most effective mode of contraception for a woman with heart disease?

- a) Oral pills
- b) Hormonal implants
- c) Condom
- d) IUCD

Question 21:

Which of the following techniques of permanent contraception does not require an abdominal incision?

- a) Pomeroy
- b) Parkland
- c) Essure
- d) Irving

Question 22:

A woman has come to your clinic for a checkup following tubal ligation by Pomeroy technique. Which portion of the tube is ligated in this technique?

- a) Infundibulum
- b) Ampulla
- c) Isthmus
- d) Interstitium

Question 23:

Fimbriectomy is also known as:

- a) Pomeroy technique
- b) Uchida's procedure
- c) Kroener's procedure
- d) Irving's procedure

Question 24:

A 40-year-old male patient underwent a vasectomy. What is the failure rate of this procedure?

- a) 0.5%
- b) 0.15%
- c) 3%
- d) 1%

Question 25:

A couple with a history of unprotected sexual intercourse requires emergency contraception. Which of the following is not a mechanism by which these drugs act?

- a) Delaying or inhibiting ovulation
- b) Inhibiting fertilization
- c) Preventing the implantation of the fertilized egg
- d) Interrupting an early pregnancy

Question 26:

A 23-year-old woman comes to the clinic with a history of unprotected intercourse 2 days ago. Which of the following will you prescribe as a single dose contraceptive?

- a) Levonorgestrel 0.75 mg
- b) Ethinyl estradiol 50 mcg
- c) Levonorgestrel 1.5 mg
- d) Norgestrel 0.25 mg

Question 27:

What is the dose of ulipristal for emergency contraception?

- a) 60 µg
- b) 3 mg
- c) 30 mg
- d) 30 µg

Answer Key

Question No.	Correct Option
1	a
2	b
3	b
4	a
5	a
6	d
7	a
8	a
9	a
10	c
11	c
12	c
13	a
14	b
15	d
16	c
17	a
18	b
19	d
20	d
21	c
22	c
23	c
24	b
25	d
26	c
27	c

Detailed Explanations

Solution to Question 1:

Anencephaly is not seen in the case of accidental conception in a woman using the rhythm method. It occurs due to deficiency of folic acid in the mother.

The calendar method or the rhythm method is one of the natural methods of contraception by fertility awareness. The couple is advised avoidance of sexual intercourse around ovulation. Thus, it is very cost-effective but it can only be used by educated couples.

According to this method, safe period in a normal 28-day cycle is from the 1st day of menstrual period to the 10th day and from the 18th to the 28th day. An alternate method is by calculating the risk period i.e., 3 days before and 3 days after ovulation. Ovulation can also be estimated by changes in cervical mucus and basal body temperature.

Another method of calculating the safe period is the Knaus–Ogino method. The fertile period is determined by subtracting 18 days from the shortest cycle and 10 days from the longest cycle. This gives the first and the last day of the fertile period, respectively.

Solution to Question 2:

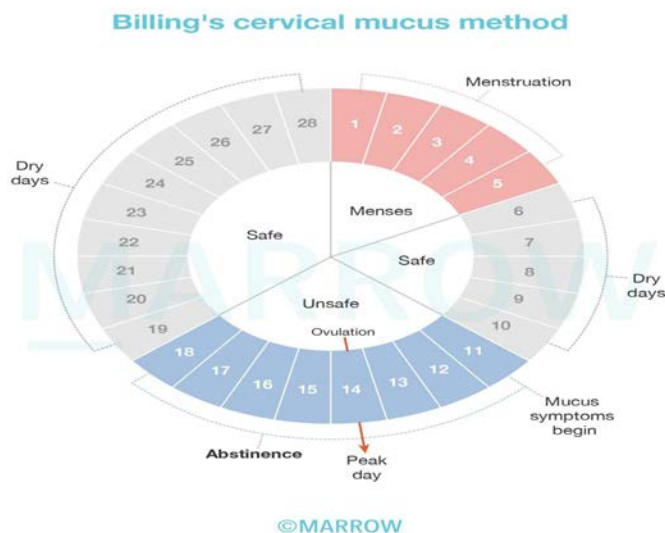
Billing's method of contraception refers to the cervical mucus method. It is also called ovulation method.

The properties of the cervical mucus change during the menstrual cycle. The female partner can predict the fertile period by examining her cervical mucus.

Under the influence of estrogen, the mucus increases in quantity and becomes more slippery and elastic i.e. wet days. This continues till it reaches a peak. After this, under the influence of progesterone, it becomes thicker, scanty i.e. dry days.

Abstinence is advised during the wet days till 4 days after the peak day. The safe period includes the dry days after menses till the appearance of mucus and from the 4th day after the peak day.

The pattern of cervical mucus secretion during a menstrual cycle is given below.



Solution to Question 3:

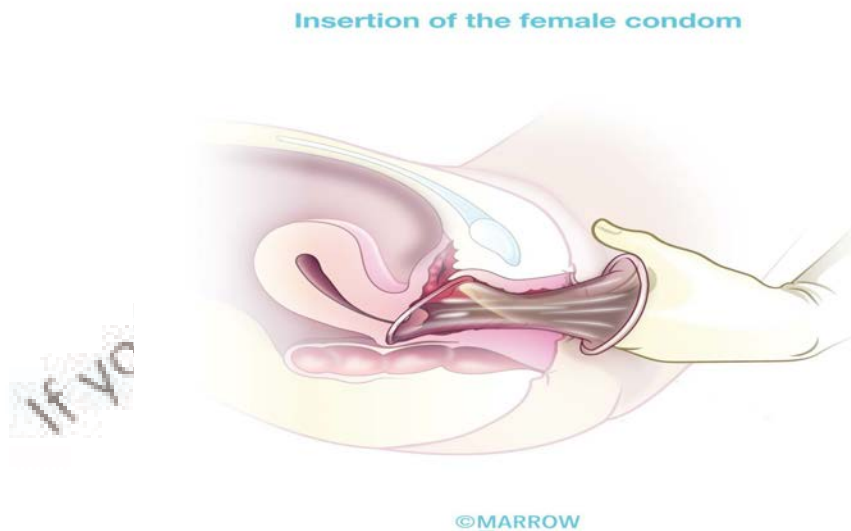
The given image shows a female condom. It consists of a pouch made of polyurethane which is 17 cm in length with a flexible ring at each end. The inner ring at the closed end is smaller than the outer ring.

The inner ring is inserted at the apex of the vagina and the outer ring remains outside. It covers the cervix, the entire vagina, and external genitalia. Hence, it provides protection against HIV and STDs. It has a failure rate of 5–15 per 100 woman-years, which is higher than that of a male condom.

It can be fitted by the patient hours before intercourse, and easily removed soon after. It covers the cervix, vagina, and external genitalia and is protective against sexually transmitted diseases. It can also be worn in the puerperal period, unlike the cervical diaphragm.

The female condom should not be used along with a male condom because if worn together they may slip, tear, or become displaced.

The given image below shows the insertion of a female condom:



Solution to Question 4:

Today sponge contains 1 g of nonoxynol 9. It is a mushroom-shaped polyurethane disposable sponge that is 2 inches wide and 1.25 inches thick.

It is provided with a loop for easy removal. It should be placed high up in the vagina with the concave side covering the cervix. It can remain effective for 24 h. It can be used only once.

It acts as a mechanical barrier preventing the entry of sperms into the cervical canal, absorbs semen, and contains a spermicidal agent. The failure rate is 9-16 per 100 woman-years.

'Today' vaginal sponge



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Solution to Question 5:

The given image shows a cervical diaphragm or Dutch cap. It can be used by women having multiple sex partners.

Contraindications of cervical diaphragm usage:

- Prolapse, cystocele, rectocele - accurate fitting is not possible
- Recurrent urinary tract infection
- Allergy to rubber or spermicidal agent

It must be removed following intercourse as it can lead to toxic shock syndrome (TSS) if left in the vagina for a long period. TSS is caused by *Streptococcus pyogenes* or *Staphylococcus aureus* infection.

Solution to Question 6:

Nonoxynol-9 is one of the spermicidal agents which act by disruption of the sperm cell membrane.

Spermicidal agents are chemical contraceptives as they contain surfactants, such as nonoxynol-9, octoxynol, menfegol and enzyme-inhibiting agents. These are available as:

- Foam tablets
- Soluble pessaries
- Creams
- Jellies
- Films

The spermicidal agent remains effective for 1–2 hours after the application. They are always used along with other contraceptives such as the diaphragm, cervical cap, and condom. When they are used alone, the failure rate is high i.e., 30 per 100 woman-years.

Chronic use of spermicides can cause vaginal irritation, abrasion, and ulceration. This increases the risk of HIV spread. Hence, spermicidal agents should not be recommended to HIV-positive couples.

Solution to Question 7:

Phexxi is a vaginal gel that contains lactic acid, citric acid, and potassium bitartrate.

Phexxi is a vaginal pH regulator which is a non-hormonal contraceptive. It works by maintaining vaginal pH within the range of 3.5 to 4.5. This acidic environment is inhospitable for sperms.

Phexxi comes in pre-filled vaginal applicators. A single dose must be applied before vaginal intercourse. If more than one act of vaginal intercourse occurs within one hour, an additional dose must be applied.

Phexxi - Applicator



Solution to Question 8:

Mala-N is a combined oral contraceptive containing 0.15 mg levonorgestrel (LNG) and 0.03 mg ethinyl estradiol. It is supplied free of cost through all PHC's, urban family welfare centers, etc.

All combined oral pills contain a mixture of ethinyl estradiol in a dose of 20-30 mcg and an orally active progestogen. Mala-D has the same composition as Mala-N but is available at a price of Rs 3/- per packet.

Combined oral pill works by suppressing pituitary hormones, FSH, and LH peak which prevents ovulation. At the same time, progestogen causes atrophic changes in the endometrium and

prevents implantation.

Progestogen also acts on the cervical mucus making it thick and tenacious and impenetrable by sperms. It also interferes with tubal motility and alters tubal transport.

, so the fertilized egg reaches the uterine cavity before the endometrium is receptive for implantation.

These tablets are taken starting on the first day of the cycle for 21 days. A new course of tablets should be commenced 7 days after the cessation of the previous course. They should be taken at bedtime, preferably after a meal.

Solution to Question 9:

Norgestimate is one of the 3rd generation progesterone agents. All OCPs have the disadvantage of increasing the risk of venous thrombosis.

Third-generation progesterone agents are less androgenic and so reduce the occurrence of acne and hirsutism. They also have a lower risk of metabolic and cardiovascular disorders compared to the previous generation progesterone agents.

Benefits of combined pills:

- Effective contraceptive
- Useful in menorrhagia and polymenorrhoea
- Relieves dysmenorrhoea and premenstrual tension
- Prevents anaemia by reducing the menstrual loss
- Lower incidence of benign breast neoplasia (fibrocystic disease), functional ovarian cyst, ovarian and uterine malignancy
- Reduced risk of PID - due to thick cervical mucus caused by progestogen, preventing the organisms from entering into the uterine cavity
- Reduced incidence of ectopic pregnancy - due to suppression of ovulation and decreased risk of PID
- Protects against rheumatoid arthritis
- Reduces the risk of anorectal cancer
- Useful in acne, PCOD, and endometriosis

Solution to Question 10:

Thyroid disorders like hypothyroidism are not a contraindication for OCP usage.

Contraindications of OCPs:

- Absolute contraindications -
- Breast and genital cancers

- Liver disease
- History of thromboembolism
- Cardiac abnormalities
- Congenital hyperlipidemia
- Undiagnosed abnormal uterine bleeding
- Relative contraindications -
- Age over 40 years, if smoker >35 years
- Mild hypertension
- Chronic renal disease
- Epilepsy
- Migraine
- First 6 months of breastfeeding
- Diabetes mellitus
- Gallbladder disease
- History of infrequent bleeding or amenorrhea

Solution to Question 11:

Sodium valproate is an inhibitor of CYP enzymes that are responsible for OCP metabolism. Thus, it does not lead to OCP failure.

Drugs leading to OCP failure are:

- Rifampicin
- Tetracyclines - interferes with enterohepatic circulation of estrogen
- Phenytoin
- Phenobarbitone
- Antifungals
- Cephalosporin
- Ritonavir for HIV - interferes with absorption of OCP

Most of the anti-convulsants are enzyme inducers and hence increase the clearance of OCPs. This leads to OCP failure.

Solution to Question 12:

Minipills or progesterone-only pills (POP) do not kill sperm in the cervical canal.

Progesterone-only pills act by thickening the cervical mucus (the main mechanism of all POPs)

except cerazette) that inhibits penetration of sperms, inhibiting tubal motility, and suppressing ovulation.

Other advantages of POPs include a reduction in the incidence of pelvic inflammatory diseases and sexually transmitted diseases (due to thickened cervical mucus).

Due to the inhibition of tubal motility, the absolute risk of all pregnancies including ectopic pregnancy decreases, while the relative risk of ectopic pregnancy increases. But a previous history of ectopic pregnancy is not a contraindication for POP.

POPs do not affect clotting, lipid metabolism, and carbohydrate metabolism. So they can safely be used by women who are obese, breastfeeding, hypertensive, have SLE or a history of thromboembolism, diabetic with vascular disease. Significant adverse effects of POPs are irregular bleeding (mostly scanty), and secondary amenorrhea after longtime usage as it causes endometrial atrophy.

Contraindications for POPs include undiagnosed vaginal bleeding, pregnancy, known or suspected breast cancer, and benign or malignant liver tumors.

Solution to Question 13:

Centchroman is a non-hormonal oral contraceptive that is started at a dose of 30 mg on the first day of the menstrual cycle. It can be given to women with a family history of breast cancer where hormonal contraceptives are contraindicated.

Centchroman or ormeloxifene is a synthetic nonsteroidal contraceptive. Following the initial dose of 30 mg on the first day of menses, it is taken twice weekly for 12 weeks and weekly thereafter. It prevents implantation through endometrial changes. The return of fertility occurs within 6 months of stopping the drug.

It can also be used as a postcoital pill, given in 60 mg dose within 24 hours of coitus. It has a failure rate of 1.83 per 100 woman years. It does not provide protection against HIV and other STDs.

Side effects of centchroman:

- Headache, nausea, vomiting
- Weight gain
- Delay in return of fertility (up to 6 months)
- Atypical endometrial hyperplasia - prolonged use

Contraindications of centchroman:

- PCOD
- Hepatic dysfunction
- Cervical dysplasia
- Allergy to the drug

This drug has been developed by Central Drug Research Institute, Lucknow. It has been released in India under the name of Chhaya (previously called Saheli) under the National Family Planning

Programme.



Solution to Question 14:

The most effective reversible method of contraception among the given options is Norplant, which is a subdermal hormonal implant.

The most effective method of contraception overall is sterilization, which is one of the permanent methods. Both methods belong to the first-tier category of contraceptives.

There are 2 types of subdermal implants:

- Implanon (which has replaced Norplant I)
- Norplant II

The implants work by inhibition of ovulation, endometrial atrophy, and thickening of cervical mucus. Implanon contains desogestrel and also reduces acne and osteoporosis.

The implants are inserted on the first day of the menstrual cycle, within 5 days of abortion, and 3 weeks after the delivery. The woman needs to use barrier contraception or abstain in the first 7 days of insertion.

The image below shows an implanon:



The image given below shows the insertion of a subdermal implant:



Advantages of implants:

- Long-acting with sustained effect
- Good compliance
- Coital-independent with no nuisance of daily oral or frequent injections
- Failure rate - 0.2 and 1.3 per 100 woman-years
- Implanon reduces acne and osteoporosis.

Disadvantages of implants:

- Breakthrough bleeding
- Irregular cycles

- Amenorrhoea
- Ectopic pregnancy
- Local infection
- Requires insertion and removal with nonbiodegradable capsules
- Expensive.

Solution to Question 15:

The given image shows Multiload Cu250, which is one of the copper containing IUCDs. Formation of cervical mucus plug is not one of its mechanisms of action. However, progesterone-containing IUCDs can alter the cervical mucus along with the local effects.

The various mechanisms of action of a copper-containing IUCD are:

- Interference with the migration of spermatozoa
- Foreign body reaction releasing prostaglandins - provokes uterine contractility and increases the tubal peristalsis which interferes with implantation
- Leucocytic infiltration in the endometrium - macrophages engulf the fertilized egg if it enters the endometrial tissue
- Enzymatic and metabolic changes in the endometrial tissue which interferes with implantation of the fertilized ovum

Some IUCDs like CuT-380A are effective for up to 10 years.

They are a good choice of method of contraception for the following groups of women:

- Having low risk of STD
- Multiparous woman
- Having a monogamous relationship
- Wanting long-term reversible method of contraception
- Unreliable users of oral contraception or barrier contraception

IUCDs are avoided in unmarried and nulliparous patients because of the risk of PID and subsequent tubal infertility. Copper IUCDs can also be used as emergency contraception following intercourse.

Solution to Question 16:

IUCDs can be used in HIV-affected women.

Intrauterine contraceptive devices (IUCDs) provide long-term contraception by preventing fertilization.

Contraindications for IUCD insertion:

- Suspected pregnancy
- Pelvic inflammatory disease (PID)
- Lower genital tract infection
- Presence of fibroids
- Severe anaemia
- Uncontrolled diabetes - due to increased pelvic infection
- Scarred uterus eg, post caesarean section
- Septate uterus
- Menorrhagia and dysmenorrhoea - if Copper T is used
- Breast cancer - if LNG IUCD is used

Complications of IUCD usage:

- Immediate -
- Vasovagal attack
- Uterine cramps
- Early -
- Expulsion of IUCD
- Perforation of uterus
- Spotting, menorrhagia
- Dysmenorrhoea
- Vaginal infection
- Actinomycosis
- Late -
- PID
- Ectopic pregnancy
- Perforation of uterus
- Menorrhagia
- Dysmenorrhoea

The failure rate of IUCD is 1–3 per 100 woman-years.

Levonorgestrel IUCD also causes hormonal adverse effects such as headache, acne, breast tenderness, and functional ovarian cysts. Amenorrhoea occurs in 35% of women after 2 years of use.

Solution to Question 17:

Nova-T contains both copper and silver. The addition of the silver increases its effective lifespan.

Copper IUCD - Nova T

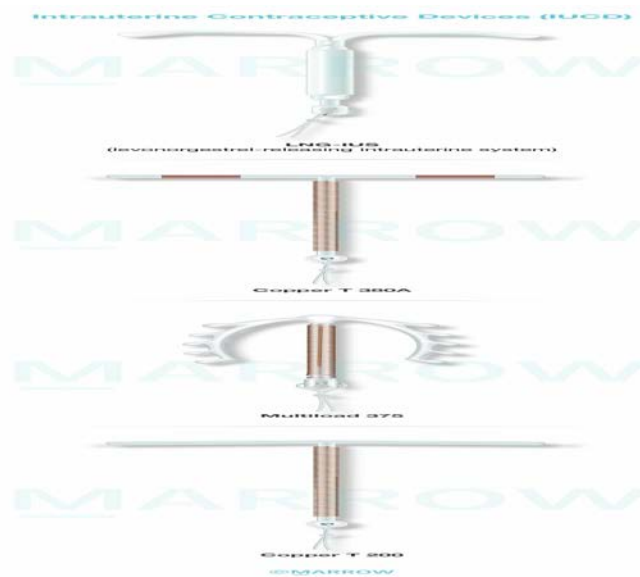


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In copper-carrying devices, copper wire is wrapped around the vertical stem of a polypropylene frame. Some of them have copper wire wrapped around the arms too. Based on the surface area, they are of various types:

- Copper T 380 A
- Multiload copper 250
- Copper T 200
- Copper 7
- Copper T 220
- Nova-T

They are effective for 3–5 years. It is estimated that about 50 mcg of copper is eluted daily in the uterus. Copper T 380A, known as Paraguard, has a lifespan of 10 years. Nova T has a lifespan of 5 years.



Solution to Question 18:

Progestasert releases 65 µg of progesterone/day.

Progestasert is a T-shaped device carrying 38 mg of progesterone in a silicon oil reservoir in the vertical stem. The progesterone released causes the formation of a thick plug of mucus at the cervical os. This prevents entry of the sperms in addition to the local contraceptive effect of IUCDs.

There is lesser incidence of menstrual problems like menorrhagia and dysmenorrhoea compared to copper-containing intrauterine devices. However, it is expensive and requires yearly replacement.

Solution to Question 19:

Copper IUCDs increase the risk of PID.

OCPs, Mirena and DMPA injectables contain progesterone which causes thickening of the cervical mucus and decreases the motility of the female genital tract. Thus, they reduce the risk of PID by preventing the ascent of bacteria from the vagina.

Solution to Question 20:

IUCDs are the most effective mode of contraception for women with heart diseases. IUCDs are safe in these patients and come under the first-tier contraceptive methods.

Condoms are not contraindicated in heart disease patients but they are third-tier contraceptives. Hence, they are not as effective as IUCDs.

OCPs and hormonal implants are contraindicated in patients with heart diseases as the constituent hormones can cause thromboembolic disorders.

Solution to Question 21:

Essure permanent device is a method of hysteroscopic sterilization. Pomeroy, Parkland, and Irving procedures are methods for tubal ligation via an abdominal approach.

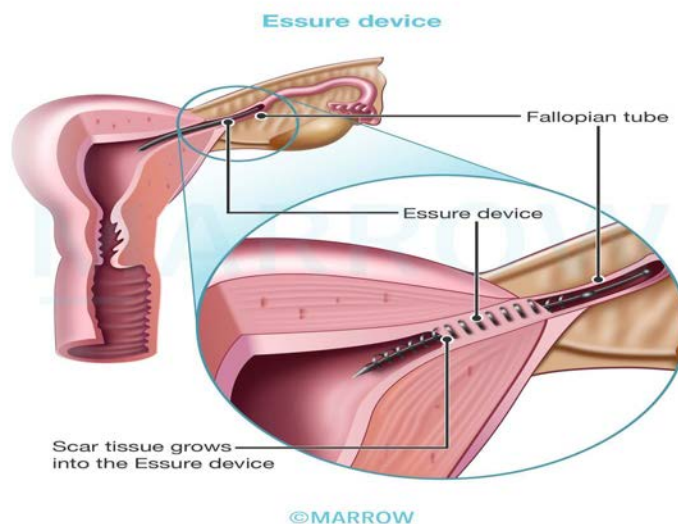
Essure device consists of a micro-inserter with a flexible inner coil made of stainless steel and a dynamic outer coil made of nickel titanium alloy (nitinol).

This device is inserted into the fallopian tube by hysteroscopy. It acts by benign local fibrous tissue growth leading to occlusion of the tube. The failure rate is 3.5%.

Its advantages are that there is no abdominal scar and the procedure can be done under local anaesthesia.

Disadvantages of Essure:

- Hysteroscopy required for the procedure and to confirm the blockage
- Expensive procedure
- Expertise required
- Irreversible - tuboplasty for reversal not possible
- 3 months waiting time to ensure sterility - HSG must be done after 3 months to confirm tubal blockage
- Bilateral insertion difficult due to spasm
- Can cause perforation of the tube



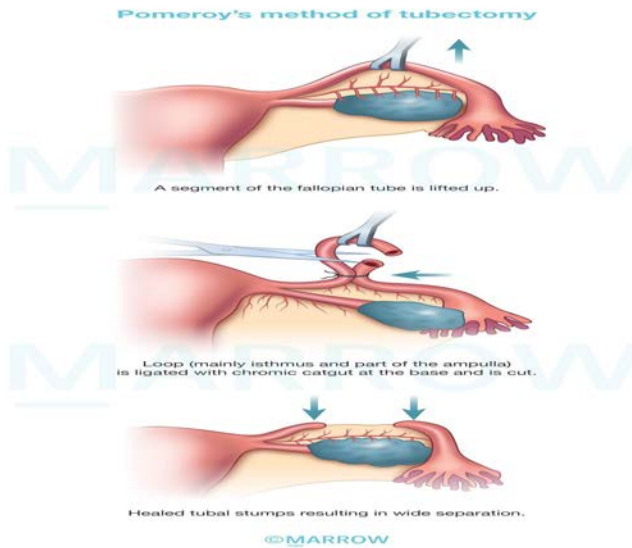
Solution to Question 22:

Isthmus of the fallopian tube is ligated in Pomeroy technique. It is the most popular technique of tubal ligation.

Midline, paramedian or transverse abdominal incision is made 2 inches above the pubic symphysis (1 inch in puerperal sterilisation). The fallopian tube is identified, made into a loop and tied at the base. The loop is then excised.

The operation is simple and requires a short period of hospitalization. The failure rate is only 0.4% and it is mainly due to spontaneous canalization. It is surgically reversible.

The below image shows Pomeroy's method of tubal ligation.



Solution to Question 23:

Fimbriectomy is also known as Kroener's procedure. This procedure involves ligation of the distal ampulla of the tube with two permanent sutures. Then, resection of the infundibulum of the tube is done.

Fimbriectomy

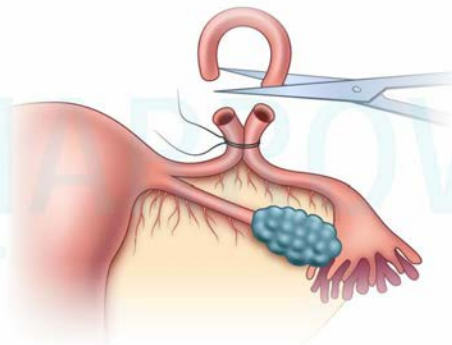


©Marrow

Some more techniques of permanent sterilization include:

Pomeroy method - The fallopian tube is identified on each side, brought out through the incision. The middle portion is formed into a loop which is tied at the base with catgut and excised. This is the most commonly used technique.

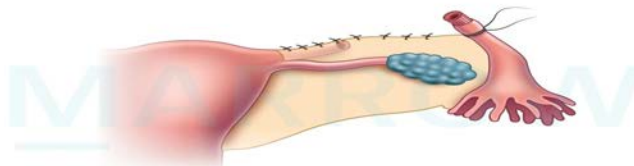
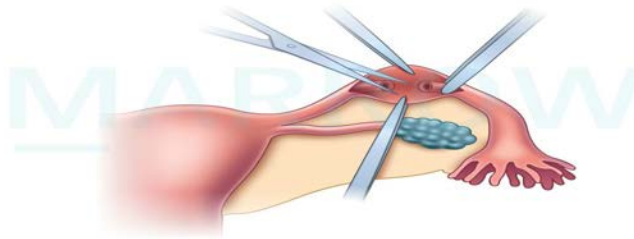
Pomeroy



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Uchida's method - The tubal serosa is stripped off the muscular layer in the midsegment of the tube. The muscle layer is excised. The proximal end is ligated and buried in the broad ligament. This minimal excision of the tube preserves the potential for tuboplasty.

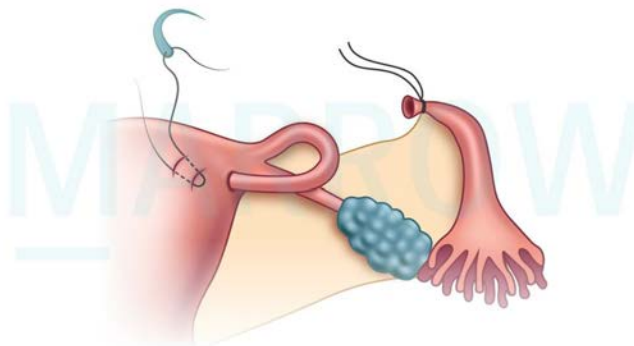
Uchida technique



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Irving's procedure - The mid-portion of the tube is ligated and the intervening portion is excised. The proximal end is buried in the myometrium and the distal end is buried in the broad ligament. It is a reliable method but irreversible and may require a laparotomy incision.

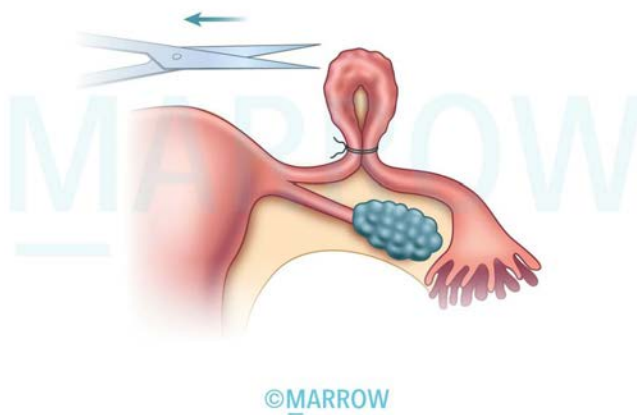
Irving procedure



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Madlener's procedure - A loop of the fallopian tube is crushed and then ligated with a nonabsorbable suture.

Madlener procedure



Solution to Question 24:

The failure rate of vasectomy is 0.15%

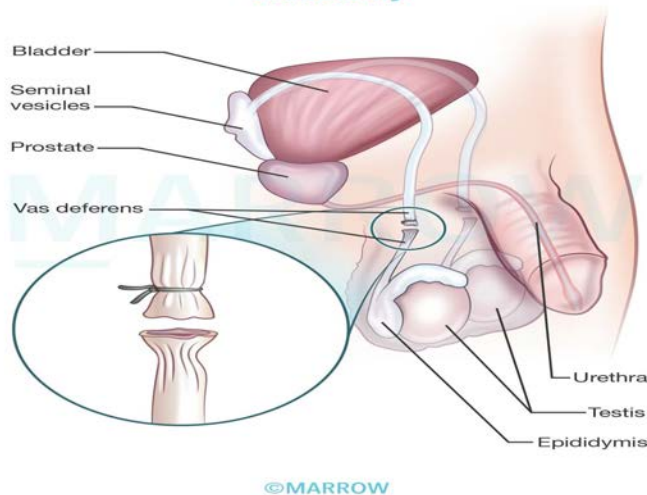
This procedure is done under local anaesthesia. A small incision is made in the scrotum. The vas deferens are divided on both sides which disrupts the passage of sperm.

The no-scalpel technique has been now adopted. One single incision is made with special forceps and skin stitch is not required. Clips and plugs can be applied over the vas instead of cutting.

The sperm can remain in the reproductive tract for up to 3 months. Hence, sterility is not immediate. The couple must abstain from intercourse during this period or use other methods of contraception.

Approximately, 20 ejaculates clear the semen of all sperm. Two semen analysis reports must confirm the absence of sperm before the man can be declared sterile.

Vasectomy



Solution to Question 25:

Emergency contraceptives do not interrupt early pregnancy, once conception has occurred. They are used to prevent pregnancy after unprotected intercourse. Hence, they are also called interceptives.

The main action of emergency contraceptives is to make the endometrium unsuitable for implantation. They can also delay or inhibit ovulation, and prevent fertilization of the egg by sperms.

Methods of emergency contraception:

Drug	Dose
Levonorgestrel	0.75mg stat and repeat dose after 12 hours or 1.5mg as a single dose
Ethinyl estradiol 50 µg + Norgestrel 0.25 mg	2 tabs stat and 2 tabs after 12 hours
RU486 (Mifepristone)	Single dose of 25-50mg
Ethinyl Estradiol	2.5 mg BD X 5 days
Copper IUDs	Insertion within 5 days of intercourse
Centchroman	2 tabs(60 mg) to be taken twice at an interval of 12 hours should be taken within 24 hours of intercourse.
Ulipristal	30 mg within 5 days of intercourse

Solution to Question 26:

Levonorgestrel (LNG) 1.5 mg i.e. 2 tablets of LNG 0.75 mg can be used as a single dose emergency contraceptive. This should be taken within 72 hours of unprotected intercourse.

Alternatively, one tablet of LNG 0.75mg can be followed by another tablet 12 hours later. The tablets can be offered up to 120 hours but the efficacy decreases with the increase in interval between coitus and drug intake.

LNG prevents ovulation and makes endometrium unsuitable for implantation. The failure rate is 1.1%. If pregnancy occurs as a result of contraceptive failure, the risk of ectopic pregnancy increases but it is not teratogenic.

Advantages of LNG:

- No oestrogen and its associated side effects

- Can be used in hypertensive, cardiac and diabetic patients
- Suitable for lactating woman
- Can be given up to 120 hours after the unprotected intercourse
- Single-dose therapy

Contraindications of LNG:

- Liver disease
- Thrombophlebitis
- Migraine
- Galactose allergy - as the tablet contains lactose

Solution to Question 27:

The dose of ulipristal is 30 mg tablet and it should be taken within 5 days of intercourse.

Ulipristal is a synthetic progesterone receptor modulator. It attaches to the progesterone receptor and acts by:

- Preventing/delaying ovulation
- Suppressing endometrial and follicular growth
- Inhibiting implantation

Side effects are headaches and mood changes. It is contraindicated in women with severe hepatic dysfunction and severe asthma.

Note: The use of ulipristal for the treatment of fibroids has been discontinued due to the increased incidence of severe liver injury.

Vaginal Infections

Question 1:

Which of the following is not a risk factor for bacterial vaginosis?

- a) Cigarette smoking
- b) Intrauterine device usage
- c) Vaginal douching
- d) Caucasian race

Question 2:

A young woman was suspected to have bacterial vaginosis. Which of the following statements is false about this condition?

- a) Gardnerella vaginalis is a causative agent
- b) Amsel's criteria is used for diagnosis
- c) Vaginal itching is present
- d) Oral metronidazole is given for treatment

Question 3:

A 30-year old woman presents to your OPD with complaints of foul-smelling vaginal discharge for the past 2 weeks. On examination, the discharge is adherent to the vaginal mucosa and has a fishy odour. What is the most likely cause?

- a) Trichomonas
- b) Candida
- c) Gardnerella
- d) Chlamydia

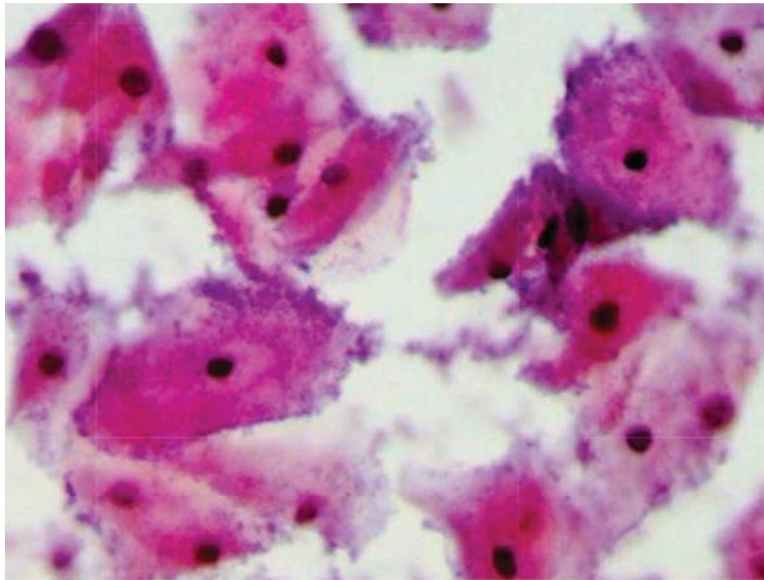
Question 4:

In a patient with bacterial vaginosis, which of the following microscopic features would not be seen?

- a) Abundance of gram variable coccobacilli
- b) Reduced number of lactobacilli
- c) Abundance of polymorphs
- d) Presence of clue cells

Question 5:

Given below is the image of a smear prepared from the vaginal discharge of a patient. What is the most likely diagnosis?



- a) Chlamydial vaginitis
- b) Gonococcal vaginitis
- c) Bacterial vaginosis
- d) Trichomonas vaginitis

Question 6:

A woman comes to the clinic with complaints of foul-smelling vaginal discharge. On examination, the discharge was profuse and had a fishy odour. Which of the following is the best criteria to confirm the diagnosis?

- a) Boer-Meisel's criteria
- b) Light's criteria
- c) Amsel's criteria

d) Nugent's criteria

Question 7:

In a patient with suspected bacterial vaginosis, Nugent's criteria is used to quantify the following bacteria except....

- a) Mycoplasma**
- b) Gardnerella**
- c) Lactobacillus**
- d) Mobiluncus**

Question 8:

A gynaecology resident noted a frothy, greenish, offensive-smelling vaginal discharge in a young lady. Whiff test was positive. Which organism is most likely implicated?

- a) Trichomonas**
- b) Candidia**
- c) Gardnerella**
- d) Chlamydia**

Question 9:

A patient came to the gynaecology OPD with complaints of foul-smelling frothy vaginal discharge and intense itching. On examination, the cervix was spotted and had the appearance of a strawberry. Identify the causative organism.

- a) Trichomonas vaginalis**
- b) Gardnerella vaginalis**
- c) Neisseria gonorrhoea**
- d) Candida albicans**

Question 10:

Which of the following organisms can be detected on a hanging drop preparation?

- a) Gardnerella vaginalis**

- b) *Trichomonas vaginalis*
- c) *Candida albicans*
- d) *Mobiluncus*

Question 11:

A 25-year-old woman came with complaints of greenish vaginal discharge. On saline microscopy, motile flagellated organisms were seen. What is the best treatment option for her?

- a) Doxycycline
- b) Amoxicillin
- c) Fluconazole
- d) Metronidazole

Question 12:

A pregnant woman in her 7th week of gestation was diagnosed with trichomoniasis. Which of the following can be used to treat her condition?

- a) Topical clotrimazole
- b) Oral metronidazole
- c) Oral cotrimoxazole
- d) Intravenous azithromycin

Question 13:

A 60-year-old diabetic woman presents to your OPD with complaints of curdy white vaginal discharge and intense itching. What is the likely causative agent for her condition?

- a) *Trichomonas vaginalis*
- b) *Mobiluncus*
- c) *Chlamydia trachomatis*
- d) *Candida albicans*

Question 14:

In an obese lady with complaints of vaginal discharge, a vaginal pH of 4.2 was detected. Which organism is implicated in this patient?

- a) Trichomonas
- b) Candida
- c) Gardnerella
- d) Mycoplasma

Question 15:

Which of the following is not a risk factor for developing a candidal infection?

- a) Oral contraceptive pills
- b) Obesity
- c) Pregnancy
- d) Hypertension

Question 16:

A 32-year old woman presents to you with complaints of vaginal discharge and itching. On examination, you note a curdy white discharge and an inflamed vagina. What is the best treatment for her?

- a) Metronidazole
- b) Fluconazole
- c) Doxycycline
- d) Clindamycin

Question 17:

Which of the following is not a primary site for gonococcal infection?

- a) Urethra
- b) Bartholin's glands
- c) Ectocervix
- d) Skene's gland

Question 18:

Which of the following presentations is unlikely to be seen in a patient with gonorrhoea?

- a) Intermenstrual bleeding
- b) Vaginal discharge
- c) Proctitis
- d) Haematuria

Question 19:

What is the most common cause of non-gonococcal urethritis?

- a) Chlamydia trachomatis
- b) Treponema pallidum
- c) Candida albicans
- d) Staphylococcus aureus

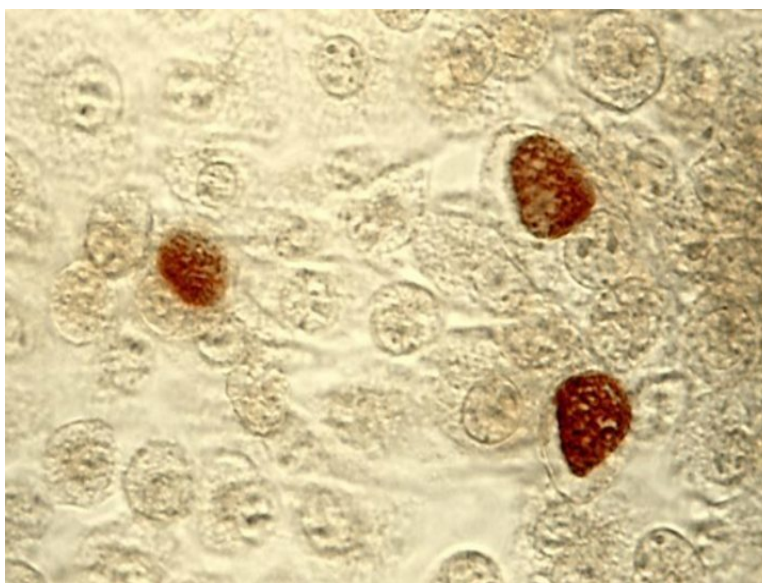
Question 20:

A 35-year-old sex worker was suspected to have chlamydial cervicitis. What is the most sensitive method for confirming the diagnosis?

- a) Direct fluorescent antibody test
- b) Enzyme immunoassay
- c) Polymerase chain reaction
- d) Culture on irradiated McCoy cells

Question 21:

A 25-year-old woman had complaints of lower abdominal pain and vaginal discharge. On microscopy, the following finding was present. What is the selective culture media used to isolate the causative organism?



- a) Culture on Thayer martin media
- b) Culture on McCoy cells
- c) Culture on bilayer human blood agar
- d) Culture on Vero cell lines

Question 22:

A patient with complaints of vaginal discharge was evaluated. On an endocervical biopsy, chlamydia was detected. How will you treat this patient?

- a) Azithromycin
- b) Azithromycin plus contact tracing
- c) Doxycycline and metronidazole
- d) Metronidazole

Question 23:

A 25-year-old lady complains of thick curdy white vaginal discharge with itching and burning. What is the route of administration of the first non-azole drug approved to treat this condition?

- a) Oral
- b) Intradermal
- c) Intravenous

d) Topical

Answer Key

Question No.	Correct Option
1	d
2	c
3	c
4	c
5	c
6	c
7	a
8	a
9	a
10	b
11	d
12	b
13	d
14	b
15	d
16	b
17	c
18	d
19	a
20	c
21	b
22	b
23	a

Detailed Explanations

Solution to Question 1:

Caucasian race is not a risk factor for bacterial vaginosis.

The risk factors for bacterial vaginosis are:

- Oral sex
- Vaginal douching
- African race
- Cigarette smoking
- Sex during menses
- Intrauterine device
- Early age of sexual intercourse
- New or multiple sexual partners
- Sexual activity with other women

Note: Bacterial vaginosis is not an infection. It most commonly affects young women. Although it is not a sexually transmitted disease, the chances of developing bacterial vaginosis seem to increase with the number of sexual partners that a woman has had.

Solution to Question 2:

Vaginal itching is not seen in bacterial vaginosis. It is a symptom of Trichomonas vaginitis.

About half of the women having bacterial vaginosis are asymptomatic. Symptomatic women mostly present with a profuse, fishy vaginal discharge. It does not cause pruritus, dysuria, and dyspareunia.

Option A: Bacterial vaginosis occurs due to the absence of lactobacilli. This causes an increase in the growth of other anaerobic bacteria namely Gardnerella vaginalis, Mobiluncus, and Mycoplasma hominis.

Option B: Amsel's criteria is used for diagnosing bacterial vaginosis. 3 out of 4 criteria are needed to make a positive diagnosis.

Option D: Oral metronidazole 500 mg twice daily for 7 days or oral clindamycin 300 mg twice daily for 7 days is given for the treatment of bacterial vaginosis. Treatment remains the same in pregnancy.

Solution to Question 3:

The given clinical scenario is suggestive of bacterial vaginosis (BV) caused by Gardnerella.

Patients with BV have profuse and yellowish vaginal discharge with a fishy odour. It is adherent to the vaginal mucosa. Dysuria, dyspareunia, and vaginal itching are absent. On microscopy, clue cells are visible.

Option A: In trichomonas vaginitis, the vaginal discharge is frothy, greenish-yellow and has an offensive smell. Dysuria, dyspareunia, and vaginal itching are generally present.

Option B: In candidal vaginitis, the discharge is thick and curdy white.

Option D: In chlamydial infection, the vaginal discharge is generally scanty.

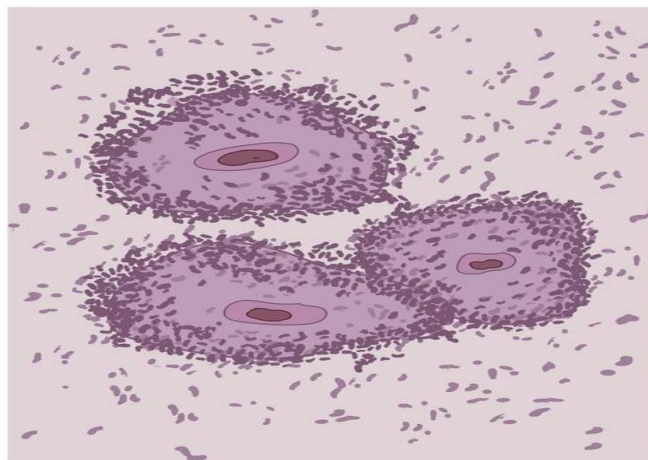
Solution to Question 4:

An abundance of polymorphs is not a microscopic finding of bacterial vaginosis because it is not an inflammatory condition. It occurs due to an alteration in the normal vaginal flora.

A swab of the discharge is taken from the vaginal wall and a smear is prepared. The microscopic evaluation is done using a gram stained smear. In bacterial vaginosis, this would reveal a clean background with an abundance of gram variable coccobacilli adhering to the epithelial cells. These are called clue cells. A few inflammatory cells, other pathogenic organisms, and a reduced number of Lactobacilli would also be seen.

Shown below is the Gram-stained smear of vaginal discharge of a patient diagnosed with bacterial vaginosis. It reveals vaginal epithelial cells that have a fuzzy border due to the adherence of bacteria. These are called clue cells.

Clue cells - Bacterial vaginosis



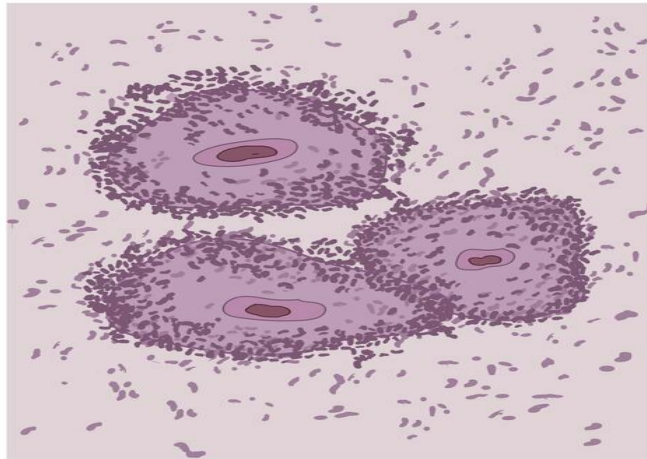
Solution to Question 5:

The presence of clue cells on microscopy is suggestive of bacterial vaginosis.

A swab of the vaginal discharge is taken from the vaginal wall and a smear is prepared. The microscopic evaluation of the smear is done on a gram stained smear. It would reveal a clean background with an abundance of gram variable coccobacilli adhering to the epithelial cells. These are called clue cells. A few inflammatory cells, other pathogenic organisms, and a reduced number of Lactobacilli would also be seen.

Shown below is the Gram-stained smear of vaginal discharge of a patient diagnosed with bacterial vaginosis. It reveals vaginal epithelial cells that have a fuzzy border due to the adherence of bacteria. These are called clue cells.

Clue cells - Bacterial vaginosis



Solution to Question 6:

The given clinical scenario is suggestive of bacterial vaginosis which is confirmed by using Amsel's criteria.

3 out of 4 of the following criteria are needed, to make a positive diagnosis of bacterial vaginosis:

- A homogeneous, thin, greyish-white discharge that smoothly coats the vaginal walls.
- A vaginal pH ≥ 4.5 .
- A positive whiff test - an amine like fishy odour is produced when a drop of 10% KOH is added to the sample of vaginal discharge.
- The presence of $\geq 20\%$ clue cells on saline wet mount. These are vaginal epithelial cells studded with adherent coccobacilli at the edge of the cell.

Option A: Boer-Miesel's criteria is used for the prognostic evaluation of pelvic inflammatory disease.

Option B: Light's criteria is used for determining the nature of the pleural fluid i.e., transudate or exudate, in the diagnosis of pleural effusion.

Option D: Nugent's criteria can also be used for diagnosing bacterial vaginosis. However, it is not the best confirmatory method.

Solution to Question 7:

Nugent's criteria is used to evaluate three types of bacteria in a case of suspected bacterial vaginosis: Mobiluncus, Lactobacillus, and Gardnerella/Bacteroides. Bacteria are graded through gram stain of vaginal samples.

They are each graded on a scale of 1+ to 4+ where:

- 1+ is $\leq$ 1 cell per field
- 2+ is 1-5 cells per field
- 3+ is 6-30 cells per field
- 4+ is $\geq$ 30 cells per field

Lactobacillus and Bacteroides/Gardnerella are given scores between 0-4, but Mobiluncus is graded only from 0-2. The total scores are then calculated and interpreted as follows:

- 0-3 - Normal
- 4-6 - Intermediate bacterial count
- 7-10 - Bacterial vaginosis

Solution to Question 8:

Trichomonas vaginalis is the likely organism implicated in this patient, causing trichomonas vaginitis.

The whiff test can be positive in both Trichomonas and Gardnerella infections. However, greenish frothy discharge is seen in trichomonas vaginitis.

Solution to Question 9:

In the given clinical scenario, Trichomonas vaginalis is the likely causative agent.

Vaginal itching and foul-smelling frothy vaginal discharge are seen in trichomonas vaginitis.

The given image shows colpitis macularis or strawberry cervix, which is a characteristic finding of trichomoniasis. This is due to microscopic, punctate haemorrhages of the cervix.



Solution to Question 10:

Trichomonas vaginalis can be detected on a hanging drop preparation. *Trichomonas vaginalis* is a protozoan with four anterior flagellae, and a fifth posterior flagellum responsible for motility.

The image given below shows trophozoites of *Trichomonas vaginalis* with flagella.

**Solution to Question 11:**

The given clinical scenario is suggestive of trichomonas vaginitis and the best treatment option for her is oral metronidazole 2 g stat.

The treatment regimen for trichomonas vaginitis is:

- An oral dose of metronidazole 500 mg twice daily for 7 days or
- A single oral dose of metronidazole 2 g

The male partner should be treated with a single dose of oral metronidazole in case of recurrent infection. Treatment in pregnancy remains the same.

Solution to Question 12:

Trichomonas vaginitis is treated with oral metronidazole in pregnant women.

According to the current guidelines, metronidazole is now considered safe for treating trichomonas vaginitis in pregnancy as it has no teratogenic effect.

The treatment regimen for trichomonas vaginitis in pregnant women include:

- An oral dose of metronidazole 500 mg twice daily for 7 days or a single oral dose of metronidazole 2 g
- Usage of trichofuran suppositories
- Vinegar douche to lower the vaginal pH
- Betadine gel

The male partner should be treated with a single dose of oral metronidazole in case of recurrent infection.

Solution to Question 13:

The given clinical scenario is suggestive of moniliasis and *Candida albicans* is the likely causative agent.

The risk factors for moniliasis are:

- Diabetes mellitus
- Obesity
- Multiple sexual partners
- Immunosuppression
- HIV
- Pregnancy
- Steroid therapy
- Following long-term broad-spectrum antibiotic therapy
- Oral contraceptive pills
- Poor personal hygiene.

Solution to Question 14:

The likely causative agent in the given clinical scenario is *Candida* which is known to flourish in an acidic environment. The pH of the vaginal discharge in candidal vaginitis is in the range of 4-4.5.

Solution to Question 15:

Hypertension does not increase the risk of developing a candidal infection.

The risk factors for developing candidal vaginitis (moniliasis) are:

- Diabetes mellitus

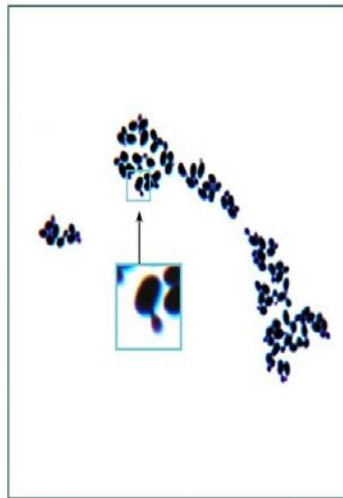
- Obesity
- Multiple sexual partners
- Immunosuppression
- HIV
- Pregnancy
- Steroid therapy
- Following long-term broad-spectrum antibiotic therapy
- Oral contraceptive pills
- Poor personal hygiene.

Solution to Question 16:

The given clinical scenario is suggestive of candidal vaginitis. A single dose of fluconazole 150 mg is the best treatment for treating candidal vaginitis. Intravaginal antifungal (imidazole/miconazole/clotrimazole) vaginal pessaries or creams can also be used for 3–6 days.



Curdy white discharge



Candida albicans

Solution to Question 17:

Ectocervix and vagina are lined by squamous epithelium which is resistant to gonococcal infection.

The principal sites of invasion for gonococcal infection are the columnar and transitional epithelium. Hence, gonococcus usually affects the following structures:

- Endocervix
- Urethra

- Skene's glands
- Bartholin's glands.

Solution to Question 18:

Haematuria is unlikely to be seen in a patient with gonorrhoea.

Gynaecological presentation of gonorrhoea is:

- Excessive vaginal discharge associated with pruritus and inflammation
- Acute unilateral pain and swelling over labia due to the involvement of Bartholin's gland
- Intermenstrual bleeding

Non-gynaecological presentation of gonorrhoea is:

- Pharyngeal infection
- Dysuria
- Rectal discomfort due to associated proctitis.

Solution to Question 19:

The most common cause of non-gonococcal urethritis is Chlamydia trachomatis.

The causes of non-gonococcal urethritis are:

- Chlamydia trachomatis (most common)
- Ureaplasma urealyticum
- Adenovirus
- Haemophilus vaginalis
- Mycoplasma genitalium
- Trichomonas vaginalis (rare)
- Herpes simplex virus (rare)

Solution to Question 20:

The most sensitive method for confirming the diagnosis of chlamydial cervicitis is the polymerase chain reaction (PCR). It is fast, highly sensitive and specific (96%).

Solution to Question 21:

Chlamydia trachomatis is the most likely causative organism. This is suggested by the presence of characteristic iodine-stained inclusion bodies in the McCoy cell culture of the patient's endocervical sample.

Chlamydia is a gram-negative coccobacillus. The gram stain of the cervical smear would show abundant leucocytes but no organisms.

The nucleic acid amplification test (NAAT) is currently done for the diagnosis of lymphogranuloma venereum.

Solution to Question 22:

A single dose of 1 g of azithromycin is given and contact tracing is done to treat a patient diagnosed with a chlamydial infection.

Treatment of chlamydial infection:

- Azithromycin 1 g single dose (or)
- Doxycycline 100 mg orally BID $\times$ 7 days (or)
- Ofloxacin 200 mg orally BID $\times$ 7 days (or)
- Erythromycin 500 mg orally BID $\times$ 7 days

Contact tracing should always be done. The sexual partner should also be treated with the same regimen.

Solution to Question 23:

The clinical history of thick curdy white vaginal discharge with itching and burning are suggestive of vulvovaginal candidiasis. The first and the only non-azole antifungal approved for the treatment of vulvovaginal candidiasis is Ibrexafungerp. The route of administration of this drug is oral.

Ibrexafungerp is a triterpenoid antifungal agent that acts by inhibiting glucan synthase. Glucan synthase is an enzyme involved in the formation of 1,3-beta-D-glucan which is an essential component of the fungal cell wall.

The recommended dosage is 600 mg orally as 2 divided doses 12 hrs apart. Contraindications include pregnancy and hypersensitivity to ibrexafungerp. The most frequent adverse reactions are diarrhea, nausea, abdominal pain, dizziness, and vomiting.

Vulval Infections

Question 1:

Which of the following is the causative organism of vulval warts?

- a) Human papilloma virus
- b) Herpes simplex virus
- c) Human immunodeficiency virus
- d) Cytomegalovirus

Question 2:

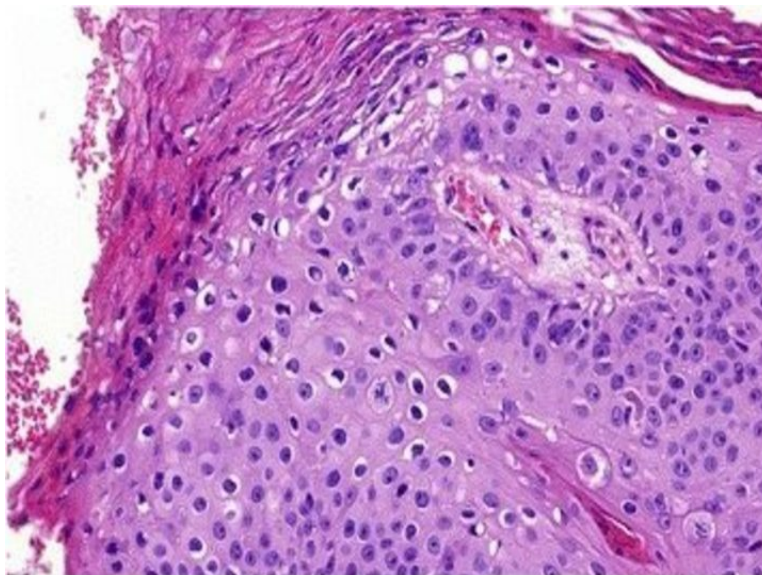
A young woman presented with the following lesion on her vulva. What is the most likely diagnosis?



- a) Condyloma acuminata
- b) Molluscum contagiosum
- c) Verruca vulgaris
- d) Condyloma lata

Question 3:

A sexually active woman presents with cauliflower-like lesions on the vulva. The biopsy of the lesion is taken and histopathology reveals the following picture. What is the most likely causative organism?



- a) Pox virus
- b) Human papilloma virus
- c) Herpes simplex virus type 1
- d) Herpes simplex virus type 2

Question 4:

Which of the following serotypes are present in the bivalent vaccine against the human papillomavirus?

- a) 6 and 11
- b) 16 and 18
- c) 11 and 16
- d) 6 and 18

Question 5:

Which of the following is not a treatment option for a patient with genital warts?

- a) Local application of 25% podophyllin
- b) Local application of imiquimod cream

- c) Intralesional interferon injection
- d) Local application of 5% trichloroacetic acid

Question 6:

A sexually active woman who presented with an ulcer over the vulva was diagnosed with primary syphilis. What is the characteristic feature associated with this condition?

- a) Non suppurative, tender, and enlarged nodes
- b) Firm, shotty, and painless nodes that do not suppurate
- c) Painful, tender, and matted nodes forming an abscess
- d) Painful, inflamed, necrotic nodes developing a chronic sinus

Question 7:

A 32-year-old woman known to have syphilis presented with the following lesions over the vulva. What is the stage of the disease?



- a) Primary
- b) Secondary
- c) Tertiary
- d) Latent

Question 8:

A sexually active patient was diagnosed with lymphogranuloma venereum. What is the causative organism for this condition?

- a) Klebsiella granulomatis
- b) Chlamydia trachomatis
- c) Treponema pallidum
- d) Herpes virus

Question 9:

A 26-year-old sexually active man comes to the STD clinic with a painful genital ulcer. On examination, the ulcer and the right inguinal lymph nodes were tender. What is the causative organism?

- a) Treponema pallidum
- b) Haemophilus ducreyi
- c) Chlamydia trachomatis
- d) Treponema pertenu

Question 10:

A sexually active man presents with complaints of painful swelling over the inguinal region. He also gives a history of a painless genital ulcer that resolved spontaneously. On examination, groove sign is seen. What is the likely diagnosis?

- a) Granuloma inguinale
- b) Lymphogranuloma venerum
- c) Syphilis
- d) Chancroid

Question 11:

A 20-year-old female sex worker presents with complaints of constipation and blood in stools for 3 months. On DRE, a circumferential constriction is felt beyond which the finger cannot be advanced. Which of the following is a probable cause for this complication?

- a) Granuloma inguinale
- b) Lymphogranuloma venerum

- c) Syphilis
- d) Chancroid

Question 12:

What is the causative agent for granuloma inguinale?

- a) Calymmatobacterium granulomatis
- b) Chlamydia trachomatis
- c) Ureaplasma urealyticum
- d) Treponema pallidum

Question 13:

A 19-year-old girl presented with multiple beefy-red ulcers over the labia. Microscopic examination of smear from the lesion showed intracytoplasmic inclusion bodies and gram-negative bacteria having a safety pin appearance. What is your likely diagnosis?

- a) Lymphogranuloma inguinale
- b) Granuloma inguinale
- c) Chancroid
- d) Syphilis

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	b
5	d
6	b
7	b
8	b
9	b

10	b
11	b
12	a
13	b

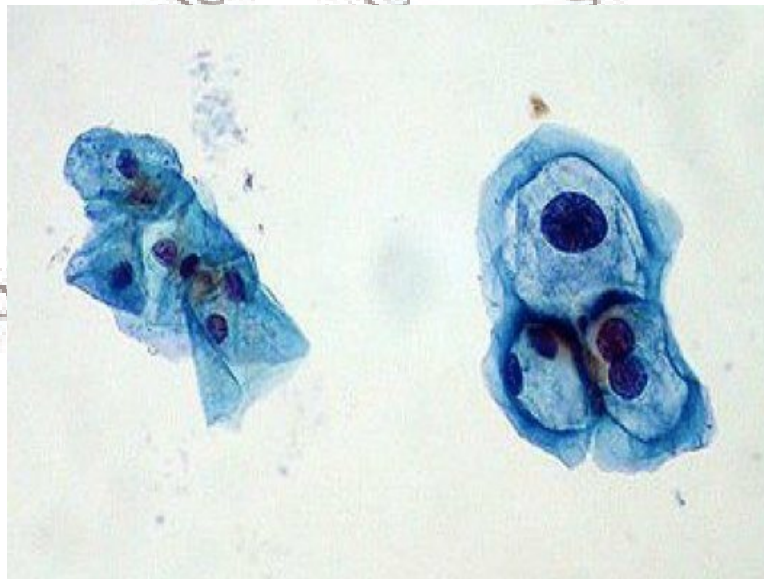
Detailed Explanations

Solution to Question 1:

The causative agent of vulval warts is the human papillomavirus (HPV).

Genital warts/venereal warts/condyloma acuminata are verrucous growths that affect the skin of the labia majora, perineum, perianal region and vulva. The lesions may be discrete or coalesce to form cauliflower-like growth. The disease is sexually transmitted. The growth of the warts is favoured by vaginal discharge, the use of oral contraceptives, and pregnancy.

HPV causes structural changes in the epithelial cells of the skin called koilocytes, and their presence is characteristic of HPV infection. The koilocytes have large hyperchromatic nuclei and a perinuclear halo, as shown below.



The diagnosis is made by colposcopic studies. The lesions on the vulva are stained by toluidine blue. The abnormal areas of the skin retain the blue dye. These areas are then biopsied for histopathological examination.

The given image below shows condyloma acuminata:



Solution to Question 2:

The given image is suggestive of condyloma acuminata that shows cauliflower-like masses of tissue on and around the female anogenital region.

Condyloma acuminatum is an epidermal manifestation of the human papillomavirus (HPV).

HPV is transmitted through contact with infected skin or mucosa. The virus invades the cells of the epidermal basal layer through microabrasions. However, anogenital HPV infection is almost always acquired through sexual contact.

Warts are not required for transmission but are highly infectious because of their high viral load.

Option B: The image given below shows shiny, pearly white, umbilicated papules with a characteristic dimple at the centre. These are seen in molluscum contagiosum caused by the pox virus.

Molluscum contagiosum



Option C: The image given below shows firm papules with a rough horny surface. These are seen in verruca vulgaris caused by the human papillomavirus type 2.



Option D: The image given below shows condyloma lata which is the characteristic feature of secondary syphilis. It is caused by *Treponema pallidum*.



Solution to Question 3:

The given image shows koilocytes, a characteristic feature of infection with human papillomavirus (HPV).

Other features that can also be seen on histology are:

- Perinuclear halo

- Multinucleation
- Orangeophilic cytoplasm
- Acanthosis
- Chronic inflammatory infiltrate

Option A: Henderson-Paterson bodies are seen in molluscum contagiosum, caused by poxvirus.

Option C and D: Cowdry bodies are seen in herpes infection.

Solution to Question 4:

Serotypes 16 and 18 are present in the bivalent vaccine against the human papillomavirus (HPV). It is called Cervarix. Cervarix is given at 0, 1 and 6 months.

Gardasil/Silgard is the quadrivalent vaccine against HPV. It has serotypes 6, 11, 16 and 18 and is given at 0, 2 and 6 months.

Gardasil 9 is the nonavalent vaccine against HPV. It has serotypes 6, 11, 16, 18, 31, 33, 45, 52 and 58. In 2020 it has received an expanded indication which now includes the prevention of oropharyngeal and head and neck cancers.

Solution to Question 5:

Genital warts are treated by a local application of 25% trichloroacetic acid + 5% fluorouracil.

Treatment of genital warts includes the local application of:

- Podophyllin 25% in alcohol/podophyllin 20% in tincture benzoin daily for 4 days (or)
- 25% trichloroacetic acid + 5% fluorouracil daily for 4 days

If this fails, the following treatment is given:

- Interferon ointment/cream or intralesional interferon injection
- Imiquimod cream thrice a week for 4 months

If medical treatment fails then cryosurgery, diathermy or laser removal is done.

Solution to Question 6:

The characteristic feature associated with primary syphilis is firm, shotty, painless nodes that do not suppurate.

Syphilis is a sexually transmitted infection caused by the motile spirochete, *Treponema pallidum*. In primary syphilis, the classical lesion i.e., chancre appears within 9–90 days from the first exposure.

The macular lesion becomes papular and then ulcerates. The ulcer(s) is painless and firm, with a punched out base and rolled edges. If left unattended, these heal within 3–9 weeks. It is accompanied by painless inguinal discrete lymphadenopathy.

Solution to Question 7:

The given image shows condyloma lata, a classic finding of secondary syphilis.

Multiple moist well-demarcated papules and plaques are seen in condyloma lata. They are highly contagious, exophytic broad excrescences that ulcerate and are commonly seen on the vulva, perianal area and upper thighs.

Solution to Question 8:

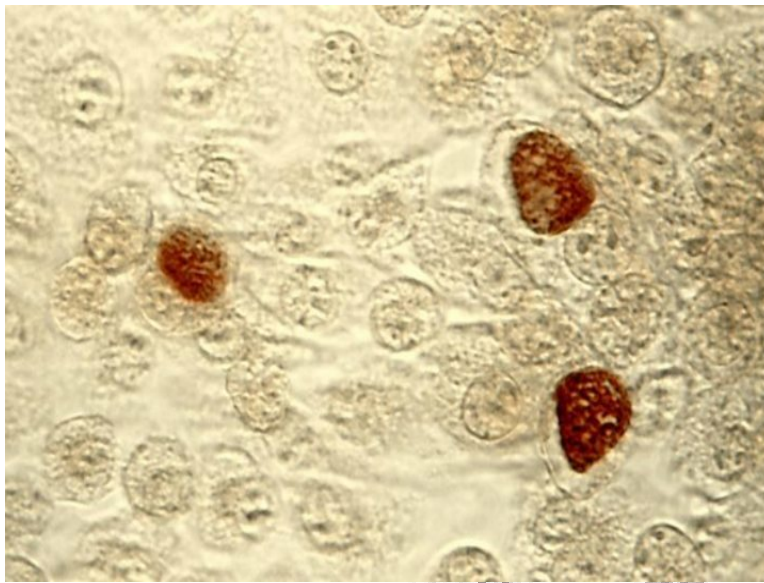
The causative agent for lymphogranuloma venereum is *Chlamydia trachomatis* (serotypes L1/L2/L3). Lymphogranuloma venereum (LGV) is a sexually transmitted disease that causes genital ulcers.

It has the following three stages of infection:

Chlamydia trachomatis is a small gram-negative bacterium and an obligate intracellular parasite that appears as an intracytoplasmic inclusion body. The gram stain of the cervical smear would show leucocytes but no organisms. Hence, selective culture media like the McCoy cells culture is used to isolate the organism.

The slide given below is an iodine-stained smear prepared from the patient's endocervical sample. It shows inclusion bodies characteristic of *Chlamydia trachomatis*.

Stage of LGV	Characteristic features
Primary stage	Painless genital ulcer Papules Lymphangitis
Secondary stage	Painful unilateral/bilateral lymphadenopathy Buboes Groove sign
Late stage	Stricture formation Fibrosis and fistulae of the anogenital region



Solution to Question 9:

The given clinical scenario is suggestive of chancroid caused by *Haemophilus ducreyi* which is a gram-negative anaerobe.

Chancroid is symptomatic in men. Both ulcer and lymphadenopathy are painful. Lymphadenopathy in chancroid is either unilateral or bilateral. Chancroid is generally asymptomatic in women but sometimes presents as painful lymphadenopathy.

The image below shows a chancroid ulcer with purulent and haemorrhagic discharge. It has a non-indurated, erythematous base.





Solution to Question 10:

The pathognomic groove sign of Greenblatt is suggestive of lymphogranuloma venereum (LGV) caused by *Chlamydia trachomatis* (serovars L1, L2, L3).

This image shows groove sign which is seen in ■ of the cases in the secondary stage of LGV. It refers to the depression between the enlarged femoral and inguinal lymph nodes on either side of the inguinal ligament.



Lymphogranuloma venereum (LGV) is a sexually transmitted disease that causes genital ulcers. It has the following three stages of infection:

Stage of LGV	Characteristic features
Primary stage	Painless genital ulcer Papules Lymphangitis
Secondary stage	Painful unilateral/bilateral lymphadenopathy Buboes Groove sign
Late stage	Stricture formation Fibrosis and fistulae of the anogenital region

Solution to Question 11:

The given clinical scenario is suggestive of rectal stricture which is a complication of lymphogranuloma venereum (LGV).

LGV is a sexually transmitted disease caused by *Chlamydia trachomatis* (serotypes L1/L2/L3). Rectal involvement is commonly seen in women as the primary disease often goes unnoticed and untreated. Anorectal involvement causes proctocolitis and causes rectal strictures. This occurs due to the contiguous spread of the primary infection from the perirectal nodes leading to scar tissue formation.

The complications of lymphogranuloma venereum include:

- Rectal stricture formation may lead to:
- Intestinal obstruction
- Rectovaginal fistulae
- Proctitis
- Vulvar cancer

Solution to Question 12:

The causative agent for granuloma inguinale (donovanosis) is *Calymmatobacterium granulomatis*. It is now called *Klebsiella granulomatis* and is a gram-negative bacillus causing chronic ulcerative infection of the vulva.

It is highly contagious and transmitted through repeated sexual or non-sexual contact. The incubation period is 1-12 weeks.

Clinical features:

It begins as a painless nodule which later ulcerates to form multiple beefy red painless ulcers that tend to coalesce; the vulva is progressively destroyed, and minimal adenopathy may occur.

Diagnosis:

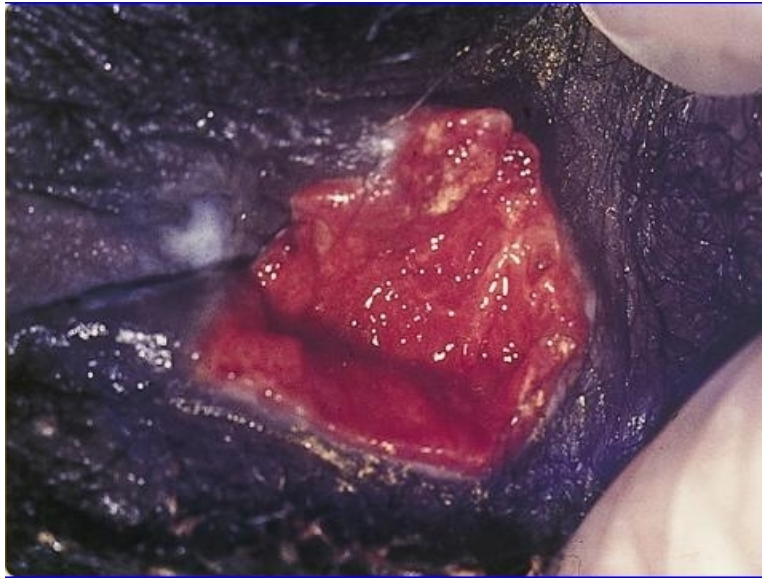
Microscopic examination of smears from the lesion/ biopsy specimens reveals pathognomonic intracytoplasmic Donovan bodies and clusters of bacteria with a bipolar (safety pin) appearance.

Blue-black stained organisms are seen in the cytoplasm of mononuclear cells.

Management:

Azithromycin 1 g orally once per week or 500 mg daily for at least 3 weeks and until all lesions have completely healed.

The given image below shows the donovanosis in a female patient with a beefy red ulcer:



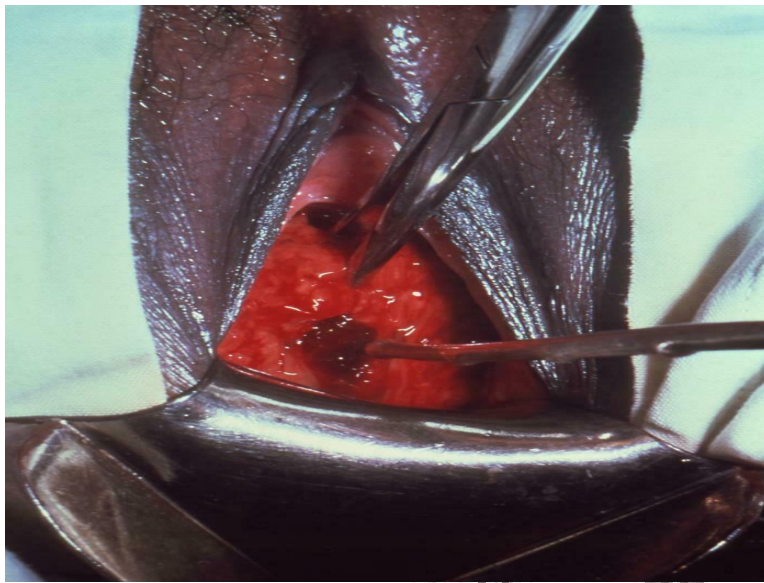
Solution to Question 13:

The given clinical scenario is suggestive of granuloma inguinale which is caused by a gram-negative bacteria, *Klebsiella granulomatis*.

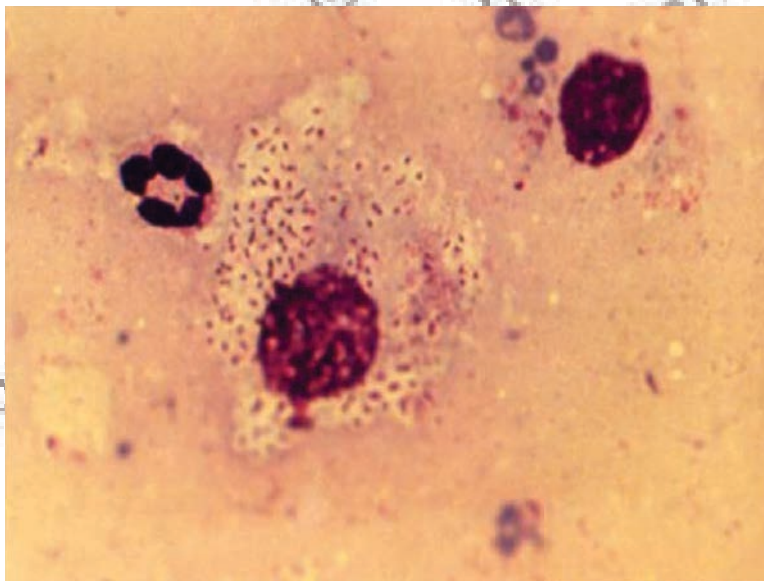
Granuloma inguinale begins as a painless nodule that later ulcerates to form multiple beefy red painless ulcers. These ulcers tend to coalesce and bleed easily on contact. If secondarily infected, they may become painful. The ulcers heal by fibrosis, which can result in scarring resembling keloids.

The lymph nodes are usually uninvolved but can become enlarged. New lesions can appear along the lymphatic drainage channels.

The gross image below shows granuloma inguinale of the cervix with a beefy red appearance of the hypervascular intravaginal tissues.



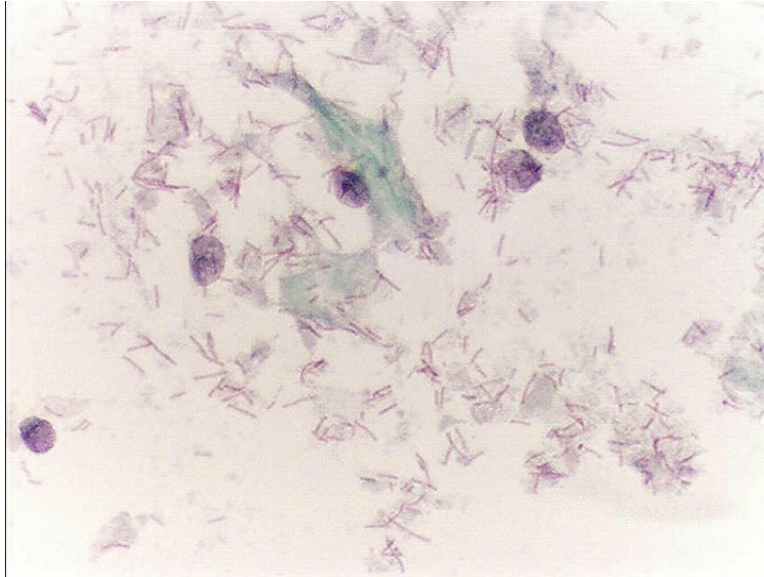
The microscopic image below shows the pathognomonic intracytoplasmic Donovan bodies. It also shows clusters of gram-negative bacteria with a bipolar/safety-pin appearance.



Pelvic Inflammatory Disease

Question 1:

The Pap smear of a healthy 30-year-old lady is given below. Identify the microorganism seen.



- a) Doderlein's bacillus
- b) Gardnerella vaginalis
- c) Trichomonas vaginalis
- d) Frisch bacillus

Question 2:

All of the following are risk factors for PID except_____.

- a) Unprotected sexual intercourse
- b) Pregnancy
- c) IUD insertion
- d) Early age of sexual intercourse

Question 3:

A 24-year-old woman is diagnosed with pelvic inflammatory disease. Which of the following is not included in this condition?

- a) Endometritis
- b) Oophoritis
- c) Cervicitis
- d) Peritonitis

Question 4:

What is the most common symptom seen in acute PID?

- a) Bilateral lower abdominal pain
- b) Unilateral lower abdominal pain
- c) High fever with vomiting
- d) Pain in the right upper quadrant of the abdomen

Question 5:

A young woman with fever and lower abdominal pain is diagnosed to have PID. Which is the most common cause for this condition?

- a) Chlamydia
- b) Mycoplasma
- c) Staphylococcus
- d) Neisseria

Question 6:

A 42-year-old woman presented to the emergency room with abdominal pain and fever for the last 48 hours. She has been using a Cu T 380A for contraception for the past 8 years. On examination, cervical motion tenderness was present. What is the most likely organism to have caused her condition?

- a) Actinomyces
- b) Chlamydia
- c) Gonococcus
- d) Staphylococcus

Question 7:

What is the mechanism of the spread of gonococci to the tubes in pelvic inflammatory disease?

- a) Hematogenous spread
- b) Lymphatic spread
- c) Parametrium spread
- d) Mucosal spread

Question 8:

Which infection causes more damage to the mucosa and the wall of the tube leading to fibrosis and tubal blockage?

- a) Chlamydial infection
- b) Gonococcal infection
- c) Mycoplasma infection
- d) Mobiluncus infection

Question 9:

Which of the following is not a differential diagnosis of acute PID?

- a) Septic abortion
- b) Cholecystitis
- c) Twisted ovarian cyst
- d) Pelvic endometriosis

Question 10:

A 26-year-old female with endometritis did not complete the course of antibiotics she was prescribed. Following this, her infection failed to resolve and she developed a tubo-ovarian mass. How can this mass manifest?

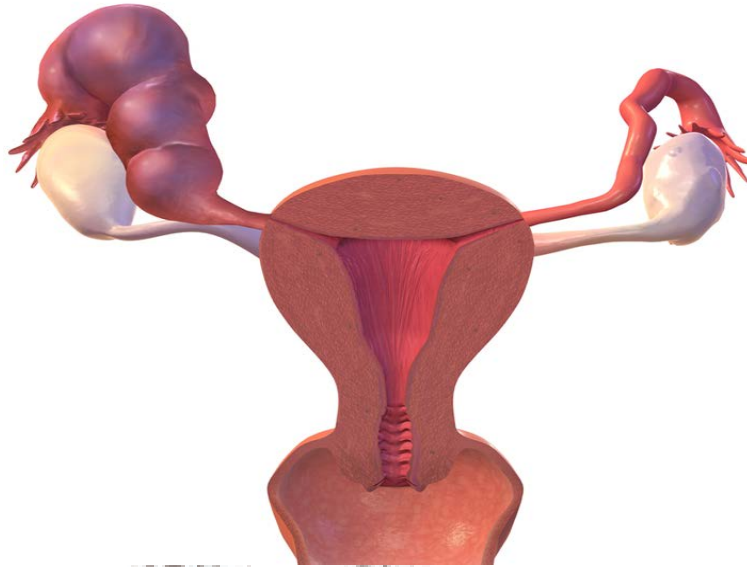
- a) 1, 3, 5
- b) 1, 3, 4

c) 2, 5

d) 1, 2, 4

Question 11:

Which of the following statements is incorrect regarding the condition seen in the image given below?



a) It is often bilateral

b) Its wall is opaque

c) Flattening of tubal plicae is seen on histology

d) Its ampullary region is filled with clear fluid

Question 12:

A patient with fever and abdominal pain presented to the emergency. On examination, you note mucopurulent vaginal discharge and cervical motion tenderness. You suspect she has PID. Which of these is not included in the criteria to diagnose this condition?

a) Oral temperature $\geq 38.3^{\circ}\text{C}$

b) Cervical motion tenderness

c) Mucopurulent vaginal discharge

d) Bilateral abdominal pain

Question 13:

Which of the following is not a definitive criterion used to diagnose PID?

- a) Histopathological evidence of endometritis
- b) Tubo-ovarian abscess on sonography
- c) Temperature $\geq 38.3^{\circ}\text{C}$
- d) Laparoscopic abnormalities consistent with PID

Question 14:

A 24-year-old woman presented with bilateral abdominal pain for 3 days and a temperature of 102°F . She also had a mucopurulent vaginal discharge, cervical motion tenderness, and a negative urine pregnancy test. Which is the gold standard investigation to confirm her likely diagnosis?

- a) Laparoscopy
- b) USG
- c) Culdocentesis
- d) Vaginal discharge microscopy

Question 15:

The laparoscopy report of a woman with pelvic inflammatory disease states she has Gainesville stage III PID. Which of the following presentations could have been seen in her?

- a) 3 and 5
- b) 3 and 4
- c) 1 and 2
- d) 1 and 4

Question 16:

In a patient with PID, laparoscopy shows the following appearance. What is true about the syndrome this patient likely has?



- a) Perihepatitis
- b) Cholecystitis
- c) Tuboovarian abscess
- d) Gonococcal urethritis

Question 17:

Identify the wrong statement with respect to silent PID.

- a) Dysmenorrhea is a common presentation
- b) Ectopic pregnancy is common
- c) Associated with infertility
- d) Mycobacterium can be grown from menstrual blood

Question 18:

A patient is admitted to the gynaecology ward for pelvic inflammatory disease. Which of the following is a part of this patient's treatment regimen?

- a) Administration of cefotetan 2 g IV 12 hourly + doxycycline 100 mg IV BD
- b) Removal of IUD irrespective of antibiotic response
- c) Administration of a single dose of ceftriaxone 250 mg IM
- d) Administration of 10 doses of Placentex 2ml IM daily or on alternate days

Question 19:

A 30-year-old female presents to the OPD with fever, lower abdominal pain and purulent vaginal discharge. On examination, her vitals are stable and per vaginal examination reveals lower abdominal and cervical motion tenderness. What is the best treatment option for her?

- a) Single IM injection of ceftriaxone 250 mg with oral doxycycline and metronidazole for 14 days
- b) Single IM injection of ceftriaxone 250 mg with oral metronidazole 400 mg for 7 days
- c) Oral doxycycline 100 mg BD with oral metronidazole 400 mg for 7 days
- d) Oral ofloxacin 200 mg BD with oral metronidazole 400 mg for 14 days

Question 20:

Which of the following would be an indication for surgery in a 30-year-old woman diagnosed with PID?

- a) A tubo-ovarian abscess of 2 cm
- b) Peritonitis
- c) Sepsis
- d) Intestinal obstruction

Question 21:

Which among the following is an indication for minimally invasive surgery for tubo-ovarian abscess?

- a) The size of the abscess is more than 5 cm
- b) The size of the abscess is more than 2cm
- c) The abscess fails to respond to antibiotics in 48 - 72 h
- d) The abscess fails to respond to antibiotics in 24 - 36 h

Question 22:

A 55-year-old multiparous woman who presented with chronic lower abdominal pain has been diagnosed with chronic PID. What is the treatment of choice for this patient?

- a) Radical hysterectomy
- b) Supracervical hysterectomy with tuboplasty

- c) Abdominal hysterectomy with bilateral salpingo-oophorectomy
- d) Supracervical hysterectomy with bilateral salpingectomy

Question 23:

Who has described the prognostic evaluation for PID?

- a) Boer–Meisel
- b) Gainesville
- c) Both A and B
- d) Westrom

Question 24:

Long-standing pelvic inflammation may lead to which of the following conditions?

- a) Pyometra
- b) Uterine polyposis
- c) Pseudocyesis
- d) Endometrial hyperplasia

Answer Key

Question No.	Correct Option
1	a
2	b
3	c
4	a
5	a
6	a
7	d
8	a
9	d
10	d
11	b

12	d
13	c
14	a
15	a
16	a
17	a
18	a
19	a
20	d
21	c
22	c
23	a
24	a

Detailed Explanations

Solution to Question 1:

The organism seen in the given image of a Pap smear is the Doderlien's bacillus, also known as Lactobacillus. This is a gram-positive, rod-shaped bacillus that grows in anaerobic conditions.

It is a part of the normal vaginal flora that is responsible for the vaginal defence mechanism.

It is an estrogen-dependent bacillus and appears at puberty and disappears after menopause.

It converts glycogen into lactic acid and maintains the acidic pH in the vagina. This prevents the growth of pathogenic bacteria in the vagina. Its absence causes a condition called bacterial vaginosis where pathogenic organisms replace the normal flora of the vagina. In this condition, clue cells are visible.

Option B: Gardnerella vaginalis is the most common organism involved in bacterial vaginosis. As the patient in the given question is healthy, this is not the correct option.

Option C: Trichomonas vaginalis is flagellated protozoa that causes trichomonas vaginitis. The organism and the nucleus of epithelial cells appear equal in size on the pap smear. This criterion is used for the identification of trichomonas as flagella are not visible on a pap smear.

Option D: Frisch bacillus is another name for Klebsiella rhinoscleromatis. It is not seen on Pap smears.

Solution to Question 2:

Pregnancy is not a risk factor for PID. Instead, it plays a protective role in this condition.

The risk factors for developing PID are:

- Unprotected sexual intercourse
- Having multiple sex partners
- Previous history of PID
- IUD insertion
- Early age of sexual intercourse (<18 years)

The protective factors for PID are:

- Pregnancy
- Using barrier contraception like condoms, diaphragms
- Oral contraceptive pills (OCPs)
- Menopause

Solution to Question 3:

Cervicitis is not included in pelvic inflammatory disease (PID).

Pelvic inflammatory disease is a spectrum of infection and inflammation of the upper genital tract, typically causing:

- Endometritis
- Salpingitis
- Oophoritis
- Peritonitis

The minimum criteria for diagnosis of PID are:

Pain in the lower abdomen + any one of the following:

- Uterine tenderness
- Cervical motion tenderness
- Adnexal tenderness

Cervicitis and vaginitis are lower genital tract infections.

Solution to Question 4:

The most common symptom of acute PID is bilateral lower abdominal pain.

The pain is severe in acute stages and it may or may not be accompanied by a high temperature (>38.3 degrees C). It may be followed by vomiting.

Some patients with PID can present with pain in the right upper quadrant of the abdomen. This is seen in Fitz-Hugh-Curtis syndrome which is a form of perihepatitis.

If PID occurs due to sexually transmitted organisms, dysuria and vaginal discharge may be seen.

Solution to Question 5:

Chlamydia trachomatis is the most common cause of PID.

PID is an infection of the upper genital tract. It is generally transmitted sexually as an ascending infection along with sperms. It is usually a polymicrobial infection,

Organisms causing PID include:

Sexually transmitted:

- Chlamydia (most common)
- Gonococcus
- Mycoplasma
- Trichomonas

Pyogenic Aerobes:

- Staphylococci
- Streptococci
- E Coli

Anaerobes:

- Bacteroides fragilis
- Peptostreptococcus
- Clostridium
- Actinomyces (most common in IUD users)

Other:

- Mycobacterium tuberculosis

Note: Genital tuberculosis is the only PID that does not spread sexually. It has a hematogenous spread. Thus, it is the most common PID in virgin girls.

Solution to Question 6:

The given clinical scenario points towards a diagnosis of PID in an IUCD user. The most common organism causing this condition is Actinomyces. It has been reported in 7% of IUCD users if the device is worn for more than 2 years.

This emphasizes the need for strict asepsis during insertion of the device and the need for strict follow up of women wearing these devices.

Organisms causing PID are:

Sexually transmitted:

- Chlamydia (most common)
- Gonococcus
- Mycoplasma
- Trichomonas

Pyogenic Aerobes:

- Staphylococci
- Streptococci
- E Coli

Anaerobes:

- Bacteroides fragilis
- Peptostreptococcus
- Clostridium
- Actinomyces (most common in IUD users)

Other:

- Mycobacterium tuberculosis

Solution to Question 7:

The mechanism of the spread of gonococci to the tubes in pelvic inflammatory disease is mucosal. Gonococci ascend to the tubes mainly through mucosal continuity. This is facilitated by sexually transmitted vectors such as sperms and Trichomonas.

The mucous membranes of the fallopian tubes are affected first. They become edematous and are infiltrated with leukocytes and plasma cells. Another route of spread is through the reflux of mucosal blood.

Option A: Hematogenous spread is seen in genital tuberculosis.

Option B: Lymphatics are involved in the spread of secondary organisms and in PID following post-abortal and puerperal infection.

Option C: Parametrium facilitates the spread of Mycoplasma. The gut flora can directly spread to tubes.

Solution to Question 8:

Among the given options, chlamydial infection causes the most amount of damage to the mucosa and the wall of the fallopian tubes leading to fibrosis and tubal blockage.

Chlamydial infection can remain asymptomatic in the endo-cervix. Hence, the infection goes unnoticed and untreated. Thus, it can lead to more damage and complications.

Gonococcal infection is more symptomatic than chlamydial infection. Hence, it is diagnosed early and causes less damage to the tubes than chlamydial infection.

Mycoplasma and Mobiluncus generally cause lower genital tract infections like cervicitis and vaginitis.

Solution to Question 9:

Pelvic endometriosis is not a differential diagnosis of acute PID. It is a differential diagnosis of chronic PID.

Differential diagnosis of acute PID:

- Acute appendicitis
- Septic abortion
- Cholecystitis
- Twisted ovarian cyst
- Ectopic pregnancy
- Diverticulitis
- Ruptured endometriotic cyst

Differential diagnosis of chronic PID:

- Uterine fibroids
- Pelvic endometriosis
- Ovarian tumour
- Tubercular tubo-ovarian mass

Solution to Question 10:

The given scenario of development of a chronic tubo-ovarian mass following acute PID points towards a diagnosis of chronic PID. A chronic tubo-ovarian mass can manifest as a tubo-ovarian cyst, hydrosalpinx or even a chronic pyosalpinx.

Failure to complete treatment of acute PID or drug resistance can lead to the development of chronic PID. This can lead to the development of chronic tubo-ovarian masses which may manifest as:

- Hydrosalpinx
- Chronic pyosalpinx
- Chronic interstitial salpingitis

- Tubo-ovarian cyst
- Tuberculous form

A twisted ovarian cyst and a ruptured endometriotic cyst are differential diagnoses of acute PID.

Solution to Question 11:

The above image depicts a hydrosalpinx. Its wall is not opaque but translucent.

If a pyosalpinx responds to antibiotics, the pus present inside becomes sterile within 6 weeks of the initial attack. However, the damage to the tube remains as chronic pyosalpinx or hydrosalpinx. A hydrosalpinx represents the end result of previous acute salpingitis and is often bilateral.

The tube becomes retort shaped due to the enormous dilatation of the ampullary region that is filled with clear fluid.

Histological examination shows a flattening of the tubal plicae and exfoliation of the lining epithelium.

Solution to Question 12:

Bilateral abdominal pain is not included in the criteria to diagnose PID. Bilateral abdominal pain is a clinical symptom of PID.

Lower abdominal tenderness is included in the criteria to diagnose PID.

CDC criteria for the diagnosis of PID:

Definitive criteria:

- Histopathologic evidence of endometritis on biopsy
- Imaging study (TVS/MRI) evidence of thickened fluid-filled tubes (tubo-ovarian abscess)
- Laparoscopic evidence of PID

Minimum criteria:

- Cervical motion tenderness
- Uterine tenderness
- Adnexal tenderness

Additional criteria:

- Oral temperature ≥ 38.3 °C
- Mucopurulent cervical or vaginal discharge, cervical friability
- Raised ESR or CRP
- Laboratory documentation of positive cervical infection with Gonorrhea or Chlamydia

Solution to Question 13:

A temperature of more than 38.3°C is an additional criterion and not a definitive criterion for the diagnosis of PID.

CDC 2006 criteria for the diagnosis of PID:

Definitive criteria:

- Histopathologic evidence of endometritis on biopsy
- Imaging studies (TVS/MRI) evidence of thickened fluid-filled tubes (Tubo-ovarian abscess)
- Laparoscopic evidence of PID

Minimum criteria:

- Lower abdominal tenderness
- Adnexal tenderness
- Cervical motion tenderness

Additional criteria:

- Oral temperature > 38.3°C
- Mucopurulent cervical or vaginal discharge and cervical friability
- Raised ESR or CRP
- Laboratory documentation of positive cervical infection with Gonorrhea or Chlamydia

Solution to Question 14:

The given clinical scenario is likely to be acute PID. Laparoscopy is the gold standard to confirm acute PID.

Laparoscopy is not done routinely as it is an invasive procedure. It is reserved for only cases where the differential diagnosis is salpingitis, appendicitis, ectopic pregnancy or a case of a non-responding pelvic mass.

The diagnosis of PID is mostly clinical. If diagnosis is unclear, the investigation of choice for PID is USG and culdocentesis.

CDC 2006 criteria for the diagnosis of PID:

Definitive criteria:

- Histopathologic evidence of endometritis on biopsy
- Imaging studies (TVS/MRI) evidence of thickened fluid-filled tubes (tubo-ovarian abscess)
- Laparoscopic evidence of PID

Minimum criteria:

- Lower abdominal tenderness
- Adnexal tenderness
- Cervical motion tenderness

Additional criteria:

- Oral temperature $\geq 38.3^{\circ}\text{C}$
- Mucopurulent cervical or vaginal discharge and cervical friability
- Raised ESR or CRP
- Laboratory documentation of positive cervical infection with Gonorrhea or Chlamydia

Solution to Question 15:

Gainesville stage III includes two forms of PID - acute salpingitis with superimposed tubal occlusion or tubo-ovarian complex.

Gainesville has described the stages of PID depending upon the severity of tubal damage as follows:

Stage	Condition
Stage I	Acute salpingitis without peritonitis - no adhesions
Stage I	Acute salpingitis with peritonitis - purulent discharge
Stage I	Acute salpingitis with superimposed tubal occlusion or tubo-ovarian complex
Stage I	Ruptured tubo-ovarian abscess
Stage V	Tubercular salpingitis

Solution to Question 16:

The image shows the laparoscopic finding of violin string adhesions that are characteristic of Fitz-Hugh-Curtis syndrome. It is a form of perihepatitis that occurs in the setting of PID.

It is characterized by right upper quadrant pain with a pleuritic component, sometimes referred to the right shoulder. It rarely occurs in women with acute PID. The pathological mechanism is as follows:

Gonococcal / Chlamydial infection $\rightarrow$ transperitoneal/vascular spread $\rightarrow$ inflammation of the liver capsule and peritoneal surfaces $\rightarrow$ adhesions and fibrosis in the perihepatic region $\rightarrow$ Fitz-Hugh-Curtis Syndrome

Cholecystitis and ruptured tubo-ovarian abscess are the differential diagnoses of Fitz-Hugh-Curtis syndrome. Violin string adhesions would not be seen in these conditions.

Solution to Question 17:

Dysmenorrhea rarely occurs in PID. Pain is uncommon and is a result of subacute PID.

Silent PID is thought to follow multiple or continuous low-grade infections in asymptomatic women. It is not a clinical diagnosis. Rather, it is an ultimate diagnosis given to women with tubal-factor infertility who lack a history of upper genital tract infection.

On laparoscopy or laparotomy, these women may have evidence of prior tubal infection such as adhesions, but for the most part, the fallopian tubes are grossly normal. They can have ectopic pregnancies due to tubal dysmotility. Silent PID can occur following infection with M. tuberculosis.

Solution to Question 18:

Administration of cefotetan 2 g IV 12hourly + doxycycline 100 mg BD orally/ IV for 48 hours forms the crux of the treatment regimen for PID. This is followed by oral doxycycline 100 mg +/-metronidazole 400 mg for 14 days in all hospitalized patients having PID.

Treatment of PID in hospitalized patients:

- Cefotetan 2 g IV 12hourly + doxycycline 100 mg BD orally/ IV for 48 hours is given followed by oral doxycycline 100 mg +/- metronidazole 400 mg for 14 days
- Cefoxitin 2 g IV 6hourly + doxycycline 100 mg IV followed by oral route.
- Azithromycin 500 mg IV 6 hourly for 2 days, then orally for Chlamydia.

Outpatient treatment of PID (according to the current RCOG guidelines):

- A single dose of IM ceftriaxone 250 mg followed by oral doxycycline 100 mg BD +/-metronidazole 400 mg BD x 14 days
- A single dose of IM ceftriaxone 250 mg followed by azithromycin 1 gm per week x 2 weeks.

IUD is not removed in a woman who responds to antibiotics. It is removed only in case of a failed antibiotic response. Barrier contraceptives and OCPs are recommended only in women unresponsive to antibiotics.

The use of Placentrex (aqueous extract of the placenta) is outdated and is not recommended currently.

Solution to Question 19:

The given clinical scenario of fever and lower abdominal pain associated with lower abdominal and cervical motion tenderness is suggestive of PID. The best treatment option for this patient is

outpatient treatment with a single dose of IM ceftriaxone 250 mg + oral doxycycline 100 mg BD +/- metronidazole 400 mg BD x 14 days.

Outpatient treatment of PID (according to the current RCOG guidelines):

- A single dose of IM ceftriaxone 250 mg + oral doxycycline 100 mg BD +/- metronidazole 400 mg BD x 14 days
- A single dose of IM ceftriaxone 250 mg followed by azithromycin 1 gm per week x 2 weeks.

Treatment of PID in hospitalized patients:

- Cefotetan 2 g IV 12hourly + doxycycline 100 mg BD orally/ IV for 48 hours is given followed by oral doxycycline 100 mg +/- metronidazole 400 mg for 14 days
- Cefoxitin 2 g IV 6hourly + doxycycline 100 mg IV followed by oral route.
- Azithromycin 500 mg IV 6 hourly for 2 days, then orally for Chlamydia.

Solution to Question 20:

Intestinal obstruction would be an indication for surgery in a patient diagnosed with PID.

Criteria for surgery in PID include:

- Drainage of pelvic abscess by colpotomy.
- Dilatation and evacuation of septic products of conception or for haemorrhage in the postabortal period.
- Acute spreading peritonitis resistant to a full course of chemotherapy. Presence of pyoperitoneum mandates laparotomy.
- Intestinal obstruction
- Suspected intestinal injury as diagnosed in criminal abortion.
- Ruptured tubo-ovarian abscess

A tubo-ovarian abscess of 2 cm is not an indication for surgery. Minimally invasive surgery is done in a tubo-ovarian abscess measuring more than 10 cm, that fails to respond to antibiotics in 48-72 hours.

Peritonitis and sepsis are not an indication for surgery. They should be managed medically first. If they do not respond to antibiotic treatment then surgical management should be done.

Solution to Question 21:

If the abscess fails to respond to antibiotics in 48–72 h, minimally invasive surgery is indicated.

Criteria for minimally invasive surgery in PID:

- The size of the abscess is more than 10 cm
- The abscess fails to respond to antibiotics in 48-72 hours

- The abscess ruptures
- Pyoperitoneum

Minimal invasive surgery includes laparoscopic drainage of the tubo ovarian abscess or posterior colpotomy.

Solution to Question 22:

The treatment of choice for an old multiparous woman with chronic PID is abdominal hysterectomy with bilateral salpingo-oophorectomy.

A radical hysterectomy involves the removal of the whole uterus, tissue from the sides of the uterus, the cervix, and the upper part of the vagina. It is generally done only in cancer patients.

A supracervical hysterectomy involves the removal of only the body of the uterus and is not done in chronic PID. It is done for the treatment of endometriosis, uterine fibroids, and uterine prolapse.

Solution to Question 23:

Boer-Meisel has described the prognostic evaluation for PID. It depends upon the following:

- Extent of adhesions
- Nature of adhesions such as flimsy or dense.
- Size of hydrosalpinx
- Macroscopic condition of hydrosalpinx.
- The thickness of the tubal wall.

Boer-Meisel system is specifically used for predicting pregnancy outcomes in patients treated for PID.

Gainesville has described the stages of PID depending upon the severity of tubal damage.

Westrom has reported that women with one previous attack of PID have a predisposition to another attack.

Stages of PID	
Stage I	Acute salpingitis without peritonitis - no adhesions
Stage I I	Acute salpingitis with peritonitis- purulent discharge
Stage I II	Acute salpingitis with superimposed tubal occlusion or tubo-ovarian complex

Stages of PID	
Stage I V	Ruptured tubo-ovarian abscess
Stage V	Tuberculous salpingitis

Solution to Question 24:

Long-standing pelvic inflammation may lead to pyometra. This refers to the collection of pus in the uterine cavity.

The initial infection involves the cervix uteri and urethra which may extend to Bartholin glands, skene glands, endometrium, and fallopian tubes. Unilateral pain and swelling over the labia may be caused by the inflammation of the Bartholin glands.

Depending on the extent of tissue involved, PID can be classified into five stages

- Stage I - Acute salpingitis without peritonitis - No adhesions
- Stage II - Acute salpingitis with peritonitis - purulent discharge
- Stage III - Acute salpingitis with superimposed tubal occlusion or tubo ovarian complex
- Stage IV - Ruptured tubo-ovarian abscess
- Stage V - Tubercular salpingitis

Long-standing pelvic inflammation may also lead to:

- Dyspareunia
- Infertility
- Ectopic pregnancy
- Formation of adhesions like hydrosalpinx, pyosalpinx, tubo-ovarian abscess
- Chronic pelvic pain

Note: Chronic pelvic inflammation leads to dense adhesions between the uterus and the appendages. Thus, it becomes difficult to define the uterus from the pelvic masses. They form a hard mass which is called the frozen pelvis.

The most common cause of the spread of PID: Ascending infection along with sperms

The only PID that does not spread with the sexual route is genital TB. It has a hematogenous spread.

The most common PID in virgin females is genital TB.

Investigation of choice in PID: Abdominal USG

The surest sign of salpingitis is pus extruding from the fimbrial end of the tube.

Genital Tuberculosis

Question 1:

Which of the following is the most common site for genital tuberculosis?

- a) Infundibulum of fallopian tubes
- b) Uterine endometrium
- c) Ampulla of fallopian tubes
- d) Endocervical part of the cervix

Question 2:

What is the most common mode of transmission of tuberculosis to the endometrium?

- a) Hematogenous
- b) Lymphatic spread
- c) Retrograde spread
- d) Local spread

Question 3:

A young woman with fertility issues is diagnosed to have genital tuberculosis. Which of the following statements is false about this condition?

- a) The causative agent is *Mycobacterium bovis*
- b) It generally occurs secondary to a primary focus
- c) It mostly spreads hematogenously
- d) It commonly presents as a unilateral condition

Question 4:

What is the most common presenting symptom of genital tuberculosis in the reproductive age group?

- a) Asymptomatic
- b) Infertility

- c) Menstrual abnormalities
- d) Fever

Question 5:

In a patient with tubercular salpingitis, which of the following statements is false?

- a) The tubes are enlarged, roughened, and may show presence of greyish tubercles
- b) The ampulla of the tube appears dilated with the fimbrial end open and pouting
- c) The granulomas and hyperplasia seen can be mistaken for carcinoma
- d) The fimbriae are commonly involved in advanced tuberculous salpingitis

Question 6:

An 18-year-old girl presents with a history of low-grade fever, weight loss, abdominal pain and amenorrhoea for six months. On examination, a pelvic mass is felt on the left side and signs of ascites are positive. What is your diagnosis?

- a) Tuberculosis with tubo-ovarian mass
- b) Ovarian malignancy
- c) Ectopic pregnancy
- d) Submucous fibroid

Question 7:

A gynaecologist suspects tuberculous salpingitis in a young woman and suggests some investigations to confirm the diagnosis. Which of the following statements is true about these investigations?

- a) A differential WBC count shows lymphocytosis in advanced disease.
- b) The ESR is frequently raised and is non-specific in early disease.
- c) The Mantoux test is positive and indicative of current exposure to tubercle bacilli.
- d) The Quantiferon test is positive and indicates latent tuberculosis.

Question 8:

Hysterosalpingography was done in a young woman with infertility and menstrual abnormalities. Which of the following findings would not point towards a diagnosis of tubercular salpingitis?

- a) A wedge shaped defect at the uterine isthmus
- b) Beading and variations in filling defect
- c) Calcification of the tube
- d) Jagged fluffiness of the tubal outline

Question 9:

The image below shows the hysterosalpingogram of a 37-year-old nulliparous woman. Identify the false statement about this finding.



- a) It is a thick walled structure on sonography
- b) It can be asymptomatic
- c) Tubal plicae are flattened on histology
- d) Sequelae of pelvic inflammatory disease

Answer Key

Question No.	Correct Option
1	c

2	d
3	d
4	b
5	d
6	a
7	d
8	a
9	a

Detailed Explanations

Solution to Question 1:

The ampulla of the fallopian tubes is the most common site for genital tuberculosis.

The distribution of genital tuberculosis is given below:

Site	Percent age
Fallopian tubes Ampulla is the most common site. Intramural part is the least common site. Fimbriae are not involved even in advanced genital TB.	90-100%
Endometrium	50-60%
Ovaries	20-30%
Cervix	5-15%
Vulva and vagina	<1%

Solution to Question 2:

Endometrial tuberculosis is most commonly acquired by direct spread from the fallopian tubes. Hematogenous spread is thought to be the process by which the tubercle bacilli implant into the mucosa of the fallopian tubes and subsequently affect the endometrium by direct extension.

Female genital tuberculosis is an important cause of significant morbidity, short and long-term sequelae especially infertility. Genital tuberculosis usually occurs secondary to pulmonary (commonest) or extrapulmonary TB like gastrointestinal, renal, skeletal, and miliary TB. However, primary genital TB can rarely occur in women whose male partners have active genitourinary TB.

(e.g., tuberculosis epididymitis) by transmission through infected semen

Most common site involved in secondary genital TB is fallopian tube (90-100%) & endometrium (50-80%) & ovaries (20-30%) & cervix 5-15%) & vulva. The most common symptom of genital TB is infertility.

Causes of infertility in genital TB:

- Tubal blockage
- If tubes are patent, then also infertility occurs due to loss of tubal function.
- Tubercular endometritis causes uterine scarring resulting in the destruction of the endometrium.
- Ovarian damage with poor ovarian reserve and volume

A laparoscopy and dye hydrotubation (lap and dye test) is the most reliable tool to diagnose genital TB, especially for tubal, ovarian and peritoneal disease. The test can be combined with hysteroscopy.

Multiple drug therapy (ATT regimen provided by DOTS) in adequate doses and for sufficient duration is the mainstay in the treatment of TB including genital tuberculosis.

Solution to Question 3:

Genital tuberculosis is a bilateral condition.

It can be caused by both *Mycobacterium tuberculosis* (95% cases) and *Mycobacterium bovis* (5% cases). It almost always occurs secondary to a primary focus elsewhere in the body. The most common primary focus is lungs & lymph nodes & abdomen & generalised miliary infection.

It mostly spreads hematogenously. Lymphatic spread is also seen sometimes. It can rarely spread by direct contact with the affected intraabdominal organs or peritoneum.

The fallopian tubes are the first to be affected. The disease subsequently spreads to other genital organs and the peritoneum. This is followed by the involvement of bilateral pelvic lymph nodes. Vulvovaginal involvement is usually secondary to uterine involvement.

The most common presentation of genital tuberculosis in the reproductive age group is infertility.

Solution to Question 4:

The most common presentation of genital tuberculosis in the reproductive age group is infertility.

Genital tuberculosis mostly presents in young women in the reproductive age group. It can present as:

- Infertility is the only complaint in 35-60 % of cases. Of these, 75% of women present with primary infertility.
- Menstrual abnormalities are seen in 10-40% of cases.
- Asymptomatic in 10% of cases.

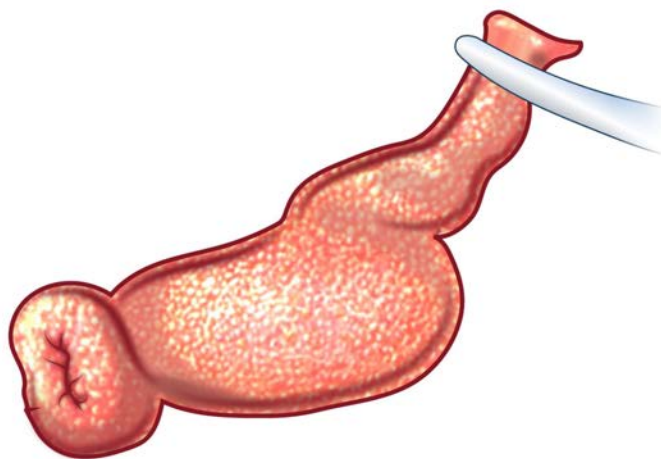
- Other symptoms like fever and malaise can be seen.

The treatment of infertility in women with genital TB depends on the condition of the endometrium. If the endometrium is receptive and the uterus is normal, then assisted reproductive techniques such as in-vitro fertilisation can be tried. If the endometrium is unfavourable then surrogacy may be considered.

Solution to Question 5:

The fimbriae are often spared even in advanced forms of tubercular salpingitis.

The involvement of fallopian tubes is 100% and bilateral, secondary to hematogenous spread. The ampullary part of the tube appears dilated with the fimbrial end open and pouting. This gives the characteristic tobacco-pouch appearance as seen in the image below.



The external surface of the tubes are enlarged, roughened, and may show the presence of greyish tubercles. The cut section shows yellowish-grey caseous matter in the lumen. Sometimes, serosanguinous fluid can also be seen in the lumen of the fallopian tube. This is called tuberculous hematosalpinx.

On histopathological examination of the tubes, granulomas and chronic inflammatory infiltrate with hyperplasia is seen. This could be mistaken for malignancy as it also produces a similar histological image.

Solution to Question 6:

The given clinical scenario with abdominal pain, amenorrhoea, pelvic mass and ascites is suggestive of genital tuberculosis.

Genital tuberculosis:

It is caused by M. tuberculosis and almost always occurs secondary to primary infection elsewhere in the extragenital states. The sites and frequency of involvement are as follows:

- Fallopian tubes (90-100%)
- Endometrium (50-60%)
- Ovaries (20-30%)
- Cervix (5-15%)
- Vulva and vagina (< 1%)

Presentation is variable and it can be asymptomatic in 10-20% of cases, infertility, menstrual irregularities, abdominal pain and mass, vaginal discharge, fistula formation, ectopic pregnancy and ascites. The mass is made up of rolled-up omentum that is adherent to the uterus and adnexa.

Constitutional symptoms such as fever, weakness, anorexia or night sweats may be present.

Investigations:

- CBC, ESR and Mantoux test
- Hysterosalpingography shows a lead-pipe appearance of the fallopian tube, calcification of the tube, beading and variation in the filling, cornual block, jagged fluffiness of tubal outline and tobacco pouch appearance of hydrosalpinx and pyosalpinx.
- Laparoscopy establishes the diagnosis of genital TB. It shows tuberculous lesions and perihepatic adhesions (if present).

Differential diagnosis:

- Ovarian cyst, broad ligament cyst or encysted fluid
- Pelvic inflammatory disease
- Ectopic pregnancy
- Carcinoma cervix
- Elephantiasis of the vulva
- Pregnancy
- Puberty menorrhagia and postmenopausal bleeding
- Fungal infection and sarcoidosis

Management:

It is treated with anti-tubercular therapy for 6 months (DOT strategy):

Indications and contraindications for surgery:

Types of surgery include vulvectomy (if vulval hypertrophy is present), removal of the adnexal mass, drainage of pyometra and fistula repair.

TAH+BSO is rarely required nowadays. Tuboplasty is contraindicated.

Option B: Ovarian tumors usually don't affect menstrual cycles. Though virilizing tumors can cause amenorrhoea, other features of virilization such as hirsutism, flattening of breasts, and enlargement of the clitoris are generally present.

Option C: Though ectopic pregnancy may be suspected here, the tube is likely to rupture with exacerbation of pain, bleeding, and peritonitis before six months of gestation. Additionally, the absence of vaginal bleeding/spotting, and the presence of weight loss makes the diagnosis unlikely.

Option D: Submucous fibroid is associated with menorrhagia.

First 2 months	Next 4 months
Isoniazid 15 mg/ kg body weight Rifampicin 450-600 mg Pyrazinamide 30 mg/body weight Ethambutol 30 mg/ kg body weight 3 times a week.	Isoniazid 15 mg/kg body weight Rifampicin 450-600 mg 3 times a week

Indications	Contraindications
Unresponsiveness of the disease in spite of adequate antitubercular chemotherapy. Tubercular pyosalpinx, ovarian abscess or pyometra. Persistent menorrhagia and/or chronic pelvic pain causing deteriorating health status	Presence of active tuberculosis in extragenital site Favorable response to antimicrobial therapy with the diminishing size of the pelvic mass The accidental discovery of tubercular tubo-ovarian mass on laparotomy in a young patient.

Solution to Question 7:

The Quantiferon test will be positive in patients having genital TB. It is an indicator of latent tuberculosis. The Quantiferon test measures the release of interferon-gamma by T cells following stimulation by antigens unique to Mycobacterium tuberculosis.

A differential WBC count shows lymphocytosis at all stages of genital tuberculosis. The ESR is frequently raised and is a non-specific investigation at any stage of the disease. None of these tests can be used for confirmation as these markers are raised in many infections.

The Mantoux test is positive in 90% of cases of genital TB. It is indicative of past exposure to the tubercle bacilli. It is not a confirmatory investigation but a negative result indicates tuberculosis to be unlikely.

Solution to Question 8:

A wedge-shaped defect is not suggestive of tubercular salpingitis. It is seen at the uterine isthmus in patients with a history of Cesarean section with a uterine scar.

Hysterosalpingography (HSG) features of genital tuberculosis are:

- Lead pipe appearance due to rigid nonperistaltic pipe-like tube

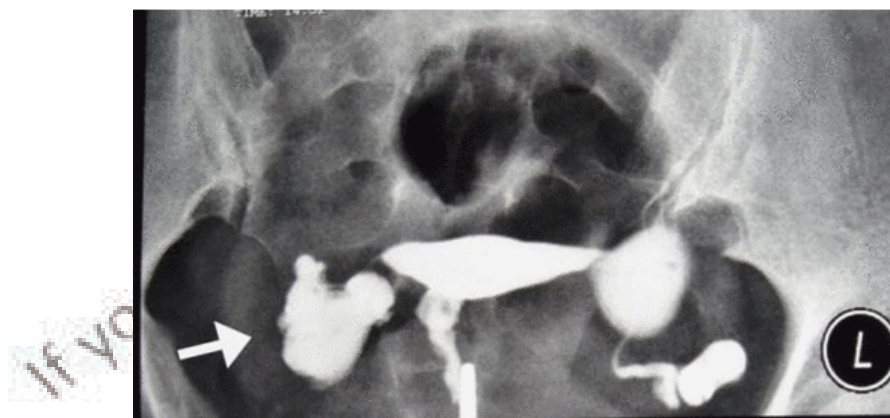
- Beading and variation in filling defect
- Calcification of the tube
- Jagged fluffiness of the tubal outline
- Tobacco-pouch appearance of hydrosalpinx and pyosalpinx
- Golf club appearance with slight or moderate dilatation of the ampullary portion of the fallopian tube.
- Rosette type appearance secondary to multiple small diverticular like outpouching surrounding the ampulla produced by caseous ulceration.

HSG is contraindicated in genital tuberculosis as it causes dissemination of the infection. This further increases the damage to the fallopian tubes.

The investigation of choice in genital tuberculosis is an endometrial biopsy. The tubercle bacilli are present in the deeper layers of the endometrium. They come to the surface just before menstruation. Hence, an endometrial biopsy is done 1 or 2 days before menstruation.

Menstrual blood can also be used for detecting mycobacteria.

An HSG showing the features of tuberculous salpingitis is given below.



©Marrow

Solution to Question 9:

The given image shows bilateral hydrosalpinx. It appears as a thin-walled structure.

A hydrosalpinx is a chronic cystic swelling of the fallopian tube that forms after a distal tubal obstruction. The tubes are enormously distended due to filling with clear serous fluid and become retort-shaped. On histology, the tubal plicae appear flattened. The interstitial end of the tube is open whereas the fimbrial end is indrawn and closed. This permits the dye to enter but not exit the tube during HSG.

It is caused due to:

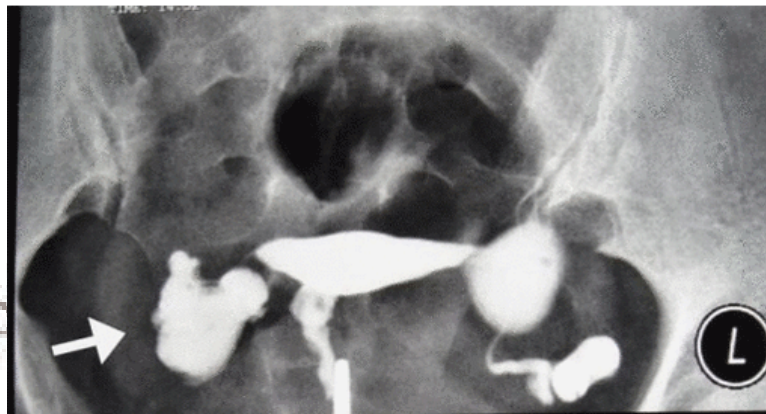
- Pelvic inflammatory disease
- Endometriosis
- Tuberculosis - tubercular salpingitis may also cause tobacco-pouch appearance, lead pipe appearance and pyosalpinx
- Fallopian tube neoplasms (rare)

It can be an incidental finding during a pelvic exam or ultrasound. The patient may be asymptomatic, or present with infertility or chronic pelvic pain.

On HSG, it appears as terminal sacculation and occlusion of both fallopian tubes with a normal uterine cavity. Usually, a thin-walled, hypoechoic, cystic structure is seen. However, it may appear as multiple hyperechoic mural nodules around the inner circumference of the tube creating a 'beads on a string' appearance.

Treatment:

- Treat the underlying cause
- Laparoscopic salpingostomy - in mild disease
- IVF recommended for women with infertility and severe hydrosalpinx



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Infertility

Question 1:

What is the accepted minimum period of sexual cohabitation required to establish the diagnosis of infertility in a couple?

- a) One year
- b) Two years
- c) Six months
- d) Nine months

Question 2:

What is the most common cause of male infertility?

- a) Disorders of spermatogenesis
- b) Obstruction of the efferent ducts
- c) Disorders of sperm motility
- d) Erectile dysfunction

Question 3:

The secretion of FSH in men is inhibited by the negative feedback effect of _____.

- a) Inhibin secreted by Sertoli cells
- b) Inhibin secreted by Leydig cells
- c) Testosterone secreted by Sertoli cells
- d) Androstenedione secreted by Leydig cells

Question 4:

A 25-year-old man underwent semen analysis as a part of an infertility workup. The results are given below. What is the most likely diagnosis?

- a) Oligospermia
- b) Normospermia

- c) Asthenospermia
- d) Teratospermia

Question 5:

While going through a patient's file at an infertility clinic, you note his semen analysis is reported to be normal. What had his semen leukocyte level likely been?

- a) 1.5 million/ml
- b) 1 million/ml
- c) 2 million/ml
- d) Traces

Question 6:

What does azoospermia refer to?

- a) No sperm in semen
- b) No semen
- c) Low sperm count
- d) Abnormal motility of sperm

Question 7:

A young couple being treated for infertility require donor semen for artificial insemination. Which of the following is the likely cause of infertility here?

- a) Oligospermia
- b) Azoospermia
- c) Impotence
- d) Presence of antisperm antibodies in the cervical mucus

Question 8:

Indications for artificial insemination with donor's semen are all except_____.

- a) Azoospermia

- b) Uncorrectable immunological factors
- c) Male with hereditary disease
- d) Idiopathic male infertility

Question 9:

A 50-year-old man with erectile dysfunction requires treatment with sildenafil. Which of the following is not a contraindication for the use of this drug?

- a) Retinitis pigmentosa
- b) Diabetic retinopathy
- c) Concurrent nitrate therapy
- d) Pulmonary hypertension

Question 10:

What is the most common cause of infertility in women?

- a) Ovarian causes
- b) Uterine causes
- c) Tubal causes
- d) Cervical causes

Question 11:

A 34-year-old woman presented with secondary infertility following radiation therapy for Hodgkin's lymphoma. You plan to test her ovarian reserve. Which of the following is the best indicator of this parameter?

- a) LH level
- b) LH/FSH ratio
- c) FSH level
- d) Oestradiol level

Question 12:

What will be the level of the anti-Mullerian hormone in low ovarian reserve?

- a) <1
- b) 1-4
- c) >7
- d) >10

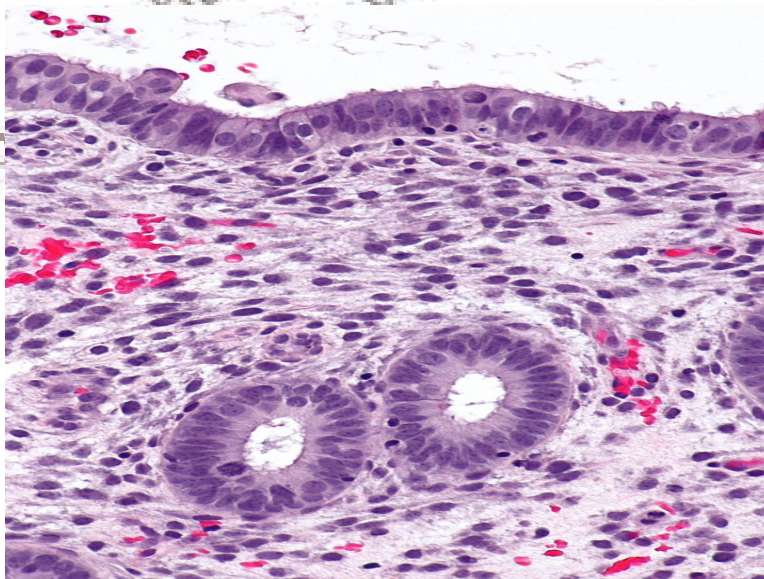
Question 13:

In a woman diagnosed with infertility, which of the following antihormonal substance can be used to induce ovulation?

- a) Mifepristone
- b) Clomiphene citrate
- c) Tamoxifen
- d) Danazol

Question 14:

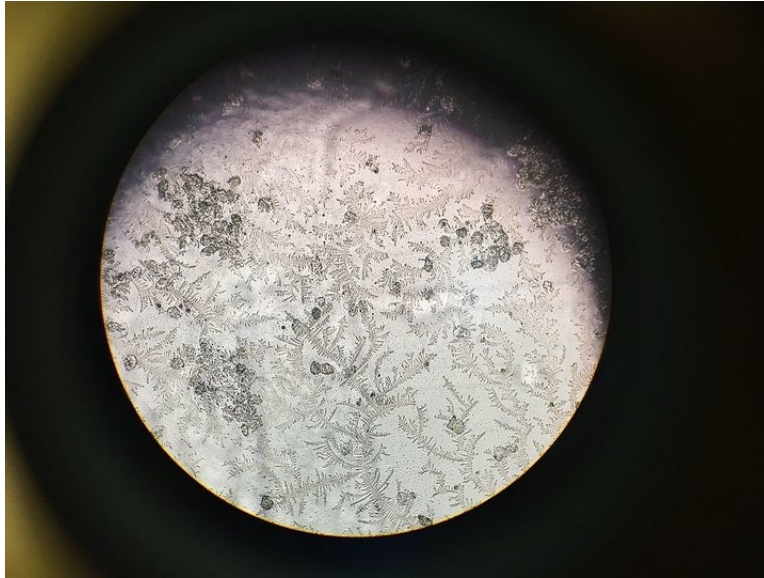
A woman with irregular periods was evaluated for infertility, and underwent an endometrial biopsy. The histology image is given below. Which phase is seen here?



- a) Secretory phase
- b) Proliferative phase
- c) Menstrual phase
- d) Ovulation phase

Question 15:

The microscopic appearance of the cervical mucus of a woman having irregular periods and infertility is shown in the image below. In which phase of the menstrual cycle was this sample likely taken?



- a) Estrogenic phase
- b) Progestogenic phase
- c) At the time of ovulation
- d) Secretory phase

Question 16:

Which of the following statements is false about the Miller-Kurzrok test?

- a) Penetration of ≥ 3 cm at 30 minutes is considered normal
- b) Cervical mucus is placed alongside husband's semen on a glass slide
- c) It is done at the time of ovulation
- d) Jerky motility of sperm is considered normal

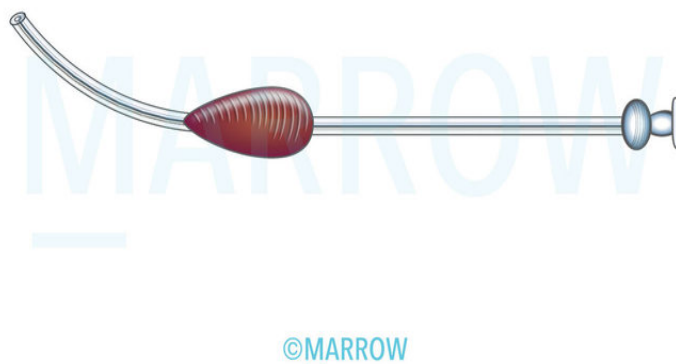
Question 17:

A post-coital test was done in a couple who presented with an inability to conceive despite staying together for 4 years and having unprotected intercourse. Which of the following conditions cannot be detected by this test?

- a) Fallopian tube block
- b) Cervical factor abnormality
- c) Abnormal sperm count
- d) Abnormal sperm morphology

Question 18:

During your posting in the infertility clinic, you notice the following instrument. Identify it.



- a) Karman cannula
- b) Spackman cannula
- c) Menstrual regulation syringe
- d) Rubin's cannula

Question 19:

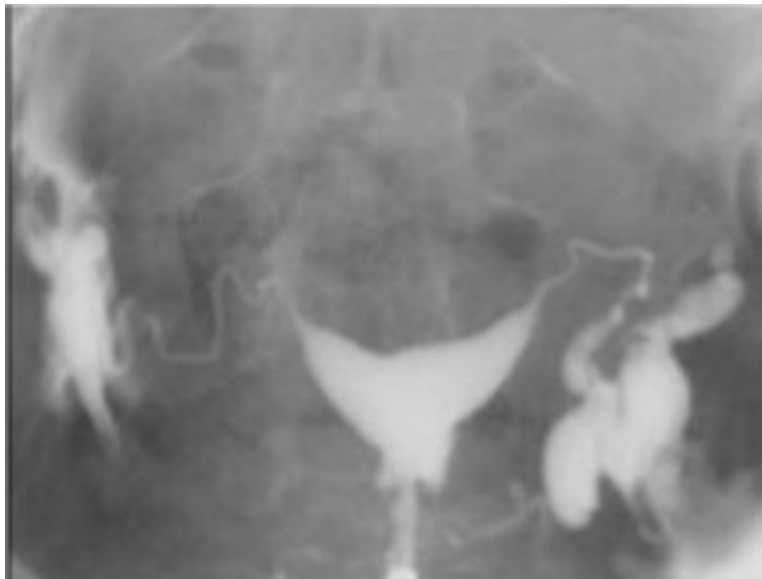
A patient with secondary amenorrhea and infertility is diagnosed to have Asherman syndrome. Hysteroscopic evaluation revealed 30% adhesions of the endometrial surfaces, but no agglutination of the uterine wall. How would you classify this patient?

- a) Mild Asherman syndrome

- b) Moderate Asherman syndrome
- c) Severe Asherman syndrome
- d) Cannot be classified with the given information

Question 20:

A 30-year-old patient with complaints of inability to conceive underwent the following investigation. Which of the following is false about the same?



- a) The best time to perform it is before ovulation
- b) It is performed to detect uterine anomalies
- c) It is the first-line investigation in genital tuberculosis
- d) Premedication can be done with atropine

Question 21:

A patient who has been trying to conceive for the past 6 months underwent hysterosalpingography (HSG). The findings are given below. What is your interpretation?



- a) Normal hysterosalpingogram
- b) HSG showing blocked tubes
- c) HSG showing a bicornuate uterus
- d) HSG showing a filling defect

Question 22:

Which the following are not the advantages of fertiloscopy over laparoscopy?

- a) No Trendelenburg position—only lithotomy position
- b) Trauma is avoided as no trocar is needed
- c) No CO₂—Hydroperitoneum with saline
- d) Good view of the anterior surface of the ovary and genital organs

Question 23:

A 26-year-old woman underwent a workup for primary infertility. On hysterosalpingography, there was a bilateral cornual block. What is the next step in management?

- a) Tuboplasty
- b) Hysteroscopy and laparoscopy-guided surgery
- c) Hydrotubation
- d) In-vitro fertilization

Question 24:

A woman came to the clinic complaining of amenorrhoea and inability to conceive. She had undergone hypophysectomy for a tumour a year ago. What would be the best line of management for her?

- a) Clomiphene citrate
- b) Letrozole
- c) GnRH analogue
- d) Human menopausal gonadotrophin, followed by hCG

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	b
5	b
6	a
7	b
8	d
9	d
10	a
11	c
12	a
13	b
14	b
15	a
16	d
17	a
18	d
19	b
20	c
21	a
22	d

23	b
24	d

Detailed Explanations

Solution to Question 1:

The accepted minimum period of sexual cohabitation to establish the diagnosis of infertility in a couple is one year of regular, unprotected sexual intercourse.

Infertility is an apparent failure of a couple to conceive while sterility is an absolute inability to conceive. Infertility is of two types:

- Primary infertility - when there have been no previous pregnancies.
- Secondary infertility - when there has been a prior pregnancy (although not necessarily a live birth).

Solution to Question 2:

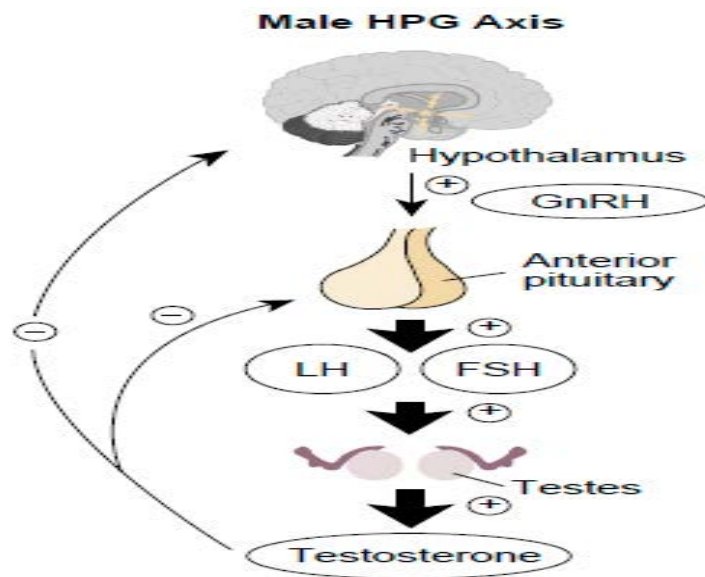
The most common cause of male infertility is disorders of spermatogenesis.

Male factor is the cause of infertility in about 20% of infertile couples. However, it is a contributing factor for infertility in 50% of cases.

Solution to Question 3:

The secretion of FSH in men is inhibited by the negative feedback effect of inhibin secreted by Sertoli cells.

GnRH stimulates the anterior pituitary gland to secrete FSH (follicle-stimulating hormone) and LH (luteinizing hormone). FSH acts on Sertoli cells while LH triggers testosterone secretion by the Leydig cells. Sertoli cells secrete inhibin B which in turn inhibits FSH but stimulate LH secretion.



Solution to Question 4:

In the given clinical scenario, the most likely diagnosis is normospermia.

Different types of disorders of sperms are as follows:

- Aspermia is defined as the absence of semen.
- Oligospermia is defined as a sperm concentration <15 million/mL.
- Severe oligospermia is defined as a sperm concentration of <5 million/mL.
- Teratospermia is defined as the presence of abnormal sperm morphology.
- Asthenospermia is defined as the absence of motile sperms or diminished sperm motility.
- Necrospermia is defined as the presence of dead sperms.

Note: Morphology $\geq$ motility is the most important marker in sperm analysis.

Solution to Question 5:

The normal leucocyte level in human semen is approximately 1 million/ml.

The semen sample should be collected after 2-7 days of ejaculatory abstinence. The patient should collect the sample by masturbation at the doctor's office. If not possible, then the sample may be collected at home and delivered to the laboratory within an hour of collection.

Solution to Question 6:

Azoospermia is defined as the complete absence of sperm in semen. Other types of sperm disorders are as follows:

- Aspermia is defined as the absence of semen.
- Oligospermia is defined as a sperm concentration <15 million/mL.
- Severe oligospermia is defined as a sperm concentration of <5 million/mL.
- Teratospermia is defined as the presence of abnormal sperm morphology.
- Asthenospermia is defined as the absence of motile sperms or diminished sperm motility.
- Necrospermia is defined as the presence of dead sperms.

Solution to Question 7:

Artificial insemination with donor semen is done in cases of infertility where the husband has azoospermia.

Azoospermia is defined as a complete absence of sperms. Hence artificial insemination with the husband's/partner's semen cannot be performed.

Indications for artificial insemination with husband's/partner's semen are:

- Chronic medical disorder
- Oligospermia after sperm washout
- Impotence
- Premature and retrograde ejaculation
- Hypospadias
- Antisperm antibodies in cervical mucus
- Unexplained infertility- direct intraperitoneal insemination (DIPI) is done
- HIV-positive male or female
- X-Y fractionation of sperms for sex selection in cases of genetic and chromosomal abnormalities

Indications for artificial insemination with donor semen are:

- Azoospermia
- Uncorrectable immunological factors
- Genetic disease in husband/partner
- Chronic ill health

Sperms can be frozen and used later for artificial insemination e.g. if the husband requires to undergo radiotherapy or chemotherapy, or if the husband is a frequent traveler and is not available at the time of ovulation.

Artificial insemination techniques include:

- Intrauterine Insemination (IUI)

- Intrafallopian-tube Insemination
- Intracervical Insemination
- Direct Intraperitoneal Insemination (DIPI)

Steps required to do artificial insemination are:

- The semen is washed to remove abnormal sperms, debris, and seminal plasma containing antibodies. It is concentrated, and its quality is improved by the 'swim up technique' or by the use of the Percoll gradient.
- Semen with normal sperms and good motility is obtained.
- Around the time of ovulation, semen obtained is inseminated into the female genital tract. About 0.5 ml of semen is injected 36 hours after hCG injection when the ovarian follicle reaches 20 mm.

Solution to Question 8:

Artificial insemination with donor's semen is not indicated in idiopathic male infertility. The preferred treatment in such cases is in vitro fertilization (IVF).

Solution to Question 9:

Pulmonary hypertension is an indication for the use of sildenafil.

The contraindications for the use of sildenafil are:

- Retinitis pigmentosa
- Diabetic retinopathy
- Patient on antihypertensive drugs, nitrates
- Cardiac disease, previous myocardial infarct, stroke

In the treatment of erectile dysfunction, sildenafil in the dose of 25–100 mg is given 1 hour before intercourse. However, it does not improve libido.

Solution to Question 10:

The most common cause for infertility in women is due to ovarian causes.

According to WHO, female factor infertility was reported in 37% of infertile couples in developed countries. Male factor infertility was reported in 8% of infertile couples while 35% of couples had both male and female factor infertility.

The major ovarian causes of infertility are:

- Anovulation (PCOS)
- Decreased ovarian reserve, e.g., premature menopause

In women with ovulatory disorders, clomifene and metformin increase ovulation and pregnancy rates.

Less common factors responsible for female infertility are:

- Endometriosis
- Pelvic adhesions
- Tubal blockage
- Other tubal abnormalities
- Hyperprolactinemia

Solution to Question 11:

Among the given options, the level of follicular stimulating hormone (FSH) is the best indicator of ovarian reserve.

Ovarian reserve is the capacity of the ovary to produce eggs. FSH levels are measured in the early follicular phase on day 2 or 3 of the menstrual cycle.

A high FSH level indicates that the ovarian reserve is getting depleted and fewer oocytes are present in the ovary. FSH is high in ovarian hypofunction and postmenopausal women.

Anti-Müllerian hormone (AMH) is a better marker than FSH for ovarian reserve. Unlike FSH, its levels can be estimated at any time of the cycle. AMH is produced by the granulosa cells in ovarian follicles.

Note: Studies show that AMH levels decrease under the current use of oral contraceptive pills and tobacco smoking.

Solution to Question 12:

In low ovarian reserve, the level of anti-Müllerian hormone (AMH) will be <1 ng/ml.

AMH is a glycoprotein secreted by the granulosa cells of the primordial follicle of the ovary. It helps in oocyte maturation and mono-follicular development. AMH is also used in assessing ovarian reserve.

In women, AMH is always absent during embryogenesis and is secreted by the granulosa cells after puberty. It helps in follicular development and oocyte maturation. Estimation of serum AMH is used in the study of ovarian reserve in female infertility and secondary amenorrhoea.

In men, AMH starts to be secreted by the Sertoli cells in the testes by the 7th week of intrauterine life. It continues until puberty. It inhibits the development of the Müllerian system. The absence of AMH results in the development of hermaphroditism.

AMH levels in ng/ml	Interpretation
2–6.8	Normal
<1	Poor ovarian reserve
>10	PCODHyperstimulation syndrome

Solution to Question 13:

Clomiphene citrate is an antihormonal substance used to induce ovulation in a patient diagnosed with infertility. Ovulation can be induced by giving clomiphene citrate 50 mg daily, starting from day 2 to day 6 of the menstrual cycle for 5 days.

Solution to Question 14:

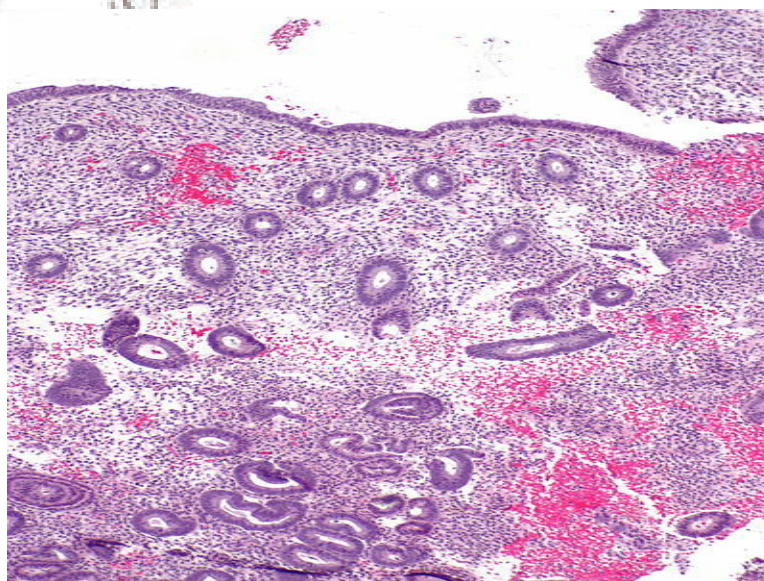
The image depicts the proliferative phase of the endometrium.

During menstruation, the endometrium goes through three phases, namely; proliferative, secretory, and menstrual phases. Their histological features are given below:

Proliferative phase

- The glands of the functional layer are simple tubules with regular epithelium.
- Arteries become more coiled and spiralled.
- The endometrium is thick, the stroma is compact and short straight glands are present.

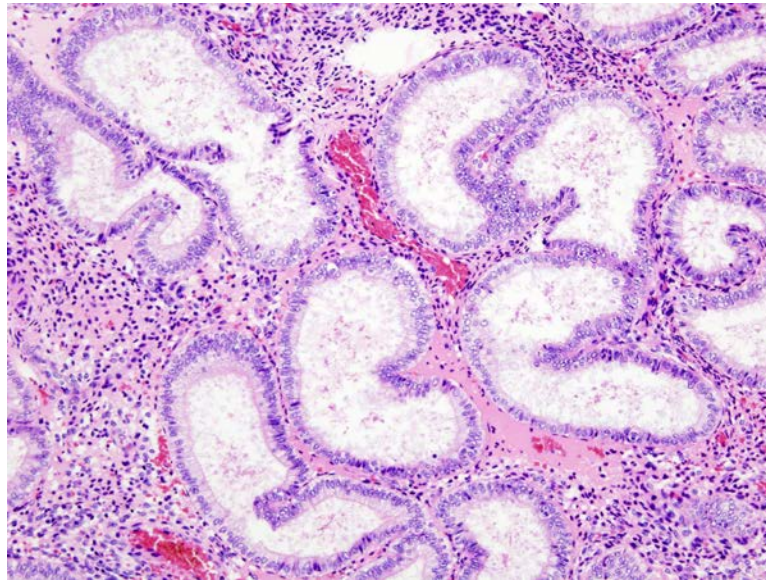
The image below shows proliferative endometrium



Secretory phase

- The secretory phase of the endometrium begins on the 15th day and persists until the onset of menstruation.
- Subnuclear vacuolation (presumptive evidence of progesterone activity)
- Corkscrew-shaped glands

The image below shows secretory endometrium



Menstrual phase: Glands shrink with prominent vessels and haemorrhagic spots.

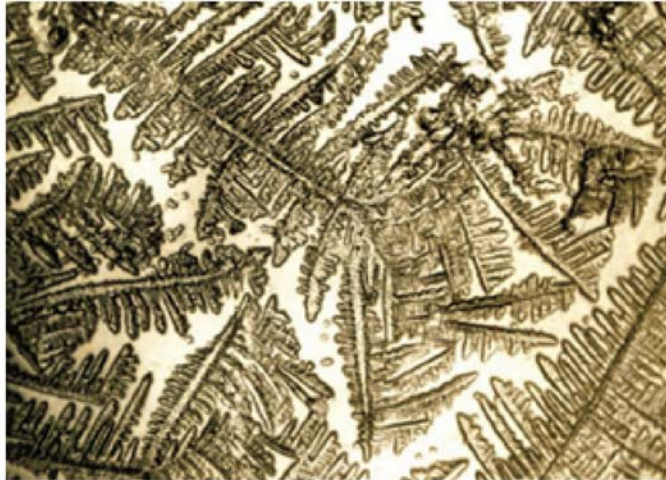
Solution to Question 15:

The image shows the characteristic ferning pattern of cervical mucus which is seen in the estrogenic phase of the menstrual cycle.

A specimen of cervical mucus is obtained using a platinum loop or pipette. This is spread on a clean glass slide and allowed to dry. When viewed under the low-power microscope, a palm leaf or fern type pattern is seen in the follicular phase. The ferning is due to the presence of sodium chloride in the mucus secreted under the effect of estrogen.

This pattern disappears after ovulation. If it was previously present, its disappearance is presumptive evidence of corpus luteum activity. Under the influence of progesterone, the cervical mucus becomes thick and tenacious and impenetrable to sperms and bacteria.

The below image shows the microscopic appearance of dried cervical mucus showing the fern appearance.



Solution to Question 16:

In the Miller–Kurzrok test, sperm acquiring jerky motility after penetrating cervical mucus is considered abnormal.

The Miller-Kurzrok test is a special test done in semen analysis to diagnose the cause of infertility. It is an in vitro test of sperm–mucus interaction.

It is based on the basic principle that a normal cervical mucus permits invasion by motile sperms. It is done at the time of ovulation. The test consists of placing cervical mucus on a glass slide alongside the specimen of the husband's semen and studying the penetration of sperms under the microscope. The penetration of sperm ≥ 3 cm into the mucus at 30 minutes is considered normal.

Other special tests in semen analysis include:

- Sperm penetration test
- Semen–cervical mucus contact test
- Postcoital test (Sims' or Huhner's test)

Solution to Question 17:

The post-coital test can not detect a fallopian tube block. It can evaluate the following conditions:

- Sperm count
- Sperm morphology
- Ferning during the estrogenic effect
- Ovulation

- The ability of the cervical mucous to allow sperm penetration
- Detection of anti-sperm antibodies in the cervical mucous

In the post-coital test, the couple is advised to have intercourse early in the morning close to the ovulation phase. The woman is asked to visit the clinic within 2 hours of having intercourse. The mucus is aspirated from the cervical canal and spread over a glass slide. Another smear made from the posterior fornix serves as a control.

On microscopy, 10–50 motile sperms are normally seen per high-power field. The cervical mucus is also examined for its quantity and viscosity. The post-coital test is useless in the presence of cervical infection.

An immunological factor causing infertility is encountered in around 5% of infertile couples.

Solution to Question 18:

The instrument in the given image is Rubin's cannula.

It is used while performing a hysterosalpingogram (HSG). The cone prevents the retrograde spillage of the dye into the vagina, while the dye is instilled into the uterine cavity. This investigation is performed to study tubal patency in women.

Another cannula that can be used to perform HSG is Spackman cannula. It allows for the intra-operative manipulation of both anteverted and retroverted uterus.

Solution to Question 19:

This patient would be classified as a case of moderate Asherman syndrome as 30% adhesions of the endometrial surfaces are present.

Asherman syndrome (uterine synechiae) is classified as follows:

- Minimal: <25% adhesion involves the uterine cavity, flimsy adhesions involving the fundus and tubal ostia.
- Moderate: 25–70% adhesion of the endometrial surfaces, but no agglutination of the uterine wall.
- Severe: 75% adhesions with agglutination and thick adhesions.

Asherman syndrome can present in the following ways:

- Secondary amenorrhea
- Oligomenorrhoea
- Dysmenorrhoea
- Habitual abortions
- Infertility

The causes of Asherman syndrome are:

- Dilatation and curettage (D&C)
- Medical termination of pregnancy (MTP), spontaneous abortions, recurrent abortions
- Uterine packing in postpartum haemorrhage
- Uterine infection
- Tubercular endometritis

Solution to Question 20:

The given image shows a hysterosalpingogram (HSG). HSG is contraindicated in genital tuberculosis.

Procedure of HSG:

- It is best performed in the period between the last day of menses to ovulation, usually on the 9th/10th day of the cycle.
- 10-15ml radioopaque dye is injected into the uterine cavity via Leech-Wilkinson cannula, Rubin's cannula, or Foley's catheter.
- Fluoroscopy is performed as the dye passes through the uterine cavity and fallopian tubes.
- If the tubes are patent, the dye will spill out of the tubal ostia into the peritoneum, and smear the adjacent bowel.

It is used for the detection of:

- Tubal block - It can be due to,
- Strictures - fibrotic block
- Inspissated amorphous material
- Tubal spasm - This can occur in apprehensive patients. It is prevented by premedication with atropine and analgesia.
- Uterine anomalies
- Uterine polyps
- Uterine synechiae
- The feasibility of tuboplasty, by studying the location and extent of tubal pathology.

Contraindications:

- Genital tract infections, such as genital tuberculosis. This is due to the risk of spreading infection.
- Bleeding
- Suspected pregnancy
- Allergy to the dye
- Premenstrual phase:
- If by chance pregnancy has occurred, it may be dislodged.

- Thick endometrium might prevent the smooth flow of the dye, at the cornual end.
- There is a risk of embolism and endometriosis if HSG is done during this phase.

Solution to Question 21:

The given image shows a normal hysterosalpingogram (HSG). Both fallopian tubes are patent with a spill into the peritoneal cavity.

The following images show the findings of an abnormal hysterosalpingogram:

The hysterosalpingogram given below shows bilateral dilated fallopian tubes with no free spill suggestive of bilateral hydrosalpinx:



The hysterosalpingogram given below shows a unicornuate uterus:



The hysterosalpingogram given below shows a bicornuate uterus/septate uterus:

There is an overlap in the appearance of the bicornuate uterus and septate/sub-septate uterus on HSG. The distinction lies in the shape of the uterine fundus, which is usually not visible on HSG. An ultrasound or MRI should be done to differentiate between these conditions if intervention is planned.



The hysterosalpingogram given below shows filling defects:



Solution to Question 22:

Fertiloscopy helps in obtaining a good view of the posterior surface (and not the anterior surface) of the ovary and genital organs. This is an advantage of fertiloscopy over laparoscopy.

The other advantages of fertiloscopy over laparoscopy are:

- The Trendelenburg position is not needed. The procedure can be done in a lithotomy position
- Trauma can be avoided as no trocar is needed
- No CO₂: Hydroperitoneum with saline
- Peritoneal spill can be seen

Fertiloscopy is a new mini-invasive diagnostic technique that allows the visualization of the posterior pelvis. In this technique, an optical device is introduced into the pouch of Douglas, through the posterior vaginal fornix, under general/local anaesthesia.

It can also be used for small interventions, such as adhesiolysis, ovarian drilling, coagulation of endometriosis spots and to perform chromosalpingoscopy and salpingoscopy. With fertiloscopy, surgical intervention can be avoided and it is a good alternative to diagnostic laparoscopy.

The image given below is of a fertiloscope.



Solution to Question 23:

In the given clinical scenario, the next step in management will be hysteroscopy and laparoscopy-guided surgery.

Tubal contractions, particularly cornual spasms, can give an incorrect impression of proximal fallopian tube obstruction (a false-positive result) on hysterosalpingography. Laparoscopy with chromotubation is therefore considered to confirm the diagnosis of cornual spasm. Laparoscopy is both diagnostic and therapeutic in this condition.

Hysteroscopy is an operative procedure that provides an endoscopic view of the endometrial cavity and tubal ostia via the cervix, performed in the preovulatory phase. It can be diagnostic or therapeutic. It is the gold standard investigation for visualizing the uterine cavity and tubal ostia.

Contraindications to hysteroscopy are as follows:

- Acute genital tract infection (Eg: genital herpes infection)

- Viable intrauterine pregnancy
- During menstruation, the view is obscured, and the infection rate is high.
- Scarred uterus and enlarged uterus of size ≥ 12 weeks are relative contraindications.
- Cervical stenosis can cause cervical tears and uterine perforation.
- Cardiopulmonary disorder due to increased risk of fluid overload and pulmonary edema.
- Known cases of cervical cancer

Laparoscopy is the gold standard for the evaluation of tubal factors in infertility, commonly done in the proliferative phase. It can be diagnostic or therapeutic.

Benefits of laparoscopic evaluation:

- Tubes: For detection of tubal patency, block (site and side), motility, hydrosalpinx change, peritubal adhesions, fimbrial agglutination
- Ovaries: PCOS changes, endometriosis, PID, adhesions
- Uterus: Anomalies, fibroids
- Peritoneal factors: Adhesions, PID, endometriosis, tuberculosis
- Therapy at the same sitting (as appropriate)

Therapeutic indications of hysteroscopy	Diagnostic indications of hysteroscopy
Uterine septum Asherman syndrome Retained IUCD Polyp ectomy Submucous fibroid Abnormal uterine bleeding Tubal sterilization using sclerosing agents	To study the endocervical mucosal lining Congenital malformations of the uterus Endometrial tuberculosis Misplaced IUCD Endometrial lesions Endometrial polyp Asherman syndrome Cornual tubal blockage

Diagnostic	Therapeutic (Operative)
Age >35 years History of pelvic pain, dyspareunia, pelvic surgery, adhesions Abnormal HSG Failure to conceive after a reasonable period (6 months) with normal HSG Women with comorbid pelvic pathology (PID, endometriosis)	Unexplained infertility Reconstructive tubal surgery Adhesiolysis Fulguration of endometriotc implants Ovarian drilling in PCOS GIFT and ZIFT procedures

Solution to Question 24:

The best line of management for the given patient is human menopausal gonadotropin (hMG) followed by hCG.

The patients who have undergone hypophysectomy i.e. removal of the pituitary gland, can not produce endogenous gonadotropins. Exogenous gonadotropins such as hMG are given to such patients. Timed administration of hCG is needed to induce ovulation after the follicle has matured.

The human menopausal gonadotropin is an extract of urine from postmenopausal women. It contains FSH and LH in the required ratio for follicular growth.

Clomiphene, letrozole, and GnRH analogs stimulate the release of gonadotropins from the pituitary gland. These are not helpful in a patient whose pituitary gland has been removed.

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Perimenopause, Menopause and Post-Menopausal Bleeding

Question 1:

Menopause is defined as cessation of menses for a period of _____.

- a) 3 months
- b) 6 months
- c) 9 months
- d) 12 months

Question 2:

A woman with menorrhagia is diagnosed with uterine fibroids. Prior to starting treatment, you want to check if she is premenopausal. On which day of the menstrual cycle will you estimate her FSH levels?

- a) On day 2-5 after last menstrual period
- b) On day 11-14 after last menstrual period
- c) On the day of menstruation
- d) Just before menstruation

Question 3:

What is the diagnostic value of FSH in menopause?

- a) > 40 mIU/mL
- b) > 30 mIU/mL
- c) > 50 mIU/mL
- d) > 20 mIU/mL

Question 4:

What is the average age of menopause?

- a) 55 years

- b) 47 years
- c) 45 years
- d) 50 years

Question 5:

A menopausal woman presents with complaints of hot flushes. Which of the following neurotransmitters are responsible for these symptoms?

- a) Serotonin and epinephrine
- b) Serotonin and norepinephrine
- c) Adrenaline and epinephrine
- d) Noradrenaline and epinephrine

Question 6:

A 53-year-old woman presents with episodes of sensation of intense heat involving the face and upper chest which rapidly spreads across the entire body. The episodes last about 2-4 minutes and are associated with profuse sweating. What is the most effective treatment for her?

- a) Combined Oral contraceptive pills
- b) Clonidine
- c) Estrogen therapy
- d) Danazol

Question 7:

A 53-year-old postmenopausal lady presented with complaints of dyspareunia and vaginal spotting. On examination, the vagina appeared pale with small petechiae. Transvaginal ultrasound revealed a 3mm thick endometrium with no abnormality. Which among the following would be the next step in management?

- a) Pipelle biopsy
- b) Estradiol-releasing vaginal ring
- c) Mirena intrauterine device
- d) Hysteroscopy

Question 8:

Reduction in breast size in post-menopausal women is due to the withdrawal of _____.

- a) Estrogen
- b) Progesterone
- c) Estrogen and progesterone
- d) FSH

Question 9:

A 67-year-old woman is being screened for osteoporosis with a DEXA scan. Which of the following sites are useful for the detection of early rapid bone loss?

- a) Lumbar vertebra
- b) Radius
- c) Shaft of femur
- d) Wrist

Question 10:

Hormone replacement therapy is not indicated in which of the following patients?

- a) 45-year-old woman with severe hot flashes
- b) 36-year-old woman with premature menopause
- c) An adolescent with gonadal dysgenesis
- d) 50-year-old woman with ischemic heart disease

Question 11:

Which of the following is not a contraindication for estrogen therapy?

- a) Breast cancer
- b) Uterine cancer
- c) Previous history of thromboembolic episode
- d) Hypertension

Question 12:

Which of the following is incorrect about hormone replacement therapy and breast cancer?

- a) Cancer risk is not increased up to 3 years of combined HRT
- b) Cancer risk is not increased up to 5 years of estrogen-only replacement therapy
- c) HRT impedes screening for breast cancer with mammography
- d) HRT may cause high grade breast cancer with poor prognosis

Question 13:

You are prescribing estrogen therapy for a woman with hot flushes. What is the minimum number of days in a month during which you should combine it with progesterone?

- a) 8 days
- b) 10 days
- c) 12 days
- d) 14 days

Question 14:

A 50-year-old woman presents with complaints of vaginal bleeding and mentions that her last menstrual period was 10 months ago. What is the minimum duration of amenorrhoea following which she should be evaluated for possible causes?

- a) 3 months
- b) 6 months
- c) 9 months
- d) 12 months

Question 15:

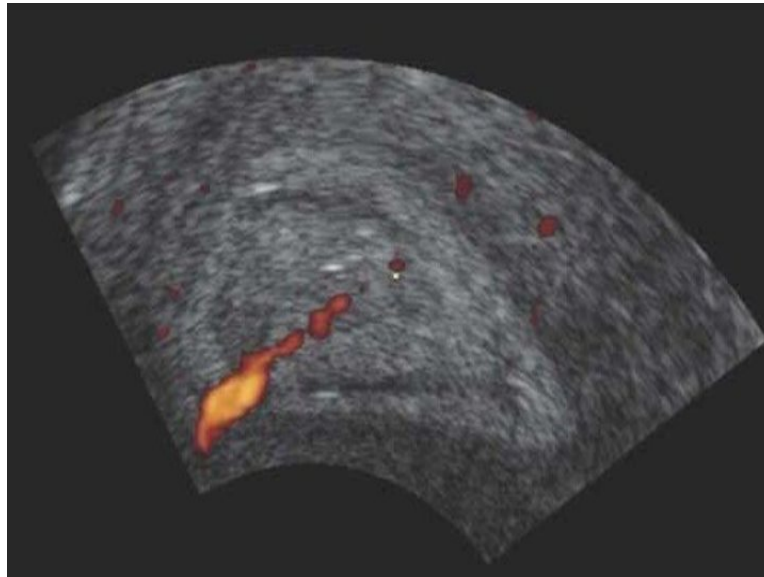
What is the investigation of choice in a 55-year-old woman who presents with post-menopausal bleed?

- a) Pap smear
- b) Endometrial biopsy
- c) Fractional curettage

d) CA-125 estimation

Question 16:

A 60-year-old woman presented with post-menopausal bleeding. Ultrasonography revealed a mass with feeding vessel as shown in the image below. What is the most likely diagnosis?



- a) Fibroid uterus
- b) Endometrial polyp
- c) Endometrial carcinoma
- d) Leiomyosarcoma

Question 17:

A patient is planned to undergo hysteroscopic polypectomy using bipolar electrocautery. What is the ideal agent for distention?

- a) Glycine
- b) Normal saline
- c) CO₂
- d) 70% dextrose

Answer Key

Question No.	Correct Option
1	d
2	a
3	a
4	b
5	b
6	c
7	b
8	c
9	a
10	d
11	d
12	d
13	c
14	b
15	b
16	b
17	b

Detailed Explanations

Solution to Question 1:

Menopause is defined as cessation of ovarian function for 12 months resulting in permanent amenorrhoea. It is a retrospective diagnosis.

Solution to Question 2:

The follicular stimulating hormone (FSH) level is studied on day 2–5 after the last menstrual period to detect the premenopausal stage.

It is a period of 3–4 years before menopause and is associated with mild ovarian hormonal deficiency leading to menstrual disorders, especially menorrhagia.

The climacteric phase is the phase of waning of ovarian activity. It begins 2 to 3 years before menopause and may continue 2-5 years after menopause. It marks the transition from active to inactive ovarian function and involves physical, sexual, and psychological adjustments.

The predictors of approaching menopause include:

- A fall in the level of inhibin B and a corresponding rise in FSH level
- A fall in the level of anti-Mullerian hormone (AMH) - This is a better marker of ovarian reserve and can be tested on any day of the cycle.

Solution to Question 3:

The diagnostic value of FSH in menopause is >40 mIU/mL.

Hormone levels in menopausal women:

Hormone	Level
Estradiol	5 to 25 pg/ml
FSH	>40 mIU/mL
LH	50-100 mIU/mL
Growth hormone, inhibin B, AMH	Decreases

Solution to Question 4:

The average age of menopause in India is 47 years, and ranges from 45 to 50 years.

Menopause sets in when the follicle number falls below 1000. If this happens before 40 years, it is termed premature menopause while attaining it after 55 years is delayed menopause.

The age at which menopause is attained is not related to age at menarche, race, number of pregnancies, socio-economic status, or oral contraceptive pills usage. However, it is directly influenced by smoking and genetic disposition.

Solution to Question 5:

Norepinephrine and serotonin are responsible for hot flushes in menopausal women.

A fall in the level of estrogen causes a reduction in hypothalamic endorphins. This causes a release of more norepinephrine and serotonin, which lower the set point in the thermoregulatory nucleus. As a result, the heat-loss mechanism is triggered by even subtle changes in core body temperature.

Risk factors for hot flushes:

- Low estradiol levels

- High BMI
- Sedentary lifestyle
- Surgical menopause has increased risk
- Low socioeconomic status
- History of premenstrual dysphoric disorder (PMDD)
- Depression.

Solution to Question 6:

The clinical features are suggestive of vasomotor symptoms (hot flushes) in a menopausal woman for which the most effective treatment is estrogen therapy.

Treatment options include:

- Hormonal
- Estrogen therapy - most effective
- Combined OC pills - given when estrogen is contraindicated
- Tibolone
- Non-hormonal
- Clonidine
- SSRIs (eg: fluoxetine)
- Dopamine antagonist
- Non-prescriptive
- Vitamin E
- Soya
- Isoflavones

Hormone therapy is only indicated for the treatment of vasomotor symptoms, osteoporosis (prevention and treatment), and vaginal atrophy.

Endometrial hyperplasia can occur after 6 months of estrogen therapy. Hence, for patients with a uterus, it is recommended to add micronized progesterone. This is not necessary for women who have undergone hysterectomy.

Note: Even though OCPs contain both estrogen and progesterone, a lesser dosage of each component is given for the treatment of hot flushes.

Solution to Question 7:

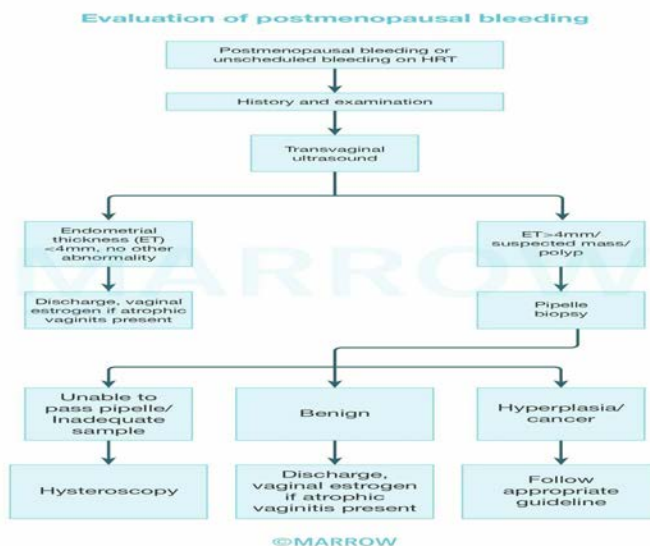
The clinical scenario is suggestive of atrophic vaginitis. The next step in management should be topical estrogen therapy such as the estradiol-releasing vaginal ring.

Atrophic vaginitis is a common cause of post-menopausal bleeding due to low estradiol levels. Patients may present with dry vagina, dyspareunia and a purulent or blood-tinged discharge. The vagina appears pale, dry, and may show petechiae.

Treatment of atrophic vaginitis involves topical estrogens in the form of creams, tablets, or rings. Estrogen replacement restores vaginal mucosal elasticity, vaginal fluid secretion levels, blood flow, and sensorimotor responses.

Systemic therapy may help prevent recurrence. Vaginal lubricants and moisturizers can be given for symptomatic relief.

FOGSI guidelines for evaluation of postmenopausal bleeding:



Solution to Question 8:

The reduction in breast size in postmenopausal women is due to the withdrawal of estrogen and progesterone.

Estrogen exerts proliferative effects on the ductal component, while progesterone exerts a proliferative effect on the glandular component.

Solution to Question 9:

Early rapid bone loss can be determined from a dual-energy x-ray absorptiometry (DEXA) scan of the hip and lumbar vertebra.

The risk for most fractures is inversely proportional to bone density. The reference standard for the diagnostic purpose is DEXA at the hip (femoral neck or total hip). These sites are superior to the spine in predicting hip as well as an overall osteoporotic fracture.

DEXA is used to assess bone mineral density (BMD). It is interpreted as follows:

Condition	T Score range
Normal	+2.5 and -1.0
Osteopenia	-1.0 and -2.5
Osteoporosis	-2.5 and below
Severe or established osteoporosis	-2.5 and below with one or more fractures OR -3.5 or below even in the absence of fractures

Solution to Question 10:

HRT is no longer recommended for the management of menopausal women with ischemic heart disease.

Indications for HRT is as follows:

- Severe hot flushes
- Premature menopause
- Surgical oophorectomy
- Turner syndrome (gonadal dysgenesis).

Solution to Question 11:

Hypertension is not a contraindication for hormone replacement therapy (HRT). HRT can be initiated in patients with a history of hypertension, diabetes, or smoking; provided they are regularly monitored.

Contraindications of estrogen therapy:

- Breast cancer, uterine cancer, or family history of cancer
- Previous history of thromboembolic episode
- Liver and gall bladder diseases
- Uterine fibroids - as the fibroids may enlarge in size.

Solution to Question 12:

Breast cancer developing subsequent to HRT is of low grade and has a good prognosis.

The risk of breast cancer is not increased up to 3 years of HRT and 5 years of estrogen-only replacement therapy. Lower risk is seen with the use of dydrogesterone in HRT.

HRT can cause recurrence of breast cancer and is, therefore, contraindicated in a woman who has been treated for breast cancer.

HRT increases the density of breast tissue and subsequently impedes the screening for breast cancer with mammography.

Solution to Question 13:

A minimum of 12 days (in a month) of progesterone must be added to estrogen therapy. This reduces the risk of endometrial cancer to 2%.

The risk of cancer with estrogen therapy is dose and duration-dependent. Combined estrogen and progesterone provides better protection against endometrial cancer.

Tibolone is a safe drug and does not cause endometrial hyperplasia.

Solution to Question 14:

Vaginal bleeding occurring after 6 months of amenorrhoea in women of menopausal age group should be considered as postmenopausal bleeding and investigated. This must be done irrespective of a previous diagnosis of menopause.

All postmenopausal women with unexpected uterine bleeding should be evaluated for endometrial carcinoma. Even in those women who are perimenopausal, endometrial carcinoma must be suspected if they fall under the above-mentioned clinical category.

Solution to Question 15:

Endometrial biopsy is the investigation of choice in a case of post-menopausal bleed, as the chances of endometrial cancer are quite high.

For those who undergo TVUS and have an EEC (endometrial echo complex) ≥ 4 mm, localized thickening, or an indistinct or nonvisible EEC, endometrial sampling is a reasonable next step.

Investigations for endometrial carcinoma include:

- Endometrial aspiration biopsy is the first investigation in abnormal uterine bleeding, especially in females ≥ 35 years.
- Dilatation and fractional curettage was the best method to diagnose but now it is not used.
- Hysteroscopy and biopsy is the investigation of choice for endometrial cancer.
- Transvaginal USG is used as an adjunct to endometrial biopsy.
- Pap smear is not reliable to diagnose endometrial carcinoma. It is more useful as a screening tool rather than a diagnostic tool.

If symptoms of postmenopausal bleeding persist, imaging of the endometrial cavity should be performed with one or a combination of the following: TVUS, contrast sonography, or

hysteroscopy.

Note: Surgical staging is the gold standard investigation for staging endometrial carcinoma.

Solution to Question 16:

Visualization of a mass with a single feeding vessel on a color Doppler is characteristic of an endometrial polyp. In contrast, submucous leiomyomas often receive blood flow from multiple vessels (option A).

Endometrial polyps are soft, fleshy growths within the uterus that consist of endometrial glands, fibrous stroma, and surface epithelium. Risk factors for endometrial polyps include increasing age, obesity, and tamoxifen use.

They are usually asymptomatic. When symptomatic, they present with heavy menstrual bleeding, prolonged bleeding, intermenstrual bleeding, or infertility.

The main diagnostic tools for endometrial polyps include transvaginal ultrasound (TVS) with applied color Doppler, saline infusion sonography, and hysteroscopy.

The risk factors for the malignant transformation include postmenopausal status, age older than 60, and abnormal bleeding. Other factors such as larger polyp size (>1.5 cm), tamoxifen use, diabetes, and obesity may also contribute to a higher risk.

Hysteroscopic polypectomy may be considered most effective for symptomatic women or those with risk factors for malignant transformation.

Solution to Question 17:

Normal saline is the ideal fluid media used for bipolar electrocautery as it can function regardless of the fluid electrolyte content.

Normal saline and lactated Ringer solutions are commonly used isotonic, electrolyte-containing fluids in operating rooms for diagnostic hysteroscopy procedures. These solutions can cause volume overload and, if rapidly absorbed in large volumes, may lead to pulmonary edema.

Isotonic, electrolyte-containing fluids are contraindicated for use with monopolar electrosurgical devices because they conduct electrical current, dissipate instrument energy and render the instrument ineffective. Low-viscosity, electrolyte-poor fluids such as 1.5-percent glycine, 3-percent sorbitol, and 5-percent mannitol are preferred as they are non-conductive and can be safely used with monopolar electrosurgical energy devices.

The primary risk of using hypotonic, hypoosmolar solutions is the intravascular absorption of these fluids, which can lead to acute hyponatremia and hypo osmolality.

The criteria for terminating the procedure for electrolyte-deficient fluid is a deficit of 1 L and electrolyte-rich fluid is a deficit of 2.5 L.

Ovarian Tumors

Question 1:

Which is the most common type of ovarian tumor?

- a) Epithelial tumor
- b) Granulosa cell tumor
- c) Gonadoblastoma
- d) Sex cord-stromal tumor

Question 2:

Which of the following is the most carcinogenic gene in ovarian cancer?

- a) BRCA 1
- b) BRCA 2
- c) MLH1
- d) MSH2

Question 3:

Which among the following reduces the risk of developing ovarian cancers?

- a) Nulliparity
- b) Polycystic ovary syndrome
- c) HRT therapy
- d) Bilateral tubectomy

Question 4:

Which among the following is the largest benign ovarian tumor?

- a) Serous cystadenoma
- b) Mucinous cystadenoma
- c) Teratoma

- d) Dermoid cyst

Question 5:

Which of the following is false regarding borderline ovarian tumors?

- a) Mitotic figures are $\leq 4/10$ high power field
- b) Stromal invasion is seen
- c) 90% survival rate
- d) Serous type is more common than mucinous type

Question 6:

Which of the following statements is false regarding ovarian cysts?

- a) Cobweb appearance is seen on USG in haemorrhagic cyst
- b) Theca lutein cyst regresses with a fall in hCG levels
- c) Leutoma of pregnancy is associated with multiple pregnancy
- d) Follicular cyst is a benign epithelial tumor

Question 7:

Meigs' syndrome was suspected in a woman with right-sided pleural effusion, ascites, and an ovarian tumor. Which of the following statements is false regarding this condition?

- a) Can be caused by ovarian fibroma
- b) The tumor would be immobile with an irregular surface
- c) Ascites and hydrothorax resolve after resection of the tumor
- d) Pleural effusion is more common on the right side

Question 8:

An elderly woman who was found to have an ovarian mass underwent staging laparotomy. Histopathological examination revealed the presence of Psammoma bodies. What is the most probable diagnosis?

- a) Serous cystadenocarcinoma
- b) Mucinous cystadenocarcinoma

- c) Embryonal cell carcinoma
- d) Dermoid tumor

Question 9:

A 49-year-old woman presented with complaints of dull pain over the right lower abdomen for the past 2 months. CT scan revealed a left-sided multiloculated adnexal mass and exploratory laparotomy showed small mucoid implants scattered in the peritoneal cavity. Which of the following is associated with this condition?

- a) Serous cystadenoma
- b) Mucinous cystadenoma
- c) Teratoma
- d) Embryonal cell carcinoma

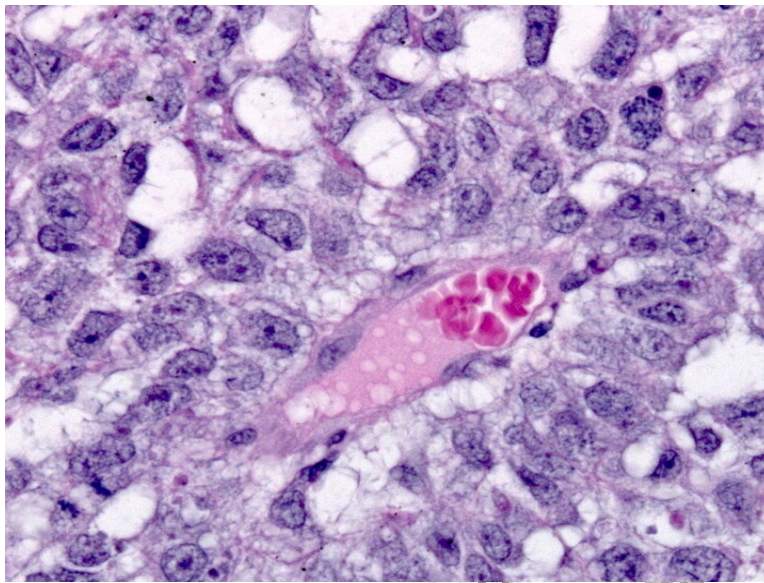
Question 10:

A 22-year-old woman at 14 weeks of gestation was rushed to the ER with complaints of severe abdominal pain, nausea, and vomiting. USG showed an edematous right ovary with a large cyst and disruption in adnexal flow on Doppler. Which of the following most commonly leads to this condition?

- a) Dysgerminoma
- b) Teratoma
- c) Thecoma
- d) Theca lutein cyst

Question 11:

Staging laparotomy was performed in a teenage girl who presented with an adnexal mass. Histopathological examination showed the following. What is the diagnosis?



- a) Mucinous cystadenoma
- b) Krukenberg tumor
- c) Endodermal sinus tumor
- d) Dysgerminoma

Question 12:

What is the counterpart of seminoma in females?

- a) Yolk sac tumor
- b) Embryonal carcinoma
- c) Endodermal sinus tumor
- d) Dysgerminoma

Question 13:

The most common ovarian tumor in women less than 20 years of age is _____.

- a) Epithelial tumor
- b) Germ cell tumor
- c) Gonadoblastoma
- d) Sex-cord stromal tumor

Question 14:

Which of the following tumors show raised CA-125 levels?

- a) Dysgerminoma
- b) Endodermal sinus tumor
- c) Teratoma
- d) Non-mucinous cystadenocarcinoma

Question 15:

A young woman with an adnexal mass is diagnosed with dysgerminoma. Which of the following tumor markers are likely to be elevated in her?

- a) Alpha fetoprotein
- b) Lactate Dehydrogenase
- c) CEA
- d) CA - 125

Question 16:

A post-menopausal woman is found to have an adnexal mass on evaluation. Above what risk of malignancy index (RMI) score is there a high risk of ovarian malignancy?

- a) 180
- b) 200
- c) 150
- d) 100

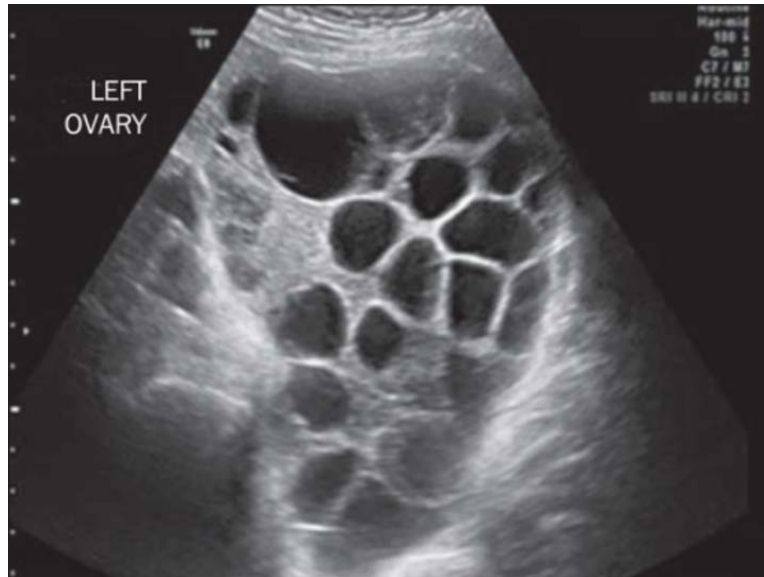
Question 17:

Which of the following statements is false regarding Brenner tumor?

- a) It resembles fibroma
- b) It is associated with Meig's syndrome
- c) Walther cell rests and coffee bean nuclei are seen
- d) It is not a malignant tumor

Question 18:

A 31-year-old woman presenting with 12 weeks of amenorrhea and passage of vesicles per vagina is suspected to have a molar pregnancy. On transvaginal ultrasonography, the left ovary appears as shown below. Which of the statements is not true regarding this condition?



- a) Regresses spontaneously after removal of molar pregnancy
- b) It is seen with clomiphene administration
- c) It is usually bilateral
- d) Oral contraceptive use is a risk factor

Question 19:

A woman noticed premature breast development in her 7-year-old daughter and brought her to the gynecology OPD. USG revealed a right-sided adnexal mass. The patient underwent laparotomy and the biopsy revealed Call-Exner bodies. Which of the following is true regarding this condition?

- a) Most common malignant tumor of ovary
- b) Associated with endometrial carcinoma
- c) It is bilateral
- d) Inhibin A is elevated more than inhibin B

Question 20:

Which among the following high-risk patients are suitable for prophylactic oophorectomy?

- a) Nulliparous aged $\geq$ 35 years
- b) History of ovarian cancer in a 2nd-degree relative
- c) Abnormal ovary in reproductive age group woman
- d) One ovary removed for benign neoplasm

Question 21:

A 63-year-old woman presented with complaints of pelvic and abdominal pain, and significant weight loss. On CT scan, a bilateral ovarian tumor with pelvic lymph node and liver capsule involvement was seen. What stage does this belong to?

- a) Stage 4A
- b) Stage 2B
- c) Stage 3
- d) Stage 4B

Question 22:

A 26-year-old nulliparous woman is diagnosed with early-stage ovarian cancer. Which surgery should be done in her case?

- a) Unilateral oophorectomy
- b) Bilateral oophorectomy
- c) Hysterectomy with bilateral oophorectomy
- d) Debulking surgery

Question 23:

Which of the following ovarian tumors is the most radiosensitive?

- a) Dysgerminoma
- b) Dermoid cyst
- c) Serous cystadenoma
- d) Teratoma

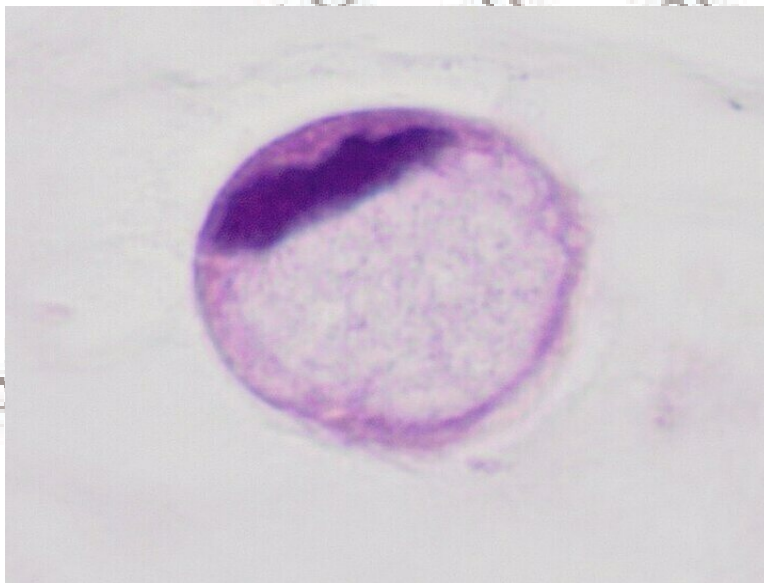
Question 24:

A 62-year-old post-menopausal woman on HRT presented with abdominal pain for the past 6 months. On examination, a non-tender, right adnexal mass was palpable. USG showed a 3 x 3 cm smooth ovarian cyst. What is the next best step in her case?

- a) Reassure and advise regular follow up once in 6 months
- b) Confirmation of USG findings by CT scan
- c) Exploratory laparotomy for evaluation of cyst
- d) CA- 125 level estimation; if normal, advise regular follow-up

Question 25:

Examination of a resected specimen of an ovarian mass reveals signet ring cells are seen as shown below. Which of the following conditions is this associated with?



- a) Mucinous cystadenoma
- b) Krukenberg tumor
- c) Serous cystadenoma
- d) Yolk sac tumor

Answer Key

Question No.	Correct Option
1	a
2	a
3	d
4	b
5	b
6	d
7	b
8	a
9	b
10	b
11	c
12	d
13	b
14	d
15	b
16	b
17	d
18	d
19	b
20	d
21	c
22	a
23	a
24	d
25	b

Detailed Explanations

Solution to Question 1:

Epithelial tumors are the most common type of ovarian tumors.

Solution to Question 2:

BRCA 1 is the most carcinogenic gene in ovarian cancer.

5–10% of malignant ovarian tumors are genetic. BRCA 1 is associated with hereditary breast-ovarian cancer syndrome.

BRCA 1 is on chromosome 17q and BRCA 2 is on chromosome 13q.

MSH2 and MLH1 gene mutation is seen in hereditary non-polyposis colorectal cancer (HNPCC). It is associated with malignancy of the colon, ovary, and endometrium.

Solution to Question 3:

Bilateral tubectomy and hysterectomy reduce the risk of ovarian cancer.

Some ovarian malignancies originate in the fallopian tubes as dysplastic lesions which later spread to the ovary (dropping down theory). Thus, bilateral tubectomy reduces the risk of ovarian malignancy.

Risk factors for ovarian cancer:

- Advancing age
- Early menarche and late menopause
- Talc exposure to the perineum
- Family history of breast / ovarian cancer
- Ovulation induction therapy by clomiphene citrate

Protective factors against ovarian cancer:

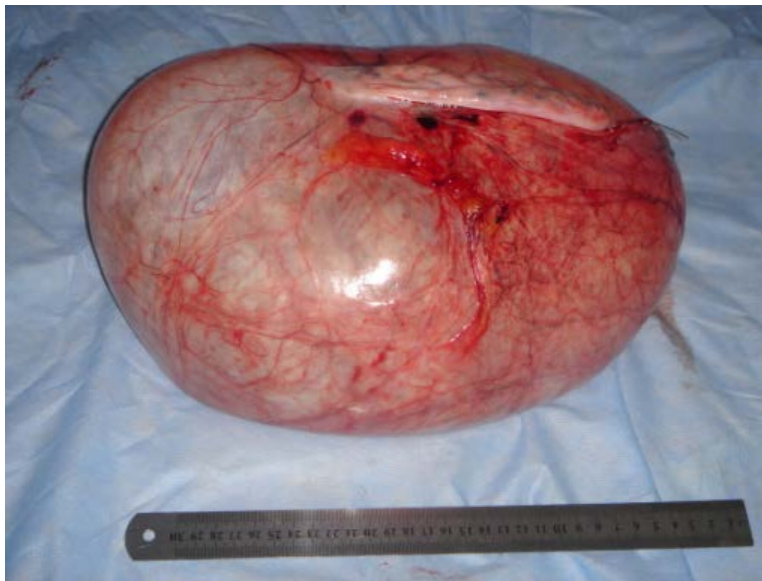
- Use of oral contraceptive pills
- Multiparity
- Breastfeeding
- Anovulation.

Solution to Question 4:

Mucinous cystadenoma is the largest benign ovarian tumor.

Mucinous cystadenoma is a mucoid-containing, thick-walled cyst. It is usually multilocular and lined by single columnar epithelium.

It is the third most common epithelial tumor of the ovary. It has a 10% chance of becoming malignant. About 40% of dermoid cysts are found in conjunction with mucinous cystadenomas.



Solution to Question 5:

Stromal invasion is not seen in borderline ovarian tumors.

Borderline tumors of the ovary (also called tumors of low malignant potential) are a heterogeneous group of lesions defined histologically by atypical epithelial proliferation without stromal invasion.

The clinical progression of these tumors is distinct from ovarian carcinoma, so they are considered a distinct clinical entity.

Features of borderline ovarian tumors:

- No stromal invasion
- Mitotic figures $\leq 4/10$ high power field
- Survival rate is 90%
- Usually occurs between 35-55 years of age
- Serous type is the most common, followed by mucinous type.

Solution to Question 6:

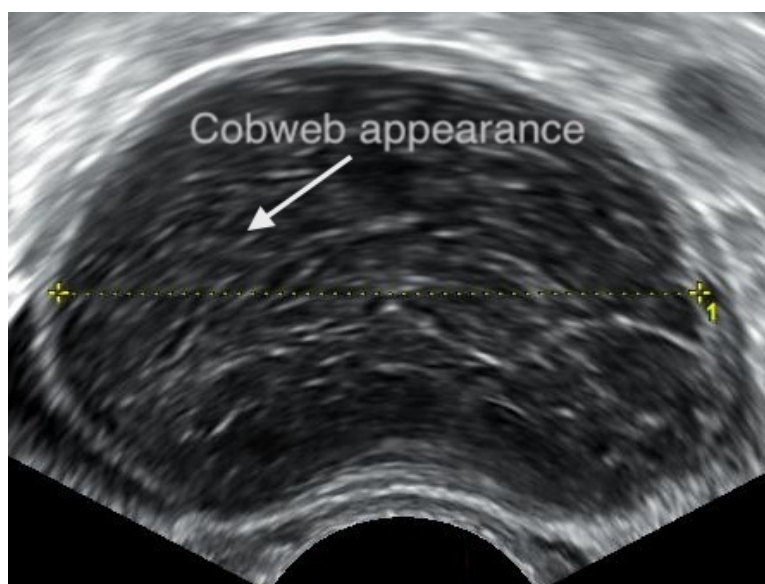
Follicular cyst is a physiological/functional cyst of the ovary.

Physiological/functional cysts of the ovary:

- Follicular cyst - Failure of rupture of the follicle during ovulation.
- Theca lutein cyst - Stimulation of atretic follicles by abnormal HCG levels. It regresses when hCG levels fall.
- Luteoma of pregnancy - Due to prolonged elevation of hCG, like in multiple pregnancies.

Hemorrhagic ovarian cysts usually occur due to bleeding into the corpus luteum or other functional cysts at the time of ovulation. Hemorrhagic cysts are normal findings in ovulatory women, generally resolving within 8 weeks.

Cobweb appearance on USG in a hemorrhagic cyst is shown below.



Solution to Question 7:

In Meig's syndrome, a benign ovarian mass is likely to be seen. Irregular surface and immobility are features of malignant tumors.

Meig's syndrome is characterized by a triad of:

- Benign ovarian tumors
- Ascites
- Right-sided pleural effusion.

The pleural effusions are usually right-sided, even though the left-sided and bilateral effusions are possible.

Tumors causing Meigs' syndrome include ovarian fibroma, granulosa cell tumor, Brenner tumor, or thecoma.

However, when the causative factor is ovarian secondaries, leiomyoma, and benign ovarian cysts (such as teratoma and mucinous cystadenoma), the condition is called Pseudo Meigs' syndrome.

Resolution of ascites and pleural effusion is seen after the removal of the tumor.

Management:

- Repeated paracentesis for ascites.
- Unilateral salpingo-oophorectomy in young patients for preserving fertility

- Total abdominal hysterectomy with bilateral salpingo-oophorectomy (TAH-BSO) in postmenopausal women.

Note: Systemic lupus erythematosus (SLE) patients may also have a combination of ascites, pleural effusion, and elevated serum levels of CA-125. This is known as Tjasma syndrome or pseudo-pseudo Meigs' syndrome.

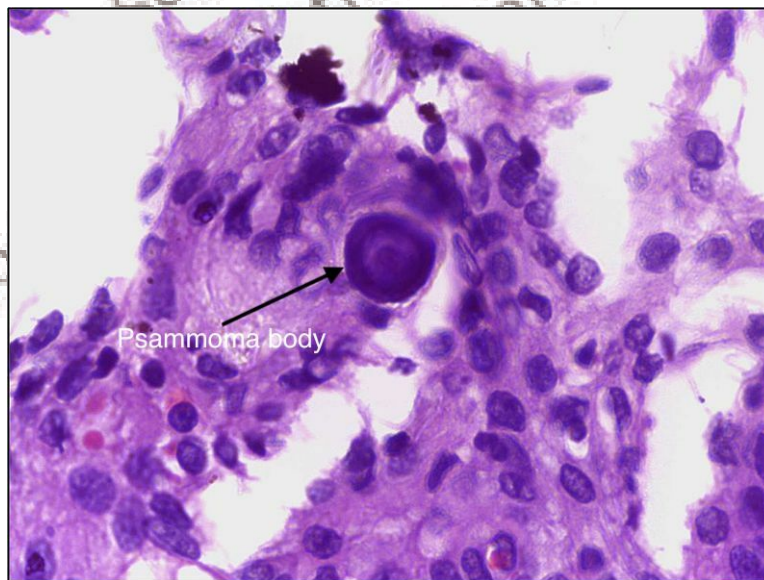
Solution to Question 8:

Psammoma bodies are a histopathological finding of serous cystadenocarcinoma of the ovary. They are calcified structures seen in areas of cellular degeneration. Their presence does not denote malignancy.

Psammoma bodies are seen in the following conditions:

- Papillary carcinoma of the thyroid
- Papillary serous carcinoma of the endometrium
- Serous cystadenocarcinoma of the ovary
- Meningioma

The image below shows Psammoma body.



Solution to Question 9:

The above clinical scenario and findings point to a diagnosis of pseudomyxoma peritonei. It is associated with mucinous cystadenoma.

Pseudomyxoma peritonei (PMP) is a rare clinical entity characterized by diffuse intra-abdominal gelatinous and mucoid material on peritoneal surfaces. Spontaneous perforation of a mucinous

cystadenoma leads to implantation of low-grade malignant tumor cells in the peritoneum.

However, the most common cause of pseudomyxoma peritonei is carcinoma of the appendix.

Diagnosis is by CT, MRI, and laparoscopy. Even after the removal of the ovarian tumor, these cells continue to secrete mucin. Recurrence rates are high.

Solution to Question 10:

The above scenario is suggestive of ovarian torsion. Mature cystic teratoma (dermoid cyst) is the commonest tumor to undergo torsion, especially at the end of the 1st trimester or during puerperium.

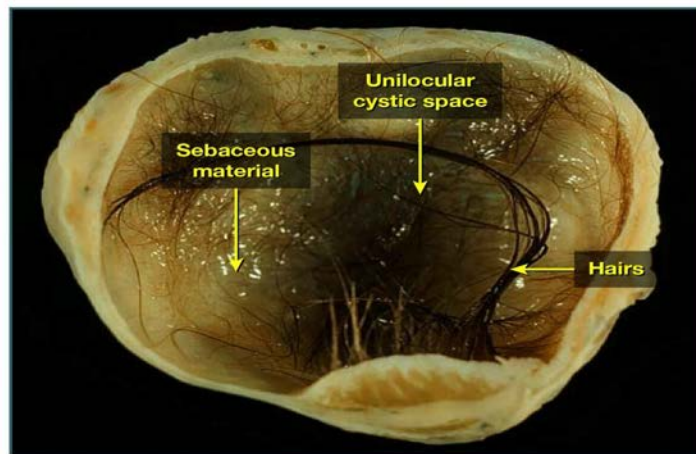
Management includes derotation and cystectomy in pregnancy irrespective of gestational age. This is also done in non-pregnant women of reproductive age to preserve ovarian function. Resection is reserved for patients in whom malignancy is suspected.

Dermoid cyst is the most common germ cell tumor. It is also the most common benign tumor of the ovary in the reproductive age group and in pregnancy. It is bilateral in 10-15% of cases.

Dermoid cyst is derived from all three germ cell layers, but ectodermal derivatives are most common. Rokitansky's protuberance is the most common site for the development of malignancy. It is seen as a projection covered by skin, sebaceous glands, sweat glands, teeth, and bones. Sometimes thyroid tissue can be seen on histopathological examination, known as struma ovarii.

The image below shows a dermoid cyst with Rokitansky protuberance.

Mature Teratoma - Gross specimen



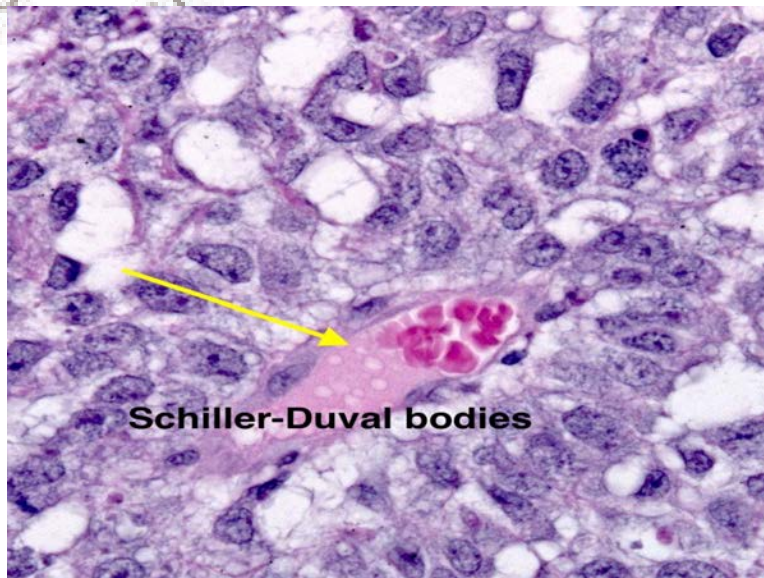


Solution to Question 11:

The given histopathological image shows Schiller-Duval bodies. It is suggestive of a yolk sac tumor (endodermal sinus tumor).

It is an aggressive malignant germ cell tumor of the ovaries. It occurs in ovaries or testes and sacrococcygeal areas in young children. It is usually unilateral. These tumors are usually associated with raised AFP and alpha 1 antitrypsin.

The image below depicts Schiller-Duval bodies. It consists of a central capillary surrounded by tumor cells, present within a cystic space that may be lined by flat to cuboidal tumor cells. 50% of yolk sac tumors have Schiller-Duval bodies and they are pathognomonic when present.



Solution to Question 12:

The counterpart of seminoma in females is dysgerminoma. It is a germ cell tumor.

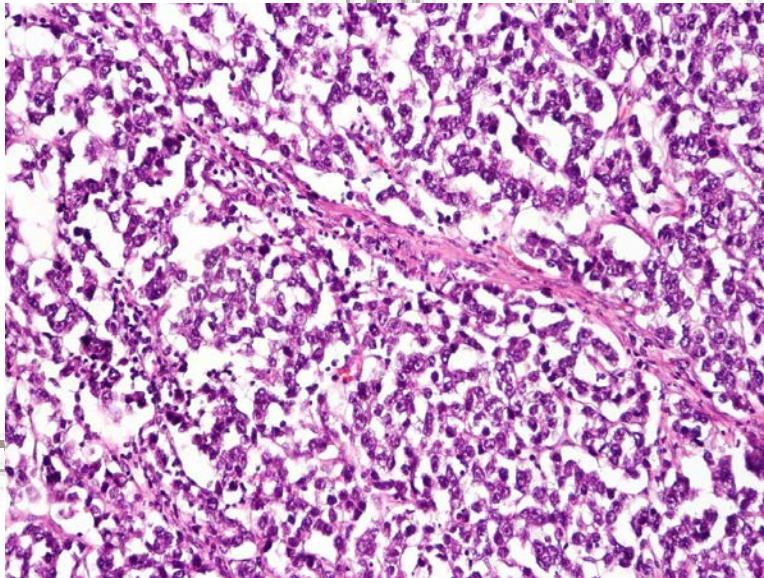
It usually arises in young women or in children. It is the most common malignant germ cell tumor diagnosed during pregnancy. They account for approximately 1/3rd of all malignant ovarian germ cell tumors.

Serum lactate dehydrogenase (LDH) and the isoenzymes LDH-1 and LDH-2 may be used in monitoring individuals or disease recurrence.

Treatment involves unilateral salpingo-oophorectomy. It is highly radiosensitive and has a good prognosis.

Dysgerminoma was considered to be the most common malignant germ cell tumor of the ovary. Recent data indicates that in the past few decades, their incidence has reduced considerably. Immature teratomas are now the most common type of malignant germ cell tumor of the ovary.

The image below shows the histopathology of a dysgerminoma, characterized by uniform cells resembling primordial germ cells separated by fibrous septa with lymphocytes.



Solution to Question 13:

Germ cell tumors are the most common ovarian tumor in women aged less than 20 years.

Epithelial tumours are the commonest tumours. It accounts for 80% of all ovarian malignant tumours.

Solution to Question 14:

Non-mucinous cystadenocarcinoma is an epithelial tumor of the ovary. It has elevated CA- 125 levels.

CA-125 value > 35 U/ml is suggestive of epithelial ovarian cancer.

Malignant conditions with raised CA 125 levels:

- Ovarian epithelial carcinoma (most common cause)
- Endometrial cancer
- Pancreatic cancer
- Colon cancer
- Breast cancer
- Lung cancer
- Lymphoma

Benign conditions with raised CA 125 levels:

- Pregnancy
- Pelvic inflammatory disease
- Endometriosis
- Uterine fibroid
- Normally elevated in 1% females
- Acute peritonitis

Solution to Question 15:

Lactate dehydrogenase is the tumor marker in dysgerminoma.

Dysgerminomas are primitive neoplasms. They are the only germ cell malignancy with a significant rate of bilateral ovarian involvement (15-20%).

It is seen in young women with an average incidence of 20 years of age. It is the most common malignant germ cell tumor diagnosed during pregnancy. They account for approximately 1/3rd of all malignant ovarian germ cell tumors.

It secretes placental alkaline phosphatase (PALP), lactate dehydrogenase (LDH), and β -hCG. Serum lactate dehydrogenase (LDH) and the isoenzymes LDH-1 and LDH-2 may be used in monitoring individuals or disease recurrence.

Treatment involves unilateral salpingo-oophorectomy. It is highly radiosensitive and has a good prognosis.

Solution to Question 16:

Risk of malignancy index (RMI) score of 200 or more indicates a high risk for ovarian cancer in a patient with an adnexal mass.

Risk of malignancy index (RMI) = $U \times M \times CA_{125}$

- U = Ultrasound score
- The ultrasound result is scored 1 point for each of the following characteristics: multilocular cysts, solid areas, metastases, ascites, and bilateral lesions.
- U = 0 (for an ultrasound score of 0)
- U = 1 (for an ultrasound score of 1)
- U = 3 (for an ultrasound score of 2–5).
- M = Menopausal status
- M= 1 (for pre-menopausal)
- M= 3 (for post-menopausal)
- CA-125 = Serum CA-125 level (in IU/ml)

For any post-menopausal woman with an ovarian cyst of size ≥ 1 cm, RMI should be calculated.

Solution to Question 17:

Brenner tumors are usually benign but rarely they can turn malignant.

Brenner tumor (transitional cell tumor) is a benign, epithelial, unilateral tumor, resembling a fibroma. Histologically, Walthard cell rests and coffee bean nuclei are seen.

Brenner tumor represent only 2-3% of all ovarian neoplasms, and $\leq 2\%$ of these tumors are malignant.

It is called Meigs' syndrome if the clinical picture is caused by an ovarian fibroma, granulosa cell tumor, Brenner tumor, or thecoma.

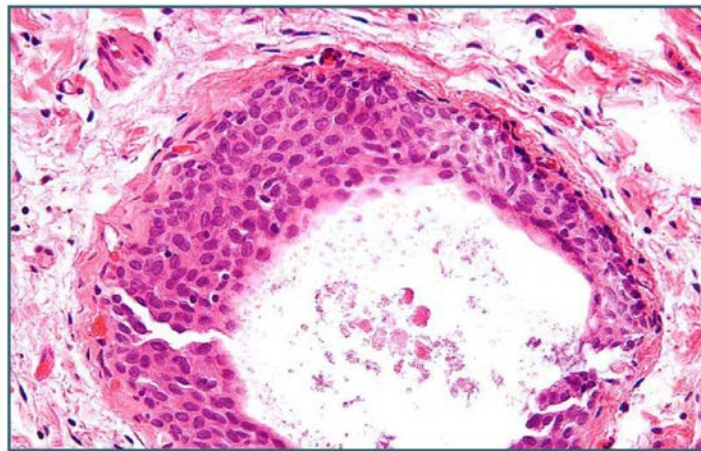
Meig's syndrome is characterized by a triad of:

- Benign ovarian tumor
- Ascites
- Right-sided pleural effusion.

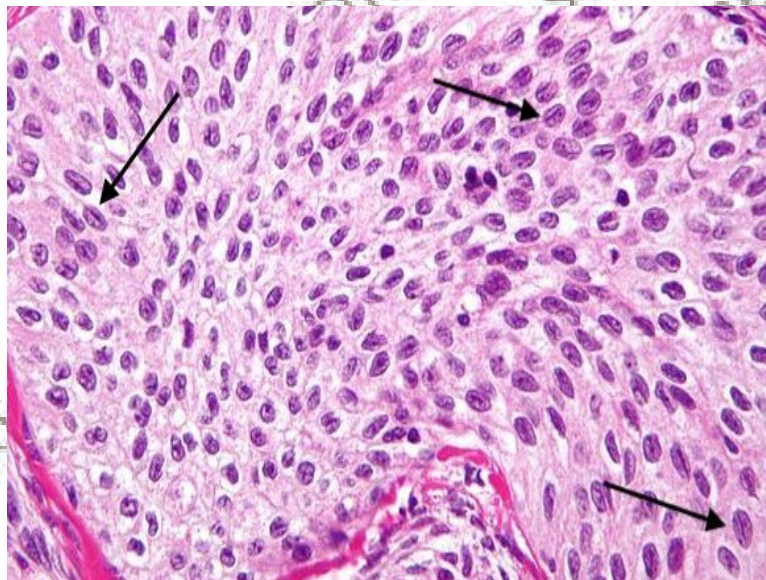
Pseudo Meigs' syndrome is when the causative factor is ovarian secondaries, leiomyoma, and benign ovarian cysts such as teratoma and mucinous cystadenoma.

The image given below shows Walthard cell rest.

Walthard cell rest



The image given below shows coffee bean nuclei



Solution to Question 18:

The above scenario points to a diagnosis of theca lutein cysts in association with molar pregnancy. Theca lutein cysts are not associated with the use of oral contraceptives.

Lutein cysts types :

- Granulosa lutein cysts - They occur due to failure of regression of corpus luteum after 14 days of ovulation.
- Theca lutein cysts - They are found in association with hydatidiform moles, choriocarcinoma, and gonadotropin (hCG), or clomiphene therapy.

Theca lutein cysts occur due to the hyperstimulation of ovaries as a result of high hCG levels in molar pregnancies and other conditions. They are usually bilateral.

These cysts usually regress spontaneously in a period of 2-4 months following the evacuation of a molar pregnancy.

Sometimes they can undergo torsion and cause marked pelvic pain. In such instances, laparoscopic untwisting of the cyst has to be done.

Solution to Question 19:

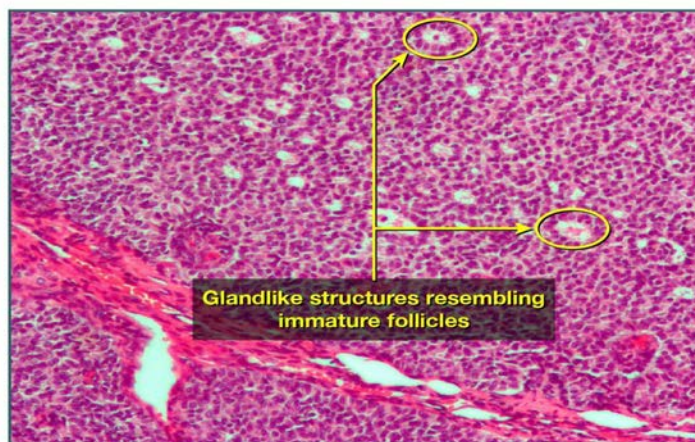
The above clinical scenario points to a diagnosis of granulosa cell tumor. It is associated with carcinoma endometrium.

Granulosa cell tumors cells are estrogen-secreting ovarian tumors. It is seen in 80% of women over 40 years and 5% of pre-pubertal girls. Precocious puberty is seen due to excess secretion of estrogen.

It is a low-grade malignancy. It is almost always unilateral. Tumor marker inhibin B is frequently elevated more than inhibin A. Surgical resection is the treatment of choice.

On histopathology, it has characteristic Call-Exner bodies. It appears as a rosette arrangement of cells around an eosinophilic fluid space, as shown below.

Call-Exner bodies - Granulosa cell tumor



Solution to Question 20:

Prophylactic oophorectomy is indicated if one ovary has already been removed for benign neoplasm.

Candidates suitable for prophylactic oophorectomy:

- Nulliparous aged > 45 years

- Any age with a history of ovarian, endometrial, or colon cancer in a 1st-degree relative
- Grossly abnormal ovary in the premenopausal age group
- One ovary removed for benign neoplasm
- Previous history of breast carcinoma
- Radiation-induced menopause

Solution to Question 21:

Stage 3 ovarian cancer is a unilateral or bilateral ovarian tumor with pelvic lymph node and/or regional lymph node involvement and liver capsule involvement.

Grading of ovarian cancer:

- Grade 1 - well-differentiated cancers have cells that closely resemble normal cells and are less likely to spread or recur.
- Grade 2 - moderately differentiated cancer cells.
- Grade 3 - poorly differentiated cancers show increasing abnormality of appearance compared to normal cells. They are also more likely to spread and recur.

Solution to Question 22:

Unilateral oophorectomy is preferred in a young woman who has not completed her family in case of early-stage disease.

Solution to Question 23:

Dysgerminoma is the most radiosensitive ovarian tumor. They are highly susceptible to radiotherapy.

Solution to Question 24:

CA-125 level measurement is the next best step according to the latest RCOG guidelines for post-menopausal ovarian cysts.

Management includes the following:

- Calculation of Risk of Malignancy Index (RMI) = $U \times M \times Ca-125$, where
- U = ultrasound score
- M = menopausal status

- Ca 125 = serum levels
- Laparotomy/laparoscopy only indicated if size ≥ 5 cm or RMI ≥ 200 .

Option B - CT scan is not recommended for initial evaluation. It is done only for further evaluation and when RMI ≥ 200 .

Solution to Question 25:

Signet ring cells are seen in Krukenberg tumor.

A Krukenberg tumor is an ovarian malignancy that metastasized from a primary site, most commonly from the stomach (70%). Carcinomas of the colon, appendix, and breast (mainly invasive lobular carcinoma) are the next most common primary sites.

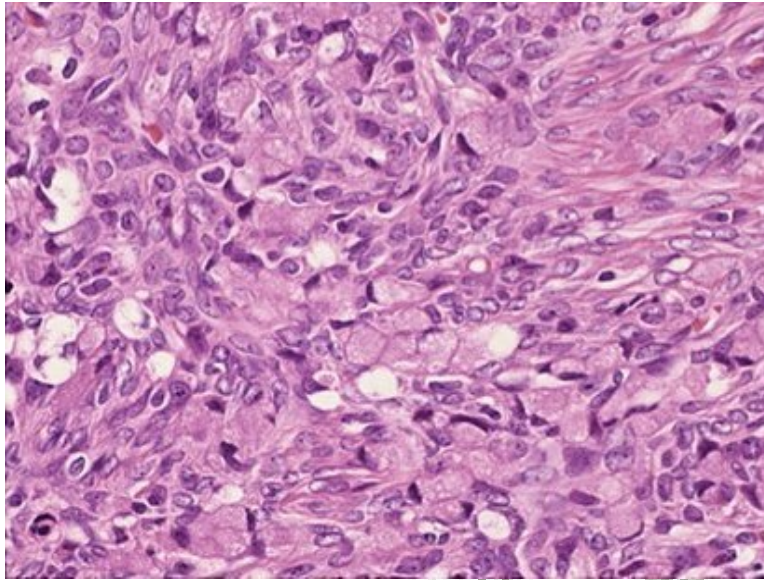
The route of spread may be:

- Lymphatic
- Hematogenous
- Transcoelomic

They are usually bilateral and have a smooth surface with a waxy consistency. They retain the shape of a normal ovary.

Given below are the images of the Krukenberg tumor's gross appearance and histological appearance with signet ring cells:





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Vulval & Vaginal Malignancy

Question 1:

Which among the following is not a premalignant lesion of vulva?

- a) Paget's disease
- b) Lichen sclerosus
- c) Condyloma lata
- d) Bowen's disease

Question 2:

Which of the following is a common differential diagnosis of verrucous carcinoma?

- a) Condyloma lata
- b) Condyloma acuminata
- c) Adenocarcinoma
- d) Chancre

Question 3:

A sexually active woman presents with multiple lesions over her genital region as shown below. Which of the following organisms are implicated in this condition?



- a) 2, 3 & 5
- b) 1, 3 & 5
- c) Only 4
- d) 2 & 4

Question 4:

What is the earliest symptom in patients with vulval intraepithelial neoplasia?

- a) Pruritus
- b) White discharge per vagina
- c) Dyspareunia
- d) Vaginal bleeding

Question 5:

A woman with vaginal itching is suspected to have vulval intraepithelial neoplasia. Which of the following is the most common cause of this condition?

- a) HPV 16
- b) HPV 18
- c) HSV 2
- d) Gardnerella vaginalis

Question 6:

An elderly woman is diagnosed with vulval intraepithelial neoplasia 3. Which of the following is false regarding this condition?

- a) Single lesion
- b) Associated with lichen sclerosis
- c) Progress to invasive cancer
- d) Human papilloma virus +ve

Question 7:

Sentinel lymph node biopsy is most useful in which of the following gynecological malignancies?

- a) Carcinoma vagina
- b) Carcinoma vulva
- c) Carcinoma cervix
- d) Carcinoma endometrium

Question 8:

Which is most common type of vulval cancer?

- a) Squamous cell carcinoma
- b) Adenocarcinoma
- c) Basal cell carcinoma
- d) Clear cell carcinoma

Question 9:

What is the most common type of vaginal cancer likely to be seen in a woman with a history of intrauterine di-ethyl stilbesterol exposure?

- a) Squamous cell carcinoma
- b) Melanoma
- c) Clear cell adenocarcinoma
- d) Fibrosarcoma

Question 10:

What is the most common site of vulval cancers?

- a) Clitoris
- b) Labia majora
- c) Labia minora
- d) Perineal area

Question 11:

Vulvar carcinoma was diagnosed in a 68-year-old woman who presented with itching and mass over the vulva for 6 months. Sentinel lymph node biopsy revealed positive inguinal nodes. What is the stage of the tumor?

- a) Stage I
- b) Stage II
- c) Stage III
- d) Stage IV

Question 12:

In patients with vulval cancer, what is the least common route through which the malignancy spreads?

- a) Direct
- b) Lymphatic
- c) Vascular
- d) Transcoelomic

Question 13:

An elderly woman is found to have abnormal cells on a pap smear. Further investigations reveal vaginal squamous cell carcinoma extending up to the pelvic wall. There is no evidence of distant metastasis. What is the stage of the tumor in this patient?

- a) Stage I
- b) Stage II
- c) Stage III
- d) Stage IV

Question 14:

A 70-year-old presented with a lesion over the vulva as shown below. Bilateral inguinal lymph nodes were palpable. Biopsy of the lesion revealed cells containing large, pleomorphic nuclei. Which of the following is false regarding this condition?



- a)** It is the 2nd most common vulval cancer
- b)** The most common site is clitoris
- c)** It may arise from a junctional naevus
- d)** Radical vulvectomy with bilateral regional lymphadenectomy is the preferred treatment

Answer Key

Question No.	Correct Option
1	c
2	b
3	d
4	a
5	a
6	d
7	b
8	a
9	c
10	b
11	c
12	c
13	c
14	b

Detailed Explanations

Solution to Question 1:

Condyloma lata is not a premalignant lesion of vulva.

Premalignant lesions of the vulva include:

- Vulval intraepithelial neoplasia (VIN) - most common
- Paget's disease
- Lichen sclerosis
- Squamous cell hyperplasia
- Condyloma acuminata
- Bowen's disease (erythroplasia of Querat)

Solution to Question 2:

Condyloma acuminata is the common differential diagnosis of verrucous carcinoma.

It refers to anogenital warts and has a verrucous appearance.

Verrucous carcinoma is a variant of squamous cell carcinoma of the cervix. It is locally invasive and rarely metastasizes.

Condyloma lata is a benign and painless cutaneous manifestation of secondary syphilis. It is characterized by gray to white lesions, typically located in the oro-genital region.

Chancre is a painless genital ulcer most commonly formed during the primary stage of syphilis.

Solution to Question 3:

The given scenario and image are suggestive of genital warts. Human papillomavirus (HPV) 6 and 11 cause nearly all genital warts.

It is usually clinically diagnosed but if co-existing neoplasia is suspected then a biopsy is required. HPV serotyping is not required for routine diagnosis.

Solution to Question 4:

Pruritus is the earliest symptom of VIN as well as vulvar cancer.

Clinical features of VIN include:

- Pruritus
- White discharge per vagina
- Dyspareunia
- Vaginal bleeding

Solution to Question 5:

HPV 16 is the most common cause of vulval intraepithelial neoplasia (VIN).

Causes of VIN include:

- HPV 16, 18, 31, 33
- HSV 2
- Syphilis
- Gonorrhea
- Gardnerella vaginalis

Solution to Question 6:

Elderly women with vulval intraepithelial neoplasia (VIN) - 3 are HPV negative.

VIN 3 is of two types:

- Young women – HPV +ve, multiple lesion, 90% regress
- Elderly women – HPV –ve, single lesion, associated with lichen sclerosis, often progress to invasive cancer

Solution to Question 7:

Among the given gynecological malignancies, sentinel lymph node biopsy (SLNB) is most useful in vulval carcinomas.

This is because women with earlier stages of vulval cancer can have diseased lymph nodes even without palpable enlargement of nodes. Hence, sentinel lymph node biopsy helps in early diagnosis lymph node spread which can be managed by lymph node dissection.

A sentinel lymph node is a physiologically, the first lymph node to receive tumor lymphatic drainage.

Malignancies where sentinel lymph node biopsy is commonly done are :

- Vulval carcinoma
- Breast carcinoma

- Penile carcinoma
- Malignant melanoma

Lymphatic mapping is similar technique to SLNB that can also be used in vulval cancers.

Solution to Question 8:

Squamous cell carcinoma is the most common type of vulval cancer.

Other types of vulval cancer include:

- Adenocarcinoma
- Melanoma
- Basal cell carcinoma
- Sarcoma

Solution to Question 9:

Clear cell adenocarcinoma of the vagina is commonly seen in adolescent girls with a history of intrauterine DES exposure.

It involves the upper third of the anterior vaginal wall. Radical hysterectomy, vaginectomy with pelvic lymphadenectomy is the treatment of choice.

The most common lesion due to DES exposure is vaginal adenosis and not clear cell adenocarcinoma. It is a benign condition.

Solution to Question 10:

The most common site of vulval cancer is labia majora & clitoris & labia minora.

Solution to Question 11:

According to FIGO staging, tumor with positive inguinal lymph node is classified as stage III.

Solution to Question 12:

Vascular route of spread is rare in vulval cancer. Lymphatic is the most common route of spread in vulval cancer.

Lymph node status is the most important prognostic factor. The sentinel lymph node biopsy is helpful for evaluating the spread of vulval cancer. Lymph nodes involved are, superficial inguinal nodes (sentinel node) & deep inguinal nodes & femoral group of nodes.

Solution to Question 13:

Vaginal cancer extending up to the pelvic wall is stage III.

Vaginal cancer commonly affects females >70 years of age. Squamous cell carcinoma is the most common histologic type. It commonly involves the upper third of the posterior wall of the vagina.

Patients are mostly asymptomatic. They may present with painless abnormal vaginal bleeding, post-coital bleed, and foul-smelling discharge per vagina.

Vaginal cancer staging:

- Stage 0 - Vaginal intraepithelial neoplasia
- Stage I - Carcinoma limited to the vaginal wall
- Stage II - Carcinoma extending beyond the vagina, but not extending to the pelvic walls
- Stage III - Carcinoma extends up to the pelvic walls
- Stage IVA - Carcinoma extending beyond the true pelvis/or involving the bladder and/or rectum, or evidence of distant metastasis
- Stage IVB - Spread to the distant metastasis

Solution to Question 14:

The given scenario is suggestive of vulval melanoma. The most commonly involved site is labia majora followed by labia minora and clitoris.

Vulval melanoma is the second most common vulval cancer. It may arise from junctional naevus. It has a high propensity for local and distant recurrence and has a poor prognosis.

Radical vulvectomy and bilateral regional lymphadenectomy is the preferred treatment.

Carcinoma Cervix

Question 1:

Which serotype of HPV is responsible for carcinoma cervix?

- a) 6, 11, 31, 32
- b) 16, 18, 31, 32
- c) 6, 11, 16, 18
- d) 16, 18, 31, 33

Question 2:

Which of the following is not true regarding HPV?

- a) E1 and E2 are viral proteins needed for replication
- b) E6 and E7 are viral proteins needed for malignant transformation
- c) E6 acts via the retinoblastoma gene
- d) HPV 16 is the most common serotype causing carcinoma cervix

Question 3:

Which is the HPV serotype most commonly associated with carcinoma cervix?

- a) 16
- b) 18
- c) 31
- d) 33

Question 4:

Which of the following is false about the Cervarix vaccine?

- a) Protects against HPV types 16 and 18
- b) Dose is 0.5ml, intramuscular
- c) Protective against genital warts

d) Dosing schedule – 0, 1 and 6 months

Question 5:

HPV nonavalent vaccine is active against which of the following serotypes?

- a)** 6, 11, 16, 18, 20, 23, 35, 42, 52
- b)** 6, 11, 16, 19, 23, 33, 45, 58
- c)** 6, 11, 16, 18, 32, 34, 45, 52, 58
- d)** 6, 11, 16, 18, 31, 33, 45, 52, 58

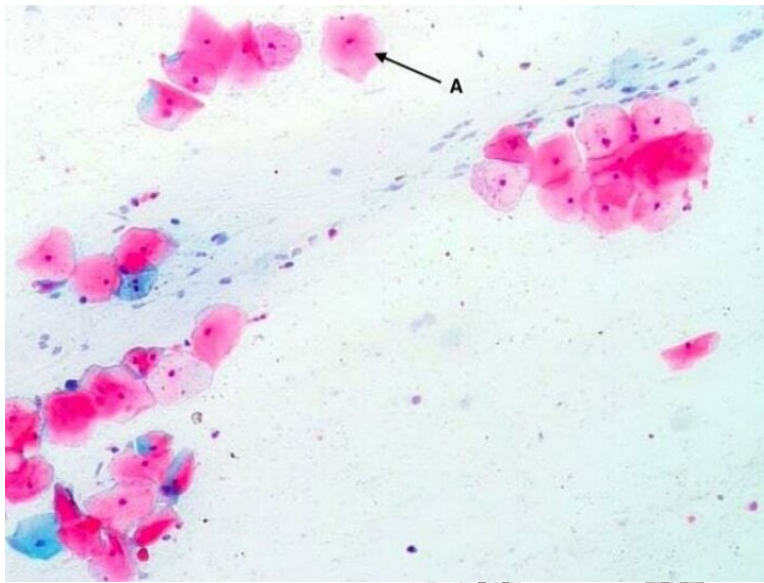
Question 6:

A 45-year-old patient comes with repeated infections of trichomonas vaginitis, her physician advises for a Pap smear. She asks you regarding the purpose of this test. What would be your appropriate response?

- a)** It is a screening test for cervical cancer
- b)** It is a diagnostic test for trichomonas infection
- c)** It helps to diagnose cancer in her genital tract
- d)** It helps to diagnose other associated genital infections in her

Question 7:

A 24-year-old woman comes to the gynecology OPD for a routine Pap smear. The histopathology slide is shown below. Identify the cell marked as "A".



- a) Superficial cell
- b) Basal cell
- c) Parabasal cell
- d) Intermediate cell

Question 8:

Match the following:

- a) A-2, B-1, C-4, D-5, E-3
- b) A-2, B-1, C-4, D-3, E-5
- c) A-1, B-2, C-4, D-5, E-3
- d) A-2, B-1, C-3, D-5, E-4

Question 9:

A 45-year-old lady complains of post-coital bleeding. She has a positive pap smear. What is the next line of management?

- a) Colposcopy directed biopsy
- b) Cone biopsy
- c) Repeat pap smear
- d) Hysterectomy

Question 10:

A 30-year-old sex worker presented with irregular menses, post-coital bleeding, and occasional foul-smelling vaginal discharge. She was advised to undergo a Pap smear. What is the best time for this investigation?

- a) Immediately after menstruation
- b) Around ovulation
- c) Secretory phase
- d) During menstruation

Question 11:

Which of the following is false about Schiller's test?

- a) Lugol's iodine is used to paint the cervix
- b) The epithelium is stained mahogany brown
- c) Abnormal vaginal epithelium is stained
- d) It detects the presence of glycogen in the epithelium

Question 12:

Which of the following would you not expect to see in the colposcopic biopsy of a patient with cervical intraepithelial neoplasia?

- a) Dysplasia
- b) Breach in basement membrane
- c) Altered N:C ratio
- d) Mitotic figures

Question 13:

Which of the following methods is most commonly used for the treatment of CIN?

- a) Cold-knife conization
- b) Large loop excision of transformation zone
- c) Electrocoagulation
- d) Needle excision of transformation zone

Question 14:

Which among the following is the treatment of choice when microinvasion of cervix neoplasia is suspected?

- a) LLTEZ
- b) LEEP
- c) Conization
- d) Hysterectomy

Question 15:

A 40-year-old woman has cervical intraepithelial neoplasia III. What would be the management?

- a) Hysterectomy
- b) Trachelectomy
- c) Conization
- d) Colposcopy with LEEP

Question 16:

Which is the most common malignancy in women in India?

- a) Cervical cancer
- b) Breast cancer
- c) Endometrial cancer
- d) Ovarian cancer

Question 17:

Which of the following is false regarding the transitional zone?

- a) CIN originates from this site
- b) It is a dynamic point
- c) Nabothian cysts mark the original site

- d) It is limited to external os

Question 18:

Identify the false statement regarding adenocarcinoma of the cervix.

- a) Most common site is transformation zone
- b) Cervical cancer is the second most common cancer among women in India
- c) Most common in OCP users
- d) Associated with HPV 18

Question 19:

You would suspect and evaluate for carcinoma cervix in patients presenting with which of the following conditions?

- a) Menorrhagia
- b) Metrorrhagia
- c) Polymenorrhea
- d) Amenorrhea

Question 20:

What is the most common site of hematogenous metastasis of cervical cancer?

- a) Lung
- b) Liver
- c) Bone
- d) Bowel

Question 21:

A 65-year-old woman comes for follow-up a few months after surgery for carcinoma cervix. On evaluation, she is found to have a recurrence of the malignancy. Which of the following is true about her condition?

- a) Most common site of recurrence is pelvic lymph node
- b) Mortality in patients with recurrence is due to pulmonary embolism

- c) Treatment involves surgery with lymph node dissection
- d) Recurrent tumor with largest diameter ≤ 3 cm has good prognosis

Question 22:

Identify the true statement about the staging of cancer cervix.

- a) Clinical examination is the first step in staging
- b) Tumor markers are important for diagnosis of metastasis
- c) CT scan is the best method to assess principal tumors
- d) MRI is the best method to assess nodal metastasis

Question 23:

The CT of a patient with cervical carcinoma revealed a lesion of 6 cm, extending to the upper part of the vagina and parametrium. What is the stage of the disease?

- a) Stage II A
- b) Stage II B
- c) Stage III A
- d) Stage III B

Question 24:

A 50-year-old woman presented to her gynecologist with postcoital bleeding. She was diagnosed with carcinoma cervix stage IB after evaluation. What is the treatment of choice for this patient?

- a) Type I hysterectomy
- b) Type II hysterectomy
- c) Type III hysterectomy
- d) Type IV hysterectomy

Question 25:

A 30-year-old woman is found to have a lesion in the cervix which is 7 mm wide and has a depth of 4 mm. She has one child and wishes to preserve fertility. What is the most

appropriate management?

- a) Cryoablation
- b) Cone biopsy
- c) Radical trachelectomy with pelvic lymphadenectomy
- d) LEEP

Question 26:

Which among the following are contraindications for radiotherapy in carcinoma cervix?

- a) 1, 2, 3
- b) 2, 4, 5
- c) 1, 2, 3, 5
- d) 1, 2, 4, 5

Question 27:

You are planning to administer radiotherapy to a 58-year-old woman diagnosed with carcinoma cervix. What is the dose to be administered at point B?

- a) 7000 cGy
- b) 6000 cGy
- c) 5000 cGy
- d) 8000 cGy

Answer Key

Question No.	Correct Option
1	d
2	c
3	a
4	c
5	d
6	a

7	a
8	a
9	a
10	b
11	c
12	b
13	b
14	c
15	d
16	b
17	d
18	a
19	b
20	a
21	d
22	a
23	b
24	c
25	c
26	d
27	b

Detailed Explanations

Solution to Question 1:

HPV serotypes 16, 18, 31, 33, 35, 39, 45, 52, 56, 58, 59, 68 are associated with carcinoma cervix.

Risk factors for carcinoma cervix include:

- Human Papilloma virus (most important)
- Sexual intercourse before 18 years
- Multiple sexual partners
- Multiparity
- Poor personal hygiene
- Smoking and drug abuse
- Women with STD, HIV, HSV
- Immune suppression

Polymerase chain reaction (PCR) or southern blot or Hybrid capture technique is used for HPV DNA detection.

Solution to Question 2:

E6 acts via the tumor suppression gene p53. It increases proteasome-dependent p53 degradation. E7 binds and acts via the retinoblastoma (Rb) gene.

The viral E6 and E7 oncoproteins encoded by the high-risk type HPVs (such as HPV 16 and HPV 18) are necessary for malignant transformation.

E1 and E2 oncoproteins are responsible for viral replication.

Solution to Question 3:

HPV 16 is the most common type associated with carcinoma cervix.

Associations between human papilloma virus (HPV) serotypes and carcinoma cervix:

- Most common serotype - HPV 16
- Most malignant serotype - HPV 18
- Most specific serotype - HPV 18
- Most common serotype associated with
- Squamous cell carcinoma - HPV 16
- Adenocarcinoma - HPV 18

Cancers caused by HPV:

- Male - penile, oral, anal
- Female - cervix, vagina, vulva

Solution to Question 4:

The Cervarix vaccine is not protective against genital warts. It is mainly protective against CIN and cervical cancer.

Vaccine Type	Cervarix	Gardasil	Gardasil 9
	Bivalent vaccine	Quadrivalent vaccine	Nonavalent vaccine
HPV types	HPV 16,18	HPV 6, 11, 16, 18	HPV 6, 11, 16, 18, 31, 33, 45, 52, and 58
Dosage	0.5 mL	0.5 mL	0.5 mL
Schedule	0, 1, 6 months	0, 2, 6 months	0, 2, 6months

Vaccine Type	Cervarix	Gardasil	Gardasil 9
Diseases prevented	CIN Carcinoma cervix	CIN Carcinoma cervix Genital warts	CIN Carcinoma cervix Genital warts Vulval cancer Vaginal epithelial cancer Oropharyngeal cancer
Adjuvant	ASo4	Alum	Alum

Solution to Question 5:

HPV nonavalent vaccine (Gardasil 9 or HPV9) is active against HPV 6,11,16,18,31,33,45,52,58.

Currently, HPV nonavalent vaccine has replaced Gardasil which is a quadrivalent vaccine acting against 6,11,16,18.

Vaccines available for HPV:

Brand name	Cervarix	Gardasil	Gardasil-9	Cervavac
Characteristic and HPV subtypes	Bivalent:16,18	Quadrivalent:6,11,16,18	Nonavalent:6,11,16,18,31,33,45,52,58	Quadrivalent:6,11,16,18
Protects against:	CINCA cervix	Anogenital warts CINCA cervix	Anogenital warts CINCA cervix Vulval intraepithelial neoplasia Vaginal intraepithelial neoplasia Anal cancer Oropharyngeal cancer Head and neck cancers	Anogenital warts CINCA cervix
Dose	0.5 ml, IM.	0.5 ml, IM.	0.5 ml, IM.	0.5 ml, IM.
Ideal Age range:	Ideally at 11-12 years or 9-26 years	Ideally at 11-12 years or 9-26 years	Ideally at 11-12 years or 9-26 years	The targeted age group is 9-26 years

Solution to Question 6:

A pap smear is a screening test for cervical cancer. It is not a diagnostic test (option C). The presence of abnormal cells/ a positive test requires further evaluation by colposcopy, cervical biopsy, or fractional curettage.

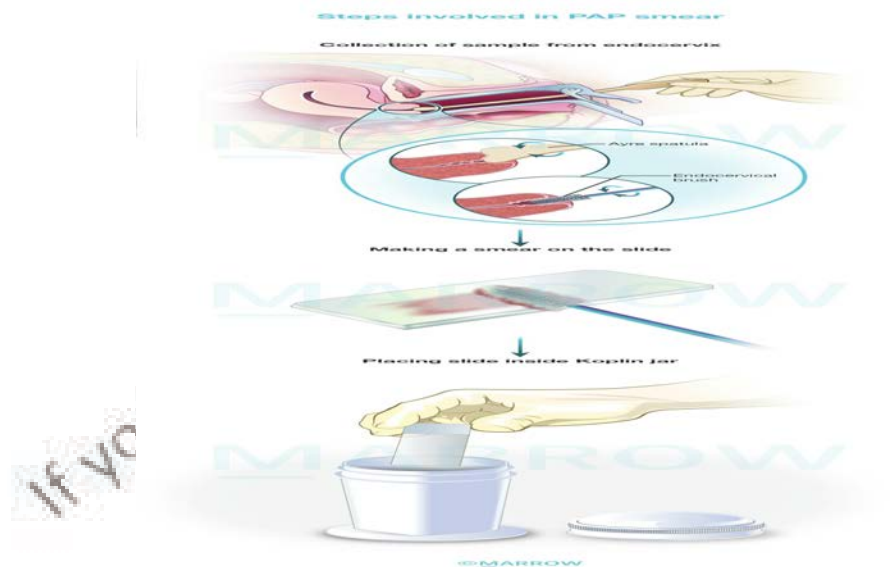
If trichomonas is reported from Pap testing, it should be confirmed by saline microscopic preparation (option B).

Cervical cancer screening should start at age 21. Women aged 21-29 should have a Pap smear every 3 years, while women aged 30-65 should have both HPV and Pap co-test every 5 years. Women should stop having cervical cancer screening after the age of 65 years. If suspicious PAP, screening till 75 years. HIV+ women should have annual checkups until three consecutive negative tests followed by testing every 3 years for life.

Pap testing should be done prior to vaginal examination. Lubricants should not be used and the patient should not be menstruating during the pap test. The patient should not have intercourse 24hrs prior to the test. The ideal time is around ovulation.

Procedure:

- The patient is put in a dorsal position, and the labia are parted
- Cusco's self-retaining speculum is introduced
- The squamocolumnar junction is scraped with Ayer's spatula:
- Bifid end - ectocervical sample
- Blunt end - vaginal cytology
- Endocervix is scraped with a cytobrush
- The sample is spread on a glass slide
- The slide is immediately fixed using 95% ethyl alcohol
- Stained with Papanicolaou stain or short-duration Papanicolaou stain (SPS).



Instruments used for Pap smear

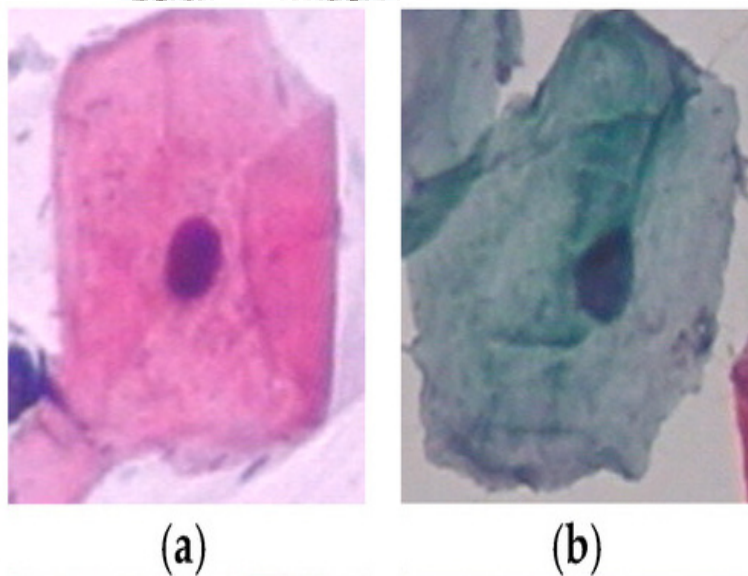


Solution to Question 7:

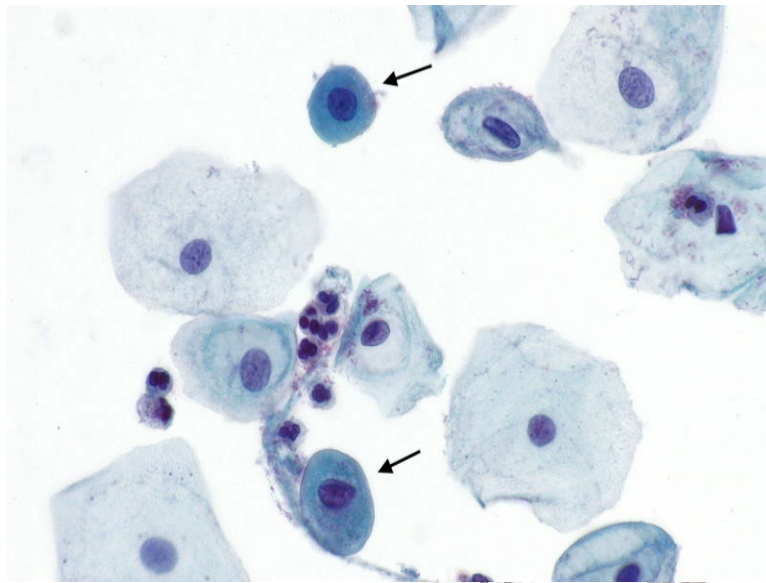
The cell marked A is the superficial cell.

ACOG guidelines recommend cervical exfoliative cytology every 3 years starting 21 years as screening for cervical cancer.

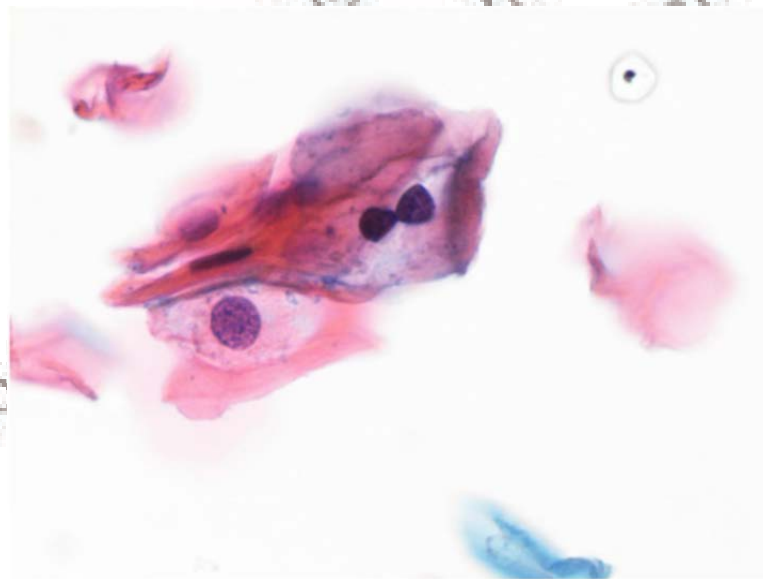
The image below shows a superficial cell (a) and an intermediate cell (b):



The image below shows parabasal cells (marked by black arrows):



Koilocytes are epithelial cells that contain eccentric, hyperchromatic, moderately enlarged nucleus that is displaced by large perinuclear vacuole. It is a feature of HPV infection.



Solution to Question 8:

Various systems of classification are used to describe cervical cytology after Pap smear such as:

- Papanicolaou classification - describes the exfoliative cytology of the cervix in a Pap smear.
- Cervical intraepithelial neoplasia (CIN) - histopathological description of how much of the thickness of the cervical epithelium is replaced by dysplastic cells
- Bethesda classification - it is another method to describe cervical dysplasia

The various systems and their corresponding classes are given below:

where, CIS: Carcinoma in situ, CIN: Cervical intraepithelial neoplasia, SCC: Squamous cell carcinoma, SIL: Squamous intraepithelial lesion, ASCUS: Atypical squamous cell of undetermined significance, HPV: Human papillomavirus.

Pap classification (1943)	Cervical intraepithelial neoplasia / CIN (WHO 1975)	Bethesda system
Grade I	Normal	Normal
Grade II	Inflammatory	Inflammatory HPV ASCUS
Grade III	CIN I	Low SIL
Grade IV	CIN II, CIN III, CIS	High SIL
Grade V	SCC	SCC

Solution to Question 9:

The appropriate management of a 45-year-old with a positive pap smear and post-coital bleeding is a colposcopy-directed biopsy.

A per speculum examination must be done whenever a female comes with post-coital bleeding in order to evaluate for carcinoma of the cervix. If growth is visible, a punch biopsy should be done, and else if no growth is visible, then a pap smear is done. A pap smear is a screening test for cervical cancer. The presence of abnormal cells/ a positive test requires further evaluation by colposcopy, cervical biopsy, or fractional curettage.

Cervical cancer screening should start at age 21. Women aged 21-29 should have a pap smear every 3 years, while women aged 30-65 should have both HPV and pap co-test every 5 years. Women should stop having cervical cancer screening after the age of 65 years. If a pap smear is suspicious, then screening is continued till 75 years. HIV+ve women should have annual checkups until three consecutive negative tests followed by testing every 3 years for life.

Pap testing should be done prior to vaginal examination. Lubricants should not be used and the patient should not be menstruating during the pap test. The patient should not have intercourse 24hrs prior to the test. The ideal time is around ovulation.

Solution to Question 10:

The given case scenario is suggestive of cervical carcinoma. The best time to do a Pap smear is around ovulation, but it can be performed at any other time as well.

Prerequisites of doing a Pap smear:

- It should be done prior to vaginal examination
- Lubricants should not be used
- The patient should not be menstruating during the pap test

- The ideal time is around ovulation
- The patient should not have intercourse 24hrs prior to the test.

Based on Pap smear findings, dysplasia of cervical cells is graded using various classification systems such as WHO grading, CIN grading, and the Bethesda system.



Solution to Question 11:

The abnormal cells remain unstained on performing Schiller's test.

Schiller's test, also known as VILI (visual inspection with Lugol's iodine), involves painting the cervix with Lugol's iodine. It is used to detect any abnormal cytology in the cervix and vagina, such as in cases of carcinoma cervix.

The normal vaginal and cervical epithelium contains glycogen, therefore stains mahogany brown on being painted with Lugol's iodine. The abnormal cells remain unstained. These areas need to be biopsied and sent for histopathological examination.

Solution to Question 12:

The basement membrane is intact in cervical intraepithelial neoplasia (CIN).

Features of CIN:

- Varying degree of dysplasia
- Altered nuclear-cytoplasmic (N:C) ratio
- Hyperchromatic nuclei
- Mitotic figures
- Intact basement membrane.

Solution to Question 13:

Large loop excision of the transformation zone (LLETZ) is the most commonly used treatment option for cervical intraepithelial neoplasia (CIN). It is also known as loop electrosurgical excision procedure (LEEP).

It is applicable for the treatment of CIN anywhere in the lower genital tract. This uses electric current through a monopolar wire electrode to either cut or coagulate cervical lesions. Like other excisional procedures, it is done in the immediate post menstrual phase.

The procedure involves minimal patient discomfort, cost, and complications. Since it can be performed using local anesthesia, it has become the primary outpatient treatment modality for high-grade cervical lesions.

Solution to Question 14:

Conization is the treatment of choice when microinvasion is suspected.

- Indications of conization
- Endocervical dysplasia
- Transformation zone is not completely visualized
- When there is discrepancy of findings between cytology, colposcopy & biopsy
- Microinvasion is suspected
- Indications of hysterectomy
- Older & parous women
- Uterus is associated with fibroid, DUB, prolapsed
- Recurrence or persistent lesion
- In-situ adenocarcinoma cervix

Solution to Question 15:

Colposcopy with LEEP would be appropriate management for CIN III.

Loop electrosurgical excision procedure (LEEP) is an opd procedure done under colposcopic view and local anesthesia. Here, a low voltage current is passed through the loop and a cone of tissue is removed from the cervix. This procedure causes minimal bleeding and the tissue removed is sent for histopathological examination (HPE). This is the treatment of choice for CIN 2&3 of any age and parity.

Solution to Question 16:

Breast cancer is the most common cancer in women in India as well as globally. Cervical cancer is the most common genital cancer in India.

Lip and oral cavity cancer is the most common malignancy in men.

Risk factors for carcinoma cervix include:

- Human Papilloma virus (most important)
- Sexual intercourse before 18 years
- Multiple sexual partners
- Multiparity
- Poor personal hygiene
- Smoking and drug abuse
- Women with STD, HIV, HSV
- Immune suppression

Solution to Question 17:

The transitional zone or squamocolumnar junction is well outside the external os during the reproductive period.

It is the site where the squamous epithelium lining the vagina merges with the columnar epithelium of the endocervix. The original squamocolumnar junction is marked by the presence of Nabothian cysts and glandular openings.

It is a dynamic part and moves with age and hormonal influence:

- During pregnancy and with oral contraceptives, it pouts out of os.
- Following menopause, it gets indrawn inside the os.

The cells here are highly sensitive to irritants, mutagens and viral agents such as HPV. Cancer screening using Pap smear involves scraping this area and studying the cells for nuclear changes.

Solution to Question 18:

The most common site of adenocarcinoma is endocervix.

It commonly occurs in young females and OCP users, and is associated with HPV 18.

Solution to Question 19:

Patients with carcinoma cervix most commonly present with metrorrhagia, now known as intermenstrual bleeding.

Post-coital bleeding is the most specific symptom of carcinoma cervix.

Solution to Question 20:

The most common site of hematogenous metastasis of cervical cancer is the lung.

About 1-2% of patients with cervical carcinomas present with lung metastases, and 5-35% eventually develop lung metastases.

Other common sites of blood-borne metastases include bone, liver and bowel.

Solution to Question 21:

A recurrent tumor with the largest diameter of ≤ 3 cm is associated with a good prognosis.

The extent of recurrent disease and involvement of pelvic lymph nodes are prognostic factors for survival. An isolated central pelvic tumour with no involvement of the pelvic sidewall also has a good prognosis.

Most recurrences are seen within 3 years and the prognosis is poor with uremia being the most common terminal event. Recurrences most commonly involve the pelvis than the para-aortic or pelvic lymph node.

All patients with a suspected recurrence should undergo PET/CT for evaluation and biopsy for confirmation.

Treatment involves either radical chemoradiation or pelvic exenteration. Concurrent chemotherapy with cisplatin and/or 5-fluorouracil may improve the outcome.

Solution to Question 22:

Clinical examination is the first step in staging carcinoma cervix.

Clinical examination was the principal method in staging until the 2018 FIGO guidelines which recommended the use of imaging and histological investigations when available. USG, CT, MRI, and PET-CT can be used to assess the extent of the tumor, nodal status, and local or systemic spread.

MRI is the best method to assess the principal tumor while PET-CT is the best method to assess nodal metastasis.

FIGO no longer mandates biochemical investigations, such as tumor markers, for diagnosis of metastasis.

Solution to Question 23:

The stage of cervical carcinoma is II B as it involves the upper part of the vagina and parametrium.

Solution to Question 24:

Type III hysterectomy is the treatment of choice in carcinoma cervix - stage IB. It involves radical hysterectomy with pelvic lymphadenectomy and para-aortic lymph node evaluation.

Stage IB is defined as a macroscopic lesion still confined to the cervix, with the depth of invasion $\geq 5\text{mm}$. It is the first stage in which lesions become clinically visible.

Hysterectomy types include:

- Type I - Simple / extrafascial
- Type II - Modified radical / Wertheim
- Type III - Radical / Meigs-Wertheim
- Type IV - Extended radical Hysterectomy
- Type V - Exenteration.

Solution to Question 25:

The woman has stage I A2 cervical cancer and the most appropriate management is radical trachelectomy with pelvic lymphadenectomy.

Radical trachelectomy is a fertility-preserving surgical procedure. This involves the removal of the entire cervix and parametrium.

Cervical conization with laparoscopic or extraperitoneal pelvic lymphadenectomy is an alternative that also preserves fertility.

Solution to Question 26:

Post-menopausal status is not a contraindication for radiotherapy in carcinoma cervix.

Contraindications for radiotherapy in carcinoma cervix include:

- Pelvic inflammatory disease
- Young patient (to preserve ovarian function)
- Associated myoma, prolapse, ovarian tumor, genital fistula
- Vaginal stenosis
- Adenocarcinoma or adenosquamous carcinoma – surgery preferred.

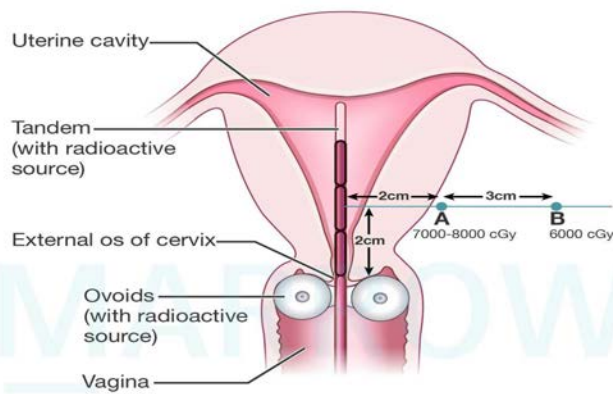
Solution to Question 27:

In the treatment of carcinoma cervix, the dose to be administered at point B is 6000 cGy.

Radiotherapy in carcinoma cervix:

	Point A	Point B
Location	2cm above and 2cm lateral to external os	2cm above and 5cm lateral to external os
Structure	Paracervical/Parametrial lymph node	Obturator lymph node
Dose	7000-8000 cGy	6000 cGy

Radiotherapy for carcinoma cervix



Carcinoma Endometrium

Question 1:

What is the most common cause of postmenopausal bleeding?

- a) Senile endometrial atrophy
- b) Cervical cancer
- c) Endometrial cancer
- d) Atypical endometrial hyperplasia

Question 2:

A 58-year-old, postmenopausal woman presented with bleeding PV. Hysteroscopic biopsy revealed complex endometrial hyperplasia with atypia. What is the risk of progression of this condition to cancer?

- a) 3%
- b) 54%
- c) 8%
- d) 29%

Question 3:

Which is the most common benign cause of pyometra?

- a) Endometrial hyperplasia
- b) Senile endometritis
- c) Pelvic inflammatory disease
- d) Cervical stenosis

Question 4:

Which among the following is an ideal treatment for a 55-year-old female with simple hyperplasia of the endometrium with atypia?

- a) Simple hysterectomy

- b) Medroxy progesterone acetate
- c) Levonorgesterol
- d) Radiotherapy

Question 5:

What is the minimum endometrial thickness on transvaginal ultrasonography, above which you perform a biopsy in a case of postmenopausal bleeding?

- a) 3mm
- b) 4mm
- c) 7mm
- d) 9mm

Question 6:

What is the most common histopathological variety of endometrial cancer?

- a) Squamous cell
- b) Adenocarcinoma
- c) Clear cell
- d) Serous

Question 7:

Which of the following predisposes a patient to develop type 2 carcinoma endometrium?

- a) KRAS mutation
- b) Premenopausal age group
- c) PTEN mutation
- d) Atrophic endometrium

Question 8:

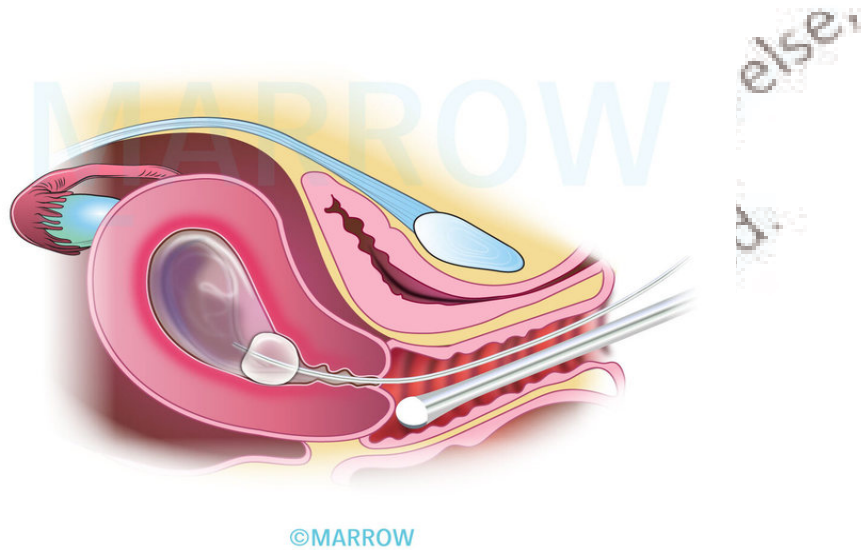
Which of the following constitute corpus cancer syndrome?

- a) 1, 2, 3, 5

- b) 1, 3, 4, 6
- c) 1, 2, 4, 5
- d) 1, 3, 4, 5

Question 9:

A gynecologist is performing the following procedure in a woman with postmenopausal bleeding. Identify it.



- a) Transvaginal sonography
- b) Balloon endometrial ablation
- c) Saline infusion sonography
- d) Hysterosalpingography

Question 10:

A menopausal patient suspected to have endometrial carcinoma is undergoing a transvaginal ultrasound. Which of the following findings would be inconsistent with her diagnosis?

- a) Endometrial thickness of 7 mm
- b) Hyperechoic endometrium
- c) Increased vascular resistance
- d) Fluid in uterine cavity

Question 11:

A 52-year-old woman with postmenopausal bleeding has an endometrial thickness of 7 mm on ultrasound. What is the next step in her management?

- a) Follow up USG after 3 months
- b) Hysterectomy
- c) Endometrial biopsy
- d) Progesterone therapy

Question 12:

A 54-year-old woman with postmenopausal bleeding is suspected to have an endometrial lesion. What is the gold standard investigation to confirm her diagnosis?

- a) Fractional curettage
- b) Dilatation and curettage
- c) Hysteroscopic biopsy
- d) Laparoscopic biopsy

Question 13:

A 62-year-old woman presents with vaginal bleeding 11 years after she had attained menopause. An endometrial biopsy reveals adenocarcinoma. What is the gold standard method for staging her condition?

- a) MRI
- b) Surgery
- c) Endometrial biopsy
- d) PET-CT

Question 14:

Which lymph nodes are most commonly involved in patients with endometrial carcinoma?

- a) Femoral
- b) Inguinal
- c) Para-aortic

d) Para-colic

Question 15:

A patient presented with carcinoma endometrium involving $>50\%$ myometrium, extending to the vagina but with no involvement of para-aortic and pre aortic nodes. What is the stage of her disease?

- a) III A**
- b) III B**
- c) III C1**
- d) III C2**

Question 16:

Which of the following is indicated for the treatment of a patient with early endometrial cancer?

- a) Hysterectomy, B/L salpingo-oophorectomy, lymph node sampling**
- b) Hysterectomy, B/L salpingo-oophorectomy, chemoradiation**
- c) Hysterectomy, radiotherapy**
- d) Chemoradiation**

Question 17:

A woman with endometrial carcinoma is undergoing radiotherapy. Which of the following is true?

- a) Intensity is inversely proportional to squared distance**
- b) Small blood vessels are most radioresistant**
- c) Rapidly proliferating cells are most radioresistant**
- d) Small bowel is most radioresistant**

Question 18:

Which among the following is the mainstay of treatment in a patient diagnosed with Stage III endometrial cancer?

- a) Surgery
- b) Chemotherapy
- c) External beam radiotherapy
- d) Brachytherapy

Answer Key

Question No.	Correct Option
1	a
2	d
3	b
4	a
5	b
6	b
7	d
8	c
9	c
10	c
11	c
12	c
13	b
14	c
15	b
16	a
17	a
18	b

Detailed Explanations

Solution to Question 1:

The most common cause of postmenopausal bleeding all over the world is senile endometrial atrophy.

In a menopausal woman, hypoestrogenism causes atrophy of the endometrium and vagina. The atrophic endometrial surfaces contain little or no fluid to prevent intracavitary friction. This

results in micro-erosions of the surface epithelium and a subsequent chronic inflammatory reaction, making the endometrium prone to light bleeding or spotting.

Hormone replacement therapy is the second most common cause of post-menopausal bleeding all over the world.

Cervical cancer is the most common cause of postmenopausal bleeding in India, while endometrial cancer is the most common cause of postmenopausal bleeding in developed countries.

Note: Endometrial cancer accounts for only 10 % of cases of post-menopausal bleeding. Still, all postmenopausal women with unexpected uterine bleeding should be evaluated for endometrial carcinoma, since this is a potentially lethal disease.

Solution to Question 2:

Complex hyperplasia with atypia has a 29% increased risk of progression to endometrial cancer. It has the highest risk of progressing to carcinoma compared to other varieties of endometrial hyperplasia.

The risk of endometrial hyperplasia progressing to cancer is tabulated below.

Type of hyperplasia	Progression to cancer
Simple without atypia	1%
Complex without atypia	3%
Simple with atypia	8%
Complex with atypia	29%

Solution to Question 3:

The most common cause of pyometra is senile endometritis.

Pyometra refers the collection of pus in the uterus. The most common cancer that causes pyometra is carcinoma cervix followed by carcinoma endometrium.

Solution to Question 4:

Simple hysterectomy is the best treatment in a case of atypical endometrial hyperplasia in a peri-menopausal and post-menopausal female.

Endometrial hyperplasia with atypia carries 29% risk of developing into endometrial carcinoma and hence, must be treated with total abdominal hysterectomy (simple hysterectomy).

The presence of any concomitant endometrial cancer must be ruled out by doing fractional curettage (or dilatation and curettage) before doing a hysterectomy.

If a premenopausal woman does not desire to undergo hysterectomy, progesterone can be given and the patient is followed up every 3-6 months. If the atypia persists during the follow-up hysterectomy should be performed.

The table below summarizes the management of patients with endometrial hyperplasia

Management of Endometrial hyperplasia	
Hyperplasia without atypia	
Pre-menopausal women	Progesterone therapy
Postmenopausal women: Simple hyperplasia without atypia Complex hyperplasia without atypia	Follow up without therapy Progesterone therapy
Hyperplasia with atypia	
Pre-menopausal women	If patient wishes to preserve uterus, give High-dose progesterone therapy. If atypia persists, then perform Total abdominal hysterectomy
Peri-menopausal women	Total abdominal hysterectomy
Post-menopausal women	Total abdominal hysterectomy + BSO (bilateral salphingo-oophorectomy)

Solution to Question 5:

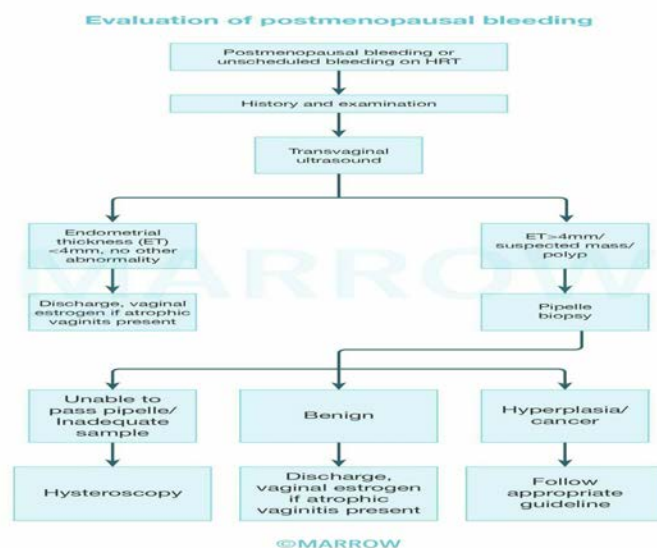
A finding of endometrial thickness ≥ 4 mm (according to ACOG) in a postmenopausal women requires further evaluation, such as endometrial biopsy.

If the endometrial thickness is ≤ 4 mm in a post menopausal female, malignancy is rare and hence, can be managed by tranexamic acid. If the bleeding still persists, the patient needs to undergo fractional curettage and hysteroscopy.

The algorithm for next step for abnormal uterine bleeding in different age groups is as follows:

Age group	Endometrial thickness on TVS	Action
Age <40 years	>12mm	Endometrial sampling/Endometrial biopsy
Age >40 years or Age >45 years	-	Endometrial sampling/Endometrial biopsy is done irrespective of usg findings

Age group	Endometrial thickness on TVS	Action
Post menopausal women	$\geq 4\text{mm}$	Endometrial sampling/Endometrial biopsy
Post menopausal women	$< 4\text{mm}$	Reassurance+tranexamic acid. If bleeding still persists then do endometrial biopsy



Solution to Question 6:

The most common variety of endometrial cancer is adenocarcinoma.

The other varieties of endometrial cancer are squamous cell, clear cell, and serous. The most malignant varieties are clear cell carcinoma and papillary serous tumor.

Solution to Question 7:

Type 2 carcinoma endometrium is commonly seen in menopausal women with atrophic endometrium.

Histological classification of endometrial cancer:

Note: Type 1 (estrogen-dependent) endometrial CA is the more common type.

Type 1 (Endometrioid)	Type 2 (Nonendometrioid)
Estrogen dependent	Estrogen independent
Occurs in Pre- and Peri-menopausal women	Occurs in post-menopausal women

Type 1 (Endometrioid)	Type 2 (Nonendometrioid)
Hyperplasia present	Hyperplasia absent
Good prognosis	Poor prognosis
Associated with PTEN/KRAS mutation	Associated with p53 mutation

Solution to Question 8:

Corpus cancer syndrome constitutes of:

- Diabetes mellitus
- Obesity
- Hypertension
- Cancer of the body of the uterus.

Etiology of endometrial carcinoma: (Mnemonic: Family Has OLD AUNT)

- Family history
- Hypertension
- Obesity
- Late menopause/Early menarche
- Diabetes
- Atypical endometrial hyperplasia
- Unopposed estrogen/increased estrogen → Hormone replacement therapy, fibroid, PCOD
- Nulliparous
- Tamoxifen therapy (Raloxifene has no genito-urinary action)
- Infertility/irregular menses.

Protective factors against endometrial carcinoma include the use of OCPs, smoking, and multiparity.

Solution to Question 9:

The image depicts saline infusion sonography (SIS), also known as sonohysterography. It is usually done after a transvaginal ultrasound, to assess the detailed anatomy of the uterine cavity.

In this procedure, a balloon catheter is inserted through the cervix and the endometrial cavity is distended with 20-40ml of sterile saline. The balloon is inflated to block the backflow of the infused sterile saline and transvaginal ultrasound is performed to visualize the cavity.

It is used:

- To determine the endometrial thickness
- To determine the cause in cases of infertility
- To identify Mullerian anomalies
- To visualize endometrial polyps and pedunculated fibroids

The image given below shows the appearance of the uterus on SIS.



Option A: Transvaginal ultrasound is the initial evaluation for postmenopausal bleeding.

Option B: In balloon endometrial ablation, a balloon is inserted into the uterus and inflated with glycine or 5% glucose. It is then heated up, and the endometrium is ablated by the tamponade effect.

Option D: In hysterosalpingography, contrast dye is inserted into the uterus via a catheter, and the uterine and fallopian tube anatomy is visualized.

Solution to Question 10:

Decreased vascular resistance with increased tumor vascularity is seen in carcinoma endometrium.

USG findings in carcinoma endometrium include:

- Endometrial thickness ≥ 4 mm in postmenopausal women and ≥ 10 mm in perimenopausal women
- Hyperechoic endometrium with irregular outlines
- Increased tumor vascularity with low vascular resistance
- Fluid in uterine cavity

Solution to Question 11:

The next line of management in a patient with increased endometrial thickness is an endometrial biopsy.

In a postmenopausal woman, the normal endometrium should not exceed 4 mm in thickness. If it exceeds 4 mm, then histopathological examination (HPE) of the endometrium is mandatory, irrespective of the presence or absence of symptoms. In a perimenopausal woman, the cutoff for normal endometrial thickness is 10mm.

Hysteroscopy and biopsy is preferred over other blind curettage methods for HPE. A biopsy is made from suspicious areas after visualization so chances of missing the lesion are reduced.

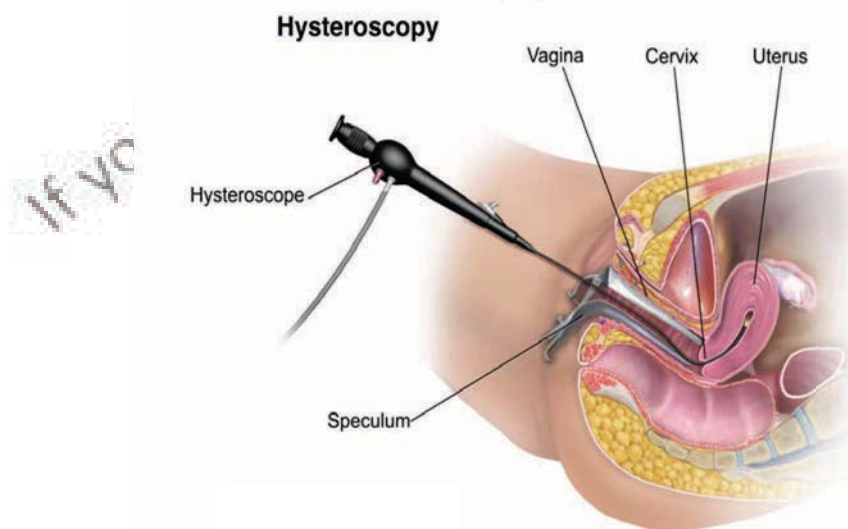
Solution to Question 12:

Hysteroscopic inspection and selective biopsy is now considered the gold standard in the diagnosis of an endometrial lesion. It is preferred for the initial evaluation of women with bleeding suspicious for malignancy.

Endometrial sampling can be done by hysteroscopic biopsy. It is more sensitive for focal endometrial lesions. Dilatation and curettage may also aid the sampling of endometrial lesions.

Ultrasound, CT, and MRI are adjuvant to other investigations. Diagnostic laparoscopy will be required to study the nature of the tumor and its spread if ultrasound picks up a pelvic tumor.

The image given below shows the procedure of hysteroscopy.



Solution to Question 13:

The given scenario depicts endometrial carcinoma and the gold standard method for staging this condition is surgery.

According to FIGO/TNM classification system, a total hysterectomy with bilateral salpingo-oophorectomy and pelvic and para-aortic lymph node dissection is the standard staging procedure for endometrial cancer.

Option A: Contrast-enhanced magnetic resonance imaging (MRI) appears to be the best radiographic modality for detecting myometrial invasion or cervical involvement.

Option C: Hysteroscopy visualizes the entire uterine lining and a biopsy is taken from suspicious areas. This is not 100% predictive, as an early lesion can be missed. Recently, the concern has been raised regarding the spilling of cancer cells into the peritoneal cavity during hysteroscopy.

Option D: PET-CT reveals metabolic activity in the tissue and is highly sensitive and specific for detecting positive pelvic and/or para-aortic lymphadenopathy as well as distant metastases in selected high-risk patients with endometrial cancer.

Solution to Question 14:

Pelvic and para-aortic lymph nodes are commonly involved in endometrial cancer.

The spread is through the infundibulopelvic ligament.

Rarely, inguinal & femoral lymph nodes are involved.

Solution to Question 15:

Carcinoma endometrium involving the vagina but having no lymph node involvement is classified as stage IIIB.

Solution to Question 16:

Early endometrial cancer (stages I & II) is primarily managed surgically. The surgery performed here is abdominal hysterectomy, B/L salpingo-oophorectomy, omentectomy and pelvic & para-aortic lymph node sampling.

Postoperative management of endometrial cancer:

In low-risk endometrial cancer, no further treatment is required.

In intermediate-risk, radiotherapy is done.

In high risk, chemotherapy + radiotherapy is done.

Solution to Question 17:

The intensity of radiation is inversely proportional to the squared distance. This is known as the inverse square law in radiotherapy.

Chemoradiation is the mainstay of treatment of stage III and IV endometrial cancers. They are treated with brachytherapy followed by external radiation. Brachytherapy is a form of radiation therapy in which the radiation source is placed close to the tumor.

Other options:

Option B: Larger blood vessels are more radioresistant than microvasculature.

Option C: Permanent non-dividing cells are more radioresistant than rapidly proliferating cells.

Option D: Small intestinal epithelial cells (intestinal crypts), which divide rapidly with no differentiation are the most radiosensitive.

Solution to Question 18:

Chemotherapy is the mainstay of treatment in Stage III endometrial cancer.

In order to stage carcinoma endometrium, total abdominal hysterectomy with bilateral salpingo-oophorectomy and lymphadenectomy is done (surgical staging).

Thus the management, once staging is done, is mainly postoperative. Up to stage II, radiotherapy can be given. Stages III & IV are managed by chemotherapy such as doxorubicin, cisplatin, and paclitaxel.

Note: Stage III is said to be inoperable but debulking surgery is sometimes attempted. However, the mainstay of treatment for stage III disease remains chemotherapy.

Instruments

Question 1:

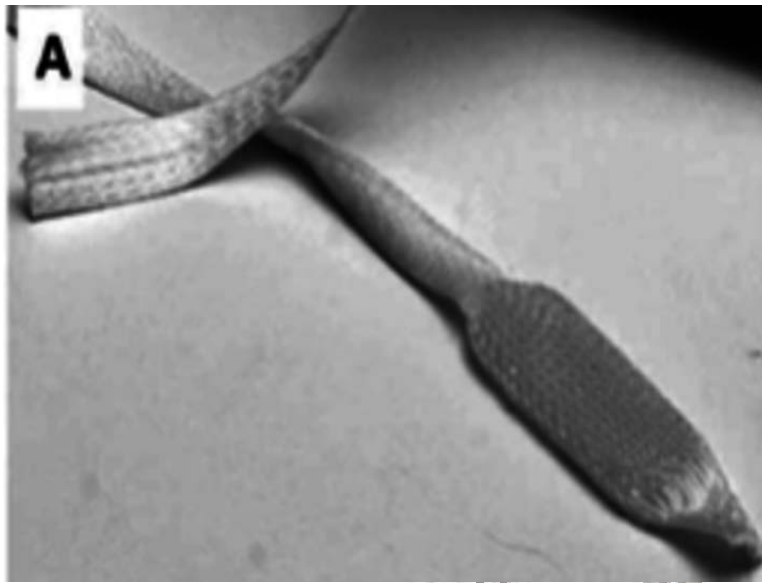
You notice the following instrument being used to monitor a primigravida in the first stage of labor. What is the investigation being performed here?



- a) Fetal doppler
- b) Nonstress test
- c) Cardiotocography
- d) Biophysical profiling

Question 2:

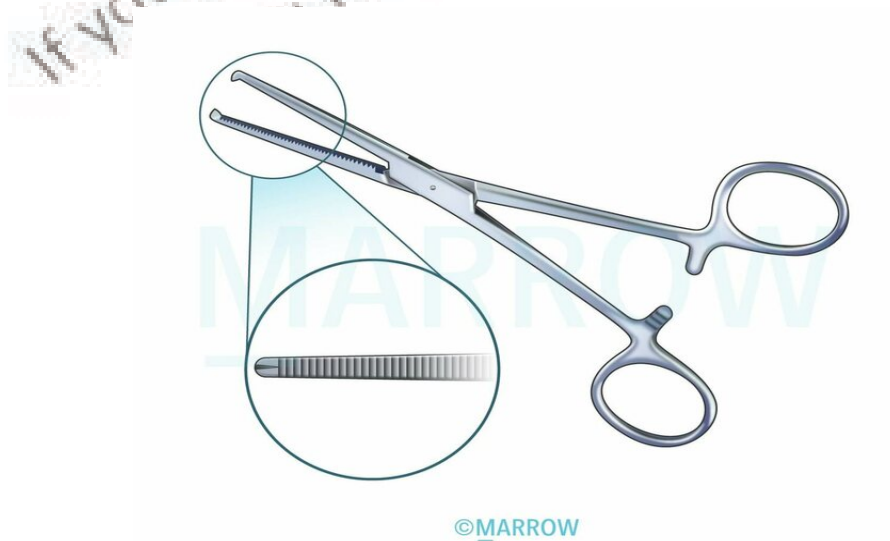
During your OBG posting, one of the senior residents asks you to induce labor in a primigravida admitted at 40 weeks of gestation. Identify the following device that you can use for this purpose.



- a) Mirena
- b) Fenton's dilator
- c) Cervidil vaginal insert
- d) Lippes loop

Question 3:

The following instrument is used to augment labour in a 28-year-old primigravida via low rupture of membranes. Identify this instrument.



- a) Crile artery forceps

- b) Kelly artery forceps
- c) Kocher's artery forceps
- d) Rochester-Pean artery forceps

Question 4:

A primigravida was diagnosed with a prolonged second stage of labor due to the fetus being in occipitoposterior position. The obstetrician decided to give a posterior mediolateral cut at the introitus to facilitate delivery. Which of the following instruments would she use for the same?



- a) A
- b) D
- c) C
- d) B

Question 5:

The following instrument is used on an exhausted primigravida during crowning of the fetal head. Which of the following is false about this instrument?



- a) It can be used in caesarean sections.
- b) It can be used to deliver aftercoming head in breech.
- c) It is a type of outlet forceps.
- d) It provides traction force.

Question 6:

While assisting in the caesarean section of a multigravida, you see the senior obstetrician using the following instrument. Identify it.

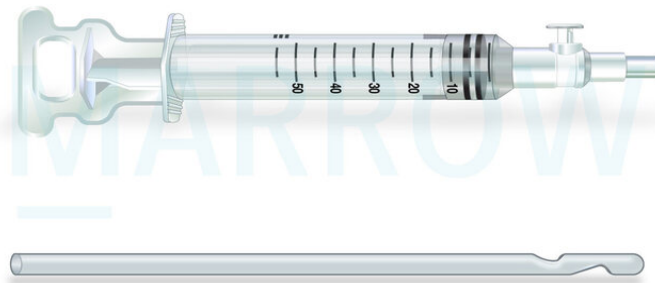


- a) Green-Armytage forceps

- b) Kocher's clamp
- c) Lahey's tissue holding forceps
- d) Allis tissue holding forceps

Question 7:

An obstetrician performs an aspiration of the endometrial cavity in a woman with complaints of a missed period 2 weeks back. Identify the following instrument. What is the procedure it is used for?



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- a) Endometrial ablation
- b) Suction and evacuation
- c) Dilatation and curettage
- d) Menstrual regulation

Question 8:

A 20-year-old woman at 9 weeks of gestation comes to the clinic for check-up following an MTP. On ultrasound, retained tissue was found and the following instrument was used for its removal. What is this instrument?



- a) Karmann syringe
- b) Suction apparatus
- c) Manual vacuum aspiration syringe
- d) Menstrual regulation syringe

Question 9:

Identify the following instrument that is commonly used in a dilatation and curettage procedure.

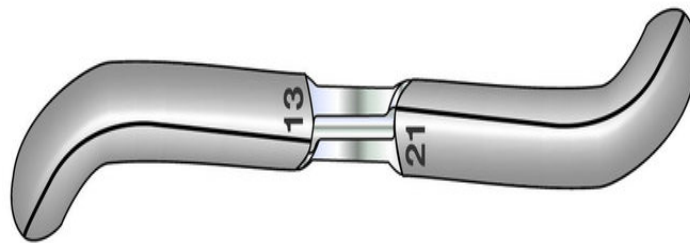


- a) Hawkin dilator

- b) Hegar dilator
- c) Bakes dilator
- d) Fenton dilator

Question 10:

A 25-year-old woman presents with dyspareunia due to scarring secondary to perineal tears from previous vaginal delivery. The gynaecologist advises the use of the following dilator to widen the introitus. Identify this instrument.



- a) Hegar dilator
- b) Hawkin dilator
- c) Pratt dilator
- d) Fenton dilator

Question 11:

You have posted a 43-year-old woman with abnormal uterine bleeding for dilatation and curettage in the minor OT. Identify the following instrument used in this procedure.



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- a) Blake's uterine curette
- b) Sim's curette
- c) Simpson's uterine sound
- d) Sim's anterior vaginal wall retractor

Question 12:

The following instrument is used for all except:



- a) Removal of embedded IUD
- b) Check curettage after MTP

- c) Uterine length measurement
- d) Blunt curettage in abortions

Question 13:

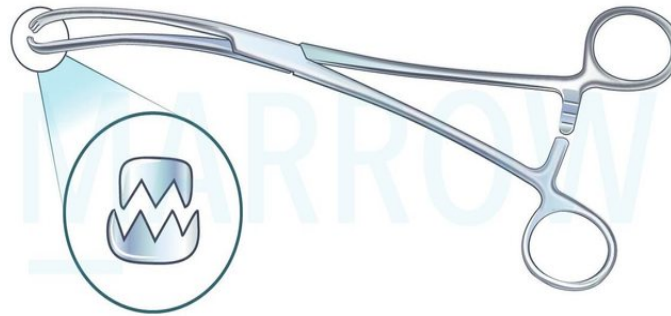
Which of the following is a false statement about the instrument given below?



- a) It measures the uterine cavity and the cervical length
- b) It differentiates between chronic inversion and fibroid polyp
- c) It retracts the posterior vaginal wall to expose the cervix
- d) It is used to diagnose cervical stenosis

Question 14:

You see a resident use the following instrument to grasp the anterior lip of the cervix in a patient posted for a vaginal hysterectomy. What is this instrument known as?



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- a) Allis forceps
- b) Vulsellum forceps
- c) Kocher's artery forceps
- d) Lahey tissue grasping forceps

Question 15:

Identify the following instrument.



- a) Sponge holding forceps
- b) Towel holding clip

- c) Filshie clip
- d) Small curved artery forceps

Question 16:

You are assisting a case of vaginal hysterectomy where your assistant professor asks you to hold the retractor shown in the image. What is this retractor known as?



- a) Doyen's retractor
- b) Simon retractor
- c) Balfour retractor
- d) Czerny retractor

Question 17:

Identify the instrument.



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- a) Sims speculum
- b) Graves vaginal speculum
- c) Cusco vaginal speculum
- d) Auvard speculum

Question 18:

Your senior resident uses the following instrument while performing a pap smear test for screening of cervical cancer in a 36-year-old woman. Identify the same.



- a) Cervex brush

- b) Endocervical brush
- c) Ayres spatula
- d) Aylesbury spatula

Question 19:

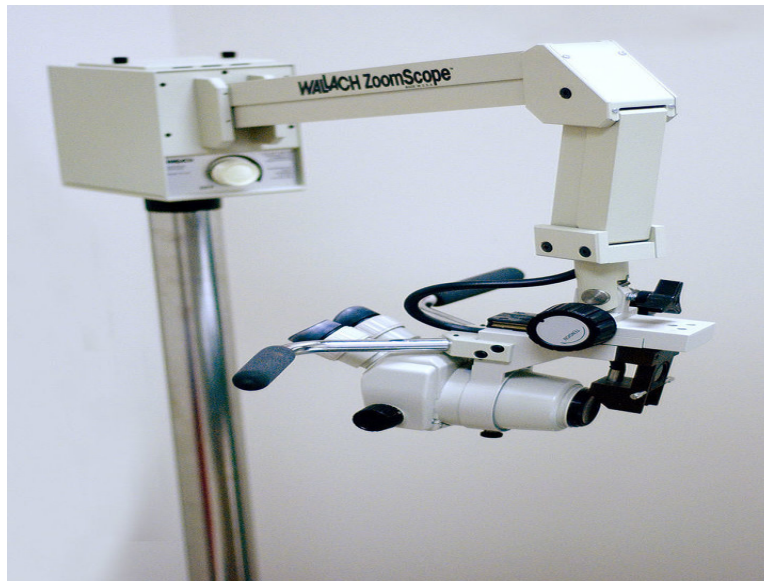
A 32-year-old working IT professional came for a mobile cervical cancer screening camp planned at her workplace. What is the following instrument used to perform this screening test?



- a) Aylesbury spatula
- b) Cervex brush
- c) Ayre's spatula
- d) Vaginal spatula

Question 20:

A 35-year-old lady who had tested positive for HPV is undergoing an investigation where the following device is used. Which of the following statements is false about this investigation?



- a) This device provides binocular vision
- b) It is used in lithotomy position
- c) Green filter is used to study vascular pattern
- d) External os is visualised at a distance of 50 cm

Question 21:

The following instrument is used for a young lady with fibroids posted for open myomectomy. Identify it.

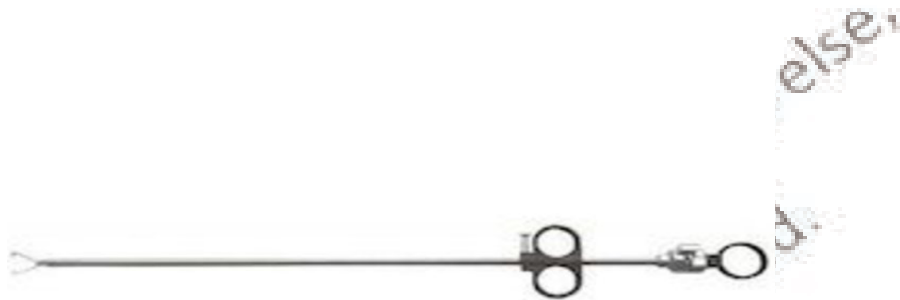


- a) Manual trocar

- b) Laparoscopic myoma screw
- c) Doyen myoma screw
- d) Auto insertion trocar

Question 22:

Identify the use of the instrument given below:



- a) For tubal ligation
- b) For ligation of uterine artery during hysterectomy
- c) Ovarian cystectomy
- d) Ectopic pregnancy

Answer Key

Question No.	Correct Option
1	c
2	c
3	c
4	b
5	b
6	a

7	d
8	c
9	b
10	d
11	b
12	c
13	c
14	b
15	b
16	b
17	d
18	b
19	c
20	d
21	c
22	a

Detailed Explanations

Solution to Question 1:

The image shows a cardiotocograph (electronic fetal monitor) that is used to perform cardiotocography in labor.

It can be used in both antenatal and intranatal mothers. It measures fetal heart rate and uterine contractions. It is used to perform the non-stress test (NST) in the antenatal period and cardiotocography during labor.

Solution to Question 2:

This is an image of cervidil vaginal insert. It is used for cervical ripening and improves the Bishop's score.

It contains 10 mg of dinoprostone designed to release at a rate of approximately 0.3 mg/hr over 10-hours.

Solution to Question 3:

The given instrument is Kocher's artery forceps and is identified by the presence of a single tooth at the tip.

Kocher's artery forceps are used to achieve hemostasis by holding bleeding vessels. They are also used in low rupture of membranes to augment labour or for the surgical induction of labour.

The difference between artery forceps and needle holders is in the edge and smoothness of corrugations. Artery forceps have smooth edges and single lined corrugations whereas, needle holders have rough edges and crossed lines forming square to diamond shaped corrugations.

Solution to Question 4:

The given scenario of prolonged second stage of labor can be managed by performing an episiotomy. In the given figure, the instrument marked as "D" corresponds to episiotomy scissors.

An episiotomy is done to prevent posterior extension leading to obstetric anal sphincteric injuries.



Solution to Question 5:

The given image shows Wrigley's forceps.

These are used as outlet forceps for the extraction of the fetal head. It is used to provide traction force. It can also be used in C-section for delivery of the baby.

The instrument is light and short due to a reduction in the length of the shanks and handles. It has a marked cephalic curve with a slight pelvic curve.

Conditions that require to be fulfilled for the use of outlet forceps are:

- The fetal scalp is visible without separating the labia (crowning)
- The fetal skull has reached the pelvic floor, or is on the perineum

- The sagittal suture is in the anteroposterior diameter or right or left occiput anterior or posterior position (rotation does not exceed 45°)

Note: Delivery of aftercoming head in breech is done with Pipers forceps.

Solution to Question 6:

The given instrument is Green-Armytage forceps.

Its tips are broad and hence can compress a wide area. During lower segment caesarean section (LSCS), these forceps are applied at the bleeding cut uterine edges for hemostasis.

Solution to Question 7:

The instrument is Karman's cannula and syringe, which is also used for menstrual regulation.

Menstrual regulation is the aspiration of the endometrial cavity within 14 days of a missed period in a woman with a previous normal cycle.

Under aseptic precautions, the cervix may be gently dilated using 4-5 mm size dilators. Then 5-6mm Karman's cannula is inserted and attached to a 50ml syringe for suction. The cannula is rotated, and pushed in and out with gentle strokes.

This procedure is contraindicated in advanced pregnancy and in the presence of local pelvic inflammation. The drawbacks of the procedure include a missed implanted zygote or an unrecognized ectopic pregnancy.

This procedure is not commonly done anymore.

Solution to Question 8:

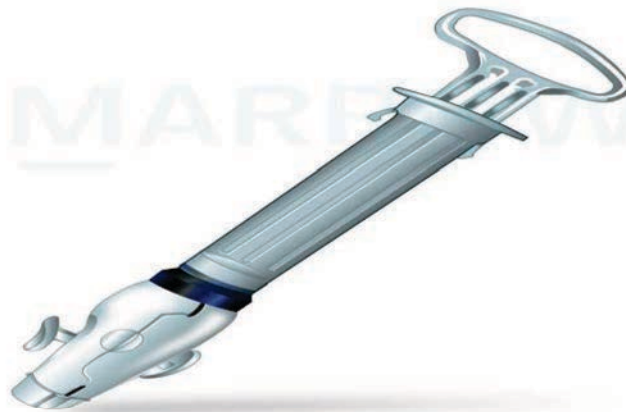
The image shows a manual vacuum aspiration (MVA) syringe.

Manual vacuum aspiration can be safely done upto 12 weeks of gestation. However, WHO recommends it as the procedure of choice for MTP only upto 10 weeks of gestation .

The indications of this procedure are given as follows:

- Medical termination of pregnancy
- Incomplete abortion or missed abortion
- Molar pregnancy
- Endometrial aspiration

Manual Vacuum Aspiration (MVA) syringe



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Solution to Question 9:

The given image shows Hegar's double-ended dilator which is used to dilate the cervix in dilation and curettage procedure.

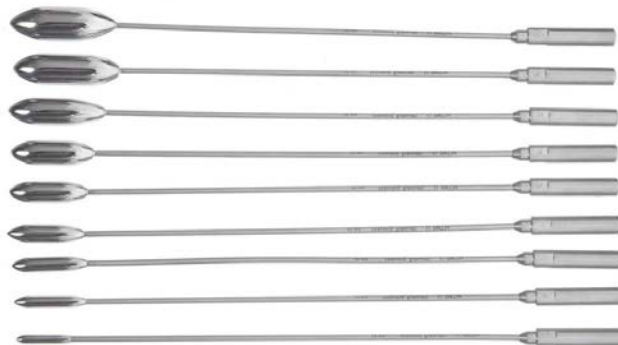
Pratt dilator is also a double-ended dilator but has a more linear shape as compared to the curved Hegar's dilators. The image given below shows a Pratt dilator.



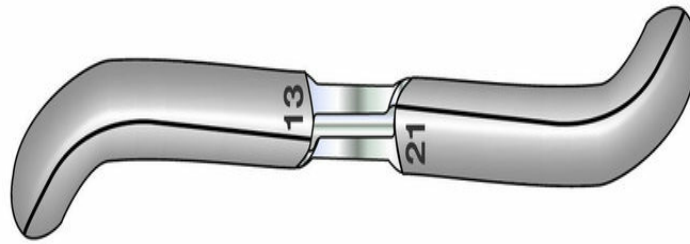
Option A: Hawkin dilator is a single-ended dilator. The image given below shows a Hawkin dilator.



Option C: Bakes rosebud dilator is a single-ended dilator used to dilate the urethra. It is called so since its tip is similar to a rosebud. The image given below shows a Bakes dilator.



Option D: Fenton dilator is also double-ended but it has a notch in the middle. The image given below shows a Fenton dilator.



Solution to Question 10:

The image shows a Fenton dilator. These can be identified by the neck/notch at the centre.

Fenton's or modified Fenton's procedure is performed to widen the introitus in women with superficial dyspareunia secondary to scarring. The narrowing of introitus could be due to scarring secondary to perineal tears during childbirth or from conditions such as lichen sclerosus or vestibulitis.

Fenton dilators come in sizes numbered from 3-10 and the sizes of the dilators range from 3mm to 26mm.

Option B: Hawkins dilator is single-ended.

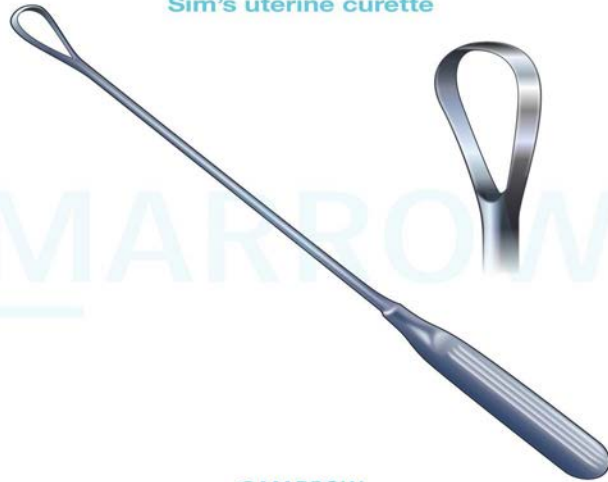
Options A and C: Hegar and Pratt dilators are also double-ended dilators but they do not have a notch at the centre. Pratt dilator is more linear in appearance compared to Hegar dilator.

Solution to Question 11:

The instrument shown in the image is Sim's curette. It is a single-ended uterine curette used for dilatation and curettage.

It is designed to remove tough tissue from the lining of the uterus for biopsy or excision. The curette features a sharp loop tip to aid in the removal of the uterine tissue.

Sim's uterine curette



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Solution to Question 12:

The image shows a double-ended uterine curette with a sharp and a blunt end. It is not used in uterine length measurement.

Uterine length measurement is done with a uterine sound which is given below.

Simpson's uterine sound



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Solution to Question 13:

The given instrument is Simpson's uterine sound. It is not used to retract the vagina for cervical examination. This is done using Sim's speculum.

The instrument is held like a pencil with the thumb and the first 2 fingers and introduced into the uterine cavity by keeping it in the midline and ensuring it is not forced.

The uses are given as follows:

- To measure uterine cavity and cervical length
- To diagnose cervical stenosis
- To sound a polyp, IUCD or uterine septum
- To break adhesions
- To differentiate between chronic inversion and fibroid polyp

Solution to Question 14:

The image shows vulsellum forceps. It is used to grasp the cervical lip and steady the cervix during vaginal surgery. It is a long instrument with a gentle curve so that the line of vision is not obstructed.

Option A: Allis forceps can be identified by continuous teeth, equal on both ends. It is used to hold structures with a firm grip.



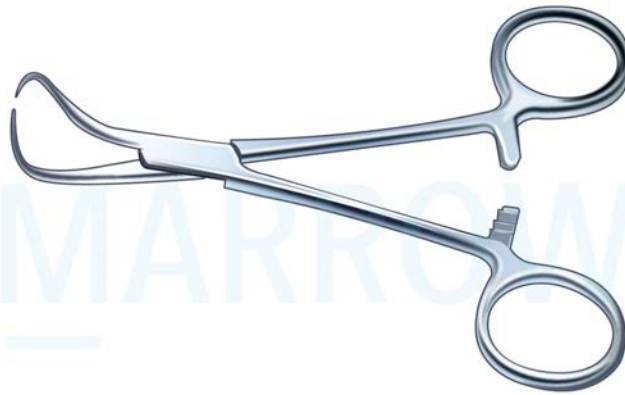
Option C: Kocher's artery forceps are non-traumatic with a single tooth at the tip. They help in achieving hemostasis by firm pressure.

Option D: Lahey's tissue grasping forceps has teeth similar to dinner forks. They are used to hold fibrous tissue and they give a firm grip without slippage.

Solution to Question 15:

The given instrument is a towel-holding clip. It is used to secure drapings and towels during surgery.

Towel holding clip

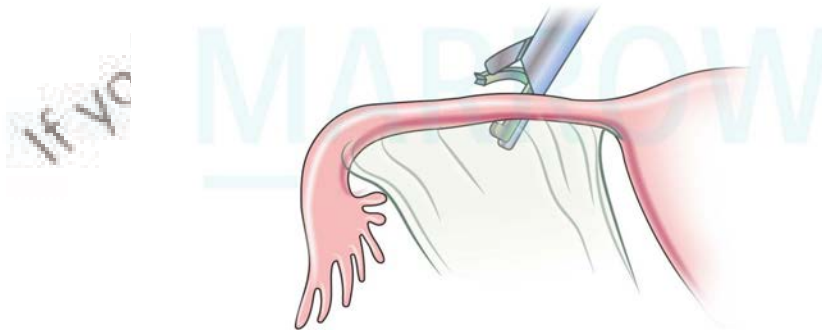


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Sponge-holding forceps are non-traumatic forceps. They are used for painting and preparation of parts preoperatively.

Filshie clips are used for laparoscopic tubal sterilisation. They are made of titanium on the outer side and sialistic material on the inner side.

Filshie clip



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Solution to Question 16:

The image shown is that of a Simon vaginal retractor, used to retract vaginal walls during examination and vaginal procedures.

The below image is of Doyen's retractor. It is used during abdominal procedures like C-section.

Doyen's retractor



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Solution to Question 17:

The instrument shown in the image is Auvard speculum. It retracts the posterior vaginal wall during vaginal hysterectomy and is self retractor due to the circular structure on its handle which is heavy.

The image given below shows Sim's speculum.



The image given below shows Cusco's self retaining bivalved speculum.



Solution to Question 18:

The image shows an endocervical brush. It is used to collect samples from the endocervix cervix, especially from deep lesions.

Option A: Cervex brush is ideal for collecting samples for liquid-based cytological studies.

Option C: Ayres spatula is used to collect samples from the ectocervix and the endocervix for cytological studies.

Option D: Aylesbury spatula has a sharper and longer tip than Ayers spatula.

Solution to Question 19:

The given instrument used in the patient's pap smear preparation is Ayre's spatula.

Sharpness on the end can differentiate between Ayre's and Aylesbury spatula. Aylesbury spatula has a more sharp and longer tip than Ayer's spatula. So, Aylesbury spatula is superior to Ayre's in collecting endocervical samples.

The image given below shows a slide holding bottle i.e. Koplirks jar with Ayre's spatula.



Solution to Question 20:

The device shown in the image is a colposcope. It is used to perform colposcopy where the cervix is visualized at a distance of around 20 cm with the patient in the lithotomy position.

This instrument provides binocular vision with a magnification of 6x-15x. A green filter is used to study vascular pattern as the vessels appear black. Abnormal areas can be identified by their peculiar vascular pattern such as mosaic, reticular, and punctate.

Solution to Question 21:

The given image shows a Doyen myoma screw.

It is used to perform an open myomectomy. The handle present in this can be used to distinguish it from a laparoscopic myoma screw which also has a similar tip.

Solution to Question 22:

The given instrument is a fallope ring applicator. It is used in tubal ligation.

Falope ring sterilization is accomplished by applying silicone rubber bands to the fallopian tubes. The major advantage is the elimination of thermocoagulation. The occlusion is only in a short segment and can allow for tubal reconstruction.